

johnston-linear-matrix-algebra: formula evidence

The page, the confidence and the picture of an inline formula are its HOST LINE’s — a formula has none of its own. A line’s confidence is not a formula’s.

Inline formulas (first occurrence)

Identifier	Page	Conf.	LaTeX source	Rendered	Image
johnston-linear-matrix-algebra_F00001	7	1.000	$\cos (\pi / 6)=\sqrt{3} / 2$	$\cos (\pi / 6)=\sqrt{3} / 2$	For example, if we make use of the fact that $\cos (\pi / 6)=\sqrt{3} / 2$ in the middle of a long calculation, we
johnston-linear-matrix-algebra_F00002	8	1.000	$\mathbb{R}^n$	$\mathbb{R}^n$	this book considers the concrete vector spaces $\mathbb{R}^n$ and $\mathbb{C}^n$ exclusively (and briefly $\mathbb{F}^n$, where $\mathbb{F}$ is a
johnston-linear-matrix-algebra_F00003	9	1.000	$\mathbb{C}^n$	$\mathbb{C}^n$	this book considers the concrete vector spaces $\mathbb{R}^n$ and $\mathbb{C}^n$ exclusively (and briefly $\mathbb{F}^n$, where $\mathbb{F}$ is a
johnston-linear-matrix-algebra_F00004	9	1.000	$\mathbb{F}^n$	$\mathbb{F}^n$	this book considers the concrete vector spaces $\mathbb{R}^n$ and $\mathbb{C}^n$ exclusively (and briefly $\mathbb{F}^n$, where $\mathbb{F}$ is a
johnston-linear-matrix-algebra_F00005	9	1.000	$\mathbb{F}$	$\mathbb{F}$	this book considers the concrete vector spaces $\mathbb{R}^n$ and $\mathbb{C}^n$ exclusively (and briefly $\mathbb{F}^n$, where $\mathbb{F}$ is a
johnston-linear-matrix-algebra_F00006	16	1.000	$a \in S$	$a \in S$	The notation $a \in S$
johnston-linear-matrix-algebra_F00007	16	1.000	a	a	object a is in the
johnston-linear-matrix-algebra_F00008	16	1.000	S	S	set S , so $\mathbf{v} \in \mathbb{R}^n$
johnston-linear-matrix-algebra_F00009	16	1.000	$\mathbf{v} \in \mathbb{R}^n$	$\mathbf{v} \in \mathbb{R}^n$	set S , so $\mathbf{v} \in \mathbb{R}^n$
johnston-linear-matrix-algebra_F00010	16	1.000	$\mathbf{v}$	$\mathbf{v}$	vector $\mathbf{v}$ is in the set
johnston-linear-matrix-algebra_F00011	16	1.000	n	n	$\mathbb{R}^n$ of n -dimensional
johnston-linear-matrix-algebra_F00012	16	1.000	$4 x-3=7$	$4 x-3=7$	single variable. For example, we learned how to solve equations like $4 x-3=7$
johnston-linear-matrix-algebra_F00013	16	1.000	$f(x)=3 x+8$	$f(x)=3 x+8$	and we learned about properties of functions like $f(x)=3 x+8$, where in each
johnston-linear-matrix-algebra_F00014	16	1.000	x	x	case the one variable was called “ x ”. One way of looking at linear algebra
johnston-linear-matrix-algebra_F00015	16	1.000	$3 x+2 y=1$	$3 x+2 y=1$	more variables. For example, we might try solving an equation like $3 x+2 y=1$,
johnston-linear-matrix-algebra_F00016	16	0.679	$\mathbf{w}$	$\mathbf{w}$	themselves by lowercase bold letters like $\mathbf{v}$ and $\mathbf{w}$, and we write their entries
johnston-linear-matrix-algebra_F00017	16	0.644	$\mathbf{v}=(2,3) \in \mathbb{R}^2$	$\mathbf{v}=(2,3) \in \mathbb{R}^2$	within parentheses. For example, $\mathbf{v}=(2,3) \in \mathbb{R}^2$ is a 2-dimensional vector and
johnston-linear-matrix-algebra_F00018	16	0.611	$\mathbf{w}=(1,3,2) \in \mathbb{R}^3$	$\mathbf{w}=(1,3,2) \in \mathbb{R}^3$	$\mathbf{w}=(1,3,2) \in \mathbb{R}^3$ is a 3-dimensional vector (just like $4 \in \mathbb{R}$ is a real number).
johnston-linear-matrix-algebra_F00019	16	0.611	$4 \in \mathbb{R}$	$4 \in \mathbb{R}$	$\mathbf{w}=(1,3,2) \in \mathbb{R}^3$ is a 3-dimensional vector (just like $4 \in \mathbb{R}$ is a real number).
johnston-linear-matrix-algebra_F00020	16	0.968	y	y	ond entry represents displacement in the y -direction, and in the 3-dimensional
johnston-linear-matrix-algebra_F00021	16	1.000	z	z	case its third entry represents displacement in the z -direction, as in Figure 1.1.
johnston-linear-matrix-algebra_F00022	17	1.000	$\vec{v}$	$\vec{v}$	with arrows like $\vec{v}$, or
johnston-linear-matrix-algebra_F00023	17	1.000	$\overrightarrow{A B}$	$\overrightarrow{A B}$	$\overrightarrow{A B}$ if they wish to
johnston-linear-matrix-algebra_F00024	17	1.000	A	A	A and its head is

Identifier	Page	Conf.	LaTeX source	Rendered	Image
johnston-linear-matrix-algebra-F00025	17	■ 1.000	B	B	located at point B .
johnston-linear-matrix-algebra-F00026	17	■ 0.968	$(-1, 1)$	$(-1, 1)$	$(-1, 1)$ to the point $(2, 2)$ is $(2, 2) - (-1, 1) = (3, 1)$. However, this is also the
johnston-linear-matrix-algebra-F00027	17	■ 0.968	$(2, 2)$	$(2, 2)$	$(-1, 1)$ to the point $(2, 2)$ is $(2, 2) - (-1, 1) = (3, 1)$. However, this is also the
johnston-linear-matrix-algebra-F00028	17	■ 0.968	$(2, 2) - (-1, 1) = (3, 1)$	$(2, 2) - (-1, 1) = (3, 1)$	$(-1, 1)$ to the point $(2, 2)$ is $(2, 2) - (-1, 1) = (3, 1)$. However, this is also the
johnston-linear-matrix-algebra-F00029	17	■ 0.395	$(4, 1) - (1, 0) = (3, 1)$	$(4, 1) - (1, 0) = (3, 1)$	same as the vector that points from $(1, 0)$ to $(4, 1)$, since $(4, 1) - (1, 0) = (3, 1)$
johnston-linear-matrix-algebra-F00030	17	■ 1.000	$\mathbf{v} = (3, 1)$	$\mathbf{v} = (3, 1)$	Figure 1.2: Three copies of the vector $\mathbf{v} = (3, 1)$ located at different positions in
johnston-linear-matrix-algebra-F00031	17	■ 1.000	$\mathbb{R}^7$	$\mathbb{R}^7$	and theorems themselves should work just as well in $\mathbb{R}^7$ (even though we can-
johnston-linear-matrix-algebra-F00032	17	■ 1.000	$\mathbb{R}^3$	$\mathbb{R}^3$	not really visualize this space) as they do in $\mathbb{R}^3$. For this reason, we typically
johnston-linear-matrix-algebra-F00033	18	■ 1.000	$\mathbf{v} + \mathbf{w}$	$\mathbf{v} + \mathbf{w}$	vector $\mathbf{v} + \mathbf{w}$ is the
johnston-linear-matrix-algebra-F00034	18	■ 1.000	$\mathbf{v} = (v_1, v_2, \dots, v_n) \in \mathbb{R}^n$	$\mathbf{v} = (v_1, v_2, \dots, v_n) \in \mathbb{R}^n$	Suppose $\mathbf{v} = (v_1, v_2, \dots, v_n) \in \mathbb{R}^n$ and $\mathbf{w} = (w_1, w_2, \dots, w_n) \in \mathbb{R}^n$ are vec-
johnston-linear-matrix-algebra-F00035	18	■ 1.000	$\mathbf{w} = (w_1, w_2, \dots, w_n) \in \mathbb{R}^n$	$\mathbf{w} = (w_1, w_2, \dots, w_n) \in \mathbb{R}^n$	Suppose $\mathbf{v} = (v_1, v_2, \dots, v_n) \in \mathbb{R}^n$ and $\mathbf{w} = (w_1, w_2, \dots, w_n) \in \mathbb{R}^n$ are vec-
johnston-linear-matrix-algebra-F00036	18	■ 1.000	$\mathbf{v} +$	$\mathbf{v} +$	If we instead work entirely with vectors in standard position, then $\mathbf{v} +$
johnston-linear-matrix-algebra-F00037	18	■ 1.000	$\mathbf{v}, \mathbf{w}, \mathbf{x} \in \mathbb{R}^n$	$\mathbf{v}, \mathbf{w}, \mathbf{x} \in \mathbb{R}^n$	Suppose $\mathbf{v}, \mathbf{w}, \mathbf{x} \in \mathbb{R}^n$ are vectors. Then the following properties hold:
johnston-linear-matrix-algebra-F00038	18	■ 1.000	$\mathbf{v} + \mathbf{w} = \mathbf{w} + \mathbf{v}$	$\mathbf{v} + \mathbf{w} = \mathbf{w} + \mathbf{v}$	a) $\mathbf{v} + \mathbf{w} = \mathbf{w} + \mathbf{v}$, and
johnston-linear-matrix-algebra-F00039	18	■ 1.000	$(\mathbf{v} + \mathbf{w}) + \mathbf{x} = \mathbf{v} + (\mathbf{w} + \mathbf{x})$	$(\mathbf{v} + \mathbf{w}) + \mathbf{x} = \mathbf{v} + (\mathbf{w} + \mathbf{x})$	b) $(\mathbf{v} + \mathbf{w}) + \mathbf{x} = \mathbf{v} + (\mathbf{w} + \mathbf{x})$.
johnston-linear-matrix-algebra-F00040	18	■ 1.000	$x + y = y + x$	$x + y = y + x$	$x + y = y + x$ for all $x, y \in \mathbb{R}$:
johnston-linear-matrix-algebra-F00041	18	■ 1.000	$x, y \in \mathbb{R}$	$x, y \in \mathbb{R}$	$x + y = y + x$ for all $x, y \in \mathbb{R}$:
johnston-linear-matrix-algebra-F00042	19	—	$\square$	$\square$	—
johnston-linear-matrix-algebra-F00043	19	■ 1.000	$\mathbf{v} + \mathbf{w} + \mathbf{x}$	$\mathbf{v} + \mathbf{w} + \mathbf{x}$	this theorem shows that expressions like $\mathbf{v} + \mathbf{w} + \mathbf{x}$ make sense, since there is
johnston-linear-matrix-algebra-F00044	19	■ 1.000	$(\mathbf{v} + \mathbf{w}) + \mathbf{x}$	$(\mathbf{v} + \mathbf{w}) + \mathbf{x}$	no need to question whether it means $(\mathbf{v} + \mathbf{w}) + \mathbf{x}$ or $\mathbf{v} + (\mathbf{w} + \mathbf{x})$.
johnston-linear-matrix-algebra-F00045	19	■ 1.000	$\mathbf{v} + (\mathbf{w} + \mathbf{x})$	$\mathbf{v} + (\mathbf{w} + \mathbf{x})$	no need to question whether it means $(\mathbf{v} + \mathbf{w}) + \mathbf{x}$ or $\mathbf{v} + (\mathbf{w} + \mathbf{x})$.
johnston-linear-matrix-algebra-F00046	19	■ 1.000	$\mathbf{v} \times \mathbf{w} \neq \mathbf{w} \times \mathbf{v}$	$\mathbf{v} \times \mathbf{w} \neq \mathbf{w} \times \mathbf{v}$	commutative (i.e., the order of “multiplication” matters since $\mathbf{v} \times \mathbf{w} \neq \mathbf{w} \times \mathbf{v}$),
johnston-linear-matrix-algebra-F00047	19	■ 1.000	$(2, 5, -1) + (1, -1, 2)$	$(2, 5, -1) + (1, -1, 2)$	a) $(2, 5, -1) + (1, -1, 2)$,
johnston-linear-matrix-algebra-F00048	19	■ 1.000	$(1, 2) + (3, 1) + (2, -1)$	$(1, 2) + (3, 1) + (2, -1)$	b) $(1, 2) + (3, 1) + (2, -1)$, and
johnston-linear-matrix-algebra-F00049	19	■ 0.672	$0, 0, 0$	$0, 0, 0$	a cube with opposite corners at $(0, 0, 0)$ and $(1, 1, 1)$, as shown:
johnston-linear-matrix-algebra-F00050	19	■ 0.672	$1, 1, 1$	$1, 1, 1$	a cube with opposite corners at $(0, 0, 0)$ and $(1, 1, 1)$, as shown:
johnston-linear-matrix-algebra-F00051	19	■ 1.000	$(2, 5, -1) + (1, -1, 2) = (2+1, 5-1, -1+2) = (3, 4, 1)$	$(2, 5, -1) + (1, -1, 2) = (2+1, 5-1, -1+2) = (3, 4, 1)$	a) $(2, 5, -1) + (1, -1, 2) = (2+1, 5-1, -1+2) = (3, 4, 1)$.

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johnston-linear-matrix-algebra_F00052	19	1.000	$(1,2)+(3,1)+(2,-1)=(1+3+2,2+1-1)=(6,2)$	$(1,2) + (3,1) + (2,-1) = (1 + 3 + 2, 2 + 1 - 1) = (6,2)$	b) $(1,2) + (3,1) + (2,-1) = (1 + 3 + 2, 2 + 1 - 1) = (6,2)$. Note that
johnston-linear-matrix-algebra_F00053	20	1.000	$c \in \mathbb{R}$	$c \in \mathbb{R}$	Suppose $\mathbf{v} = (v_1, v_2, \dots, v_n) \in \mathbb{R}^n$ is a vector and $c \in \mathbb{R}$ is a scalar. Then
johnston-linear-matrix-algebra_F00054	20	1.000	$c \mathbf{v}$	$c\mathbf{v}$	their scalar multiplication , denoted by $c\mathbf{v}$, is the vector
johnston-linear-matrix-algebra_F00055	20	0.531	c	c	of $\mathbf{v}$ by c seems like a rather natural operation, and it has the simple geometric
johnston-linear-matrix-algebra_F00056	20	0.967	$ c > 1$	$ c > 1$	if $ c > 1$ then scalar multiplication stretches $\mathbf{v}$, but if $ c < 1$ then it shrinks $\mathbf{v}$.
johnston-linear-matrix-algebra_F00057	20	0.967	$ c < 1$	$ c < 1$	if $ c > 1$ then scalar multiplication stretches $\mathbf{v}$, but if $ c < 1$ then it shrinks $\mathbf{v}$.
johnston-linear-matrix-algebra_F00058	20	0.967	$c < 0$	$c < 0$	When $c < 0$ then this operation also reverses the direction of $\mathbf{v}$, in addition to
johnston-linear-matrix-algebra_F00059	20	1.000	$ c \neq 1$	$ c \neq 1$	any stretching or shrinking that it does if $ c \neq 1$.
johnston-linear-matrix-algebra_F00060	20	0.772	$c = 0$	$c = 0$	• If $c = 0$ then $c\mathbf{v}$ is the zero vector , all of whose entries are 0, which we
johnston-linear-matrix-algebra_F00061	20	1.000	$\mathbf{0}$	$\mathbf{0}$	denote by $\mathbf{0}$.
johnston-linear-matrix-algebra_F00062	20	0.957	$c = -1$	$c = -1$	• If $c = -1$ then $c\mathbf{v}$ is the vector whose entries are the negatives of $\mathbf{v}$'s
johnston-linear-matrix-algebra_F00063	20	0.637	$\mathrm{via} \mathbf{v} - \mathbf{w} \stackrel{\text{def}}{=} \mathbf{v} + (-\mathbf{w})$	via $\mathbf{v} - \mathbf{w} \stackrel{\text{def}}{=} \mathbf{v} + (-\mathbf{w})$	We also define vector subtraction via $\mathbf{v} - \mathbf{w} \stackrel{\text{def}}{=} \mathbf{v} + (-\mathbf{w})$, and we note that
johnston-linear-matrix-algebra_F00064	20	1.000	$\mathbf{v} - \mathbf{w}$	$\mathbf{v} - \mathbf{w}$	it has the geometric interpretation that $\mathbf{v} - \mathbf{w}$ is the vector pointing from the
johnston-linear-matrix-algebra_F00065	21	1.000	$\mathbf{v} - \mathbf{v} = \mathbf{0}$	$\mathbf{v} - \mathbf{v} = \mathbf{0}$	such as the facts that $\mathbf{v} - \mathbf{v} = \mathbf{0}$ and $\mathbf{v} + \mathbf{0} = \mathbf{v}$ for every vector $\mathbf{v} \in \mathbb{R}^n$, by
johnston-linear-matrix-algebra_F00066	21	1.000	$\mathbf{v} + \mathbf{0} = \mathbf{v}$	$\mathbf{v} + \mathbf{0} = \mathbf{v}$	such as the facts that $\mathbf{v} - \mathbf{v} = \mathbf{0}$ and $\mathbf{v} + \mathbf{0} = \mathbf{v}$ for every vector $\mathbf{v} \in \mathbb{R}^n$, by
johnston-linear-matrix-algebra_F00067	21	1.000	$\mathbf{v}, \mathbf{w} \in \mathbb{R}^n$	$\mathbf{v}, \mathbf{w} \in \mathbb{R}^n$	Suppose $\mathbf{v}, \mathbf{w} \in \mathbb{R}^n$ are vectors and $c, d \in \mathbb{R}$ are scalars. Then the following
johnston-linear-matrix-algebra_F00068	21	1.000	$c, d \in \mathbb{R}$	$c, d \in \mathbb{R}$	Suppose $\mathbf{v}, \mathbf{w} \in \mathbb{R}^n$ are vectors and $c, d \in \mathbb{R}$ are scalars. Then the following
johnston-linear-matrix-algebra_F00069	21	1.000	$c(\mathbf{v} + \mathbf{w}) = c\mathbf{v} + c\mathbf{w}$	$c(\mathbf{v} + \mathbf{w}) = c\mathbf{v} + c\mathbf{w}$	a) $c(\mathbf{v} + \mathbf{w}) = c\mathbf{v} + c\mathbf{w}$,
johnston-linear-matrix-algebra_F00070	21	1.000	$(c + d)\mathbf{v} = c\mathbf{v} + d\mathbf{v}$	$(c + d)\mathbf{v} = c\mathbf{v} + d\mathbf{v}$	b) $(c + d)\mathbf{v} = c\mathbf{v} + d\mathbf{v}$, and
johnston-linear-matrix-algebra_F00071	21	1.000	$c(d\mathbf{v}) = (cd)\mathbf{v}$	$c(d\mathbf{v}) = (cd)\mathbf{v}$	c) $c(d\mathbf{v}) = (cd)\mathbf{v}$.
johnston-linear-matrix-algebra_F00072	21	1.000	$3\mathbf{v} - 2\mathbf{w}$	$3\mathbf{v} - 2\mathbf{w}$	a) $3\mathbf{v} - 2\mathbf{w}$, where $\mathbf{v} = (2, 1, -1)$ and $\mathbf{w} = (-1, 0, 3)$, and
johnston-linear-matrix-algebra_F00073	21	1.000	$\mathbf{v} = (2, 1, -1)$	$\mathbf{v} = (2, 1, -1)$	a) $3\mathbf{v} - 2\mathbf{w}$, where $\mathbf{v} = (2, 1, -1)$ and $\mathbf{w} = (-1, 0, 3)$, and
johnston-linear-matrix-algebra_F00074	21	1.000	$\mathbf{w} = (-1, 0, 3)$	$\mathbf{w} = (-1, 0, 3)$	a) $3\mathbf{v} - 2\mathbf{w}$, where $\mathbf{v} = (2, 1, -1)$ and $\mathbf{w} = (-1, 0, 3)$, and
johnston-linear-matrix-algebra_F00075	21	1.000	$(0, 0)$	$(0, 0)$	b) the sum of the 6 vectors that point from the center $(0, 0)$ of a regu-
johnston-linear-matrix-algebra_F00076	22	1.000	$\mathbf{x} = (1, 1, 1)$	$\mathbf{x} = (1, 1, 1)$	$\mathbf{x} = (1, 1, 1)$ follows
johnston-linear-matrix-algebra_F00077	22	0.968	$4\mathbf{x} = (4, 4, 4)$	$4\mathbf{x} = (4, 4, 4)$	$4\mathbf{x} = (4, 4, 4)$.
johnston-linear-matrix-algebra_F00078	22	1.000	$3\mathbf{v} - 2\mathbf{w} = (6, 3, -3) - (-2, 0, 6) = (8, 3, -9)$	$3\mathbf{v} - 2\mathbf{w} = (6, 3, -3) - (-2, 0, 6) = (8, 3, -9)$	a) $3\mathbf{v} - 2\mathbf{w} = (6, 3, -3) - (-2, 0, 6) = (8, 3, -9)$.

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johnston-linear-matrix-algebra-F00079	22	■ 0.980	$\mathbf{v}, \mathbf{w}, \mathbf{x}, -\mathbf{v}, -\mathbf{w}, -\mathbf{x}$	$\mathbf{v}, \mathbf{w}, \mathbf{x}, -\mathbf{v}, -\mathbf{w}, -\mathbf{x}$	as $\mathbf{v}, \mathbf{w}, \mathbf{x}, -\mathbf{v}, -\mathbf{w}, -\mathbf{x}$ (since the final 3 vectors point in the op-
johnston-linear-matrix-algebra-F00080	22	■ 1.000	$\mathbf{v} + \mathbf{w} + \mathbf{x} - \mathbf{v} - \mathbf{w} - \mathbf{x} = \mathbf{0}$	$\mathbf{v} + \mathbf{w} + \mathbf{x} - \mathbf{v} - \mathbf{w} - \mathbf{x} = \mathbf{0}$	$\mathbf{v} + \mathbf{w} + \mathbf{x} - \mathbf{v} - \mathbf{w} - \mathbf{x} = \mathbf{0}$.
johnston-linear-matrix-algebra-F00081	22	■ 0.972	$\mathbf{x}$	$\mathbf{x}$	Solve the following equations for the vector $\mathbf{x}$:
johnston-linear-matrix-algebra-F00082	22	■ 1.000	$\mathbf{x} - (3, 2, 1) = (1, 2, 3) - 3\mathbf{x}$	$\mathbf{x} - (3, 2, 1) = (1, 2, 3) - 3\mathbf{x}$	a) $\mathbf{x} - (3, 2, 1) = (1, 2, 3) - 3\mathbf{x}$, and
johnston-linear-matrix-algebra-F00083	22	■ 1.000	$\mathbf{x} + 2(\mathbf{v} + \mathbf{w}) = -\mathbf{v} - 3(\mathbf{x} - \mathbf{w})$	$\mathbf{x} + 2(\mathbf{v} + \mathbf{w}) = -\mathbf{v} - 3(\mathbf{x} - \mathbf{w})$	b) $\mathbf{x} + 2(\mathbf{v} + \mathbf{w}) = -\mathbf{v} - 3(\mathbf{x} - \mathbf{w})$.
johnston-linear-matrix-algebra-F00084	22	■ 1.000	$\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k$	$\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k$	vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k$ and then use vector addition and scalar multiplication to
johnston-linear-matrix-algebra-F00085	23	■ 1.000	$\mathbf{e}_j$	$\mathbf{e}_j$	dimension of $\mathbf{e}_j$ will
johnston-linear-matrix-algebra-F00086	23	■ 0.574	$\mathbf{e}_3 \in \mathbb{R}^7$	$\mathbf{e}_3 \in \mathbb{R}^7$	$\mathbf{e}_3 \in \mathbb{R}^7$.
johnston-linear-matrix-algebra-F00087	23	■ 0.999	$v_1, v_2, \dots, v_n$	$v_1, v_2, \dots, v_n$	object: $v_1, v_2, \dots, v_n$
johnston-linear-matrix-algebra-F00088	23	■ 1.000	$\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n$	$\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n$	$\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n$ are
johnston-linear-matrix-algebra-F00089	23	■ 0.997	$\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k \in \mathbb{R}^n$	$\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k \in \mathbb{R}^n$	A linear combination of the vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k \in \mathbb{R}^n$ is any vector of
johnston-linear-matrix-algebra-F00090	23	■ 0.999	$c_1, c_2, \dots, c_k \in \mathbb{R}$	$c_1, c_2, \dots, c_k \in \mathbb{R}$	where $c_1, c_2, \dots, c_k \in \mathbb{R}$.
johnston-linear-matrix-algebra-F00091	23	■ 0.647	$(-1, 0, 1)$	$(-1, 0, 1)$	$(-1, 0, 1)$ since $(1, 2, 3) = 2(1, 1, 1) + (-1, 0, 1)$. On the other hand, $(1, 2, 3)$ is
johnston-linear-matrix-algebra-F00092	23	■ 0.647	$(1, 2, 3) = 2(1, 1, 1) + (-1, 0, 1)$	$(1, 2, 3) = 2(1, 1, 1) + (-1, 0, 1)$	$(-1, 0, 1)$ since $(1, 2, 3) = 2(1, 1, 1) + (-1, 0, 1)$. On the other hand, $(1, 2, 3)$ is
johnston-linear-matrix-algebra-F00093	23	■ 0.647	$(1, 2, 3)$	$(1, 2, 3)$	$(-1, 0, 1)$ since $(1, 2, 3) = 2(1, 1, 1) + (-1, 0, 1)$. On the other hand, $(1, 2, 3)$ is
johnston-linear-matrix-algebra-F00094	23	■ 1.000	$c_1(1, 1, 0) + c_2(2, 1, 0)$	$c_1(1, 1, 0) + c_2(2, 1, 0)$	of the form $c_1(1, 1, 0) + c_2(2, 1, 0)$ has a 0 in its third entry, and thus cannot
johnston-linear-matrix-algebra-F00095	23	■ 1.000	$j=1, 2, \dots, n$	$j = 1, 2, \dots, n$	Specifically, for each $j = 1, 2, \dots, n$, we define the vector $\mathbf{e}_j \in \mathbb{R}^n$ by
johnston-linear-matrix-algebra-F00096	23	■ 1.000	$\mathbf{e}_j \in \mathbb{R}^n$	$\mathbf{e}_j \in \mathbb{R}^n$	Specifically, for each $j = 1, 2, \dots, n$, we define the vector $\mathbf{e}_j \in \mathbb{R}^n$ by
johnston-linear-matrix-algebra-F00097	23	■ 1.000	$\mathbb{R}^2$	$\mathbb{R}^2$	For example, in $\mathbb{R}^2$ there are two such vectors: $\mathbf{e}_1 = (1, 0)$ and $\mathbf{e}_2 = (0, 1)$.
johnston-linear-matrix-algebra-F00098	23	■ 1.000	$\mathbf{e}_1 = (1, 0)$	$\mathbf{e}_1 = (1, 0)$	For example, in $\mathbb{R}^2$ there are two such vectors: $\mathbf{e}_1 = (1, 0)$ and $\mathbf{e}_2 = (0, 1)$.
johnston-linear-matrix-algebra-F00099	23	■ 1.000	$\mathbf{e}_2 = (0, 1)$	$\mathbf{e}_2 = (0, 1)$	For example, in $\mathbb{R}^2$ there are two such vectors: $\mathbf{e}_1 = (1, 0)$ and $\mathbf{e}_2 = (0, 1)$.
johnston-linear-matrix-algebra-F00100	23	■ 1.000	$\mathbf{e}_1 = (1, 0, 0), \mathbf{e}_2 = (0, 1, 0)$	$\mathbf{e}_1 = (1, 0, 0), \mathbf{e}_2 = (0, 1, 0)$	Similarly, in $\mathbb{R}^3$ there are three such vectors: $\mathbf{e}_1 = (1, 0, 0), \mathbf{e}_2 = (0, 1, 0)$, and
johnston-linear-matrix-algebra-F00101	23	■ 1.000	$\mathbf{e}_3 = (0, 0, 1)$	$\mathbf{e}_3 = (0, 0, 1)$	$\mathbf{e}_3 = (0, 0, 1)$. In general, in $\mathbb{R}^n$ there are n of these vectors, $\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n$, and
johnston-linear-matrix-algebra-F00102	23	■ 0.540	$x - y$	$x - y$	Figure 1.6: The standard basis vectors point a distance of 1 along the x -, y -, and
johnston-linear-matrix-algebra-F00103	23	■ 1.000	$\mathbf{v} = (v_1, v_2, \dots, v_n)$	$\mathbf{v} = (v_1, v_2, \dots, v_n)$	if $\mathbf{v} = (v_1, v_2, \dots, v_n)$ then
johnston-linear-matrix-algebra-F00104	24	■ 1.000	$3\mathbf{e}_1 - 2\mathbf{e}_2 + \mathbf{e}_3 \in \mathbb{R}^3$	$3\mathbf{e}_1 - 2\mathbf{e}_2 + \mathbf{e}_3 \in \mathbb{R}^3$	a) Compute $3\mathbf{e}_1 - 2\mathbf{e}_2 + \mathbf{e}_3 \in \mathbb{R}^3$, and
johnston-linear-matrix-algebra-F00105	24	■ 0.973	$\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3, \mathbf{e}_4 \in \mathbb{R}^4$	$\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3, \mathbf{e}_4 \in \mathbb{R}^4$	b) Write $(3, 5, -2, -1)$ as a linear combination of $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3, \mathbf{e}_4 \in \mathbb{R}^4$.

Identifier	Page	Conf.	LaTeX source	Rendered	Image
johnston-linear-matrix-algebra-F00106	24	1.000	$3 \mathbf{e}_1 - 2 \mathbf{e}_2 + \mathbf{e}_3 = 3(1, 0, 0) - 2(0, 1, 0) + (0, 0, 1) = (3, -2, 1)$	$3\mathbf{e}_1 - 2\mathbf{e}_2 + \mathbf{e}_3 = 3(1, 0, 0) - 2(0, 1, 0) + (0, 0, 1) = (3, -2, 1)$	a) $3\mathbf{e}_1 - 2\mathbf{e}_2 + \mathbf{e}_3 = 3(1, 0, 0) - 2(0, 1, 0) + (0, 0, 1) = (3, -2, 1)$. In gen-
johnston-linear-matrix-algebra-F00107	24	1.000	$\mathbf{e}_1$	$\mathbf{e}_1$	ing vector has the coefficient of $\mathbf{e}_1$ in its first entry, the coefficient
johnston-linear-matrix-algebra-F00108	24	1.000	$\mathbf{e}_2$	$\mathbf{e}_2$	of $\mathbf{e}_2$ in its second entry, and so on.
johnston-linear-matrix-algebra-F00109	24	1.000	$(3, 5, -2, -1) = 3 \mathbf{e}_1 + 5 \mathbf{e}_2 - 2 \mathbf{e}_3 - \mathbf{e}_4$	$(3, 5, -2, -1) = 3\mathbf{e}_1 + 5\mathbf{e}_2 - 2\mathbf{e}_3 - \mathbf{e}_4$	linear combination: $(3, 5, -2, -1) = 3\mathbf{e}_1 + 5\mathbf{e}_2 - 2\mathbf{e}_3 - \mathbf{e}_4$.
johnston-linear-matrix-algebra-F00110	24	1.000	$\mathbf{v} = (3, 2)$	$\mathbf{v} = (3, 2)$	*(a) $\mathbf{v} = (3, 2)$
johnston-linear-matrix-algebra-F00111	24	1.000	$\mathbf{w} = (-0.5, 3)$	$\mathbf{w} = (-0.5, 3)$	(b) $\mathbf{w} = (-0.5, 3)$
johnston-linear-matrix-algebra-F00112	24	0.998	$\mathbf{x} = (1, -3)$	$\mathbf{x} = (1, -3)$	*(c) $\mathbf{x} = (1, -3)$
johnston-linear-matrix-algebra-F00113	24	1.000	$\mathbf{y} = (-2, -1)$	$\mathbf{y} = (-2, -1)$	(d) $\mathbf{y} = (-2, -1)$
johnston-linear-matrix-algebra-F00114	24	1.000	$\mathbf{v} = (0, 0, 2)$	$\mathbf{v} = (0, 0, 2)$	*(a) $\mathbf{v} = (0, 0, 2)$
johnston-linear-matrix-algebra-F00115	24	1.000	$\mathbf{w} = (-1, 2, 1)$	$\mathbf{w} = (-1, 2, 1)$	(b) $\mathbf{w} = (-1, 2, 1)$
johnston-linear-matrix-algebra-F00116	24	0.999	$\mathbf{x} = (1, 2, 0)$	$\mathbf{x} = (1, 2, 0)$	*(c) $\mathbf{x} = (1, 2, 0)$
johnston-linear-matrix-algebra-F00117	24	1.000	$\mathbf{y} = (3, 2, -1)$	$\mathbf{y} = (3, 2, -1)$	(d) $\mathbf{y} = (3, 2, -1)$
johnston-linear-matrix-algebra-F00118	24	1.000	$\mathbf{v}, \mathbf{w}, \mathbf{x}$	$\mathbf{v}, \mathbf{w}, \mathbf{x}$	1.1.5 If the vectors $\mathbf{v}, \mathbf{w}, \mathbf{x}$, and $\mathbf{y}$ are as in Exercise 1.1.1,
johnston-linear-matrix-algebra-F00119	24	1.000	$\mathbf{y}$	$\mathbf{y}$	1.1.5 If the vectors $\mathbf{v}, \mathbf{w}, \mathbf{x}$, and $\mathbf{y}$ are as in Exercise 1.1.1,
johnston-linear-matrix-algebra-F00120	24	1.000	$\mathbf{v} + \mathbf{w} + \mathbf{y}$	$\mathbf{v} + \mathbf{w} + \mathbf{y}$	(b) $\mathbf{v} + \mathbf{w} + \mathbf{y}$
johnston-linear-matrix-algebra-F00121	24	0.967	$\mathbf{y} - 2\mathbf{x}$	$\mathbf{y} - 2\mathbf{x}$	*(c) $\mathbf{y} - 2\mathbf{x}$
johnston-linear-matrix-algebra-F00122	24	1.000	$\mathbf{v} + 2\mathbf{w} + 2\mathbf{x} + 2\mathbf{y}$	$\mathbf{v} + 2\mathbf{w} + 2\mathbf{x} + 2\mathbf{y}$	(d) $\mathbf{v} + 2\mathbf{w} + 2\mathbf{x} + 2\mathbf{y}$
johnston-linear-matrix-algebra-F00123	24	1.000	$\mathbf{v} + \mathbf{y}$	$\mathbf{v} + \mathbf{y}$	*(a) $\mathbf{v} + \mathbf{y}$
johnston-linear-matrix-algebra-F00124	24	1.000	$4\mathbf{w} + 3\mathbf{w} - (2\mathbf{w} + 6\mathbf{w})$	$4\mathbf{w} + 3\mathbf{w} - (2\mathbf{w} + 6\mathbf{w})$	(b) $4\mathbf{w} + 3\mathbf{w} - (2\mathbf{w} + 6\mathbf{w})$
johnston-linear-matrix-algebra-F00125	24	0.979	$4\mathbf{x} - 2\mathbf{w}$	$4\mathbf{x} - 2\mathbf{w}$	*(c) $4\mathbf{x} - 2\mathbf{w}$
johnston-linear-matrix-algebra-F00126	24	0.992	$2\mathbf{x} - \mathbf{w} - \mathbf{y}$	$2\mathbf{x} - \mathbf{w} - \mathbf{y}$	(d) $2\mathbf{x} - \mathbf{w} - \mathbf{y}$
johnston-linear-matrix-algebra-F00127	24	1.000	$\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3 \in \mathbb{R}^3$	$\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3 \in \mathbb{R}^3$	vectors $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3 \in \mathbb{R}^3$.
johnston-linear-matrix-algebra-F00128	24	1.000	$\mathbf{v} = (1, 4)$	$\mathbf{v} = (1, 4)$	$\mathbf{v} = (1, 4)$ and $\mathbf{w} = (-2, 1)$. Find vectors describing both of
johnston-linear-matrix-algebra-F00129	24	1.000	$\mathbf{w} = (-2, 1)$	$\mathbf{w} = (-2, 1)$	$\mathbf{v} = (1, 4)$ and $\mathbf{w} = (-2, 1)$. Find vectors describing both of
johnston-linear-matrix-algebra-F00130	24	1.000	$\mathbf{x} = (3, -2)$	$\mathbf{x} = (3, -2)$	gram are $\mathbf{x} = (3, -2)$ and $\mathbf{y} = (1, 4)$. Find vectors describing
johnston-linear-matrix-algebra-F00131	24	1.000	$\mathbf{y} = (1, 4)$	$\mathbf{y} = (1, 4)$	gram are $\mathbf{x} = (3, -2)$ and $\mathbf{y} = (1, 4)$. Find vectors describing
johnston-linear-matrix-algebra-F00132	24	0.630	$(1, 2) - \mathbf{x} = (3, 4) - 2\mathbf{x}$	$(1, 2) - \mathbf{x} = (3, 4) - 2\mathbf{x}$	*(a) $(1, 2) - \mathbf{x} = (3, 4) - 2\mathbf{x}$
johnston-linear-matrix-algebra-F00133	24	1.000	$3((1, -1) + \mathbf{x}) = 2\mathbf{x}$	$3((1, -1) + \mathbf{x}) = 2\mathbf{x}$	(b) $3((1, -1) + \mathbf{x}) = 2\mathbf{x}$
johnston-linear-matrix-algebra-F00134	24	1.000	$2(\mathbf{x} + 2(\mathbf{x} + 2\mathbf{x})) = 3(\mathbf{x} + 3(\mathbf{x} + 3\mathbf{x}))$	$2(\mathbf{x} + 2(\mathbf{x} + 2\mathbf{x})) = 3(\mathbf{x} + 3(\mathbf{x} + 3\mathbf{x}))$	*(c) $2(\mathbf{x} + 2(\mathbf{x} + 2\mathbf{x})) = 3(\mathbf{x} + 3(\mathbf{x} + 3\mathbf{x}))$

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johnston-linear-matrix-algebra-F00135	24	■ 1.000	$-2(\mathbf{x} - (1, -2)) = \mathbf{x} + 2(\mathbf{x} + (1, 1))$	$-2(\mathbf{x} - (1, -2)) = \mathbf{x} + 2(\mathbf{x} + (1, 1))$	(d) $-2(\mathbf{x} - (1, -2)) = \mathbf{x} + 2(\mathbf{x} + (1, 1))$
johnston-linear-matrix-algebra-F00136	25	■ 1.000	$\mathbf{v} - \mathbf{x} = \mathbf{w} + \mathbf{x}$	$\mathbf{v} - \mathbf{x} = \mathbf{w} + \mathbf{x}$	*(a) $\mathbf{v} - \mathbf{x} = \mathbf{w} + \mathbf{x}$
johnston-linear-matrix-algebra-F00137	25	■ 1.000	$2\mathbf{v} - 3\mathbf{x} = 4\mathbf{x} - 5\mathbf{w}$	$2\mathbf{v} - 3\mathbf{x} = 4\mathbf{x} - 5\mathbf{w}$	(b) $2\mathbf{v} - 3\mathbf{x} = 4\mathbf{x} - 5\mathbf{w}$
johnston-linear-matrix-algebra-F00138	25	■ 1.000	$4(\mathbf{x} + \mathbf{v}) - \mathbf{x} = 2(\mathbf{w} + \mathbf{x})$	$4(\mathbf{x} + \mathbf{v}) - \mathbf{x} = 2(\mathbf{w} + \mathbf{x})$	*(c) $4(\mathbf{x} + \mathbf{v}) - \mathbf{x} = 2(\mathbf{w} + \mathbf{x})$
johnston-linear-matrix-algebra-F00139	25	■ 1.000	$2(\mathbf{x} + 2(\mathbf{x} + 2\mathbf{x})) = 2(\mathbf{v} + 2\mathbf{v})$	$2(\mathbf{x} + 2(\mathbf{x} + 2\mathbf{x})) = 2(\mathbf{v} + 2\mathbf{v})$	(d) $2(\mathbf{x} + 2(\mathbf{x} + 2\mathbf{x})) = 2(\mathbf{v} + 2\mathbf{v})$
johnston-linear-matrix-algebra-F00140	25	■ 0.856	$c(1, 2) =$	$c(1, 2) =$	*(1.1.12) Does there exist a scalar $c \in \mathbb{R}$ such that $c(1, 2) =$
johnston-linear-matrix-algebra-F00141	25	■ 1.000	$n \geq 3$	$n \geq 3$	1.1.13 Let $n \geq 3$ be an integer and consider the set of n
johnston-linear-matrix-algebra-F00142	25	■ 0.844	$n=6$	$n = 6$	tors is 0 . [Hint: We solved the $n = 6$ case in Exam-
johnston-linear-matrix-algebra-F00143	25	■ 0.985	$\mathbf{v} \cdot \mathbf{w}$	$\mathbf{v} \cdot \mathbf{w}$	tors. Then their dot product , denoted by $\mathbf{v} \cdot \mathbf{w}$, is the quantity
johnston-linear-matrix-algebra-F00144	25	■ 1.000	$\mathbf{v} \cdot (\mathbf{w} \cdot \mathbf{x})$	$\mathbf{v} \cdot (\mathbf{w} \cdot \mathbf{x})$	not a vector. So, for example, the expression $\mathbf{v} \cdot (\mathbf{w} \cdot \mathbf{x})$ does not make sense,
johnston-linear-matrix-algebra-F00145	25	■ 0.995	$\mathbf{w} \cdot \mathbf{x}$	$\mathbf{w} \cdot \mathbf{x}$	since $\mathbf{w} \cdot \mathbf{x}$ is a number, and so we cannot take its dot product with $\mathbf{v}$. On the
johnston-linear-matrix-algebra-F00146	25	■ 1.000	$\mathbf{v} / (\mathbf{w} \cdot \mathbf{x})$	$\mathbf{v} / (\mathbf{w} \cdot \mathbf{x})$	other hand, the expression $\mathbf{v} / (\mathbf{w} \cdot \mathbf{x})$ <i>does</i> make sense, since dividing a vector by
johnston-linear-matrix-algebra-F00147	25	■ 1.000	$(1, 2, 3) \cdot (4, -3, 2)$	$(1, 2, 3) \cdot (4, -3, 2)$	a) $(1, 2, 3) \cdot (4, -3, 2)$,
johnston-linear-matrix-algebra-F00148	25	■ 0.945	$(3, 6, 2) \cdot (-1, 5, 2, 1)$	$(3, 6, 2) \cdot (-1, 5, 2, 1)$	b) $(3, 6, 2) \cdot (-1, 5, 2, 1)$, and
johnston-linear-matrix-algebra-F00149	26	■ 1.000	j	j	j -th entry and 0s
johnston-linear-matrix-algebra-F00150	26	■ 1.000	$(v_1, v_2, \dots, v_n) \cdot \mathbf{e}_j$	$(v_1, v_2, \dots, v_n) \cdot \mathbf{e}_j$	c) $(v_1, v_2, \dots, v_n) \cdot \mathbf{e}_j$, where $1 \leq j \leq n$.
johnston-linear-matrix-algebra-F00151	26	■ 1.000	$1 \leq j \leq n$	$1 \leq j \leq n$	c) $(v_1, v_2, \dots, v_n) \cdot \mathbf{e}_j$, where $1 \leq j \leq n$.
johnston-linear-matrix-algebra-F00152	26	■ 1.000	$(1, 2, 3) \cdot (4, -3, 2) = 1 \cdot 4 + 2 \cdot (-3) + 3 \cdot 2 = 4 - 6 + 6 = 4$	$(1, 2, 3) \cdot (4, -3, 2) = 1 \cdot 4 + 2 \cdot (-3) + 3 \cdot 2 = 4 - 6 + 6 = 4$	a) $(1, 2, 3) \cdot (4, -3, 2) = 1 \cdot 4 + 2 \cdot (-3) + 3 \cdot 2 = 4 - 6 + 6 = 4$.
johnston-linear-matrix-algebra-F00153	26	■ 0.997	$(v_1, v_2, \dots, v_n)$	$(v_1, v_2, \dots, v_n)$	$\mathbf{e}_j$ has n entries (the same number of entries as $(v_1, v_2, \dots, v_n)$). Then
johnston-linear-matrix-algebra-F00154	26	■ 0.996	$\mathbf{v} = \mathbf{w} = (1, 0)$	$\mathbf{v} = \mathbf{w} = (1, 0)$	the amount of overlap between $\mathbf{v}$ and $\mathbf{w}$. For example, if $\mathbf{v} = \mathbf{w} = (1, 0)$ then
johnston-linear-matrix-algebra-F00155	26	■ 0.755	$\mathbf{v} \cdot \mathbf{w} = 1$	$\mathbf{v} \cdot \mathbf{w} = 1$	$\mathbf{v} \cdot \mathbf{w} = 1$, but as we rotate $\mathbf{w}$ away from $\mathbf{v}$, their dot product decreases down
johnston-linear-matrix-algebra-F00156	26	■ 0.982	$\mathbf{w} = (0, 1)$	$\mathbf{w} = (0, 1)$	to 0 when $\mathbf{v}$ and $\mathbf{w}$ are perpendicular (i.e., when $\mathbf{w} = (0, 1)$ or $\mathbf{w} = (0, -1)$),
johnston-linear-matrix-algebra-F00157	26	■ 0.982	$\mathbf{w} = (0, -1)$	$\mathbf{w} = (0, -1)$	to 0 when $\mathbf{v}$ and $\mathbf{w}$ are perpendicular (i.e., when $\mathbf{w} = (0, 1)$ or $\mathbf{w} = (0, -1)$),
johnston-linear-matrix-algebra-F00158	26	■ 0.963	$\mathbf{w} = (-1, 0)$	$\mathbf{w} = (-1, 0)$	points in the opposite direction of $\mathbf{v}$ (i.e., when $\mathbf{w} = (-1, 0)$).
johnston-linear-matrix-algebra-F00159	26	■ 1.000	θ	θ	θ then its coordinates become $\mathbf{w} = (\cos(\theta), \sin(\theta))$. The dot product between
johnston-linear-matrix-algebra-F00160	26	■ 1.000	$\mathbf{w} = (\cos(\theta), \sin(\theta))$	$\mathbf{w} = (\cos(\theta), \sin(\theta))$	θ then its coordinates become $\mathbf{w} = (\cos(\theta), \sin(\theta))$. The dot product between
johnston-linear-matrix-algebra-F00161	26	■ 1.000	$\mathbf{v} \cdot \mathbf{w} = 1 \cos(\theta) + 0 \sin(\theta) = \cos(\theta)$	$\mathbf{v} \cdot \mathbf{w} = 1 \cos(\theta) + 0 \sin(\theta) = \cos(\theta)$	$\mathbf{v}$ and $\mathbf{w}$ is then $\mathbf{v} \cdot \mathbf{w} = 1 \cos(\theta) + 0 \sin(\theta) = \cos(\theta)$, which is largest when θ
johnston-linear-matrix-algebra-F00162	26	■ 1.000	$\mathbf{v} \cdot \mathbf{w} = 1 \cos(\theta) + 0 \sin(\theta) =$	$\mathbf{v} \cdot \mathbf{w} = 1 \cos(\theta) + 0 \sin(\theta) =$	each other. Here, the dot product between $\mathbf{v}$ and $\mathbf{w}$ is $\mathbf{v} \cdot \mathbf{w} = 1 \cos(\theta) + 0 \sin(\theta) =$

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johnston-linear-matrix-algebra-F00163	26	■ 1.000	$\backslash\cos (\backslash\theta)$	$\cos(\theta)$	$\cos(\theta)$, which is largest when θ is small.
johnston-linear-matrix-algebra-F00164	26	■ 1.000	$\backslash\mathbf{v} \cdot \backslash\mathbf{w}=\backslash\mathbf{w} \cdot \backslash\mathbf{v}$	$\mathbf{v} \cdot \mathbf{w} = \mathbf{w} \cdot \mathbf{v}$	a) $\mathbf{v} \cdot \mathbf{w} = \mathbf{w} \cdot \mathbf{v}$,
johnston-linear-matrix-algebra-F00165	26	■ 1.000	$\backslash\mathbf{v} \cdot \backslash\mathbf{w}+\backslash\mathbf{v} \cdot \backslash\mathbf{x}=\backslash\mathbf{v} \cdot \backslash\mathbf{w}+\backslash\mathbf{v} \cdot \backslash\mathbf{x}$	$\mathbf{v} \cdot (\mathbf{w} + \mathbf{x}) = \mathbf{v} \cdot \mathbf{w} + \mathbf{v} \cdot \mathbf{x}$	b) $\mathbf{v} \cdot (\mathbf{w} + \mathbf{x}) = \mathbf{v} \cdot \mathbf{w} + \mathbf{v} \cdot \mathbf{x}$, and
johnston-linear-matrix-algebra-F00166	26	■ 1.000	$\backslash\mathbf{v} \cdot (c \backslash\mathbf{w})=c(\backslash\mathbf{v} \cdot \backslash\mathbf{w})$	$\mathbf{v} \cdot (c\mathbf{w}) = c(\mathbf{v} \cdot \mathbf{w})$	c) $\mathbf{v} \cdot (c\mathbf{w}) = c(\mathbf{v} \cdot \mathbf{w})$.
johnston-linear-matrix-algebra-F00167	27	■ 1.000	$(x+2)\backslash\left(x^{\wedge}\{2\}+3 x\right)$	$(x+2)\left(x^2+3 x\right)$	$(x+2)(x^2+3x)$, the
johnston-linear-matrix-algebra-F00168	27	■ 1.000	b	b	and b , then
johnston-linear-matrix-algebra-F00169	27	■ 1.000	$c^{\wedge}\{2\}=\mathbf{a}^{\wedge}\{2\}+\mathbf{b}^{\wedge}\{2\}$	$c^2=a^2+b^2$	$c^2=a^2+b^2$ (so
johnston-linear-matrix-algebra-F00170	27	■ 1.000	$\backslash\left.c=\backslash\sqrt{\mathbf{a}^{\wedge}\{2\}+\mathbf{b}^{\wedge}\{2\}}\right)$	$c=\sqrt{a^2+b^2}$	$c=\sqrt{a^2+b^2}$.
johnston-linear-matrix-algebra-F00171	27	■ 0.992	$\backslash\left \backslash\mathbf{v}\right $	$\ \mathbf{v}\ $	of a vector $\mathbf{v}$, which we represent by $\ \mathbf{v}\ $. The length of a vector $\mathbf{v} = (v_1, v_2) \in$
johnston-linear-matrix-algebra-F00172	27	■ 0.992	$\backslash\mathbf{v}=\backslash\left(v_{\{1\}}, v_{\{2\}}\right) \backslash\text {in}$	$\mathbf{v} = (v_1, v_2) \in$	of a vector $\mathbf{v}$, which we represent by $\ \mathbf{v}\ $. The length of a vector $\mathbf{v} = (v_1, v_2) \in$
johnston-linear-matrix-algebra-F00173	27	■ 1.000	$\backslash\mathbf{v}=\backslash\left(v_{\{1\}}, 0\right)+\backslash\left(0, v_{\{2\}}\right)$	$\mathbf{v} = (v_1, 0) + (0, v_2)$	$\mathbb{R}^2$ can be computed by noticing that $\mathbf{v} = (v_1, 0) + (0, v_2)$, so $\mathbf{v}$ forms the
johnston-linear-matrix-algebra-F00174	27	■ 1.000	$\backslash\left(v_{\{1\}}, 0\right)$	$(v_1, 0)$	$(v_1, 0)$ and $(0, v_2)$, as illustrated in Figure 1.8(a). Since the length of $(v_1, 0)$ is
johnston-linear-matrix-algebra-F00175	27	■ 1.000	$\backslash\left(0, v_{\{2\}}\right)$	$(0, v_2)$	$(v_1, 0)$ and $(0, v_2)$, as illustrated in Figure 1.8(a). Since the length of $(v_1, 0)$ is
johnston-linear-matrix-algebra-F00176	27	■ 1.000	$\backslash\left v_{\{1\}}\right $	$ v_1 $	$ v_1 $ and the length of $(0, v_2)$ is $ v_2 $, the Pythagorean theorem tells us that
johnston-linear-matrix-algebra-F00177	27	■ 1.000	$\backslash\left v_{\{2\}}\right $	$ v_2 $	$ v_1 $ and the length of $(0, v_2)$ is $ v_2 $, the Pythagorean theorem tells us that
johnston-linear-matrix-algebra-F00178	27	■ 1.000	$\backslash\mathbf{v}=\backslash\left(v_{\{1\}}, v_{\{2\}}, v_{\{3\}}\right) \backslash\text {in} \backslash\mathbb{R}^{\wedge}\{3\}$	$\mathbf{v} = (v_1, v_2, v_3) \in \mathbb{R}^3$	$\mathbf{v} = (v_1, v_2, v_3) \in \mathbb{R}^3$. In this case, we instead write $\mathbf{v} = (v_1, v_2, 0) + (0, 0, v_3)$,
johnston-linear-matrix-algebra-F00179	27	■ 1.000	$\backslash\mathbf{v}=\backslash\left(v_{\{1\}}, v_{\{2\}}, 0\right)+\backslash\left(0,0, v_{\{3\}}\right)$	$\mathbf{v} = (v_1, v_2, 0) + (0, 0, v_3)$	$\mathbf{v} = (v_1, v_2, v_3) \in \mathbb{R}^3$. In this case, we instead write $\mathbf{v} = (v_1, v_2, 0) + (0, 0, v_3)$,
johnston-linear-matrix-algebra-F00180	27	■ 0.995	$\backslash\left(v_{\{1\}}, v_{\{2\}}, 0\right)$	$(v_1, v_2, 0)$	given by the vectors $(v_1, v_2, 0)$ and $(0, 0, v_3)$, as in Figure 1.8(b). Since the
johnston-linear-matrix-algebra-F00181	27	■ 0.995	$\backslash\left(0,0, v_{\{3\}}\right)$	$(0, 0, v_3)$	given by the vectors $(v_1, v_2, 0)$ and $(0, 0, v_3)$, as in Figure 1.8(b). Since the
johnston-linear-matrix-algebra-F00182	27	■ 1.000	$\backslash\sqrt{v_{\{1\}}^{\wedge}\{2\}+v_{\{2\}}^{\wedge}\{2\}}$	$\sqrt{v_1^2+v_2^2}$	length of $(v_1, v_2, 0)$ is $\sqrt{v_1^2+v_2^2}$ (it is just a vector in $\mathbb{R}^2$ with an extra “0”
johnston-linear-matrix-algebra-F00183	28	■ 1.000	$\backslash\left v_{\{3\}}\right $	$ v_3 $	entry tacked on) and the length of $(0, 0, v_3)$ is $ v_3 $, the Pythagorean theorem
johnston-linear-matrix-algebra-F00184	28	■ 1.000	$\backslash\mathbf{v} \cdot \backslash\mathbf{v}=\mathbf{v}_{\{1\}}^{\wedge}\{2\}+\mathbf{v}_{\{2\}}^{\wedge}\{2\}+\backslash\cdots+\mathbf{v}_{\{n\}}^{\wedge}\{2\}$	$\mathbf{v} \cdot \mathbf{v} = v_1^2 + v_2^2 + \cdots + v_n^2$	quantity $\mathbf{v} \cdot \mathbf{v} = v_1^2 + v_2^2 + \cdots + v_n^2$ is non-negative, so we can take its square root.
johnston-linear-matrix-algebra-F00185	28	■ 0.963	$(\backslash\cos (\backslash\theta), \backslash\sin (\backslash\theta))$	$(\cos(\theta), \sin(\theta))$	b) $(\cos(\theta), \sin(\theta))$, and
johnston-linear-matrix-algebra-F00186	28	■ 1.000	$\backslash\left (2,-5,4,6)\right =\backslash\sqrt{2^{\wedge}\{2\}+(-5)^{\wedge}\{2\}+4^{\wedge}\{2\}+6^{\wedge}\{2\}}=\backslash\sqrt{81}=9$	$\ (2,-5,4,6)\ = \sqrt{2^2+(-5)^2+4^2+6^2} = \sqrt{81} = 9$	a) $\ (2,-5,4,6)\ = \sqrt{2^2+(-5)^2+4^2+6^2} = \sqrt{81} = 9$.
johnston-linear-matrix-algebra-F00187	28	■ 1.000	$\backslash\left (\backslash\cos (\backslash\theta), \backslash\sin (\backslash\theta))\right =\backslash\sqrt{\cos ^{\wedge}\{2\}(\backslash\theta)+\sin ^{\wedge}\{2\}(\backslash\theta)}=\backslash\sqrt{1}=1$	$\ (\cos(\theta), \sin(\theta))\ = \sqrt{\cos^2(\theta) + \sin^2(\theta)} = \sqrt{1} = 1$	b) $\ (\cos(\theta), \sin(\theta))\ = \sqrt{\cos^2(\theta) + \sin^2(\theta)} = \sqrt{1} = 1$.
johnston-linear-matrix-algebra-F00188	29	■ 1.000	$\backslash\sqrt{c^{\wedge}\{2\}}= c $	$\sqrt{c^2} = c $	fact that $\sqrt{c^2} = c $.
johnston-linear-matrix-algebra-F00189	29	■ 1.000	$\mathbf{v}_{\{1\}}^{\wedge}\{2\}+\mathbf{v}_{\{2\}}^{\wedge}\{2\}+\backslash\cdots+\mathbf{v}_{\{n\}}^{\wedge}\{2\}$	$v_1^2+v_2^2+\cdots+v_n^2$	$v_1^2+v_2^2+\cdots+v_n^2$

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johnston-linear-matrix-algebra_F00190	29	■ 1.000	$\mathbf{v}=\mathbf{0}$	$\mathbf{v} = \mathbf{0}$	If $\mathbf{v} = \mathbf{0}$ then we
johnston-linear-matrix-algebra_F00191	29	■ 1.000	$\mathbf{v}=\ \mathbf{v}\ \mathbf{u}$	$\mathbf{v} = \ \mathbf{v}\ \mathbf{u}$	$\mathbf{v} = \ \mathbf{v}\ \mathbf{u}$ where $\mathbf{u}$ is
johnston-linear-matrix-algebra_F00192	29	■ 1.000	$\mathbf{u}$	$\mathbf{u}$	$\mathbf{v} = \ \mathbf{v}\ \mathbf{u}$ where $\mathbf{u}$ is
johnston-linear-matrix-algebra_F00193	29	■ 1.000	$\mathbf{v}=(1,1,1)$	$\mathbf{v} = (1, 1, 1)$	The main diagonal of this cube is the vector $\mathbf{v} = (1, 1, 1)$, which has
johnston-linear-matrix-algebra_F00194	29	■ 1.000	$\ \mathbf{v}\ =\sqrt{1^2+1^2+1^2}=\sqrt{3}$	$\ \mathbf{v}\ = \sqrt{1^2 + 1^2 + 1^2} = \sqrt{3}$	length $\ \mathbf{v}\ = \sqrt{1^2 + 1^2 + 1^2} = \sqrt{3}$.
johnston-linear-matrix-algebra_F00195	29	■ 0.998	$ c \mathbf{v} = c \ \mathbf{v}\ $	$\ c\mathbf{v}\ = c \ \mathbf{v}\ $	a) $\ c\mathbf{v}\ = c \ \mathbf{v}\ $, and
johnston-linear-matrix-algebra_F00196	29	■ 1.000	$\ \mathbf{v}\ \geq 0$	$\ \mathbf{v}\ \geq 0$	b) $\ \mathbf{v}\ \geq 0$, with equality if and only if $\mathbf{v} = \mathbf{0}$.
johnston-linear-matrix-algebra_F00197	29	■ 1.000	$\ \mathbf{0}\ =0$	$\ \mathbf{0}\ = 0$	straightforward to show that $\ \mathbf{0}\ = 0$, so to complete the proof we just need to
johnston-linear-matrix-algebra_F00198	29	■ 1.000	$\ \mathbf{v}\ =0$	$\ \mathbf{v}\ = 0$	show that if $\ \mathbf{v}\ = 0$ then $\mathbf{v} = \mathbf{0}$. Well, if $\ \mathbf{v}\ = 0$ then $v_1^2 + v_2^2 + \cdots + v_n^2 = 0$,
johnston-linear-matrix-algebra_F00199	29	■ 1.000	$v_1^2+v_2^2+\cdots+v_n^2=0$	$v_1^2 + v_2^2 + \cdots + v_n^2 = 0$	show that if $\ \mathbf{v}\ = 0$ then $\mathbf{v} = \mathbf{0}$. Well, if $\ \mathbf{v}\ = 0$ then $v_1^2 + v_2^2 + \cdots + v_n^2 = 0$,
johnston-linear-matrix-algebra_F00200	29	■ 1.000	$v_j^2 \geq 0$	$v_j^2 \geq 0$	and since $v_j^2 \geq 0$ for each $1 \leq j \leq n$, with equality if and only if $v_j = 0$, we see
johnston-linear-matrix-algebra_F00201	29	■ 1.000	$v_j=0$	$v_j = 0$	and since $v_j^2 \geq 0$ for each $1 \leq j \leq n$, with equality if and only if $v_j = 0$, we see
johnston-linear-matrix-algebra_F00202	29	■ 1.000	$v_1=v_2=\cdots=v_n=0$	$v_1 = v_2 = \cdots = v_n = 0$	that it must be the case that $v_1 = v_2 = \cdots = v_n = 0$ (i.e., $\mathbf{v} = \mathbf{0}$).
johnston-linear-matrix-algebra_F00203	30	■ 1.000	$\mathbf{u}=\mathbf{v}/\ \mathbf{v}\ $	$\mathbf{u} = \mathbf{v}/\ \mathbf{v}\ $	in the form $\mathbf{v} = \ \mathbf{v}\ \mathbf{u}$, where $\mathbf{u} = \mathbf{v}/\ \mathbf{v}\ $ is the unique unit vector pointing in
johnston-linear-matrix-algebra_F00204	30	■ 1.000	$\mathbf{v}=(1,0)$	$\mathbf{v} = (1, 0)$	Figure 1.7, where we noticed that the dot product of the vector $\mathbf{v} = (1, 0)$ with
johnston-linear-matrix-algebra_F00205	30	■ 1.000	$ \mathbf{v} \cdot \mathbf{w} \leq \ \mathbf{v}\ \ \mathbf{w}\ $	$ \mathbf{v} \cdot \mathbf{w} \leq \ \mathbf{v}\ \ \mathbf{w}\ $	Suppose that $\mathbf{v}, \mathbf{w} \in \mathbb{R}^n$ are vectors. Then $ \mathbf{v} \cdot \mathbf{w} \leq \ \mathbf{v}\ \ \mathbf{w}\ $.
johnston-linear-matrix-algebra_F00206	30	■ 0.994	$\ c\mathbf{v}+d\mathbf{w}\ ^2$	$\ c\mathbf{v} + d\mathbf{w}\ ^2$	tion of $\mathbf{v}$ and $\mathbf{w}$. Specifically, if $c, d \in \mathbb{R}$ are any real numbers then $\ c\mathbf{v} + d\mathbf{w}\ ^2$
johnston-linear-matrix-algebra_F00207	30	■ 1.000	d	d	for all real numbers c and d . Well, if $\mathbf{w} = \mathbf{0}$ then the Cauchy–Schwarz inequality
johnston-linear-matrix-algebra_F00208	30	■ 1.000	$\mathbf{w}=\mathbf{0}$	$\mathbf{w} = \mathbf{0}$	for all real numbers c and d . Well, if $\mathbf{w} = \mathbf{0}$ then the Cauchy–Schwarz inequality
johnston-linear-matrix-algebra_F00209	30	■ 1.000	$0 \leq 0$	$0 \leq 0$	follows trivially since it just says that $0 \leq 0$, and otherwise we can choose
johnston-linear-matrix-algebra_F00210	30	■ 1.000	$c=\ \mathbf{w}\ $	$c = \ \mathbf{w}\ $	$c = \ \mathbf{w}\ $ and $d = -(\mathbf{v} \cdot \mathbf{w})/\ \mathbf{w}\ $ in the above inequality to see that
johnston-linear-matrix-algebra_F00211	30	■ 1.000	$d=-(\mathbf{v} \cdot \mathbf{w})/\ \mathbf{w}\ $	$d = -(\mathbf{v} \cdot \mathbf{w})/\ \mathbf{w}\ $	$c = \ \mathbf{w}\ $ and $d = -(\mathbf{v} \cdot \mathbf{w})/\ \mathbf{w}\ $ in the above inequality to see that
johnston-linear-matrix-algebra_F00212	30	■ 1.000	$\ \mathbf{v}+\mathbf{w}\ $	$\ \mathbf{v} + \mathbf{w}\ $	lengths—the fact that $\ \mathbf{v} + \mathbf{w}\ $ is never larger than $\ \mathbf{v}\ + \ \mathbf{w}\ $. To get some
johnston-linear-matrix-algebra_F00213	30	■ 1.000	$\ \mathbf{v}\ +\ \mathbf{w}\ $	$\ \mathbf{v}\ + \ \mathbf{w}\ $	lengths—the fact that $\ \mathbf{v} + \mathbf{w}\ $ is never larger than $\ \mathbf{v}\ + \ \mathbf{w}\ $. To get some
johnston-linear-matrix-algebra_F00214	30	■ 1.000	$\ \mathbf{v}+\mathbf{w}\ \leq \ \mathbf{v}\ +\ \mathbf{w}\ $	$\ \mathbf{v} + \mathbf{w}\ \leq \ \mathbf{v}\ + \ \mathbf{w}\ $	The inequality $\ \mathbf{v} + \mathbf{w}\ \leq \ \mathbf{v}\ + \ \mathbf{w}\ $ thus simply says that the length of one
johnston-linear-matrix-algebra_F00215	31	■ 1.000	a, b	a, b	triangle are a, b
johnston-linear-matrix-algebra_F00216	31	■ 1.000	$\ \mathbf{v}+\mathbf{w}\ ^2$	$\ \mathbf{v} + \mathbf{w}\ ^2$	<i>Proof.</i> We start by expanding $\ \mathbf{v} + \mathbf{w}\ ^2$ in terms of the dot product:

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johnston-linear-matrix-algebra_F00217	31	■ 0.971	$\mathbf{v}, \mathbf{w} \in \mathbb{R}^2$	$\mathbf{v}, \mathbf{w} \in \mathbb{R}^2$	the angle between two vectors $\mathbf{v}$ and $\mathbf{w}$. If $\mathbf{v}, \mathbf{w} \in \mathbb{R}^2$ then we can place $\mathbf{v}$ and
johnston-linear-matrix-algebra_F00218	31	■ 1.000	$\mathbf{v}, \mathbf{w}$	$\mathbf{v}, \mathbf{w}$	$\mathbf{w}$ in standard position, so that the vectors $\mathbf{v}$, $\mathbf{w}$, and $\mathbf{v} - \mathbf{w}$ form the sides of a
johnston-linear-matrix-algebra_F00219	31	■ 1.000	$\ \mathbf{v}\ , \ \mathbf{w}\ , \ \mathbf{v} - \mathbf{w}\ $	$\ \mathbf{v}\ , \ \mathbf{w}\ , \ \mathbf{v} - \mathbf{w}\ $	We can then use the law of cosines to relate $\ \mathbf{v}\ $, $\ \mathbf{w}\ $, $\ \mathbf{v} - \mathbf{w}\ $, and the
johnston-linear-matrix-algebra_F00220	32	■ 1.000	$0 \leq \theta \leq \pi$	$0 \leq \theta \leq \pi$	$0 \leq \theta \leq \pi$, then
johnston-linear-matrix-algebra_F00221	32	■ 1.000	$\arccos(x) = \theta$	$\arccos(x) = \theta$	$\arccos(x) = \theta$ is
johnston-linear-matrix-algebra_F00222	32	■ 1.000	$\cos(\theta) = x$	$\cos(\theta) = x$	$\cos(\theta) = x$. It is
johnston-linear-matrix-algebra_F00223	32	■ 1.000	$\cos^{-1}$	$\cos^{-1}$	as $\cos^{-1}$ or $\arccos$.
johnston-linear-matrix-algebra_F00224	32	■ 1.000	$\ \mathbf{v}\ , \ \mathbf{w}\ $	$\ \mathbf{v}\ , \ \mathbf{w}\ $	$\ \mathbf{v}\ $, $\ \mathbf{w}\ $, and $\ \mathbf{v} - \mathbf{w}\ $ via the law of cosines.
johnston-linear-matrix-algebra_F00225	32	■ 1.000	$\ \mathbf{v} - \mathbf{w}\ $	$\ \mathbf{v} - \mathbf{w}\ $	$\ \mathbf{v}\ $, $\ \mathbf{w}\ $, and $\ \mathbf{v} - \mathbf{w}\ $ via the law of cosines.
johnston-linear-matrix-algebra_F00226	32	■ 1.000	$\ \mathbf{v} - \mathbf{w}\ ^2$	$\ \mathbf{v} - \mathbf{w}\ ^2$	By setting these two expressions for $\ \mathbf{v} - \mathbf{w}\ ^2$ equal to each other, we see that
johnston-linear-matrix-algebra_F00227	32	■ 0.651	$\mathbf{v}, \mathbf{w} \in \mathbb{R}^3$	$\mathbf{v}, \mathbf{w} \in \mathbb{R}^3$	with vector $\mathbf{v}, \mathbf{w} \in \mathbb{R}^3$ that are 3-dimensional. In this case, we can still arrange
johnston-linear-matrix-algebra_F00228	32	■ 1.000	$(\mathbf{v} \cdot \mathbf{w}) / (\ \mathbf{v}\ \ \mathbf{w}\)$	$(\mathbf{v} \cdot \mathbf{w}) / (\ \mathbf{v}\ \ \mathbf{w}\)$	between vectors in this way, since it ensures that the fraction $(\mathbf{v} \cdot \mathbf{w}) / (\ \mathbf{v}\ \ \mathbf{w}\)$
johnston-linear-matrix-algebra_F00229	32	■ 1.000	$\mathbf{v} = (1, 2)$	$\mathbf{v} = (1, 2)$	a) $\mathbf{v} = (1, 2)$ and $\mathbf{w} = (3, 4)$,
johnston-linear-matrix-algebra_F00230	32	■ 1.000	$\mathbf{w} = (3, 4)$	$\mathbf{w} = (3, 4)$	a) $\mathbf{v} = (1, 2)$ and $\mathbf{w} = (3, 4)$,
johnston-linear-matrix-algebra_F00231	32	■ 1.000	$\mathbf{v} = (1, 2, -1, -2)$	$\mathbf{v} = (1, 2, -1, -2)$	b) $\mathbf{v} = (1, 2, -1, -2)$ and $\mathbf{w} = (1, -1, 1, -1)$, and
johnston-linear-matrix-algebra_F00232	32	■ 1.000	$\mathbf{w} = (1, -1, 1, -1)$	$\mathbf{w} = (1, -1, 1, -1)$	b) $\mathbf{v} = (1, 2, -1, -2)$ and $\mathbf{w} = (1, -1, 1, -1)$, and
johnston-linear-matrix-algebra_F00233	33	■ 1.000	$\arccos(11 / (5\sqrt{5}))$	$\arccos(11 / (5\sqrt{5}))$	$\arccos(11 / (5\sqrt{5}))$.
johnston-linear-matrix-algebra_F00234	33	■ 1.000	$\arccos(x)$	$\arccos(x)$	values of $\arccos(x)$
johnston-linear-matrix-algebra_F00235	33	■ 1.000	$\arccos(\sqrt{0}/2) = \pi/2$	$\arccos(\sqrt{0}/2) = \pi/2$	$\arccos(\sqrt{0}/2) = \pi/2$
johnston-linear-matrix-algebra_F00236	33	■ 1.000	$\arccos(\sqrt{1}/2) = \pi/3$	$\arccos(\sqrt{1}/2) = \pi/3$	$\arccos(\sqrt{1}/2) = \pi/3$
johnston-linear-matrix-algebra_F00237	33	■ 1.000	$\arccos(\sqrt{2}/2) = \pi/4$	$\arccos(\sqrt{2}/2) = \pi/4$	$\arccos(\sqrt{2}/2) = \pi/4$
johnston-linear-matrix-algebra_F00238	33	■ 1.000	$\arccos(\sqrt{3}/2) = \pi/6$	$\arccos(\sqrt{3}/2) = \pi/6$	$\arccos(\sqrt{3}/2) = \pi/6$
johnston-linear-matrix-algebra_F00239	33	■ 1.000	$\arccos(\sqrt{4}/2) = 0$	$\arccos(\sqrt{4}/2) = 0$	$\arccos(\sqrt{4}/2) = 0$
johnston-linear-matrix-algebra_F00240	33	■ 1.000	$\mathbf{v} \cdot \mathbf{w} = 3 + 8 = 11, \ \mathbf{v}\ = \sqrt{5}$	$\mathbf{v} \cdot \mathbf{w} = 3 + 8 = 11, \ \mathbf{v}\ = \sqrt{5}$	a) $\mathbf{v} \cdot \mathbf{w} = 3 + 8 = 11$, $\ \mathbf{v}\ = \sqrt{5}$, and $\ \mathbf{w}\ = 5$, so the angle between
johnston-linear-matrix-algebra_F00241	33	■ 1.000	$\ \mathbf{w}\ = 5$	$\ \mathbf{w}\ = 5$	a) $\mathbf{v} \cdot \mathbf{w} = 3 + 8 = 11$, $\ \mathbf{v}\ = \sqrt{5}$, and $\ \mathbf{w}\ = 5$, so the angle between
johnston-linear-matrix-algebra_F00242	33	■ 1.000	$\mathbf{v} \cdot \mathbf{w} = 1 - 2 - 1 + 2 = 0$	$\mathbf{v} \cdot \mathbf{w} = 1 - 2 - 1 + 2 = 0$	b) $\mathbf{v} \cdot \mathbf{w} = 1 - 2 - 1 + 2 = 0$, so the angle between $\mathbf{v}$ and $\mathbf{w}$ is
johnston-linear-matrix-algebra_F00243	33	■ 1.000	$\ \mathbf{w}\ $	$\ \mathbf{w}\ $	$\mathbf{v}$ and $\mathbf{w}$ without even computing $\ \mathbf{v}\ $ or $\ \mathbf{w}\ $. For this reason, it is a
johnston-linear-matrix-algebra_F00244	33	■ 0.999	$\mathbf{v} \cdot \mathbf{w} = 0$	$\mathbf{v} \cdot \mathbf{w} = 0$	good idea to compute $\mathbf{v} \cdot \mathbf{w}$ first (as we did here)—if $\mathbf{v} \cdot \mathbf{w} = 0$ then

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johnston-linear-matrix-algebra_F00245	33	■ 1.000	$\theta = \pi / 2$	$\theta = \pi/2$	we know right away that the angle is $\theta = \pi/2$.
johnston-linear-matrix-algebra_F00246	33	■ 0.503	$(0,0,0)$	$(0,0,0)$	vertex at $(0,0,0)$ and its opposite vertex at $(1,1,1)$. There are lots
johnston-linear-matrix-algebra_F00247	33	■ 0.503	$(1,1,1)$	$(1,1,1)$	vertex at $(0,0,0)$ and its opposite vertex at $(1,1,1)$. There are lots
johnston-linear-matrix-algebra_F00248	33	■ 1.000	$\mathbf{v}=(1,0,1)-(1,1,0)=(0,-1,1)$	$\mathbf{v} = (1,0,1) - (1,1,0) = (0,-1,1)$	choose the face diagonals $\mathbf{v} = (1,0,1) - (1,1,0) = (0,-1,1)$ and
johnston-linear-matrix-algebra_F00249	33	■ 1.000	$\mathbf{w}=(0,1,1)-(1,1,0)=(-1,0,1)$	$\mathbf{w} = (0,1,1) - (1,1,0) = (-1,0,1)$	$\mathbf{w} = (0,1,1) - (1,1,0) = (-1,0,1)$, as shown below.
johnston-linear-matrix-algebra_F00250	33	■ 1.000	$\mathbf{v} \cdot \mathbf{w} = 0+0+1=1, \ \mathbf{v}\ = \sqrt{2}$	$\mathbf{v} \cdot \mathbf{w} = 0 + 0 + 1 = 1, \ \mathbf{v}\ = \sqrt{2}$	Then $\mathbf{v} \cdot \mathbf{w} = 0 + 0 + 1 = 1, \ \mathbf{v}\ = \sqrt{2}$, and $\ \mathbf{w}\ = \sqrt{2}$, so the angle
johnston-linear-matrix-algebra_F00251	33	■ 1.000	$\ \mathbf{w}\ = \sqrt{2}$	$\ \mathbf{w}\ = \sqrt{2}$	Then $\mathbf{v} \cdot \mathbf{w} = 0 + 0 + 1 = 1, \ \mathbf{v}\ = \sqrt{2}$, and $\ \mathbf{w}\ = \sqrt{2}$, so the angle
johnston-linear-matrix-algebra_F00252	33	■ 0.988	$\pi / 2$	$\pi/2$	gle between $\mathbf{v}$ and $\mathbf{w}$ was $\pi/2$ based only on the fact that $\mathbf{v} \cdot \mathbf{w} = 0$ (since
johnston-linear-matrix-algebra_F00253	33	■ 0.953	$\arccos(0) = \pi / 2$	$\arccos(0) = \pi/2$	$\arccos(0) = \pi/2$). This implication goes both ways (i.e., if the angle between
johnston-linear-matrix-algebra_F00254	33	■ 1.000	$\mathbf{v}, \mathbf{w} \in \mathbb{R}^4$	$\mathbf{v}, \mathbf{w} \in \mathbb{R}^4$	also applies to higher-dimensional situations (e.g., the two vectors $\mathbf{v}, \mathbf{w} \in \mathbb{R}^4$
johnston-linear-matrix-algebra_F00255	34	■ 0.998	$\mathbf{v}=(-2,4), \mathbf{w}=(2,1)$	$\mathbf{v} = (-2,4), \mathbf{w} = (2,1)$	*(a) $\mathbf{v} = (-2,4), \mathbf{w} = (2,1)$
johnston-linear-matrix-algebra_F00256	34	■ 1.000	$\mathbf{v}=(1,2,3), \mathbf{w}=(-3,2,-1)$	$\mathbf{v} = (1,2,3), \mathbf{w} = (-3,2,-1)$	(b) $\mathbf{v} = (1,2,3), \mathbf{w} = (-3,2,-1)$
johnston-linear-matrix-algebra_F00257	34	■ 0.998	$\mathbf{v}=(3,-1,0,1), \mathbf{w}=(0,2,1,3)$	$\mathbf{v} = (3,-1,0,1), \mathbf{w} = (0,2,1,3)$	*(c) $\mathbf{v} = (3,-1,0,1), \mathbf{w} = (0,2,1,3)$
johnston-linear-matrix-algebra_F00258	34	■ 1.000	$\mathbf{v}=(\sqrt{2}, \sqrt{3}, \sqrt{5}), \mathbf{w}=(\sqrt{2}, \sqrt{3}, \sqrt{5})$	$\mathbf{v} = (\sqrt{2}, \sqrt{3}, \sqrt{5}), \mathbf{w} = (\sqrt{2}, \sqrt{3}, \sqrt{5})$	(d) $\mathbf{v} = (\sqrt{2}, \sqrt{3}, \sqrt{5}), \mathbf{w} = (\sqrt{2}, \sqrt{3}, \sqrt{5})$
johnston-linear-matrix-algebra_F00259	34	■ 0.999	$\mathbf{v}=\mathbf{0} \in \mathbb{R}^9, \mathbf{w}=(8,1,5,-7,3,9,1,-3,2)$	$\mathbf{v} = \mathbf{0} \in \mathbb{R}^9, \mathbf{w} = (8,1,5,-7,3,9,1,-3,2)$	*(e) $\mathbf{v} = \mathbf{0} \in \mathbb{R}^9, \mathbf{w} = (8,1,5,-7,3,9,1,-3,2)$
johnston-linear-matrix-algebra_F00260	34	■ 0.999	$\mathbf{v}=(3,4)$	$\mathbf{v} = (3,4)$	*(a) $\mathbf{v} = (3,4)$
johnston-linear-matrix-algebra_F00261	34	■ 1.000	$\mathbf{v}=(2,1,-2)$	$\mathbf{v} = (2,1,-2)$	(b) $\mathbf{v} = (2,1,-2)$
johnston-linear-matrix-algebra_F00262	34	■ 1.000	$\mathbf{v}=(-2 \sqrt{2},-3, \sqrt{10}, 3)$	$\mathbf{v} = (-2\sqrt{2}, -3, \sqrt{10}, 3)$	*(c) $\mathbf{v} = (-2\sqrt{2}, -3, \sqrt{10}, 3)$
johnston-linear-matrix-algebra_F00263	34	■ 1.000	$\mathbf{v}=(\cos (\theta), \sin (\theta))$	$\mathbf{v} = (\cos(\theta), \sin(\theta))$	(d) $\mathbf{v} = (\cos(\theta), \sin(\theta))$
johnston-linear-matrix-algebra_F00264	34	■ 1.000	$\mathbf{v}=(1, \sqrt{3}), \mathbf{w}=(\sqrt{3}, 1)$	$\mathbf{v} = (1, \sqrt{3}), \mathbf{w} = (\sqrt{3}, 1)$	*(a) $\mathbf{v} = (1, \sqrt{3}), \mathbf{w} = (\sqrt{3}, 1)$
johnston-linear-matrix-algebra_F00265	34	■ 1.000	$\mathbf{v}=(0,-2,2), \mathbf{w}=(1,0,1)$	$\mathbf{v} = (0,-2,2), \mathbf{w} = (1,0,1)$	(b) $\mathbf{v} = (0,-2,2), \mathbf{w} = (1,0,1)$
johnston-linear-matrix-algebra_F00266	34	■ 1.000	$\mathbf{v}=(1,1,1,1), \mathbf{w}=(-2,0,-2,0)$	$\mathbf{v} = (1,1,1,1), \mathbf{w} = (-2,0,-2,0)$	*(c) $\mathbf{v} = (1,1,1,1), \mathbf{w} = (-2,0,-2,0)$
johnston-linear-matrix-algebra_F00267	34	■ 1.000	$\mathbf{v}=(2,1,-3), \mathbf{w}=(-1,2,3)$	$\mathbf{v} = (2,1,-3), \mathbf{w} = (-1,2,3)$	(d) $\mathbf{v} = (2,1,-3), \mathbf{w} = (-1,2,3)$
johnston-linear-matrix-algebra_F00268	34	■ 0.842	$\mathbf{v}=(\cos (\boldsymbol{\theta}), \sin (\boldsymbol{\theta})), \mathbf{w}=(-\sin (\boldsymbol{\theta}), \cos (\boldsymbol{\theta}))$	$\mathbf{v} = (\cos(\theta), \sin(\theta)), \mathbf{w} = (-\sin(\theta), \cos(\theta))$	*(e) $\mathbf{v} = (\cos(\theta), \sin(\theta)), \mathbf{w} = (-\sin(\theta), \cos(\theta))$
johnston-linear-matrix-algebra_F00269	34	■ 1.000	$\mathbf{v} \cdot \mathbf{w}=\mathbf{v} \cdot \mathbf{x}$	$\mathbf{v} \cdot \mathbf{w} = \mathbf{v} \cdot \mathbf{x}$	*(a) If $\mathbf{v}, \mathbf{w}, \mathbf{x} \in \mathbb{R}^n$ are vectors with $\mathbf{v} \cdot \mathbf{w} = \mathbf{v} \cdot \mathbf{x}$, then
johnston-linear-matrix-algebra_F00270	34	■ 0.785	$\mathbf{w}=\mathbf{x}$	$\mathbf{w} = \mathbf{x}$	$\mathbf{w} = \mathbf{x}$.
johnston-linear-matrix-algebra_F00271	34	■ 1.000	$\mathbf{w} \cdot \mathbf{x}=0$	$\mathbf{w} \cdot \mathbf{x} = 0$	(b) If $\mathbf{v}, \mathbf{w}, \mathbf{x} \in \mathbb{R}^n$ are vectors with $\mathbf{v} \cdot \mathbf{w} = 0$ and $\mathbf{w} \cdot \mathbf{x} = 0$,
johnston-linear-matrix-algebra_F00272	34	■ 1.000	$\mathbf{v} \cdot \mathbf{x}=0$	$\mathbf{v} \cdot \mathbf{x} = 0$	then $\mathbf{v} \cdot \mathbf{x} = 0$ too.

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johnston-linear-matrix-algebra_F00273	34	0.760	$\ \mathbf{v}\ + \ \mathbf{w}\ \leq 2$	$\ \mathbf{v}\ + \ \mathbf{w}\ \leq 2$	* (c) If $\mathbf{v}, \mathbf{w} \in \mathbb{R}^n$ are vectors with $\ \mathbf{v}\ + \ \mathbf{w}\ \leq 2$, then
johnston-linear-matrix-algebra_F00274	34	1.000	$\ \mathbf{v} + \mathbf{w}\ \leq 2$	$\ \mathbf{v} + \mathbf{w}\ \leq 2$	$\ \mathbf{v} + \mathbf{w}\ \leq 2$ too.
johnston-linear-matrix-algebra_F00275	34	1.000	$\mathbf{v}, \mathbf{w} \in \mathbb{R}^5$	$\mathbf{v}, \mathbf{w} \in \mathbb{R}^5$	(d) There exist vectors $\mathbf{v}, \mathbf{w} \in \mathbb{R}^5$ such that $\ \mathbf{v}\ = 2$,
johnston-linear-matrix-algebra_F00276	34	1.000	$\ \mathbf{v}\ = 2$	$\ \mathbf{v}\ = 2$	(d) There exist vectors $\mathbf{v}, \mathbf{w} \in \mathbb{R}^5$ such that $\ \mathbf{v}\ = 2$,
johnston-linear-matrix-algebra_F00277	34	1.000	$\ \mathbf{w}\ = 4$	$\ \mathbf{w}\ = 4$	$\ \mathbf{w}\ = 4$, and $\ \mathbf{v} - \mathbf{w}\ = 1$.
johnston-linear-matrix-algebra_F00278	34	1.000	$\ \mathbf{v} - \mathbf{w}\ = 1$	$\ \mathbf{v} - \mathbf{w}\ = 1$	$\ \mathbf{w}\ = 4$, and $\ \mathbf{v} - \mathbf{w}\ = 1$.
johnston-linear-matrix-algebra_F00279	34	0.995	$\ \mathbf{v}\ = 1$	$\ \mathbf{v}\ = 1$	* (e) There exist vectors $\mathbf{v}, \mathbf{w} \in \mathbb{R}^3$ such that $\ \mathbf{v}\ = 1$,
johnston-linear-matrix-algebra_F00280	34	1.000	$\ \mathbf{w}\ = 2$	$\ \mathbf{w}\ = 2$	$\ \mathbf{w}\ = 2$, and $\mathbf{v} \cdot \mathbf{w} = -1$.
johnston-linear-matrix-algebra_F00281	34	1.000	$\mathbf{v} \cdot \mathbf{w} = -1$	$\mathbf{v} \cdot \mathbf{w} = -1$	$\ \mathbf{w}\ = 2$, and $\mathbf{v} \cdot \mathbf{w} = -1$.
johnston-linear-matrix-algebra_F00282	34	1.000	$ \mathbf{v} \cdot \mathbf{w} \leq 1$	$ \mathbf{v} \cdot \mathbf{w} \leq 1$	(f) If $\mathbf{v}, \mathbf{w} \in \mathbb{R}^n$ are unit vectors, then $ \mathbf{v} \cdot \mathbf{w} \leq 1$.
johnston-linear-matrix-algebra_F00283	34	0.598	$\ \mathbf{v}\ \leq 1$	$\ \mathbf{v}\ \leq 1$	* (g) If $\mathbf{v}, \mathbf{w} \in \mathbb{R}^n$ are vectors with $ \mathbf{v} \cdot \mathbf{w} \leq 1$, then $\ \mathbf{v}\ \leq 1$
johnston-linear-matrix-algebra_F00284	34	1.000	$\ \mathbf{w}\ \leq 1$	$\ \mathbf{w}\ \leq 1$	or $\ \mathbf{w}\ \leq 1$ (or both).
johnston-linear-matrix-algebra_F00285	34	0.965	$\mathbf{v} = (3, \sqrt{3})$	$\mathbf{v} = (3, \sqrt{3})$	*1.2.6 Suppose $\mathbf{v} = (3, \sqrt{3})$. Find all vectors $\mathbf{w} \in \mathbb{R}^2$ such
johnston-linear-matrix-algebra_F00286	34	0.965	$\mathbf{w} \in \mathbb{R}^2$	$\mathbf{w} \in \mathbb{R}^2$	*1.2.6 Suppose $\mathbf{v} = (3, \sqrt{3})$. Find all vectors $\mathbf{w} \in \mathbb{R}^2$ such
johnston-linear-matrix-algebra_F00287	34	1.000	$\theta = \pi/3$	$\theta = \pi/3$	that $\ \mathbf{w}\ = 2$ and the angle between $\mathbf{v}$ and $\mathbf{w}$ is $\theta = \pi/3$.
johnston-linear-matrix-algebra_F00288	34	0.997	$\mathbf{v} \cdot (\mathbf{w} - \mathbf{x})$	$\mathbf{v} \cdot (\mathbf{w} - \mathbf{x})$	* (a) $\mathbf{v} \cdot (\mathbf{w} - \mathbf{x})$
johnston-linear-matrix-algebra_F00289	34	1.000	$(\mathbf{v} \cdot \mathbf{w})\mathbf{x}$	$(\mathbf{v} \cdot \mathbf{w})\mathbf{x}$	(b) $(\mathbf{v} \cdot \mathbf{w})\mathbf{x}$
johnston-linear-matrix-algebra_F00290	34	1.000	$\mathbf{v} + (\mathbf{w} \cdot \mathbf{x})$	$\mathbf{v} + (\mathbf{w} \cdot \mathbf{x})$	* (c) $\mathbf{v} + (\mathbf{w} \cdot \mathbf{x})$
johnston-linear-matrix-algebra_F00291	34	1.000	$\mathbf{v} / \ \mathbf{v}\ $	$\mathbf{v} / \ \mathbf{v}\ $	(d) $\mathbf{v} / \ \mathbf{v}\ $
johnston-linear-matrix-algebra_F00292	34	0.647	$\mathbf{v}^2$	$\mathbf{v}^2$	* (e) $\mathbf{v}^2$
johnston-linear-matrix-algebra_F00293	34	1.000	$(\mathbf{v} + \mathbf{w})/\mathbf{x}$	$(\mathbf{v} + \mathbf{w})/\mathbf{x}$	(f) $(\mathbf{v} + \mathbf{w})/\mathbf{x}$
johnston-linear-matrix-algebra_F00294	34	1.000	$\ \mathbf{e}_j\ $	$\ \mathbf{e}_j\ $	(a) Compute $\ \mathbf{e}_j\ $.
johnston-linear-matrix-algebra_F00295	34	1.000	$1 \leq i, j \leq n$	$1 \leq i, j \leq n$	(b) Suppose $1 \leq i, j \leq n$. Compute $\mathbf{e}_i \cdot \mathbf{e}_j$. [Side note: You
johnston-linear-matrix-algebra_F00296	34	1.000	$\mathbf{e}_i \cdot \mathbf{e}_j$	$\mathbf{e}_i \cdot \mathbf{e}_j$	(b) Suppose $1 \leq i, j \leq n$. Compute $\mathbf{e}_i \cdot \mathbf{e}_j$. [Side note: You
johnston-linear-matrix-algebra_F00297	34	1.000	$i = j$	$i = j$	$i = j$ or $i \neq j$.]
johnston-linear-matrix-algebra_F00298	34	1.000	$i \neq j$	$i \neq j$	$i = j$ or $i \neq j$.]
johnston-linear-matrix-algebra_F00299	34	1.000	$\mathbf{v} = (1, 2) \in \mathbb{R}^2$	$\mathbf{v} = (1, 2) \in \mathbb{R}^2$	*1.2.9 Let $\mathbf{v} = (1, 2) \in \mathbb{R}^2$.
johnston-linear-matrix-algebra_F00300	34	1.000	$\mathbf{v} = (1, 2, 3) \in \mathbb{R}^3$	$\mathbf{v} = (1, 2, 3) \in \mathbb{R}^3$	1.2.10 Let $\mathbf{v} = (1, 2, 3) \in \mathbb{R}^3$.
johnston-linear-matrix-algebra_F00301	35	1.000	$\mathbf{v}, \mathbf{w} \in \mathbb{C}^n$	$\mathbf{v}, \mathbf{w} \in \mathbb{C}^n$	(instead of real) entries, we write $\mathbf{v}, \mathbf{w} \in \mathbb{C}^n$ and we define

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johnston-linear-matrix-algebra_F00302	35	■ 1.000	$\overline{a+ib}=a-i b$	$a+ib=a-ib$	where $\overline{a+ib}=a-ib$ is the complex conjugate (see Ap-
johnston-linear-matrix-algebra_F00303	35	■ 1.000	$\mathbf{v} \cdot \mathbf{w}=\overline{\mathbf{v}} \cdot \mathbf{w}$	$\mathbf{v} \cdot \mathbf{w}=\overline{\mathbf{w}} \cdot \mathbf{v}$	(a) Show that $\mathbf{v} \cdot \mathbf{w}=\overline{\mathbf{w}} \cdot \mathbf{v}$.
johnston-linear-matrix-algebra_F00304	35	■ 0.988	$c \in \mathbb{C}$	$c \in \mathbb{C}$	$c \in \mathbb{C}$, but $(c \mathbf{v}) \cdot \mathbf{w}=\overline{c}(\mathbf{v} \cdot \mathbf{w})$.
johnston-linear-matrix-algebra_F00305	35	■ 0.988	$(c \mathbf{v}) \cdot \mathbf{w}=\overline{c}(\mathbf{v} \cdot \mathbf{w})$	$(c \mathbf{v}) \cdot \mathbf{w}=\overline{c}(\mathbf{v} \cdot \mathbf{w})$	$c \in \mathbb{C}$, but $(c \mathbf{v}) \cdot \mathbf{w}=\overline{c}(\mathbf{v} \cdot \mathbf{w})$.
johnston-linear-matrix-algebra_F00306	35	■ 1.000	$\mathbf{x} \in \mathbb{R}^n$	$\mathbf{x} \in \mathbb{R}^n$	1.2.16 Suppose that $\mathbf{x} \in \mathbb{R}^n$. Show that $\mathbf{x} \cdot \mathbf{y}=0$ for all
johnston-linear-matrix-algebra_F00307	35	■ 1.000	$\mathbf{x} \cdot \mathbf{y}=0$	$\mathbf{x} \cdot \mathbf{y}=0$	1.2.16 Suppose that $\mathbf{x} \in \mathbb{R}^n$. Show that $\mathbf{x} \cdot \mathbf{y}=0$ for all
johnston-linear-matrix-algebra_F00308	35	■ 1.000	$\mathbf{y} \in \mathbb{R}^n$	$\mathbf{y} \in \mathbb{R}^n$	$\mathbf{y} \in \mathbb{R}^n$ if and only if $\mathbf{x}=\mathbf{0}$.
johnston-linear-matrix-algebra_F00309	35	■ 1.000	$\mathbf{x}=\mathbf{0}$	$\mathbf{x}=\mathbf{0}$	$\mathbf{y} \in \mathbb{R}^n$ if and only if $\mathbf{x}=\mathbf{0}$.
johnston-linear-matrix-algebra_F00310	35	■ 1.000	$ \mathbf{v} \cdot \mathbf{w} = \mathbf{v} \mathbf{w} $	$ \mathbf{v} \cdot \mathbf{w} = \mathbf{v} \mathbf{w} $	(a) Prove that $ \mathbf{v} \cdot \mathbf{w} = \mathbf{v} \mathbf{w} $ if and only if either $\mathbf{w}=\mathbf{0}$
johnston-linear-matrix-algebra_F00311	35	■ 1.000	$\mathbf{v}=c \mathbf{w}$	$\mathbf{v}=c \mathbf{w}$	or there exists a scalar $c \in \mathbb{R}$ such that $\mathbf{v}=c \mathbf{w}$.
johnston-linear-matrix-algebra_F00312	35	■ 1.000	$ \mathbf{v}+\mathbf{w} = \mathbf{v} + \mathbf{w} $	$ \mathbf{v}+\mathbf{w} = \mathbf{v} + \mathbf{w} $	(b) Prove that $ \mathbf{v}+\mathbf{w} = \mathbf{v} + \mathbf{w} $ if and only if either
johnston-linear-matrix-algebra_F00313	35	■ 1.000	$0 \leq c \in \mathbb{R}$	$0 \leq c \in \mathbb{R}$	$\mathbf{w}=\mathbf{0}$ or there exists a scalar $0 \leq c \in \mathbb{R}$ such that
johnston-linear-matrix-algebra_F00314	35	■ 1.000	$ \mathbf{v}+\mathbf{w} ^2= \mathbf{v} ^2+ \mathbf{w} ^2$	$ \mathbf{v}+\mathbf{w} ^2= \mathbf{v} ^2+ \mathbf{w} ^2$	other. Prove that $ \mathbf{v}+\mathbf{w} ^2= \mathbf{v} ^2+ \mathbf{w} ^2$.
johnston-linear-matrix-algebra_F00315	35	■ 1.000	$ \mathbf{v}+\mathbf{w} ^2+ \mathbf{v}-\mathbf{w} ^2=2 \mathbf{v} ^2+2 \mathbf{w} ^2$	$ \mathbf{v}+\mathbf{w} ^2+ \mathbf{v}-\mathbf{w} ^2=2 \mathbf{v} ^2+2 \mathbf{w} ^2$	(a) Show that $ \mathbf{v}+\mathbf{w} ^2+ \mathbf{v}-\mathbf{w} ^2=2 \mathbf{v} ^2+2 \mathbf{w} ^2$.
johnston-linear-matrix-algebra_F00316	35	■ 1.000	$\mathbf{v} \cdot \mathbf{w}=\frac{1}{4}\left(\mathbf{v}+\mathbf{w} ^2- \mathbf{v}-\mathbf{w} ^2\right)$	$\mathbf{v} \cdot \mathbf{w}=\frac{1}{4}\left(\mathbf{v}+\mathbf{w} ^2- \mathbf{v}-\mathbf{w} ^2\right)$	1.2.20 Let $\mathbf{v}, \mathbf{w} \in \mathbb{R}^n$ be vectors. Prove that $\mathbf{v} \cdot \mathbf{w}=\frac{1}{4}\left(\mathbf{v}+\mathbf{w} ^2- \mathbf{v}-\mathbf{w} ^2\right)$.
johnston-linear-matrix-algebra_F00317	35	■ 1.000	$ \mathbf{v}-\mathbf{w} \geq$	$ \mathbf{v}-\mathbf{w} \geq$	*1.2.21 Let $\mathbf{v}, \mathbf{w} \in \mathbb{R}^n$ be vectors. Prove that $ \mathbf{v}-\mathbf{w} \geq$
johnston-linear-matrix-algebra_F00318	35	■ 1.000	$ \mathbf{v} = \mathbf{w} $	$ \mathbf{v} = \mathbf{w} $	$ \mathbf{v} = \mathbf{w} $.
johnston-linear-matrix-algebra_F00319	35	■ 1.000	$d= \mathbf{v} $	$d= \mathbf{v} $	$d= \mathbf{v} $ in the proof?
johnston-linear-matrix-algebra_F00320	35	■ 1.000	$d=- \mathbf{v} $	$d=- \mathbf{v} $	$d=- \mathbf{v} $ in the proof?
johnston-linear-matrix-algebra_F00321	35	■ 1.000	$f(x)= \mathbf{v}-x \mathbf{w} ^2$	$f(x)= \mathbf{v}-x \mathbf{w} ^2$	and consider the function $f(x)= \mathbf{v}-x \mathbf{w} ^2$.
johnston-linear-matrix-algebra_F00322	35	■ 1.000	$a x^2+b x+c$	$a x^2+b x+c$	that the discriminant of the quadratic $a x^2+b x+c$ is
johnston-linear-matrix-algebra_F00323	35	■ 1.000	$b^2-4 a c$	$b^2-4 a c$	$b^2-4 a c$.
johnston-linear-matrix-algebra_F00324	35	■ 1.000	≤ 0	≤ 0	(c) Why must the discriminant from part (b) be ≤ 0 ?
johnston-linear-matrix-algebra_F00325	35	■ 1.000	f	f	[Hint: How many roots does f have?]
johnston-linear-matrix-algebra_F00326	35	■ 1.000	$ x+y \leq x + y $	$ x+y \leq x + y $	(a) Show that if $x, y \in \mathbb{R}$ then $ x+y \leq x + y $. Do <i>not</i>
johnston-linear-matrix-algebra_F00327	35	■ 1.000	$x_1, \ldots, x_n \in \mathbb{R}$	$x_1, \ldots, x_n \in \mathbb{R}$	(b) Show that if $x_1, \ldots, x_n \in \mathbb{R}$ then $ x_1+\cdots+x_n \leq$
johnston-linear-matrix-algebra_F00328	35	■ 1.000	$ x_1+\cdots+x_n \leq$	$ x_1+\cdots+x_n \leq$	(b) Show that if $x_1, \ldots, x_n \in \mathbb{R}$ then $ x_1+\cdots+x_n \leq$
johnston-linear-matrix-algebra_F00329	35	■ 1.000	$ x_1 +\cdots+ x_n $	$ x_1 +\cdots+ x_n $	$ x_1 +\cdots+ x_n $.
johnston-linear-matrix-algebra_F00330	35	■ 1.000			

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johnston-linear-matrix-algebra-F00331	35	1.000	$x y \leq \frac{1}{2} \left(x^2 + y^2 \right)$	$xy \leq \frac{1}{2} (x^2 + y^2)$	(c) Show that if $x, y \in \mathbb{R}$ then $xy \leq \frac{1}{2} (x^2 + y^2)$.
johnston-linear-matrix-algebra-F00332	35	1.000	$ \mathbf{v} \cdot \mathbf{w} / (\mathbf{v} \mathbf{w}) \leq 1$	$ \mathbf{v} \cdot \mathbf{w} / (\mathbf{v} \mathbf{w}) \leq 1$	$ \mathbf{v} \cdot \mathbf{w} / (\mathbf{v} \mathbf{w}) \leq 1$. [Hint: Start by writing out
johnston-linear-matrix-algebra-F00333	35	1.000	$\mathbf{v} \cdot \mathbf{w} = v_1 w_1 + \cdots + v_n w_n$	$\mathbf{v} \cdot \mathbf{w} = v_1 w_1 + \cdots + v_n w_n$	$\mathbf{v} \cdot \mathbf{w} = v_1 w_1 + \cdots + v_n w_n$ in the fraction on the left.
johnston-linear-matrix-algebra-F00334	36	1.000	$a_{i,j}$	$a_{i,j}$	columns: $a_{i,j}$
johnston-linear-matrix-algebra-F00335	36	1.000	i	i	the i -th row and
johnston-linear-matrix-algebra-F00336	36	0.623	3×4	3×4	3×4 matrix is one
johnston-linear-matrix-algebra-F00337	36	1.000	$A, B, C, \ldots$	$A, B, C, \ldots$	We typically denote matrices by uppercase letters like $A, B, C, \ldots$, and
johnston-linear-matrix-algebra-F00338	36	1.000	$a_{1,2} = 3$	$a_{1,2} = 3$	from. For example, if A and B are as above, then $a_{1,2} = 3$, since the entry of
johnston-linear-matrix-algebra-F00339	36	0.998	$a_{2,1} = 2$	$a_{2,1} = 2$	A in the 1st row and 2nd column is 3. Similarly, $a_{2,1} = 2$ and $b_{2,3} = 1$. The
johnston-linear-matrix-algebra-F00340	36	0.998	$b_{2,3} = 1$	$b_{2,3} = 1$	A in the 1st row and 2nd column is 3. Similarly, $a_{2,1} = 2$ and $b_{2,3} = 1$. The
johnston-linear-matrix-algebra-F00341	36	0.999	(i, j)	(i, j)	entry in the i -th row and j -th column is also called its “ (i, j) -entry”, and we
johnston-linear-matrix-algebra-F00342	36	1.000	$[A]_{i,j}$	$[A]_{i,j}$	sometimes alternatively denote it using square brackets like $[A]_{i,j}$. For example,
johnston-linear-matrix-algebra-F00343	36	0.633	$b_{2,1} = [B]_{2,1} = 0$	$b_{2,1} = [B]_{2,1} = 0$	the $(2, 1)$ -entry of B is $b_{2,1} = [B]_{2,1} = 0$.
johnston-linear-matrix-algebra-F00344	36	1.000	m	m	matrices with m rows and n columns is denoted by $\mathcal{M}_{m,n}$, or simply by $\mathcal{M}_n$ if
johnston-linear-matrix-algebra-F00345	36	1.000	$\mathcal{M}_{m,n}$	$\mathcal{M}_{m,n}$	matrices with m rows and n columns is denoted by $\mathcal{M}_{m,n}$, or simply by $\mathcal{M}_n$ if
johnston-linear-matrix-algebra-F00346	36	1.000	$\mathcal{M}_n$	$\mathcal{M}_n$	matrices with m rows and n columns is denoted by $\mathcal{M}_{m,n}$, or simply by $\mathcal{M}_n$ if
johnston-linear-matrix-algebra-F00347	36	1.000	$m = n$	$m = n$	$m = n$ (in which case the matrix is called square). So if A and B again refer to
johnston-linear-matrix-algebra-F00348	36	1.000	$A \in \mathcal{M}_2$	$A \in \mathcal{M}_2$	the matrices displayed above, then $A \in \mathcal{M}_2$ and $B \in \mathcal{M}_{2,3}$.
johnston-linear-matrix-algebra-F00349	36	1.000	$B \in \mathcal{M}_{2,3}$	$B \in \mathcal{M}_{2,3}$	the matrices displayed above, then $A \in \mathcal{M}_2$ and $B \in \mathcal{M}_{2,3}$.
johnston-linear-matrix-algebra-F00350	36	1.000	$A, B \in \mathcal{M}_{m,n}$	$A, B \in \mathcal{M}_{m,n}$	Suppose $A, B \in \mathcal{M}_{m,n}$ are matrices and $c \in \mathbb{R}$ is a scalar. Then the sum
johnston-linear-matrix-algebra-F00351	36	0.985	$A + B$	$A + B$	$A + B$ and scalar multiplication cA are the $m \times n$ matrices whose (i, j) -
johnston-linear-matrix-algebra-F00352	36	0.985	cA	cA	$A + B$ and scalar multiplication cA are the $m \times n$ matrices whose (i, j) -
johnston-linear-matrix-algebra-F00353	36	0.985	$m \times n$	$m \times n$	$A + B$ and scalar multiplication cA are the $m \times n$ matrices whose (i, j) -
johnston-linear-matrix-algebra-F00354	36	1.000	$1 \leq i \leq m$	$1 \leq i \leq m$	entries, for each $1 \leq i \leq m$ and $1 \leq j \leq n$, are
johnston-linear-matrix-algebra-F00355	36	1.000	O	O	We also use O to denote the zero matrix whose entries all equal 0 (or
johnston-linear-matrix-algebra-F00356	36	1.000	$O_{m,n}$	$O_{m,n}$	$O_{m,n}$ if we wish to emphasize or clarify that it is $m \times n$, or O_n if it is $n \times n$).
johnston-linear-matrix-algebra-F00357	36	1.000	O_n	O_n	$O_{m,n}$ if we wish to emphasize or clarify that it is $m \times n$, or O_n if it is $n \times n$).

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johnston-linear-matrix-algebra_F00358	36	■ 1.000	$n \times n$	$n \times n$	$O_{m,n}$ if we wish to emphasize or clarify that it is $m \times n$, or O_n if it is $n \times n$.
johnston-linear-matrix-algebra_F00359	36	■ 0.971	$(A - B = A + (-1) B)$	$(A - B = A + (-1)B)$	Similarly, we define matrix subtraction $(A - B = A + (-1)B)$ and the negative of
johnston-linear-matrix-algebra_F00360	36	■ 1.000	$(-A = (-1) A)$	$(-A = (-1)A)$	a matrix $(-A = (-1)A)$ in the obvious entrywise ways. There is nothing fancy
johnston-linear-matrix-algebra_F00361	36	■ 0.755	$A = \begin{bmatrix} 1 & 3 \\ 2 & -1 \end{bmatrix}, B = \begin{bmatrix} 2 & 1 \\ 0 & 1 \end{bmatrix}$	$A = \begin{bmatrix} 1 & 3 \\ 2 & -1 \end{bmatrix}, B = \begin{bmatrix} 2 & 1 \\ 0 & 1 \end{bmatrix}$	Suppose $A = \begin{bmatrix} 1 & 3 \\ 2 & -1 \end{bmatrix}, B = \begin{bmatrix} 2 & 1 \\ 0 & 1 \end{bmatrix}$, and $C = \begin{bmatrix} 1 & 0 & 1 \\ 0 & -1 & 1 \end{bmatrix}$. Compute
johnston-linear-matrix-algebra_F00362	36	■ 0.755	$C = \begin{bmatrix} 1 & 0 & 1 \\ 0 & -1 & 1 \end{bmatrix}$	$C = \begin{bmatrix} 1 & 0 & 1 \\ 0 & -1 & 1 \end{bmatrix}$	Suppose $A = \begin{bmatrix} 1 & 3 \\ 2 & -1 \end{bmatrix}, B = \begin{bmatrix} 2 & 1 \\ 0 & 1 \end{bmatrix}$, and $C = \begin{bmatrix} 1 & 0 & 1 \\ 0 & -1 & 1 \end{bmatrix}$. Compute
johnston-linear-matrix-algebra_F00363	36	■ 1.000	$2 A - 3 B$	$2A - 3B$	b) $2A - 3B$, and
johnston-linear-matrix-algebra_F00364	36	■ 1.000	$A + 2 C$	$A + 2C$	c) $A + 2C$.
johnston-linear-matrix-algebra_F00365	37	■ 1.000	$A + B = \begin{bmatrix} 1 & 3 \\ 2 & -1 \end{bmatrix} + \begin{bmatrix} 2 & 1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 3 & 4 \\ 2 & 0 \end{bmatrix}$	$A + B = \begin{bmatrix} 1 & 3 \\ 2 & -1 \end{bmatrix} + \begin{bmatrix} 2 & 1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 3 & 4 \\ 2 & 0 \end{bmatrix}$	a) $A + B = \begin{bmatrix} 1 & 3 \\ 2 & -1 \end{bmatrix} + \begin{bmatrix} 2 & 1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 3 & 4 \\ 2 & 0 \end{bmatrix}$.
johnston-linear-matrix-algebra_F00366	37	■ 0.994	$2 A - 3 B = \begin{bmatrix} 2 & 6 \\ 4 & -2 \end{bmatrix} - \begin{bmatrix} 6 & 3 \\ 0 & 3 \end{bmatrix} = \begin{bmatrix} -4 & 3 \\ 4 & -5 \end{bmatrix}$	$2A - 3B = \begin{bmatrix} 2 & 6 \\ 4 & -2 \end{bmatrix} - \begin{bmatrix} 6 & 3 \\ 0 & 3 \end{bmatrix} = \begin{bmatrix} -4 & 3 \\ 4 & -5 \end{bmatrix}$	b) $2A - 3B = \begin{bmatrix} 2 & 6 \\ 4 & -2 \end{bmatrix} - \begin{bmatrix} 6 & 3 \\ 0 & 3 \end{bmatrix} = \begin{bmatrix} -4 & 3 \\ 4 & -5 \end{bmatrix}$.
johnston-linear-matrix-algebra_F00367	37	■ 0.999	2×2	2×2	matrices that have different sizes (A is a 2×2 matrix, but $2C$ is a
johnston-linear-matrix-algebra_F00368	37	■ 0.999	$2 C$	$2C$	matrices that have different sizes (A is a 2×2 matrix, but $2C$ is a
johnston-linear-matrix-algebra_F00369	37	■ 0.991	2×3	2×3	2×3 matrix).
johnston-linear-matrix-algebra_F00370	37	■ 1.000	$A, B, C \in \mathcal{M}_{m,n}$	$A, B, C \in \mathcal{M}_{m,n}$	Suppose $A, B, C \in \mathcal{M}_{m,n}$ are $m \times n$ matrices and $c, d \in \mathbb{R}$ are scalars. Then
johnston-linear-matrix-algebra_F00371	37	■ 1.000	$A + B = B + A$	$A + B = B + A$	a) $A + B = B + A$,
johnston-linear-matrix-algebra_F00372	37	■ 1.000	$(A + B) + C = A + (B + C)$	$(A + B) + C = A + (B + C)$	b) $(A + B) + C = A + (B + C)$,
johnston-linear-matrix-algebra_F00373	37	■ 1.000	$c(A + B) = cA + cB$	$c(A + B) = cA + cB$	c) $c(A + B) = cA + cB$,
johnston-linear-matrix-algebra_F00374	37	■ 1.000	$(c + d) A = cA + dA$	$(c + d)A = cA + dA$	d) $(c + d)A = cA + dA$, and
johnston-linear-matrix-algebra_F00375	37	■ 1.000	$c(d A) = (cd) A$	$c(dA) = (cd)A$	e) $c(dA) = (cd)A$.
johnston-linear-matrix-algebra_F00376	37	■ 1.000	$a + b = b + a$	$a + b = b + a$	that $a + b = b + a$
johnston-linear-matrix-algebra_F00377	37	■ 1.000	$a, b \in \mathbb{R}$	$a, b \in \mathbb{R}$	when $a, b \in \mathbb{R}$. We
johnston-linear-matrix-algebra_F00378	37	■ 1.000	$m n$	mn	each of the mn
johnston-linear-matrix-algebra_F00379	37	■ 1.000	$A + 0 = A$	$A + O = A$	are the facts that $A + O = A$ and $A - A = O$ for all matrices $A \in \mathcal{M}_{m,n}$,
johnston-linear-matrix-algebra_F00380	37	■ 1.000	$A - A = 0$	$A - A = O$	are the facts that $A + O = A$ and $A - A = O$ for all matrices $A \in \mathcal{M}_{m,n}$.

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johnston-linear-matrix-algebra_F00381	37	■ 1.000	$A \in \mathcal{M}_{m,n}$	$A \in \mathcal{M}_{m,n}$	are the facts that $A + O = A$ and $A - A = O$ for all matrices $A \in \mathcal{M}_{m,n}$.
johnston-linear-matrix-algebra_F00382	38	■ 1.000	$r \times p$	$r \times p$	matrix by an $r \times p$
johnston-linear-matrix-algebra_F00383	38	■ 0.910	$n=r$	$n = r$	$n = r$.
johnston-linear-matrix-algebra_F00384	38	■ 1.000	$B \in \mathcal{M}_{n,p}$	$B \in \mathcal{M}_{n,p}$	If $A \in \mathcal{M}_{m,n}$ and $B \in \mathcal{M}_{n,p}$ are matrices, then their product AB is the
johnston-linear-matrix-algebra_F00385	38	■ 1.000	$A B$	AB	If $A \in \mathcal{M}_{m,n}$ and $B \in \mathcal{M}_{n,p}$ are matrices, then their product AB is the
johnston-linear-matrix-algebra_F00386	38	■ 0.918	$m \times p$	$m \times p$	$m \times p$ matrix whose (i,j) -entry, for each $1 \leq i \leq m$ and $1 \leq j \leq p$, is
johnston-linear-matrix-algebra_F00387	38	■ 0.918	$1 \leq j \leq p$	$1 \leq j \leq p$	$m \times p$ matrix whose (i,j) -entry, for each $1 \leq i \leq m$ and $1 \leq j \leq p$, is
johnston-linear-matrix-algebra_F00388	38	■ 0.981	3×7	3×7	3×7 matrix.
johnston-linear-matrix-algebra_F00389	38	■ 0.554	3×2	3×2	Figure 1.13: A visualization of the multiplication of a 3×2 matrix A and a 2×4 matrix
johnston-linear-matrix-algebra_F00390	38	■ 0.554	2×4	2×4	Figure 1.13: A visualization of the multiplication of a 3×2 matrix A and a 2×4 matrix
johnston-linear-matrix-algebra_F00391	38	■ 1.000	$A C$	AC	b) AC ,
johnston-linear-matrix-algebra_F00392	38	■ 1.000	$B A$	BA	c) BA , and
johnston-linear-matrix-algebra_F00393	38	■ 1.000	$B C$	BC	d) BC .
johnston-linear-matrix-algebra_F00394	38	■ 1.000	$B : (1,2) \cdot (5,8) = 1 \cdot 5 + 2 \cdot 8 = 21$	$B : (1,2) \cdot (5,8) = 1 \cdot 5 + 2 \cdot 8 = 21$	the first column of B : $(1,2) \cdot (5,8) = 1 \cdot 5 + 2 \cdot 8 = 21$. Similarly
johnston-linear-matrix-algebra_F00395	38	■ 0.998	$A B^{\prime}$	AB'	AB' 's size consists of
johnston-linear-matrix-algebra_F00396	39	■ 1.000	$n \times p$	$n \times p$	and B is $n \times p$ then
johnston-linear-matrix-algebra_F00397	39	■ 1.000	C	C	b) AC does not exist since A has 2 columns but C has 3 rows, and those
johnston-linear-matrix-algebra_F00398	39	■ 1.000	$C : (5,6,7) \cdot (1,0,2) = 5 \cdot 1 + 6 \cdot 0 + 7 \cdot 2 = 19$	$C : (5,6,7) \cdot (1,0,2) = 5 \cdot 1 + 6 \cdot 0 + 7 \cdot 2 = 19$	the first column of C : $(5,6,7) \cdot (1,0,2) = 5 \cdot 1 + 6 \cdot 0 + 7 \cdot 2 = 19$.
johnston-linear-matrix-algebra_F00399	40	■ 1.000	A, B	A, B	Let A, B , and C be matrices (with sizes such that the multiplications and
johnston-linear-matrix-algebra_F00400	40	■ 1.000	$(A B) C = A(B C)$	$(AB)C = A(BC)$	a) $(AB)C = A(BC)$,
johnston-linear-matrix-algebra_F00401	40	■ 1.000	$A(B+C) = A B + A C$	$A(B+C) = AB + AC$	b) $A(B+C) = AB + AC$,
johnston-linear-matrix-algebra_F00402	40	■ 1.000	$(A+B) C = A C + B C$	$(A+B)C = AC + BC$	c) $(A+B)C = AC + BC$, and
johnston-linear-matrix-algebra_F00403	40	■ 1.000	$c(A B) = (c A) B$	$c(AB) = (cA)B$	d) $c(AB) = (cA)B$.
johnston-linear-matrix-algebra_F00404	40	■ 0.619	$A(B+C)$	$A(B+C)$	To this end we note that, for each i and j , the (i,j) -entry of $A(B+C)$ is
johnston-linear-matrix-algebra_F00405	40	■ 0.998	$A B + A C$	$AB + AC$	whereas the (i,j) -entry of $AB + AC$ is
johnston-linear-matrix-algebra_F00406	40	■ 1.000	$A B =$	$AB =$	It is worth stressing the fact that we did <i>not</i> list commutativity (i.e., $AB =$
johnston-linear-matrix-algebra_F00407	40	■ 1.000	$I_{\{n\}}$	I_n	size $n \times n$ then it is denoted by I_n (or if its size is clear from context or irrelevant,
johnston-linear-matrix-algebra_F00408	40	■ 1.000	I	I	we denote it just by I). For example, the identity matrices in $\mathcal{M}_2$ and $\mathcal{M}_3$ are

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johnston-linear-matrix-algebra_F00409	40	■ 1.000	$\mathcal{M}_{\{2\}}$	$\mathcal{M}_2$	we denote it just by I). For example, the identity matrices in $\mathcal{M}_2$ and $\mathcal{M}_3$ are
johnston-linear-matrix-algebra_F00410	40	■ 1.000	$\mathcal{M}_{\{3\}}$	$\mathcal{M}_3$	we denote it just by I). For example, the identity matrices in $\mathcal{M}_2$ and $\mathcal{M}_3$ are
johnston-linear-matrix-algebra_F00411	41	■ 1.000	$x \in \mathbb{R}$	$x \in \mathbb{R}$	Recall that if $x \in \mathbb{R}$
johnston-linear-matrix-algebra_F00412	41	■ 1.000	$x^{\{0\}}=1$	$x^0 = 1$	then $x^0 = 1$. The
johnston-linear-matrix-algebra_F00413	41	■ 1.000	$A^{\{0\}}=I$	$A^0 = I$	definition $A^0 = I$ is
johnston-linear-matrix-algebra_F00414	41	■ 1.000	$A I_{\{n\}}=A=I_{\{m\}} A$	$AI_n = A = I_m A$	If $A \in \mathcal{M}_{m,n}$ then $AI_n = A = I_m A$.
johnston-linear-matrix-algebra_F00415	41	■ 0.996	$A I_{\{n\}}$	AI_n	<i>Proof.</i> For the first equality, we recall that the (i, j) -entry of AI_n is the dot
johnston-linear-matrix-algebra_F00416	41	■ 1.000	$A I_{\{n\}}=A$	$AI_n = A$	and A coincide, so $AI_n = A$.
johnston-linear-matrix-algebra_F00417	41	■ 1.000	$A^{\{2\}}=A A, A^{\{3\}}=A A A$	$A^2 = AA, A^3 = AAA$	obtain powers of A . That is, we define $A^2 = AA, A^3 = AAA$, and in general
johnston-linear-matrix-algebra_F00418	41	■ 1.000	$A^{\{k+\ell\}}=A^{\{k\}} A^{\{\ell\}}$	$A^{k+\ell} = A^k A^\ell$	to, like $A^{k+\ell} = A^k A^\ell$ and $(A^k)^\ell = A^{k\ell}$ for all $k, \ell \geq 1$ (see Exercise 1.3.22). We
johnston-linear-matrix-algebra_F00419	41	■ 1.000	$\left(A^{\{k\}}\right)^{\{\ell\}}=A^{\{k \ell\}}$	$(A^k)^\ell = A^{k\ell}$	to, like $A^{k+\ell} = A^k A^\ell$ and $(A^k)^\ell = A^{k\ell}$ for all $k, \ell \geq 1$ (see Exercise 1.3.22). We
johnston-linear-matrix-algebra_F00420	41	■ 1.000	$k, \ell \geq 1$	$k, \ell \geq 1$	to, like $A^{k+\ell} = A^k A^\ell$ and $(A^k)^\ell = A^{k\ell}$ for all $k, \ell \geq 1$ (see Exercise 1.3.22). We
johnston-linear-matrix-algebra_F00421	41	■ 1.000	$k=0$	$k = 0$	also define $A^0 = I$ so that these same properties even hold if $k = 0$ or $\ell = 0$.
johnston-linear-matrix-algebra_F00422	41	■ 1.000	$\ell=0$	$\ell = 0$	also define $A^0 = I$ so that these same properties even hold if $k = 0$ or $\ell = 0$.
johnston-linear-matrix-algebra_F00423	41	■ 1.000	$A=\left[\begin{array}{cc} 2 & 1 \\ -1 & 3 \end{array}\right]$	$A = \begin{bmatrix} 2 & 1 \\ -1 & 3 \end{bmatrix}$	Compute each of the indicated matrix powers if $A = \begin{bmatrix} 2 & 1 \\ -1 & 3 \end{bmatrix}$.
johnston-linear-matrix-algebra_F00424	41	■ 1.000	$A^{\{2\}}$	A^2	a) A^2 ,
johnston-linear-matrix-algebra_F00425	41	■ 1.000	$A^{\{4\}}$	A^4	b) A^4 , and
johnston-linear-matrix-algebra_F00426	41	■ 1.000	$I^{\{7\}}$	I^7	c) I^7 .
johnston-linear-matrix-algebra_F00427	41	■ 1.000	$A^{\{4\}}=\left(A^{\{2\}}\right)^{\{2\}}$	$A^4 = (A^2)^2$	that $A^4 = (A^2)^2$, so we just need to square our answer from part (a):
johnston-linear-matrix-algebra_F00428	41	■ 1.000	$I^{\{k\}}=I$	$I^k = I$	c) We can repeatedly use Theorem 1.3.3 to see that $I^k = I$ for all non-
johnston-linear-matrix-algebra_F00429	41	■ 1.000	k	k	negative integers k , so $I^7 = I$.
johnston-linear-matrix-algebra_F00430	41	■ 1.000	$I^{\{7\}}=I$	$I^7 = I$	negative integers k , so $I^7 = I$.
johnston-linear-matrix-algebra_F00431	41	■ 1.000	$(A+B)^{\{2\}}$	$(A+B)^2$	when expanding the expression $(A+B)^2$, the reader might be tempted to write
johnston-linear-matrix-algebra_F00432	41	■ 1.000	$(A+B)^{\{2\}}=A^{\{2\}}+2 A B+B^{\{2\}}$	$(A+B)^2 = A^2 + 2AB + B^2$	$(A+B)^2 = A^2 + 2AB + B^2$. However, if we go through the calculation a bit
johnston-linear-matrix-algebra_F00433	42	■ 1.000	$(A+B)^{\{2\}}=$	$(A+B)^2 =$	$(A+B)^2 =$
johnston-linear-matrix-algebra_F00434	42	■ 1.000	$A^{\{2\}}+2 A B+B^{\{2\}}$	$A^2 + 2AB + B^2$	$A^2 + 2AB + B^2$ in

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johnston-linear-matrix-algebra_F00435	42	■ 1.000	$(A+I)^2=A^2+2A+I$	$(A+I)^2 = A^2 + 2A + I$	$(A+I)^2 = A^2 + 2A + I$,
johnston-linear-matrix-algebra_F00436	42	■ 0.994	$1 \times n$	$1 \times n$	vectors as $1 \times n$ or
johnston-linear-matrix-algebra_F00437	42	■ 1.000	$m \times 1$	$m \times 1$	$m \times 1$ matrices.
johnston-linear-matrix-algebra_F00438	42	■ 1.000	$A, B \in \mathcal{M}_n$	$A, B \in \mathcal{M}_n$	Suppose $A, B \in \mathcal{M}_n$. Expand and simplify the following expressions:
johnston-linear-matrix-algebra_F00439	42	■ 1.000	$(A+I)^2(A-I)$	$(A+I)^2(A-I)$	a) $(A+I)^2(A-I)$, and
johnston-linear-matrix-algebra_F00440	42	■ 1.000	$(A-A)(A+B-I)^7$	$(A-A)(A+B-I)^7$	b) $(A-A)(A+B-I)^7$.
johnston-linear-matrix-algebra_F00441	42	■ 1.000	$A+I$	$A+I$	a) We first square the $A+I$ and then multiply it by $A-I$:
johnston-linear-matrix-algebra_F00442	42	■ 1.000	$A-I$	$A-I$	a) We first square the $A+I$ and then multiply it by $A-I$:
johnston-linear-matrix-algebra_F00443	42	■ 1.000	$(A+B-I)^7$	$(A+B-I)^7$	b) We could expand $(A+B-I)^7$ (which would be ugly!) and then
johnston-linear-matrix-algebra_F00444	42	■ 1.000	$(A-A)(A+$	$(A-A)(A+$	that O times <i>any</i> matrix is equal to O . It follows that $(A-A)(A+$
johnston-linear-matrix-algebra_F00445	42	■ 0.950	$B-I)^7=O(A+B-I)^7=O$	$B-I)^7 = O(A+B-I)^7 = O$	$B-I)^7 = O(A+B-I)^7 = O$.
johnston-linear-matrix-algebra_F00446	42	■ 1.000	$\mathbf{v} \in \mathcal{M}_{n,1}$	$\mathbf{v} \in \mathcal{M}_{n,1}$	move vectors around. In particular, if $A \in \mathcal{M}_{m,n}$ and $\mathbf{v} \in \mathcal{M}_{n,1}$ then $A\mathbf{v} \in \mathcal{M}_{m,1}$,
johnston-linear-matrix-algebra_F00447	42	■ 1.000	$A \mathbf{v} \in \mathcal{M}_{m,1}$	$A\mathbf{v} \in \mathcal{M}_{m,1}$	move vectors around. In particular, if $A \in \mathcal{M}_{m,n}$ and $\mathbf{v} \in \mathcal{M}_{n,1}$ then $A\mathbf{v} \in \mathcal{M}_{m,1}$,
johnston-linear-matrix-algebra_F00448	42	■ 0.942	$A \mathbf{v}$	$A\mathbf{v}$	and since each of $\mathbf{v}$ and $A\mathbf{v}$ have just one column, we can think of them as vectors
johnston-linear-matrix-algebra_F00449	42	■ 1.000	$\mathbf{v}=(2,-1)$	$\mathbf{v} = (2, -1)$	so we think of the matrix A as transforming $\mathbf{v} = (2, -1)$ into $A\mathbf{v} = (0, 2)$, but
johnston-linear-matrix-algebra_F00450	42	■ 1.000	$A \mathbf{v}=(0,2)$	$A\mathbf{v} = (0, 2)$	so we think of the matrix A as transforming $\mathbf{v} = (2, -1)$ into $A\mathbf{v} = (0, 2)$, but
johnston-linear-matrix-algebra_F00451	43	■ 1.000	$[1,2,3] \in \mathbb{R}^3$	$[1, 2, 3] \in \mathbb{R}^3$	$[1, 2, 3] \in \mathbb{R}^3$ (even
johnston-linear-matrix-algebra_F00452	43	■ 1.000	$\mathcal{M}_{1,3}$	$\mathcal{M}_{1,3}$	lives in $\mathcal{M}_{1,3}$).
johnston-linear-matrix-algebra_F00453	43	■ 1.000	$\mathbf{v}=[1,2,3] \in \mathcal{M}_{1,3}$	$\mathbf{v} = [1, 2, 3] \in \mathcal{M}_{1,3}$	(as we did in the previous two sections). For example, $\mathbf{v} = [1, 2, 3] \in \mathcal{M}_{1,3}$ is a
johnston-linear-matrix-algebra_F00454	43	■ 1.000	$\mathbf{x}=(1,2,3) \in \mathbb{R}^3$	$\mathbf{x} = (1, 2, 3) \in \mathbb{R}^3$	is a column vector, and $\mathbf{x} = (1, 2, 3) \in \mathbb{R}^3$ is a vector for which we do not care
johnston-linear-matrix-algebra_F00455	43	■ 1.000	$\mathbf{v} \mapsto A \mathbf{v}$	$\mathbf{v} \mapsto A\mathbf{v}$	used as a function $\mathbf{v} \mapsto A\mathbf{v}$ that moves vectors around $\mathbb{R}^n$. We introduce this
johnston-linear-matrix-algebra_F00456	43	■ 0.774	A^T	A^T	denote by A^T , is the $n \times m$ matrix whose (i, j) -entry is $a_{j,i}$.
johnston-linear-matrix-algebra_F00457	43	■ 0.774	$n \times m$	$n \times m$	denote by A^T , is the $n \times m$ matrix whose (i, j) -entry is $a_{j,i}$.
johnston-linear-matrix-algebra_F00458	43	■ 0.774	$a_{j,i}$	$a_{j,i}$	denote by A^T , is the $n \times m$ matrix whose (i, j) -entry is $a_{j,i}$.
johnston-linear-matrix-algebra_F00459	43	■ 1.000	B^T	B^T	b) B^T ,
johnston-linear-matrix-algebra_F00460	43	■ 1.000	$(AB)^T$	$(AB)^T$	c) $(AB)^T$, and
johnston-linear-matrix-algebra_F00461	43	■ 1.000	$B^T A^T$	$B^T A^T$	d) $B^T A^T$.

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johnston-linear-matrix-algebra_F00462	43	■ 1.000	$A^T=\left[\begin{array}{ll}1 & 3 \\ 2 & 4\end{array}\right]$	$A^T = \begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix}$	a) $A^T = \begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix}$.
johnston-linear-matrix-algebra_F00463	44	■ 1.000	$\left[(A\ B)^T\right]_{i,j}$	$[(AB)^T]_{i,j}$	$[(AB)^T]_{i,j}$ means the
johnston-linear-matrix-algebra_F00464	44	■ 0.994	$\mathbf{v}^T \mathbf{w}$	$\mathbf{v}^T \mathbf{w}$	$\mathbf{v}^T \mathbf{w}$ is a 1×1 matrix,
johnston-linear-matrix-algebra_F00465	44	■ 0.994	1×1	1×1	$\mathbf{v}^T \mathbf{w}$ is a 1×1 matrix,
johnston-linear-matrix-algebra_F00466	44	■ 1.000	$B^T=\left[\begin{array}{cc}-1 & 0 \\ 1 & 1 \\ 1 & 0\end{array}\right]$	$B^T = \begin{bmatrix} -1 & 0 \\ 1 & 1 \\ 1 & 0 \end{bmatrix}$	b) $B^T = \begin{bmatrix} -1 & 0 \\ 1 & 1 \\ 1 & 0 \end{bmatrix}$.
johnston-linear-matrix-algebra_F00467	44	■ 0.971	$(A\ B)^T=\left(\left[\begin{array}{ll}1 & 2 \\ 3 & 4\end{array}\right]\left[\begin{array}{ccc}-1 & 1 & 1 \\ 0 & 1 & 0\end{array}\right]\right)^T=\left[\begin{array}{lll}-1 & 3 & 1 \\ -3 & 7 & 3\end{array}\right]$	$(AB)^T = \left(\left[\begin{array}{cc} 1 & 2 \\ 3 & 4 \end{array}\right] \left[\begin{array}{ccc} -1 & 1 & 1 \\ 0 & 1 & 0 \end{array}\right]\right)^T = \begin{bmatrix} -1 & 3 & 1 \\ -3 & 7 & 3 \end{bmatrix}^T = \begin{bmatrix} -1 & -3 \\ 3 & 7 \\ 1 & 3 \end{bmatrix}$	c) $(AB)^T = \left(\left[\begin{array}{cc} 1 & 2 \\ 3 & 4 \end{array}\right] \left[\begin{array}{ccc} -1 & 1 & 1 \\ 0 & 1 & 0 \end{array}\right]\right)^T = \begin{bmatrix} -1 & 3 & 1 \\ -3 & 7 & 3 \end{bmatrix}^T = \begin{bmatrix} -1 & -3 \\ 3 & 7 \\ 1 & 3 \end{bmatrix}$.
johnston-linear-matrix-algebra_F00468	44	■ 1.000	$B^T\ A^T=\left[\begin{array}{cc}-1 & 0 \\ 1 & 1 \\ 1 & 0\end{array}\right]\left[\begin{array}{ll}1 & 3 \\ 2 & 4\end{array}\right]=\left[\begin{array}{cc}-1 & -3 \\ 3 & 7 \\ 1 & 3\end{array}\right]$	$B^T A^T = \begin{bmatrix} -1 & 0 \\ 1 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix} = \begin{bmatrix} -1 & -3 \\ 3 & 7 \\ 1 & 3 \end{bmatrix}$	d) $B^T A^T = \begin{bmatrix} -1 & 0 \\ 1 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix} = \begin{bmatrix} -1 & -3 \\ 3 & 7 \\ 1 & 3 \end{bmatrix}$.
johnston-linear-matrix-algebra_F00469	44	■ 1.000	$(A\ B)^T=B^T\ A^T$	$(AB)^T = B^T A^T$	Parts (c) and (d) of the previous example suggest that $(AB)^T = B^T A^T$ might
johnston-linear-matrix-algebra_F00470	44	■ 1.000	$\left(A^T\right)^T=A$	$(A^T)^T = A$	a) $(A^T)^T = A$,
johnston-linear-matrix-algebra_F00471	44	■ 1.000	$(A+B)^T=A^T+B^T$	$(A+B)^T = A^T + B^T$	b) $(A+B)^T = A^T + B^T$,
johnston-linear-matrix-algebra_F00472	44	■ 1.000	$(c\ A)^T=c\ A^T$	$(cA)^T = cA^T$	d) $(cA)^T = cA^T$.
johnston-linear-matrix-algebra_F00473	44	■ 1.000	$\left[(A\ B)^T\right]_{i,j}=\left[B^T\ A^T\right]_{i,j}$	$[(AB)^T]_{i,j} = [B^T A^T]_{i,j}$	Since these two quantities are the same, we conclude that $[(AB)^T]_{i,j} = [B^T A^T]_{i,j}$
johnston-linear-matrix-algebra_F00474	45	■ 1.000	$C\ D$	CD	is CD , not DC .
johnston-linear-matrix-algebra_F00475	45	■ 1.000	$D\ C$	DC	is CD , not DC .
johnston-linear-matrix-algebra_F00476	45	■ 0.722	I_3	I_3	We then recognize two of the blocks of A as I_3 (the 3×3 identity matrix)
johnston-linear-matrix-algebra_F00477	45	■ 0.722	3×3	3×3	We then recognize two of the blocks of A as I_3 (the 3×3 identity matrix)
johnston-linear-matrix-algebra_F00478	45	■ 1.000	D	D	then we can write A and B in terms of C and D as follows:
johnston-linear-matrix-algebra_F00479	45	■ 1.000	$4 \cdot 5=20$	$4 \cdot 5 = 20$	the long way (i.e., computing $4 \cdot 5 = 20$ dot products of 6-dimensional vectors),
johnston-linear-matrix-algebra_F00480	46	■ 1.000	I_2	I_2	Recall that I_2 and
johnston-linear-matrix-algebra_F00481	46	■ 0.926	$I_2=\left[\begin{array}{ll}1 & 0 \\ 0 & 1\end{array}\right]$	$I_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$	$I_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$.

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johnston-linear-matrix-algebra_F00482	46	■ 1.000	$C D=\left[\begin{array}{cc} 5 & 4 \\ -7 & 1 \end{array}\right]$	$CD = \begin{bmatrix} 5 & 4 \\ -7 & 1 \end{bmatrix}$	It is not difficult to compute $CD = \begin{bmatrix} 5 & 4 \\ -7 & 1 \end{bmatrix}$, so it follows that
johnston-linear-matrix-algebra_F00483	46	■ 0.996	$\left[\begin{array}{ll} A & B \\ B & A \end{array}\right]^2$	$\begin{bmatrix} A & B \\ B & A \end{bmatrix}^2$	a) $\begin{bmatrix} A & B \\ B & A \end{bmatrix}^2$,
johnston-linear-matrix-algebra_F00484	46	■ 0.863	$\left[\begin{array}{ll} A & B \\ 0 & I_3 \end{array}\right]\left[\begin{array}{ll} A & A \\ 0 & A \\ I_2 & 0 \end{array}\right]$	$\begin{bmatrix} A & B \\ O & I_3 \end{bmatrix} \begin{bmatrix} A & A \\ O & A \\ I_2 & O \end{bmatrix}$	b) $\begin{bmatrix} A & B \\ O & I_3 \end{bmatrix} \begin{bmatrix} A & A \\ O & A \\ I_2 & O \end{bmatrix}$,
johnston-linear-matrix-algebra_F00485	46	■ 0.434	$\left[\begin{array}{ll} A & B \\ 0 & I_3 \end{array}\right]\left[\begin{array}{ll} A & A \\ 0 & A \end{array}\right]$	$\begin{bmatrix} A & B \\ O & I_3 \end{bmatrix} \begin{bmatrix} A & A \\ O & A \end{bmatrix}$	c) $\begin{bmatrix} A & B \\ O & I_3 \end{bmatrix} \begin{bmatrix} A & A \\ O & A \end{bmatrix}$, and
johnston-linear-matrix-algebra_F00486	46	■ 0.995	$\left[\begin{array}{ll} A & B \\ 0 & I_3 \end{array}\right]\left[\begin{array}{ll} B & 0 \\ I_3 & I_3 \end{array}\right]$	$\begin{bmatrix} A & B \\ O & I_3 \end{bmatrix} \begin{bmatrix} B & O \\ I_3 & I_3 \end{bmatrix}$	d) $\begin{bmatrix} A & B \\ O & I_3 \end{bmatrix} \begin{bmatrix} B & O \\ I_3 & I_3 \end{bmatrix}$.
johnston-linear-matrix-algebra_F00487	46	■ 0.969	$\left[\begin{array}{ll} A & B \\ B & A \end{array}\right]$	$\begin{bmatrix} A & B \\ B & A \end{bmatrix}$	a) The block matrix $\begin{bmatrix} A & B \\ B & A \end{bmatrix}$ itself makes no sense, since A has two
johnston-linear-matrix-algebra_F00488	47	■ 1.000	$A+B$	$AB+B$	The computation of the top-left block $AB+B$ did take some effort,
johnston-linear-matrix-algebra_F00489	47	■ 1.000	$\mathbf{a}_1, \mathbf{a}_2, \ldots, \mathbf{a}_n$	$\mathbf{a}_1, \mathbf{a}_2, \ldots, \mathbf{a}_n$	Suppose $A \in \mathcal{M}_{m,n}$ has columns $\mathbf{a}_1, \mathbf{a}_2, \ldots, \mathbf{a}_n$ (in that order) and $\mathbf{v} \in \mathbb{R}^n$
johnston-linear-matrix-algebra_F00490	47	■ 0.813	$n \times 1$	$n \times 1$	$\mathbf{v}$ as an $n \times 1$ matrix.
johnston-linear-matrix-algebra_F00491	47	■ 1.000	$\mathbf{b}_1, \mathbf{b}_2, \ldots, \mathbf{b}_p$	$\mathbf{b}_1, \mathbf{b}_2, \ldots, \mathbf{b}_p$	$\mathbf{b}_1, \mathbf{b}_2, \ldots, \mathbf{b}_p$ (in that order), then
johnston-linear-matrix-algebra_F00492	47	■ 1.000	$1 \times p$	$1 \times p$	B as a $1 \times p$ block matrix.
johnston-linear-matrix-algebra_F00493	48	■ 1.000	$A=\left[\begin{array}{ll} 1 & 2 \\ 3 & 4 \end{array}\right]$	$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$	Suppose $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} -1 & 1 & 1 \\ 0 & 1 & 0 \end{bmatrix}$. Compute AB via
johnston-linear-matrix-algebra_F00494	48	■ 1.000	$B=\left[\begin{array}{ccc} -1 & 1 & 1 \\ 0 & 1 & 0 \end{array}\right]$	$B = \begin{bmatrix} -1 & 1 & 1 \\ 0 & 1 & 0 \end{bmatrix}$	Suppose $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} -1 & 1 & 1 \\ 0 & 1 & 0 \end{bmatrix}$. Compute AB via
johnston-linear-matrix-algebra_F00495	48	■ 1.000	$2 B-3 A$	$2B-3A$	(b) $2B-3A$
johnston-linear-matrix-algebra_F00496	48	■ 0.881	$A-B$	$A-B$	*(c) $A-B$
johnston-linear-matrix-algebra_F00497	48	■ 1.000	$2(A+B)-A$	$2(A+B)-A$	(d) $2(A+B)-A$
johnston-linear-matrix-algebra_F00498	48	■ 1.000	$C D^T$	CD^T	(b) CD^T
johnston-linear-matrix-algebra_F00499	48	■ 1.000	$C^T D$	$C^T D$	(d) $C^T D$
johnston-linear-matrix-algebra_F00500	48	■ 1.000	$\left(C^T D\right)^2$	$(C^T D)^2$	(f) $(C^T D)^2$
johnston-linear-matrix-algebra_F00501	48	■ 1.000	$(C-D)^T\left(A-B^T\right)$	$(C-D)^T(A-B^T)$	(h) $(C-D)^T(A-B^T)$

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johnston-linear-matrix-algebra_F00502	48	■ 0.997	$(A+B)(C+D)$	$(A+B)(C+D)$	*(i) $(A+B)(C+D)$
johnston-linear-matrix-algebra_F00503	48	■ 1.000	$D^T(A^T+B^T)C$	$D^T(A^T+B)^TC$	(j) $D^T(A^T+B)^TC$
johnston-linear-matrix-algebra_F00504	49	■ 1.000	$A^T A$	$A^T A$	(b) The matrices $A^T A$ and AA^T are always defined, re-
johnston-linear-matrix-algebra_F00505	49	■ 1.000	$A A^T$	AA^T	(b) The matrices $A^T A$ and AA^T are always defined, re-
johnston-linear-matrix-algebra_F00506	49	■ 0.749	$(AB)^2 = A^2 B^2$	$(AB)^2 = A^2 B^2$	$(AB)^2 = A^2 B^2$.
johnston-linear-matrix-algebra_F00507	49	■ 1.000	$AB=0$	$AB=O$	(f) If A and B are matrices such that $AB=O$ and $A \neq O$,
johnston-linear-matrix-algebra_F00508	49	■ 1.000	$A \neq 0$	$A \neq O$	(f) If A and B are matrices such that $AB=O$ and $A \neq O$,
johnston-linear-matrix-algebra_F00509	49	■ 1.000	$B=0$	$B=O$	then $B=O$.
johnston-linear-matrix-algebra_F00510	49	■ 0.998	$A^2=I$	$A^2=I$	*(g) If A is a square matrix for which $A^2=I$, then $A=I$
johnston-linear-matrix-algebra_F00511	49	■ 0.998	$A=I$	$A=I$	*(g) If A is a square matrix for which $A^2=I$, then $A=I$
johnston-linear-matrix-algebra_F00512	49	■ 1.000	$A=-I$	$A=-I$	or $A=-I$.
johnston-linear-matrix-algebra_F00513	49	■ 1.000	$A^2=A$	$A^2=A$	(h) If A is a square matrix for which $A^2=A$, then either
johnston-linear-matrix-algebra_F00514	49	■ 1.000	$A=0$	$A=O$	$A=I$ or $A=O$.
johnston-linear-matrix-algebra_F00515	49	■ 1.000	$A \in \mathcal{M}_{n,m}$	$A \in \mathcal{M}_{n,m}$	*(i) If $A \in \mathcal{M}_{n,m}$ and $\mathbf{b} \in \mathbb{R}^m$ is a column vector, then
johnston-linear-matrix-algebra_F00516	49	■ 1.000	$\mathbf{b} \in \mathbb{R}^m$	$\mathbf{b} \in \mathbb{R}^m$	*(i) If $A \in \mathcal{M}_{n,m}$ and $\mathbf{b} \in \mathbb{R}^m$ is a column vector, then
johnston-linear-matrix-algebra_F00517	49	■ 1.000	$A \mathbf{b}$	$A\mathbf{b}$	$A\mathbf{b}$ is a linear combination of the rows of A .
johnston-linear-matrix-algebra_F00518	49	■ 1.000	$A \in \mathcal{M}_{2,2}, B \in \mathcal{M}_{2,5}$	$A \in \mathcal{M}_{2,2}, B \in \mathcal{M}_{2,5}$	1.3.4 Suppose $A \in \mathcal{M}_{2,2}, B \in \mathcal{M}_{2,5}$, and $C \in \mathcal{M}_{5,2}$ are
johnston-linear-matrix-algebra_F00519	49	■ 1.000	$C \in \mathcal{M}_{5,2}$	$C \in \mathcal{M}_{5,2}$	1.3.4 Suppose $A \in \mathcal{M}_{2,2}, B \in \mathcal{M}_{2,5}$, and $C \in \mathcal{M}_{5,2}$ are
johnston-linear-matrix-algebra_F00520	49	■ 1.000	A^7	A^7	(b) A^7
johnston-linear-matrix-algebra_F00521	49	■ 1.000	B^2	B^2	(d) B^2
johnston-linear-matrix-algebra_F00522	49	■ 1.000	$B B^T$	BB^T	(f) BB^T
johnston-linear-matrix-algebra_F00523	49	■ 0.999	$A B C$	ABC	*(g) ABC
johnston-linear-matrix-algebra_F00524	49	■ 1.000	$B^T B + C^T C$	$B^T B + C^T C$	(h) $B^T B + C^T C$
johnston-linear-matrix-algebra_F00525	49	■ 0.963	$A+B C$	$A+BC$	*(i) $A+BC$
johnston-linear-matrix-algebra_F00526	49	■ 1.000	$(A+C^T C)(B+C^T)$	$(A+C^T C)(B+C^T)$	(j) $(A+C^T C)(B+C^T)$
johnston-linear-matrix-algebra_F00527	49	■ 0.830	$A=\begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$	$A=\begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$	**1.3.5 Let $A=\begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$. Compute A^{1000} .
johnston-linear-matrix-algebra_F00528	49	■ 0.830	A^{1000}	A^{1000}	**1.3.5 Let $A=\begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$. Compute A^{1000} .
johnston-linear-matrix-algebra_F00529	49	■ 1.000	$A^2, A^3, \ldots$	$A^2, A^3, \ldots$	[Hint: Start by computing $A^2, A^3, \ldots$, and see if you notice

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johnston-linear-matrix-algebra_F00530	49	■ 1.000	$B=\left[\begin{array}{cc}1 & 2 \\ 2 & -1\end{array}\right]$	$B = \begin{bmatrix} 1 & 2 \\ 2 & -1 \end{bmatrix}$	1.3.6 Let $B = \begin{bmatrix} 1 & 2 \\ 2 & -1 \end{bmatrix}$. Compute B^{1000} .
johnston-linear-matrix-algebra_F00531	49	■ 1.000	$B^{\{1000\}}$	B^{1000}	1.3.6 Let $B = \begin{bmatrix} 1 & 2 \\ 2 & -1 \end{bmatrix}$. Compute B^{1000} .
johnston-linear-matrix-algebra_F00532	49	■ 1.000	$B^{\{2\}}, B^{\{3\}}, \ldots$	$B^2, B^3, \dots$	[Hint: Start by computing $B^2, B^3, \dots$, and see if you notice
johnston-linear-matrix-algebra_F00533	49	■ 0.972	$A=\left[\begin{array}{ll}1 & 1 \\ 0 & 1\end{array}\right]$	$A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$	*1.3.7 Let $A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$.
johnston-linear-matrix-algebra_F00534	49	■ 1.000	$A^{\{3\}}$	A^3	(b) Compute A^3 .
johnston-linear-matrix-algebra_F00535	49	■ 1.000	$A^{\{k\}}$	A^k	(c) Find a general formula for A^k (where $k \geq 0$ is an
johnston-linear-matrix-algebra_F00536	49	■ 1.000	$k \geq 0$	$k \geq 0$	(c) Find a general formula for A^k (where $k \geq 0$ is an
johnston-linear-matrix-algebra_F00537	49	■ 1.000	$B=\left[\begin{array}{lll}1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1\end{array}\right]$	$B = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$	1.3.8 Let $B = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$.
johnston-linear-matrix-algebra_F00538	49	■ 1.000	$B^{\{3\}}$	B^3	(b) Compute B^3 .
johnston-linear-matrix-algebra_F00539	49	■ 1.000	$B^{\{k\}}$	B^k	(c) Find a general formula for B^k (where $k \geq 0$ is an
johnston-linear-matrix-algebra_F00540	49	■ 1.000	$C=\left[\begin{array}{cc}1 & -1 \\ -1 & 1\end{array}\right]$	$C = \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$	1.3.9 Let $C = \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$.
johnston-linear-matrix-algebra_F00541	49	■ 1.000	$C^{\{2\}}$	C^2	(a) Compute C^2 .
johnston-linear-matrix-algebra_F00542	49	■ 1.000	$C^{\{3\}}$	C^3	(b) Compute C^3 .
johnston-linear-matrix-algebra_F00543	49	■ 1.000	$C^{\{k\}}$	C^k	(c) Find a general formula for C^k (where $k \geq 0$ is an
johnston-linear-matrix-algebra_F00544	49	■ 0.342	$h \in \mathbb{R}$	$h \in \mathbb{R}$	 1.3.10 Let $h \in \mathbb{R}$ and
johnston-linear-matrix-algebra_F00545	49	■ 1.000	$A^{\{2500\}}$	A^{2500}	Use computer software to compute A^{2500} when $h = 0.39$,
johnston-linear-matrix-algebra_F00546	49	■ 1.000	$h=0.39$	$h = 0.39$	Use computer software to compute A^{2500} when $h = 0.39$,
johnston-linear-matrix-algebra_F00547	49	■ 1.000	$h=0.40$	$h = 0.40$	$h = 0.40$, and $h = 0.41$. What do you think happens as
johnston-linear-matrix-algebra_F00548	49	■ 1.000	$h=0.41$	$h = 0.41$	$h = 0.40$, and $h = 0.41$. What do you think happens as
johnston-linear-matrix-algebra_F00549	49	■ 0.948	$J_{\{n\}}$	J_n	*1.3.11 Suppose J_n is the $n \times n$ matrix all of whose entries
johnston-linear-matrix-algebra_F00550	49	■ 1.000	$J_{\{n\}}^{\{2\}}$	J_n^2	Compute J_n^2 .
johnston-linear-matrix-algebra_F00551	49	■ 1.000	$A \in \mathcal{M}_{\{2,2\}}, B \in \mathcal{M}_{\{2,3\}}$	$A \in \mathcal{M}_{2,2}, B \in \mathcal{M}_{2,3}$	1.3.12 Suppose $A \in \mathcal{M}_{2,2}, B \in \mathcal{M}_{2,3}$, and $C \in \mathcal{M}_{3,4}$ are
johnston-linear-matrix-algebra_F00552	49	■ 1.000	$C \in \mathcal{M}_{\{3,4\}}$	$C \in \mathcal{M}_{3,4}$	1.3.12 Suppose $A \in \mathcal{M}_{2,2}, B \in \mathcal{M}_{2,3}$, and $C \in \mathcal{M}_{3,4}$ are
johnston-linear-matrix-algebra_F00553	49	■ 0.885	$\left[\begin{array}{cc}A & A \\ A & A\end{array}\right]^{\{2\}}$	$\begin{bmatrix} A & A \\ A & A \end{bmatrix}^2$	*(a) $\begin{bmatrix} A & A \\ A & A \end{bmatrix}^2$

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johnston-linear-matrix-algebra-F00554	48	■ 0.999	$\left[\begin{array}{cc}C & I_3 \\ I_4 & O\end{array}\right]\left[\begin{array}{cc}O & I_3 \\ I_4 & C^T\end{array}\right]$	$\left[\begin{array}{cc}C & I_3 \\ I_4 & O\end{array}\right]\left[\begin{array}{cc}O & I_3 \\ I_4 & C^T\end{array}\right]$	(b) $\left[\begin{array}{cc}C & I_3 \\ I_4 & O\end{array}\right]\left[\begin{array}{cc}O & I_3 \\ I_4 & C^T\end{array}\right]$
johnston-linear-matrix-algebra-F00555	49	■ 1.000	$\left[\begin{array}{cc}A & B \\ B^T & I_3\end{array}\right]\left[\begin{array}{cc}O & C^T \\ B^T & I_3\end{array}\right]$	$\left[\begin{array}{cc}A & B \\ B^T & I_3\end{array}\right]\left[\begin{array}{cc}O & C^T \\ B^T & I_3\end{array}\right]$	*(c) $\left[\begin{array}{cc}A & B \\ B^T & I_3\end{array}\right]\left[\begin{array}{cc}O & C^T \\ B^T & I_3\end{array}\right]$
johnston-linear-matrix-algebra-F00556	49	■ 1.000	$\left[\begin{array}{cc}A & B \\ O & C^T\end{array}\right]\left[\begin{array}{cc}I_2 & B \\ B^T & O\end{array}\right]$	$\left[\begin{array}{cc}A & B \\ O & C^T\end{array}\right]\left[\begin{array}{cc}I_2 & B \\ B^T & O\end{array}\right]$	(d) $\left[\begin{array}{cc}A & B \\ O & C^T\end{array}\right]\left[\begin{array}{cc}I_2 & B \\ B^T & O\end{array}\right]$
johnston-linear-matrix-algebra-F00557	49	■ 1.000	$A=\left[\begin{array}{ll l}1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ \hline 2 & 1 & 2 & 1 \\ 2 & 1 & 2 & 1\end{array}\right], B=\left[\begin{array}{ll l}1 & 2 & 2 & 1 \\ 2 & 1 & 1 & 2 \\ \hline 2 & 1 & 1 & 2 \\ 2 & 1 & 2 & 1\end{array}\right]$	$A=\left[\begin{array}{ll l}1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ \hline 2 & 1 & 1 & 2 \\ 1 & 2 & 2 & 1\end{array}\right], B=\left[\begin{array}{ll l}1 & 2 & 2 & 1 \\ 2 & 1 & 1 & 2 \\ \hline 2 & 1 & 1 & 2 \\ 1 & 2 & 2 & 1\end{array}\right]$	*(a) $A=\left[\begin{array}{ll l}1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ \hline 2 & 1 & 1 & 2 \\ 1 & 2 & 2 & 1\end{array}\right], B=\left[\begin{array}{ll l}1 & 2 & 2 & 1 \\ 2 & 1 & 1 & 2 \\ \hline 2 & 1 & 1 & 2 \\ 1 & 2 & 2 & 1\end{array}\right]$
johnston-linear-matrix-algebra-F00558	49	■ 0.999	$A=\left[\begin{array}{ll l}1 & 2 & 0 \\ 3 & 1 & 0 \\ \hline 0 & 0 & 2\end{array}\right], B=\left[\begin{array}{ll l}-1 & 1 & 0 & 0 \\ 2 & 2 & 0 & 0 \\ \hline 0 & 0 & 1 & 1\end{array}\right]$	$A=\left[\begin{array}{ll l}1 & 2 & 0 \\ 3 & 1 & 0 \\ \hline 0 & 0 & 2\end{array}\right], B=\left[\begin{array}{ll l}-1 & 1 & 0 & 0 \\ 2 & 2 & 0 & 0 \\ \hline 0 & 0 & 1 & 1\end{array}\right]$	(b) $A=\left[\begin{array}{ll l}1 & 2 & 0 \\ 3 & 1 & 0 \\ \hline 0 & 0 & 2\end{array}\right], B=\left[\begin{array}{ll l}-1 & 1 & 0 & 0 \\ 2 & 2 & 0 & 0 \\ \hline 0 & 0 & 1 & 1\end{array}\right]$
johnston-linear-matrix-algebra-F00559	49	■ 1.000	$A=\left[\begin{array}{lll l}1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \\ \hline 0 & 0 & 0 & 2\end{array}\right], B=\left[\begin{array}{lll l}1 & 2 & 3 & 4 \\ 3 & 4 & 5 & 1 \\ \hline 1 & 2 & 3 & 4 \\ 4 & 5 & 1 & 2\end{array}\right]$	$A=\left[\begin{array}{lll l}1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \\ \hline 0 & 0 & 0 & 2\end{array}\right], B=\left[\begin{array}{lll l}1 & 2 & 3 \\ 2 & 3 & 4 \\ 3 & 4 & 5 \\ \hline 1 & 2 & 3\end{array}\right]$	*(c) $A=\left[\begin{array}{lll l}1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \\ \hline 0 & 0 & 0 & 2\end{array}\right], B=\left[\begin{array}{lll l}1 & 2 & 3 \\ 2 & 3 & 4 \\ 3 & 4 & 5 \\ \hline 1 & 2 & 3\end{array}\right]$
johnston-linear-matrix-algebra-F00560	50	■ 0.991	7×7	7×7	1.3.14 Consider the 7×7 matrix
johnston-linear-matrix-algebra-F00561	50	■ 0.999	$B \in \mathcal{M}_{r,p}$	$B \in \mathcal{M}_{r,p}$	**1.3.15 Suppose $A \in \mathcal{M}_{m,n}$ and $B \in \mathcal{M}_{r,p}$ are matrices.
johnston-linear-matrix-algebra-F00562	50	■ 1.000	m, n, r	m, n, r	(a) What restrictions must be placed on m, n, r , and p to
johnston-linear-matrix-algebra-F00563	50	■ 1.000	p	p	(a) What restrictions must be placed on m, n, r , and p to
johnston-linear-matrix-algebra-F00564	50	■ 1.000	$\mathbf{v}^T A \mathbf{w} = \mathbf{v}^T B \mathbf{w}$	$\mathbf{v}^T A \mathbf{w} = \mathbf{v}^T B \mathbf{w}$	1.3.16 Let $A, B \in \mathcal{M}_{m,n}$. Show that $\mathbf{v}^T A \mathbf{w} = \mathbf{v}^T B \mathbf{w}$ for
johnston-linear-matrix-algebra-F00565	50	■ 1.000	$\mathbf{v} \in \mathbb{R}^m$	$\mathbf{v} \in \mathbb{R}^m$	all $\mathbf{v} \in \mathbb{R}^m$ and $\mathbf{w} \in \mathbb{R}^n$ if and only if $A = B$.
johnston-linear-matrix-algebra-F00566	50	■ 1.000	$\mathbf{w} \in \mathbb{R}^n$	$\mathbf{w} \in \mathbb{R}^n$	all $\mathbf{v} \in \mathbb{R}^m$ and $\mathbf{w} \in \mathbb{R}^n$ if and only if $A = B$.
johnston-linear-matrix-algebra-F00567	50	■ 1.000	$A = B$	$A = B$	all $\mathbf{v} \in \mathbb{R}^m$ and $\mathbf{w} \in \mathbb{R}^n$ if and only if $A = B$.
johnston-linear-matrix-algebra-F00568	50	■ 1.000	$\mathbf{e}_i^T A \mathbf{e}_j$	$\mathbf{e}_i^T A \mathbf{e}_j$	[Hint: What is $\mathbf{e}_i^T A \mathbf{e}_j$?]
johnston-linear-matrix-algebra-F00569	50	■ 1.000	$\mathbf{x} \cdot (A \mathbf{y}) = (A^T \mathbf{x}) \cdot \mathbf{y}$	$\mathbf{x} \cdot (A \mathbf{y}) = (A^T \mathbf{x}) \cdot \mathbf{y}$	(a) Show that $\mathbf{x} \cdot (A \mathbf{y}) = (A^T \mathbf{x}) \cdot \mathbf{y}$ for all $\mathbf{x} \in \mathbb{R}^m$ and
johnston-linear-matrix-algebra-F00570	50	■ 1.000	$\mathbf{x} \in \mathbb{R}^m$	$\mathbf{x} \in \mathbb{R}^m$	(a) Show that $\mathbf{x} \cdot (A \mathbf{y}) = (A^T \mathbf{x}) \cdot \mathbf{y}$ for all $\mathbf{x} \in \mathbb{R}^m$ and
johnston-linear-matrix-algebra-F00571	50	■ 1.000	$\mathbf{x} \cdot (A \mathbf{y}) = (B \mathbf{x}) \cdot \mathbf{y}$	$\mathbf{x} \cdot (A \mathbf{y}) = (B \mathbf{x}) \cdot \mathbf{y}$	(b) Show that if $\mathbf{x} \cdot (A \mathbf{y}) = (B \mathbf{x}) \cdot \mathbf{y}$ for all $\mathbf{x} \in \mathbb{R}^m$ and
johnston-linear-matrix-algebra-F00572	50	■ 1.000	$B = A^T$	$B = A^T$	$\mathbf{y} \in \mathbb{R}^n$ then $B = A^T$.

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johnston-linear-matrix-algebra-F00573	50	■ 0.919	$\mathbf{e}_i$	$\mathbf{e}_i$	**1.3.18 Let $\mathbf{e}_i$ be the standard basis vector with 1 in its
johnston-linear-matrix-algebra-F00574	50	■ 1.000	$A \mathbf{e}_i$	$A\mathbf{e}_i$	(a) Show that $A\mathbf{e}_i$ is the i -th column of A .
johnston-linear-matrix-algebra-F00575	50	■ 1.000	$\mathbf{e}_i^T A$	$\mathbf{e}_i^T A$	(b) Show that $\mathbf{e}_i^T A$ is the i -th row of A .
johnston-linear-matrix-algebra-F00576	50	■ 1.000	$A \mathbf{v} = B \mathbf{v}$	$A\mathbf{v} = B\mathbf{v}$	(d) Use part (a) to show that $A\mathbf{v} = B\mathbf{v}$ for all $\mathbf{v} \in \mathbb{R}^n$ if
johnston-linear-matrix-algebra-F00577	50	■ 1.000	$I_m A = A$	$I_m A = A$	That is, show that if $A \in \mathcal{M}_{m,n}$ then $I_m A = A$.
johnston-linear-matrix-algebra-F00578	50	■ 1.000	$A \in \mathcal{M}_n$	$A \in \mathcal{M}_n$	**1.3.22 Suppose $A \in \mathcal{M}_n$ and k and ℓ are non-negative
johnston-linear-matrix-algebra-F00579	50	■ 1.000	ℓ	ℓ	**1.3.22 Suppose $A \in \mathcal{M}_n$ and k and ℓ are non-negative
johnston-linear-matrix-algebra-F00580	50	■ 1.000	$A_1, A_2, \dots, A_k$	$A_1, A_2, \dots, A_k$	1.3.24 Suppose that $A_1, A_2, \dots, A_k$ are matrices whose
johnston-linear-matrix-algebra-F00581	50	■ 1.000	$A_1 A_2 \dots A_k$	$A_1 A_2 \dots A_k$	sizes are such that the product $A_1 A_2 \dots A_k$ makes sense.
johnston-linear-matrix-algebra-F00582	50	■ 1.000	$(A_1 A_2 \dots A_k)^T = A_1^T A_2^T \dots A_k^T$	$(A_1 A_2 \dots A_k)^T = A_1^T A_2^T \dots A_k^T$	Prove that $(A_1 A_2 \dots A_k)^T = A_k^T \dots A_2^T A_1^T$.
johnston-linear-matrix-algebra-F00583	50	■ 1.000	$(A^n)^T = (A^T)^n$	$(A^n)^T = (A^T)^n$	1.3.25 Suppose that $A \in \mathcal{M}_n$. Show that $(A^n)^T = (A^T)^n$
johnston-linear-matrix-algebra-F00584	50	■ 1.000	$n \geq 0$	$n \geq 0$	for all integers $n \geq 0$.
johnston-linear-matrix-algebra-F00585	50	■ 1.000	$\mathbf{a}_j^T$	$\mathbf{a}_j^T$	and let $\mathbf{a}_j^T$ be the j -th row of A . Show that
johnston-linear-matrix-algebra-F00586	51	■ 1.000	$T: \mathbb{R}^n \rightarrow \mathbb{R}^m$	$T: \mathbb{R}^n \rightarrow \mathbb{R}^m$	$T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ means
johnston-linear-matrix-algebra-F00587	51	■ 1.000	T	T	that T is a function
johnston-linear-matrix-algebra-F00588	51	■ 1.000	$\mathbb{R}^m$	$\mathbb{R}^m$	$\mathbb{R}^m$. More generally,
johnston-linear-matrix-algebra-F00589	51	■ 1.000	$f: X \rightarrow Y$	$f: X \rightarrow Y$	$f: X \rightarrow Y$ means
johnston-linear-matrix-algebra-F00590	51	■ 1.000	X	X	the set X to the set Y .
johnston-linear-matrix-algebra-F00591	51	■ 1.000	Y	Y	the set X to the set Y .
johnston-linear-matrix-algebra-F00592	51	■ 1.000	$T(\mathbf{0}) = \mathbf{0}$	$T(\mathbf{0}) = \mathbf{0}$	case that $T(\mathbf{0}) = \mathbf{0}$
johnston-linear-matrix-algebra-F00593	51	■ 1.000	$T(\mathbf{v} + \mathbf{w}) = T(\mathbf{v}) + T(\mathbf{w})$	$T(\mathbf{v} + \mathbf{w}) = T(\mathbf{v}) + T(\mathbf{w})$	a) $T(\mathbf{v} + \mathbf{w}) = T(\mathbf{v}) + T(\mathbf{w})$ for all vectors $\mathbf{v}, \mathbf{w} \in \mathbb{R}^n$, and
johnston-linear-matrix-algebra-F00594	51	■ 1.000	$T(c \mathbf{v}) = c T(\mathbf{v})$	$T(c\mathbf{v}) = cT(\mathbf{v})$	b) $T(c\mathbf{v}) = cT(\mathbf{v})$ for all vectors $\mathbf{v} \in \mathbb{R}^n$ and all scalars $c \in \mathbb{R}$.
johnston-linear-matrix-algebra-F00595	51	■ 1.000	$A \mathbf{v} \in \mathbb{R}^m$	$A\mathbf{v} \in \mathbb{R}^m$	sends $\mathbf{v} \in \mathbb{R}^n$ to $A\mathbf{v} \in \mathbb{R}^m$ preserves vector addition (i.e., $A(\mathbf{v} + \mathbf{w}) = A\mathbf{v} + A\mathbf{w}$)
johnston-linear-matrix-algebra-F00596	51	■ 1.000	$A(\mathbf{v} + \mathbf{w}) = A\mathbf{v} + A\mathbf{w}$	$A(\mathbf{v} + \mathbf{w}) = A\mathbf{v} + A\mathbf{w}$	sends $\mathbf{v} \in \mathbb{R}^n$ to $A\mathbf{v} \in \mathbb{R}^m$ preserves vector addition (i.e., $A(\mathbf{v} + \mathbf{w}) = A\mathbf{v} + A\mathbf{w}$)
johnston-linear-matrix-algebra-F00597	51	■ 0.999	$A(c \mathbf{v}) = c(A\mathbf{v})$	$A(c\mathbf{v}) = c(A\mathbf{v})$	and scalar multiplication (i.e., $A(c\mathbf{v}) = c(A\mathbf{v})$).
johnston-linear-matrix-algebra-F00598	51	■ 1.000	$T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$	$T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$	transformation $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ can rotate, stretch, shrink, and/or reflect that grid,
johnston-linear-matrix-algebra-F00599	51	■ 1.000	$T(\mathbf{v})$	$T(\mathbf{v})$	then $T(\mathbf{v})$ is located in the x -th cell in the direction of $T(\mathbf{e}_1)$ and the y -th cell
johnston-linear-matrix-algebra-F00600	51	■ 1.000	$T(\mathbf{e}_1)$	$T(\mathbf{e}_1)$	then $T(\mathbf{v})$ is located in the x -th cell in the direction of $T(\mathbf{e}_1)$ and the y -th cell

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johnston-linear-matrix-algebra-F00601	51	■ 1.000	$T(\mathbf{e}_2)$	$T(\mathbf{e}_2)$	in the direction of $T(\mathbf{e}_2)$ (again, see Figure 1.14).
johnston-linear-matrix-algebra-F00602	51	■ 1.000	$T(v_1, v_2) = (1 + v_1, 2 + v_2)$	$T(v_1, v_2) = (1 + v_1, 2 + v_2)$	a) $T(v_1, v_2) = (1 + v_1, 2 + v_2)$,
johnston-linear-matrix-algebra-F00603	51	■ 1.000	$T(v_1, v_2) = (v_1 - v_2, v_1 v_2)$	$T(v_1, v_2) = (v_1 - v_2, v_1 v_2)$	b) $T(v_1, v_2) = (v_1 - v_2, v_1 v_2)$, and
johnston-linear-matrix-algebra-F00604	51	■ 1.000	$T(v_1, v_2) = (v_1 - v_2, v_1 + v_2)$	$T(v_1, v_2) = (v_1 - v_2, v_1 + v_2)$	c) $T(v_1, v_2) = (v_1 - v_2, v_1 + v_2)$.
johnston-linear-matrix-algebra-F00605	51	■ 1.000	$2T(0, 0) = 2(1, 2) = (2, 4)$	$2T(0, 0) = 2(1, 2) = (2, 4)$	that $2T(0, 0) = 2(1, 2) = (2, 4)$, but $T(2(0, 0)) = T(0, 0) = (1, 2)$.
johnston-linear-matrix-algebra-F00606	51	■ 1.000	$T(2(0, 0)) = T(0, 0) = (1, 2)$	$T(2(0, 0)) = T(0, 0) = (1, 2)$	that $2T(0, 0) = 2(1, 2) = (2, 4)$, but $T(2(0, 0)) = T(0, 0) = (1, 2)$.
johnston-linear-matrix-algebra-F00607	52	■ 1.000	$2T(1, 1) = 2(0, 1) = (0, 2)$	$2T(1, 1) = 2(0, 1) = (0, 2)$	that $2T(1, 1) = 2(0, 1) = (0, 2)$, but $T(2(1, 1)) = T(2, 2) = (0, 4)$.
johnston-linear-matrix-algebra-F00608	52	■ 1.000	$T(2(1, 1)) = T(2, 2) = (0, 4)$	$T(2(1, 1)) = T(2, 2) = (0, 4)$	that $2T(1, 1) = 2(0, 1) = (0, 2)$, but $T(2(1, 1)) = T(2, 2) = (0, 4)$.
johnston-linear-matrix-algebra-F00609	52	■ 1.000	$\sqrt{2}$	$\sqrt{2}$	factor of $\sqrt{2}$:
johnston-linear-matrix-algebra-F00610	53	■ 1.000	$\mathbf{v} = v_1 \mathbf{e}_1 + v_2 \mathbf{e}_2 + \dots +$	$\mathbf{v} = v_1 \mathbf{e}_1 + v_2 \mathbf{e}_2 + \dots +$	for every vector $\mathbf{v} = (v_1, v_2, \dots, v_n) \in \mathbb{R}^n$, we can write $\mathbf{v} = v_1 \mathbf{e}_1 + v_2 \mathbf{e}_2 + \dots +$
johnston-linear-matrix-algebra-F00611	53	■ 1.000	$v_n \mathbf{e}_n$	$v_n \mathbf{e}_n$	$v_n \mathbf{e}_n$, where $\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n$ are the standard basis vectors. By using the fact that
johnston-linear-matrix-algebra-F00612	53	■ 1.000	$\mathbf{v} \in \mathbb{R}^2$	$\mathbf{v} \in \mathbb{R}^2$	which is exactly what we said before: if $\mathbf{v} \in \mathbb{R}^2$ extends a distance of v_1 in
johnston-linear-matrix-algebra-F00613	53	■ 1.000	v_1	v_1	which is exactly what we said before: if $\mathbf{v} \in \mathbb{R}^2$ extends a distance of v_1 in
johnston-linear-matrix-algebra-F00614	53	■ 1.000	v_2	v_2	the direction of $\mathbf{e}_1$ and a distance of v_2 in the direction of $\mathbf{e}_2$, then $T(\mathbf{v})$ ex-
johnston-linear-matrix-algebra-F00615	53	■ 1.000	$\mathbf{e}_3, \dots, \mathbf{e}_n$	$\mathbf{e}_3, \dots, \mathbf{e}_n$	similarly for $\mathbf{e}_3, \dots, \mathbf{e}_n$ in higher dimensions). We thus learn the very impor-
johnston-linear-matrix-algebra-F00616	53	■ 1.000	$T(\mathbf{e}_1), T(\mathbf{e}_2), \dots, T(\mathbf{e}_n)$	$T(\mathbf{e}_1), T(\mathbf{e}_2), \dots, T(\mathbf{e}_n)$	$T(\mathbf{e}_1), T(\mathbf{e}_2), \dots, T(\mathbf{e}_n)$: if we know what T does to the standard basis vectors,
johnston-linear-matrix-algebra-F00617	53	■ 1.000	$T(\mathbf{e}_1) = (1, 1)$	$T(\mathbf{e}_1) = (1, 1)$	Suppose $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ is a linear transformation for which $T(\mathbf{e}_1) = (1, 1)$
johnston-linear-matrix-algebra-F00618	53	■ 1.000	$T(\mathbf{e}_2) = (-1, 1)$	$T(\mathbf{e}_2) = (-1, 1)$	and $T(\mathbf{e}_2) = (-1, 1)$.
johnston-linear-matrix-algebra-F00619	53	■ 1.000	$T(2, 3)$	$T(2, 3)$	a) Compute $T(2, 3)$, and
johnston-linear-matrix-algebra-F00620	53	■ 1.000	$T(v_1, v_2)$	$T(v_1, v_2)$	b) Find a general formula for $T(v_1, v_2)$.
johnston-linear-matrix-algebra-F00621	53	■ 1.000	$(2, 3) = 2\mathbf{e}_1 + 3\mathbf{e}_2$	$(2, 3) = 2\mathbf{e}_1 + 3\mathbf{e}_2$	a) Since $(2, 3) = 2\mathbf{e}_1 + 3\mathbf{e}_2$, we know that $T(2, 3) = T(2\mathbf{e}_1 + 3\mathbf{e}_2) =$
johnston-linear-matrix-algebra-F00622	53	■ 1.000	$T(2, 3) = T(2\mathbf{e}_1 + 3\mathbf{e}_2) =$	$T(2, 3) = T(2\mathbf{e}_1 + 3\mathbf{e}_2) =$	a) Since $(2, 3) = 2\mathbf{e}_1 + 3\mathbf{e}_2$, we know that $T(2, 3) = T(2\mathbf{e}_1 + 3\mathbf{e}_2) =$
johnston-linear-matrix-algebra-F00623	53	■ 1.000	$2T(\mathbf{e}_1) + 3T(\mathbf{e}_2) = 2(1, 1) + 3(-1, 1) = (-1, 5)$	$2T(\mathbf{e}_1) + 3T(\mathbf{e}_2) = 2(1, 1) + 3(-1, 1) = (-1, 5)$	$2T(\mathbf{e}_1) + 3T(\mathbf{e}_2) = 2(1, 1) + 3(-1, 1) = (-1, 5)$.
johnston-linear-matrix-algebra-F00624	53	■ 1.000	$(v_1, v_2) = v_1 \mathbf{e}_1 + v_2 \mathbf{e}_2$	$(v_1, v_2) = v_1 \mathbf{e}_1 + v_2 \mathbf{e}_2$	b) We can mimic the computation from part (a): $(v_1, v_2) = v_1 \mathbf{e}_1 + v_2 \mathbf{e}_2$,
johnston-linear-matrix-algebra-F00625	54	■ 1.000	$[T]$	$[T]$	In other words, $[T]$
johnston-linear-matrix-algebra-F00626	54	■ 1.000	$T(\mathbf{e}_2), \dots, T(\mathbf{e}_n)$	$T(\mathbf{e}_2), \dots, T(\mathbf{e}_n)$	$T(\mathbf{e}_2), \dots, T(\mathbf{e}_n)$ as its
johnston-linear-matrix-algebra-F00627	54	■ 1.000	$T(v_1, v_2) = (v_1 -$	$T(v_1, v_2) = (v_1 -$	effect as the linear transformation $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ defined by $T(v_1, v_2) = (v_1 -$

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johnston-linear-matrix-algebra-F00628	54	■ 0.951	$\mathbf{v}_2, \mathbf{v}_1 + \mathbf{v}_2$	$v_2, v_1 + v_2$	$v_2, v_1 + v_2$) (i.e., the linear transformation from Examples 1.4.1(c) and 1.4.2(b)).
johnston-linear-matrix-algebra-F00629	54	■ 1.000	$[T] \in \mathcal{M}_{m,n}$	$[T] \in \mathcal{M}_{m,n}$	exists a matrix $[T] \in \mathcal{M}_{m,n}$ such that
johnston-linear-matrix-algebra-F00630	54	■ 0.623	$[T] \mathbf{v}$	$[T]\mathbf{v}$	function that sends $\mathbf{v}$ to $[T]\mathbf{v}$ is a linear transformation. We thus only have to
johnston-linear-matrix-algebra-F00631	54	■ 1.000	$[T] \mathbf{v} = T(\mathbf{v})$	$[T]\mathbf{v} = T(\mathbf{v})$	satisfies $[T]\mathbf{v} = T(\mathbf{v})$, and no other matrix has this property.
johnston-linear-matrix-algebra-F00632	54	■ 1.000	$T(\mathbf{v}) = A \mathbf{v}$	$T(\mathbf{v}) = A\mathbf{v}$	the property that $T(\mathbf{v}) = A\mathbf{v}$ for all $\mathbf{v} \in \mathbb{R}^n$. Then $T(\mathbf{v}) = [T]\mathbf{v}$ and $T(\mathbf{v}) = A\mathbf{v}$,
johnston-linear-matrix-algebra-F00633	54	■ 1.000	$T(\mathbf{v}) = [T] \mathbf{v}$	$T(\mathbf{v}) = [T]\mathbf{v}$	the property that $T(\mathbf{v}) = A\mathbf{v}$ for all $\mathbf{v} \in \mathbb{R}^n$. Then $T(\mathbf{v}) = [T]\mathbf{v}$ and $T(\mathbf{v}) = A\mathbf{v}$,
johnston-linear-matrix-algebra-F00634	54	■ 1.000	$[T] \mathbf{v} = A \mathbf{v}$	$[T]\mathbf{v} = A\mathbf{v}$	so $[T]\mathbf{v} = A\mathbf{v}$ for all $\mathbf{v} \in \mathbb{R}^n$ as well. It follows from Exercise 1.3.18(d) that
johnston-linear-matrix-algebra-F00635	54	■ 1.000	$A = [T]$	$A = [T]$	$A = [T]$, which completes the proof.
johnston-linear-matrix-algebra-F00636	55	■ 1.000	$T(\mathbf{v}_1, \mathbf{v}_2) = (\mathbf{v}_1 + 2\mathbf{v}_2, 3\mathbf{v}_1 + 4\mathbf{v}_2)$	$T(v_1, v_2) = (v_1 + 2v_2, 3v_1 + 4v_2)$	a) $T(v_1, v_2) = (v_1 + 2v_2, 3v_1 + 4v_2)$, and
johnston-linear-matrix-algebra-F00637	55	■ 1.000	$T(\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3) = (3\mathbf{v}_1 - \mathbf{v}_2 + \mathbf{v}_3, 2\mathbf{v}_1 + 4\mathbf{v}_2 - 2\mathbf{v}_3)$	$T(v_1, v_2, v_3) = (3v_1 - v_2 + v_3, 2v_1 + 4v_2 - 2v_3)$	b) $T(v_1, v_2, v_3) = (3v_1 - v_2 + v_3, 2v_1 + 4v_2 - 2v_3)$.
johnston-linear-matrix-algebra-F00638	55	■ 1.000	$T(\mathbf{e}_1) = (1, 3)$	$T(\mathbf{e}_1) = (1, 3)$	a) We use Theorem 1.4.1, which tells us to compute $T(\mathbf{e}_1) = (1, 3)$
johnston-linear-matrix-algebra-F00639	55	■ 1.000	$T(\mathbf{e}_2) = (2, 4)$	$T(\mathbf{e}_2) = (2, 4)$	and $T(\mathbf{e}_2) = (2, 4)$ and place these as columns into a matrix, in that
johnston-linear-matrix-algebra-F00640	55	■ 1.000	$T(\mathbf{e}_1), T(\mathbf{e}_2)$	$T(\mathbf{e}_1), T(\mathbf{e}_2)$	b) We could explicitly compute $T(\mathbf{e}_1)$, $T(\mathbf{e}_2)$, and $T(\mathbf{e}_3)$ and place
johnston-linear-matrix-algebra-F00641	55	■ 1.000	$T(\mathbf{e}_3)$	$T(\mathbf{e}_3)$	b) We could explicitly compute $T(\mathbf{e}_1)$, $T(\mathbf{e}_2)$, and $T(\mathbf{e}_3)$ and place
johnston-linear-matrix-algebra-F00642	55	■ 1.000	$\mathbf{v}_1, \mathbf{v}_2$	v_1, v_2	simply place the coefficients of v_1, v_2 , and v_3 in the output of T , in
johnston-linear-matrix-algebra-F00643	55	■ 1.000	$\mathbf{v}_3$	v_3	simply place the coefficients of v_1, v_2 , and v_3 in the output of T , in
johnston-linear-matrix-algebra-F00644	55	■ 1.000	$A = \begin{bmatrix} 2.1 & 0.3 \\ 0.2 & 1.2 \end{bmatrix}$	$A = \begin{bmatrix} 2.1 & 0.3 \\ 0.2 & 1.2 \end{bmatrix}$	a) $A = \begin{bmatrix} 2.1 & 0.3 \\ 0.2 & 1.2 \end{bmatrix}$
johnston-linear-matrix-algebra-F00645	55	■ 1.000	$B = \begin{bmatrix} 2 & 1 \\ 1 & 2 \\ 1 & 1 \end{bmatrix}$	$B = \begin{bmatrix} 2 & 1 \\ 1 & 2 \\ 1 & 1 \end{bmatrix}$	b) $B = \begin{bmatrix} 2 & 1 \\ 1 & 2 \\ 1 & 1 \end{bmatrix}$
johnston-linear-matrix-algebra-F00646	55	■ 0.508	$A \mathbf{e}_1$	$A\mathbf{e}_1$	a) Since $A\mathbf{e}_1$ (i.e., the first column of A) equals $(2.1, 0.2)$ and $A\mathbf{e}_2$ (i.e.,
johnston-linear-matrix-algebra-F00647	55	■ 0.508	$A \mathbf{e}_2$	$A\mathbf{e}_2$	a) Since $A\mathbf{e}_1$ (i.e., the first column of A) equals $(2.1, 0.2)$ and $A\mathbf{e}_2$ (i.e.,
johnston-linear-matrix-algebra-F00648	55	■ 0.530	$B \mathbf{e}_1$	$B\mathbf{e}_1$	b) Since $B\mathbf{e}_1$ (i.e., the first column of B) equals $(2, 1, 1)$ and $B\mathbf{e}_2$ (i.e.,
johnston-linear-matrix-algebra-F00649	55	■ 0.530	$2, 1, 1$	$2, 1, 1$	b) Since $B\mathbf{e}_1$ (i.e., the first column of B) equals $(2, 1, 1)$ and $B\mathbf{e}_2$ (i.e.,
johnston-linear-matrix-algebra-F00650	55	■ 0.530	$B \mathbf{e}_2$	$B\mathbf{e}_2$	b) Since $B\mathbf{e}_1$ (i.e., the first column of B) equals $(2, 1, 1)$ and $B\mathbf{e}_2$ (i.e.,
johnston-linear-matrix-algebra-F00651	55	■ 0.637	$(1, 2, 1)$	$(1, 2, 1)$	its second column) equals $(1, 2, 1)$, we can think of this matrix as
johnston-linear-matrix-algebra-F00652	56	■ 1.000	$S, T: \mathbb{R}^n \rightarrow \mathbb{R}^m$	$S, T: \mathbb{R}^n \rightarrow \mathbb{R}^m$	$S, T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ and $c \in \mathbb{R}$ then $S + T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ and $cT: \mathbb{R}^n \rightarrow \mathbb{R}^m$ are
johnston-linear-matrix-algebra-F00653	56	■ 1.000	$S + T: \mathbb{R}^n \rightarrow \mathbb{R}^m$	$S + T: \mathbb{R}^n \rightarrow \mathbb{R}^m$	$S, T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ and $c \in \mathbb{R}$ then $S + T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ and $cT: \mathbb{R}^n \rightarrow \mathbb{R}^m$ are

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johnston-linear-matrix-algebra_F00654	56	■ 1.000	$\mathbf{T} : \mathbb{R}^n \rightarrow \mathbb{R}^m$	$cT : \mathbb{R}^n \rightarrow \mathbb{R}^m$	$S, T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ and $c \in \mathbb{R}$ then $S + T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ and $cT : \mathbb{R}^n \rightarrow \mathbb{R}^m$ are
johnston-linear-matrix-algebra_F00655	56	■ 1.000	$S + T$	$S + T$	ticular, the standard matrix of $S + T$ is $[S + T] = [S] + [T]$ and similarly
johnston-linear-matrix-algebra_F00656	56	■ 1.000	$[S + T] = [S] + [T]$	$[S + T] = [S] + [T]$	ticular, the standard matrix of $S + T$ is $[S + T] = [S] + [T]$ and similarly
johnston-linear-matrix-algebra_F00657	56	■ 1.000	$c \mathbf{T}$	cT	the standard matrix of cT is $[cT] = c[T]$ (see Exercise 1.4.11), so we can
johnston-linear-matrix-algebra_F00658	56	■ 1.000	$[c \mathbf{T}] = c[T]$	$[cT] = c[T]$	the standard matrix of cT is $[cT] = c[T]$ (see Exercise 1.4.11), so we can
johnston-linear-matrix-algebra_F00659	56	■ 1.000	$\mathbf{O} : \mathbb{R}^n \rightarrow \mathbb{R}^m$	$O : \mathbb{R}^n \rightarrow \mathbb{R}^m$	mation $O : \mathbb{R}^n \rightarrow \mathbb{R}^m$, defined by $O(\mathbf{v}) = \mathbf{0}$ for all $\mathbf{v} \in \mathbb{R}^n$, and the identity
johnston-linear-matrix-algebra_F00660	56	■ 1.000	$\mathbf{O}(\mathbf{v}) = \mathbf{0}$	$O(\mathbf{v}) = \mathbf{0}$	mation $O : \mathbb{R}^n \rightarrow \mathbb{R}^m$, defined by $O(\mathbf{v}) = \mathbf{0}$ for all $\mathbf{v} \in \mathbb{R}^n$, and the identity
johnston-linear-matrix-algebra_F00661	56	■ 1.000	$\mathbf{I} : \mathbb{R}^n \rightarrow \mathbb{R}^n$	$I : \mathbb{R}^n \rightarrow \mathbb{R}^n$	transformation $I : \mathbb{R}^n \rightarrow \mathbb{R}^n$, defined by $I(\mathbf{v}) = \mathbf{v}$ for all $\mathbf{v} \in \mathbb{R}^n$. It is perhaps
johnston-linear-matrix-algebra_F00662	56	■ 1.000	$\mathbf{I}(\mathbf{v}) = \mathbf{v}$	$I(\mathbf{v}) = \mathbf{v}$	transformation $I : \mathbb{R}^n \rightarrow \mathbb{R}^n$, defined by $I(\mathbf{v}) = \mathbf{v}$ for all $\mathbf{v} \in \mathbb{R}^n$. It is perhaps
johnston-linear-matrix-algebra_F00663	56	■ 1.000	$\mathbf{O} \in \mathcal{M}_{m,n}$	$O \in \mathcal{M}_{m,n}$	that if $O \in \mathcal{M}_{m,n}$ and $I \in \mathcal{M}_n$ are the zero matrix and the identity matrix, then
johnston-linear-matrix-algebra_F00664	56	■ 1.000	$\mathbf{I} \in \mathcal{M}_n$	$I \in \mathcal{M}_n$	that if $O \in \mathcal{M}_{m,n}$ and $I \in \mathcal{M}_n$ are the zero matrix and the identity matrix, then
johnston-linear-matrix-algebra_F00665	56	■ 1.000	$\mathbf{O} \mathbf{v} = \mathbf{0}$	$O\mathbf{v} = \mathbf{0}$	$O\mathbf{v} = \mathbf{0}$ and $I\mathbf{v} = \mathbf{v}$ for all $\mathbf{v} \in \mathbb{R}^n$ too.
johnston-linear-matrix-algebra_F00666	56	■ 1.000	$\mathbf{I} \mathbf{v} = \mathbf{v}$	$I\mathbf{v} = \mathbf{v}$	$O\mathbf{v} = \mathbf{0}$ and $I\mathbf{v} = \mathbf{v}$ for all $\mathbf{v} \in \mathbb{R}^n$ too.
johnston-linear-matrix-algebra_F00667	56	■ 1.000	$\mathbf{T} : \mathbb{R}^n \rightarrow \mathbb{R}^n$	$T : \mathbb{R}^n \rightarrow \mathbb{R}^n$	The next simplest type of linear transformation $T : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is one that does
johnston-linear-matrix-algebra_F00668	56	■ 1.000	$c_1, c_2, \dots, c_n \in \mathbb{R}^n$	$c_1, c_2, \dots, c_n \in \mathbb{R}^n$	formations are the ones for which there exist scalars $c_1, c_2, \dots, c_n \in \mathbb{R}^n$ such
johnston-linear-matrix-algebra_F00669	56	■ 0.995	$T(\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n) = (c_1 \mathbf{v}_1, c_2 \mathbf{v}_2, \dots, c_n \mathbf{v}_n)$	$T(v_1, v_2, \dots, v_n) = (c_1 v_1, c_2 v_2, \dots, c_n v_n)$	that $T(v_1, v_2, \dots, v_n) = (c_1 v_1, c_2 v_2, \dots, c_n v_n)$.
johnston-linear-matrix-algebra_F00670	57	■ 1.000	$\mathbf{D} \mathbf{A}$	DA	matrix, then the matrix multiplication DA simply multiplies each row of A by
johnston-linear-matrix-algebra_F00671	57	■ 1.000	$\mathbf{A} \mathbf{D}$	AD	the corresponding diagonal entry of D . Similarly, the product AD multiplies
johnston-linear-matrix-algebra_F00672	57	■ 0.999	$\mathbf{v} = (v_1, v_2) \in \mathbb{R}^2$	$\mathbf{v} = (v_1, v_2) \in \mathbb{R}^2$	the projection onto the x-axis sends every vector $\mathbf{v} = (v_1, v_2) \in \mathbb{R}^2$ to $(v_1, 0)$
johnston-linear-matrix-algebra_F00673	58	■ 1.000	$\mathbf{u}^T \mathbf{u}$	$\mathbf{u}^T \mathbf{u}$	Be careful: $\mathbf{u}^T \mathbf{u}$ is a
johnston-linear-matrix-algebra_F00674	58	■ 1.000	$\mathbf{u} \cdot \mathbf{u} = \ \mathbf{u}\ ^2 = 1$	$\mathbf{u} \cdot \mathbf{u} = \ \mathbf{u}\ ^2 = 1$	$\mathbf{u} \cdot \mathbf{u} = \ \mathbf{u}\ ^2 = 1$), but
johnston-linear-matrix-algebra_F00675	58	■ 1.000	$\mathbf{u} \mathbf{u}^T$	$\mathbf{u} \mathbf{u}^T$	$\mathbf{u} \mathbf{u}^T$ is a matrix!
johnston-linear-matrix-algebra_F00676	58	■ 1.000	$\mathbf{u}^T$	$\mathbf{u}^T$	$\mathbf{u}^T$ is $1 \times n$, $\mathbf{u} \mathbf{u}^T$ is
johnston-linear-matrix-algebra_F00677	58	■ 1.000	$1 \times n, \mathbf{u} \mathbf{u}^T$	$1 \times n, \mathbf{u} \mathbf{u}^T$	$\mathbf{u}^T$ is $1 \times n$, $\mathbf{u} \mathbf{u}^T$ is
johnston-linear-matrix-algebra_F00678	58	■ 1.000	$P_{\mathbf{u}}$	$P_{\mathbf{u}}$	think of unit vectors as specifying directions) and let $P_{\mathbf{u}}$ denote the projection
johnston-linear-matrix-algebra_F00679	58	■ 0.999	$P_{\mathbf{u}}(\mathbf{v})$	$P_{\mathbf{u}}(\mathbf{v})$	denoted by $P_{\mathbf{u}}(\mathbf{v})$, and it points a distance of $\ \mathbf{v}\ \cos(\theta)$ in the direction of $\mathbf{u}$, where
johnston-linear-matrix-algebra_F00680	58	■ 0.999	$\ \mathbf{v}\ \cos(\theta)$	$\ \mathbf{v}\ \cos(\theta)$	denoted by $P_{\mathbf{u}}(\mathbf{v})$, and it points a distance of $\ \mathbf{v}\ \cos(\theta)$ in the direction of $\mathbf{u}$, where
johnston-linear-matrix-algebra_F00681	58	■ 1.000	$P_{\mathbf{u}}(\mathbf{v}) = \mathbf{u}(\ \mathbf{v}\ \cos(\theta))$	$P_{\mathbf{u}}(\mathbf{v}) = \mathbf{u}(\ \mathbf{v}\ \cos(\theta))$	words, $P_{\mathbf{u}}(\mathbf{v}) = \mathbf{u}(\ \mathbf{v}\ \cos(\theta))$.

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johnston-linear-matrix-algebra_F00682	58	1.000	$\cos(\theta) = \mathbf{u} \cdot \mathbf{v} / (\ \mathbf{u}\ \ \mathbf{v}\)$	$\cos(\theta) = \mathbf{u} \cdot \mathbf{v} / (\ \mathbf{u}\ \ \mathbf{v}\)$	We then recall from Definition 1.2.3 that $\cos(\theta) = \mathbf{u} \cdot \mathbf{v} / (\ \mathbf{u}\ \ \mathbf{v}\)$. By
johnston-linear-matrix-algebra_F00683	58	1.000	$\ \mathbf{u}\ = 1$	$\ \mathbf{u}\ = 1$	plugging this into our formula for $P_{\mathbf{u}}(\mathbf{v})$ and using the fact that $\ \mathbf{u}\ = 1$, we
johnston-linear-matrix-algebra_F00684	58	1.000	$\mathbf{u} \cdot \mathbf{v} = \mathbf{u}^T \mathbf{v}$	$\mathbf{u} \cdot \mathbf{v} = \mathbf{u}^T \mathbf{v}$	Finally, if we recall from Section 1.3.3 that $\mathbf{u} \cdot \mathbf{v} = \mathbf{u}^T \mathbf{v}$, it follows that $P_{\mathbf{u}}(\mathbf{v}) =$
johnston-linear-matrix-algebra_F00685	58	1.000	$P_{\mathbf{u}}(\mathbf{v}) =$	$P_{\mathbf{u}}(\mathbf{v}) =$	Finally, if we recall from Section 1.3.3 that $\mathbf{u} \cdot \mathbf{v} = \mathbf{u}^T \mathbf{v}$, it follows that $P_{\mathbf{u}}(\mathbf{v}) =$
johnston-linear-matrix-algebra_F00686	58	1.000	$\mathbf{u}(\mathbf{u}^T \mathbf{v}) = (\mathbf{u} \mathbf{u}^T) \mathbf{v}$	$\mathbf{u}(\mathbf{u}^T \mathbf{v}) = (\mathbf{u} \mathbf{u}^T) \mathbf{v}$	$\mathbf{u}(\mathbf{u}^T \mathbf{v}) = (\mathbf{u} \mathbf{u}^T) \mathbf{v}$. In other words, $P_{\mathbf{u}}$ is indeed a linear transformation, and it
johnston-linear-matrix-algebra_F00687	58	1.000	$\mathbf{u} = (1, 0) \in \mathbb{R}^2$	$\mathbf{u} = (1, 0) \in \mathbb{R}^2$	a) $\mathbf{u} = (1, 0) \in \mathbb{R}^2$, and
johnston-linear-matrix-algebra_F00688	58	1.000	$\mathbf{w} = (1, 2, 3) \in \mathbb{R}^3$	$\mathbf{w} = (1, 2, 3) \in \mathbb{R}^3$	b) $\mathbf{w} = (1, 2, 3) \in \mathbb{R}^3$.
johnston-linear-matrix-algebra_F00689	59	1.000	$\mathbf{u} \in \mathbb{R}^n$	$\mathbf{u} \in \mathbb{R}^n$	In general, if $\mathbf{u} \in \mathbb{R}^n$
johnston-linear-matrix-algebra_F00690	59	1.000	$P_{\mathbf{u}}(\mathbf{u}) = \mathbf{u} : P_{\mathbf{u}}$	$P_{\mathbf{u}}(\mathbf{u}) = \mathbf{u} : P_{\mathbf{u}}$	then $P_{\mathbf{u}}(\mathbf{u}) = \mathbf{u} : P_{\mathbf{u}}$
johnston-linear-matrix-algebra_F00691	59	1.000	F	F	The “ F ” in $F_{\mathbf{u}}$ stands
johnston-linear-matrix-algebra_F00692	59	1.000	$F_{\mathbf{u}}$	$F_{\mathbf{u}}$	The “ F ” in $F_{\mathbf{u}}$ stands
johnston-linear-matrix-algebra_F00693	59	1.000	$[P_{\mathbf{u}}] \mathbf{e}_1 = \mathbf{e}_1$	$[P_{\mathbf{u}}] \mathbf{e}_1 = \mathbf{e}_1$	onto the x -axis (in particular, $[P_{\mathbf{u}}] \mathbf{e}_1 = \mathbf{e}_1$ and $[P_{\mathbf{u}}] \mathbf{e}_2 = \mathbf{0}$):
johnston-linear-matrix-algebra_F00694	59	1.000	$[P_{\mathbf{u}}] \mathbf{e}_2 = \mathbf{0}$	$[P_{\mathbf{u}}] \mathbf{e}_2 = \mathbf{0}$	onto the x -axis (in particular, $[P_{\mathbf{u}}] \mathbf{e}_1 = \mathbf{e}_1$ and $[P_{\mathbf{u}}] \mathbf{e}_2 = \mathbf{0}$):
johnston-linear-matrix-algebra_F00695	59	1.000	$\ \mathbf{w}\ = \sqrt{1^2 + 2^2 + 3^2} = \sqrt{14}$	$\ \mathbf{w}\ = \sqrt{1^2 + 2^2 + 3^2} = \sqrt{14}$	same direction. Well, $\ \mathbf{w}\ = \sqrt{1^2 + 2^2 + 3^2} = \sqrt{14}$, so we let $\mathbf{u} =$
johnston-linear-matrix-algebra_F00696	59	1.000	$\mathbf{u} =$	$\mathbf{u} =$	same direction. Well, $\ \mathbf{w}\ = \sqrt{1^2 + 2^2 + 3^2} = \sqrt{14}$, so we let $\mathbf{u} =$
johnston-linear-matrix-algebra_F00697	59	0.987	$\mathbf{w} / \ \mathbf{w}\ = (1, 2, 3) / \sqrt{14}$	$\mathbf{w} / \ \mathbf{w}\ = (1, 2, 3) / \sqrt{14}$	$\mathbf{w} / \ \mathbf{w}\ = (1, 2, 3) / \sqrt{14}$. The standard matrix of $P_{\mathbf{u}}$ is then
johnston-linear-matrix-algebra_F00698	59	1.000	$\mathbf{e}_1, \mathbf{e}_2$	$\mathbf{e}_1, \mathbf{e}_2$	This projection sends each of $\mathbf{e}_1$, $\mathbf{e}_2$, and $\mathbf{e}_3$ to different multiples of
johnston-linear-matrix-algebra_F00699	59	1.000	$\mathbf{e}_3$	$\mathbf{e}_3$	This projection sends each of $\mathbf{e}_1$, $\mathbf{e}_2$, and $\mathbf{e}_3$ to different multiples of
johnston-linear-matrix-algebra_F00700	59	0.999	$F_{\mathbf{u}} : \mathbb{R}^n \rightarrow \mathbb{R}^n$	$F_{\mathbf{u}} : \mathbb{R}^n \rightarrow \mathbb{R}^n$	The function $F_{\mathbf{u}} : \mathbb{R}^n \rightarrow \mathbb{R}^n$ that reflects space through the line in the direc-
johnston-linear-matrix-algebra_F00701	59	1.000	$F_{\mathbf{u}}(\mathbf{v}) = \mathbf{v} + 2(P_{\mathbf{u}}(\mathbf{v}) - \mathbf{v})$	$F_{\mathbf{u}}(\mathbf{v}) = \mathbf{v} + 2(P_{\mathbf{u}}(\mathbf{v}) - \mathbf{v})$	and determine its standard matrix, we note that $F_{\mathbf{u}}(\mathbf{v}) = \mathbf{v} + 2(P_{\mathbf{u}}(\mathbf{v}) - \mathbf{v})$, as
johnston-linear-matrix-algebra_F00702	60	1.000	$F_{\mathbf{u}}(\mathbf{v})$	$F_{\mathbf{u}}(\mathbf{v})$	denoted by $F_{\mathbf{u}}(\mathbf{v})$ and is equal to $\mathbf{v} + 2(P_{\mathbf{u}}(\mathbf{v}) - \mathbf{v})$.
johnston-linear-matrix-algebra_F00703	60	1.000	$\mathbf{v} + 2(P_{\mathbf{u}}(\mathbf{v}) - \mathbf{v})$	$\mathbf{v} + 2(P_{\mathbf{u}}(\mathbf{v}) - \mathbf{v})$	denoted by $F_{\mathbf{u}}(\mathbf{v})$ and is equal to $\mathbf{v} + 2(P_{\mathbf{u}}(\mathbf{v}) - \mathbf{v})$.
johnston-linear-matrix-algebra_F00704	60	1.000	$\mathbf{u} = (0, 1) \in \mathbb{R}^2$	$\mathbf{u} = (0, 1) \in \mathbb{R}^2$	a) $\mathbf{u} = (0, 1) \in \mathbb{R}^2$, and
johnston-linear-matrix-algebra_F00705	60	1.000	$\mathbf{w} = (1, 1, 1) \in \mathbb{R}^3$	$\mathbf{w} = (1, 1, 1) \in \mathbb{R}^3$	b) $\mathbf{w} = (1, 1, 1) \in \mathbb{R}^3$.
johnston-linear-matrix-algebra_F00706	60	1.000	$[F_{\mathbf{u}}]$	$[F_{\mathbf{u}}]$	We can verify our computation of $[F_{\mathbf{u}}]$ by noting that multiplication
johnston-linear-matrix-algebra_F00707	61	1.000	$F_{\mathbf{u}}(\mathbf{u}) = \mathbf{u} : F_{\mathbf{u}}$	$F_{\mathbf{u}}(\mathbf{u}) = \mathbf{u} : F_{\mathbf{u}}$	then $F_{\mathbf{u}}(\mathbf{u}) = \mathbf{u} : F_{\mathbf{u}}$
johnston-linear-matrix-algebra_F00708	61	1.000	$\ \mathbf{w}\ = \sqrt{1^2 + 1^2 + 1^2} = \sqrt{3}$	$\ \mathbf{w}\ = \sqrt{1^2 + 1^2 + 1^2} = \sqrt{3}$	rection. Well, $\ \mathbf{w}\ = \sqrt{1^2 + 1^2 + 1^2} = \sqrt{3}$, so we let $\mathbf{u} = \mathbf{w} / \ \mathbf{w}\ =$

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johnston-linear-matrix-algebra-F00709	61	1.000	$\mathbf{u}=\mathbf{w} / \ \mathbf{w}\ =$	$\mathbf{u} = \mathbf{w} / \ \mathbf{w}\ =$	rection. Well, $\ \mathbf{w}\ = \sqrt{1^2 + 1^2 + 1^2} = \sqrt{3}$, so we let $\mathbf{u} = \mathbf{w} / \ \mathbf{w}\ =$
johnston-linear-matrix-algebra-F00710	61	0.999	$(1,1,1) / \sqrt{3}$	$(1,1,1) / \sqrt{3}$	$(1,1,1) / \sqrt{3}$. The standard matrix of $F_{\mathbf{u}}$ is then
johnston-linear-matrix-algebra-F00711	61	1.000	$\mathbf{v}=(-1,3)$	$\mathbf{v} = (-1,3)$	Find the entries of the vector that is obtained by reflecting $\mathbf{v} = (-1,3)$
johnston-linear-matrix-algebra-F00712	61	1.000	$\pi / 3$	$\pi / 3$	through the line going through the origin at an angle of $\pi / 3$ counter-
johnston-linear-matrix-algebra-F00713	62	1.000	R^{θ}	R^{θ}	angle θ in R^{θ} might
johnston-linear-matrix-algebra-F00714	62	1.000	$\mathbf{u}=(\cos (\pi / 3), \sin (\pi / 3))=(1, \sqrt{3}) / 2$	$\mathbf{u} = (\cos(\pi/3), \sin(\pi/3)) = (1, \sqrt{3})/2$	unit vector is $\mathbf{u} = (\cos(\pi/3), \sin(\pi/3)) = (1, \sqrt{3})/2$, so
johnston-linear-matrix-algebra-F00715	62	1.000	$R^{\theta}: \mathbb{R}^2 \rightarrow$	$R^{\theta}: \mathbb{R}^2 \rightarrow$	(counter-clockwise) rotations by an angle θ , which we denote by $R^{\theta}: \mathbb{R}^2 \rightarrow$
johnston-linear-matrix-algebra-F00716	62	1.000	$R^{\theta}(\mathbf{0})=\mathbf{0}$	$R^{\theta}(\mathbf{0}) = \mathbf{0}$	origin so that $R^{\theta}(\mathbf{0}) = \mathbf{0}$. To see that R^{θ} then is indeed linear, we simply note
johnston-linear-matrix-algebra-F00717	62	1.000	$R^{\theta}(\mathbf{v}+\mathbf{w})=R^{\theta}(\mathbf{v})+R^{\theta}(\mathbf{w})$	$R^{\theta}(\mathbf{v} + \mathbf{w}) = R^{\theta}(\mathbf{v}) + R^{\theta}(\mathbf{w})$	that it is geometrically clear that $R^{\theta}(\mathbf{v} + \mathbf{w}) = R^{\theta}(\mathbf{v}) + R^{\theta}(\mathbf{w})$ for all $\mathbf{v}, \mathbf{w} \in \mathbb{R}^2$,
johnston-linear-matrix-algebra-F00718	62	1.000	$R^{\theta}(c \mathbf{v})=c R^{\theta}(\mathbf{v})$	$R^{\theta}(c \mathbf{v}) = c R^{\theta}(\mathbf{v})$	the requirement that $R^{\theta}(c \mathbf{v}) = c R^{\theta}(\mathbf{v})$ for all $\mathbf{v} \in \mathbb{R}^2$ and all $c \in \mathbb{R}$ follows from
johnston-linear-matrix-algebra-F00719	62	1.000	$R^{\theta}\left(\mathbf{e}_1\right)$	$R^{\theta}\left(\mathbf{e}_1\right)$	tion of how it acts on arbitrary vectors), it is enough to compute $R^{\theta}\left(\mathbf{e}_1\right)$ and
johnston-linear-matrix-algebra-F00720	63	1.000	$\sin (-x)=-\sin (x)$	$\sin(-x) = -\sin(x)$	$\sin(-x) = -\sin(x)$
johnston-linear-matrix-algebra-F00721	63	1.000	$\cos (-x)=\cos (x)$	$\cos(-x) = \cos(x)$	and $\cos(-x) = \cos(x)$
johnston-linear-matrix-algebra-F00722	63	1.000	$R^{\theta}\left(\mathbf{e}_2\right)$	$R^{\theta}\left(\mathbf{e}_2\right)$	$R^{\theta}\left(\mathbf{e}_2\right)$. By regarding $R^{\theta}\left(\mathbf{e}_1\right)$ and $R^{\theta}\left(\mathbf{e}_2\right)$ as the hypotenuses (with length 1) of
johnston-linear-matrix-algebra-F00723	63	1.000	$R^{\theta}\left(\mathbf{e}_1\right)=(\cos (\theta), \sin (\theta))$	$R^{\theta}\left(\mathbf{e}_1\right) = (\cos(\theta), \sin(\theta))$	right-angled triangles as in Figure 1.19, we see that $R^{\theta}\left(\mathbf{e}_1\right) = (\cos(\theta), \sin(\theta))$
johnston-linear-matrix-algebra-F00724	63	1.000	$R^{\theta}\left(\mathbf{e}_2\right)=(-\sin (\theta), \cos (\theta))$	$R^{\theta}\left(\mathbf{e}_2\right) = (-\sin(\theta), \cos(\theta))$	and $R^{\theta}\left(\mathbf{e}_2\right) = (-\sin(\theta), \cos(\theta))$.
johnston-linear-matrix-algebra-F00725	63	1.000	$\pi / 4$	$\pi / 4$	a) $\pi / 4$ radians counter-clockwise, and
johnston-linear-matrix-algebra-F00726	63	1.000	$\pi / 6$	$\pi / 6$	b) $\pi / 6$ radians clockwise.
johnston-linear-matrix-algebra-F00727	63	1.000	$-\pi / 6$	$-\pi / 6$	clockwise rotation by $-\pi / 6$ radians, so we compute
johnston-linear-matrix-algebra-F00728	63	1.000	$\mathbf{v}=(1,3)$	$\mathbf{v} = (1,3)$	a) Rotate $\mathbf{v} = (1,3)$ by $\pi / 4$ radians counter-clockwise, and
johnston-linear-matrix-algebra-F00729	63	1.000	$\mathbf{w}=(\sqrt{3}, 3)$	$\mathbf{w} = (\sqrt{3}, 3)$	b) rotate $\mathbf{w} = (\sqrt{3}, 3)$ by $\pi / 6$ radians clockwise.
johnston-linear-matrix-algebra-F00730	64	1.000	$R^{\pi / 6}(\mathbf{w})$	$R^{\pi / 6}(\mathbf{w})$	ample 1.4.8(b), so we can simply compute $R^{\pi / 6}(\mathbf{w})$ via matrix mul-
johnston-linear-matrix-algebra-F00731	64	1.000	$\mathbf{v}=(1,2,3)$	$\mathbf{v} = (1,2,3)$	“rotate $\mathbf{v} = (1,2,3)$ counter-clockwise by an angle of $\pi / 4$ ”—this instruction is
johnston-linear-matrix-algebra-F00732	64	1.000	$R_{y z}^{\theta}$	$R_{y z}^{\theta}$	To illustrate how to do this, consider the linear transformation $R_{y z}^{\theta}$ that
johnston-linear-matrix-algebra-F00733	65	1.000	$\left[R_{z x}^{\theta}\right]$	$\left[R_{z x}^{\theta}\right]$	We list $\left[R_{z x}^{\theta}\right]$ here
johnston-linear-matrix-algebra-F00734	65	1.000	$\left[R_{x z}^{\theta}\right]$	$\left[R_{x z}^{\theta}\right]$	instead of $\left[R_{x z}^{\theta}\right]$ to
johnston-linear-matrix-algebra-F00735	65	1.000	$-\sin (\theta)$	$-\sin(\theta)$	keep $-\sin(\theta)$ in the

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johnston-linear-matrix-algebra-F00736	65	1.000	$R_{yz}(\mathbf{e}_1) = \mathbf{e}_1$	$R_{yz}^\theta(\mathbf{e}_1) = \mathbf{e}_1$	has no effect at all (i.e., $R_{yz}^\theta(\mathbf{e}_1) = \mathbf{e}_1$), as shown in Figure 1.20. Similarly,
johnston-linear-matrix-algebra-F00737	65	1.000	yz	yz	to rotate $\mathbf{e}_2$ or $\mathbf{e}_3$ we can just treat the yz -plane as if it were $\mathbb{R}^2$ and repeat
johnston-linear-matrix-algebra-F00738	65	1.000	$R_{yz}(\mathbf{e}_2) = (0, \cos(\theta), \sin(\theta))$	$R_{yz}^\theta(\mathbf{e}_2) = (0, \cos(\theta), \sin(\theta))$	the derivation from Figure 1.18 to see that $R_{yz}^\theta(\mathbf{e}_2) = (0, \cos(\theta), \sin(\theta))$ and
johnston-linear-matrix-algebra-F00739	65	0.551	$R_{yz}(\mathbf{e}_3) = (0, -\sin(\theta), \cos(\theta))$	$R_{yz}^\theta(\mathbf{e}_3) = (0, -\sin(\theta), \cos(\theta))$	$R_{yz}^\theta(\mathbf{e}_3) = (0, -\sin(\theta), \cos(\theta))$.
johnston-linear-matrix-algebra-F00740	65	0.999	$\mathbf{v} = (3, -1, 2)$	$\mathbf{v} = (3, -1, 2)$	Rotate $\mathbf{v} = (3, -1, 2)$ around the z -axis by an angle of $\theta = 2\pi/3$ in the
johnston-linear-matrix-algebra-F00741	65	0.999	$\theta = 2\pi/3$	$\theta = 2\pi/3$	Rotate $\mathbf{v} = (3, -1, 2)$ around the z -axis by an angle of $\theta = 2\pi/3$ in the
johnston-linear-matrix-algebra-F00742	65	1.000	$R_{xy}^{2\pi/3}(\mathbf{v})$	$R_{xy}^{2\pi/3}(\mathbf{v})$	Since our goal is to compute $R_{xy}^{2\pi/3}(\mathbf{v})$, we start by constructing the
johnston-linear-matrix-algebra-F00743	65	1.000	$R_{xy}^{2\pi/3}$	$R_{xy}^{2\pi/3}$	standard matrix of $R_{xy}^{2\pi/3}$:
johnston-linear-matrix-algebra-F00744	66	1.000	$S \circ T$	$S \circ T$	Be careful: $S \circ T$ is
johnston-linear-matrix-algebra-F00745	66	0.907	$(S \circ T)(\mathbf{v}) = S(T(\mathbf{v}))$	$(S \circ T)(\mathbf{v}) = S(T(\mathbf{v}))$	$(S \circ T)(\mathbf{v}) = S(T(\mathbf{v}))$.
johnston-linear-matrix-algebra-F00746	66	0.999	$S, T: \mathbb{R}^n \rightarrow$	$S, T: \mathbb{R}^n \rightarrow$	transformations to create new ones. In particular, we noted that if $S, T: \mathbb{R}^n \rightarrow$
johnston-linear-matrix-algebra-F00747	66	1.000	$S + cT: \mathbb{R}^n \rightarrow \mathbb{R}^m$	$S + cT: \mathbb{R}^n \rightarrow \mathbb{R}^m$	$\mathbb{R}^m$ are linear transformations and $c \in \mathbb{R}$ then $S + cT: \mathbb{R}^n \rightarrow \mathbb{R}^m$ is also a linear
johnston-linear-matrix-algebra-F00748	66	1.000	$S: \mathbb{R}^m \rightarrow \mathbb{R}^p$	$S: \mathbb{R}^m \rightarrow \mathbb{R}^p$	Suppose $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ and $S: \mathbb{R}^m \rightarrow \mathbb{R}^p$ are linear transformations. Then
johnston-linear-matrix-algebra-F00749	66	1.000	$S \circ T: \mathbb{R}^n \rightarrow \mathbb{R}^p$	$S \circ T: \mathbb{R}^n \rightarrow \mathbb{R}^p$	their composition is the function $S \circ T: \mathbb{R}^n \rightarrow \mathbb{R}^p$ defined by
johnston-linear-matrix-algebra-F00750	66	1.000	$\mathbb{R}^p$	$\mathbb{R}^p$	plying S . In other words, while T sends $\mathbb{R}^n$ to $\mathbb{R}^m$ and S sends $\mathbb{R}^m$ to $\mathbb{R}^p$, the
johnston-linear-matrix-algebra-F00751	67	1.000	$[S \circ T] = [S][T]$	$[S \circ T] = [S][T]$	$[S \circ T] = [S][T]$ is a
johnston-linear-matrix-algebra-F00752	67	1.000	$p \times n$	$p \times n$	$p \times n$ matrix.
johnston-linear-matrix-algebra-F00753	67	1.000	$[D][F_{\mathbf{u}}]$	$[D][F_{\mathbf{u}}]$	are after is $[D][F_{\mathbf{u}}]$.
johnston-linear-matrix-algebra-F00754	67	0.980	$\text{not}[F_{\mathbf{u}}][D]$	$\text{not}[F_{\mathbf{u}}][D]$	not $[F_{\mathbf{u}}][D]$.
johnston-linear-matrix-algebra-F00755	67	0.998	$S(T(\mathbf{v}))$	$S(T(\mathbf{v}))$	mation that sends $\mathbf{v}$ to $S(T(\mathbf{v}))$.
johnston-linear-matrix-algebra-F00756	67	1.000	$[S] \in \mathcal{M}_{p,m}$	$[S] \in \mathcal{M}_{p,m}$	standard matrices $[T] \in \mathcal{M}_{m,n}$ and $[S] \in \mathcal{M}_{p,m}$, respectively. Then $S \circ T:$
johnston-linear-matrix-algebra-F00757	67	1.000	$\mathbb{R}^n \rightarrow \mathbb{R}^p$	$\mathbb{R}^n \rightarrow \mathbb{R}^p$	$\mathbb{R}^n \rightarrow \mathbb{R}^p$ is a linear transformation, and its standard matrix is
johnston-linear-matrix-algebra-F00758	67	1.000	$(S \circ T)(\mathbf{v})$	$(S \circ T)(\mathbf{v})$	<i>Proof.</i> Let $\mathbf{v} \in \mathbb{R}^n$ and compute $(S \circ T)(\mathbf{v})$:
johnston-linear-matrix-algebra-F00759	67	1.000	$[S][T]$	$[S][T]$	multiplication by the matrix $[S][T]$. It thus follows from Theorem 1.4.1 that
johnston-linear-matrix-algebra-F00760	67	1.000	$y = \frac{4}{3}x$	$y = \frac{4}{3}x$	through the line $y = \frac{4}{3}x$ and then stretches it in the x -direction by a factor
johnston-linear-matrix-algebra-F00761	67	0.960	$\mathbf{u} = (3/5, 4/5)$	$\mathbf{u} = (3/5, 4/5)$	them together. A unit vector on the line $y = \frac{4}{3}x$ is $\mathbf{u} = (3/5, 4/5)$, and the
johnston-linear-matrix-algebra-F00762	67	1.000	$T = D \circ F_{\mathbf{u}}$	$T = D \circ F_{\mathbf{u}}$	The standard matrix of the composite linear transformation $T = D \circ F_{\mathbf{u}}$ is

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johnston-linear-matrix-algebra_F00763	68	■ 0.999	ϕ	ϕ	Explain why, if θ and ϕ are any two angles, then $R^\theta \circ R^\phi = R^{\theta+\phi}$. Use
johnston-linear-matrix-algebra_F00764	68	■ 0.999	$R^{\theta} \circ R^{\phi} = R^{\theta+\phi}$	$R^\theta \circ R^\phi = R^{\theta+\phi}$	Explain why, if θ and ϕ are any two angles, then $R^\theta \circ R^\phi = R^{\theta+\phi}$. Use
johnston-linear-matrix-algebra_F00765	69	■ 1.000	$\phi = \theta$	$\phi = \theta$	If we choose $\phi = \theta$
johnston-linear-matrix-algebra_F00766	69	■ 1.000	$\sin(2\theta) =$	$\sin(2\theta) =$	that $\sin(2\theta) =$
johnston-linear-matrix-algebra_F00767	69	■ 1.000	$2 \sin(\theta) \cos(\theta)$	$2 \sin(\theta) \cos(\theta)$	$2 \sin(\theta) \cos(\theta)$ and
johnston-linear-matrix-algebra_F00768	69	■ 1.000	$\cos(2\theta) =$	$\cos(2\theta) =$	$\cos(2\theta) =$
johnston-linear-matrix-algebra_F00769	69	■ 0.980	$\cos^2(\theta) - \sin^2(\theta)$	$\cos^2(\theta) - \sin^2(\theta)$	$\cos^2(\theta) - \sin^2(\theta)$.
johnston-linear-matrix-algebra_F00770	69	■ 0.995	$\theta + \phi$	$\theta + \phi$	by $\theta + \phi$ radians:
johnston-linear-matrix-algebra_F00771	69	■ 1.000	$R^{\theta} \circ R^{\phi}$	$R^\theta \circ R^\phi$	of $R^\theta \circ R^\phi$ and $R^{\theta+\phi}$:
johnston-linear-matrix-algebra_F00772	69	■ 1.000	$R^{\theta+\phi}$	$R^{\theta+\phi}$	of $R^\theta \circ R^\phi$ and $R^{\theta+\phi}$:
johnston-linear-matrix-algebra_F00773	69	■ 1.000	$\left[R^{\theta} \circ R^{\phi}\right]$	$[R^\theta \circ R^\phi]$	The matrices $[R^\theta \circ R^\phi]$ and $[R^{\theta+\phi}]$ must be equal (since they are standard
johnston-linear-matrix-algebra_F00774	69	■ 1.000	$\left[R^{\theta+\phi}\right]$	$[R^{\theta+\phi}]$	The matrices $[R^\theta \circ R^\phi]$ and $[R^{\theta+\phi}]$ must be equal (since they are standard
johnston-linear-matrix-algebra_F00775	69	■ 0.983	$(2, 1)$	$(2, 1)$	entries must be equal. By looking at their $(2, 1)$ -entries, we see that
johnston-linear-matrix-algebra_F00776	69	■ 0.831	$\left[R^{\theta}\right]$	$[R^\theta]$	at all— $[R^\theta]$ really is the θ -th power of the matrix
johnston-linear-matrix-algebra_F00777	69	■ 1.000	R	R	For now, this interpretation of R^θ as the θ -th power of R only makes
johnston-linear-matrix-algebra_F00778	70	■ 1.000	R, S	R, S	two of them: if R, S , and T are linear transformations then their composition
johnston-linear-matrix-algebra_F00779	70	■ 1.000	$R \circ S \circ T$	$R \circ S \circ T$	$R \circ S \circ T$ is the linear transformation that applies T , then S , then R , and its
johnston-linear-matrix-algebra_F00780	70	■ 1.000	$[R \circ S \circ T] = [R][S][T]$	$[R \circ S \circ T] = [R][S][T]$	standard matrix is $[R \circ S \circ T] = [R][S][T]$.
johnston-linear-matrix-algebra_F00781	70	■ 0.999	$T(v_1, v_2) = (v_1 + 2v_2, 3v_1 - v_2)$	$T(v_1, v_2) = (v_1 + 2v_2, 3v_1 - v_2)$	* (a) $T(v_1, v_2) = (v_1 + 2v_2, 3v_1 - v_2)$.
johnston-linear-matrix-algebra_F00782	70	■ 1.000	$T(v_1, v_2) = (v_1 + v_2, 2v_1 - v_2, -v_1 + 3v_2)$	$T(v_1, v_2) = (v_1 + v_2, 2v_1 - v_2, -v_1 + 3v_2)$	(b) $T(v_1, v_2) = (v_1 + v_2, 2v_1 - v_2, -v_1 + 3v_2)$.
johnston-linear-matrix-algebra_F00783	70	■ 0.924	$T(v_1, v_2, v_3) = (v_1 + v_2, v_1 + v_2 - v_3)$	$T(v_1, v_2, v_3) = (v_1 + v_2, v_1 + v_2 - v_3)$	* (c) $T(v_1, v_2, v_3) = (v_1 + v_2, v_1 + v_2 - v_3)$.
johnston-linear-matrix-algebra_F00784	70	■ 1.000	$T(v_1, v_2, v_3) = (v_2, 2v_1 + v_3, v_2 - v_3)$	$T(v_1, v_2, v_3) = (v_2, 2v_1 + v_3, v_2 - v_3)$	(d) $T(v_1, v_2, v_3) = (v_2, 2v_1 + v_3, v_2 - v_3)$.
johnston-linear-matrix-algebra_F00785	70	■ 0.669	$T(1, 0) = (3, -1)$	$T(1, 0) = (3, -1)$	* (a) $T(1, 0) = (3, -1)$ and $T(0, 1) = (1, 2)$.
johnston-linear-matrix-algebra_F00786	70	■ 0.669	$T(0, 1) = (1, 2)$	$T(0, 1) = (1, 2)$	* (a) $T(1, 0) = (3, -1)$ and $T(0, 1) = (1, 2)$.
johnston-linear-matrix-algebra_F00787	70	■ 1.000	$T(1, 0) = (1, 3)$	$T(1, 0) = (1, 3)$	(b) $T(1, 0) = (1, 3)$ and $T(1, 1) = (3, 7)$.
johnston-linear-matrix-algebra_F00788	70	■ 1.000	$T(1, 1) = (3, 7)$	$T(1, 1) = (3, 7)$	(b) $T(1, 0) = (1, 3)$ and $T(1, 1) = (3, 7)$.
johnston-linear-matrix-algebra_F00789	70	■ 0.971	$T(1, 0) = (-1, 0, 1)$	$T(1, 0) = (-1, 0, 1)$	* (c) $T(1, 0) = (-1, 0, 1)$ and $T(0, 1) = (2, 3, 0)$.
johnston-linear-matrix-algebra_F00790	70	■ 0.971	$T(0, 1) = (2, 3, 0)$	$T(0, 1) = (2, 3, 0)$	* (c) $T(1, 0) = (-1, 0, 1)$ and $T(0, 1) = (2, 3, 0)$.

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johnston-linear-matrix-algebra-F00791	70	■ 0.975	$T(1,0,0)=(2,1), \quad T(0,1,0)=(-1,1)$	$T(1,0,0) = (2,1), \quad T(0,1,0) = (-1,1)$	(d) $T(1,0,0) = (2,1), \quad T(0,1,0) = (-1,1),$ and
johnston-linear-matrix-algebra-F00792	70	■ 1.000	$T(0,0,1)=(0,3)$	$T(0,0,1) = (0,3)$	$T(0,0,1) = (0,3).$
johnston-linear-matrix-algebra-F00793	70	■ 0.539	$T(1,0,0)=(1,2,3), \quad T(1,1,0)=(0,1,2)$	$T(1,0,0) = (1,2,3), T(1,1,0) = (0,1,2)$	*(e) $T(1,0,0) = (1,2,3), \quad T(1,1,0) = (0,1,2),$ and
johnston-linear-matrix-algebra-F00794	70	■ 0.976	$T(1,1,1)=(0,0,1)$	$T(1,1,1) = (0,0,1)$	$T(1,1,1) = (0,0,1).$
johnston-linear-matrix-algebra-F00795	70	■ 1.000	$T: \mathbb{R}^2 \rightarrow \mathbb{R}^4$	$T: \mathbb{R}^2 \rightarrow \mathbb{R}^4$	*(c) Every linear transformation $T: \mathbb{R}^2 \rightarrow \mathbb{R}^4$ is com-
johnston-linear-matrix-algebra-F00796	70	■ 1.000	$\mathbb{R}^4 \rightarrow \mathbb{R}^3$	$\mathbb{R}^4 \rightarrow \mathbb{R}^3$	$\mathbb{R}^4 \rightarrow \mathbb{R}^3$ has size $4 \times 3.$
johnston-linear-matrix-algebra-F00797	70	■ 1.000	4×3	4×3	$\mathbb{R}^4 \rightarrow \mathbb{R}^3$ has size $4 \times 3.$
johnston-linear-matrix-algebra-F00798	70	■ 0.999	$T: \mathbb{R}^2 \rightarrow$	$T: \mathbb{R}^2 \rightarrow$	*(e) There exists a linear transformation $T: \mathbb{R}^2 \rightarrow$
johnston-linear-matrix-algebra-F00799	70	■ 1.000	$T(\mathbf{e}_1)=(2,1), \quad T(\mathbf{e}_2)=(1,3)$	$T(\mathbf{e}_1) = (2,1), T(\mathbf{e}_2) = (1,3)$	$\mathbb{R}^2$ such that $T(\mathbf{e}_1) = (2,1), \quad T(\mathbf{e}_2) = (1,3),$ and
johnston-linear-matrix-algebra-F00800	70	■ 1.000	$T(1,1)=(3,3)$	$T(1,1) = (3,3)$	$T(1,1) = (3,3).$
johnston-linear-matrix-algebra-F00801	70	■ 0.966	$*(\mathbf{g})$	$*(\mathbf{g})$	*(g) If $R_{xy}^\theta, R_{yz}^\theta,$ and R_{xz}^θ are rotations around the co-
johnston-linear-matrix-algebra-F00802	70	■ 0.966	$R_{xy}^\theta, R_{yz}^\theta$	$R_{xy}^\theta, R_{yz}^\theta$	*(g) If $R_{xy}^\theta, R_{yz}^\theta,$ and R_{xz}^θ are rotations around the co-
johnston-linear-matrix-algebra-F00803	70	■ 0.966	R_{xz}^θ	R_{xz}^θ	*(g) If $R_{xy}^\theta, R_{yz}^\theta,$ and R_{xz}^θ are rotations around the co-
johnston-linear-matrix-algebra-F00804	70	■ 0.860	$R_{xy}^\theta \circ R_{yz}^\theta = R_{xz}^\theta$	$R_{xy}^\theta \circ R_{yz}^\theta = R_{xz}^\theta$	$R_{xy}^\theta \circ R_{yz}^\theta = R_{xz}^\theta.$
johnston-linear-matrix-algebra-F00805	70	■ 1.000	$\mathbb{R}^2 \rightarrow \mathbb{R}^2$	$\mathbb{R}^2 \rightarrow \mathbb{R}^2$	$\mathbb{R}^2 \rightarrow \mathbb{R}^2$ are and are not linear transformations. If T is a
johnston-linear-matrix-algebra-F00806	70	■ 0.972	$T(v_1, v_2) = (v_1^2, v_2)$	$T(v_1, v_2) = (v_1^2, v_2)$	*(a) $T(v_1, v_2) = (v_1^2, v_2).$
johnston-linear-matrix-algebra-F00807	70	■ 1.000	$T(v_1, v_2) = (v_1 + 2v_2, v_2 - v_1)$	$T(v_1, v_2) = (v_1 + 2v_2, v_2 - v_1)$	(b) $T(v_1, v_2) = (v_1 + 2v_2, v_2 - v_1).$
johnston-linear-matrix-algebra-F00808	70	■ 1.000	$T(v_1, v_2) = (\sin(v_1) + v_2, v_1 - \cos(v_2))$	$T(v_1, v_2) = (\sin(v_1) + v_2, v_1 - \cos(v_2))$	*(c) $T(v_1, v_2) = (\sin(v_1) + v_2, v_1 - \cos(v_2)).$
johnston-linear-matrix-algebra-F00809	70	■ 1.000	$T(v_1, v_2) = (\sin(3)v_1 + v_2, v_1 - \cos(2)v_2)$	$T(v_1, v_2) = (\sin(3)v_1 + v_2, v_1 - \cos(2)v_2)$	(d) $T(v_1, v_2) = (\sin(3)v_1 + v_2, v_1 - \cos(2)v_2).$
johnston-linear-matrix-algebra-F00810	70	■ 1.000	$T(v_1, v_2) = (\sqrt{v_1} + \sqrt{v_2}, \sqrt{v_1 + v_2})$	$T(v_1, v_2) = (\sqrt{v_1} + \sqrt{v_2}, \sqrt{v_1 + v_2})$	*(e) $T(v_1, v_2) = (\sqrt{v_1} + \sqrt{v_2}, \sqrt{v_1 + v_2}).$
johnston-linear-matrix-algebra-F00811	70	■ 1.000	$T(v_1, v_2) = (\sqrt{3}v_1, \sqrt{2}v_2 - \sqrt{5}v_1)$	$T(v_1, v_2) = (\sqrt{3}v_1, \sqrt{2}v_2 - \sqrt{5}v_1)$	(f) $T(v_1, v_2) = (\sqrt{3}v_1, \sqrt{2}v_2 - \sqrt{5}v_1).$
johnston-linear-matrix-algebra-F00812	70	■ 0.993	$T(v_1, v_2) = (v_1 , v_2)$	$T(v_1, v_2) = (v_1 , v_2)$	*(g) $T(v_1, v_2) = (v_1 , v_2).$
johnston-linear-matrix-algebra-F00813	70	■ 1.000	$y = 3x$	$y = 3x$	*(a) The projection onto the line $y = 3x.$
johnston-linear-matrix-algebra-F00814	70	■ 1.000	$y = 3x - 2$	$y = 3x - 2$	(b) The projection onto the line $y = 3x - 2.$
johnston-linear-matrix-algebra-F00815	70	■ 1.000	$y = 2x$	$y = 2x$	*(c) The reflection across the line $y = 2x.$
johnston-linear-matrix-algebra-F00816	70	■ 1.000	$y = 2x + 1$	$y = 2x + 1$	(d) The reflection across the line $y = 2x + 1.$
johnston-linear-matrix-algebra-F00817	70	■ 0.999	$\pi/5$	$\pi/5$	$(0,0)$ by $\pi/5$ radians.
johnston-linear-matrix-algebra-F00818	70	■ 1.000	$(0,1)$	$(0,1)$	$(0,1)$ by $\pi/5$ radians.
johnston-linear-matrix-algebra-F00819	70	■ 0.981	$S(v_1, v_2) = (2v_2, v_1 + v_2)$	$S(v_1, v_2) = (2v_2, v_1 + v_2)$	*(a) $S(v_1, v_2) = (2v_2, v_1 + v_2),$

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johnston-linear-matrix-algebra_F00820	70	■ 1.000	$S(\mathbf{v}_1, \mathbf{v}_2) = (\mathbf{v}_1 - 2\mathbf{v}_2, 3\mathbf{v}_1 + \mathbf{v}_2)$	$S(v_1, v_2) = (v_1 - 2v_2, 3v_1 + v_2)$	(b) $S(v_1, v_2) = (v_1 - 2v_2, 3v_1 + v_2)$,
johnston-linear-matrix-algebra_F00821	70	■ 0.999	$S(\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3) = (\mathbf{v}_1, \mathbf{v}_1 + \mathbf{v}_2, \mathbf{v}_1 + \mathbf{v}_2 + \mathbf{v}_3)$	$S(v_1, v_2, v_3) = (v_1, v_1 + v_2, v_1 + v_2 + v_3)$	(c) $S(v_1, v_2, v_3) = (v_1, v_1 + v_2, v_1 + v_2 + v_3)$,
johnston-linear-matrix-algebra_F00822	70	■ 0.999	$y = x$	$y = x$	(b) Projects $\mathbb{R}^2$ onto the line $y = x$ and then rotates
johnston-linear-matrix-algebra_F00823	70	■ 1.000	$y = x / 2$	$y = x/2$	$\mathbb{R}^2$ through the line $y = x/2$.
johnston-linear-matrix-algebra_F00824	70	■ 1.000	$\mathbf{v}_1, \dots, \mathbf{v}_k \in \mathbb{R}^n$	$\mathbf{v}_1, \dots, \mathbf{v}_k \in \mathbb{R}^n$	for all $\mathbf{v}_1, \dots, \mathbf{v}_k \in \mathbb{R}^n$ and all $c_1, \dots, c_k \in \mathbb{R}$.
johnston-linear-matrix-algebra_F00825	70	■ 1.000	$c_1, \dots, c_k \in \mathbb{R}$	$c_1, \dots, c_k \in \mathbb{R}$	for all $\mathbf{v}_1, \dots, \mathbf{v}_k \in \mathbb{R}^n$ and all $c_1, \dots, c_k \in \mathbb{R}$.
johnston-linear-matrix-algebra_F00826	70	■ 1.000	$D \in \mathcal{M}_n$	$D \in \mathcal{M}_n$	** 1.4.10 Let $D \in \mathcal{M}_n$ be a diagonal matrix and let
johnston-linear-matrix-algebra_F00827	70	■ 1.000	$1 \leq i \leq n, \dots$	$1 \leq i \leq n, \dots$	tiplication to show that, for all $1 \leq i \leq n$, ...
johnston-linear-matrix-algebra_F00828	70	■ 1.000	$d_{i,i}$	$d_{i,i}$	(a) the product DA multiplies the i -th row of A by $d_{i,i}$,
johnston-linear-matrix-algebra_F00829	71	■ 1.000	$Q, S, T: \mathbb{R}^n \rightarrow \mathbb{R}^m$	$Q, S, T: \mathbb{R}^n \rightarrow \mathbb{R}^m$	1.4.12 Let $Q, S, T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ be linear transformations.
johnston-linear-matrix-algebra_F00830	71	■ 1.000	$S + T = T + S$	$S + T = T + S$	(a) $S + T = T + S$, and
johnston-linear-matrix-algebra_F00831	71	■ 1.000	$Q + (S + T) = (Q + S) + T$	$Q + (S + T) = (Q + S) + T$	(b) $Q + (S + T) = (Q + S) + T$.
johnston-linear-matrix-algebra_F00832	71	■ 1.000	$c(S + T) = cS + cT$	$c(S + T) = cS + cT$	(a) $c(S + T) = cS + cT$,
johnston-linear-matrix-algebra_F00833	71	■ 1.000	$(c + d)T = cT + dT$	$(c + d)T = cT + dT$	(b) $(c + d)T = cT + dT$, and
johnston-linear-matrix-algebra_F00834	71	■ 1.000	$c(dT) = (cd)T$	$c(dT) = (cd)T$	(c) $c(dT) = (cd)T$.
johnston-linear-matrix-algebra_F00835	71	■ 1.000	A^{160}	A^{160}	Compute the matrix A^{160} .
johnston-linear-matrix-algebra_F00836	71	■ 1.000	$\theta \in \mathbb{R}$	$\theta \in \mathbb{R}$	for some $\theta \in \mathbb{R}$. Show that $A^T A = I$.
johnston-linear-matrix-algebra_F00837	71	■ 1.000	$A^T A = I$	$A^T A = I$	for some $\theta \in \mathbb{R}$. Show that $A^T A = I$.
johnston-linear-matrix-algebra_F00838	71	■ 0.999	$A = \mathbf{u}\mathbf{u}^T \in$	$A = \mathbf{u}\mathbf{u}^T \in$	1.4.16 Suppose $\mathbf{u} \in \mathbb{R}^n$ is a unit vector and $A = \mathbf{u}\mathbf{u}^T \in$
johnston-linear-matrix-algebra_F00839	71	■ 1.000	$A =$	$A =$	** 1.4.17 Suppose $\mathbf{u} \in \mathbb{R}^n$ is a unit vector and $A =$
johnston-linear-matrix-algebra_F00840	71	■ 0.481	$2\mathbf{u}\mathbf{u}^T - I \in \mathcal{M}_n$	$2\mathbf{u}\mathbf{u}^T - I \in \mathcal{M}_n$	$2\mathbf{u}\mathbf{u}^T - I \in \mathcal{M}_n$. Show that $A^2 = I$. Provide a geometric
johnston-linear-matrix-algebra_F00841	71	■ 1.000	$P_{\mathbf{u}}: \mathbb{R}^n \rightarrow \mathbb{R}^n$	$P_{\mathbf{u}}: \mathbb{R}^n \rightarrow \mathbb{R}^n$	1.4.18 Show that if $P_{\mathbf{u}}: \mathbb{R}^n \rightarrow \mathbb{R}^n$ is a projection onto a
johnston-linear-matrix-algebra_F00842	71	■ 1.000	$\ P_{\mathbf{u}}(\mathbf{v})\ \leq \ \mathbf{v}\ $	$\ P_{\mathbf{u}}(\mathbf{v})\ \leq \ \mathbf{v}\ $	line and $\mathbf{v} \in \mathbb{R}^n$ then $\ P_{\mathbf{u}}(\mathbf{v})\ \leq \ \mathbf{v}\ $.
johnston-linear-matrix-algebra_F00843	71	■ 1.000	$y = mx$	$y = mx$	$\mathbb{R}^2$ through the line $y = mx$ has standard matrix
johnston-linear-matrix-algebra_F00844	71	■ 1.000	$\mathbf{u}, \mathbf{v} \in \mathbb{R}^2$	$\mathbf{u}, \mathbf{v} \in \mathbb{R}^2$	1.4.21 Suppose $\mathbf{u}, \mathbf{v} \in \mathbb{R}^2$ are vectors with an angle of θ
johnston-linear-matrix-algebra_F00845	71	■ 1.000	$F_{\mathbf{u}} \circ F_{\mathbf{v}} = R^{2\theta}$	$F_{\mathbf{u}} \circ F_{\mathbf{v}} = R^{2\theta}$	between them. Show that $F_{\mathbf{u}} \circ F_{\mathbf{v}} = R^{2\theta}$.
johnston-linear-matrix-algebra_F00846	71	■ 1.000	$S_{i,j}^c$	$S_{i,j}^c$	for some non-zero scalar $c \in \mathbb{R}$. Let $S_{i,j}^c$ denote the shear
johnston-linear-matrix-algebra_F00847	71	■ 1.000	$S_{1,2}^c$	$S_{1,2}^c$	matrices $S_{1,2}^c$ and $S_{2,1}^c$ shown above as linear trans-
johnston-linear-matrix-algebra_F00848	71	■ 1.000	$S_{2,1}^c$	$S_{2,1}^c$	matrices $S_{1,2}^c$ and $S_{2,1}^c$ shown above as linear trans-

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johnston-linear-matrix-algebra_F00849	71	1.000	$\left(S_{\{i, j\}^c}^c\right)^n=S_{\{i, j\}^c}^n$	$\left(S_{i,j}^c\right)^n=S_{i,j}^{nc}$	(b) Show that $\left(S_{i,j}^c\right)^n=S_{i,j}^{nc}$ for all integers $n \geq 1$ (and
johnston-linear-matrix-algebra_F00850	71	1.000	$n \geq 1$	$n \geq 1$	(b) Show that $\left(S_{i,j}^c\right)^n=S_{i,j}^{nc}$ for all integers $n \geq 1$ (and
johnston-linear-matrix-algebra_F00851	71	1.000	$S_{\{i, j\}^c}=I+c E_{\{i, j\}}$	$S_{i,j}^c=I+c E_{i,j}$	$S_{i,j}^c=I+c E_{i,j}$, where $E_{i,j}$ is the matrix with a 1 in
johnston-linear-matrix-algebra_F00852	71	1.000	$E_{\{i, j\}}$	$E_{i,j}$	$S_{i,j}^c=I+c E_{i,j}$, where $E_{i,j}$ is the matrix with a 1 in
johnston-linear-matrix-algebra_F00853	71	1.000	$E_{\{i, j\}^{\{2\}}}$	$E_{i,j}^2$	its (i, j) -entry and zeros elsewhere. What is $E_{i,j}^2$?
johnston-linear-matrix-algebra_F00854	71	0.999	$R_{\{\mathbf{u}\}^{\{\theta\}}}$	$R_{\mathbf{u}}^{\theta}$	**1.4.24 Let $R_{\mathbf{u}}^{\theta}$ be the linear transformation that rotates
johnston-linear-matrix-algebra_F00855	71	1.000	$\left[R_{\{\mathbf{u}\}^{\{\theta\}}}\right]$	$\left[R_{\mathbf{u}}^{\theta}\right]$	(a) Compute $\left[R_{\mathbf{u}}^{\theta}\right]$ if $\mathbf{u}=(1,0,0)$. Which matrix that we
johnston-linear-matrix-algebra_F00856	71	1.000	$\mathbf{u}=(1,0,0)$	$\mathbf{u}=(1,0,0)$	(a) Compute $\left[R_{\mathbf{u}}^{\theta}\right]$ if $\mathbf{u}=(1,0,0)$. Which matrix that we
johnston-linear-matrix-algebra_F00857	71	1.000	$R_{\{\mathbf{u}\}^{\{\pi / 4\}}}\left(\mathbf{v}\right)$	$R_{\mathbf{u}}^{\pi / 4}(\mathbf{v})$	(b) Compute $R_{\mathbf{u}}^{\pi / 4}(\mathbf{v})$ if $\mathbf{u}=(1,2,2) / 3$ and $\mathbf{v}=(3,2,1)$.
johnston-linear-matrix-algebra_F00858	71	1.000	$\mathbf{u}=(1,2,2) / 3$	$\mathbf{u}=(1,2,2) / 3$	(b) Compute $R_{\mathbf{u}}^{\pi / 4}(\mathbf{v})$ if $\mathbf{u}=(1,2,2) / 3$ and $\mathbf{v}=(3,2,1)$.
johnston-linear-matrix-algebra_F00859	71	1.000	$\mathbf{v}=(3,2,1)$	$\mathbf{v}=(3,2,1)$	(b) Compute $R_{\mathbf{u}}^{\pi / 4}(\mathbf{v})$ if $\mathbf{u}=(1,2,2) / 3$ and $\mathbf{v}=(3,2,1)$.
johnston-linear-matrix-algebra_F00860	71	1.000	$R_{\{y z\}^{\{\pi / 3\}}} \circ R_{\{x y\}^{\{\pi / 3\}}}$	$R_{yz}^{\pi / 3} \circ R_{xy}^{\pi / 3}$	(c) Show that $R_{yz}^{\pi / 3} \circ R_{xy}^{\pi / 3}$ is a rotation matrix. Further-
johnston-linear-matrix-algebra_F00861	71	1.000	$R_{\{\mathbf{u}\}^{\{\theta\}}} = R_{\{y z\}^{\{\pi / 3\}}} \circ R_{\{x y\}^{\{\pi / 3\}}}$	$R_{\mathbf{u}}^{\theta}=R_{yz}^{\pi / 3} \circ R_{xy}^{\pi / 3}$	$R_{\mathbf{u}}^{\theta}=R_{yz}^{\pi / 3} \circ R_{xy}^{\pi / 3}$.
johnston-linear-matrix-algebra_F00862	73	1.000	$(A B)^T=A^T B^T$	$(A B)^T=A^T B^T$	then $(A B)^T=A^T B^T$.
johnston-linear-matrix-algebra_F00863	73	1.000	$(A B)^3=A^3 B^3$	$(A B)^3=A^3 B^3$	then $(A B)^3=A^3 B^3$.
johnston-linear-matrix-algebra_F00864	73	1.000	$(A+B)(A-B)=A^2-B^2$	$(A+B)(A-B)=A^2-B^2$	then $(A+B)(A-B)=A^2-B^2$.
johnston-linear-matrix-algebra_F00865	73	0.993	$A^2=O$	$A^2=O$	*(c) If A is a square matrix for which $A^2=O$, then
johnston-linear-matrix-algebra_F00866	73	0.998	$R, S, T: \mathbb{R}^n \rightarrow \mathbb{R}^n$	$R, S, T: \mathbb{R}^n \rightarrow \mathbb{R}^n$	*(g) If $R, S, T: \mathbb{R}^n \rightarrow \mathbb{R}^n$ are linear transformations, then
johnston-linear-matrix-algebra_F00867	73	1.000	$R \circ(S \circ T)=(R \circ S) \circ T$	$R \circ(S \circ T)=(R \circ S) \circ T$	$R \circ(S \circ T)=(R \circ S) \circ T$.
johnston-linear-matrix-algebra_F00868	73	1.000	$\ A \mathbf{v}\ =\ \mathbf{v}\ $	$\ A \mathbf{v}\ =\ \mathbf{v}\ $	(a) Show (algebraically) that $\ A \mathbf{v}\ =\ \mathbf{v}\ $ for all $\mathbf{v} \in \mathbb{R}^2$.
johnston-linear-matrix-algebra_F00869	73	0.717	$\left[\begin{array}{ll} 1 & 0 \\ 0 & 0 \end{array}\right]$	$\left[\begin{array}{cc} 1 & 0 \\ 0 & 0 \end{array}\right]$	*(a) $\left[\begin{array}{cc} 1 & 0 \\ 0 & 0 \end{array}\right]$
johnston-linear-matrix-algebra_F00870	73	1.000	$\left[\begin{array}{cc} -1 & 0 \\ 0 & 1 \end{array}\right]$	$\left[\begin{array}{cc} -1 & 0 \\ 0 & 1 \end{array}\right]$	(b) $\left[\begin{array}{cc} -1 & 0 \\ 0 & 1 \end{array}\right]$
johnston-linear-matrix-algebra_F00871	73	0.822	$\frac{1}{\sqrt{2}}\left[\begin{array}{cc} 1 & -1 \\ 1 & 1 \end{array}\right]$	$\frac{1}{\sqrt{2}}\left[\begin{array}{cc} 1 & -1 \\ 1 & 1 \end{array}\right]$	*(c) $\frac{1}{\sqrt{2}}\left[\begin{array}{cc} 1 & -1 \\ 1 & 1 \end{array}\right]$
johnston-linear-matrix-algebra_F00872	73	1.000	$\left[\begin{array}{ll} 1 & 0 \\ 0 & 1 \end{array}\right]$	$\left[\begin{array}{cc} 1 & 0 \\ 0 & 1 \end{array}\right]$	(d) $\left[\begin{array}{cc} 1 & 0 \\ 0 & 1 \end{array}\right]$
johnston-linear-matrix-algebra_F00873	73	0.902	$\left[\begin{array}{ll} 1 & 2 \\ 3 & 4 \end{array}\right]$	$\left[\begin{array}{cc} 1 & 2 \\ 3 & 4 \end{array}\right]$	*(e) $\left[\begin{array}{cc} 1 & 2 \\ 3 & 4 \end{array}\right]$

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johnston-linear-matrix-algebra_F00874	73	1.000	$\frac{1}{2}\left[\begin{array}{cc}1 & \sqrt{3} \\ \sqrt{3} & 1\end{array}\right]$	$\frac{1}{2}\begin{bmatrix}1 & \sqrt{3} \\ -\sqrt{3} & 1\end{bmatrix}$	
johnston-linear-matrix-algebra_F00875	73	0.931	$\frac{1}{9}\left[\begin{array}{ccc}1 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 4\end{array}\right]$	$\frac{1}{9}\begin{bmatrix}1 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 4\end{bmatrix}$	
johnston-linear-matrix-algebra_F00876	73	1.000	$\frac{1}{9}\left[\begin{array}{ccc}-1 & 4 & 8 \\ 4 & -7 & 4 \\ 8 & 4 & -1\end{array}\right]$	$\frac{1}{9}\begin{bmatrix}-1 & 4 & 8 \\ 4 & -7 & 4 \\ 8 & 4 & -1\end{bmatrix}$	
johnston-linear-matrix-algebra_F00877	73	1.000	$A^{500} \mathbf{v}$	$A^{500}\mathbf{v}$	Compute $A^{500}\mathbf{v}$.
johnston-linear-matrix-algebra_F00878	73	1.000	$\mathbf{w}=\left(v_2,-v_1\right)$	$\mathbf{w}=(v_2,-v_1)$	vector that works is $\mathbf{w}=(v_2,-v_1)$, since then $\mathbf{v}\cdot\mathbf{w}=v_1v_2-v_2v_1=0$ (see
johnston-linear-matrix-algebra_F00879	73	1.000	$\mathbf{v}\cdot\mathbf{w}=v_1v_2-v_2v_1=0$	$\mathbf{v}\cdot\mathbf{w}=v_1v_2-v_2v_1=0$	vector that works is $\mathbf{w}=(v_2,-v_1)$, since then $\mathbf{v}\cdot\mathbf{w}=v_1v_2-v_2v_1=0$ (see
johnston-linear-matrix-algebra_F00880	73	1.000	$\mathbf{x}\in\mathbb{R}^3$	$\mathbf{x}\in\mathbb{R}^3$	ask for a vector $\mathbf{x}\in\mathbb{R}^3$ that is orthogonal to <i>two</i> vectors $\mathbf{v}=(v_1,v_2,v_3)\in\mathbb{R}^3$
johnston-linear-matrix-algebra_F00881	73	1.000	$\mathbf{w}=\left(w_1,w_2,w_3\right)\in\mathbb{R}^3$	$\mathbf{w}=(w_1,w_2,w_3)\in\mathbb{R}^3$	and $\mathbf{w}=(w_1,w_2,w_3)\in\mathbb{R}^3$. It is much more difficult to eyeball a solution in
johnston-linear-matrix-algebra_F00882	73	1.000	$\mathbf{v}\times\mathbf{w}$	$\mathbf{v}\times\mathbf{w}$	cross product , denoted by $\mathbf{v}\times\mathbf{w}$, is defined by
johnston-linear-matrix-algebra_F00883	74	0.999	$\mathbf{w}\cdot(\mathbf{v}\times\mathbf{w})$	$\mathbf{w}\cdot(\mathbf{v}\times\mathbf{w})$	The dot product $\mathbf{w}\cdot(\mathbf{v}\times\mathbf{w})$ can similarly be shown to equal 0 (see Exer-
johnston-linear-matrix-algebra_F00884	74	0.717	$v_1, v_2, v_3, v_1, v_2, v_3$	$v_1, v_2, v_3, v_1, v_2, v_3$	write the entries of $\mathbf{v}$ in order twice, like $v_1, v_2, v_3, v_1, v_2, v_3$, and then similarly
johnston-linear-matrix-algebra_F00885	74	0.539	2×6	2×6	write the entries of $\mathbf{w}$ twice underneath, creating a 2×6 array. We completely
johnston-linear-matrix-algebra_F00886	75	0.986	$(2,-1)$	$(2,-1)$	of one of them: $(2,-1)$ works, since $(1,2)\cdot(2,-1)=2-2=0$.
johnston-linear-matrix-algebra_F00887	75	0.986	$(1,2)\cdot(2,-1)=2-2=0$	$(1,2)\cdot(2,-1)=2-2=0$	of one of them: $(2,-1)$ works, since $(1,2)\cdot(2,-1)=2-2=0$.
johnston-linear-matrix-algebra_F00888	75	1.000	$(1,2)$	$(1,2)$	Alternatively, we could think of $(1,2)$ as living in $\mathbb{R}^3$ (with z -entry
johnston-linear-matrix-algebra_F00889	75	0.999	$(1,2,0)\times(0,0,1)=(2,-1,0)$	$(1,2,0)\times(0,0,1)=(2,-1,0)$	the direction of the z -axis: $(1,2,0)\times(0,0,1)=(2,-1,0)$, which is
johnston-linear-matrix-algebra_F00890	75	1.000	$(1,2,3)\times(-1,0,1)=(2,-4,2)$	$(1,2,3)\times(-1,0,1)=(2,-4,2)$	b) $(1,2,3)\times(-1,0,1)=(2,-4,2)$ is orthogonal to both of these vec-
johnston-linear-matrix-algebra_F00891	75	1.000	$(2,-4,2)=0$	$(2,-4,2)=0$	$(2,-4,2)=0$ and $(-1,0,1)\cdot(2,-4,2)=0$.
johnston-linear-matrix-algebra_F00892	75	1.000	$(-1,0,1)\cdot(2,-4,2)=0$	$(-1,0,1)\cdot(2,-4,2)=0$	$(2,-4,2)=0$ and $(-1,0,1)\cdot(2,-4,2)=0$.
johnston-linear-matrix-algebra_F00893	75	1.000	$\mathbf{v}, \mathbf{w}, \mathbf{x}\in\mathbb{R}^3$	$\mathbf{v}, \mathbf{w}, \mathbf{x}\in\mathbb{R}^3$	Let $\mathbf{v}, \mathbf{w}, \mathbf{x}\in\mathbb{R}^3$ be vectors and let $c\in\mathbb{R}$ be a scalar. Then
johnston-linear-matrix-algebra_F00894	75	1.000	$\mathbf{v}\times\mathbf{w}=-(\mathbf{w}\times\mathbf{v})$	$\mathbf{v}\times\mathbf{w}=-(\mathbf{w}\times\mathbf{v})$	a) $\mathbf{v}\times\mathbf{w}=-(\mathbf{w}\times\mathbf{v})$,
johnston-linear-matrix-algebra_F00895	75	1.000	$\mathbf{v}\times(\mathbf{w}+\mathbf{x})=\mathbf{v}\times\mathbf{w}+\mathbf{v}\times\mathbf{x}$	$\mathbf{v}\times(\mathbf{w}+\mathbf{x})=\mathbf{v}\times\mathbf{w}+\mathbf{v}\times\mathbf{x}$	b) $\mathbf{v}\times(\mathbf{w}+\mathbf{x})=\mathbf{v}\times\mathbf{w}+\mathbf{v}\times\mathbf{x}$,
johnston-linear-matrix-algebra_F00896	75	1.000	$\mathbf{v}\times\mathbf{v}=\mathbf{0}$	$\mathbf{v}\times\mathbf{v}=\mathbf{0}$	c) $\mathbf{v}\times\mathbf{v}=\mathbf{0}$, and
johnston-linear-matrix-algebra_F00897	75	1.000	$(c\mathbf{v})\times\mathbf{w}=c(\mathbf{v}\times\mathbf{w})$	$(c\mathbf{v})\times\mathbf{w}=c(\mathbf{v}\times\mathbf{w})$	d) $(c\mathbf{v})\times\mathbf{w}=c(\mathbf{v}\times\mathbf{w})$.
johnston-linear-matrix-algebra_F00898	75	1.000	$\mathbf{w}\times\mathbf{v}$	$\mathbf{w}\times\mathbf{v}$	For property (a), we just compute $\mathbf{v}\times\mathbf{w}$ and $\mathbf{w}\times\mathbf{v}$ directly from the defini-

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johnston-linear-matrix-algebra_F00899	76	■ 1.000	v_1^2	v_1^2	The terms v_1^2
johnston-linear-matrix-algebra_F00900	76	■ 1.000	$w_1^2, v_2^2 w_2^2$	$w_1^2, v_2^2 w_2^2$	$w_1^2, v_2^2 w_2^2$, and $v_3^2 w_3^2$
johnston-linear-matrix-algebra_F00901	76	■ 1.000	$v_3^2 w_3^2$	$v_3^2 w_3^2$	$w_1^2, v_2^2 w_2^2$, and $v_3^2 w_3^2$
johnston-linear-matrix-algebra_F00902	76	■ 1.000	$\ \mathbf{v}\ \times \ \mathbf{w}\ $	$\ \mathbf{v} \times \mathbf{w}\ $	a) $\ \mathbf{v} \times \mathbf{w}\ $,
johnston-linear-matrix-algebra_F00903	76	■ 1.000	$\sqrt{\ \mathbf{v}\ ^2 \ \mathbf{w}\ ^2 - (\mathbf{v} \cdot \mathbf{w})^2}$	$\sqrt{\ \mathbf{v}\ ^2 \ \mathbf{w}\ ^2 - (\mathbf{v} \cdot \mathbf{w})^2}$	b) $\sqrt{\ \mathbf{v}\ ^2 \ \mathbf{w}\ ^2 - (\mathbf{v} \cdot \mathbf{w})^2}$,
johnston-linear-matrix-algebra_F00904	76	■ 0.934	$\ \mathbf{v}\ \ \mathbf{w}\ \sin(\theta)$	$\ \mathbf{v}\ \ \mathbf{w}\ \sin(\theta)$	c) $\ \mathbf{v}\ \ \mathbf{w}\ \sin(\theta)$, and
johnston-linear-matrix-algebra_F00905	76	■ 0.990	$\ \mathbf{w}\ \sin(\theta)$	$\ \mathbf{w}\ \sin(\theta)$	$\ \mathbf{w}\ \sin(\theta)$, and thus area equal to $\ \mathbf{v}\ \ \mathbf{w}\ \sin(\theta)$.
johnston-linear-matrix-algebra_F00906	76	■ 1.000	$\ \mathbf{w}\ \cos(\theta)$	$\ \mathbf{w}\ \cos(\theta)$	This process creates a right-angled triangle whose legs have lengths $\ \mathbf{w}\ \cos(\theta)$
johnston-linear-matrix-algebra_F00907	76	■ 1.000	$\ \mathbf{v} \times \mathbf{w}\ ^2 = \ \mathbf{v}\ ^2 \ \mathbf{w}\ ^2 - (\mathbf{v} \cdot \mathbf{w})^2$	$\ \mathbf{v} \times \mathbf{w}\ ^2 = \ \mathbf{v}\ ^2 \ \mathbf{w}\ ^2 - (\mathbf{v} \cdot \mathbf{w})^2$	that $\ \mathbf{v} \times \mathbf{w}\ ^2 = \ \mathbf{v}\ ^2 \ \mathbf{w}\ ^2 - (\mathbf{v} \cdot \mathbf{w})^2$ and then taking the square root of both
johnston-linear-matrix-algebra_F00908	76	■ 1.000	$\ \mathbf{v}\ ^2 \ \mathbf{w}\ ^2 - (\mathbf{v} \cdot \mathbf{w})^2$	$\ \mathbf{v}\ ^2 \ \mathbf{w}\ ^2 - (\mathbf{v} \cdot \mathbf{w})^2$	On the other hand, we can also expand $\ \mathbf{v}\ ^2 \ \mathbf{w}\ ^2 - (\mathbf{v} \cdot \mathbf{w})^2$ to get
johnston-linear-matrix-algebra_F00909	76	■ 1.000	$\ \mathbf{v} \times \mathbf{w}\ ^2$	$\ \mathbf{v} \times \mathbf{w}\ ^2$	which is the exact same as the expanded form of $\ \mathbf{v} \times \mathbf{w}\ ^2$ that we computed
johnston-linear-matrix-algebra_F00910	77	■ 1.000	$\sin^2(\theta) + \cos^2(\theta) = 1$	$\sin^2(\theta) + \cos^2(\theta) = 1$	$\sin^2(\theta) + \cos^2(\theta) = 1$,
johnston-linear-matrix-algebra_F00911	77	■ 0.718	$1 - \cos^2(\theta) = \sin^2(\theta)$	$1 - \cos^2(\theta) = \sin^2(\theta)$	$1 - \cos^2(\theta) = \sin^2(\theta)$.
johnston-linear-matrix-algebra_F00912	77	■ 0.754	$\theta = \arccos(\mathbf{v} \cdot \mathbf{w} / (\ \mathbf{v}\ \ \mathbf{w}\))$	$\theta = \arccos(\mathbf{v} \cdot \mathbf{w} / (\ \mathbf{v}\ \ \mathbf{w}\))$	fact that the angle between $\mathbf{v}$ and $\mathbf{w}$ is $\theta = \arccos(\mathbf{v} \cdot \mathbf{w} / (\ \mathbf{v}\ \ \mathbf{w}\))$ (recall
johnston-linear-matrix-algebra_F00913	77	■ 1.000	$\mathbf{v} \cdot \mathbf{w} = \ \mathbf{v}\ \ \mathbf{w}\ \cos(\theta)$	$\mathbf{v} \cdot \mathbf{w} = \ \mathbf{v}\ \ \mathbf{w}\ \cos(\theta)$	Definition 1.2.3), which can be rearranged into the form $\mathbf{v} \cdot \mathbf{w} = \ \mathbf{v}\ \ \mathbf{w}\ \cos(\theta)$.
johnston-linear-matrix-algebra_F00914	77	■ 1.000	$\sqrt{\sin^2(\theta)} = \sin(\theta) = \sin(\theta)$	$\sqrt{\sin^2(\theta)} = \sin(\theta) = \sin(\theta)$	where the final equality makes use of the fact that $\sqrt{\sin^2(\theta)} = \sin(\theta) = \sin(\theta)$
johnston-linear-matrix-algebra_F00915	77	■ 1.000	$\sin(\theta) \geq 0$	$\sin(\theta) \geq 0$	since $0 \leq \theta \leq \pi$, so $\sin(\theta) \geq 0$. This completes the proof.
johnston-linear-matrix-algebra_F00916	77	■ 0.883	$(3, 1)$	$(3, 1)$	b) the parallelogram in $\mathbb{R}^2$ with sides $(1, 2)$ and $(3, 1)$,
johnston-linear-matrix-algebra_F00917	77	■ 0.882	$\mathbb{R}^4$	$\mathbb{R}^4$	c) the parallelogram in $\mathbb{R}^4$ with sides $(2, 2, -1, 1)$ and $(-1, 1, 2, 1)$, and
johnston-linear-matrix-algebra_F00918	77	■ 1.000	$(1, 2, -1) \times (2, 1, 2) =$	$(1, 2, -1) \times (2, 1, 2) =$	a) We compute the length of the cross product: $(1, 2, -1) \times (2, 1, 2) =$
johnston-linear-matrix-algebra_F00919	77	■ 0.450	$\sqrt{5^2 + (-4)^2 + (-3)^2} = 5\sqrt{2}$	$\sqrt{5^2 + (-4)^2 + (-3)^2} = 5\sqrt{2}$	$(5, -4, -3)$, which has length $\sqrt{5^2 + (-4)^2 + (-3)^2} = 5\sqrt{2}$.
johnston-linear-matrix-algebra_F00920	77	■ 1.000	$(1, 2, 0) \times$	$(1, 2, 0) \times$	a z-entry equal to 0 and then using the cross product: $(1, 2, 0) \times$
johnston-linear-matrix-algebra_F00921	77	■ 0.991	$(3, 1, 0) = (0, 0, -5)$	$(3, 1, 0) = (0, 0, -5)$	$(3, 1, 0) = (0, 0, -5)$, which has length 5.
johnston-linear-matrix-algebra_F00922	78	■ 1.000	$ \cos(\theta) $	$ \cos(\theta) $	$ \cos(\theta) $ rather than
johnston-linear-matrix-algebra_F00923	78	■ 1.000	$\mathbf{w} \times \mathbf{x}$	$\mathbf{w} \times \mathbf{x}$	$\mathbf{w} \times \mathbf{x}$ might point
johnston-linear-matrix-algebra_F00924	78	■ 1.000	$\mathbf{v} \cdot (\mathbf{w} \times \mathbf{x})$	$\mathbf{v} \cdot (\mathbf{w} \times \mathbf{x})$	$\mathbf{v} \cdot (\mathbf{w} \times \mathbf{x})$ is called
johnston-linear-matrix-algebra_F00925	78	■ 0.960	$ \mathbf{v} \cdot (\mathbf{w} \times \mathbf{x}) $	$ \mathbf{v} \cdot (\mathbf{w} \times \mathbf{x}) $	sides $\mathbf{v}$, $\mathbf{w}$, and $\mathbf{x}$ is $ \mathbf{v} \cdot (\mathbf{w} \times \mathbf{x}) $.

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johnston-linear-matrix-algebra_F00926	78	■ 0.935	$ \mathbf{v} \cdot (\mathbf{w} \times \mathbf{x}) = \ \mathbf{v}\ \ \mathbf{w} \times \mathbf{x}\ \cos(\theta) $	$ \mathbf{v} \cdot (\mathbf{w} \times \mathbf{x}) = \ \mathbf{v}\ \ \mathbf{w} \times \mathbf{x}\ \cos(\theta)$	that $ \mathbf{v} \cdot (\mathbf{w} \times \mathbf{x}) = \ \mathbf{v}\ \ \mathbf{w} \times \mathbf{x}\ \cos(\theta)$.
johnston-linear-matrix-algebra_F00927	78	■ 0.999	$\ \mathbf{w} \times \mathbf{x}\ $	$\ \mathbf{w} \times \mathbf{x}\ $	Figure 1.26) is a parallelogram with sides $\mathbf{w}$ and $\mathbf{x}$, and thus has area $\ \mathbf{w} \times \mathbf{x}\ $
johnston-linear-matrix-algebra_F00928	78	■ 0.878	$ \cos(\theta) $	$\ \mathbf{v}\ \cos(\theta)$	its base $\ \mathbf{w} \times \mathbf{x}\ $ times its height $\ \mathbf{v}\ \cos(\theta)$, where θ is the angle between $\mathbf{w} \times \mathbf{x}$
johnston-linear-matrix-algebra_F00929	79	■ 0.998	$\mathbf{w} = (3, 2, 1)$	$\mathbf{w} = (3, 2, 1)$	*(a) $\mathbf{v} = (1, 2, 3)$ and $\mathbf{w} = (3, 2, 1)$
johnston-linear-matrix-algebra_F00930	79	■ 1.000	$\mathbf{v} = (87, 17, -43)$	$\mathbf{v} = (87, 17, -43)$	(b) $\mathbf{v} = (87, 17, -43)$ and $\mathbf{w} = (87, 17, -43)$
johnston-linear-matrix-algebra_F00931	79	■ 1.000	$\mathbf{w} = (87, 17, -43)$	$\mathbf{w} = (87, 17, -43)$	(b) $\mathbf{v} = (87, 17, -43)$ and $\mathbf{w} = (87, 17, -43)$
johnston-linear-matrix-algebra_F00932	78	■ 1.000	$\mathbf{v} = (-1, 2, 0)$	$\mathbf{v} = (-1, 2, 0)$	*(c) $\mathbf{v} = (-1, 2, 0)$ and $\mathbf{w} = (-4, 1, -2)$
johnston-linear-matrix-algebra_F00933	79	■ 1.000	$\mathbf{w} = (-4, 1, -2)$	$\mathbf{w} = (-4, 1, -2)$	*(c) $\mathbf{v} = (-1, 2, 0)$ and $\mathbf{w} = (-4, 1, -2)$
johnston-linear-matrix-algebra_F00934	79	■ 1.000	$\mathbf{v} = (1, 0, 0)$	$\mathbf{v} = (1, 0, 0)$	(d) $\mathbf{v} = (1, 0, 0)$ and $\mathbf{w} = (0, 1, 0)$
johnston-linear-matrix-algebra_F00935	79	■ 1.000	$\mathbf{w} = (0, 1, 0)$	$\mathbf{w} = (0, 1, 0)$	(d) $\mathbf{v} = (1, 0, 0)$ and $\mathbf{w} = (0, 1, 0)$
johnston-linear-matrix-algebra_F00936	79	■ 0.994	$\mathbf{v} = (2, 1)$	$\mathbf{v} = (2, 1)$	*(a) $\mathbf{v} = (2, 1)$ and $\mathbf{w} = (-2, 3)$
johnston-linear-matrix-algebra_F00937	78	■ 0.994	$\mathbf{w} = (-2, 3)$	$\mathbf{w} = (-2, 3)$	*(a) $\mathbf{v} = (2, 1)$ and $\mathbf{w} = (-2, 3)$
johnston-linear-matrix-algebra_F00938	79	■ 1.000	$\mathbf{v} = (3, 0, 0)$	$\mathbf{v} = (3, 0, 0)$	(b) $\mathbf{v} = (3, 0, 0)$ and $\mathbf{w} = (0, 4, 0)$
johnston-linear-matrix-algebra_F00939	79	■ 1.000	$\mathbf{w} = (0, 4, 0)$	$\mathbf{w} = (0, 4, 0)$	(b) $\mathbf{v} = (3, 0, 0)$ and $\mathbf{w} = (0, 4, 0)$
johnston-linear-matrix-algebra_F00940	79	■ 1.000	$\mathbf{w} = (3, -1, 2)$	$\mathbf{w} = (3, -1, 2)$	*(c) $\mathbf{v} = (1, 2, 3)$ and $\mathbf{w} = (3, -1, 2)$
johnston-linear-matrix-algebra_F00941	79	■ 1.000	$\mathbf{v} = (1, 0, 1, -2)$	$\mathbf{v} = (1, 0, 1, -2)$	(d) $\mathbf{v} = (1, 0, 1, -2)$ and $\mathbf{w} = (-2, 1, 3, 1)$
johnston-linear-matrix-algebra_F00942	79	■ 1.000	$\mathbf{w} = (-2, 1, 3, 1)$	$\mathbf{w} = (-2, 1, 3, 1)$	(d) $\mathbf{v} = (1, 0, 1, -2)$ and $\mathbf{w} = (-2, 1, 3, 1)$
johnston-linear-matrix-algebra_F00943	79	■ 0.954	$\mathbf{v} = (0, 4)$	$\mathbf{v} = (0, 4)$	* (a) $\mathbf{v} = (0, 4)$ and $\mathbf{w} = (1, 1)$
johnston-linear-matrix-algebra_F00944	79	■ 0.954	$\mathbf{w} = (1, 1)$	$\mathbf{w} = (1, 1)$	* (a) $\mathbf{v} = (0, 4)$ and $\mathbf{w} = (1, 1)$
johnston-linear-matrix-algebra_F00945	79	■ 1.000	$\mathbf{v} = (0, 2, 2)$	$\mathbf{v} = (0, 2, 2)$	(b) $\mathbf{v} = (0, 2, 2)$ and $\mathbf{w} = (1, -2, 1)$
johnston-linear-matrix-algebra_F00946	79	■ 1.000	$\mathbf{w} = (1, -2, 1)$	$\mathbf{w} = (1, -2, 1)$	(b) $\mathbf{v} = (0, 2, 2)$ and $\mathbf{w} = (1, -2, 1)$
johnston-linear-matrix-algebra_F00947	79	■ 0.650	$\mathbf{v} = (-1, 1, -1)$	$\mathbf{v} = (-1, 1, -1)$	* (c) $\mathbf{v} = (-1, 1, -1)$ and $\mathbf{w} = (3, 2, 1)$
johnston-linear-matrix-algebra_F00948	78	■ 1.000	$\mathbf{v} = (1, 2, 1, 2)$	$\mathbf{v} = (1, 2, 1, 2)$	(d) $\mathbf{v} = (1, 2, 1, 2)$ and $\mathbf{w} = (0, -1, 2, 1)$
johnston-linear-matrix-algebra_F00949	79	■ 1.000	$\mathbf{w} = (0, -1, 2, 1)$	$\mathbf{w} = (0, -1, 2, 1)$	(d) $\mathbf{v} = (1, 2, 1, 2)$ and $\mathbf{w} = (0, -1, 2, 1)$
johnston-linear-matrix-algebra_F00950	79	■ 0.988	$\mathbf{v} = (1, 0, 0), \mathbf{w} = (0, 2, 0)$	$\mathbf{v} = (1, 0, 0), \mathbf{w} = (0, 2, 0)$	* (a) $\mathbf{v} = (1, 0, 0), \mathbf{w} = (0, 2, 0),$ and $\mathbf{x} = (0, 0, 3)$
johnston-linear-matrix-algebra_F00951	79	■ 0.988	$\mathbf{x} = (0, 0, 3)$	$\mathbf{x} = (0, 0, 3)$	* (a) $\mathbf{v} = (1, 0, 0), \mathbf{w} = (0, 2, 0),$ and $\mathbf{x} = (0, 0, 3)$
johnston-linear-matrix-algebra_F00952	79	■ 1.000	$\mathbf{v} = (0, 4, 1), \mathbf{w} = (1, 1, 0)$	$\mathbf{v} = (0, 4, 1), \mathbf{w} = (1, 1, 0)$	(b) $\mathbf{v} = (0, 4, 1), \mathbf{w} = (1, 1, 0),$ and $\mathbf{x} = (2, 0, -1)$
johnston-linear-matrix-algebra_F00953	78	■ 1.000	$\mathbf{x} = (2, 0, -1)$	$\mathbf{x} = (2, 0, -1)$	(b) $\mathbf{v} = (0, 4, 1), \mathbf{w} = (1, 1, 0),$ and $\mathbf{x} = (2, 0, -1)$
johnston-linear-matrix-algebra_F00954	79	■ 0.913	$\mathbf{v} = (1, 1, 1), \mathbf{w} = (2, -1, 1)$	$\mathbf{v} = (1, 1, 1), \mathbf{w} = (2, -1, 1)$	* (c) $\mathbf{v} = (1, 1, 1), \mathbf{w} = (2, -1, 1),$ and $\mathbf{x} = (2, -2, -1)$

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johnston-linear-matrix-algebra_F00955	79	■ 0.913	$\mathbf{x}=(2,-2,-1)$	$\mathbf{x} = (2, -2, -1)$	* (c) $\mathbf{v} = (1, 1, 1)$, $\mathbf{w} = (2, -1, 1)$, and $\mathbf{x} = (2, -2, -1)$
johnston-linear-matrix-algebra_F00956	79	■ 1.000	$\mathbf{v}=(-2,1,3)$, $\mathbf{w}=(1,-4,2)$	$\mathbf{v} = (-2, 1, 3), \mathbf{w} = (1, -4, 2)$	(d) $\mathbf{v} = (-2, 1, 3)$, $\mathbf{w} = (1, -4, 2)$, and $\mathbf{x} = (3, 3, -2)$
johnston-linear-matrix-algebra_F00957	79	■ 1.000	$\mathbf{x}=(3,3,-2)$	$\mathbf{x} = (3, 3, -2)$	(d) $\mathbf{v} = (-2, 1, 3)$, $\mathbf{w} = (1, -4, 2)$, and $\mathbf{x} = (3, 3, -2)$
johnston-linear-matrix-algebra_F00958	79	■ 1.000	$\left.\begin{array}{c} -2 \\ 3 \end{array}\right]$	$\left.\begin{array}{c} -2 \\ 3 \end{array}\right]$	$\left.\begin{array}{c} -2 \\ 3 \end{array}\right]$
johnston-linear-matrix-algebra_F00959	79	■ 0.455	$\begin{aligned} & \left[\begin{array}{c} 1 & 7 \\ 0 & 1 \end{array} \right] \text{ (b) } \left[\begin{array}{c} 1 & 2 \\ 3 & 4 \end{array} \right] \end{aligned}$	$\left[\begin{array}{c} 1 & 7 \\ 0 & 1 \end{array} \right]$ (b) $\left[\begin{array}{c} 1 & 2 \\ 3 & 4 \end{array} \right]$	(b) $\left[\begin{array}{c} 1 & 7 \\ 0 & 1 \end{array} \right]$ (b) $\left[\begin{array}{c} 1 & 2 \\ 3 & 4 \end{array} \right]$
johnston-linear-matrix-algebra_F00960	79	■ 1.000	$\left[\begin{array}{ccc} 1 & -3 & 2 \\ 0 & 2 & -7 \\ 0 & 0 & 3 \end{array} \right]$	$\left[\begin{array}{ccc} 1 & -3 & 2 \\ 0 & 2 & -7 \\ 0 & 0 & 3 \end{array} \right]$	$\left[\begin{array}{ccc} 1 & -3 & 2 \\ 0 & 2 & -7 \\ 0 & 0 & 3 \end{array} \right]$
johnston-linear-matrix-algebra_F00961	79	■ 1.000	$\left[\begin{array}{ccc} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{array} \right]$	$\left[\begin{array}{ccc} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{array} \right]$	$\left[\begin{array}{ccc} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{array} \right]$
johnston-linear-matrix-algebra_F00962	79	■ 1.000	$\ \mathbf{v} \times \mathbf{w}\ = \ \mathbf{w} \times \mathbf{v}\ $	$\ \mathbf{v} \times \mathbf{w}\ = \ \mathbf{w} \times \mathbf{v}\ $	(b) If $\mathbf{v}, \mathbf{w} \in \mathbb{R}^3$ then $\ \mathbf{v} \times \mathbf{w}\ = \ \mathbf{w} \times \mathbf{v}\ $.
johnston-linear-matrix-algebra_F00963	79	■ 0.991	$\mathbf{v} \in \mathbb{R}^3$	$\mathbf{v} \in \mathbb{R}^3$	* (c) If $\mathbf{v} \in \mathbb{R}^3$ then $\mathbf{v} \times \mathbf{v} = \mathbf{v}^2$.
johnston-linear-matrix-algebra_F00964	79	■ 0.991	$\mathbf{v} \times \mathbf{v} = \mathbf{v}^2$	$\mathbf{v} \times \mathbf{v} = \mathbf{v}^2$	* (c) If $\mathbf{v} \in \mathbb{R}^3$ then $\mathbf{v} \times \mathbf{v} = \mathbf{v}^2$.
johnston-linear-matrix-algebra_F00965	79	■ 1.000	$(\mathbf{v} + \mathbf{v}) \times \mathbf{w} = 2(\mathbf{v} \times \mathbf{w})$	$(\mathbf{v} + \mathbf{v}) \times \mathbf{w} = 2(\mathbf{v} \times \mathbf{w})$	(d) If $\mathbf{v}, \mathbf{w} \in \mathbb{R}^3$ then $(\mathbf{v} + \mathbf{v}) \times \mathbf{w} = 2(\mathbf{v} \times \mathbf{w})$.
johnston-linear-matrix-algebra_F00966	79	■ 1.000	$\ \mathbf{v}\ = 1, \ \mathbf{w}\ = 1$	$\ \mathbf{v}\ = 1, \ \mathbf{w}\ = 1$	(f) There exist vectors $\mathbf{v}, \mathbf{w} \in \mathbb{R}^3$ with $\ \mathbf{v}\ = 1, \ \mathbf{w}\ = 1$,
johnston-linear-matrix-algebra_F00967	79	■ 1.000	$\mathbf{v} \times \mathbf{w} = (1, 1, 1)$	$\mathbf{v} \times \mathbf{w} = (1, 1, 1)$	and $\mathbf{v} \times \mathbf{w} = (1, 1, 1)$.
johnston-linear-matrix-algebra_F00968	79	■ 1.000	$\mathbf{e}_1 \times \mathbf{e}_2, \mathbf{e}_2 \times \mathbf{e}_3, \mathbf{e}_3 \times \mathbf{e}_1$	$\mathbf{e}_1 \times \mathbf{e}_2, \mathbf{e}_2 \times \mathbf{e}_3$	products: $\mathbf{e}_1 \times \mathbf{e}_2$, $\mathbf{e}_2 \times \mathbf{e}_3$, and $\mathbf{e}_3 \times \mathbf{e}_1$.
johnston-linear-matrix-algebra_F00969	79	■ 1.000	$\mathbf{e}_3 \times \mathbf{e}_1$	$\mathbf{e}_3 \times \mathbf{e}_1$	products: $\mathbf{e}_1 \times \mathbf{e}_2$, $\mathbf{e}_2 \times \mathbf{e}_3$, and $\mathbf{e}_3 \times \mathbf{e}_1$.
johnston-linear-matrix-algebra_F00970	79	■ 1.000	$\mathbf{v} \times \mathbf{w} = (1/3, 2/3, 2/3)$	$\mathbf{v} \times \mathbf{w} = (1/3, 2/3, 2/3)$	$\mathbf{v} \times \mathbf{w} = (1/3, 2/3, 2/3)$. What is the angle between $\mathbf{v}$ and $\mathbf{w}$?
johnston-linear-matrix-algebra_F00971	80	■ 0.999	$\ \mathbf{v} \times \mathbf{w}\ = \ \mathbf{v}\ \ \mathbf{w}\ $	$\ \mathbf{v} \times \mathbf{w}\ = \ \mathbf{v}\ \ \mathbf{w}\ $	$\ \mathbf{v} \times \mathbf{w}\ = \ \mathbf{v}\ \ \mathbf{w}\ $ if and only if $\mathbf{v}$ and $\mathbf{w}$ are orthogonal.
johnston-linear-matrix-algebra_F00972	80	■ 0.560	$\mathbf{v} \cdot (\mathbf{v} \times \mathbf{w}) = 0$	$\mathbf{v} \cdot (\mathbf{v} \times \mathbf{w}) = 0$	**1.A.10 We showed that if $\mathbf{v}, \mathbf{w} \in \mathbb{R}^3$ then $\mathbf{v} \cdot (\mathbf{v} \times \mathbf{w}) = 0$.
johnston-linear-matrix-algebra_F00973	80	■ 0.560	$\mathbf{w} \cdot (\mathbf{v} \times \mathbf{w}) = 0$	$\mathbf{w} \cdot (\mathbf{v} \times \mathbf{w}) = 0$	**1.A.10 We showed that if $\mathbf{v}, \mathbf{w} \in \mathbb{R}^3$ then $\mathbf{v} \cdot (\mathbf{v} \times \mathbf{w}) = 0$.
johnston-linear-matrix-algebra_F00974	80	■ 1.000	$\mathbf{v} \times (c\mathbf{w}) = c(\mathbf{v} \times \mathbf{w})$	$\mathbf{v} \times (c\mathbf{w}) = c(\mathbf{v} \times \mathbf{w})$	Show that $\mathbf{w} \cdot (\mathbf{v} \times \mathbf{w}) = 0$ too.
johnston-linear-matrix-algebra_F00975	80	■ 1.000	$(\mathbf{v} \times \mathbf{w}) \times \mathbf{x} = \mathbf{v} \times (\mathbf{w} \times \mathbf{x})$	$(\mathbf{v} \times \mathbf{w}) \times \mathbf{x} = \mathbf{v} \times (\mathbf{w} \times \mathbf{x})$	a scalar. Show that $\mathbf{v} \times (c\mathbf{w}) = c(\mathbf{v} \times \mathbf{w})$.
johnston-linear-matrix-algebra_F00976	80	■ 1.000	$\mathbf{v} \times \mathbf{w} = \mathbf{0}$	$\mathbf{v} \times \mathbf{w} = \mathbf{0}$	is it true that $(\mathbf{v} \times \mathbf{w}) \times \mathbf{x} = \mathbf{v} \times (\mathbf{w} \times \mathbf{x})$ for all $\mathbf{v}, \mathbf{w}, \mathbf{x} \in \mathbb{R}^3$?
johnston-linear-matrix-algebra_F00977	80	■ 1.000	$\mathbf{v} \times \mathbf{w} = \mathbf{0}$	$\mathbf{v} \times \mathbf{w} = \mathbf{0}$	the stronger statement that $\mathbf{v} \times \mathbf{w} = \mathbf{0}$ if and only if $\mathbf{w} = \mathbf{0}$
johnston-linear-matrix-algebra_F00978	80	■ 1.000	$\mathbf{v} \times \mathbf{w} = \mathbf{v} \times \mathbf{x}$	$\mathbf{v} \times \mathbf{w} = \mathbf{v} \times \mathbf{x}$	$\mathbf{v} \times \mathbf{w} = \mathbf{v} \times \mathbf{x}$ even if $\mathbf{w} \neq \mathbf{x}$.

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johnston-linear-matrix-algebra-F00979	80	■ 1.000	$\mathbf{w} \neq \mathbf{x}$	$\mathbf{w} \neq \mathbf{x}$	$\mathbf{v} \times \mathbf{w} = \mathbf{v} \times \mathbf{x}$ even if $\mathbf{w} \neq \mathbf{x}$.
johnston-linear-matrix-algebra-F00980	80	■ 1.000	$\mathbf{v} \cdot (\mathbf{w} \times \mathbf{x}) = \mathbf{w} \cdot (\mathbf{x} \times \mathbf{v}) = \mathbf{x} \cdot (\mathbf{v} \times \mathbf{w})$	$\mathbf{v} \cdot (\mathbf{w} \times \mathbf{x}) = \mathbf{w} \cdot (\mathbf{x} \times \mathbf{v}) = \mathbf{x} \cdot (\mathbf{v} \times \mathbf{w})$	(a) Show that $\mathbf{v} \cdot (\mathbf{w} \times \mathbf{x}) = \mathbf{w} \cdot (\mathbf{x} \times \mathbf{v}) = \mathbf{x} \cdot (\mathbf{v} \times \mathbf{w})$.
johnston-linear-matrix-algebra-F00981	80	■ 1.000	$\mathbf{v} \times (\mathbf{w} \times \mathbf{x}) = (\mathbf{v} \cdot \mathbf{x})\mathbf{w} - (\mathbf{v} \cdot \mathbf{w})\mathbf{x}$	$\mathbf{v} \times (\mathbf{w} \times \mathbf{x}) = (\mathbf{v} \cdot \mathbf{x})\mathbf{w} - (\mathbf{v} \cdot \mathbf{w})\mathbf{x}$	$\mathbf{v} \times (\mathbf{w} \times \mathbf{x}) = (\mathbf{v} \cdot \mathbf{x})\mathbf{w} - (\mathbf{v} \cdot \mathbf{w})\mathbf{x}$. [Side note: This is called
johnston-linear-matrix-algebra-F00982	81	■ 1.000	$A - D - B$	$A - D - B$	path: $A - D - B$.
johnston-linear-matrix-algebra-F00983	82	■ 1.000	$a_{i,j} = a_{j,i}$	$a_{i,j} = a_{j,i}$	$a_{i,j} = a_{j,i}$ for all i, j
johnston-linear-matrix-algebra-F00984	82	■ 1.000	i, j	i, j	$a_{i,j} = a_{j,i}$ for all i, j
johnston-linear-matrix-algebra-F00985	82	■ 0.999	$\left(A = A^T\right)$	$A = A^T$	$A = A^T$.
johnston-linear-matrix-algebra-F00986	82	■ 1.000	$a_{i,j} = 1$	$a_{i,j} = 1$	the number of paths of length 1 between vertices i and j (since $a_{i,j} = 1$ if there
johnston-linear-matrix-algebra-F00987	82	■ 1.000	$a_{i,j} = 0$	$a_{i,j} = 0$	is such a path and $a_{i,j} = 0$ otherwise). Remarkably, the entries of higher powers
johnston-linear-matrix-algebra-F00988	83	■ 1.000	$\left[A^k\right]_{i,j}$	$\left[A^k\right]_{i,j}$	notation $\left[A^k\right]_{i,j}$
johnston-linear-matrix-algebra-F00989	83	■ 1.000	$k \geq 1$	$k \geq 1$	Suppose $A \in \mathcal{M}_n$ is the adjacency matrix of a graph, and let $k \geq 1$ be
johnston-linear-matrix-algebra-F00990	83	■ 1.000	$k=1$	$k = 1$	<i>Proof.</i> We already noted why this theorem is true when $k = 1$. To show that it
johnston-linear-matrix-algebra-F00991	83	■ 1.000	$k=2$	$k = 2$	is true when $k = 2$, recall from the definition of matrix multiplication that
johnston-linear-matrix-algebra-F00992	83	■ 0.981	$a_{i,1} a_{1,j}$	$a_{i,1} a_{1,j}$	The first term in the sum above ($a_{i,1} a_{1,j}$) equals 1 exactly if there is an edge
johnston-linear-matrix-algebra-F00993	83	■ 1.000	$a_{i,2} a_{2,j}$	$a_{i,2} a_{2,j}$	Similarly, $a_{i,2} a_{2,j}$ counts the number of paths of length 2 between vertices
johnston-linear-matrix-algebra-F00994	83	■ 1.000	$\left[A^2\right]_{i,j}$	$\left[A^2\right]_{i,j}$	terms up, we see that $\left[A^2\right]_{i,j}$ counts the number of paths of length 2 between
johnston-linear-matrix-algebra-F00995	84	■ 1.000	A^6	A^6	entire matrix A^6
johnston-linear-matrix-algebra-F00996	84	■ 0.968	(A)	(A)	from the first vertex (A) to the fourth vertex (D). Notice that this
johnston-linear-matrix-algebra-F00997	84	■ 0.968	(D)	(D)	from the first vertex (A) to the fourth vertex (D). Notice that this
johnston-linear-matrix-algebra-F00998	84	■ 1.000	$A^6 = A A A A A A$	$A^6 = A A A A A A$	laboriously multiply five times to get $A^6 = A A A A A A$, it is quicker
johnston-linear-matrix-algebra-F00999	84	■ 1.000	$A^6 = \left(A^3\right)^2$	$A^6 = \left(A^3\right)^2$	to recall that $A^6 = \left(A^3\right)^2$, so we can just square the matrix that we
johnston-linear-matrix-algebra-F01000	84	■ 1.000	(B)	(B)	length 6 from the second vertex (B) to the third vertex (C).
johnston-linear-matrix-algebra-F01001	84	■ 1.000	(C)	(C)	length 6 from the second vertex (B) to the third vertex (C).
johnston-linear-matrix-algebra-F01002	85	■ 1.000	$\left[A^2\right]_{5,1}$	$\left[A^2\right]_{5,1}$	stopover in between is then $\left[A^2\right]_{5,1}$, so we compute A^2 :
johnston-linear-matrix-algebra-F01003	85	■ 1.000	$\left[A^3\right]_{4,6} + \left[A^2\right]_{4,6} + a_{4,6}$	$\left[A^3\right]_{4,6} + \left[A^2\right]_{4,6} + a_{4,6}$	$\left[A^3\right]_{4,6} + \left[A^2\right]_{4,6} + a_{4,6}$, so we start by computing $\left[A^3\right]_{4,6}$, which is
johnston-linear-matrix-algebra-F01004	85	■ 1.000	$\left[A^3\right]_{4,6}$	$\left[A^3\right]_{4,6}$	$\left[A^3\right]_{4,6} + \left[A^2\right]_{4,6} + a_{4,6}$, so we start by computing $\left[A^3\right]_{4,6}$, which is

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johnston-linear-matrix-algebra-F01005	86	■ 1.000	E	E	b) of length 4 from C to E , and
johnston-linear-matrix-algebra-F01006	86	■ 1.000	$\left[A^4\right]_{3,5}$	$[A^4]_{3,5}$	b) We need to compute $[A^4]_{3,5}$, which is the dot product of the 3rd row
johnston-linear-matrix-algebra-F01007	86	■ 0.995	A^{87}	A^{87}	c) We could compute A^{87} and look at its $(1,3)$ -entry (ugh!), but an
johnston-linear-matrix-algebra-F01008	87	■ 0.995	$0,0,1,0,0$	$0,0,1,0,0$	path of length k from B to C) and $(0,0,1,0,0)$ when k is even (and
johnston-linear-matrix-algebra-F01009	87	■ 1.000	$\left[A^2\right]_{1,4}$	$[A^2]_{1,4}$	D , we compute $[A^2]_{1,4}$, which is the dot product of the 1st row of
johnston-linear-matrix-algebra-F01010	88	■ 1.000	$\left[A^3\right]_{2,3}$	$[A^3]_{2,3}$	$[A^3]_{2,3}$ is also the
johnston-linear-matrix-algebra-F01011	88	■ 1.000	$(0,1,1,1) \cdot (1,1,2,1)=4$	$(0,1,1,1) \cdot (1,1,2,1) = 4$	$(0,1,1,1) \cdot (1,1,2,1) = 4$
johnston-linear-matrix-algebra-F01012	88	■ 1.000	$\sum_{j=1}^n w_j$	$\sum_{j=1}^n w_j$	$\sum_{j=1}^n w_j$ means
johnston-linear-matrix-algebra-F01013	88	■ 1.000	$w_1+w_2+\cdots+w_n$	$w_1 + w_2 + \cdots + w_n$	$w_1 + w_2 + \cdots + w_n$.
johnston-linear-matrix-algebra-F01014	88	■ 0.964	$n \approx 7.5$	$n \approx 7.5$	in the world ($n \approx 7.5$ billion) and let $\mathbf{w} = (w_1, w_2, \dots, w_n) \in \mathbb{R}^n$ be such
johnston-linear-matrix-algebra-F01015	88	■ 1.000	$1 \leq j \leq n, w_j$	$1 \leq j \leq n, w_j$	that, for each $1 \leq j \leq n$, w_j is the number of friends that the j -th person
johnston-linear-matrix-algebra-F01016	89	■ 1.000	$(3+2+1+2) / 4=2$	$(3 + 2 + 1 + 2)/4 = 2$	$(3 + 2 + 1 + 2)/4 = 2$.
johnston-linear-matrix-algebra-F01017	89	■ 1.000	$2 \leq 2.25$	$2 \leq 2.25$	Since $2 \leq 2.25$, we
johnston-linear-matrix-algebra-F01018	89	■ 0.999	$2 \times 2=4$	$2 \times 2 = 4$	$2 \times 2 = 4$ names,
johnston-linear-matrix-algebra-F01019	89	■ 0.835	$1 \times 1=1$	$1 \times 1 = 1$	$1 \times 1 = 1$.
johnston-linear-matrix-algebra-F01020	89	■ 1.000	$(2+1+2+3+2+3+3+2) / 8=$	$(2 + 1 + 2 + 3 + 2 + 3 + 3 + 2)/8 =$	(excluding the bold headers), which is $(2 + 1 + 2 + 3 + 2 + 3 + 3 + 2)/8 =$
johnston-linear-matrix-algebra-F01021	89	■ 1.000	w_j	w_j	In general, the j -th person creates w_j lists—one for each of their
johnston-linear-matrix-algebra-F01022	89	■ 1.000	w_j^2	w_j^2	of those lists contain w_j entries. It follows that person j contributes w_j^2
johnston-linear-matrix-algebra-F01023	89	■ 1.000	$\sum_{j=1}^n w_j^2$	$\sum_{j=1}^n w_j^2$	entries to these lists, so there are $\sum_{j=1}^n w_j^2$ total friends of friends listed.
johnston-linear-matrix-algebra-F01024	89	■ 1.000	$\mathbf{v}=(1,1,\ldots,1) \in \mathbb{R}^n$	$\mathbf{v} = (1,1,\dots,1) \in \mathbb{R}^n$	If we let $\mathbf{v} = (1,1,\dots,1) \in \mathbb{R}^n$ and then apply the Cauchy–Schwarz
johnston-linear-matrix-algebra-F01025	89	■ 1.000	$n \sum_{j=1}^n w_j$	$n \sum_{j=1}^n w_j$	by $n \sum_{j=1}^n w_j$, results in the inequality
johnston-linear-matrix-algebra-F01026	90	■ 1.000	$A=A^T$	$A = A^T$	is symmetric (i.e., has the property $A = A^T$).
johnston-linear-matrix-algebra-F01027	90	■ 1.000	$\left[\begin{array}{llll}0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0\end{array}\right]$	$\left[\begin{array}{cccc}0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0\end{array}\right]$	$\left[\begin{array}{cccc}0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0\end{array}\right]$

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johnston-linear-matrix-algebra-F01028	90	1.000	$\left[\begin{array}{lllll}0 & 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 & 1 \\ 1 & 1 & 0 & 1 & 1 \\ 1 & 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 1 & 0\end{array}\right]$	$\begin{bmatrix} 0 & 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 & 1 \\ 1 & 1 & 0 & 1 & 1 \\ 1 & 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 1 & 0 \end{bmatrix}$	
johnston-linear-matrix-algebra-F01029	90	1.000	$\left[\begin{array}{llll}2 & 0 & 0 & 1 \\ 0 & 1 & 3 & 0 \\ 0 & 0 & 1 & 1 \\ 3 & 0 & 1 & 1\end{array}\right]$	$\begin{bmatrix} 2 & 0 & 0 & 1 \\ 0 & 1 & 3 & 0 \\ 0 & 0 & 1 & 1 \\ 3 & 0 & 1 & 1 \end{bmatrix}$	
johnston-linear-matrix-algebra-F01030	91	1.000	A^n	A^n	(c) Use part (b) to find a formula for A^n . [Hint: Your
johnston-linear-matrix-algebra-F01031	92	1.000	x, y	x, y	use x, y , and z if
johnston-linear-matrix-algebra-F01032	92	1.000	$x_{\{1\}}, x_{\{2\}}, \ldots, x_{\{n\}}$	$x_1, x_2, \ldots, x_n$	use $x_1, x_2, \ldots, x_n$ if
johnston-linear-matrix-algebra-F01033	92	1.000	$a_{\{1\}}, a_{\{2\}}, \ldots, a_{\{n\}}$	$a_1, a_2, \ldots, a_n$	where $a_1, a_2, \ldots, a_n$ and b are constants called the coefficients of the linear
johnston-linear-matrix-algebra-F01034	92	1.000	$4x - 6y = -3$	$4x - 6y = -3$	is linear. In particular, it is equivalent to the equation $4x - 6y = -3$, which is in
johnston-linear-matrix-algebra-F01035	93	1.000	$y = mx + b$	$y = mx + b$	the form $y = mx + b$,
johnston-linear-matrix-algebra-F01036	93	1.000	$x = 2$	$x = 2$	Try plugging $x = 2$,
johnston-linear-matrix-algebra-F01037	93	1.000	$y = 1$	$y = 1$	$y = 1$ back into the
johnston-linear-matrix-algebra-F01038	93	1.000	$ax + by = c$	$ax + by = c$	example, the general equation of a line is $ax + by = c$, and the general equation
johnston-linear-matrix-algebra-F01039	93	1.000	$ax + by + cz = d$	$ax + by + cz = d$	of a plane is $ax + by + cz = d$, both of which are linear equations (see Figure 2.1).
johnston-linear-matrix-algebra-F01040	93	1.000	$\mathbf{x} =$	$\mathbf{x} =$	• a solution of a system of linear equations is a vector $\mathbf{x} =$
johnston-linear-matrix-algebra-F01041	93	1.000	$\left(x_{\{1\}}, x_{\{2\}}, \ldots, x_{\{n\}}\right)$	$(x_1, x_2, \ldots, x_n)$	$(x_1, x_2, \ldots, x_n)$ whose entries satisfy all of the linear equations in the
johnston-linear-matrix-algebra-F01042	94	0.834	$\mathbf{x} = (2, 1)$	$\mathbf{x} = (2, 1)$	it is located at the point $(2, 1)$. The vector $\mathbf{x} = (2, 1)$ thus seems to be the
johnston-linear-matrix-algebra-F01043	94	1.000	$3y = 3$	$3y = 3$	new equation $3y = 3$, which tells us that $y = 1$. Plugging $y = 1$ back into the
johnston-linear-matrix-algebra-F01044	94	1.000	$x + 2y = 4$	$x + 2y = 4$	original equation $x + 2y = 4$ then tells us that $x = 2$.
johnston-linear-matrix-algebra-F01045	94	1.000	$x + 2y = 4, -x + y = -1$	$x + 2y = 4, -x + y = -1$	Figure 2.2: The system of linear equations $x + 2y = 4, -x + y = -1$ is represented graph-
johnston-linear-matrix-algebra-F01046	94	0.943	$x + 2y$	$x + 2y$	$x + 2y$ cannot simultaneously equal 4 and 3. Geometrically, these two equations
johnston-linear-matrix-algebra-F01047	95	1.000	$A \mathbf{x} = \mathbf{b}$	$A\mathbf{x} = \mathbf{b}$	can be rewritten as the single matrix equation $A\mathbf{x} = \mathbf{b}$, where $A \in \mathcal{M}_{m,n}$ is the co-
johnston-linear-matrix-algebra-F01048	95	0.950	$a_{\{i, j\}}, \mathbf{b} = \left(b_{\{1\}}, b_{\{2\}}, \ldots, b_{\{m\}}\right) \in \mathbb{R}^m$	$a_{i,j}, \mathbf{b} = (b_1, b_2, \ldots, b_m) \in \mathbb{R}^m$	efficient matrix whose (i, j) -entry is $a_{i,j}$, $\mathbf{b} = (b_1, b_2, \ldots, b_m) \in \mathbb{R}^m$ is a vector
johnston-linear-matrix-algebra-F01049	95	0.997	$\mathbf{x} = \left(x_{\{1\}}, x_{\{2\}}, \ldots, x_{\{n\}}\right) \in \mathbb{R}^n$	$\mathbf{x} = (x_1, x_2, \ldots, x_n) \in \mathbb{R}^n$	containing the constants from the right-hand side, and $\mathbf{x} = (x_1, x_2, \ldots, x_n) \in \mathbb{R}^n$
johnston-linear-matrix-algebra-F01050	96	0.743	$\mathbf{x}_1$	$\mathbf{x}_1$	heads of $\mathbf{x}_1$ and $\mathbf{x}_2$
johnston-linear-matrix-algebra-F01051	96	0.743	$\mathbf{x}_2$	$\mathbf{x}_2$	heads of $\mathbf{x}_1$ and $\mathbf{x}_2$

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johnston-linear-matrix-algebra_F01052	96	1.000	$A \in \mathcal{M}_{m,n}, \mathbf{x} \in \mathbb{R}^n$	$A \in \mathcal{M}_{m,n}, \mathbf{x} \in \mathbb{R}^n$	If $A\mathbf{x} = \mathbf{b}$ is the matrix form of the linear system (where $A \in \mathcal{M}_{m,n}, \mathbf{x} \in \mathbb{R}^n$,
johnston-linear-matrix-algebra_F01053	96	1.000	$\mathbf{x}_1 \neq \mathbf{x}_2 \in \mathbb{R}^n$	$\mathbf{x}_1 \neq \mathbf{x}_2 \in \mathbb{R}^n$	means that there exist vectors $\mathbf{x}_1 \neq \mathbf{x}_2 \in \mathbb{R}^n$ such that $A\mathbf{x}_1 = \mathbf{b}$ and $A\mathbf{x}_2 = \mathbf{b}$.
johnston-linear-matrix-algebra_F01054	96	1.000	$A \mathbf{x}_1 = \mathbf{b}$	$A\mathbf{x}_1 = \mathbf{b}$	means that there exist vectors $\mathbf{x}_1 \neq \mathbf{x}_2 \in \mathbb{R}^n$ such that $A\mathbf{x}_1 = \mathbf{b}$ and $A\mathbf{x}_2 = \mathbf{b}$.
johnston-linear-matrix-algebra_F01055	96	1.000	$A \mathbf{x}_2 = \mathbf{b}$	$A\mathbf{x}_2 = \mathbf{b}$	means that there exist vectors $\mathbf{x}_1 \neq \mathbf{x}_2 \in \mathbb{R}^n$ such that $A\mathbf{x}_1 = \mathbf{b}$ and $A\mathbf{x}_2 = \mathbf{b}$.
johnston-linear-matrix-algebra_F01056	96	1.000	$(1-c) \mathbf{x}_1 + c \mathbf{x}_2$	$(1-c)\mathbf{x}_1 + c\mathbf{x}_2$	so every vector of the form $(1-c)\mathbf{x}_1 + c\mathbf{x}_2$ is a solution of the linear system.
johnston-linear-matrix-algebra_F01057	96	1.000	$3z=6$	$3z = 6$	The final equation in this system tells us that $3z = 6$, so $z = 2$. Plugging $z = 2$
johnston-linear-matrix-algebra_F01058	96	1.000	$z=2$	$z = 2$	The final equation in this system tells us that $3z = 6$, so $z = 2$. Plugging $z = 2$
johnston-linear-matrix-algebra_F01059	97	1.000	$2y=16$	$2y = 16$	equation: $2y = 16$, so $y = 8$. Finally, plugging $y = 8$ into the first equation then
johnston-linear-matrix-algebra_F01060	97	1.000	$y=8$	$y = 8$	equation: $2y = 16$, so $y = 8$. Finally, plugging $y = 8$ into the first equation then
johnston-linear-matrix-algebra_F01061	97	1.000	$x+24=9$	$x + 24 = 9$	tells us that $x + 24 = 9$, so $x = -15$. This system of linear equations thus has a
johnston-linear-matrix-algebra_F01062	97	1.000	$x=-15$	$x = -15$	tells us that $x + 24 = 9$, so $x = -15$. This system of linear equations thus has a
johnston-linear-matrix-algebra_F01063	97	1.000	$(x, y, z) = (-15, 8, 2)$	$(x, y, z) = (-15, 8, 2)$	unique solution, and it is $(x, y, z) = (-15, 8, 2)$.
johnston-linear-matrix-algebra_F01064	98	1.000	$y-3z=2$	$y - 3z = 2$	$y - 3z = 2$ if we
johnston-linear-matrix-algebra_F01065	98	1.000	$[A \mid \mathbf{b}]$	$[A \mid \mathbf{b}]$	$[A \mid \mathbf{b}]$, so that we only have to write down the coefficients in the linear system
johnston-linear-matrix-algebra_F01066	99	1.000	cR_j	cR_j	Multiplication. Multiplying row j by a non-zero scalar $c \in \mathbb{R}$ is denoted by cR_j .
johnston-linear-matrix-algebra_F01067	99	1.000	$R_i \leftrightarrow R_j$	$R_i \leftrightarrow R_j$	Swap. Swapping rows i and j is denoted by $R_i \leftrightarrow R_j$.
johnston-linear-matrix-algebra_F01068	99	0.974	$) + c($	$) + c($	Addition. For any scalar $c \in \mathbb{R}$, replacing row i by (row i) $+ c$ (row j) is denoted by
johnston-linear-matrix-algebra_F01069	99	0.974	$)$	$)$	Addition. For any scalar $c \in \mathbb{R}$, replacing row i by (row i) $+ c$ (row j) is denoted by
johnston-linear-matrix-algebra_F01070	99	0.996	$R_i + cR_j$	$R_i + cR_j$	$R_i + cR_j$.
johnston-linear-matrix-algebra_F01071	100	0.997	$(x, y, z) =$	$(x, y, z) =$	off from the right-hand side of the bottom augmented matrix: $(x, y, z) =$
johnston-linear-matrix-algebra_F01072	100	1.000	$\frac{1}{c}R_j$	$\frac{1}{c}R_j$	particular, the multiplication operation cR_j is reversed by $\frac{1}{c}R_j$, the swap
johnston-linear-matrix-algebra_F01073	100	1.000	$R_i - cR_j$	$R_i - cR_j$	is undone by $R_i - cR_j$.
johnston-linear-matrix-algebra_F01074	100	1.000	$c \neq 0$	$c \neq 0$	undoing the row operations). In fact, this is exactly why we require $c \neq 0$
johnston-linear-matrix-algebra_F01075	100	1.000	$\mathbf{x} = (1, 1)$	$\mathbf{x} = (1, 1)$	the system on the left has a unique solution $\mathbf{x} = (1, 1)$, but the system on
johnston-linear-matrix-algebra_F01076	100	1.000	$\mathbf{x} = (3, 0)$	$\mathbf{x} = (3, 0)$	the right has infinitely many others, such as $\mathbf{x} = (3, 0)$.
johnston-linear-matrix-algebra_F01077	101	1.000	$\left[\begin{array}{lll} 1 & 2 & 3 \\ 0 & 1 & 0 \end{array} \right]$	$\left[\begin{array}{ccc} 1 & 2 & 3 \\ 0 & 1 & 0 \end{array} \right]$	a) $\left[\begin{array}{ccc} 1 & 2 & 3 \\ 0 & 1 & 0 \end{array} \right]$

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johnston-linear-matrix-algebra F01078	101	1.000	$\left[\begin{array}{lll}1 & 0 & 2 \\ 0 & 1 & 4\end{array}\right]$	$\begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 4 \end{bmatrix}$	b) $\begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 4 \end{bmatrix}$
johnston-linear-matrix-algebra F01079	101	1.000	$\left[\begin{array}{cccc}2 & 0 & -1 & 5 \\ 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 0\end{array}\right]$	$\begin{bmatrix} 2 & 0 & -1 & 5 \\ 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$	c) $\begin{bmatrix} 2 & 0 & -1 & 5 \\ 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$
johnston-linear-matrix-algebra F01080	101	1.000	$\left[\begin{array}{cccc}1 & 2 & 3 & 4 \\ 0 & 0 & 2 & -1 \\ 0 & 0 & 1 & 0\end{array}\right]$	$\begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & 0 & 2 & -1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$	d) $\begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & 0 & 2 & -1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$
johnston-linear-matrix-algebra F01081	104	1.000	$0=1$	$0 = 1$	Of particular note is the third equation, which says $0 = 1$. This tells us that the
johnston-linear-matrix-algebra F01082	104	0.513	$\left[\begin{array}{llll}0 & 0 & \cdots & 0\end{array}\right] \left[\begin{array}{c} b \end{array}\right]$	$\left[\begin{array}{cccc c} 0 & 0 & \cdots & 0 & b \end{array} \right]$	forms have a row that looks like $\left[\begin{array}{cccc c} 0 & 0 & \cdots & 0 & b \end{array} \right]$ for some
johnston-linear-matrix-algebra F01083	104	1.000	$b \neq 0$	$b \neq 0$	scalar $b \neq 0$.
johnston-linear-matrix-algebra F01084	105	1.000	$x=3-4y$	$x = 3 - 4y$	that $x = 3 - 4y$ and
johnston-linear-matrix-algebra F01085	105	1.000	$y=y$	$y = y$	$y = y$.
johnston-linear-matrix-algebra F01086	106	0.996	$x+4y=3$	$x + 4y = 3$	Figure 2.5: Geometrically, the solution set to the system of linear equations $x + 4y = 3$,
johnston-linear-matrix-algebra F01087	107	1.000	$\left[\begin{array}{cccc c}1 & -1 & 0 & 1 & 2 \\ 0 & 0 & 1 & -1 & 1 \\ 0 & 0 & 0 & 0 & 0\end{array}\right]$	$\left[\begin{array}{cccc c} 1 & -1 & 0 & 1 & 2 \\ 0 & 0 & 1 & -1 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right]$	a) $\left[\begin{array}{cccc c} 1 & -1 & 0 & 1 & 2 \\ 0 & 0 & 1 & -1 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right]$
johnston-linear-matrix-algebra F01088	107	0.997	$\left[\begin{array}{ccc c}1 & 2 & -4 & -4 \\ 0 & 3 & -1 & 2 \\ 0 & 0 & 8 & 8\end{array}\right]$	$\left[\begin{array}{ccc c} 1 & 2 & -4 & -4 \\ 0 & 3 & -1 & 2 \\ 0 & 0 & 8 & 8 \end{array} \right]$	b) $\left[\begin{array}{ccc c} 1 & 2 & -4 & -4 \\ 0 & 3 & -1 & 2 \\ 0 & 0 & 8 & 8 \end{array} \right]$
johnston-linear-matrix-algebra F01089	107	1.000	w, x, y, z	w, x, y, z	corresponding to the columns from left-to-right as w, x, y, z , then w
johnston-linear-matrix-algebra F01090	107	1.000	w	w	corresponding to the columns from left-to-right as w, x, y, z , then w
johnston-linear-matrix-algebra F01091	107	1.000	$w=2+x-z$	$w = 2 + x - z$	hand-side) $w = 2 + x - z$, and the second equation says $y = 1 + z$
johnston-linear-matrix-algebra F01092	107	1.000	$y=1+z$	$y = 1 + z$	hand-side) $w = 2 + x - z$, and the second equation says $y = 1 + z$
johnston-linear-matrix-algebra F01093	107	0.818	$\left[\begin{array}{lll}0 & 0 & 0\end{array}\right] \left[\begin{array}{c} b \end{array}\right]$	$\left[\begin{array}{ccc c} 0 & 0 & 0 & b \end{array} \right]$	the form $\left[\begin{array}{ccc c} 0 & 0 & 0 & b \end{array} \right]$, we know the system has a unique solution. To
johnston-linear-matrix-algebra F01094	107	1.000	x, y, z	x, y, z	as x, y, z , then the third equation says $8z = 8$, so $z = 1$. Plugging that
johnston-linear-matrix-algebra F01095	107	1.000	$8z=8$	$8z = 8$	as x, y, z , then the third equation says $8z = 8$, so $z = 1$. Plugging that
johnston-linear-matrix-algebra F01096	107	1.000	$z=1$	$z = 1$	as x, y, z , then the third equation says $8z = 8$, so $z = 1$. Plugging that
johnston-linear-matrix-algebra F01097	107	1.000	$3y-1=2$	$3y - 1 = 2$	into the second equation tells us $3y - 1 = 2$, so $y = 1$. Plugging those
johnston-linear-matrix-algebra F01098	107	1.000	$x+2-4=-4$	$x + 2 - 4 = -4$	values into the first equation tells us $x + 2 - 4 = -4$, so $x = -2$. The
johnston-linear-matrix-algebra F01099	107	1.000	$x=-2$	$x = -2$	values into the first equation tells us $x + 2 - 4 = -4$, so $x = -2$. The

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johnston-linear-matrix-algebra F01100	107	1.000	$(x, y, z) = (-2, 1, 1)$	$(x, y, z) = (-2, 1, 1)$	unique solution is thus $(x, y, z) = (-2, 1, 1)$.
johnston-linear-matrix-algebra F01101	107	1.000	$0x + 0y + 0z = -42/5$	$0x + 0y + 0z = -42/5$	equation $0x + 0y + 0z = -42/5$, we see that this system of equations has
johnston-linear-matrix-algebra F01102	107	1.000	$A \mathbf{x} = \mathbf{0}$	$A\mathbf{x} = \mathbf{0}$	if the linear system has a zero right-hand side (i.e., it is of the form $A\mathbf{x} = \mathbf{0}$)
johnston-linear-matrix-algebra F01103	108	1.000	$m < n$	$m < n$	Suppose $A \in \mathcal{M}_{m,n}$ and $m < n$. Then the linear system $A\mathbf{x} = \mathbf{b}$ has either
johnston-linear-matrix-algebra F01104	108	1.000	$v_1 = -2v_2 - 3v_3$	$v_1 = -2v_2 - 3v_3$	form $v_1 = -2v_2 - 3v_3$,
johnston-linear-matrix-algebra F01105	108	1.000	$v_2 = v_3$	$v_2 = v_3$	$v_2 = v_3$ to replace
johnston-linear-matrix-algebra F01106	108	1.000	$\mathbf{v} = (v_1, v_2, v_3)$	$\mathbf{v} = (v_1, v_2, v_3)$	vector $\mathbf{v} = (v_1, v_2, v_3)$ such that
johnston-linear-matrix-algebra F01107	109	1.000	v_4	v_4	can choose v_4 to
johnston-linear-matrix-algebra F01108	109	1.000	(v_1, v_2, v_3)	(v_1, v_2, v_3)	the vectors (v_1, v_2, v_3) with $v_2 = v_3$ and $v_1 = -2v_2 - 3v_3 = -5v_3$,
johnston-linear-matrix-algebra F01109	109	1.000	$v_1 = -2v_2 - 3v_3 = -5v_3$	$v_1 = -2v_2 - 3v_3 = -5v_3$	the vectors (v_1, v_2, v_3) with $v_2 = v_3$ and $v_1 = -2v_2 - 3v_3 = -5v_3$,
johnston-linear-matrix-algebra F01110	109	1.000	$v_3 = 1$	$v_3 = 1$	choose $v_3 = 1$ to get $\mathbf{v} = (-5, 1, 1)$.
johnston-linear-matrix-algebra F01111	109	1.000	$\mathbf{v} = (-5, 1, 1)$	$\mathbf{v} = (-5, 1, 1)$	choose $v_3 = 1$ to get $\mathbf{v} = (-5, 1, 1)$.
johnston-linear-matrix-algebra F01112	109	0.999	v_1, v_2, v_3	v_1, v_2, v_3	We thus see that v_4 is a free variable, and v_1, v_2, v_3 are all leading
johnston-linear-matrix-algebra F01113	109	1.000	(v_1, v_2, v_3, v_4)	(v_1, v_2, v_3, v_4)	variables. Solutions (v_1, v_2, v_3, v_4) to this system have the form $v_1 =$
johnston-linear-matrix-algebra F01114	109	1.000	$v_1 =$	$v_1 =$	variables. Solutions (v_1, v_2, v_3, v_4) to this system have the form $v_1 =$
johnston-linear-matrix-algebra F01115	109	0.991	$0, v_2 = -v_4/2$	$0, v_2 = -v_4/2$	$0, v_2 = -v_4/2$, and $v_3 = -v_4/2$, with v_4 arbitrary. If we choose
johnston-linear-matrix-algebra F01116	109	0.991	$v_3 = -v_4/2$	$v_3 = -v_4/2$	$0, v_2 = -v_4/2$, and $v_3 = -v_4/2$, with v_4 arbitrary. If we choose
johnston-linear-matrix-algebra F01117	109	1.000	$v_4 = 2$	$v_4 = 2$	$v_4 = 2$ then we find the specific vector $\mathbf{v} = (0, -1, -1, 2)$, which is
johnston-linear-matrix-algebra F01118	109	1.000	$\mathbf{v} = (0, -1, -1, 2)$	$\mathbf{v} = (0, -1, -1, 2)$	$v_4 = 2$ then we find the specific vector $\mathbf{v} = (0, -1, -1, 2)$, which is
johnston-linear-matrix-algebra F01119	109	1.000	$c_1, c_2 \in \mathbb{R}$	$c_1, c_2 \in \mathbb{R}$	$c_1, c_2 \in \mathbb{R}$ such that
johnston-linear-matrix-algebra F01120	110	1.000	c_1, c_2	c_1, c_2	coefficients c_1, c_2
johnston-linear-matrix-algebra F01121	110	1.000	$(1, 3, 1) = \frac{4}{5}(-1, 3, 2) +$	$(1, 3, 1) = \frac{4}{5}(-1, 3, 2) +$	$(1, 3, 1) = \frac{4}{5}(-1, 3, 2) +$
johnston-linear-matrix-algebra F01122	110	1.000	$\frac{3}{5}(3, 1, -1)$	$\frac{3}{5}(3, 1, -1)$	$\frac{3}{5}(3, 1, -1)$. The hard
johnston-linear-matrix-algebra F01123	110	1.000	c_1	c_1	part is finding c_1
johnston-linear-matrix-algebra F01124	110	1.000	c_2	c_2	and c_2 in the first
johnston-linear-matrix-algebra F01125	110	1.000	$0 = 3$	$0 = 3$	the final row in this row echelon form says that $0 = 3$. It follows
johnston-linear-matrix-algebra F01126	110	1.000	$c_2 = 3/5$	$c_2 = 3/5$	This linear system can now be solved via back substitution: $c_2 = 3/5$,
johnston-linear-matrix-algebra F01127	110	1.000	$-c_1 + 3c_2 = 1$	$-c_1 + 3c_2 = 1$	so $-c_1 + 3c_2 = 1$ implies $c_1 = 4/5$. We thus conclude that $(1, 3, 1)$

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johnston-linear-matrix-augmented-F01128	110	1.000	$c_1=4 / 5$	$c_1 = 4/5$	so $-c_1 + 3c_2 = 1$ implies $c_1 = 4/5$. We thus conclude that $(1, 3, 1)$
johnston-linear-matrix-augmented-F01129	110	1.000	$0.02 \times 500=10$	$0.02 \times 500 = 10$	We similarly want there to be a total of $0.02 \times 500 = 10$ liters of butterfat
johnston-linear-matrix-augmented-F01130	111	1.000	$x=200$	$x = 200$	We thus see that the farmer should mix together $x = 200$ liters of the
johnston-linear-matrix-augmented-F01131	111	0.996	3.5%	3.5%	3.5% butterfat milk and $y = 300$ liters of the 1% butterfat milk.
johnston-linear-matrix-augmented-F01132	111	0.996	$y=300$	$y = 300$	3.5% butterfat milk and $y = 300$ liters of the 1% butterfat milk.
johnston-linear-matrix-augmented-F01133	111	0.996	1%	1%	3.5% butterfat milk and $y = 300$ liters of the 1% butterfat milk.
johnston-linear-matrix-augmented-F01134	111	1.000	$(x, y)=(200,300)$	$(x, y) = (200, 300)$	changes considerably, from $(x, y) = (200, 300)$ to $(x, y) = (125, 375)$. For
johnston-linear-matrix-augmented-F01135	111	1.000	$(x, y)=(125,375)$	$(x, y) = (125, 375)$	changes considerably, from $(x, y) = (200, 300)$ to $(x, y) = (125, 375)$. For
johnston-linear-matrix-augmented-F01136	112	1.000	$\mathrm{H}_2 \mathrm{O}$	H ₂ O	formula H ₂ O for
johnston-linear-matrix-augmented-F01137	112	0.996	$\left(\mathrm{C}_4 \mathrm{H}_{10}\right)$	(C ₄ H ₁₀)	When butane (C ₄ H ₁₀) burns in the presence of oxygen gas (O ₂), it pro-
johnston-linear-matrix-augmented-F01138	112	0.996	$\left(\mathrm{O}_2\right)$	(O ₂)	When butane (C ₄ H ₁₀) burns in the presence of oxygen gas (O ₂), it pro-
johnston-linear-matrix-augmented-F01139	112	0.890	$\left(\mathrm{CO}_2\right)$	(CO ₂)	duces carbon dioxide (CO ₂) and water vapor (H ₂ O). We thus say that the
johnston-linear-matrix-augmented-F01140	112	0.890	$\left(\mathrm{H}_2 \mathrm{O}\right)$	(H ₂ O)	duces carbon dioxide (CO ₂) and water vapor (H ₂ O). We thus say that the
johnston-linear-matrix-augmented-F01141	112	1.000	w, x, y	w, x, y	Find the corresponding “balanced” equation. That is, find integers w, x, y ,
johnston-linear-matrix-augmented-F01142	112	1.000	$2 x$	$2x$	On the left-hand side of this chemical equation, we have $2x$ atoms of
johnston-linear-matrix-augmented-F01143	112	1.000	$2 y+z$	$2y + z$	oxygen (O) and on the right-hand side we have $2y + z$ atoms of oxygen, so
johnston-linear-matrix-augmented-F01144	112	1.000	$z=10$	$z = 10$	In particular, if we choose $z = 10$ then we get $w = 2, x = 13$, and $y = 8$,
johnston-linear-matrix-augmented-F01145	112	1.000	$w=2, x=13$	$w = 2, x = 13$	In particular, if we choose $z = 10$ then we get $w = 2, x = 13$, and $y = 8$,
johnston-linear-matrix-augmented-F01146	113	1.000	$\left[\begin{array}{ccc} 4 & 0 & 8 \\ 4 & 1 & 11 \\ 1 & 2 & 8 \\ -1 & 0 & -2 \end{array}\right]$	$\left[\begin{array}{ccc} 4 & 0 & 8 \\ 4 & 1 & 11 \\ 1 & 2 & 8 \\ -1 & 0 & -2 \end{array}\right]$	$\left[\begin{array}{ccc} 4 & 0 & 8 \\ 4 & 1 & 11 \\ 1 & 2 & 8 \\ -1 & 0 & -2 \end{array}\right]$
johnston-linear-matrix-augmented-F01147	113	1.000	$4 x-\sin (1) y+2 z=\sqrt[3]{5}$	$4x - \sin(1)y + 2z = \sqrt[3]{5}$	*(a) The equation $4x - \sin(1)y + 2z = \sqrt[3]{5}$ is linear.
johnston-linear-matrix-augmented-F01148	113	1.000	$x y+4 z=4$	$xy + 4z = 4$	(b) The equation $xy + 4z = 4$ is linear.
johnston-linear-matrix-augmented-F01149	114	1.000	$t \in \mathbb{R}$	$t \in \mathbb{R}$	*2.1.8 Let $t \in \mathbb{R}$ and consider the linear system described
johnston-linear-matrix-augmented-F01150	114	1.000	t	t	For which values of t does this system have (i) no solutions,
johnston-linear-matrix-augmented-F01151	114	1.000	$h, k \in \mathbb{R}$	$h, k \in \mathbb{R}$	2.1.9 Let $h, k \in \mathbb{R}$ and consider the following system of
johnston-linear-matrix-augmented-F01152	114	1.000	h	h	For which values of h and k does this system have (i) no
johnston-linear-matrix-augmented-F01153	114	1.000	$h=0.95, h=1.00$	$h = 0.95, h = 1.00$	$h = 0.95, h = 1.00$, and $h = 1.05$. Do you notice anything

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johnston-linear-matrix-augmented-F01154	114	1.000	$h=1.05$	$h = 1.05$	$h = 0.95$, $h = 1.00$, and $h = 1.05$. Do you notice anything
johnston-linear-matrix-augmented-F01155	114	1.000	$a, b, c, d \in \mathbb{R}$	$a, b, c, d \in \mathbb{R}$	*2.1.11 Let $a, b, c, d \in \mathbb{R}$ be scalars and consider the system
johnston-linear-matrix-augmented-F01156	114	1.000	$a \neq b \neq c \neq d$	$ad - bc \neq 0$	tion whenever $ad - bc \neq 0$.
johnston-linear-matrix-augmented-F01157	114	1.000	$A, B, R \in \mathcal{M}_{m,n}$	$A, B, R \in \mathcal{M}_{m,n}$	**2.1.13 Suppose $A, B, R \in \mathcal{M}_{m,n}$.
johnston-linear-matrix-augmented-F01158	114	1.000	$R_2 - R_1$	$R_2 - R_1$	(a) Perform the row operation $R_2 - R_1$, followed by the
johnston-linear-matrix-augmented-F01159	114	1.000	$R_1 - R_2$	$R_1 - R_2$	row operation $R_1 - R_2$, on this augmented matrix.
johnston-linear-matrix-augmented-F01160	114	0.998	$\mathbf{b}$	$\mathbf{b}$	2.1.17 Determine whether or not $\mathbf{b}$ is a linear combination
johnston-linear-matrix-augmented-F01161	114	0.999	$\mathbf{b} = (3, 4)$, $\mathbf{v}_1 = (6, 8)$	$\mathbf{b} = (3, 4), \mathbf{v}_1 = (6, 8)$	*(a) $\mathbf{b} = (3, 4), \mathbf{v}_1 = (6, 8)$
johnston-linear-matrix-augmented-F01162	114	1.000	$\mathbf{b} = (2, -3)$, $\mathbf{v}_1 = (1, 2)$, $\mathbf{v}_2 = (4, -1)$	$\mathbf{b} = (2, -3), \mathbf{v}_1 = (1, 2), \mathbf{v}_2 = (4, -1)$	(b) $\mathbf{b} = (2, -3), \mathbf{v}_1 = (1, 2), \mathbf{v}_2 = (4, -1)$
johnston-linear-matrix-augmented-F01163	114	1.000	$\mathbf{b} = (1, 5, -6)$, $\mathbf{v}_1 = (1, 0, 0)$, $\mathbf{v}_2 = (0, 1, 0)$, $\mathbf{v}_3 = (0, 1, 0)$	$\mathbf{b} = (1, 5, -6), \mathbf{v}_1 = (1, 0, 0), \mathbf{v}_2 = (0, 1, 0), \mathbf{v}_3 = (0, 1, 0)$	*(c) $\mathbf{b} = (1, 5, -6), \mathbf{v}_1 = (1, 0, 0), \mathbf{v}_2 = (0, 1, 0), \mathbf{v}_3 = (0, 1, 0)$
johnston-linear-matrix-augmented-F01164	114	1.000	$\mathbf{b} = (2, 1, 2)$, $\mathbf{v}_1 = (-2, 2, 1)$, $\mathbf{v}_2 = (1, 2, 3)$	$\mathbf{b} = (2, 1, 2), \mathbf{v}_1 = (-2, 2, 1), \mathbf{v}_2 = (1, 2, 3)$	(d) $\mathbf{b} = (2, 1, 2), \mathbf{v}_1 = (-2, 2, 1), \mathbf{v}_2 = (1, 2, 3)$
johnston-linear-matrix-augmented-F01165	114	1.000	$\mathbf{b} = (2, 1, 2)$, $\mathbf{v}_1 = (-4, 4, -1)$, $\mathbf{v}_2 = (2, -1, 1)$	$\mathbf{b} = (2, 1, 2), \mathbf{v}_1 = (-4, 4, -1), \mathbf{v}_2 = (2, -1, 1)$	*(e) $\mathbf{b} = (2, 1, 2), \mathbf{v}_1 = (-4, 4, -1), \mathbf{v}_2 = (2, -1, 1)$
johnston-linear-matrix-augmented-F01166	114	1.000	$\mathbf{b} = (2, 1, 2, 3)$, $\mathbf{v}_1 = (1, 2, 3, 4)$, $\mathbf{v}_2 = (4, 3, 2, 1)$	$\mathbf{b} = (2, 1, 2, 3), \mathbf{v}_1 = (1, 2, 3, 4), \mathbf{v}_2 = (4, 3, 2, 1)$	(f) $\mathbf{b} = (2, 1, 2, 3), \mathbf{v}_1 = (1, 2, 3, 4), \mathbf{v}_2 = (4, 3, 2, 1)$
johnston-linear-matrix-augmented-F01167	114	1.000	$\mathbf{v}_3 = (1, -1, 1, -1)$	$\mathbf{v}_3 = (1, -1, 1, -1)$	$\mathbf{v}_3 = (1, -1, 1, -1)$
johnston-linear-matrix-augmented-F01168	114	0.831	$\mathbf{b} = (1, 3, -3, -1)$, $\mathbf{v}_1 = (1, 2, 3, 4)$, $\mathbf{v}_2 = (4, 3, 2, 1)$	$\mathbf{b} = (1, 3, -3, -1), \mathbf{v}_1 = (1, 2, 3, 4), \mathbf{v}_2 = (4, 3, 2, 1)$	*(g) $\mathbf{b} = (1, 3, -3, -1), \mathbf{v}_1 = (1, 2, 3, 4), \mathbf{v}_2 = (4, 3, 2, 1)$
johnston-linear-matrix-augmented-F01169	114	1.000	$\mathbf{w} = (w_1, w_2, w_3)$	$\mathbf{w} = (w_1, w_2, w_3)$	*2.1.18 Let $\mathbf{v} = (v_1, v_2, v_3)$ and $\mathbf{w} = (w_1, w_2, w_3)$ be non-
johnston-linear-matrix-augmented-F01170	115	1.000	$\left(\mathrm{SO}_2 \right)$	(SO_2)	fur dioxide (SO_2). Determine how many of each of these
johnston-linear-matrix-augmented-F01171	115	0.996	$\left(\mathrm{C}_2\mathrm{H}_6 \right)$	$(\mathrm{C}_2\mathrm{H}_6)$	2.1.22 When ethane ($\mathrm{C}_2\mathrm{H}_6$) burns in the presence of oxy-
johnston-linear-matrix-augmented-F01172	115	1.000	$P_{\mathbf{u}}(\mathbf{v}) = (2, 4, 4)$	$P_{\mathbf{u}}(\mathbf{v}) = (2, 4, 4)$	property that $P_{\mathbf{u}}(\mathbf{v}) = (2, 4, 4)$.
johnston-linear-matrix-augmented-F01173	115	1.000	$(3, -2, 1) = c_1(1, 4, 2) +$	$(3, -2, 1) = c_1(1, 4, 2) +$	[Hint: Rewrite the equation $(3, -2, 1) = c_1(1, 4, 2) +$
johnston-linear-matrix-augmented-F01174	115	1.000	$c_2(2, -1, 1)$	$c_2(2, -1, 1)$	$c_2(2, -1, 1)$ as a system of linear equations.]
johnston-linear-matrix-augmented-F01175	115	0.566	$T(1, 1) = (3, 7)$, $T(1, -1) = (-1, -1)$	$T(1, 1) = (3, 7), T(1, -1) = (-1, -1)$	*(a) $T(1, 1) = (3, 7), T(1, -1) = (-1, -1)$
johnston-linear-matrix-augmented-F01176	115	1.000	$T(1, 2) = (5, 3, 4)$, $T(2, -1) = (0, 1, 3)$	$T(1, 2) = (5, 3, 4), T(2, -1) = (0, 1, 3)$	(b) $T(1, 2) = (5, 3, 4), T(2, -1) = (0, 1, 3)$
johnston-linear-matrix-augmented-F01177	115	0.992	$T(1, 1, 1) = (4, 6, 1)$, $T(2, -1, 1) = (1, 1, -4)$	$T(1, 1, 1) = (4, 6, 1), T(2, -1, 1) = (1, 1, -4)$	*(c) $T(1, 1, 1) = (4, 6, 1), T(2, -1, 1) = (1, 1, -4)$
johnston-linear-matrix-augmented-F01178	115	1.000	$T(0, 0, 1) = (1, 2, 0)$	$T(0, 0, 1) = (1, 2, 0)$	$T(0, 0, 1) = (1, 2, 0)$
johnston-linear-matrix-augmented-F01179	115	1.000	$AB = BA$	$AB = BA$	$AB = BA$, and we can find matrices with this property by
johnston-linear-matrix-augmented-F01180	115	1.000	$B \in \mathcal{M}_2$	$B \in \mathcal{M}_2$	Find all $B \in \mathcal{M}_2$ with the property that $AB = BA$.
johnston-linear-matrix-augmented-F01181	115	1.000	$D \in \mathcal{M}_2$	$D \in \mathcal{M}_2$	Find all $D \in \mathcal{M}_2$ with the property that $CD = DC$.
johnston-linear-matrix-augmented-F01182	115	1.000	$CD = DC$	$CD = DC$	Find all $D \in \mathcal{M}_2$ with the property that $CD = DC$.

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johnston-linear-matrix-algebra F01183	115	1.000	$c \ I$	cI	ces of the form cI , where $c \in \mathbb{R}$. [Side note: These
johnston-linear-matrix-algebra F01184	115	1.000	$2=m+b$	$2 = m + b$	us that $2 = m + b$ and the fact that the line goes
johnston-linear-matrix-algebra F01185	115	1.000	$(3,8)$	$(3,8)$	through $(3,8)$ tells us that $8 = 3m + b$.]
johnston-linear-matrix-algebra F01186	115	1.000	$8=3 \ m+b$	$8 = 3m + b$	through $(3,8)$ tells us that $8 = 3m + b$.]
johnston-linear-matrix-algebra F01187	115	0.977	$y=a \ x^{\{2\}}+b \ x+c$	$y = ax^2 + bx + c$	$y = ax^2 + bx + c$ goes through the points $(1,3)$, $(2,6)$,
johnston-linear-matrix-algebra F01188	115	1.000	$(2,6)$	$(2,6)$	[Hint: The fact that the parabola goes through $(2,6)$
johnston-linear-matrix-algebra F01189	115	1.000	$6=4 \ a+2 \ b+c$	$6 = 4a + 2b + c$	tells us that $6 = 4a + 2b + c$, and so on.]
johnston-linear-matrix-algebra F01190	115	1.000	$y=a \ x^{\{3\}}+b \ x^{\{2\}}+c \ x+d$	$y = ax^3 + bx^2 + cx + d$	$y = ax^3 + bx^2 + cx + d$ that goes through the 3 points?
johnston-linear-matrix-algebra F01191	116	1.000	$a, \ b, \ c, \ d$	a, b, c, d	nomial for this data. That is, find scalars a, b, c, d ,
johnston-linear-matrix-algebra F01192	116	1.000	e	e	e , and f such that if x is the number of decades that
johnston-linear-matrix-algebra F01193	116	1.000	$p(0)=3.70, \ p(1)=4.46$	$p(0) = 3.70, p(1) = 4.46$	the decade (i.e., $p(0) = 3.70$, $p(1) = 4.46$, and so
johnston-linear-matrix-algebra F01194	117	1.000	$R_{\{1\}} \ \leftrightsquigarrow R_{\{2\}}$	$R_1 \leftrightarrow R_2$	For example, we carried out the “swap” row operation $R_1 \leftrightarrow R_2$ in Equa-
johnston-linear-matrix-algebra F01195	117	1.000	$R_{\{3\}}-3 \ R_{\{1\}}$	$R_3 - 3R_1$	row operations $R_3 - 3R_1$ and $\frac{1}{2}R_2$ to the identity matrix. We will make extensive
johnston-linear-matrix-algebra F01196	117	1.000	$\frac{1}{2} \ R_{\{2\}}$	$\frac{1}{2}R_2$	row operations $R_3 - 3R_1$ and $\frac{1}{2}R_2$ to the identity matrix. We will make extensive
johnston-linear-matrix-algebra F01197	117	1.000	$R_{\{i\}} \ \leftrightsquigarrow$	$R_i \leftrightarrow$	The elementary matrices corresponding to the “swap” row operation $R_i \leftrightarrow$
johnston-linear-matrix-algebra F01198	117	1.000	$R_{\{j\}}$	R_j	R_j look like
johnston-linear-matrix-algebra F01199	118	0.844	$A=\left[\begin{array}{ll}1 & c \\ 0 & 1\end{array}\right]$	$A = \begin{bmatrix} 1 & c \\ 0 & 1 \end{bmatrix}$	Figure 2.10: The “addition” matrix $A = \begin{bmatrix} 1 & c \\ 0 & 1 \end{bmatrix}$ shears space in the x -direction. A
johnston-linear-matrix-algebra F01200	119	0.512	$E \ A=R$	$EA = R$	Find a 3×3 matrix E such that $EA = R$, where R is the reduced row
johnston-linear-matrix-algebra F01201	119	1.000	$R_{\{3\}}-R_{\{2\}}$	$R_3 - R_2$	elementary row operations we perform are $R_1 - R_2$ and $R_3 - R_2$, which
johnston-linear-matrix-algebra F01202	119	1.000	$\frac{1}{4} \ R_{\{3\}}$	$\frac{1}{4}R_3$	Finally, we apply the elementary row operations $\frac{1}{4}R_3$ and then $R_1 + R_3$,
johnston-linear-matrix-algebra F01203	119	1.000	$R_{\{1\}}+R_{\{3\}}$	$R_1 + R_3$	Finally, we apply the elementary row operations $\frac{1}{4}R_3$ and then $R_1 + R_3$,
johnston-linear-matrix-algebra F01204	119	1.000	$E=E_{\{7\}} \ E_{\{6\}} \ E_{\{5\}} \ E_{\{4\}} \ E_{\{3\}} \ E_{\{2\}} \ E_{\{1\}}$	$E = E_7E_6E_5E_4E_3E_2E_1$	$E = E_7E_6E_5E_4E_3E_2E_1$, which can be computed by straightforward (albeit
johnston-linear-matrix-algebra F01205	120	0.999	$E_{\{1\}}, \ E_{\{2\}}, \ \ldots, \ E_{\{7\}}$	$E_1, E_2, \dots, E_7$	The matrices $E_1, E_2, \dots, E_7$ in the previous example can be thought of like
johnston-linear-matrix-algebra F01206	120	1.000	$E_{\{1\}}$	E_1	its reduced row echelon form R —first the row operation corresponding to E_1
johnston-linear-matrix-algebra F01207	120	1.000	$E_{\{2\}}$	E_2	was performed, then the one corresponding to E_2 , and so on. Similarly, the
johnston-linear-matrix-algebra F01208	120	0.516	$-5 \ / \ 8$	$-5/8$	$-5/8$ of A ’s first row, $2/8 = 1/4$ of its second row, and $1/4$ of its third row.
johnston-linear-matrix-algebra F01209	120	0.516	$2 \ / \ 8=1 \ / \ 4$	$2/8 = 1/4$	$-5/8$ of A ’s first row, $2/8 = 1/4$ of its second row, and $1/4$ of its third row.

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johnston-linear-matrix-algebra-F01210	120	0.516	$1 / 4$	$1/4$	$-5/8$ of A 's first row, $2/8 = 1/4$ of its second row, and $1/4$ of its third row.
johnston-linear-matrix-algebra-F01211	120	1.000	$E =$	$E =$	To actually compute the product of all of those elementary matrices $E =$
johnston-linear-matrix-algebra-F01212	120	0.984	$E_{\{7\}} E_{\{6\}} E_{\{5\}} E_{\{4\}} E_{\{3\}} E_{\{2\}} E_{\{1\}}$	$E_7 E_6 E_5 E_4 E_3 E_2 E_1$	$E_7 E_6 E_5 E_4 E_3 E_2 E_1$, we could perform six matrix multiplications (ugh!), but a
johnston-linear-matrix-algebra-F01213	120	0.913	$[A \mid I]$	$[A \mid I]$	Compute the reduced row echelon form of the 1×2 block matrix $[A \mid I]$.
johnston-linear-matrix-algebra-F01214	121	1.000	$[R \mid E]$	$[R \mid E]$	to some other form $[R \mid E]$ then these blocks must satisfy the property $EA = R$.
johnston-linear-matrix-algebra-F01215	121	1.000	$A, R \in \mathcal{M}_{m,n}$	$A, R \in \mathcal{M}_{m,n}$	If $A, R \in \mathcal{M}_{m,n}$ and $E \in \mathcal{M}_m$ are matrices such that the block matrix $[A \mid I]$
johnston-linear-matrix-algebra-F01216	121	1.000	$E \in \mathcal{M}_m$	$E \in \mathcal{M}_m$	If $A, R \in \mathcal{M}_{m,n}$ and $E \in \mathcal{M}_m$ are matrices such that the block matrix $[A \mid I]$
johnston-linear-matrix-algebra-F01217	121	1.000	$R = E A$	$R = EA$	can be row-reduced to $[R \mid E]$, then $R = EA$.
johnston-linear-matrix-algebra-F01218	121	0.706	$E_{\{1\}}, E_{\{2\}}, \ldots, E_{\{k\}}$	$E_1, E_2, \dots, E_k$	corresponding to elementary matrices $E_1, E_2, \dots, E_k$, in that order, then
johnston-linear-matrix-algebra-F01219	121	1.000	$E = E_{\{k\}} \cdots E_{\{2\}} E_{\{1\}}$	$E = E_k \cdots E_2 E_1$	$E = E_k \cdots E_2 E_1$, which then implies (by looking at the left half of the block
johnston-linear-matrix-algebra-F01220	121	1.000	$E_{\{1\}} E_{\{2\}} = I$	$E_1 E_2 = I$	then it is straightforward to check that $E_1 E_2 = I$ and $E_2 E_1 = I$. We thus say that
johnston-linear-matrix-algebra-F01221	121	1.000	$E_{\{2\}} E_{\{1\}} = I$	$E_2 E_1 = I$	then it is straightforward to check that $E_1 E_2 = I$ and $E_2 E_1 = I$. We thus say that
johnston-linear-matrix-algebra-F01222	122	1.000	$R_{\{3\}} + 3 R_{\{1\}}$	$R_3 + 3R_1$	operation $R_3 + 3R_1$, and performing these row operations one after another has
johnston-linear-matrix-algebra-F01223	122	1.000	A^{-1}	A^{-1}	we denote by A^{-1} and call the inverse of A , such that
johnston-linear-matrix-algebra-F01224	122	1.000	$B, C \in \mathcal{M}_n$	$B, C \in \mathcal{M}_n$	had <i>two</i> inverses $B, C \in \mathcal{M}_n$ (i.e., $AB = BA = I$ and $AC = CA = I$). It would
johnston-linear-matrix-algebra-F01225	122	1.000	$AB = BA = I$	$AB = BA = I$	had <i>two</i> inverses $B, C \in \mathcal{M}_n$ (i.e., $AB = BA = I$ and $AC = CA = I$). It would
johnston-linear-matrix-algebra-F01226	122	1.000	$AC = CA = I$	$AC = CA = I$	had <i>two</i> inverses $B, C \in \mathcal{M}_n$ (i.e., $AB = BA = I$ and $AC = CA = I$). It would
johnston-linear-matrix-algebra-F01227	122	1.000	$\frac{1}{2} \left[\begin{array}{cc} -4 & 2 \\ 3 & -1 \end{array} \right]$	$\frac{1}{2} \begin{bmatrix} -4 & 2 \\ 3 & -1 \end{bmatrix}$	a) $\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ and $\frac{1}{2} \begin{bmatrix} -4 & 2 \\ 3 & -1 \end{bmatrix}$,
johnston-linear-matrix-algebra-F01228	122	1.000	$\left[\begin{array}{ll} 1 & 2 \\ 2 & 4 \end{array} \right]$	$\begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$	b) $\begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$ and $\begin{bmatrix} -4 & 2 \\ 2 & -1 \end{bmatrix}$, and
johnston-linear-matrix-algebra-F01229	122	1.000	$\left[\begin{array}{cc} -4 & 2 \\ 2 & -1 \end{array} \right]$	$\begin{bmatrix} -4 & 2 \\ 2 & -1 \end{bmatrix}$	b) $\begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$ and $\begin{bmatrix} -4 & 2 \\ 2 & -1 \end{bmatrix}$, and
johnston-linear-matrix-algebra-F01230	122	0.991	$\left[\begin{array}{lll} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{array} \right]$	$\begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$	c) $\begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$ and $\frac{1}{2} \begin{bmatrix} -1 & 1 & 1 \\ 1 & -1 & 1 \\ 1 & 1 & -1 \end{bmatrix}$.
johnston-linear-matrix-algebra-F01231	122	0.991	$\frac{1}{2} \left[\begin{array}{ccc} -1 & 1 & 1 \\ 1 & -1 & 1 \\ 1 & 1 & -1 \end{array} \right]$	$\frac{1}{2} \begin{bmatrix} -1 & 1 & 1 \\ 1 & -1 & 1 \\ 1 & 1 & -1 \end{bmatrix}$	c) $\begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$ and $\frac{1}{2} \begin{bmatrix} -1 & 1 & 1 \\ 1 & -1 & 1 \\ 1 & 1 & -1 \end{bmatrix}$.
johnston-linear-matrix-algebra-F01232	123	1.000	T^{-1}	T^{-1}	denote by T^{-1} , such that $T \circ T^{-1} = T^{-1} \circ T = I$. In other words, T^{-1} is the

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johnston-linear-matrix-algebra F01233	123	1.000	$T \circ T^{-1} = T^{-1} \circ T = I$	$T \circ T^{-1} = T^{-1} \circ T = I$	denote by T^{-1} , such that $T \circ T^{-1} = T^{-1} \circ T = I$. In other words, T^{-1} is the
johnston-linear-matrix-algebra F01234	123	1.000	$T^{-1}(T(\mathbf{v})) = \mathbf{v}$	$T^{-1}(T(\mathbf{v})) = \mathbf{v}$	$T^{-1}(T(\mathbf{v})) = \mathbf{v}$ for all $\mathbf{v} \in \mathbb{R}^n$ (and $T(T^{-1}(\mathbf{v})) = \mathbf{v}$ too).
johnston-linear-matrix-algebra F01235	123	1.000	$T(\text{left}(T^{-1}(\mathbf{v}))) = \mathbf{v}$	$T(T^{-1}(\mathbf{v})) = \mathbf{v}$	$T^{-1}(T(\mathbf{v})) = \mathbf{v}$ for all $\mathbf{v} \in \mathbb{R}^n$ (and $T(T^{-1}(\mathbf{v})) = \mathbf{v}$ too).
johnston-linear-matrix-algebra F01236	124	1.000	$P_{\{\mathbf{u}\}}(\mathbf{v}_1) = P_{\{\mathbf{u}\}}(\mathbf{v}_2)$	$P_{\mathbf{u}}(\mathbf{v}_1) = P_{\mathbf{u}}(\mathbf{v}_2)$	single vector on the dashed line on the right is sent to $P_{\mathbf{u}}(\mathbf{v}_1) = P_{\mathbf{u}}(\mathbf{v}_2)$.
johnston-linear-matrix-algebra F01237	124	0.966	$0x$	$0x$	$0x$, we cannot determine x : it could be absolutely anything. Multiplication by 0
johnston-linear-matrix-algebra F01238	124	1.000	$\text{left}(A^{-1})^{-1} = A$	$(A^{-1})^{-1} = A$	a) A^{-1} is invertible and $(A^{-1})^{-1} = A$.
johnston-linear-matrix-algebra F01239	124	1.000	$(cA)^{-1} = \frac{1}{c}A^{-1}$	$(cA)^{-1} = \frac{1}{c}A^{-1}$	b) cA is invertible and $(cA)^{-1} = \frac{1}{c}A^{-1}$.
johnston-linear-matrix-algebra F01240	124	1.000	$\text{left}(A^T)^{-1} = \text{left}(A^{-1})^T$	$(A^T)^{-1} = (A^{-1})^T$	c) A^T is invertible and $(A^T)^{-1} = (A^{-1})^T$.
johnston-linear-matrix-algebra F01241	124	1.000	$\text{left}(A^k)^{-1} = \text{left}(A^{-1})^k$	$(A^k)^{-1} = (A^{-1})^k$	d) A^k is invertible and $(A^k)^{-1} = (A^{-1})^k$.
johnston-linear-matrix-algebra F01242	124	1.000	$\frac{1}{c}A^{-1}$	$\frac{1}{c}A^{-1}$	which shows that cA has $\frac{1}{c}A^{-1}$ as its inverse, as desired.
johnston-linear-matrix-algebra F01243	124	1.000	$A^{k\ell} =$	$A^{k\ell} =$	then matrix powers still have the “nice” properties like $A^{k+\ell} = A^k A^\ell$ and $A^{k\ell} =$
johnston-linear-matrix-algebra F01244	124	1.000	$\text{left}(A^k)^{\ell}$	$(A^k)^\ell$	$(A^k)^\ell$ even when k and ℓ are allowed to be negative (see Exercise 2.2.18).
johnston-linear-matrix-algebra F01245	124	0.965	A^{-2}	A^{-2}	Compute A^{-2} if $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$.
johnston-linear-matrix-algebra F01246	125	1.000	cR_i	cR_i	• The row operation cR_i is undone by the row operation $\frac{1}{c}R_j$.
johnston-linear-matrix-algebra F01247	125	1.000	$\text{left}[\begin{array}{ll} 1 & 0 \\ 0 & 3 \end{array}\right]$	$\begin{bmatrix} 1 & 0 \\ 0 & 3 \end{bmatrix}$	a) $\begin{bmatrix} 1 & 0 \\ 0 & 3 \end{bmatrix}$
johnston-linear-matrix-algebra F01248	125	1.000	$\text{left}[\begin{array}{ll} 1 & 2 \\ 0 & 1 \end{array}\right]$	$\begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$	b) $\begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$
johnston-linear-matrix-algebra F01249	125	1.000	$\text{left}[\begin{array}{lll} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{array}\right]$	$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$	c) $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$
johnston-linear-matrix-algebra F01250	125	1.000	$\text{left}[\begin{array}{ccc} 1 & 0 & 0 \\ -5 & 1 & 0 \\ 0 & 0 & 1 \end{array}\right]$	$\begin{bmatrix} 1 & 0 & 0 \\ -5 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$	d) $\begin{bmatrix} 1 & 0 & 0 \\ -5 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$
johnston-linear-matrix-algebra F01251	125	1.000	$3R_2$	$3R_2$	$3R_2$, so its inverse is the elementary matrix for the row operation
johnston-linear-matrix-algebra F01252	125	1.000	$\frac{1}{3}R_2$	$\frac{1}{3}R_2$	$\frac{1}{3}R_2$:
johnston-linear-matrix-algebra F01253	125	1.000	$R_1 + 2R_2$	$R_1 + 2R_2$	b) This is the elementary matrix for the row operation $R_1 + 2R_2$, so its
johnston-linear-matrix-algebra F01254	125	1.000	$R_1 - 2R_2$	$R_1 - 2R_2$	inverse is the elementary matrix for the row operation $R_1 - 2R_2$:
johnston-linear-matrix-algebra F01255	125	1.000	$R_2 \leftrightarrow R_3$	$R_2 \leftrightarrow R_3$	c) This is the elementary matrix for the row operation $R_2 \leftrightarrow R_3$, so its

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johnston-linear-matrix-algebra F01256	125	■ 1.000	$R_{\{2\}} - 5 R_{\{1\}}$	$R_2 - 5R_1$	d) This is the elementary matrix for to the row operation $R_2 - 5R_1$, so
johnston-linear-matrix-algebra F01257	126	■ 1.000	$R_{\{2\}} + 5 R_{\{1\}}$	$R_2 + 5R_1$	its inverse is the elementary matrix for the row operation $R_2 + 5R_1$:
johnston-linear-matrix-algebra F01258	126	■ 1.000	$B^{-1} A^{-1}$	$B^{-1}A^{-1}$	invertible, and its inverse is $B^{-1}A^{-1}$, as claimed.
johnston-linear-matrix-algebra F01259	126	■ 1.000	$\left[\begin{array}{ll} 1 & 2 \\ 0 & 1 \end{array}\right] \left[\begin{array}{ll} 1 & 0 \\ 0 & 3 \end{array}\right] = \left[\begin{array}{ll} 1 & 6 \\ 0 & 3 \end{array}\right]$	$\left[\begin{array}{ll} 1 & 2 \\ 0 & 1 \end{array}\right] \left[\begin{array}{ll} 1 & 0 \\ 0 & 3 \end{array}\right] = \left[\begin{array}{ll} 1 & 6 \\ 0 & 3 \end{array}\right]$	Find the inverse of the matrix $\left[\begin{array}{ll} 1 & 2 \\ 0 & 1 \end{array}\right] \left[\begin{array}{ll} 1 & 0 \\ 0 & 3 \end{array}\right] = \left[\begin{array}{ll} 1 & 6 \\ 0 & 3 \end{array}\right]$.
johnston-linear-matrix-algebra F01260	127	■ 1.000	$\mathbf{b} \in \mathbb{R}^n$	$\mathbf{b} \in \mathbb{R}^n$	b) The linear system $A\mathbf{x} = \mathbf{b}$ has a solution for all $\mathbf{b} \in \mathbb{R}^n$.
johnston-linear-matrix-algebra F01261	127	■ 1.000	$\mathbf{x} = A^{-1}\mathbf{b}$	$\mathbf{x} = A^{-1}\mathbf{b}$	To see that (a) $\implies$ (c), we note that if A is invertible then $\mathbf{x} = A^{-1}\mathbf{b}$ is a
johnston-linear-matrix-algebra F01262	127	■ 1.000	$\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$	$\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$	$\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$. Then $A\mathbf{x} = \mathbf{b}$ and $A\mathbf{y} = \mathbf{b}$, so subtracting gives us $A(\mathbf{x} - \mathbf{y}) = \mathbf{0}$. It
johnston-linear-matrix-algebra F01263	127	■ 1.000	$A\mathbf{y} = \mathbf{b}$	$A\mathbf{y} = \mathbf{b}$	$\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$. Then $A\mathbf{x} = \mathbf{b}$ and $A\mathbf{y} = \mathbf{b}$, so subtracting gives us $A(\mathbf{x} - \mathbf{y}) = \mathbf{0}$. It
johnston-linear-matrix-algebra F01264	127	■ 1.000	$A(\mathbf{x} - \mathbf{y}) = \mathbf{0}$	$A(\mathbf{x} - \mathbf{y}) = \mathbf{0}$	$\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$. Then $A\mathbf{x} = \mathbf{b}$ and $A\mathbf{y} = \mathbf{b}$, so subtracting gives us $A(\mathbf{x} - \mathbf{y}) = \mathbf{0}$. It
johnston-linear-matrix-algebra F01265	127	■ 1.000	$\mathbf{x} = \mathbf{y}$	$\mathbf{x} = \mathbf{y}$	so $\mathbf{x} = \mathbf{y}$ (i.e., the solution is unique).
johnston-linear-matrix-algebra F01266	127	■ 0.842	$\mathbf{b} = \mathbf{0}$	$\mathbf{b} = \mathbf{0}$	The implication (c) $\implies$ (d) follows simply by choosing $\mathbf{b} = \mathbf{0}$, so we jump
johnston-linear-matrix-algebra F01267	127	■ 1.000	$[A \mid \mathbf{0}]$	$[A \mid \mathbf{0}]$	in augmented matrix form $[A \mid \mathbf{0}]$ and then apply Gauss–Jordan elimination,
johnston-linear-matrix-algebra F01268	127	■ 1.000	$[R \mid \mathbf{0}]$	$[R \mid \mathbf{0}]$	we get the augmented matrix $[R \mid \mathbf{0}]$, where R is the reduced row echelon form
johnston-linear-matrix-algebra F01269	127	■ 1.000	$R \neq I$	$R \neq I$	of A . Since R is square, if $R \neq I$ then it must have a row of zeros at the bottom,
johnston-linear-matrix-algebra F01270	128	■ 0.999	$[R \mid \mathbf{e}_n]$	$[R \mid \mathbf{e}_n]$	produces $[R \mid \mathbf{e}_n]$,
johnston-linear-matrix-algebra F01271	128	■ 1.000	$\mathbf{e}_n$	$\mathbf{e}_n$	row operations to $\mathbf{e}_n$.
johnston-linear-matrix-algebra F01272	128	■ 1.000	$R = I$	$R = I$	a unique solution (i.e., (d) holds) then $R = I$ (i.e., (e) holds).
johnston-linear-matrix-algebra F01273	128	■ 0.996	$E_{\{k\}} \cdots E_{\{2\}} E_{\{1\}} A = I$	$E_k \cdots E_2 E_1 A = I$	sequence of elementary matrices $E_1, E_2, \dots, E_k$ such that $E_k \cdots E_2 E_1 A = I$. By
johnston-linear-matrix-algebra F01274	128	■ 1.000	$E_{\{k\}}^{-1}, E_{\{k-1\}}^{-1}$	E_k^{-1}, E_{k-1}^{-1}	multiplying on the left by E_k^{-1}, E_{k-1}^{-1} , and so on to E_1^{-1} , we see that
johnston-linear-matrix-algebra F01275	128	■ 1.000	$E_{\{1\}}^{-1}$	E_1^{-1}	multiplying on the left by E_k^{-1}, E_{k-1}^{-1} , and so on to E_1^{-1} , we see that
johnston-linear-matrix-algebra F01276	128	■ 1.000	$\left[\begin{array}{ccc} 1 & 2 & -2 \\ 0 & 1 & -2 \\ 1 & 1 & 1 \end{array}\right]$	$\left[\begin{array}{ccc} 1 & 2 & -2 \\ 0 & 1 & -2 \\ 1 & 1 & 1 \end{array}\right]$	b) $\left[\begin{array}{ccc} 1 & 2 & -2 \\ 0 & 1 & -2 \\ 1 & 1 & 1 \end{array}\right]$
johnston-linear-matrix-algebra F01277	129	■ 1.000	$A^{-1}[A \mid I] = [I \mid A^{-1}]$	$A^{-1}[A \mid I] = [I \mid A^{-1}]$	tiplication shows that $A^{-1}[A \mid I] = [I \mid A^{-1}]$. However, multiplication on
johnston-linear-matrix-algebra F01278	129	■ 1.000	$[I \mid A^{-1}]$	$[I \mid A^{-1}]$	$[A \mid I]$, so we conclude that $[A \mid I]$ can be row-reduced to $[I \mid A^{-1}]$. This leads
johnston-linear-matrix-algebra F01279	129	■ 1.000	$E \in \mathcal{M}_n$	$E \in \mathcal{M}_n$	$E \in \mathcal{M}_n$ such that the RREF of the block matrix $[A \mid I]$ is $[I \mid E]$.
johnston-linear-matrix-algebra F01280	129	■ 1.000	$[I \mid E]$	$[I \mid E]$	$E \in \mathcal{M}_n$ such that the RREF of the block matrix $[A \mid I]$ is $[I \mid E]$.

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johnston-linear-matrix-algebra-F01281	129	1.000	$A^{-1}=E$	$A^{-1} = E$	Furthermore, if A is invertible then it is necessarily the case that $A^{-1} = E$.
johnston-linear-matrix-algebra-F01282	129	1.000	$I=E \ A$	$I = EA$	$I = EA$ (and thus $E = A^{-1}$).
johnston-linear-matrix-algebra-F01283	129	1.000	$E=A^{-1}$	$E = A^{-1}$	$I = EA$ (and thus $E = A^{-1}$).
johnston-linear-matrix-algebra-F01284	129	0.995	$\left[\begin{array}{ll}2 & 2 \\ 4 & 5\end{array}\right]$	$\begin{bmatrix} 2 & 2 \\ 4 & 5 \end{bmatrix}$	a) $\begin{bmatrix} 2 & 2 \\ 4 & 5 \end{bmatrix}$
johnston-linear-matrix-algebra-F01285	129	1.000	$\left[\begin{array}{cc}1 & -2 \\ -3 & 6\end{array}\right]$	$\begin{bmatrix} 1 & -2 \\ -3 & 6 \end{bmatrix}$	b) $\begin{bmatrix} 1 & -2 \\ -3 & 6 \end{bmatrix}$
johnston-linear-matrix-algebra-F01286	129	1.000	$\left[\begin{array}{lll}1 & 1 & 1 \\ 1 & 2 & 4 \\ 1 & 3 & 9\end{array}\right]$	$\begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 4 \\ 1 & 3 & 9 \end{bmatrix}$	d) $\begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 4 \\ 1 & 3 & 9 \end{bmatrix}$
johnston-linear-matrix-algebra-F01287	131	0.500	$\mathbf{2} \times \mathbf{2}$	2×2	2×2 Matrix
johnston-linear-matrix-algebra-F01288	131	1.000	$a \ d \ b \ c$	$ad - bc$	The quantity $ad - bc$
johnston-linear-matrix-algebra-F01289	131	0.924	$A=\left[\begin{array}{ll}a & b \\ c & d\end{array}\right]$	$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$	The matrix $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ is invertible if and only if $ad - bc \neq 0$, and
johnston-linear-matrix-algebra-F01290	131	1.000	$a \ d \ b \ c=0$	$ad - bc = 0$	On the other hand, if $ad - bc = 0$ then $ad = bc$. From here we split into
johnston-linear-matrix-algebra-F01291	131	1.000	$a \ d=b \ c$	$ad = bc$	On the other hand, if $ad - bc = 0$ then $ad = bc$. From here we split into
johnston-linear-matrix-algebra-F01292	131	1.000	$a=0$	$a = 0$	Case 1: If either $a = 0$ or $b = 0$ then $ad = bc$ implies that two of the entries
johnston-linear-matrix-algebra-F01293	131	1.000	$b=0$	$b = 0$	Case 1: If either $a = 0$ or $b = 0$ then $ad = bc$ implies that two of the entries
johnston-linear-matrix-algebra-F01294	131	1.000	$a, \ b \neq 0$	$a, b \neq 0$	Case 2: If $a, b \neq 0$ then $ad = bc$ implies $d/b = c/a$, so the second row of A is
johnston-linear-matrix-algebra-F01295	131	1.000	$d \ / \ b=c \ / \ a$	$d/b = c/a$	Case 2: If $a, b \neq 0$ then $ad = bc$ implies $d/b = c/a$, so the second row of A is
johnston-linear-matrix-algebra-F01296	131	1.000	$a \ d \ b \ c=1 \times 4 \ 2 \times 3=4 \cdot 6=2 \neq 0$	$ad - bc = 1 \times 4 - 2 \times 3 = 4 - 6 = -2 \neq 0$	a) We first compute $ad - bc = 1 \times 4 - 2 \times 3 = 4 - 6 = -2 \neq 0$, so this
johnston-linear-matrix-algebra-F01297	131	1.000	$a \ d \ b \ c=1 \times 4 \ 2 \times 2=4 \cdot 4=0$	$ad - bc = 1 \times 4 - 2 \times 2 = 4 - 4 = 0$	b) Again, we start by computing $ad - bc = 1 \times 4 - 2 \times 2 = 4 - 4 = 0$.
johnston-linear-matrix-algebra-F01298	131	1.000	$a \ d \ b \ c=2 \times 5 \ 2 \times 4=10 \cdot 8=2 \neq 0$	$ad - bc = 2 \times 5 - 2 \times 4 = 10 - 8 = 2 \neq 0$	c) Yet again, we first compute $ad - bc = 2 \times 5 - 2 \times 4 = 10 - 8 = 2 \neq 0$,
johnston-linear-matrix-algebra-F01299	132	1.000	$a \ d \ b \ c=1 \times 6 \ -(-2) \times (-3)=6 \cdot 6=0$	$ad - bc = 1 \times 6 - (-2) \times (-3) = 6 - 6 = 0$	d) Once again, computing $ad - bc = 1 \times 6 - (-2) \times (-3) = 6 - 6 = 0$
johnston-linear-matrix-algebra-F01300	133	1.000	$A \ A^{-1}=A^{-1} \ A=I$	$AA^{-1} = A^{-1}A = I$	A^{-1} as being inverses of each other if $AA^{-1} = A^{-1}A = I$, it is actually enough
johnston-linear-matrix-algebra-F01301	133	1.000	$B \ \text{in} \ \mathcal{M}_n$	$B \in \mathcal{M}_n$	Suppose $A \in \mathcal{M}_n$. If there exists a matrix $B \in \mathcal{M}_n$ such that $AB = I$ or
johnston-linear-matrix-algebra-F01302	133	1.000	$A \ B=I$	$AB = I$	Suppose $A \in \mathcal{M}_n$. If there exists a matrix $B \in \mathcal{M}_n$ such that $AB = I$ or
johnston-linear-matrix-algebra-F01303	133	1.000	$B \ A=I$	$BA = I$	$BA = I$ then A is invertible and $A^{-1} = B$.
johnston-linear-matrix-algebra-F01304	133	1.000	$A^{-1}=B$	$A^{-1} = B$	$BA = I$ then A is invertible and $A^{-1} = B$.

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johnston-linear-matrix-align=left F01305	133	■ 0.974	$B \ A \ \mathbf{x} = B \ \mathbf{0} = \mathbf{0}$	$BA\mathbf{x} = B\mathbf{0} = \mathbf{0}$	shows that $BA\mathbf{x} = B\mathbf{0} = \mathbf{0}$, but also $BA\mathbf{x} = I\mathbf{x} = \mathbf{x}$, so $\mathbf{x} = \mathbf{0}$. The linear system
johnston-linear-matrix-align=left F01306	133	■ 0.974	$B \ A \ \mathbf{x} = I \ \mathbf{x} = \mathbf{x}$	$BA\mathbf{x} = I\mathbf{x} = \mathbf{x}$	shows that $BA\mathbf{x} = B\mathbf{0} = \mathbf{0}$, but also $BA\mathbf{x} = I\mathbf{x} = \mathbf{x}$, so $\mathbf{x} = \mathbf{0}$. The linear system
johnston-linear-matrix-align=left F01307	133	■ 1.000	$B = A^{-1}$	$B = A^{-1}$	that $B = A^{-1}$, as desired.
johnston-linear-matrix-align=left F01308	133	■ 0.966	$A \ C = I$	$AC = I$	$AC = I$, since the only way that AC could be square is if C is 2×3 , in
johnston-linear-matrix-align=left F01309	133	■ 1.000	$x \ y$	xy	that sends $\mathbb{R}^2$ to the xy -plane in $\mathbb{R}^3$, and there is another linear transforma-
johnston-linear-matrix-align=left F01310	134	■ 1.000	$E \ \text{in} \ \mathcal{M}_3$	$E \in \mathcal{M}_3$	2.2.1 Write down the elementary matrix $E \in \mathcal{M}_3$ corre-
johnston-linear-matrix-align=left F01311	134	■ 0.975	$R_{\{1\}} \ \text{\textbackslash leftrightarrow} \ R_{\{3\}}$	$R_1 \leftrightarrow R_3$	*(a) $R_1 \leftrightarrow R_3$
johnston-linear-matrix-align=left F01312	134	■ 1.000	$R_{\{1\}} + 3 \ R_{\{2\}}$	$R_1 + 3R_2$	*(c) $R_1 + 3R_2$
johnston-linear-matrix-align=left F01313	134	■ 1.000	$R_{\{3\}} - 4 \ R_{\{1\}}$	$R_3 - 4R_1$	(d) $R_3 - 4R_1$
johnston-linear-matrix-align=left F01314	134	■ 1.000	$3 \ R_{\{1\}}$	$3R_1$	*(e) $3R_1$
johnston-linear-matrix-align=left F01315	134	■ 1.000	$6 \ R_{\{3\}}$	$6R_3$	(f) $6R_3$
johnston-linear-matrix-align=left F01316	134	■ 0.995	$R_{\{2\}} - 2 \ R_{\{3\}}$	$R_2 - 2R_3$	*(g) $R_2 - 2R_3$
johnston-linear-matrix-align=left F01317	134	■ 1.000	$R_{\{2\}} + 3 \ R_{\{1\}}$	$R_2 + 3R_1$	(h) $R_2 + 3R_1$
johnston-linear-matrix-align=left F01318	134	■ 1.000	$\left[\begin{array}{ll} 6 & 3 \\ 2 & 1 \end{array} \right]$	$\left[\begin{array}{cc} 6 & 3 \\ 2 & 1 \end{array} \right]$	(b) $\left[\begin{array}{cc} 6 & 3 \\ 2 & 1 \end{array} \right]$
johnston-linear-matrix-align=left F01319	134	■ 0.892	$\left[\begin{array}{ll} 2 & 3 \\ 3 & 2 \end{array} \right]$	$\left[\begin{array}{cc} 2 & 3 \\ 3 & 2 \end{array} \right]$	*(c) $\left[\begin{array}{cc} 2 & 3 \\ 3 & 2 \end{array} \right]$
johnston-linear-matrix-align=left F01320	134	■ 0.995	$\left[\begin{array}{lll} 1 & 2 & 1 \\ 0 & 1 & 2 \end{array} \right]$	$\left[\begin{array}{ccc} 1 & 2 & 1 \\ 0 & 1 & 2 \end{array} \right]$	(d) $\left[\begin{array}{ccc} 1 & 2 & 1 \\ 0 & 1 & 2 \end{array} \right]$
johnston-linear-matrix-align=left F01321	134	■ 0.887	$\left[\begin{array}{ccc} 2 & 4 & 0 \\ 1 & -2 & 0 \\ 2 & 0 & -1 \end{array} \right]$	$\left[\begin{array}{ccc} 2 & 4 & 0 \\ 1 & -2 & 0 \\ 2 & 0 & -1 \end{array} \right]$	*(e) $\left[\begin{array}{ccc} 2 & 4 & 0 \\ 1 & -2 & 0 \\ 2 & 0 & -1 \end{array} \right]$
johnston-linear-matrix-align=left F01322	134	■ 0.997	$\left[\begin{array}{ccc} 2 & 6 & 1 \\ 0 & 0 & 0 \\ 3 & -2 & 7 \end{array} \right]$	$\left[\begin{array}{ccc} 2 & 6 & 1 \\ 0 & 0 & 0 \\ 3 & -2 & 7 \end{array} \right]$	(f) $\left[\begin{array}{ccc} 2 & 6 & 1 \\ 0 & 0 & 0 \\ 3 & -2 & 7 \end{array} \right]$
johnston-linear-matrix-align=left F01323	134	■ 0.980	$\left[\begin{array}{ll} 1 & 0 \\ 0 & 2 \\ 0 & 1 \end{array} \right]$	$\left[\begin{array}{cc} 1 & 0 \\ 0 & 2 \\ 0 & 1 \end{array} \right]$	*(g) $\left[\begin{array}{cc} 1 & 0 \\ 0 & 2 \\ 0 & 1 \end{array} \right]$
johnston-linear-matrix-align=left F01324	134	■ 1.000	$\left[\begin{array}{ccc} 1 & -2 & -2 \\ -2 & 1 & -2 \\ -2 & -2 & 1 \end{array} \right]$	$\left[\begin{array}{ccc} 1 & -2 & -2 \\ -2 & 1 & -2 \\ -2 & -2 & 1 \end{array} \right]$	(h) $\left[\begin{array}{ccc} 1 & -2 & -2 \\ -2 & 1 & -2 \\ -2 & -2 & 1 \end{array} \right]$
johnston-linear-matrix-align=left F01325	134	■ 0.997	$\left[\begin{array}{llll} 0 & 1 & 1 & 2 \\ 4 & 2 & 3 & 0 \\ 3 & 1 & 5 & 5 \\ 5 & 0 & 5 & 2 \end{array} \right]$	$\left[\begin{array}{cccc} 0 & 1 & 1 & 2 \\ 4 & 2 & 3 & 0 \\ 3 & 1 & 5 & 5 \\ 5 & 0 & 5 & 2 \end{array} \right]$	$\left[\begin{array}{cccc} 0 & 1 & 1 & 2 \\ 4 & 2 & 3 & 0 \\ 3 & 1 & 5 & 5 \\ 5 & 0 & 5 & 2 \end{array} \right]$

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johnston-linear-matrix-aligned-F01326	134	1.000	$\left[\begin{array}{cccc}-1 & -2 & 2 & 2 \\ -1 & 1 & 1 & 0 \\ 3 & 3 & 5 & 1 \\ 5 & 2 & 2 & 0\end{array}\right]$	$\begin{bmatrix} -1 & -2 & 2 & 2 \\ -1 & 1 & 1 & 0 \\ 3 & 3 & 5 & 1 \\ 5 & 2 & 2 & 0 \end{bmatrix}$	(b) $\begin{bmatrix} -1 & -2 & 2 & 2 \\ -1 & 1 & 1 & 0 \\ 3 & 3 & 5 & 1 \\ 5 & 2 & 2 & 0 \end{bmatrix}$
johnston-linear-matrix-aligned-F01327	134	1.000	$\left[\begin{array}{ccccc}-1 & -3 & 3 & 5 & 4 \\ 1 & 5 & -2 & 4 & 5 \\ 5 & -3 & 0 & 1 & 0 \\ 4 & 1 & -2 & 1 & 1 \\ 4 & 4 & 5 & 3 & 5\end{array}\right]$	$\begin{bmatrix} -1 & -3 & 3 & 5 & 4 \\ 1 & 5 & -2 & 4 & 5 \\ 5 & -3 & 0 & 1 & 0 \\ 4 & 1 & -2 & 1 & 1 \\ 4 & 4 & 5 & 3 & 5 \end{bmatrix}$	$\begin{bmatrix} -1 & -3 & 3 & 5 & 4 \\ 1 & 5 & -2 & 4 & 5 \\ 5 & -3 & 0 & 1 & 0 \\ 4 & 1 & -2 & 1 & 1 \\ 4 & 4 & 5 & 3 & 5 \end{bmatrix}$
johnston-linear-matrix-aligned-F01328	134	1.000	$\left[\begin{array}{cccccc}-1 & -2 & -1 & 1 & 2 & 4 \\ -1 & 0 & 0 & -1 & 0 \\ 2 & 4 & 1 & 1 & -4 & -6 \\ 1 & 2 & 1 & 0 & -2 & -5 \\ -3 & -6 & -1 & -4 & 7 & 9 \\ 0 & -1 & 1 & -5 & 3 & 1\end{array}\right]$	$\begin{bmatrix} -1 & -2 & -1 & 1 & 2 & 4 \\ -1 & -1 & 0 & 0 & -1 & 0 \\ 2 & 4 & 1 & 1 & -4 & -6 \\ 1 & 2 & 1 & 0 & -2 & -5 \\ -3 & -6 & -1 & -4 & 7 & 9 \\ 0 & -1 & 1 & -5 & 3 & 1 \end{bmatrix}$	$\begin{bmatrix} -1 & -2 & -1 & 1 & 2 & 4 \\ -1 & -1 & 0 & 0 & -1 & 0 \\ 2 & 4 & 1 & 1 & -4 & -6 \\ 1 & 2 & 1 & 0 & -2 & -5 \\ -3 & -6 & -1 & -4 & 7 & 9 \\ 0 & -1 & 1 & -5 & 3 & 1 \end{bmatrix}$
johnston-linear-matrix-aligned-F01329	134	1.000	E A	EA	a matrix $E \in \mathcal{M}_m$ such that EA equals the reduced row
johnston-linear-matrix-aligned-F01330	134	1.000	$\left[\begin{array}{cc}2 & -1 \\ -4 & 2\end{array}\right]$	$\begin{bmatrix} 2 & -1 \\ -4 & 2 \end{bmatrix}$	(b) $\begin{bmatrix} 2 & -1 \\ -4 & 2 \end{bmatrix}$
johnston-linear-matrix-aligned-F01331	134	0.897	$\left[\begin{array}{lll}1 & 2 & 3 \\ 4 & 5 & 6\end{array}\right]$	$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$	*(c) $\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$
johnston-linear-matrix-aligned-F01332	134	0.992	$\left[\begin{array}{ccc}0 & -1 & 3 \\ 0 & 2 & 1\end{array}\right]$	$\begin{bmatrix} 0 & -1 & 3 \\ 0 & 2 & 1 \end{bmatrix}$	(d) $\begin{bmatrix} 0 & -1 & 3 \\ 0 & 2 & 1 \end{bmatrix}$
johnston-linear-matrix-aligned-F01333	134	0.876	$\left[\begin{array}{cc}2 & 1 \\ -1 & 2 \\ 1 & 1\end{array}\right]$	$\begin{bmatrix} 2 & 1 \\ -1 & 2 \\ 1 & 1 \end{bmatrix}$	*(e) $\begin{bmatrix} 2 & 1 \\ -1 & 2 \\ 1 & 1 \end{bmatrix}$
johnston-linear-matrix-aligned-F01334	134	1.000	$\left[\begin{array}{ccc}1 & 2 & 1 \\ -1 & 0 & -1 \\ 0 & 1 & 2\end{array}\right]$	$\begin{bmatrix} 1 & 2 & 1 \\ -1 & 0 & -1 \\ 0 & 1 & 2 \end{bmatrix}$	(f) $\begin{bmatrix} 1 & 2 & 1 \\ -1 & 0 & -1 \\ 0 & 1 & 2 \end{bmatrix}$
johnston-linear-matrix-aligned-F01335	134	0.999	$\left[\begin{array}{ccc}0 & 1 & 3 \\ 3 & -2 & 3 \\ 1 & 0 & 3\end{array}\right]$	$\begin{bmatrix} 0 & 1 & 3 \\ 3 & -2 & 3 \\ 1 & 0 & 3 \end{bmatrix}$	*(g) $\begin{bmatrix} 0 & 1 & 3 \\ 3 & -2 & 3 \\ 1 & 0 & 3 \end{bmatrix}$
johnston-linear-matrix-aligned-F01336	134	1.000	$\left[\begin{array}{cccc}4 & 0 & 2 & 4 \\ -1 & -1 & 1 & 2 \\ 5 & 0 & 0 & 2\end{array}\right]$	$\begin{bmatrix} 4 & 0 & 2 & 4 \\ -1 & -1 & 1 & 2 \\ 5 & 0 & 0 & 2 \end{bmatrix}$	(h) $\begin{bmatrix} 4 & 0 & 2 & 4 \\ -1 & -1 & 1 & 2 \\ 5 & 0 & 0 & 2 \end{bmatrix}$
johnston-linear-matrix-aligned-F01337	134	1.000	$A^6=I$	$A^6 = I$	(h) If $A^6 = I$ then A is invertible.
johnston-linear-matrix-aligned-F01338	134	1.000	$A^7=0$	$A^7 = O$	*(i) If $A^7 = O$ then A is not invertible.
johnston-linear-matrix-aligned-F01339	134	1.000	$X A=B$	$XA = B$	$XA = B$, then $X = A^{-1}B$.
johnston-linear-matrix-aligned-F01340	134	1.000	$X=A^{-1} B$	$X = A^{-1}B$	$XA = B$, then $X = A^{-1}B$.
johnston-linear-matrix-aligned-F01341	134	1.000	$a \neq 0$	$a \neq 0$	2.2.6 Let $a \neq 0$ be a real number. Show that the following

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johnston-linear-matrix-augmented F01342	134	■ 1.000	$a=b$	$a = b$	(a) Show that A is not invertible if $a = b$.
johnston-linear-matrix-augmented F01343	134	■ 1.000	$a \neq b$	$a \neq b$	vertible) if $a \neq b$. [Hint: Try some small examples
johnston-linear-matrix-augmented F01344	134	■ 1.000	A^{-4}	A^{-4}	(d) A^{-4}
johnston-linear-matrix-augmented F01345	134	■ 1.000	$A^k=0$	$A^k = O$	some positive integer k such that $A^k = O$. In this exercise,
johnston-linear-matrix-augmented F01346	135	■ 1.000	$I-A$	$I - A$	(a) Suppose $A^2 = O$. Prove that $I - A$ is invertible, and
johnston-linear-matrix-augmented F01347	135	■ 1.000	$I+A$	$I + A$	its inverse is $I + A$.
johnston-linear-matrix-augmented F01348	135	■ 1.000	$A^3=0$	$A^3 = O$	(b) Suppose $A^3 = O$. Prove that $I - A$ is invertible, and
johnston-linear-matrix-augmented F01349	135	■ 1.000	$I+A+A^2$	$I + A + A^2$	its inverse is $I + A + A^2$.
johnston-linear-matrix-augmented F01350	135	■ 1.000	$P=A\left(A^T A\right)^{-1} A^T$	$P = A \left(A^T A\right)^{-1} A^T$	and let $P = A \left(A^T A\right)^{-1} A^T$.
johnston-linear-matrix-augmented F01351	135	■ 1.000	P	P	(a) What is the size of P ?
johnston-linear-matrix-augmented F01352	135	■ 1.000	$P=P^T$	$P = P^T$	(b) Show that $P = P^T$. [Side note: Matrices with this
johnston-linear-matrix-augmented F01353	135	■ 0.999	$P^2=P$	$P^2 = P$	(c) Show that $P^2 = P$. [Side note: Matrices such that
johnston-linear-matrix-augmented F01354	135	■ 1.000	$P^T=P^2=P$	$P^T = P^2 = P$	$P^T = P^2 = P$ are called orthogonal projections .]
johnston-linear-matrix-augmented F01355	135	■ 0.997	$P, Q \in \mathcal{M}_n$	$P, Q \in \mathcal{M}_n$	*2.2.12 Suppose that $P, Q \in \mathcal{M}_n$ and P is invertible. Show
johnston-linear-matrix-augmented F01356	135	■ 0.996	$[P \mid Q]$	$[P \mid Q]$	that row-reducing the augmented matrix $[P \mid Q]$ produces
johnston-linear-matrix-augmented F01357	135	■ 1.000	$\left[I \mid P^{-1} Q\right]$	$[I \mid P^{-1} Q]$	the matrix $[I \mid P^{-1} Q]$.
johnston-linear-matrix-augmented F01358	135	■ 1.000	$A X=B$	$AX = B$	*(a) $AX = B$
johnston-linear-matrix-augmented F01359	135	■ 1.000	$A X B=0$	$AXB = O$	(b) $AXB = O$
johnston-linear-matrix-augmented F01360	135	■ 1.000	$A X B=I$	$AXB = I$	*(c) $AXB = I$
johnston-linear-matrix-augmented F01361	135	■ 1.000	$A X+B X=A-B$	$AX + BX = A - B$	(d) $AX + BX = A - B$
johnston-linear-matrix-augmented F01362	135	■ 1.000	B^{-1}	B^{-1}	tained by swapping two of the rows of A . Explain why B^{-1}
johnston-linear-matrix-augmented F01363	135	■ 0.996	$A_1, A_2, \ldots, A_n$	$A_1, A_2, \ldots, A_n$	is invertible if and only if each of $A_1, A_2, \ldots, A_n$ are
johnston-linear-matrix-augmented F01364	135	■ 0.959	$\left[\begin{array}{cc} A & I \\ I & O \end{array}\right]^{-1}=\left[\begin{array}{cc} 0 & I \\ I & -A \end{array}\right]$	$\left[\begin{array}{cc} A & I \\ I & O \end{array}\right]^{-1}=\left[\begin{array}{cc} O & I \\ I & -A \end{array}\right]$	*(a) $\left[\begin{array}{cc} A & I \\ I & O \end{array}\right]^{-1}=\left[\begin{array}{cc} O & I \\ I & -A \end{array}\right]$
johnston-linear-matrix-augmented F01365	135	■ 0.998	$\left[\begin{array}{cc} I & B \\ O & I \end{array}\right]^{-1}=\left[\begin{array}{cc} I & -B \\ O & I \end{array}\right]$	$\left[\begin{array}{cc} I & B \\ O & I \end{array}\right]^{-1}=\left[\begin{array}{cc} I & -B \\ O & I \end{array}\right]$	(b) $\left[\begin{array}{cc} I & B \\ O & I \end{array}\right]^{-1}=\left[\begin{array}{cc} I & -B \\ O & I \end{array}\right]$
johnston-linear-matrix-augmented F01366	135	■ 0.946	$\left[\begin{array}{ll} A & B \\ O & D \end{array}\right]^{-1}=\left[\begin{array}{cc} A^{-1} & -A^{-1} B D^{-1} \\ O & D^{-1} \end{array}\right]$	$\left[\begin{array}{cc} A & B \\ O & D \end{array}\right]^{-1}=\left[\begin{array}{cc} A^{-1} & -A^{-1} B D^{-1} \\ O & D^{-1} \end{array}\right]$	*(c) $\left[\begin{array}{cc} A & B \\ O & D \end{array}\right]^{-1}=\left[\begin{array}{cc} A^{-1} & -A^{-1} B D^{-1} \\ O & D^{-1} \end{array}\right]$

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johnston-linear-matrix-align=0 F01367	135	■ 1.000	$S=D-C A^{-1} B$	$S = D - CA^{-1}B$	(d) If we define $S = D - CA^{-1}B$, then
johnston-linear-matrix-align=0 F01368	135	■ 1.000	$\left[\begin{array}{llll}1 & 2 & 1 & 0 \\ 2 & 3 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0\end{array}\right]$	$\begin{bmatrix} 1 & 2 & 1 & 0 \\ 2 & 3 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}$	$\begin{bmatrix} 1 & 2 & 1 & 0 \\ 2 & 3 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}$
johnston-linear-matrix-align=0 F01369	135	■ 1.000	$\left[\begin{array}{llll}1 & 0 & 4 & 6 \\ 0 & 1 & 1 & 5 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1\end{array}\right]$	$\begin{bmatrix} 1 & 0 & 4 & 6 \\ 0 & 1 & 1 & 5 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} 1 & 0 & 4 & 6 \\ 0 & 1 & 1 & 5 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$
johnston-linear-matrix-align=0 F01370	135	■ 0.999	$\left[\begin{array}{ccccc}1 & 1 & 2 & -1 & 0 \\ 1 & 2 & 2 & 0 & 1 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 0 & 3\end{array}\right]$	$\begin{bmatrix} 1 & 1 & 2 & -1 & 0 \\ 1 & 2 & 2 & 0 & 1 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 0 & 3 \end{bmatrix}$	$\begin{bmatrix} 1 & 1 & 2 & -1 & 0 \\ 1 & 2 & 2 & 0 & 1 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 0 & 3 \end{bmatrix}$
johnston-linear-matrix-align=0 F01371	135	■ 1.000	$\left[\begin{array}{llll}2 & 1 & 1 & 0 \\ 1 & 2 & 1 & 1 \\ 1 & 1 & 2 & 1 \\ 0 & 1 & 1 & 2\end{array}\right]$	$\begin{bmatrix} 2 & 1 & 1 & 0 \\ 1 & 2 & 1 & 1 \\ 1 & 1 & 2 & 1 \\ 0 & 1 & 1 & 2 \end{bmatrix}$	$\begin{bmatrix} 2 & 1 & 1 & 0 \\ 1 & 2 & 1 & 1 \\ 1 & 1 & 2 & 1 \\ 0 & 1 & 1 & 2 \end{bmatrix}$
johnston-linear-matrix-align=0 F01372	136	■ 1.000	$A+\mathbf{vw}^T$	$A + \mathbf{vw}^T$	that this formula holds by multiplying $A + \mathbf{vw}^T$ by its pro-
johnston-linear-matrix-align=0 F01373	136	■ 0.520	$A_{\{n\}}$	A_n	□ 2.2.24 Let A_n be the $n \times n$ matrix
johnston-linear-matrix-align=0 F01374	136	■ 1.000	$A_{\{n\}}^{-1}$	A_n^{-1}	(a) Use computer software to compute A_n^{-1} when $n =$
johnston-linear-matrix-align=0 F01375	136	■ 1.000	$n=$	$n =$	(a) Use computer software to compute A_n^{-1} when $n =$
johnston-linear-matrix-align=0 F01376	136	■ 0.998	$\mathcal{S}$	$\mathcal{S}$	A subspace of $\mathbb{R}^n$ is a non-empty set $\mathcal{S}$ of vectors in $\mathbb{R}^n$ with the properties
johnston-linear-matrix-align=0 F01377	136	■ 1.000	$\mathbf{v}, \mathbf{w} \in \mathcal{S}$	$\mathbf{v}, \mathbf{w} \in \mathcal{S}$	a) if $\mathbf{v}, \mathbf{w} \in \mathcal{S}$ then $\mathbf{v} + \mathbf{w} \in \mathcal{S}$, and
johnston-linear-matrix-align=0 F01378	136	■ 1.000	$\mathbf{v} + \mathbf{w} \in \mathcal{S}$	$\mathbf{v} + \mathbf{w} \in \mathcal{S}$	a) if $\mathbf{v}, \mathbf{w} \in \mathcal{S}$ then $\mathbf{v} + \mathbf{w} \in \mathcal{S}$, and
johnston-linear-matrix-align=0 F01379	136	■ 1.000	$\mathbf{v} \in \mathcal{S}$	$\mathbf{v} \in \mathcal{S}$	b) if $\mathbf{v} \in \mathcal{S}$ and $c \in \mathbb{R}$ then $c\mathbf{v} \in \mathcal{S}$.
johnston-linear-matrix-align=0 F01380	136	■ 1.000	$c \mathbf{v} \in \mathcal{S}$	$c\mathbf{v} \in \mathcal{S}$	b) if $\mathbf{v} \in \mathcal{S}$ and $c \in \mathbb{R}$ then $c\mathbf{v} \in \mathcal{S}$.
johnston-linear-matrix-align=0 F01381	137	■ 1.000	$y=x / 2+1$	$y = x/2 + 1$	origin, so the line (a) $y = x/2$ is a subspace but the line (b) $y = x/2 + 1$ is not.
johnston-linear-matrix-align=0 F01382	137	■ 0.993	$1 / 2$	$1/2$	b) The line in $\mathbb{R}^2$ with slope $1/2$ and y -intercept -1 .
johnston-linear-matrix-align=0 F01383	137	■ 1.000	$x+y-3 z=0$	$x + y - 3z = 0$	c) The plane in $\mathbb{R}^3$ with equation $x + y - 3z = 0$.
johnston-linear-matrix-align=0 F01384	137	■ 0.995	$(2,0)$	$(2,0)$	by noticing that $(2,0)$ is on the line, but $2(2,0) = (4,0)$ is not, so it
johnston-linear-matrix-align=0 F01385	137	■ 0.995	$2(2,0)=(4,0)$	$2(2,0) = (4,0)$	by noticing that $(2,0)$ is on the line, but $2(2,0) = (4,0)$ is not, so it
johnston-linear-matrix-align=0 F01386	138	■ 1.000	$\mathbf{v}_1, \mathbf{v}_2, \ldots$	$\mathbf{v}_1, \mathbf{v}_2, \dots$	Be careful: $\mathbf{v}_1, \mathbf{v}_2, \dots,$
johnston-linear-matrix-align=0 F01387	138	■ 1.000	$\mathbf{v}_k$	$\mathbf{v}_k$	$\mathbf{v}_k$ are vectors, not

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johnston-linear-matrix-algebra-F01388	138	0.600	$\ldots, v_n$	$\dots, v_n$	$\dots, v_n$.
johnston-linear-matrix-algebra-F01389	138	0.999	$\{(x, y) \in$	$\{(x, y) \in$	The notation $\{(x, y) \in$
johnston-linear-matrix-algebra-F01390	138	1.000	$\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_k \in \mathcal{S}$	$\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_k \in \mathcal{S}$	for all $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_k \in \mathcal{S}$ and all $c_1, c_2, \ldots, c_k \in \mathbb{R}$ (see Exercise 2.3.15).
johnston-linear-matrix-algebra-F01391	138	1.000	$m \leq n$	$m \leq n$	of $\mathbb{R}^m$ (for some $m \leq n$) going through the origin. For example, lines look like
johnston-linear-matrix-algebra-F01392	138	0.993	$\mathbb{R}^1$	$\mathbb{R}^1$	copies or $\mathbb{R}^1$, planes look like copies of $\mathbb{R}^2$, and so on. The following example
johnston-linear-matrix-algebra-F01393	138	1.000	$y=x^2$	$y = x^2$	a) The graph of the function $y = x^2$ in $\mathbb{R}^2$.
johnston-linear-matrix-algebra-F01394	138	1.000	$\left\{(x, y) \in \mathbb{R}^2 : x \geq 0, y \geq 0\right\}$	$\{(x, y) \in \mathbb{R}^2 : x \geq 0, y \geq 0\}$	b) The set $\{(x, y) \in \mathbb{R}^2 : x \geq 0, y \geq 0\}$.
johnston-linear-matrix-algebra-F01395	138	1.000	$\mathbf{v} =$	$\mathbf{v} =$	a) This set is <i>not</i> a subspace, since (for example) the vectors $\mathbf{v} =$
johnston-linear-matrix-algebra-F01396	138	0.867	$-1, 1$	$-1, 1$	$(-1, 1)$ and $\mathbf{w} = (1, 1)$ are in the set, but $\mathbf{v} + \mathbf{w} = (0, 2)$ is not, since
johnston-linear-matrix-algebra-F01397	138	0.867	$\mathbf{v} + \mathbf{w} = (0, 2)$	$\mathbf{v} + \mathbf{w} = (0, 2)$	$(-1, 1)$ and $\mathbf{w} = (1, 1)$ are in the set, but $\mathbf{v} + \mathbf{w} = (0, 2)$ is not, since
johnston-linear-matrix-algebra-F01398	138	1.000	$2=0^2$	$2 = 0^2$	it is not the case that $2 = 0^2$:
johnston-linear-matrix-algebra-F01399	139	1.000	$\{\mathbf{0}\}$	$\{\mathbf{0}\}$	origin, $\{\mathbf{0}\}$, and all
johnston-linear-matrix-algebra-F01400	139	1.000	$\mathbf{b} \neq \mathbf{0}$	$\mathbf{b} \neq \mathbf{0}$	when $\mathbf{b} \neq \mathbf{0}$ (see
johnston-linear-matrix-algebra-F01401	139	1.000	$\ker(A)$	$\ker(A)$	denoted by $\ker(A)$.
johnston-linear-matrix-algebra-F01402	139	1.000	$\mathbf{v} = (1, 1)$	$\mathbf{v} = (1, 1)$	b) This set is <i>not</i> a subspace, since (for example) the vector $\mathbf{v} = (1, 1)$
johnston-linear-matrix-algebra-F01403	139	1.000	$-\mathbf{v} = (-1, -1)$	$-\mathbf{v} = (-1, -1)$	is in the set, but $-\mathbf{v} = (-1, -1)$ is not, since the set only contains
johnston-linear-matrix-algebra-F01404	139	1.000	$v_1 + v_2 - 3v_3 = 0$	$v_1 + v_2 - 3v_3 = 0$	are both on the plane, so $v_1 + v_2 - 3v_3 = 0$ and $w_1 + w_2 - 3w_3 = 0$.
johnston-linear-matrix-algebra-F01405	139	1.000	$w_1 + w_2 - 3w_3 = 0$	$w_1 + w_2 - 3w_3 = 0$	are both on the plane, so $v_1 + v_2 - 3v_3 = 0$ and $w_1 + w_2 - 3w_3 = 0$.
johnston-linear-matrix-algebra-F01406	139	1.000	$A \mathbf{x} \in \mathbb{R}^m$	$A\mathbf{x} \in \mathbb{R}^m$	transformation that sends $\mathbf{x} \in \mathbb{R}^n$ to $A\mathbf{x} \in \mathbb{R}^m$. The set of possible outputs of
johnston-linear-matrix-algebra-F01407	139	1.000	$A \mathbf{x}$	$A\mathbf{x}$	this linear transformation—that is, the set of vectors of the form $A\mathbf{x}$ —is also a
johnston-linear-matrix-algebra-F01408	139	0.751	$\text{range}(A)$	$\text{range}(A)$	a) The range of A is the subspace of $\mathbb{R}^m$, denoted by $\text{range}(A)$, that
johnston-linear-matrix-algebra-F01409	139	1.000	$\text{null}(A)$	$\text{null}(A)$	b) The null space of A is the subspace of $\mathbb{R}^n$, denoted by $\text{null}(A)$, that
johnston-linear-matrix-algebra-F01410	139	1.000	$A \mathbf{0} = \mathbf{0}$	$A\mathbf{0} = \mathbf{0}$	Definition 2.3.1. It is non-empty because $A\mathbf{0} = \mathbf{0}$, so $\mathbf{0} \in \text{null}(A)$. To see that it
johnston-linear-matrix-algebra-F01411	139	1.000	$\mathbf{0} \in \text{null}(A)$	$\mathbf{0} \in \text{null}(A)$	Definition 2.3.1. It is non-empty because $A\mathbf{0} = \mathbf{0}$, so $\mathbf{0} \in \text{null}(A)$. To see that it
johnston-linear-matrix-algebra-F01412	140	1.000	$\mathbf{v}, \mathbf{w} \in \text{null}(A)$	$\mathbf{v}, \mathbf{w} \in \text{null}(A)$	satisfies property (a), let $\mathbf{v}, \mathbf{w} \in \text{null}(A)$, so that $A\mathbf{v} = \mathbf{0}$ and $A\mathbf{w} = \mathbf{0}$. Then
johnston-linear-matrix-algebra-F01413	140	1.000	$A \mathbf{v} = \mathbf{0}$	$A\mathbf{v} = \mathbf{0}$	satisfies property (a), let $\mathbf{v}, \mathbf{w} \in \text{null}(A)$, so that $A\mathbf{v} = \mathbf{0}$ and $A\mathbf{w} = \mathbf{0}$. Then
johnston-linear-matrix-algebra-F01414	140	1.000	$A \mathbf{w} = \mathbf{0}$	$A\mathbf{w} = \mathbf{0}$	satisfies property (a), let $\mathbf{v}, \mathbf{w} \in \text{null}(A)$, so that $A\mathbf{v} = \mathbf{0}$ and $A\mathbf{w} = \mathbf{0}$. Then

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johnston-linear-matrix-algebra-F01415	140	1.000	$\mathbf{v} + \mathbf{w} \in \operatorname{null}(A)$	$\mathbf{v} + \mathbf{w} \in \operatorname{null}(A)$	so $\mathbf{v} + \mathbf{w} \in \operatorname{null}(A)$ too. Similarly, for property (b) we note that if $\mathbf{v} \in \operatorname{null}(A)$
johnston-linear-matrix-algebra-F01416	140	1.000	$\mathbf{v} \in \operatorname{null}(A)$	$\mathbf{v} \in \operatorname{null}(A)$	so $\mathbf{v} + \mathbf{w} \in \operatorname{null}(A)$ too. Similarly, for property (b) we note that if $\mathbf{v} \in \operatorname{null}(A)$
johnston-linear-matrix-algebra-F01417	140	0.984	$c \mathbf{v} \in \operatorname{null}(A)$	$c\mathbf{v} \in \operatorname{null}(A)$	so $c\mathbf{v} \in \operatorname{null}(A)$ too. It follows that $\operatorname{null}(A)$ is indeed a subspace.
johnston-linear-matrix-algebra-F01418	140	1.000	$A \mathbf{0} = \mathbf{0} \in \operatorname{range}(A)$	$A\mathbf{0} = \mathbf{0} \in \operatorname{range}(A)$	empty since $A\mathbf{0} = \mathbf{0} \in \operatorname{range}(A)$. Furthermore, if $A\mathbf{x}, A\mathbf{y} \in \operatorname{range}(A)$ and $c \in \mathbb{R}$
johnston-linear-matrix-algebra-F01419	140	1.000	$A \mathbf{x}, A \mathbf{y} \in \operatorname{range}(A)$	$A\mathbf{x}, A\mathbf{y} \in \operatorname{range}(A)$	empty since $A\mathbf{0} = \mathbf{0} \in \operatorname{range}(A)$. Furthermore, if $A\mathbf{x}, A\mathbf{y} \in \operatorname{range}(A)$ and $c \in \mathbb{R}$
johnston-linear-matrix-algebra-F01420	140	1.000	$A \mathbf{x} + A \mathbf{y} = A(\mathbf{x} + \mathbf{y}) \in \operatorname{range}(A)$	$A\mathbf{x} + A\mathbf{y} = A(\mathbf{x} + \mathbf{y}) \in \operatorname{range}(A)$	then $A\mathbf{x} + A\mathbf{y} = A(\mathbf{x} + \mathbf{y}) \in \operatorname{range}(A)$ as well, as is $A(c\mathbf{x}) = c(A\mathbf{x})$.
johnston-linear-matrix-algebra-F01421	140	1.000	$A(c \mathbf{x}) = c(A \mathbf{x})$	$A(c\mathbf{x}) = c(A\mathbf{x})$	then $A\mathbf{x} + A\mathbf{y} = A(\mathbf{x} + \mathbf{y}) \in \operatorname{range}(A)$ as well, as is $A(c\mathbf{x}) = c(A\mathbf{x})$.
johnston-linear-matrix-algebra-F01422	140	0.899	$A = \begin{bmatrix} 2 & -2 \\ 1 & -1 \end{bmatrix}$	$A = \begin{bmatrix} 2 & -2 \\ 1 & -1 \end{bmatrix}$	Describe and plot the range and null space of the matrix $A = \begin{bmatrix} 2 & -2 \\ 1 & -1 \end{bmatrix}$.
johnston-linear-matrix-algebra-F01423	140	1.000	$\mathbf{x} = (x, y)$	$\mathbf{x} = (x, y)$	To get our hands on the range of A , we let $\mathbf{x} = (x, y)$ and compute
johnston-linear-matrix-algebra-F01424	140	1.000	$x - y$	$x - y$	Notice that $A\mathbf{x}$ only depends on $x - y$, not x and y individually. We can
johnston-linear-matrix-algebra-F01425	140	1.000	$z = x - y$	$z = x - y$	thus define a new variable $z = x - y$ and see that $A\mathbf{x} = (2z, z) = z(2, 1)$, so
johnston-linear-matrix-algebra-F01426	140	1.000	$A \mathbf{x} = (2z, z) = z(2, 1)$	$A\mathbf{x} = (2z, z) = z(2, 1)$	thus define a new variable $z = x - y$ and see that $A\mathbf{x} = (2z, z) = z(2, 1)$, so
johnston-linear-matrix-algebra-F01427	140	1.000	$x = y$	$x = y$	$x = y$. It follows that $\mathbf{x} = (x, x) = x(1, 1)$, so the null space is the line going
johnston-linear-matrix-algebra-F01428	140	1.000	$\mathbf{x} = (x, x) = x(1, 1)$	$\mathbf{x} = (x, x) = x(1, 1)$	$x = y$. It follows that $\mathbf{x} = (x, x) = x(1, 1)$, so the null space is the line going
johnston-linear-matrix-algebra-F01429	141	1.000	$\mathbf{v}_1, \dots, \mathbf{v}_k$	$\mathbf{v}_1, \dots, \mathbf{v}_k$	vectors $\mathbf{v}_1, \dots, \mathbf{v}_k$ is
johnston-linear-matrix-algebra-F01430	141	1.000	$c_1 \mathbf{v}_1 + \dots + c_k \mathbf{v}_k$	$c_1 \mathbf{v}_1 + \dots + c_k \mathbf{v}_k$	form $c_1 \mathbf{v}_1 + \dots + c_k \mathbf{v}_k$.
johnston-linear-matrix-algebra-F01431	141	0.724	$2(2, 1) = (4, 2)$	$2(2, 1) = (4, 2)$	vector $(2, 1)$ is not a subspace of $\mathbb{R}^2$ because, for example, $2(2, 1) = (4, 2)$ is
johnston-linear-matrix-algebra-F01432	141	1.000	$(4, 2)$	$(4, 2)$	not in the set. To fix this problem, we could simply add $(4, 2)$ to the set and ask
johnston-linear-matrix-algebra-F01433	141	1.000	$\{(2, 1), (4, 2)\}$	$\{(2, 1), (4, 2)\}$	whether or not $\{(2, 1), (4, 2)\}$ is a subspace. This larger set also fails to be a
johnston-linear-matrix-algebra-F01434	141	1.000	$\{(2, 1)\}$	$\{(2, 1)\}$	Figure 2.14: To turn the set $\{(2, 1)\}$ into a subspace, we have to add all of the scalar
johnston-linear-matrix-algebra-F01435	141	1.000	$B = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}$	$B = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}$	If $B = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}$ is a set of vectors in $\mathbb{R}^n$, then the set of all linear
johnston-linear-matrix-algebra-F01436	141	0.511	$((2, 1))$	$((2, 1))$	For example, $\operatorname{span}((2, 1))$ is the line through the origin and the point $(2, 1)$,
johnston-linear-matrix-algebra-F01437	141	1.000	$\operatorname{span}(\mathbf{e}_1, \mathbf{e}_2) = \mathbb{R}^2$	$\operatorname{span}(\mathbf{e}_1, \mathbf{e}_2) = \mathbb{R}^2$	$\operatorname{span}(\mathbf{e}_1, \mathbf{e}_2) = \mathbb{R}^2$. To verify this claim, we just need to show that every vector
johnston-linear-matrix-algebra-F01438	141	1.000	$\operatorname{span}(\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n) = \mathbb{R}^n$	$\operatorname{span}(\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n) = \mathbb{R}^n$	dimensions also holds: $\operatorname{span}(\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n) = \mathbb{R}^n$ for all $n \geq 1$, since we can
johnston-linear-matrix-algebra-F01439	142	1.000	$\operatorname{span}((1, 2), (2, 1))$	$\operatorname{span}((1, 2), (2, 1))$	a) $\operatorname{span}((1, 2), (2, 1))$
johnston-linear-matrix-algebra-F01440	142	1.000	$\operatorname{span}((1, 2, 1), (2, 1, 1))$	$\operatorname{span}((1, 2, 1), (2, 1, 1))$	b) $\operatorname{span}((1, 2, 1), (2, 1, 1))$

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johnston-linear-matrix-algebra-F01441	142	1.000	$\operatorname{span}((1,2),(2,1))=\mathbb{R}^2$	$\operatorname{span}((1,2),(2,1))=\mathbb{R}^2$	To verify that $\operatorname{span}((1,2),(2,1))=\mathbb{R}^2$ algebraically, we show that
johnston-linear-matrix-algebra-F01442	142	1.000	$(x,y)\in\mathbb{R}^2$	$(x,y)\in\mathbb{R}^2$	for every $(x,y)\in\mathbb{R}^2$ we can find coefficients $c_1,c_2\in\mathbb{R}$ such that
johnston-linear-matrix-algebra-F01443	142	0.726	$\left[\begin{array}{cc c}0&0&b\end{array}\right]$	$\left[\begin{array}{cc c}0&0&b\end{array}\right]$	no row of the form $\left[\begin{array}{cc c}0&0&b\end{array}\right]$, so it has a solution regardless of
johnston-linear-matrix-algebra-F01444	142	1.000	(x,y)	(x,y)	the values of x and y . That is, we can always write (x,y) as a linear
johnston-linear-matrix-algebra-F01445	143	1.000	$\operatorname{span}(\mathbf{v}_1,\mathbf{v}_2,\ldots,\mathbf{v}_k)$	$\operatorname{span}(\mathbf{v}_1,\mathbf{v}_2,\ldots,\mathbf{v}_k)$	Of $\operatorname{span}(\mathbf{v}_1,\mathbf{v}_2,\ldots,\mathbf{v}_k)$
johnston-linear-matrix-algebra-F01446	143	1.000	$(x,y,z)\in\mathbb{R}^3$	$(x,y,z)\in\mathbb{R}^3$	$(x,y,z)\in\mathbb{R}^3$ and we try to find $c_1,c_2\in\mathbb{R}$ such that
johnston-linear-matrix-algebra-F01447	143	1.000	$z=\frac{1}{3}x+\frac{1}{3}y$	$z=\frac{1}{3}x+\frac{1}{3}y$	a solution if and only if the bottom-right entry, $z=\frac{1}{3}x+\frac{1}{3}y$, equals
johnston-linear-matrix-algebra-F01448	143	0.955	(x,y,z)	(x,y,z)	$x+y-3z=0$, so the vector (x,y,z) is a linear combination of
johnston-linear-matrix-algebra-F01449	143	0.703	$(2,1,1)$	$(2,1,1)$	$(1,2,1)$ and $(2,1,1)$ if and only if $x+y-3z=0$.
johnston-linear-matrix-algebra-F01450	143	1.000	$\mathbf{v},\mathbf{w}\in\operatorname{span}(\mathbf{v}_1,\mathbf{v}_2,\ldots,\mathbf{v}_k)$	$\mathbf{v},\mathbf{w}\in\operatorname{span}(\mathbf{v}_1,\mathbf{v}_2,\ldots,\mathbf{v}_k)$	of subspaces (Definition 2.3.1) holds, suppose that $\mathbf{v},\mathbf{w}\in\operatorname{span}(\mathbf{v}_1,\mathbf{v}_2,\ldots,\mathbf{v}_k)$.
johnston-linear-matrix-algebra-F01451	143	0.991	$c_1,c_2,\ldots,c_k,d_1,d_2,\ldots,d_k\in\mathbb{R}$	$c_1,c_2,\ldots,c_k,d_1,d_2,\ldots,d_k\in\mathbb{R}$	Then there exist $c_1,c_2,\ldots,c_k,d_1,d_2,\ldots,d_k\in\mathbb{R}$ such that
johnston-linear-matrix-algebra-F01452	143	1.000	$\mathbf{v}+\mathbf{w}\in\operatorname{span}(\mathbf{v}_1,\mathbf{v}_2,\ldots,\mathbf{v}_k)$	$\mathbf{v}+\mathbf{w}\in\operatorname{span}(\mathbf{v}_1,\mathbf{v}_2,\ldots,\mathbf{v}_k)$	$\mathbf{v}+\mathbf{w}\in\operatorname{span}(\mathbf{v}_1,\mathbf{v}_2,\ldots,\mathbf{v}_k)$ too.
johnston-linear-matrix-algebra-F01453	144	1.000	$\mathcal{S}=\operatorname{span}(\mathbf{v}_1,\mathbf{v}_2,\ldots,\mathbf{v}_k)$	$\mathcal{S}=\operatorname{span}(\mathbf{v}_1,\mathbf{v}_2,\ldots,\mathbf{v}_k)$	$\mathcal{S}=\operatorname{span}(\mathbf{v}_1,\mathbf{v}_2,\ldots,\mathbf{v}_k)$
johnston-linear-matrix-algebra-F01454	144	1.000	$\{\mathbf{v}_1,\mathbf{v}_2,\ldots,\mathbf{v}_k\}$	$\{\mathbf{v}_1,\mathbf{v}_2,\ldots,\mathbf{v}_k\}$	$\{\mathbf{v}_1,\mathbf{v}_2,\ldots,\mathbf{v}_k\}$ spans
johnston-linear-matrix-algebra-F01455	144	1.000	$\operatorname{col}(A)$	$\operatorname{col}(A)$	denoted by $\operatorname{col}(A)$.
johnston-linear-matrix-algebra-F01456	144	0.997	$c\mathbf{v}\in\operatorname{span}(\mathbf{v}_1,\mathbf{v}_2,\ldots,\mathbf{v}_k)$	$c\mathbf{v}\in\operatorname{span}(\mathbf{v}_1,\mathbf{v}_2,\ldots,\mathbf{v}_k)$	so $c\mathbf{v}\in\operatorname{span}(\mathbf{v}_1,\mathbf{v}_2,\ldots,\mathbf{v}_k)$, which shows that $\operatorname{span}(\mathbf{v}_1,\mathbf{v}_2,\ldots,\mathbf{v}_k)$ is indeed a
johnston-linear-matrix-algebra-F01457	144	0.982	$(2,3)$	$(2,3)$	things like “the line going through the origin and $(2,3)$ ” or “the plane going
johnston-linear-matrix-algebra-F01458	144	1.000	$\operatorname{span}((1,2,3),(3,-1,2))$	$\operatorname{span}((1,2,3),(3,-1,2))$	$\operatorname{span}((1,2,3),(3,-1,2))$, respectively. Being able to describe subspaces like
johnston-linear-matrix-algebra-F01459	144	1.000	$\operatorname{span}(\mathbf{a}_1,\mathbf{a}_2,\ldots,\mathbf{a}_n)$	$\operatorname{span}(\mathbf{a}_1,\mathbf{a}_2,\ldots,\mathbf{a}_n)$	of all linear combinations of $\mathbf{a}_1,\mathbf{a}_2,\ldots,\mathbf{a}_n$, which is $\operatorname{span}(\mathbf{a}_1,\mathbf{a}_2,\ldots,\mathbf{a}_n)$.
johnston-linear-matrix-algebra-F01460	145	0.958	$A\operatorname{span}\mathbb{R}^n$	$A\operatorname{span}\mathbb{R}^n$	c) The rows of A $\operatorname{span}\mathbb{R}^n$.
johnston-linear-matrix-algebra-F01461	145	0.755	$\operatorname{span}\mathbb{R}^n$	$\operatorname{span}\mathbb{R}^n$	$\operatorname{span}\mathbb{R}^n$.
johnston-linear-matrix-algebra-F01462	146	1.000	$\{\mathbf{e}_1,\mathbf{e}_2,\ldots,\mathbf{e}_n\}\subseteq\mathbb{R}^n$	$\{\mathbf{e}_1,\mathbf{e}_2,\ldots,\mathbf{e}_n\}\subseteq\mathbb{R}^n$	$\{\mathbf{e}_1,\mathbf{e}_2,\ldots,\mathbf{e}_n\}\subseteq\mathbb{R}^n$ is
johnston-linear-matrix-algebra-F01463	146	1.000	$-x+y=-2$	$-x+y=-2$	above, the second equation of the associated linear system says $-x+y=-2$,
johnston-linear-matrix-algebra-F01464	146	1.000	$x-y=2$	$x-y=2$	which is redundant since the first equation already said that $x-y=2$, and these
johnston-linear-matrix-algebra-F01465	146	1.000	$\{(2,3),(1,0),(0,1)\}$	$\{(2,3),(1,0),(0,1)\}$	For example, the set of vectors $\{(2,3),(1,0),(0,1)\}$ is linearly dependent
johnston-linear-matrix-algebra-F01466	146	1.000	$\{(1,0,0),(0,1,0),(0,0,1)\}$	$\{(1,0,0),(0,1,0),(0,0,1)\}$	On the other hand, the set of vectors $\{(1,0,0),(0,1,0),(0,0,1)\}$ is linearly

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johnston-linear-matrix-algebra-F01467	146	1.000	$c_{\{1\}}, c_{\{2\}}, \ldots, c_{\{k\}}$	$c_1, c_2, \dots, c_k$	and then try to solve for the scalars $c_1, c_2, \dots, c_k$. If they must all equal 0
johnston-linear-matrix-algebra-F01468	146	0.998	$\{(1, -1, 0), (-2, 1, 2), (1, 1, -4)\}$	$\{(1, -1, 0), (-2, 1, 2), (1, 1, -4)\}$	a) $\{(1, -1, 0), (-2, 1, 2), (1, 1, -4)\}$
johnston-linear-matrix-algebra-F01469	146	1.000	$\{(1, 2, 3), (1, 0, 1), (0, -1, 2)\}$	$\{(1, 2, 3), (1, 0, 1), (0, -1, 2)\}$	b) $\{(1, 2, 3), (1, 0, 1), (0, -1, 2)\}$
johnston-linear-matrix-algebra-F01470	147	1.000	$c_{\{3\}}$	c_3	chosen c_3 to be any
johnston-linear-matrix-algebra-F01471	147	1.000	$c_{\{1\}}, c_{\{2\}}, c_{\{3\}} \in \mathbb{R}$	$c_1, c_2, c_3 \in \mathbb{R}$	all zero) scalars $c_1, c_2, c_3 \in \mathbb{R}$ such that
johnston-linear-matrix-algebra-F01472	147	1.000	$c_{\{3\}}=1$	$c_3 = 1$	$c_3 = 1$ and then solve by back substitution, we find that $c_2 = 2$ and
johnston-linear-matrix-algebra-F01473	147	1.000	$c_{\{2\}}=2$	$c_2 = 2$	$c_3 = 1$ and then solve by back substitution, we find that $c_2 = 2$ and
johnston-linear-matrix-algebra-F01474	147	1.000	$c_{\{1\}}=3$	$c_1 = 3$	$c_1 = 3$, which tells us that
johnston-linear-matrix-algebra-F01475	147	1.000	$c_{\{1\}}=c_{\{2\}}=c_{\{3\}}=0$	$c_1 = c_2 = c_3 = 0$	$c_1 = c_2 = c_3 = 0$ is a solution, it must be the only solution, so
johnston-linear-matrix-algebra-F01476	148	0.997	$\left\{\mathbf{a}_{\{1\}}, \mathbf{a}_{\{2\}}, \ldots, \mathbf{a}_{\{n\}}\right\}$	$\{\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_n\}$	a) The set $\{\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_n\}$ is linearly dependent.
johnston-linear-matrix-algebra-F01477	148	0.998	$A=\left[\mathbf{a}_{\{1\}}\left \mathbf{a}_{\{2\}}\right \cdots\right]\mid\mathbf{a}_{\{n\}}\right]$	$A = [\mathbf{a}_1 \mathbf{a}_2 \cdots \mathbf{a}_n]$	<i>Proof.</i> We use block matrix multiplication. If $A = [\mathbf{a}_1 \mathbf{a}_2 \cdots \mathbf{a}_n]$ then
johnston-linear-matrix-algebra-F01478	148	0.778	$x_{\{1\}}, x_{\{2\}}, \ldots, x_{\{n\}}($	$x_1, x_2, \dots, x_n($	there exist constants $x_1, x_2, \dots, x_n$ (not all 0) such that $x_1 \mathbf{a}_1 + x_2 \mathbf{a}_2 + \cdots + x_n \mathbf{a}_n =$
johnston-linear-matrix-algebra-F01479	148	0.778	$x_{\{1\}} \mathbf{a}_{\{1\}}+x_{\{2\}} \mathbf{a}_{\{2\}}+\cdots+x_{\{n\}} \mathbf{a}_{\{n\}}=$	$x_1 \mathbf{a}_1 + x_2 \mathbf{a}_2 + \cdots + x_n \mathbf{a}_n =$	there exist constants $x_1, x_2, \dots, x_n$ (not all 0) such that $x_1 \mathbf{a}_1 + x_2 \mathbf{a}_2 + \cdots + x_n \mathbf{a}_n =$
johnston-linear-matrix-algebra-F01480	148	0.999	$\mathbf{v}_{\{1\}}$	$\mathbf{v}_1$	so as to solve for one of the vectors (say $\mathbf{v}_1$) in terms of the others. In other
johnston-linear-matrix-algebra-F01481	148	1.000	$\{(1, 2, 1), (2, 4, 2), (2, 1, 1)\}$	$\{(1, 2, 1), (2, 4, 2), (2, 1, 1)\}$	set. To illustrate why, consider the set $\{(1, 2, 1), (2, 4, 2), (2, 1, 1)\}$, which is
johnston-linear-matrix-algebra-F01482	149	0.863	$\{(1, 2), (3, 1)\}$	$\{(1, 2), (3, 1)\}$	a) $\{(1, 2), (3, 1)\}$
johnston-linear-matrix-algebra-F01483	149	0.985	$\{(1, 2, 3), (2, 4, 6)\}$	$\{(1, 2, 3), (2, 4, 6)\}$	b) $\{(1, 2, 3), (2, 4, 6)\}$
johnston-linear-matrix-algebra-F01484	149	0.861	$(3, 1)=c(1, 2)$	$(3, 1) = c(1, 2)$	is no $c \in \mathbb{R}$ such that $(3, 1) = c(1, 2)$), we conclude that this set
johnston-linear-matrix-algebra-F01485	150	1.000	$(2, 4, 6)=2(1, 2, 3)$	$(2, 4, 6) = 2(1, 2, 3)$	b) Since $(2, 4, 6) = 2(1, 2, 3)$, these two vectors are multiples of each
johnston-linear-matrix-algebra-F01486	150	1.000	$(n-1)$	$(n - 1)$	only if they lie on a common $(n - 1)$ -dimensional hyperplane. More generally,
johnston-linear-matrix-algebra-F01487	151	1.000	$a_{\{0\}}=1$	$a_0 = 1$	matrix with $a_0 = 1$,
johnston-linear-matrix-algebra-F01488	151	1.000	$a_{\{1\}}=2$	$a_1 = 2$	$a_1 = 2$, and $a_2 = 3$.
johnston-linear-matrix-algebra-F01489	151	1.000	$a_{\{2\}}=3$	$a_2 = 3$	$a_1 = 2$, and $a_2 = 3$.
johnston-linear-matrix-algebra-F01490	151	1.000	V	V	Be careful: V is an
johnston-linear-matrix-algebra-F01491	151	1.000	$(n+1) \times (n+1)$	$(n + 1) \times (n + 1)$	$(n + 1) \times (n + 1)$
johnston-linear-matrix-algebra-F01492	151	1.000	$x_{\{0\}}, x_{\{1\}}, \ldots, x_{\{n\}}$	$x_0, x_1, \dots, x_n$	$x_0, x_1, \dots, x_n$ are
johnston-linear-matrix-algebra-F01493	151	1.000	$a_{\{0\}}, a_{\{1\}}, \ldots, a_{\{n\}}$	$a_0, a_1, \dots, a_n$	where $a_0, a_1, \dots, a_n$ are real numbers. Show that V is invertible if and only

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johnston-linear-matrix-algebra-F01494	151	0.998	$a_{\{i\}} \neq a_{\{j\}}$	$a_i \neq a_j$	if $a_0, a_1, \dots, a_n$ are distinct (i.e., $a_i \neq a_j$ whenever $i \neq j$).
johnston-linear-matrix-algebra-F01495	151	0.989	$c_{\{0\}}=c_{\{1\}}=\cdots=c_{\{n\}}=0$	$c_0 = c_1 = \cdots = c_n = 0$	implies $c_0 = c_1 = \cdots = c_n = 0$.
johnston-linear-matrix-algebra-F01496	151	1.000	$p(x)=c_{\{0\}}+c_{\{1\}} x+c_{\{2\}} x^2+\cdots+c_{\{n\}} x^n$	$p(x) = c_0 + c_1x + c_2x^2 + \cdots + c_nx^n$	If we define the polynomial $p(x) = c_0 + c_1x + c_2x^2 + \cdots + c_nx^n$, then
johnston-linear-matrix-algebra-F01497	151	1.000	$p\left(a_{\{0\}}\right)=0$	$p(a_0) = 0$	the first entry of Equation (2.3.1) says that $p(a_0) = 0$, the next entry says
johnston-linear-matrix-algebra-F01498	151	1.000	$p\left(a_{\{1\}}\right)=0$	$p(a_1) = 0$	that $p(a_1) = 0$, and so on to $p(a_n) = 0$. Since p is a degree- n polynomial,
johnston-linear-matrix-algebra-F01499	151	1.000	$p\left(a_{\{n\}}\right)=0$	$p(a_n) = 0$	that $p(a_1) = 0$, and so on to $p(a_n) = 0$. Since p is a degree- n polynomial,
johnston-linear-matrix-algebra-F01500	151	1.000	$n+1$	$n + 1$	and we just showed that it has $n + 1$ distinct roots (since $a_0, a_1, \dots, a_n$ are
johnston-linear-matrix-algebra-F01501	151	0.982	$c_{\{0\}}=c_{\{1\}}=$	$c_0 = c_1 =$	zero polynomial. That is, its coefficients must all equal 0, so $c_0 = c_1 =$
johnston-linear-matrix-algebra-F01502	151	1.000	$\cdots=c_{\{n\}}=0$	$\cdots = c_n = 0$	$\cdots = c_n = 0$, so the columns of A form a linearly independent set, which
johnston-linear-matrix-algebra-F01503	151	1.000	$\left(x_{\{0\}}, y_{\{0\}}\right), \left(x_{\{1\}}, y_{\{1\}}\right), \ldots, \left(x_{\{n\}}, y_{\{n\}}\right)$	$(x_0, y_0), (x_1, y_1), \dots, (x_n, y_n)$	Suppose we are given a set of points $(x_0, y_0), (x_1, y_1), \dots, (x_n, y_n)$ in $\mathbb{R}^2$
johnston-linear-matrix-algebra-F01504	151	0.960	$p\left(x_{\{0\}}\right)=y_{\{0\}}, p\left(x_{\{1\}}\right)=y_{\{1\}}, \ldots, p\left(x_{\{n\}}\right)=y_{\{n\}}$	$p(x_0) = y_0, p(x_1) = y_1, \dots, p(x_n) = y_n$	$p(x_0) = y_0, p(x_1) = y_1, \dots, p(x_n) = y_n$.
johnston-linear-matrix-algebra-F01505	151	1.000	$n=1$	$n = 1$	Since we can find a line going through any two points, the $n = 1$ case
johnston-linear-matrix-algebra-F01506	152	1.000	$p\left(x_{\{0\}}\right)=y_{\{0\}}$	$p(x_0) = y_0$	$p(x_0) = y_0$ can be
johnston-linear-matrix-algebra-F01507	152	0.623	$c_{\{n\}} x_{\{0\}}^n+\cdots+c_{\{0\}}=y_{\{0\}}$	$c_nx_0^n + \cdots + c_0 = y_0$	$c_nx_0^n + \cdots + c_0 = y_0$,
johnston-linear-matrix-algebra-F01508	152	1.000	$c_{\{0\}}, c_{\{1\}}, \ldots, c_{\{n\}}$	$c_0, c_1, \dots, c_n$	for some scalars $c_0, c_1, \dots, c_n$. The fact that we are requiring $p(x_0) = y_0$,
johnston-linear-matrix-algebra-F01509	152	1.000	$p\left(x_{\{1\}}\right)=y_{\{1\}}, \ldots, p\left(x_{\{n\}}\right)=y_{\{n\}}$	$p(x_1) = y_1, \dots, p(x_n) = y_n$	$p(x_1) = y_1, \dots, p(x_n) = y_n$, can be written explicitly as the linear system
johnston-linear-matrix-algebra-F01510	153	1.000	$\left(c_{\{0\}}, c_{\{1\}}, c_{\{2\}}\right)=(-1,-3,2)$	$(c_0, c_1, c_2) = (-1, -3, 2)$	which has unique solution $(c_0, c_1, c_2) = (-1, -3, 2)$. The desired quadratic
johnston-linear-matrix-algebra-F01511	153	1.000	$p(-2)=13$	$p(-2) = 13$	we could plug values into p to see that it is indeed the case that $p(-2) = 13$,
johnston-linear-matrix-algebra-F01512	153	1.000	$p(1)=-2$	$p(1) = -2$	$p(1) = -2$, and $p(3) = 8$.
johnston-linear-matrix-algebra-F01513	153	1.000	$p(3)=8$	$p(3) = 8$	$p(1) = -2$, and $p(3) = 8$.
johnston-linear-matrix-algebra-F01514	153	1.000	$y=x+1$	$y = x + 1$	*(a) The graph of the function $y = x + 1$ in $\mathbb{R}^2$.
johnston-linear-matrix-algebra-F01515	153	1.000	$y=\sin (x)$	$y = \sin(x)$	*(c) The graph of the function $y = \sin(x)$ in $\mathbb{R}^2$.
johnston-linear-matrix-algebra-F01516	153	0.995	$x+2 y+3 z=4$	$x + 2y + 3z = 4$	*(e) The set of solutions (x, y, z) to $x + 2y + 3z = 4$.
johnston-linear-matrix-algebra-F01517	153	1.000	$x-y+8 z=0$	$x - y + 8z = 0$	(f) The set of solutions (x, y, z) to $x - y + 8z = 0$.
johnston-linear-matrix-algebra-F01518	153	1.000	$\left\{(x, y) \in \mathbb{R}^2: x+2 y=0\right\}$	$\{(x, y) \in \mathbb{R}^2 : x + 2y = 0\}$	*(g) $\{(x, y) \in \mathbb{R}^2 : x + 2y = 0\}$
johnston-linear-matrix-algebra-F01519	153	1.000	$\left\{(x, y) \in \mathbb{R}^2: x+y \geq 0\right\}$	$\{(x, y) \in \mathbb{R}^2 : x + y \geq 0\}$	(h) $\{(x, y) \in \mathbb{R}^2 : x + y \geq 0\}$
johnston-linear-matrix-algebra-F01520	153	1.000	$\left\{(x, y) \in \mathbb{R}^2: x y \geq 0\right\}$	$\{(x, y) \in \mathbb{R}^2 : xy \geq 0\}$	*(i) $\{(x, y) \in \mathbb{R}^2 : xy \geq 0\}$
johnston-linear-matrix-algebra-F01521	153	1.000	$\left\{(x, y, z) \in \mathbb{R}^3: x y+y z=0\right\}$	$\{(x, y, z) \in \mathbb{R}^3 : xy + yz = 0\}$	(j) $\{(x, y, z) \in \mathbb{R}^3 : xy + yz = 0\}$

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johnston-linear-matrix-align=0 F01522	153	■ 0.995	$\set{(2,3),(0,0)}$	$\{(2,3),(0,0)\}$	(b) $\{(2,3),(0,0)\}$
johnston-linear-matrix-align=0 F01523	153	■ 1.000	$\set{(0,1,-1),(1,2,1),(3,-1,4)}$	$\{(0,1,-1),(1,2,1),(3,-1,4)\}$	(b) $\{(0,1,-1),(1,2,1),(3,-1,4)\}$
johnston-linear-matrix-align=0 F01524	153	■ 1.000	$\set{(1,2,1),(0,1,-1),(2,5,1)}$	$\{(1,2,1),(0,1,-1),(2,5,1)\}$	*(c) $\{(1,2,1),(0,1,-1),(2,5,1)\}$
johnston-linear-matrix-align=0 F01525	153	■ 1.000	$\set{(1,1,0),(1,0,-1),(0,1,1),(1,2,1)}$	$\{(1,1,0),(1,0,-1),(0,1,1),(1,2,1)\}$	(d) $\{(1,1,0),(1,0,-1),(0,1,1),(1,2,1)\}$
johnston-linear-matrix-align=0 F01526	153	■ 0.359	$\text{*(\mathbf{a})}\left[\begin{array}{rr}1&1\\1&1\end{array}\right]$	$\text{*(\mathbf{a})}\left[\begin{array}{cc}1&1\\1&1\end{array}\right]$	*(a) $\left[\begin{array}{cc}1&1\\1&1\end{array}\right]$ *(c) $\left[\begin{array}{cc}0&1\\2&3\end{array}\right]$
johnston-linear-matrix-align=0 F01527	153	■ 0.359	$\text{*(\mathbf{c})}\quad\left[\begin{array}{l}0&1\\2&3\end{array}\right]$	$\text{*(\mathbf{c})}\quad\left[\begin{array}{cc}0&1\\2&3\end{array}\right]$	*(a) $\left[\begin{array}{cc}1&1\\1&1\end{array}\right]$ *(c) $\left[\begin{array}{cc}0&1\\2&3\end{array}\right]$
johnston-linear-matrix-align=0 F01528 [conf 0.083]	153	■ 0.083	$\left[\begin{array}{r}\text{\texttt{11}}\\22\end{array}\right]\text{\texttt{\texttt{ (d) }}}$	$\left[\begin{array}{c}11\\22\\(\mathbf{d})\end{array}\right]$	(b) $\left[\begin{array}{cc}1&2\\2&4\end{array}\right]$ (d) $\left[\begin{array}{cc}0&0\\0&0\end{array}\right]$
johnston-linear-matrix-align=0 F01529	153	■ 0.420	$\text{*(\mathbf{a})}\quad\left[\begin{array}{lll}1&1&2\\1&1&2\\1&2&2\end{array}\right]$	$\text{*(\mathbf{a})}\quad\left[\begin{array}{ccc}1&1&2\\1&1&2\\2&2&4\end{array}\right]$	*(a) $\left[\begin{array}{ccc}1&1&2\\1&1&2\\2&2&4\end{array}\right]$ *(c) $\left[\begin{array}{ccc}1&1&1\\1&1&0\\1&0&0\end{array}\right]$
johnston-linear-matrix-align=0 F01530	153	■ 0.420	$\text{*(\mathbf{c})}\quad\left[\begin{array}{lll}1&1&1\\1&1&0\\1&0&0\end{array}\right]$	$\text{*(\mathbf{c})}\quad\left[\begin{array}{ccc}1&1&1\\1&1&0\\1&0&0\end{array}\right]$	*(a) $\left[\begin{array}{ccc}1&1&2\\1&1&2\\2&2&4\end{array}\right]$ *(c) $\left[\begin{array}{ccc}1&1&1\\1&1&0\\1&0&0\end{array}\right]$
johnston-linear-matrix-align=0 F01531	153	■ 1.000	$\left[\begin{array}{lll}1&0&0\\0&1&1\\0&1&1\end{array}\right]$	$\left[\begin{array}{ccc}1&0&0\\0&1&1\\0&1&1\end{array}\right]$	(b) $\left[\begin{array}{ccc}1&0&0\\0&1&1\\0&1&1\end{array}\right]$
johnston-linear-matrix-align=0 F01532	153	■ 0.999	$\left[\begin{array}{lll}1&2&3\\2&3&4\\3&4&6\end{array}\right]$	$\left[\begin{array}{ccc}1&2&3\\2&3&4\\3&4&6\end{array}\right]$	(d) $\left[\begin{array}{ccc}1&2&3\\2&3&4\\3&4&6\end{array}\right]$
johnston-linear-matrix-align=0 F01533	153	■ 1.000	$\set{(1,0,1),(1,1,1)}$	$\{(1,0,1),(1,1,1)\}$	(b) $\{(1,0,1),(1,1,1)\}$
johnston-linear-matrix-align=0 F01534	153	■ 0.640	$\set{(1,0,-1),(1,1,1),(1,2,-1)}$	$\{(1,0,-1),(1,1,1),(1,2,-1)\}$	*(c) $\{(1,0,-1),(1,1,1),(1,2,-1)\}$
johnston-linear-matrix-align=0 F01535	153	■ 1.000	$\set{(1,2,3),(4,5,6),(7,8,9)}$	$\{(1,2,3),(4,5,6),(7,8,9)\}$	(d) $\{(1,2,3),(4,5,6),(7,8,9)\}$
johnston-linear-matrix-align=0 F01536	153	■ 0.969	$\set{(1,1),(2,1),(3,-2)}$	$\{(1,1),(2,1),(3,-2)\}$	*(e) $\{(1,1),(2,1),(3,-2)\}$
johnston-linear-matrix-align=0 F01537	153	■ 0.990	$\set{(2,1,0),(0,0,0),(1,1,2)}$	$\{(2,1,0),(0,0,0),(1,1,2)\}$	(f) $\{(2,1,0),(0,0,0),(1,1,2)\}$
johnston-linear-matrix-align=0 F01538	153	■ 0.516	$\set{(1,2,4,1),(2,4,-1,3),(-1,1,1,-1)}$	$\{(1,2,4,1),(2,4,-1,3),(-1,1,1,-1)\}$	*(g) $\{(1,2,4,1),(2,4,-1,3),(-1,1,1,-1)\}$

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johnston-linear-matrix-algebra-F01539	153	1.000	$\set{(0,1,1,1),(1,0,1,1),(1,1,0,1),(1,1,1,0)}$	$\set{(0,1,1,1),(1,0,1,1),(1,1,0,1),(1,1,1,0)}$	(h) $\set{(0,1,1,1),(1,0,1,1),(1,1,0,1),(1,1,1,0)}$
johnston-linear-matrix-algebra-F01540	153	0.963	$\set{(4,2,5,2),(3,1,2,4),(1,4,2,3),(3,1,4,2)}$	$\set{(4,2,5,2),(3,1,2,4),(1,4,2,3),(3,1,4,2)}$	(b) $\set{(4,2,5,2),(3,1,2,4),(1,4,2,3),(3,1,4,2)}$
johnston-linear-matrix-algebra-F01541	153	0.998	$\set{(4,4,4,3),(3,3,-1,1),(-1,2,1,2)}$	$\set{(4,4,4,3),(3,3,-1,1),(-1,2,1,2)}$	*(c) $\set{(4,4,4,3),(3,3,-1,1),(-1,2,1,2),}$
johnston-linear-matrix-algebra-F01542	153	0.999	$\set{(2,-1,4,-1),(3,1,2,1),(0,1,3,4)}$	$\set{(2,-1,4,-1),(3,1,2,1),(0,1,3,4)}$	(d) $\set{(2,-1,4,-1),(3,1,2,1),(0,1,3,4),}$
johnston-linear-matrix-algebra-F01543	153	1.000	$\set{(3,5,1,4),(4,4,5,5),(5,0,4,3),(1,1,5,-1)}$	$\set{(3,5,1,4),(4,4,5,5),(5,0,4,3),(1,1,5,-1)}$	(b) $\set{(3,5,1,4),(4,4,5,5),(5,0,4,3),(1,1,5,-1)}$
johnston-linear-matrix-algebra-F01544	153	1.000	$\set{(5,4,5,1,5),(4,3,3,0,4),(-1,0,3,-1,4)}$	$\set{(5,4,5,1,5),(4,3,3,0,4),(-1,0,3,-1,4)}$	*(c) $\set{(5,4,5,1,5),(4,3,3,0,4),(-1,0,3,-1,4)}$
johnston-linear-matrix-algebra-F01545	153	0.999	$\set{(5,-1,2,4,3),(4,-8,1,-4,-9),(2,2,1,4,5)}$	$\set{(5,-1,2,4,3),(4,-8,1,-4,-9),(2,2,1,4,5)}$	(d) $\set{(5,-1,2,4,3),(4,-8,1,-4,-9),(2,2,1,4,5)}$
johnston-linear-matrix-algebra-F01546	154	1.000	$\mathbf{v}_2$	$\mathbf{v}_2$	head-to-tail (i.e., the head of $\mathbf{v}_1$ at the tail of $\mathbf{v}_2$, the
johnston-linear-matrix-algebra-F01547	154	1.000	$\mathbf{v}_3$	$\mathbf{v}_3$	head of $\mathbf{v}_2$ at the tail of $\mathbf{v}_3$, and so on). If the vectors
johnston-linear-matrix-algebra-F01548	154	0.999	$\operatorname{span}(\mathbf{v}, \mathbf{w})$	$\operatorname{span}(\mathbf{v}, \mathbf{w})$	*(e) If $\mathbf{v}, \mathbf{w} \in \mathbb{R}^3$ then $\operatorname{span}(\mathbf{v}, \mathbf{w})$ is a plane through the
johnston-linear-matrix-algebra-F01549	154	1.000	$k \geq n$	$k \geq n$	(f) Let $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k$ be vectors in $\mathbb{R}^n$. If $k \geq n$ then it
johnston-linear-matrix-algebra-F01550	154	1.000	$\operatorname{span}\left(\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k, \ldots, \mathbf{v}_k\right)=\mathbb{R}^n$	$\operatorname{span}(\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k) = \mathbb{R}^n$	must be the case that $\operatorname{span}(\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k) = \mathbb{R}^n$.
johnston-linear-matrix-algebra-F01551	154	0.534	$(k, -1, 4)$	$(k, -1, 4)$	*(a) $(0, 1, -1), (1, 2, 1), (k, -1, 4)$
johnston-linear-matrix-algebra-F01552	154	0.976	$(1, 2, 3), (3, k, k+3), (2, 4, k)$	$(1, 2, 3), (3, k, k+3), (2, 4, k)$	(b) $(1, 2, 3), (3, k, k+3), (2, 4, k)$
johnston-linear-matrix-algebra-F01553	154	0.870	$(5, 7)$	$(5, 7)$	*(a) $(2, 1), (5, 7)$
johnston-linear-matrix-algebra-F01554	154	0.721	$(4, 4)$	$(4, 4)$	(b) $(1, 1), (2, 2), (4, 4)$
johnston-linear-matrix-algebra-F01555	154	0.772	$(4, 10)$	$(4, 10)$	*(c) $(1, 1), (2, 2), (4, 10)$
johnston-linear-matrix-algebra-F01556	154	0.999	$1, 2, \ldots, n^2$	$1, 2, \dots, n^2$	**2.3.14 Let A_n be the $n \times n$ matrix with entries $1, 2, \dots, n^2$
johnston-linear-matrix-algebra-F01557	154	1.000	$n \leq 2$	$n \leq 2$	Show that A_n is invertible if and only if $n \leq 2$.
johnston-linear-matrix-algebra-F01558	154	1.000	$\mathbf{v}_1, \ldots, \mathbf{v}_k \in \mathcal{S}$	$\mathbf{v}_1, \dots, \mathbf{v}_k \in \mathcal{S}$	for all $\mathbf{v}_1, \dots, \mathbf{v}_k \in \mathcal{S}$ and all $c_1, \dots, c_k \in \mathbb{R}$.
johnston-linear-matrix-algebra-F01559	154	1.000	$A=\left[\begin{array}{ll}0 & 1 \\ 1 & 0\end{array}\right]$	$A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$	2.3.16 Suppose $A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$.
johnston-linear-matrix-algebra-F01560	154	1.000	$\set{\mathbf{v}, A \mathbf{v}}$	$\set{\mathbf{v}, A\mathbf{v}}$	(a) Find all vectors $\mathbf{v} \in \mathbb{R}^2$ with the property that $\set{\mathbf{v}, A\mathbf{v}}$
johnston-linear-matrix-algebra-F01561	154	1.000	$\mathbf{v}=(1,2,3,4,5)$	$\mathbf{v} = (1, 2, 3, 4, 5)$	not $\mathbf{v} = (1, 2, 3, 4, 5)$ is in the range of the given matrix.
johnston-linear-matrix-algebra-F01562	154	1.000	$A \mathbf{v}=\mathbf{v}$	$A\mathbf{v} = \mathbf{v}$	satisfying $A\mathbf{v} = \mathbf{v}$ is a subspace of $\mathbb{R}^n$.
johnston-linear-matrix-algebra-F01563	154	1.000	$\set{\mathbf{v}, \mathbf{w}} \subset \mathbb{R}^n$	$\set{\mathbf{v}, \mathbf{w}} \subset \mathbb{R}^n$	**2.3.20 Prove that a set of two vectors $\set{\mathbf{v}, \mathbf{w}} \subset \mathbb{R}^n$ is
johnston-linear-matrix-algebra-F01564	154	1.000	$\set{\mathbf{v}, \mathbf{w}, \mathbf{x}} \subseteq \mathbb{R}^n$	$\set{\mathbf{v}, \mathbf{w}, \mathbf{x}} \subseteq \mathbb{R}^n$	2.3.22 Suppose that the set of vectors $\set{\mathbf{v}, \mathbf{w}, \mathbf{x}} \subseteq \mathbb{R}^n$ is
johnston-linear-matrix-algebra-F01565	154	1.000	$\set{\mathbf{v}+\mathbf{w}, \mathbf{v}+\mathbf{x}, \mathbf{w}-\mathbf{x}}$	$\set{\mathbf{v} + \mathbf{w}, \mathbf{v} + \mathbf{x}, \mathbf{w} - \mathbf{x}}$	(a) Show that the set $\set{\mathbf{v} + \mathbf{w}, \mathbf{v} + \mathbf{x}, \mathbf{w} - \mathbf{x}}$ is linearly
johnston-linear-matrix-algebra-F01566	154	1.000	$\set{\mathbf{v}+\mathbf{w}, \mathbf{v}+\mathbf{x}, \mathbf{w}+\mathbf{x}}$	$\set{\mathbf{v} + \mathbf{w}, \mathbf{v} + \mathbf{x}, \mathbf{w} + \mathbf{x}}$	(b) Show that the set $\set{\mathbf{v} + \mathbf{w}, \mathbf{v} + \mathbf{x}, \mathbf{w} + \mathbf{x}}$ is linearly
johnston-linear-matrix-algebra-F01567	154	1.000	$\set{\mathbf{v}, \mathbf{v}+\mathbf{w}, \mathbf{v}+\mathbf{w}+\mathbf{x}}$	$\set{\mathbf{v}, \mathbf{v} + \mathbf{w}, \mathbf{v} + \mathbf{w} + \mathbf{x}}$	(c) Determine whether or not $\set{\mathbf{v}, \mathbf{v} + \mathbf{w}, \mathbf{v} + \mathbf{w} + \mathbf{x}}$ is

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johnston-linear-matrix-algebra-F01568	154	■ 1.000	$B \subseteq \mathbb{R}^n$	$B \subseteq \mathbb{R}^n$	**2.3.23 Suppose that $B \subseteq \mathbb{R}^n$ is a finite set of vectors.
johnston-linear-matrix-algebra-F01569	154	■ 0.822	$\mathbf{0} \in B$	$\mathbf{0} \in B$	Show that if $\mathbf{0} \in B$ then B is linearly dependent.
johnston-linear-matrix-algebra-F01570	154	■ 1.000	$\mathbf{w}, \mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k \in \mathbb{R}^n$	$\mathbf{w}, \mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k \in \mathbb{R}^n$	2.3.24 Suppose that $\mathbf{w}, \mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k \in \mathbb{R}^n$ are vectors such
johnston-linear-matrix-algebra-F01571	154	■ 1.000	$B \subseteq C \subseteq \mathbb{R}^n$	$B \subseteq C \subseteq \mathbb{R}^n$	*2.3.25 Suppose that $B \subseteq C \subseteq \mathbb{R}^n$ are finite sets of vectors.
johnston-linear-matrix-algebra-F01572	154	■ 1.000	$n > m$	$n > m$	pendent whenever $n > m$.
johnston-linear-matrix-algebra-F01573	155	■ 0.943	$(AB) \subseteq \text{range}(A)$	$(AB) \subseteq \text{range}(A)$	(a) Show that $\text{range}(AB) \subseteq \text{range}(A)$.
johnston-linear-matrix-algebra-F01574	155	■ 1.000	$\text{null}(B) \subseteq \text{null}(AB)$	$\text{null}(B) \subseteq \text{null}(AB)$	(b) Show that $\text{null}(B) \subseteq \text{null}(AB)$.
johnston-linear-matrix-algebra-F01575	155	■ 1.000	(AB)	(AB)	(c) Provide an example to show that $\text{range}(AB)$ might
johnston-linear-matrix-algebra-F01576	155	■ 1.000	$\text{range}(B)$	$\text{range}(B)$	not be contained in $\text{range}(B)$, and $\text{null}(A)$ might not
johnston-linear-matrix-algebra-F01577	155	■ 0.988	$\text{null}(AB)$	$\text{null}(AB)$	be contained in $\text{null}(AB)$.
johnston-linear-matrix-algebra-F01578	155	■ 1.000	$\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n \in \mathbb{R}^n$	$\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n \in \mathbb{R}^n$	** 2.3.28 Suppose $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n \in \mathbb{R}^n$ are vectors
johnston-linear-matrix-algebra-F01579	155	■ 1.000	$c_1, c_2, \dots, c_n \in \mathbb{R}$	$c_1, c_2, \dots, c_n \in \mathbb{R}$	and $c_1, c_2, \dots, c_n \in \mathbb{R}$ are non-zero scalars. Show that
johnston-linear-matrix-algebra-F01580	155	■ 1.000	$\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$	$\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$	$\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ is linearly independent if and only if
johnston-linear-matrix-algebra-F01581	155	■ 1.000	$\{c_1 \mathbf{v}_1, c_2 \mathbf{v}_2, \dots, c_n \mathbf{v}_n\}$	$\{c_1 \mathbf{v}_1, c_2 \mathbf{v}_2, \dots, c_n \mathbf{v}_n\}$	$\{c_1 \mathbf{v}_1, c_2 \mathbf{v}_2, \dots, c_n \mathbf{v}_n\}$ is linearly independent.
johnston-linear-matrix-algebra-F01582	155	■ 0.402	V_n	V_n	 2.3.29 Let V_n be the $n \times n$ Vandermonde matrix (see
johnston-linear-matrix-algebra-F01583	155	■ 1.000	V_n^{-1}	V_n^{-1}	(a) Use computer software to compute V_n^{-1} when $n =$
johnston-linear-matrix-algebra-F01584	155	■ 0.812	$(3, 3, 2) =$	$(3, 3, 2) =$	$(3, 3, 2) =$
johnston-linear-matrix-algebra-F01585	155	■ 0.898	$(1, 2, 1) + (2, 1, 1)$	$(1, 2, 1) + (2, 1, 1)$	$(1, 2, 1) + (2, 1, 1)$.
johnston-linear-matrix-algebra-F01586	155	■ 0.993	$\text{span}((1, 2, 1), (2, 1, 1), (3, 3, 2))$	$\text{span}((1, 2, 1), (2, 1, 1), (3, 3, 2))$	$x + y - 3z = 0$ in $\mathbb{R}^3$ (see Figure 2.17) as $\text{span}((1, 2, 1), (2, 1, 1), (3, 3, 2))$, since
johnston-linear-matrix-algebra-F01587	155	■ 0.972	$(1, 2, 1), (2, 1, 1)$	$(1, 2, 1), (2, 1, 1)$	Figure 2.17: The plane $x + y - 3z = 0$ is the span of the three vectors $(1, 2, 1), (2, 1, 1),$
johnston-linear-matrix-algebra-F01588	156	■ 0.988	$\{\mathbf{v}, \mathbf{w}\}$	$\{\mathbf{v}, \mathbf{w}\}$	$\{\mathbf{v}, \mathbf{w}\}$ with two
johnston-linear-matrix-algebra-F01589	156	■ 1.000	$S \subseteq \mathbb{R}^n$	$S \subseteq \mathbb{R}^n$	A basis of a subspace $S \subseteq \mathbb{R}^n$ is a set of vectors in S that
johnston-linear-matrix-algebra-F01590	156	■ 1.000	$\{\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n\} \subset \mathbb{R}^n$	$\{\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n\} \subset \mathbb{R}^n$	example, the set $\{\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n\} \subset \mathbb{R}^n$ of standard basis vectors is (as their name
johnston-linear-matrix-algebra-F01591	156	■ 1.000	$\{\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n\}$	$\{\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n\}$	$\{\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n\}$ is linearly independent. To see linear independence, suppose
johnston-linear-matrix-algebra-F01592	156	■ 0.742	$(c_1, c_2, \dots, c_n)$	$(c_1, c_2, \dots, c_n)$	The vector on the left is just $(c_1, c_2, \dots, c_n)$, and if that vector equals $\mathbf{0}$ then
johnston-linear-matrix-algebra-F01593	156	■ 0.999	$c_1 = c_2 = \dots = c_n = 0$	$c_1 = c_2 = \dots = c_n = 0$	$c_1 = c_2 = \dots = c_n = 0$, which means exactly that $\{\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n\}$ is linearly
johnston-linear-matrix-algebra-F01594	156	■ 0.999	$B = \{(1, 2, 1), (2, 1, 1)\}, S$	$B = \{(1, 2, 1), (2, 1, 1)\}, S$	a) $B = \{(1, 2, 1), (2, 1, 1)\}, S$ is the plane $x + y - 3z = 0$ in $\mathbb{R}^3$.
johnston-linear-matrix-algebra-F01595	156	■ 1.000	$B = \{(1, 1, 2), (1, 2, 1), (2, 1, 1)\}, S = \mathbb{R}^3$	$B = \{(1, 1, 2), (1, 2, 1), (2, 1, 1)\}, S = \mathbb{R}^3$	b) $B = \{(1, 1, 2), (1, 2, 1), (2, 1, 1)\}, S = \mathbb{R}^3$.

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johnston-linear-matrix-algebra-F01596	156	1.000	$\operatorname{span}(B)=\mathbb{R}^3$	$\operatorname{span}(B) = \mathbb{R}^3$	b) We have to show that $\operatorname{span}(B) = \mathbb{R}^3$ and that B is linearly indepen-
johnston-linear-matrix-algebra-F01597	157	0.973	$\setminus$	$\}$	The only subsets of $\{\mathbf{0}\}$ are $\{\}$ and $\{\mathbf{0}\}$ itself, so these are the only sets
johnston-linear-matrix-algebra-F01598	157	0.999	$B=\setminus \setminus$	$B = \{\}$	independent, notice that if we choose $B = \{\}$ (i.e., $k = 0$) in Definition 2.3.4
johnston-linear-matrix-algebra-F01599	157	1.000	$\left\{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\right\}$	$\{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\}$	$\mathbb{R}^3$ has at least two bases: $\{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\}$ and $\{(1, 1, 2), (1, 2, 1), (2, 1, 1)\}$ (from
johnston-linear-matrix-algebra-F01600	157	1.000	$\{(1, 1, 2), (1, 2, 1), (2, 1, 1)\}$	$\{(1, 1, 2), (1, 2, 1), (2, 1, 1)\}$	$\mathbb{R}^3$ has at least two bases: $\{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\}$ and $\{(1, 1, 2), (1, 2, 1), (2, 1, 1)\}$ (from
johnston-linear-matrix-algebra-F01601	158	1.000	$ B $	$ B $	The notation $ B $
johnston-linear-matrix-algebra-F01602	158	0.520	$(1, 1, 2)$	$(1, 1, 2)$	of $\mathbb{R}^3$ even if we change $(1, 1, 2)$ to any vector that is not a linear combination of
johnston-linear-matrix-algebra-F01603	158	1.000	$B, C \subseteq \mathcal{S}$	$B, C \subseteq \mathcal{S}$	Let $\mathcal{S}$ be a subspace of $\mathbb{R}^n$, and suppose that $B, C \subseteq \mathcal{S}$ are finite sets with
johnston-linear-matrix-algebra-F01604	158	1.000	$\operatorname{span}(C)=\mathcal{S}$	$\operatorname{span}(C) = \mathcal{S}$	the properties that B is linearly independent and $\operatorname{span}(C) = \mathcal{S}$. Then
johnston-linear-matrix-algebra-F01605	158	1.000	$B=$	$B =$	<i>Proof.</i> We begin by giving names to the vectors in B and C : we write $B =$
johnston-linear-matrix-algebra-F01606	158	1.000	$C=\left\{\mathbf{w}_1, \mathbf{w}_2, \ldots, \mathbf{w}_\ell\right\}$	$C = \{\mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_\ell\}$	$\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}$ and $C = \{\mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_\ell\}$, where $k = B $ and $\ell = C $.
johnston-linear-matrix-algebra-F01607	158	1.000	$k= B $	$k = B $	$\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}$ and $C = \{\mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_\ell\}$, where $k = B $ and $\ell = C $.
johnston-linear-matrix-algebra-F01608	158	1.000	$\ell= C $	$\ell = C $	$\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}$ and $C = \{\mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_\ell\}$, where $k = B $ and $\ell = C $.
johnston-linear-matrix-algebra-F01609	158	1.000	$\ell < k$	$\ell < k$	We prove the result by showing that if $\ell < k$ and C spans $\mathcal{S}$, then B must in
johnston-linear-matrix-algebra-F01610	158	1.000	$\mathbf{v}_i \in B \subseteq \mathcal{S}$	$\mathbf{v}_i \in B \subseteq \mathcal{S}$	$\mathcal{S}$, and $\mathbf{v}_i \in B \subseteq \mathcal{S}$ for all $1 \leq i \leq k$, we can write each $\mathbf{v}_i$ as a linear combination
johnston-linear-matrix-algebra-F01611	158	1.000	$1 \leq i \leq k$	$1 \leq i \leq k$	$\mathcal{S}$, and $\mathbf{v}_i \in B \subseteq \mathcal{S}$ for all $1 \leq i \leq k$, we can write each $\mathbf{v}_i$ as a linear combination
johnston-linear-matrix-algebra-F01612	158	1.000	$\mathbf{v}_i$	$\mathbf{v}_i$	$\mathcal{S}$, and $\mathbf{v}_i \in B \subseteq \mathcal{S}$ for all $1 \leq i \leq k$, we can write each $\mathbf{v}_i$ as a linear combination
johnston-linear-matrix-algebra-F01613	158	1.000	$\mathbf{w}_1, \mathbf{w}_2, \ldots, \mathbf{w}_\ell$	$\mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_\ell$	of $\mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_\ell$:
johnston-linear-matrix-algebra-F01614	158	0.999	$(1 \leq i \leq k, 1 \leq j \leq \ell)$	$(1 \leq i \leq k, 1 \leq j \leq \ell)$	where each of the $a_{i,j}$ ’s ($1 \leq i \leq k, 1 \leq j \leq \ell$) is a scalar.
johnston-linear-matrix-algebra-F01615	158	1.000	$A \in \mathcal{M}_{k, \ell}$	$A \in \mathcal{M}_{k, \ell}$	and matrices in the “usual” way—we define $A \in \mathcal{M}_{k, \ell}$ to be the matrix whose
johnston-linear-matrix-algebra-F01616	158	0.997	$a_{i, j}, V \in \mathcal{M}_{n, k}$	$a_{i, j}, V \in \mathcal{M}_{n, k}$	(i, j) -entry is $a_{i, j}$, $V \in \mathcal{M}_{n, k}$ to be the matrix whose columns are $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k$,
johnston-linear-matrix-algebra-F01617	158	1.000	$W \in \mathcal{M}_{n, \ell}$	$W \in \mathcal{M}_{n, \ell}$	and $W \in \mathcal{M}_{n, \ell}$ to be the matrix whose columns are $\mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_\ell$. We can
johnston-linear-matrix-algebra-F01618	158	1.000	$V=W A^T$	$V = W A^T$	$V = W A^T$, since block matrix multiplication shows that the i -th column of $W A^T$
johnston-linear-matrix-algebra-F01619	158	1.000	$W A^T$	$W A^T$	$V = W A^T$, since block matrix multiplication shows that the i -th column of $W A^T$
johnston-linear-matrix-algebra-F01620	158	1.000	$\mathbf{x} \in \mathbb{R}^k$	$\mathbf{x} \in \mathbb{R}^k$	Next, Corollary 2.1.4 tells us that there is a non-zero solution $\mathbf{x} \in \mathbb{R}^k$ to the
johnston-linear-matrix-algebra-F01621	158	1.000	$A^T \mathbf{x}=\mathbf{0}$	$A^T \mathbf{x} = \mathbf{0}$	linear system $A^T \mathbf{x} = \mathbf{0}$, since A^T has more columns than rows. It follows that
johnston-linear-matrix-algebra-F01622	159	1.000	$V \mathbf{x}=\mathbf{0}$	$V \mathbf{x} = \mathbf{0}$	Finally, since the linear system $V \mathbf{x} = \mathbf{0}$ has a non-zero solution, Theorem 2.3.4

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johnston-linear-matrix-algebra-F01623	159	■ 1.000	$B \subseteq \mathcal{S}$	$B \subseteq \mathcal{S}$	<i>Proof.</i> This result follows immediately from Theorem 2.4.1: if $B \subseteq \mathcal{S}$ and
johnston-linear-matrix-algebra-F01624	159	■ 1.000	$C \subseteq \mathcal{S}$	$C \subseteq \mathcal{S}$	$C \subseteq \mathcal{S}$ are bases of $\mathcal{S}$ then $ B \leq C $ since B is linearly independent and C spans
johnston-linear-matrix-algebra-F01625	159	■ 1.000	$ B \leq C $	$ B \leq C $	$C \subseteq \mathcal{S}$ are bases of $\mathcal{S}$ then $ B \leq C $ since B is linearly independent and C spans
johnston-linear-matrix-algebra-F01626	159	■ 1.000	$ C \leq B $	$ C \leq B $	$\mathcal{S}$. On the other hand, $ C \leq B $ since C is linearly independent and B spans $\mathcal{S}$.
johnston-linear-matrix-algebra-F01627	159	■ 1.000	$ B = C $	$ B = C $	It follows that $ B = C $.
johnston-linear-matrix-algebra-F01628	159	■ 1.000	$\operatorname{dim}(\mathcal{S})$	$\dim(\mathcal{S})$	called the dimension of $\mathcal{S}$ and is denoted by $\dim(\mathcal{S})$.
johnston-linear-matrix-algebra-F01629	159	■ 1.000	$\operatorname{dim}(\mathbb{R}^n) = n$	$\dim(\mathbb{R}^n) = n$	$\dim(\mathbb{R}^n) = n$ since the standard basis of $\mathbb{R}^n$ (i.e., the set $\{\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n\}$) con-
johnston-linear-matrix-algebra-F01630	160	■ 0.994	$\leq \operatorname{dim}(\mathcal{S}) \leq$	$\leq \dim(\mathcal{S}) \leq$	 size of a linearly independent set $\leq \dim(\mathcal{S}) \leq$ size of a spanning set
johnston-linear-matrix-algebra-F01631	160	■ 1.000	$\mathcal{S} = \operatorname{span}((1, 1, 1), (1, 2, 3), (3, 2, 1))$	$\mathcal{S} = \operatorname{span}((1, 1, 1), (1, 2, 3), (3, 2, 1))$	a) $\mathcal{S} = \operatorname{span}((1, 1, 1), (1, 2, 3), (3, 2, 1))$.
johnston-linear-matrix-algebra-F01632	160	■ 0.998	$2x - y + 3z = 0$	$2x - y + 3z = 0$	b) The set $\mathcal{S}$ of vectors (x, y, z) satisfying the equation $2x - y + 3z = 0$.
johnston-linear-matrix-algebra-F01633	160	■ 1.000	$B = \{(1, 1, 1), (1, 2, 3), (3, 2, 1)\}$	$B = \{(1, 1, 1), (1, 2, 3), (3, 2, 1)\}$	a) It seems natural to first guess that $B = \{(1, 1, 1), (1, 2, 3), (3, 2, 1)\}$
johnston-linear-matrix-algebra-F01634	160	■ 1.000	$\mathcal{S} = \operatorname{span}(B)$	$\mathcal{S} = \operatorname{span}(B)$	know that $\mathcal{S} = \operatorname{span}(B)$, so we just need to check linear indepen-
johnston-linear-matrix-algebra-F01635	160	■ 0.671	$\operatorname{span}(B)$	$\operatorname{span}(B)$	$\operatorname{span}(B)$. If we (arbitrarily) decide to discard $(1, 1, 1)$ then we arrive
johnston-linear-matrix-algebra-F01636	160	■ 1.000	$C = \{(1, 2, 3), (3, 2, 1)\}$	$C = \{(1, 2, 3), (3, 2, 1)\}$	at the set $C = \{(1, 2, 3), (3, 2, 1)\}$, which has the same span as B .
johnston-linear-matrix-algebra-F01637	161	■ 0.856	$(3, 3, -1) \in \mathcal{S}$	$(3, 3, -1) \in \mathcal{S}$	$(3, 3, -1) \in \mathcal{S}$ we
johnston-linear-matrix-algebra-F01638	161	■ 0.992	$2(3) - (3) + 3(-1) = 0$	$2(3) - (3) + 3(-1) = 0$	$2(3) - (3) + 3(-1) = 0$.
johnston-linear-matrix-algebra-F01639	161	■ 0.889	$(1, -1, -1)$	$(1, -1, -1)$	the vector $(1, -1, -1)$, which we can see is in $\mathcal{S}$ since $2(1) - (-1) +$
johnston-linear-matrix-algebra-F01640	161	■ 0.889	$2(1) - (-1) +$	$2(1) - (-1) +$	the vector $(1, -1, -1)$, which we can see is in $\mathcal{S}$ since $2(1) - (-1) +$
johnston-linear-matrix-algebra-F01641	161	■ 1.000	$3(-1) = 0$	$3(-1) = 0$	$3(-1) = 0$. However, this vector by itself does not span $\mathcal{S}$ since we
johnston-linear-matrix-algebra-F01642	161	■ 0.809	$\operatorname{span}((1, -1, -1))$	$\operatorname{span}((1, -1, -1))$	can find a vector in $\mathcal{S}$ but not in $\operatorname{span}((1, -1, -1))$ —in particular,
johnston-linear-matrix-algebra-F01643	161	■ 0.979	$(3, 3, -1)$	$(3, 3, -1)$	$(3, 3, -1)$ is one such vector. If we set $B = \{(1, -1, -1), (3, 3, -1)\}$,
johnston-linear-matrix-algebra-F01644	161	■ 0.979	$B = \{(1, -1, -1), (3, 3, -1)\}$	$B = \{(1, -1, -1), (3, 3, -1)\}$	$(3, 3, -1)$ is one such vector. If we set $B = \{(1, -1, -1), (3, 3, -1)\}$,
johnston-linear-matrix-algebra-F01645	161	■ 0.655	$\operatorname{span}(B) = \mathcal{S}$	$\operatorname{span}(B) = \mathcal{S}$	$\operatorname{span}(B) = \mathcal{S}$, we check which vectors (x, y, z) are linear combina-
johnston-linear-matrix-algebra-F01646	161	■ 1.000	$\frac{2}{3}x - \frac{1}{3}y + z = 0$	$\frac{2}{3}x - \frac{1}{3}y + z = 0$	is in $\operatorname{span}(B)$) if and only if $\frac{2}{3}x - \frac{1}{3}y + z = 0$. Multiplying this equa-
johnston-linear-matrix-algebra-F01647	161	■ 1.000	$\operatorname{dim}(\mathcal{S}) = 2$	$\dim(\mathcal{S}) = 2$	subspace $\mathcal{S}$ and B is a basis of $\mathcal{S}$. Furthermore, $\dim(\mathcal{S}) = 2$ since B
johnston-linear-matrix-algebra-F01648	161	■ 1.000	$B \subseteq C$	$B \subseteq C$	a) If B is linearly independent then there is a basis C of $\mathcal{S}$ with $B \subseteq C$.
johnston-linear-matrix-algebra-F01649	161	■ 1.000	$C \subseteq B$	$C \subseteq B$	b) If B spans $\mathcal{S}$ then there is a basis C of $\mathcal{S}$ with $C \subseteq B$.

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johnston-linear-matrix-algebra-F01650	161	1.000	$\operatorname{span}(B)=$	$\operatorname{span}(B) =$	<i>Proof.</i> Suppose $B = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}$. To prove part (a), notice that if $\operatorname{span}(B) =$
johnston-linear-matrix-algebra-F01651	162	1.000	$\mathbf{w} \in \mathcal{S}$	$\mathbf{w} \in \mathcal{S}$	of it. On the other hand, if it does not $\operatorname{span} \mathcal{S}$ then there exists a vector $\mathbf{w} \in \mathcal{S}$
johnston-linear-matrix-algebra-F01652	162	1.000	$\left\{\mathbf{w}, \mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\right\}$	$\{\mathbf{w}, \mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}$	Our next step is to show that the set $\{\mathbf{w}, \mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}$ is linearly indepen-
johnston-linear-matrix-algebra-F01653	162	1.000	$d=c_1=c_2=\cdots=c_k=0$	$d = c_1 = c_2 = \cdots = c_k = 0$	with the goal of showing that $d = c_1 = c_2 = \cdots = c_k = 0$. Well, $d = 0$ since oth-
johnston-linear-matrix-algebra-F01654	162	1.000	$d=0$	$d = 0$	with the goal of showing that $d = c_1 = c_2 = \cdots = c_k = 0$. Well, $d = 0$ since oth-
johnston-linear-matrix-algebra-F01655	162	0.999	$c_1=c_2=\cdots=c_k=0$	$c_1 = c_2 = \cdots = c_k = 0$	$c_1 = c_2 = \cdots = c_k = 0$ too, so $\{\mathbf{w}, \mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}$ is indeed linearly independent.
johnston-linear-matrix-algebra-F01656	162	1.000	$k \neq \operatorname{dim}(\mathcal{S})$	$k \neq \operatorname{dim}(\mathcal{S})$	a) If $k \neq \operatorname{dim}(\mathcal{S})$ then B is not a basis of $\mathcal{S}$.
johnston-linear-matrix-algebra-F01657	162	1.000	$k=\operatorname{dim}(\mathcal{S})$	$k = \operatorname{dim}(\mathcal{S})$	b) If $k = \operatorname{dim}(\mathcal{S})$ then the following are equivalent:
johnston-linear-matrix-algebra-F01658	162	1.000	$\operatorname{dim}(\mathcal{S})=k$	$\operatorname{dim}(\mathcal{S}) = k$	create a basis of $\mathcal{S}$. However, B already has k vectors and $\operatorname{dim}(\mathcal{S}) = k$, so all
johnston-linear-matrix-algebra-F01659	163	1.000	$B=\{(1,2,3),(4,1,2)\}$	$B = \{(1, 2, 3), (4, 1, 2)\}$	a) $B = \{(1, 2, 3), (4, 1, 2)\}$
johnston-linear-matrix-algebra-F01660	163	1.000	$B=\{(1,-1,-1),(2,2,-1),(1,1,0)\}$	$B = \{(1, -1, -1), (2, 2, -1), (1, 1, 0)\}$	b) $B = \{(1, -1, -1), (2, 2, -1), (1, 1, 0)\}$
johnston-linear-matrix-algebra-F01661	163	0.913	$B=\{(1,-1,-1),(2,2,-1)\}$	$B = \{(1, -1, -1), (2, 2, -1)\}$	Determine whether or not $B = \{(1, -1, -1), (2, 2, -1)\}$ is a basis of the
johnston-linear-matrix-algebra-F01662	163	1.000	$\mathcal{S} \subset \mathbb{R}^3$	$\mathcal{S} \subset \mathbb{R}^3$	plane $\mathcal{S} \subset \mathbb{R}^3$ defined by the equation $3x - y + 4z = 0$.
johnston-linear-matrix-algebra-F01663	163	1.000	$3x - y + 4z = 0$	$3x - y + 4z = 0$	plane $\mathcal{S} \subset \mathbb{R}^3$ defined by the equation $3x - y + 4z = 0$.
johnston-linear-matrix-algebra-F01664	163	0.624	$(2, 2, -1)$	$(2, 2, -1)$	$(1, -1, -1)$ and $(2, 2, -1)$ are not multiples of each other, so B is indeed a
johnston-linear-matrix-algebra-F01665	164	0.993	$\mathcal{S} = \{\mathbf{0}\}$	$\mathcal{S} = \{\mathbf{0}\}$	If $\mathcal{S} = \{\mathbf{0}\}$ is the zero
johnston-linear-matrix-algebra-F01666	164	1.000	$(2, 1, 1) = c_1(1, 2, 1) +$	$(2, 1, 1) = c_1(1, 2, 1) +$	$(2, 1, 1) = c_1(1, 2, 1) +$
johnston-linear-matrix-algebra-F01667	164	0.975	$c_2(1, -1, 0)$	$c_2(1, -1, 0)$	$c_2(1, -1, 0)$.
johnston-linear-matrix-algebra-F01668	164	1.000	$B = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_m\}$	$B = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_m\}$	vectors in the subspace $\mathcal{S}$, and calling it $B = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_m\}$ (a largest such set
johnston-linear-matrix-algebra-F01669	164	1.000	$\mathbf{w} \in \operatorname{span}(B)$	$\mathbf{w} \in \operatorname{span}(B)$	$\mathbf{w} \in \operatorname{span}(B)$. We know that the set $\{\mathbf{w}, \mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_m\}$ is linearly dependent
johnston-linear-matrix-algebra-F01670	164	1.000	$\left\{\mathbf{w}, \mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_m\right\}$	$\{\mathbf{w}, \mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_m\}$	$\mathbf{w} \in \operatorname{span}(B)$. We know that the set $\{\mathbf{w}, \mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_m\}$ is linearly dependent
johnston-linear-matrix-algebra-F01671	164	1.000	$m+1$	$m + 1$	since it contains $m + 1$ vectors, and the largest linearly independent set in $\mathcal{S}$
johnston-linear-matrix-algebra-F01672	164	1.000	$d, c_1, c_2, \dots, c_m$	$d, c_1, c_2, \dots, c_m$	contains m vectors. We can thus find scalars $d, c_1, c_2, \dots, c_m$ (not all equal to
johnston-linear-matrix-algebra-F01673	164	1.000	$c_1 = c_2 = \cdots =$	$c_1 = c_2 = \cdots =$	Since B is linearly independent, we know that if $d = 0$ then $c_1 = c_2 = \cdots =$
johnston-linear-matrix-algebra-F01674	164	1.000	$c_m = 0$	$c_m = 0$	$c_m = 0$ too, which contradicts our assumption that not all of these scalars equal
johnston-linear-matrix-algebra-F01675	164	0.991	$d \neq 0$	$d \neq 0$	0. We thus conclude that $d \neq 0$, so we can rearrange this equation into the form

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johnston-linear-matrix-augmented-F01676	164	1.000	$A=\left[\begin{array}{ccc}1 & 1 & 2 \\ 2 & -1 & 1 \\ 0 & 1 & 0\end{array}\right]$	$A = \begin{bmatrix} 1 & 1 & 2 \\ 2 & -1 & 1 \\ 1 & 0 & 1 \end{bmatrix}$	Find bases of the range and null space of $A = \begin{bmatrix} 1 & 1 & 2 \\ 2 & -1 & 1 \\ 1 & 0 & 1 \end{bmatrix}$.
johnston-linear-matrix-augmented-F01677	164	1.000	$(2,1,1)=(1,2,1)+(1,-1,0)$	$(2,1,1) = (1,2,1) + (1,-1,0)$	To this end, we note that $(2,1,1) = (1,2,1) + (1,-1,0)$, so we can
johnston-linear-matrix-augmented-F01678	164	1.000	$\{(1,2,1),(1,-1,0)\}$	$\{(1,2,1), (1,-1,0)\}$	indeed, it is straightforward to see that $\{(1,2,1), (1,-1,0)\}$ is linearly
johnston-linear-matrix-augmented-F01679	165	1.000	$2+1=3$	$2 + 1 = 3$	$2 + 1 = 3$ for $\alpha \ 3 \times 3$
johnston-linear-matrix-augmented-F01680	165	1.000	$x_{\{3\}}$	x_3	From here we see that x_3 is a free variable, and we can solve for x_1 and x_2
johnston-linear-matrix-augmented-F01681	165	1.000	$x_{\{1\}}$	x_1	From here we see that x_3 is a free variable, and we can solve for x_1 and x_2
johnston-linear-matrix-augmented-F01682	165	1.000	$x_{\{2\}}$	x_2	From here we see that x_3 is a free variable, and we can solve for x_1 and x_2
johnston-linear-matrix-augmented-F01683	165	1.000	$x_{\{1\}}+x_{\{3\}}=0$	$x_1 + x_3 = 0$	(the leading variables) in terms of x_3 : the first row tells us that $x_1 + x_3 = 0$
johnston-linear-matrix-augmented-F01684	165	1.000	$x_{\{1\}}=-x_{\{3\}}$	$x_1 = -x_3$	(i.e., $x_1 = -x_3$) and the second row tells us that $x_2 + x_3 = 0$ (i.e., $x_2 = -x_3$).
johnston-linear-matrix-augmented-F01685	165	1.000	$x_{\{2\}}+x_{\{3\}}=0$	$x_2 + x_3 = 0$	(i.e., $x_1 = -x_3$) and the second row tells us that $x_2 + x_3 = 0$ (i.e., $x_2 = -x_3$).
johnston-linear-matrix-augmented-F01686	165	1.000	$x_{\{2\}}=-x_{\{3\}}$	$x_2 = -x_3$	(i.e., $x_1 = -x_3$) and the second row tells us that $x_2 + x_3 = 0$ (i.e., $x_2 = -x_3$).
johnston-linear-matrix-augmented-F01687	165	1.000	$\left(x_{\{1\}}, x_{\{2\}}, x_{\{3\}}\right)=$	$(x_1, x_2, x_3) =$	It follows that $\text{null}(A)$ consists of the vectors of the form $(x_1, x_2, x_3) =$
johnston-linear-matrix-augmented-F01688	165	1.000	$\left(-x_{\{3\}}, -x_{\{3\}}, x_{\{3\}}\right)=x_{\{3\}}(-1,-1,1)$	$(-x_3, -x_3, x_3) = x_3(-1, -1, 1)$	$(-x_3, -x_3, x_3) = x_3(-1, -1, 1)$, so $\{(-1, -1, 1)\}$ is a basis of $\text{null}(A)$.
johnston-linear-matrix-augmented-F01689	165	1.000	$\{(-1,-1,1)\}$	$\{(-1, -1, 1)\}$	$(-x_3, -x_3, x_3) = x_3(-1, -1, 1)$, so $\{(-1, -1, 1)\}$ is a basis of $\text{null}(A)$.
johnston-linear-matrix-augmented-F01690	165	1.000	$R \ \mathbf{x}=\mathbf{0}$	$R\mathbf{x} = \mathbf{0}$	the linear system $R\mathbf{x} = \mathbf{0}$ (this is why we started looking at row echelon forms
johnston-linear-matrix-augmented-F01691	165	0.837	$\text{range}(R)$	$\text{range}(R)$	if and only if that same subset of columns of R is a basis of $\text{range}(R)$. Since
johnston-linear-matrix-augmented-F01692	165	0.572	(R)	(R)	the leading columns of R form a basis of $\text{range}(R)$ (after all, the non-leading
johnston-linear-matrix-augmented-F01693	166	0.984	$\{(1,1,-1,2),(0,1,0,1),(0,0,1,-1)\}$	$\{(1,1,-1,2), (0,1,0,1), (0,0,1,-1)\}$	$\{(1,1,-1,2), (0,1,0,1), (0,0,1,-1)\}$ is a basis of $\text{range}(A)$.
johnston-linear-matrix-augmented-F01694	166	1.000	$x_{\{5\}}$	x_5	that the RREF tells us that x_3 and x_5 are free variables, so we solve for
johnston-linear-matrix-augmented-F01695	166	1.000	$x_{\{1\}}, x_{\{2\}}$	x_1, x_2	x_1, x_2 , and x_4 (the leading variables) in terms of them. Explicitly, we get
johnston-linear-matrix-augmented-F01696	166	1.000	$x_{\{4\}}$	x_4	x_1, x_2 , and x_4 (the leading variables) in terms of them. Explicitly, we get
johnston-linear-matrix-augmented-F01697	167	1.000	$\{(-1,1,1,0,0),(1,-2,0,-3,1)\}$	$\{(-1,1,1,0,0), (1,-2,0,-3,1)\}$	We thus conclude that $\{(-1,1,1,0,0), (1,-2,0,-3,1)\}$ is a basis of
johnston-linear-matrix-augmented-F01698	167	0.977	$\text{range}(A^T)$	$\text{range}(A^T)$	It turns out that the subspaces $\text{range}(A^T)$ and $\text{null}(A^T)$ play just as important
johnston-linear-matrix-augmented-F01699	167	0.977	$\text{null}(A^T)$	$\text{null}(A^T)$	It turns out that the subspaces $\text{range}(A^T)$ and $\text{null}(A^T)$ play just as important
johnston-linear-matrix-augmented-F01700	168	0.719	(A^T)	(A^T)	why $\text{range}(A^T)$ and
johnston-linear-matrix-augmented-F01701	168	1.000	$\{(0,-1,1,1)\}$	$\{(0,-1,1,1)\}$	so $\{(0,-1,1,1)\}$ is a basis of $\text{null}(A^T)$.

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johnston-linear-matrix-algebra-F01702	169	■ 1.000	$\operatorname{row}(A)$	$\operatorname{row}(A)$	$\operatorname{row}(A)$.
johnston-linear-matrix-algebra-F01703	169	■ 1.000	$\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_m$	$\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_m$	Suppose $A \in \mathcal{M}_{m,n}$ has rows $\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_m$. Then
johnston-linear-matrix-algebra-F01704	169	■ 1.000	$\operatorname{range}(A^T) = \operatorname{range}(R^T)$	$\operatorname{range}(A^T) = \operatorname{range}(R^T)$	It follows that $\operatorname{range}(A^T) = \operatorname{range}(R^T)$, which is the span of the rows of R . Since
johnston-linear-matrix-algebra-F01705	169	■ 0.999	$A^T E^T = R^T$	$A^T E^T = R^T$	Taking the transpose of both sides of that equation gives $A^T E^T = R^T$. If we
johnston-linear-matrix-algebra-F01706	169	■ 1.000	E^T	E^T	write E^T in terms of its columns $\mathbf{v}_1, \dots, \mathbf{v}_m$ (i.e., the rows of E), then we see
johnston-linear-matrix-algebra-F01707	169	■ 1.000	$\mathbf{v}_1, \dots, \mathbf{v}_m$	$\mathbf{v}_1, \dots, \mathbf{v}_m$	write E^T in terms of its columns $\mathbf{v}_1, \dots, \mathbf{v}_m$ (i.e., the rows of E), then we see
johnston-linear-matrix-algebra-F01708	169	■ 1.000	R^T	R^T	k columns of R^T are $\mathbf{0}$ as well, so $A^T \mathbf{v}_{m-k+1} = A^T \mathbf{v}_{m-k+2} = \dots = A^T \mathbf{v}_m = \mathbf{0}$.
johnston-linear-matrix-algebra-F01709	169	■ 1.000	$A^T \mathbf{v}_{m-k+1} = A^T \mathbf{v}_{m-k+2} = \dots = A^T \mathbf{v}_m = \mathbf{0}$	$A^T \mathbf{v}_{m-k+1} = A^T \mathbf{v}_{m-k+2} = \dots = A^T \mathbf{v}_m = \mathbf{0}$	k columns of R^T are $\mathbf{0}$ as well, so $A^T \mathbf{v}_{m-k+1} = A^T \mathbf{v}_{m-k+2} = \dots = A^T \mathbf{v}_m = \mathbf{0}$.
johnston-linear-matrix-algebra-F01710	170	■ 1.000	$A = \begin{bmatrix} 1 & 1 & 1 & -1 \\ 0 & 1 & 1 & 0 \\ -1 & 1 & 1 & 1 \end{bmatrix}$	$A = \begin{bmatrix} 1 & 1 & 1 & -1 \\ 0 & 1 & 1 & 0 \\ -1 & 1 & 1 & 1 \end{bmatrix}$	Find bases of the four fundamental subspaces of $A = \begin{bmatrix} 1 & 1 & 1 & -1 \\ 0 & 1 & 1 & 0 \\ -1 & 1 & 1 & 1 \end{bmatrix}$.
johnston-linear-matrix-algebra-F01711	170	■ 1.000	$\operatorname{range}(A) : \{(1, 0, -1), (1, 1, 1)\}$	$\operatorname{range}(A) : \{(1, 0, -1), (1, 1, 1)\}$	• $\operatorname{range}(A) : \{(1, 0, -1), (1, 1, 1)\}$,
johnston-linear-matrix-algebra-F01712	170	■ 1.000	$\operatorname{null}(A^T) : \{(1, -2, 1)\}$	$\operatorname{null}(A^T) : \{(1, -2, 1)\}$	• $\operatorname{null}(A^T) : \{(1, -2, 1)\}$, and
johnston-linear-matrix-algebra-F01713	170	■ 1.000	$\operatorname{range}(A^T) : \{(1, 0, 0, -1), (0, 1, 1, 0)\}$	$\operatorname{range}(A^T) : \{(1, 0, 0, -1), (0, 1, 1, 0)\}$	• $\operatorname{range}(A^T) : \{(1, 0, 0, -1), (0, 1, 1, 0)\}$.
johnston-linear-matrix-algebra-F01714	170	■ 1.000	$x_1 - x_4 = 0$	$x_1 - x_4 = 0$	Furthermore, $x_1 - x_4 = 0$ and $x_2 + x_3 = 0$, so writing the solution in vector
johnston-linear-matrix-algebra-F01715	170	■ 0.995	$\{(0, -1, 1, 0), (1, 0, 0, 1)\}$	$\{(0, -1, 1, 0), (1, 0, 0, 1)\}$	It follows that one basis of $\operatorname{null}(A)$ is $\{(0, -1, 1, 0), (1, 0, 0, 1)\}$.
johnston-linear-matrix-algebra-F01716	171	■ 1.000	$\operatorname{range}(A) = \mathbb{R}^n$	$\operatorname{range}(A) = \mathbb{R}^n$	b) $\operatorname{range}(A) = \mathbb{R}^n$.
johnston-linear-matrix-algebra-F01717	171	■ 1.000	$\operatorname{null}(A) = \{\mathbf{0}\}$	$\operatorname{null}(A) = \{\mathbf{0}\}$	c) $\operatorname{null}(A) = \{\mathbf{0}\}$.
johnston-linear-matrix-algebra-F01718	171	■ 0.990	$\left(A^T\right)=\mathbb{R}^n$	$(A^T) = \mathbb{R}^n$	We could also add “ $\operatorname{range}(A^T) = \mathbb{R}^n$ ” and “ $\operatorname{null}(A^T) = \{\mathbf{0}\}$ ” as additional
johnston-linear-matrix-algebra-F01719	171	■ 0.990	$\left(A^T\right)=\{\mathbf{0}\}$	$(A^T) = \{\mathbf{0}\}$	We could also add “ $\operatorname{range}(A^T) = \mathbb{R}^n$ ” and “ $\operatorname{null}(A^T) = \{\mathbf{0}\}$ ” as additional
johnston-linear-matrix-algebra-F01720	171	■ 0.531	$\operatorname{rank}(A)$	$\operatorname{rank}(A)$	Suppose A is a matrix. Its rank , denoted by $\operatorname{rank}(A)$, is the dimension of
johnston-linear-matrix-algebra-F01721	171	■ 1.000	$A = \mathbf{u} \mathbf{u}^T$	$A = \mathbf{u} \mathbf{u}^T$	For example, if $\mathbf{u} \in \mathbb{R}^n$ is a unit vector and $A = \mathbf{u} \mathbf{u}^T$ is the standard matrix
johnston-linear-matrix-algebra-F01722	171	■ 0.823	$\operatorname{rank}(A) = 1$	$\operatorname{rank}(A) = 1$	1-dimensional, so $\operatorname{rank}(A) = 1$. Similarly, we showed that the matrix
johnston-linear-matrix-algebra-F01723	171	■ 0.406	$((1, 2, 1), (1, -1, 0))$	$((1, 2, 1), (1, -1, 0))$	from Example 2.4.5 has range equal to $\operatorname{span}((1, 2, 1), (1, -1, 0))$. Since this
johnston-linear-matrix-algebra-F01724	172	■ 1.000	$\operatorname{rank}(A) \leq \min\{m, n\}$	$\operatorname{rank}(A) \leq \min\{m, n\}$	$\operatorname{rank}(A) \leq \min\{m, n\}$
johnston-linear-matrix-algebra-F01725	172	■ 1.000	$\operatorname{rank}(A) = \min\{m, n\}$	$\operatorname{rank}(A) = \min\{m, n\}$	$\operatorname{rank}(A) = \min\{m, n\}$.
johnston-linear-matrix-algebra-F01726	172	■ 0.994	$\operatorname{rank}(A) \leq n$	$\operatorname{rank}(A) \leq n$	we conclude that A is invertible if and only if its rank is n . Since $\operatorname{rank}(A) \leq n$ for

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johnston-linear-matrix-algebra F01727	172	1.000	$\operatorname{rank}(A^T)$	$\operatorname{rank}(A^T)$	b) $\operatorname{rank}(A^T)$
johnston-linear-matrix-algebra F01728	172	1.000	$C=\left[\begin{array}{ccccc}0 & 0 & -2 & 2 & -2 \\ 2 & -2 & -1 & 3 & 3 \\ -1 & 1 & -1 & 0 & -3\end{array}\right]$	$C = \begin{bmatrix} 0 & 0 & -2 & 2 & -2 \\ 2 & -2 & -1 & 3 & 3 \\ -1 & 1 & -1 & 0 & -3 \end{bmatrix}$	c) $C = \begin{bmatrix} 0 & 0 & -2 & 2 & -2 \\ 2 & -2 & -1 & 3 & 3 \\ -1 & 1 & -1 & 0 & -3 \end{bmatrix}$
johnston-linear-matrix-algebra F01729	173	1.000	$\operatorname{rank}(A)=n$	$\operatorname{rank}(A) = n$	that $\operatorname{rank}(A) = n$.
johnston-linear-matrix-algebra F01730	173	1.000	$\operatorname{rank}(B)=2$	$\operatorname{rank}(B) = 2$	dimensional. It follows that $\operatorname{rank}(B) = 2$.
johnston-linear-matrix-algebra F01731	173	1.000	$\operatorname{rank}(C)=2$	$\operatorname{rank}(C) = 2$	$\operatorname{rank}(C) = 2$.
johnston-linear-matrix-algebra F01732	173	1.000	$\operatorname{nullity}(A)=0$	$\operatorname{nullity}(A) = 0$	c) $\operatorname{nullity}(A) = 0$.
johnston-linear-matrix-algebra F01733	173	1.000	$\operatorname{rank}(A)+\operatorname{nullity}(A)=n$	$\operatorname{rank}(A) + \operatorname{nullity}(A) = n$	Suppose $A \in \mathcal{M}_{m,n}$ is a matrix. Then $\operatorname{rank}(A) + \operatorname{nullity}(A) = n$.
johnston-linear-matrix-algebra F01734	173	1.000	$r=\operatorname{rank}(A)$	$r = \operatorname{rank}(A)$	Let $r = \operatorname{rank}(A)$ and consider the linear system $A\mathbf{x} = \mathbf{0}$, which has r leading
johnston-linear-matrix-algebra F01735	173	1.000	r	r	Let $r = \operatorname{rank}(A)$ and consider the linear system $A\mathbf{x} = \mathbf{0}$, which has r leading
johnston-linear-matrix-algebra F01736	173	1.000	$n-r$	$n - r$	variables. Since this linear system has n variables total, it must have $n - r$ free
johnston-linear-matrix-algebra F01737	173	0.996	$\operatorname{nullity}(A)=\operatorname{dim}(\operatorname{null}(A))$ $\operatorname{nullity}(A)=n-r$	$\operatorname{nullity}(A) = \operatorname{dim}(\operatorname{null}(A)) = n - r$	of a basis of $\operatorname{null}(A)$, so $\operatorname{nullity}(A) = \operatorname{dim}(\operatorname{null}(A)) = n - r$. It follows that
johnston-linear-matrix-algebra F01738	174	1.000	$\operatorname{rank}(A)=3$	$\operatorname{rank}(A) = 3$	$\operatorname{rank}(A) = 3$ and $\operatorname{nullity}(A) = 5 - 3 = 2$.
johnston-linear-matrix-algebra F01739	174	1.000	$\operatorname{nullity}(A)=5-3=2$	$\operatorname{nullity}(A) = 5 - 3 = 2$	$\operatorname{rank}(A) = 3$ and $\operatorname{nullity}(A) = 5 - 3 = 2$.
johnston-linear-matrix-algebra F01740	174	1.000	$\mathbb{R}^5$	$\mathbb{R}^5$	transformation from $\mathbb{R}^5$ to $\mathbb{R}^4$. Since its rank is 3, it sends a 3-dimensional
johnston-linear-matrix-algebra F01741	175	1.000	$\operatorname{rank}(AB)=\operatorname{rank}(A)$	$\operatorname{rank}(AB) = \operatorname{rank}(A)$	$\operatorname{rank}(AB) = \operatorname{rank}(A)$
johnston-linear-matrix-algebra F01742	175	1.000	$\sum_{j=1}^n$	$\sum_{j=1}^n$	The notation $\sum_{j=1}^n$
johnston-linear-matrix-algebra F01743	175	1.000	$\sum_{j=1}^n b_{j,1} \mathbf{a}_j =$	$\sum_{j=1}^n b_{j,1} \mathbf{a}_j =$	$\sum_{j=1}^n b_{j,1} \mathbf{a}_j =$
johnston-linear-matrix-algebra F01744	175	1.000	$b_{1,1} \mathbf{a}_1 + \cdots + b_{n,1} \mathbf{a}_n$	$b_{1,1} \mathbf{a}_1 + \cdots + b_{n,1} \mathbf{a}_n$	$b_{1,1} \mathbf{a}_1 + \cdots + b_{n,1} \mathbf{a}_n$.
johnston-linear-matrix-algebra F01745	175	1.000	$A \in \mathcal{M}_{3,2}$	$A \in \mathcal{M}_{3,2}$	Figure 2.25: A non-zero matrix $A \in \mathcal{M}_{3,2}$ either has rank 1 and nullity 1 (as in (a)) or
johnston-linear-matrix-algebra F01746	175	1.000	$\operatorname{rank}(A+B)=\operatorname{rank}(A)+\operatorname{rank}(B)$	$\operatorname{rank}(A+B) = \operatorname{rank}(A) + \operatorname{rank}(B)$	it is not the case that $\operatorname{rank}(A+B) = \operatorname{rank}(A) + \operatorname{rank}(B)$, nor is it the case that
johnston-linear-matrix-algebra F01747	175	0.813	$\operatorname{rank}(AB)=\operatorname{rank}(A) \times \operatorname{rank}(B)$	$\operatorname{rank}(AB) = \operatorname{rank}(A) \times \operatorname{rank}(B)$	$\operatorname{rank}(AB) = \operatorname{rank}(A) \times \operatorname{rank}(B)$). However, there are some <i>inequalities</i> that at
johnston-linear-matrix-algebra F01748	175	0.994	$\operatorname{rank}(A+B)$	$\operatorname{rank}(A+B)$	least loosely relate $\operatorname{rank}(A+B)$ and $\operatorname{rank}(AB)$ to the ranks of A and B :
johnston-linear-matrix-algebra F01749	175	0.994	$\operatorname{rank}(AB)$	$\operatorname{rank}(AB)$	least loosely relate $\operatorname{rank}(A+B)$ and $\operatorname{rank}(AB)$ to the ranks of A and B :
johnston-linear-matrix-algebra F01750	175	1.000	$\operatorname{rank}(A+B) \leq \operatorname{rank}(A)+\operatorname{rank}(B)$	$\operatorname{rank}(A+B) \leq \operatorname{rank}(A) + \operatorname{rank}(B)$	a) $\operatorname{rank}(A+B) \leq \operatorname{rank}(A) + \operatorname{rank}(B)$
johnston-linear-matrix-algebra F01751	175	1.000	$\operatorname{rank}(AB) \leq \min \{ \operatorname{rank}(A), \operatorname{rank}(B) \}$	$\operatorname{rank}(AB) \leq \min \{ \operatorname{rank}(A), \operatorname{rank}(B) \}$	b) $\operatorname{rank}(AB) \leq \min \{ \operatorname{rank}(A), \operatorname{rank}(B) \}$
johnston-linear-matrix-algebra F01752	175	0.996	$\mathbf{a}_1, \mathbf{a}_2, \ldots$	$\mathbf{a}_1, \mathbf{a}_2, \ldots$	<i>Proof.</i> Denote the columns of A and B by $\mathbf{a}_1, \mathbf{a}_2, \ldots$ and $\mathbf{b}_1, \mathbf{b}_2, \ldots$, respectively.

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johnston-linear-matrix-algebra-F01753	175	■ 0.996	$\mathbf{b}_1, \mathbf{b}_2, \ldots$	$\mathbf{b}_1, \mathbf{b}_2, \ldots$	<i>Proof.</i> Denote the columns of A and B by $\mathbf{a}_1, \mathbf{a}_2, \ldots$ and $\mathbf{b}_1, \mathbf{b}_2, \ldots$, respectively.
johnston-linear-matrix-algebra-F01754	175	■ 1.000	$\mathbf{a}_1 + \mathbf{b}_1, \mathbf{a}_2 + \mathbf{b}_2, \ldots, \mathbf{a}_n + \mathbf{b}_n$	$\mathbf{a}_1 + \mathbf{b}_1, \mathbf{a}_2 + \mathbf{b}_2, \ldots, \mathbf{a}_n + \mathbf{b}_n$	of $A + B$ are $\mathbf{a}_1 + \mathbf{b}_1, \mathbf{a}_2 + \mathbf{b}_2, \ldots, \mathbf{a}_n + \mathbf{b}_n$, which are each contained within the
johnston-linear-matrix-algebra-F01755	175	■ 0.999	$\mathbf{a}_1, \mathbf{b}_1, \mathbf{a}_2, \mathbf{b}_2, \ldots, \mathbf{a}_n, \mathbf{b}_n$	$\mathbf{a}_1, \mathbf{b}_1, \mathbf{a}_2, \mathbf{b}_2, \ldots, \mathbf{a}_n, \mathbf{b}_n$	span of $\mathbf{a}_1, \mathbf{b}_1, \mathbf{a}_2, \mathbf{b}_2, \ldots, \mathbf{a}_n, \mathbf{b}_n$. It follows that
johnston-linear-matrix-algebra-F01756	175	■ 0.955	$\operatorname{range}(A+B)$	$\operatorname{range}(A+B)$	as well. The span on the left equals $\operatorname{range}(A+B)$, so its dimension is $\operatorname{rank}(A+$
johnston-linear-matrix-algebra-F01757	175	■ 0.955	$\operatorname{rank}(A+$	$\operatorname{rank}(A+$	as well. The span on the left equals $\operatorname{range}(A+B)$, so its dimension is $\operatorname{rank}(A+$
johnston-linear-matrix-algebra-F01758	175	■ 0.863	$\operatorname{rank}(A) + \operatorname{rank}(B)$	$\operatorname{rank}(A) + \operatorname{rank}(B)$	$B)$. The span on the right cannot have dimension larger than $\operatorname{rank}(A) + \operatorname{rank}(B)$,
johnston-linear-matrix-algebra-F01759	176	■ 1.000	$\operatorname{range}(A\ B) \subseteq \operatorname{range}(A)$	$\operatorname{range}(AB) \subseteq \operatorname{range}(A)$	$\operatorname{range}(AB) \subseteq \operatorname{range}(A)$
johnston-linear-matrix-algebra-F01760	176	■ 0.998	$\operatorname{rank}(A\ B) \leq \operatorname{rank}(A)$	$\operatorname{rank}(AB) \leq \operatorname{rank}(A)$	the dimension of its range, it follows that $\operatorname{rank}(AB) \leq \operatorname{rank}(A)$. To see that
johnston-linear-matrix-algebra-F01761	176	■ 1.000	$\operatorname{rank}(A\ B) \leq \operatorname{rank}(B)$	$\operatorname{rank}(AB) \leq \operatorname{rank}(B)$	$\operatorname{rank}(AB) \leq \operatorname{rank}(B)$ as well (thus completing the proof), we could apply this
johnston-linear-matrix-algebra-F01762	176	■ 0.999	$B = \{(1, 1), (1, 2), (3, 4)\}, \mathcal{S} = \mathbb{R}^2$	$B = \{(1, 1), (1, 2), (3, 4)\}, \mathcal{S} = \mathbb{R}^2$	*(a) $B = \{(1, 1), (1, 2), (3, 4)\}, \mathcal{S} = \mathbb{R}^2$
johnston-linear-matrix-algebra-F01763	176	■ 1.000	$B = \{(1, 1), (1, -1)\}, \mathcal{S} = \mathbb{R}^2$	$B = \{(1, 1), (1, -1)\}, \mathcal{S} = \mathbb{R}^2$	(b) $B = \{(1, 1), (1, -1)\}, \mathcal{S} = \mathbb{R}^2$
johnston-linear-matrix-algebra-F01764	176	■ 0.963	$B = \{(1, 1, 1), (2, 1, -1)\}, \mathcal{S} = \mathbb{R}^3$	$B = \{(1, 1, 1), (2, 1, -1)\}, \mathcal{S} = \mathbb{R}^3$	*(c) $B = \{(1, 1, 1), (2, 1, -1)\}, \mathcal{S} = \mathbb{R}^3$
johnston-linear-matrix-algebra-F01765	176	■ 0.993	$B = \{(2, 2, 1), (0, 1, 1)\}, \mathcal{S}$	$B = \{(2, 2, 1), (0, 1, 1)\}, \mathcal{S}$	(d) $B = \{(2, 2, 1), (0, 1, 1)\}, \mathcal{S}$ is the plane in $\mathbb{R}^3$ with
johnston-linear-matrix-algebra-F01766	176	■ 1.000	$x+y-z=0$	$x+y-z=0$	equation $x+y-z=0$.
johnston-linear-matrix-algebra-F01767	176	■ 0.997	$B = \{(1, 0, 1), (2, 1, 3)\}, \mathcal{S}$	$B = \{(1, 0, 1), (2, 1, 3)\}, \mathcal{S}$	*(e) $B = \{(1, 0, 1), (2, 1, 3)\}, \mathcal{S}$ is the plane in $\mathbb{R}^3$ with
johnston-linear-matrix-algebra-F01768	176	■ 1.000	$B = \{(1, 3, 2), (-1, -2, 1), (1, 4, 5)\}, \mathcal{S} = \operatorname{span}(B)$	$B = \{(1, 3, 2), (-1, -2, 1), (1, 4, 5)\}, \mathcal{S} = \operatorname{span}(B)$	(f) $B = \{(1, 3, 2), (-1, -2, 1), (1, 4, 5)\}, \mathcal{S} = \operatorname{span}(B)$.
johnston-linear-matrix-algebra-F01769	176	■ 0.958	$2\ x+y+z=0$	$2x+y+z=0$	*(c) The plane in $\mathbb{R}^3$ with equation $2x+y+z=0$.
johnston-linear-matrix-algebra-F01770	176	■ 1.000	$z=4\ x-3\ y$	$z=4x-3y$	(d) The plane in $\mathbb{R}^3$ with equation $z=4x-3y$.
johnston-linear-matrix-algebra-F01771	176	■ 1.000	$2\ x-y+2\ z=0$	$2x-y+2z=0$	$x+y-z=0$ and $2x-y+2z=0$.
johnston-linear-matrix-algebra-F01772	176	■ 1.000	$\operatorname{span}(\{(1, 3, 2, 1), (-1, -2, 1, 0), (1, 4, -1, 1)\})$	$\operatorname{span}(\{(1, 3, 2, 1), (-1, -2, 1, 0), (1, 4, -1, 1)\})$	(f) $\operatorname{span}(\{(1, 3, 2, 1), (-1, -2, 1, 0), (1, 4, -1, 1)\})$
johnston-linear-matrix-algebra-F01773	176	■ 1.000	$\{(5, 5, 5, 1), (3, 6, 3, 2), (4, 1, 4, 4), (1, 4, 1, 4)\}$	$\{(5, 5, 5, 1), (3, 6, 3, 2), (4, 1, 4, 4), (1, 4, 1, 4)\}$	(b) $\{(5, 5, 5, 1), (3, 6, 3, 2), (4, 1, 4, 4), (1, 4, 1, 4)\}$
johnston-linear-matrix-algebra-F01774	176	■ 0.942	$\{(1, 1, 4, 5, 1), (2, -7, 4, 2, 2), (5, 2, 4, 5, 3)\}$	$\{(1, 1, 4, 5, 1), (2, -7, 4, 2, 2), (5, 2, 4, 5, 3)\}$	*(c) $\{(1, 1, 4, 5, 1), (2, -7, 4, 2, 2), (5, 2, 4, 5, 3),$
johnston-linear-matrix-algebra-F01775	176	■ 1.000	$\{(3, 4, -1, 1, 3), (-1, 1, 3, 0, 1), (4, 6, -5, 3, 6)\}$	$\{(3, 4, -1, 1, 3), (-1, 1, 3, 0, 1), (4, 6, -5, 3, 6)\}$	(d) $\{(3, 4, -1, 1, 3), (-1, 1, 3, 0, 1), (4, 6, -5, 3, 6),$
johnston-linear-matrix-algebra-F01776	176	■ 0.811	$\{(2, 3, 1, 1), (3, 3, 2, 3), (2, 1, 2, 3)\}$	$\{(2, 3, 1, 1), (3, 3, 2, 3), (2, 1, 2, 3)\}$	*(a) $\{(2, 3, 1, 1), (3, 3, 2, 3), (2, 1, 2, 3)\}$
johnston-linear-matrix-algebra-F01777	176	■ 0.993	$\{(3, 2, 0, 2), (0, 1, 3, 3), (2, 1, 1, 1), (0, 2, 3, 2)\}$	$\{(3, 2, 0, 2), (0, 1, 3, 3), (2, 1, 1, 1), (0, 2, 3, 2)\}$	(b) $\{(3, 2, 0, 2), (0, 1, 3, 3), (2, 1, 1, 1), (0, 2, 3, 2)\}$
johnston-linear-matrix-algebra-F01778	176	■ 1.000	$\{(2, 4, 3, 4), (3, 2, -1, -1)\}$	$\{(2, 4, 3, 4), (3, 2, -1, -1)\}$	*(c) $\{(2, 4, 3, 4), (3, 2, -1, -1)\}$
johnston-linear-matrix-algebra-F01779	176	■ 1.000	$\{(-1, 3, 6, 3), (1, 3, 5, 0), (3, 0, -1, -3)\}$	$\{(-1, 3, 6, 3), (1, 3, 5, 0), (3, 0, -1, -3)\}$	(d) $\{(-1, 3, 6, 3), (1, 3, 5, 0), (3, 0, -1, -3)\}$
johnston-linear-matrix-algebra-F01780	176	■ 1.000	$\operatorname{range}(A), \operatorname{null}(A), \operatorname{range}(A^T), \operatorname{null}(A^T)$	$\operatorname{range}(A), \operatorname{null}(A), \operatorname{range}(A^T)$	each of $\operatorname{range}(A), \operatorname{null}(A), \operatorname{range}(A^T)$, and $\operatorname{null}(A^T)$.

Identifier	Page	Conf.	LaTeX source	Rendered	Image
johnston-linear-matrix-algebra-F01781	176	1.000	$\left[\begin{array}{lll}0 & 0 & 0 \\ 0 & 1 & 2 \\ 0 & 0 & 0\end{array}\right]$	$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix}$	$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix}$
johnston-linear-matrix-algebra-F01782	176	1.000	$\left[\begin{array}{ccccc}1 & 2 & 0 & 3 & 0 \\ 0 & 0 & 1 & -1 & 0 \\ -1 & 0 & 0 & 0 & 1\end{array}\right]$	$\begin{bmatrix} 1 & 2 & 0 & 3 & 0 \\ 0 & 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} 1 & 2 & 0 & 3 & 0 \\ 0 & 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$
johnston-linear-matrix-algebra-F01783	176	1.000	$\left[\begin{array}{ccccc}0 & -4 & 0 & 2 & 1 \\ -1 & 2 & 1 & 2 & 1 \\ 1 & 2 & 1 & -2 & 0 \\ 2 & 2 & 6 & 3\end{array}\right]$	$\begin{bmatrix} 0 & -4 & 0 & 2 & 1 \\ -1 & 2 & 1 & 2 & 1 \\ -2 & 0 & 2 & 6 & 3 \end{bmatrix}$	$\begin{bmatrix} 0 & -4 & 0 & 2 & 1 \\ -1 & 2 & 1 & 2 & 1 \\ -2 & 0 & 2 & 6 & 3 \end{bmatrix}$
johnston-linear-matrix-algebra-F01784	176	1.000	$\left[\begin{array}{ccccc}2 & -1 & 2 & 3 & 5 \\ -1 & 1 & -2 & -2 & -2 \\ 1 & -1 & 2 & 1 & 2 \\ 1 & -2 & 4 & 2 & 1\end{array}\right]$	$\begin{bmatrix} 2 & -1 & 2 & 3 & 5 \\ -1 & 1 & -2 & -2 & -2 \\ 1 & -1 & 2 & 1 & 2 \\ 1 & -2 & 4 & 2 & 1 \end{bmatrix}$	$\begin{bmatrix} 2 & -1 & 2 & 3 & 5 \\ -1 & 1 & -2 & -2 & -2 \\ 1 & -1 & 2 & 1 & 2 \\ 1 & -2 & 4 & 2 & 1 \end{bmatrix}$
johnston-linear-matrix-algebra-F01785	176	0.651	$\left[\begin{array}{cc}1 & -1 \\ -1 & 1\end{array}\right]$	$\begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$	$\ast(\mathbf{a}) \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$
johnston-linear-matrix-algebra-F01786	176	1.000	$\left[\begin{array}{cccc}5 & 4 & 8 & 1 \\ 3 & -2 & 7 & 2\end{array}\right]$	$\begin{bmatrix} 5 & 4 & 8 & 1 \\ 3 & -2 & 7 & 2 \end{bmatrix}$	$\begin{bmatrix} 5 & 4 & 8 & 1 \\ 3 & -2 & 7 & 2 \end{bmatrix}$
johnston-linear-matrix-algebra-F01787	176	1.000	$\left[\begin{array}{ll}1 & 0 \\ 0 & 2 \\ 3 & 1\end{array}\right]$	$\begin{bmatrix} 1 & 0 \\ 0 & 2 \\ 3 & 1 \end{bmatrix}$	$\begin{bmatrix} 1 & 0 \\ 0 & 2 \\ 3 & 1 \end{bmatrix}$
johnston-linear-matrix-algebra-F01788	176	1.000	$\left[\begin{array}{cccc}1 & 2 & 1 & 0 \\ 2 & 1 & 2 & -3 \\ 0 & 3 & 1 & 2 \\ 0 & 0 & 2 & -2\end{array}\right]$	$\begin{bmatrix} 1 & 2 & 1 & 0 \\ 2 & 1 & 2 & -3 \\ 0 & 3 & 1 & 2 \\ 0 & 0 & 2 & -2 \end{bmatrix}$	$\begin{bmatrix} 1 & 2 & 1 & 0 \\ 2 & 1 & 2 & -3 \\ 0 & 3 & 1 & 2 \\ 0 & 0 & 2 & -2 \end{bmatrix}$
johnston-linear-matrix-algebra-F01789	177	0.902	$A \in \mathcal{M}_{3,7}$	$A \in \mathcal{M}_{3,7}$	$(\mathbf{h})$ If $A \in \mathcal{M}_{3,7}$ then $\text{rank}(A) \leq 3$.
johnston-linear-matrix-algebra-F01790	177	0.902	$(A) \leq 3$	$(A) \leq 3$	$(\mathbf{h})$ If $A \in \mathcal{M}_{3,7}$ then $\text{rank}(A) \leq 3$.
johnston-linear-matrix-algebra-F01791	177	1.000	$\text{rank}(A \cdot B) = \text{rank}(A) \cdot \text{rank}(B)$	$\text{rank}(AB) = \text{rank}(A) \cdot \text{rank}(B)$	$\text{rank}(AB) = \text{rank}(A) \cdot \text{rank}(B)$.
johnston-linear-matrix-algebra-F01792	177	1.000	$\text{rank}(A \cdot B) = \min\{\text{rank}(A), \text{rank}(B)\}$	$\text{rank}(AB) = \min\{\text{rank}(A), \text{rank}(B)\}$	$\text{rank}(AB) = \min\{\text{rank}(A), \text{rank}(B)\}$.
johnston-linear-matrix-algebra-F01793	177	1.000	$\text{null}(A) =$	$\text{null}(A) =$	$(\mathbf{n})$ If $A, B \in \mathcal{M}_{m,n}$ are row equivalent then $\text{null}(A) =$
johnston-linear-matrix-algebra-F01794	177	1.000	$\text{null}(B)$	$\text{null}(B)$	$\text{null}(B)$.
johnston-linear-matrix-algebra-F01795	177	0.603	$\text{nullity}(A) = \text{nullity}(A^T)$	$\text{nullity}(A) = \text{nullity}(A^T)$	$\ast(\mathbf{o})$ If $A \in \mathcal{M}_{m,n}$ then $\text{nullity}(A) = \text{nullity}(A^T)$.
johnston-linear-matrix-algebra-F01796	177	1.000	$A \in \mathcal{M}_4$	$A \in \mathcal{M}_4$	$(\mathbf{a})$ Find a matrix $A \in \mathcal{M}_4$ whose range is the same as
johnston-linear-matrix-algebra-F01797	177	1.000	$A \in \mathcal{M}_3$	$A \in \mathcal{M}_3$	$(\mathbf{b})$ Show that there does not exist a matrix $A \in \mathcal{M}_3$
johnston-linear-matrix-algebra-F01798	177	0.715	$\text{rank}(J_n)$	$\text{rank}(J_n)$	1. Compute $\text{rank}(J_n)$.
johnston-linear-matrix-algebra-F01799	177	1.000	$A \mathbf{v} \neq B \mathbf{v}$	$A \mathbf{v} \neq B \mathbf{v}$	erty that $A \mathbf{v} \neq B \mathbf{v}$ for all $\mathbf{v} \neq \mathbf{0}$. What is the rank of the
johnston-linear-matrix-algebra-F01800	177	1.000	$\mathbf{v} \neq \mathbf{0}$	$\mathbf{v} \neq \mathbf{0}$	erty that $A \mathbf{v} \neq B \mathbf{v}$ for all $\mathbf{v} \neq \mathbf{0}$. What is the rank of the

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johnston-linear-matrix-algebra-F01801	177	0.884	$\operatorname{rank}(A_n)=2$	$\operatorname{rank}(A_n) = 2$	Show that $\operatorname{rank}(A_n) = 2$ for all $n \geq 2$. [Hint: Write each row
johnston-linear-matrix-algebra-F01802	177	0.884	$n \geq 2$	$n \geq 2$	Show that $\operatorname{rank}(A_n) = 2$ for all $n \geq 2$. [Hint: Write each row
johnston-linear-matrix-algebra-F01803	177	0.814	$\left[\begin{array}{ll}1 & x \\ 2 & 4\end{array}\right]$	$\begin{bmatrix} 1 & x \\ 2 & 4 \end{bmatrix}$	(a) $\begin{bmatrix} 1 & x \\ 2 & 4 \end{bmatrix}$
johnston-linear-matrix-algebra-F01804	177	1.000	$\left[\begin{array}{cc}x-1 & -1 \\ 1 & 1-x\end{array}\right]$	$\begin{bmatrix} x-1 & -1 \\ 1 & 1-x \end{bmatrix}$	(b) $\begin{bmatrix} x-1 & -1 \\ 1 & 1-x \end{bmatrix}$
johnston-linear-matrix-algebra-F01805	177	1.000	$\left[\begin{array}{ccc}1 & 2 & x \\ 2 & 3-x & 3x+1 \\ -x & -2x & -1\end{array}\right]$	$\begin{bmatrix} 1 & 2 & x \\ 2 & 3-x & 3x+1 \\ -x & -2x & -1 \end{bmatrix}$	$\begin{bmatrix} 1 & 2 & x \\ 2 & 3-x & 3x+1 \\ -x & -2x & -1 \end{bmatrix}$
johnston-linear-matrix-algebra-F01806	177	0.987	$\left[\begin{array}{lll}1 & 1 & 1 \\ 1 & 2 & 2 \\ 1 & 2 & x\end{array}\right]$	$\begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 2 \\ 1 & 2 & x \end{bmatrix}$	(d) $\begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 2 \\ 1 & 2 & x \end{bmatrix}$
johnston-linear-matrix-algebra-F01807	177	0.931	$\mid \operatorname{rank}(A) -$	$\mid \operatorname{rank}(A) -$	of their other entries are the same) then $\mid \operatorname{rank}(A) -$
johnston-linear-matrix-algebra-F01808	177	0.967	$\operatorname{rank}(B) \mid \leq 1$	$\operatorname{rank}(B) \mid \leq 1$	$\operatorname{rank}(B) \mid \leq 1$. That is, show that $\operatorname{rank}(A)$ and $\operatorname{rank}(B)$
johnston-linear-matrix-algebra-F01809	177	0.967	$\operatorname{rank}(B)$	$\operatorname{rank}(B)$	$\operatorname{rank}(B) \mid \leq 1$. That is, show that $\operatorname{rank}(A)$ and $\operatorname{rank}(B)$
johnston-linear-matrix-algebra-F01810	177	1.000	$\mid \operatorname{rank}(A) - \operatorname{rank}(B) \mid \leq k$	$\mid \operatorname{rank}(A) - \operatorname{rank}(B) \mid \leq k$	k entries then $\mid \operatorname{rank}(A) - \operatorname{rank}(B) \mid \leq k$.
johnston-linear-matrix-algebra-F01811	177	1.000	$x=1$	$x = 1$	(a) Show that if $x = 1$ then $\operatorname{rank}(A) = 1$.
johnston-linear-matrix-algebra-F01812	177	1.000	$x \neq 1$	$x \neq 1$	(b) Show that if $x \neq 1$ then $\operatorname{rank}(A) = n$.
johnston-linear-matrix-algebra-F01813	177	1.000	$\operatorname{rank}(A B)=\operatorname{rank}(B)$	$\operatorname{rank}(AB) = \operatorname{rank}(B)$	$\operatorname{rank}(AB) = \operatorname{rank}(B)$. [Hint: Try to make use of The-
johnston-linear-matrix-algebra-F01814	177	1.000	$\operatorname{rank}(A B)=$	$\operatorname{rank}(AB) =$	(b) Show that if $B \in \mathcal{M}_n$ is invertible then $\operatorname{rank}(AB) =$
johnston-linear-matrix-algebra-F01815	177	1.000	$\operatorname{rank}(A) \leq n / 2$	$\operatorname{rank}(A) \leq n / 2$	$A^2 = O$. Show that $\operatorname{rank}(A) \leq n / 2$.
johnston-linear-matrix-algebra-F01816	178	1.000	$\mathcal{S}_1$	$\mathcal{S}_1$	2.4.20 Suppose that $\mathcal{S}_1$ and $\mathcal{S}_2$ are subspaces of $\mathbb{R}^n$ with
johnston-linear-matrix-algebra-F01817	178	1.000	$\mathcal{S}_2$	$\mathcal{S}_2$	2.4.20 Suppose that $\mathcal{S}_1$ and $\mathcal{S}_2$ are subspaces of $\mathbb{R}^n$ with
johnston-linear-matrix-algebra-F01818	178	1.000	$\mathcal{S}_1 \subseteq \mathcal{S}_2$	$\mathcal{S}_1 \subseteq \mathcal{S}_2$	the property that $\mathcal{S}_1 \subseteq \mathcal{S}_2$. Show that $\dim(\mathcal{S}_1) \leq \dim(\mathcal{S}_2)$.
johnston-linear-matrix-algebra-F01819	178	1.000	$\operatorname{dim}(\mathcal{S}_1) \leq \operatorname{dim}(\mathcal{S}_2)$	$\dim(\mathcal{S}_1) \leq \dim(\mathcal{S}_2)$	the property that $\mathcal{S}_1 \subseteq \mathcal{S}_2$. Show that $\dim(\mathcal{S}_1) \leq \dim(\mathcal{S}_2)$.
johnston-linear-matrix-algebra-F01820	178	1.000	$\operatorname{dim}(\mathcal{S}_1)=\operatorname{dim}(\mathcal{S}_2)$	$\dim(\mathcal{S}_1) = \dim(\mathcal{S}_2)$	the property that $\mathcal{S}_1 \subseteq \mathcal{S}_2$ and $\dim(\mathcal{S}_1) = \dim(\mathcal{S}_2)$. Show
johnston-linear-matrix-algebra-F01821	178	1.000	$\mathcal{S}_1=\mathcal{S}_2$	$\mathcal{S}_1 = \mathcal{S}_2$	that $\mathcal{S}_1 = \mathcal{S}_2$.
johnston-linear-matrix-algebra-F01822	178	0.893	$-1,2,3$	$-1, 2, 3$	(b) Why does removing $(-1, 2, 3)$ from B not result in a
johnston-linear-matrix-algebra-F01823	178	1.000	$\mathbf{v} \in \operatorname{range}(A)$	$\mathbf{v} \in \operatorname{range}(A)$	(a) Show that if $\mathbf{v} \in \operatorname{range}(A)$ and $\mathbf{w} \in \operatorname{null}(A^T)$ then $\mathbf{v}$
johnston-linear-matrix-algebra-F01824	178	1.000	$\mathbf{w} \in \operatorname{null}(A^T)$	$\mathbf{w} \in \operatorname{null}(A^T)$	(a) Show that if $\mathbf{v} \in \operatorname{range}(A)$ and $\mathbf{w} \in \operatorname{null}(A^T)$ then $\mathbf{v}$
johnston-linear-matrix-algebra-F01825	178	0.999	$\mathbf{w} \in \operatorname{range}(A^T)$	$\mathbf{w} \in \operatorname{range}(A^T)$	(b) Show that if $\mathbf{v} \in \operatorname{null}(A)$ and $\mathbf{w} \in \operatorname{range}(A^T)$ then $\mathbf{v}$
johnston-linear-matrix-algebra-F01826	178	1.000	$\operatorname{null}(A)=\operatorname{null}(B)$	$\operatorname{null}(A) = \operatorname{null}(B)$	$\operatorname{null}(A) = \operatorname{null}(B)$.

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johnston-linear-matrix-algebra F01827	178	■ 1.000	$\operatorname{range}(A^T) = \operatorname{range}(B^T)$	$\operatorname{range}(A^T) = \operatorname{range}(B^T)$	$\operatorname{range}(A^T) = \operatorname{range}(B^T)$.
johnston-linear-matrix-algebra F01828	178	■ 0.981	$\operatorname{nullity}(A \ B) \leq \operatorname{nullity}(A) + \operatorname{nullity}(B)$	$\operatorname{nullity}(AB) \leq \operatorname{nullity}(A) + \operatorname{nullity}(B)$	(a) Show that $\operatorname{nullity}(AB) \leq \operatorname{nullity}(A) + \operatorname{nullity}(B)$.
johnston-linear-matrix-algebra F01829	178	■ 1.000	$\operatorname{rank}(A \ B) \geq \operatorname{rank}(A) + \operatorname{rank}(B) - n$	$\operatorname{rank}(AB) \geq \operatorname{rank}(A) + \operatorname{rank}(B) - n$	(b) Show that $\operatorname{rank}(AB) \geq \operatorname{rank}(A) + \operatorname{rank}(B) - n$.
johnston-linear-matrix-algebra F01830	178	■ 1.000	$A = \mathbf{v} \ \mathbf{w}^T$	$A = \mathbf{v} \mathbf{w}^T$	such that $A = \mathbf{v} \mathbf{w}^T$.
johnston-linear-matrix-algebra F01831	178	■ 1.000	$m \geq n$	$m \geq n$	(a) Suppose $m \geq n$ and $\operatorname{rank}(A) = n$. Show that there
johnston-linear-matrix-algebra F01832	178	■ 1.000	$B \in \mathcal{M}_{n, m}$	$B \in \mathcal{M}_{n, m}$	exists a matrix $B \in \mathcal{M}_{n, m}$ such that $BA = I_n$.
johnston-linear-matrix-algebra F01833	178	■ 1.000	$BA = I_n$	$BA = I_n$	exists a matrix $B \in \mathcal{M}_{n, m}$ such that $BA = I_n$.
johnston-linear-matrix-algebra F01834	178	■ 0.999	$n \geq m$	$n \geq m$	(b) Suppose $n \geq m$ and $\operatorname{rank}(A) = m$. Show that there
johnston-linear-matrix-algebra F01835	178	■ 0.999	$\operatorname{rank}(A) = m$	$\operatorname{rank}(A) = m$	(b) Suppose $n \geq m$ and $\operatorname{rank}(A) = m$. Show that there
johnston-linear-matrix-algebra F01836	178	■ 1.000	$C \in \mathcal{M}_{n, m}$	$C \in \mathcal{M}_{n, m}$	exists a matrix $C \in \mathcal{M}_{n, m}$ such that $AC = I_m$.
johnston-linear-matrix-algebra F01837	178	■ 1.000	$AC = I_m$	$AC = I_m$	exists a matrix $C \in \mathcal{M}_{n, m}$ such that $AC = I_m$.
johnston-linear-matrix-algebra F01838	178	■ 1.000	$R \in \mathcal{M}_{m, n}$	$R \in \mathcal{M}_{m, n}$	concept of the reduced row echelon form $R \in \mathcal{M}_{m, n}$ of a matrix $A \in \mathcal{M}_{m, n}$,
johnston-linear-matrix-algebra F01839	178	■ 1.000	$A^{-1} \in \mathcal{M}_n$	$A^{-1} \in \mathcal{M}_n$	$A^{-1} \in \mathcal{M}_n$ with the property that $AA^{-1} = A^{-1}A = I$, and we think of A^{-1} as a
johnston-linear-matrix-algebra F01840	179	■ 0.896	$(\mathbf{x} = \mathbf{0})$	$(\mathbf{x} = \mathbf{0})$	g) The linear system $A\mathbf{x} = \mathbf{0}$ has a unique solution ($\mathbf{x} = \mathbf{0}$).
johnston-linear-matrix-algebra F01841	179	■ 1.000	$P \in \mathcal{M}_m$	$P \in \mathcal{M}_m$	e) There is an invertible matrix $P \in \mathcal{M}_m$ such that $A = PB$.
johnston-linear-matrix-algebra F01842	179	■ 1.000	$A = P \ B$	$A = PB$	e) There is an invertible matrix $P \in \mathcal{M}_m$ such that $A = PB$.
johnston-linear-matrix-algebra F01843	179	■ 1.000	$A = P \ R$	$A = PR$	exists an invertible matrix $P \in \mathcal{M}_m$ such that $A = PR$.
johnston-linear-matrix-algebra F01844	180	■ 0.984	$A = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$	$A = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$	*(a) $A = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}$
johnston-linear-matrix-algebra F01845	180	■ 0.984	$B = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}$	$B = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}$	*(a) $A = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}$
johnston-linear-matrix-algebra F01846	180	■ 1.000	$A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \end{bmatrix}$	$A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \end{bmatrix}$	(b) $A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \end{bmatrix}$ and $B = \begin{bmatrix} 0 & 1 & 2 \\ 1 & 1 & 1 \end{bmatrix}$
johnston-linear-matrix-algebra F01847	180	■ 1.000	$B = \begin{bmatrix} 0 & 1 & 2 \\ 1 & 1 & 1 \end{bmatrix}$	$B = \begin{bmatrix} 0 & 1 & 2 \\ 1 & 1 & 1 \end{bmatrix}$	(b) $A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \end{bmatrix}$ and $B = \begin{bmatrix} 0 & 1 & 2 \\ 1 & 1 & 1 \end{bmatrix}$
johnston-linear-matrix-algebra F01848	180	■ 0.993	$A = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 1 & 1 \\ 1 & -1 & 0 \end{bmatrix}$	$A = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 1 & 1 \\ 1 & -1 & 0 \end{bmatrix}$	*(c) $A = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 1 & 1 \\ 1 & -1 & 0 \end{bmatrix}$ and $B = \begin{bmatrix} 3 & 3 & 2 \\ 0 & 0 & 0 \\ 0 & 3 & 1 \end{bmatrix}$
johnston-linear-matrix-algebra F01849	180	■ 0.993	$B = \begin{bmatrix} 3 & 3 & 2 \\ 0 & 0 & 0 \\ 0 & 3 & 1 \end{bmatrix}$	$B = \begin{bmatrix} 3 & 3 & 2 \\ 0 & 0 & 0 \\ 0 & 3 & 1 \end{bmatrix}$	*(c) $A = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 1 & 1 \\ 1 & -1 & 0 \end{bmatrix}$ and $B = \begin{bmatrix} 3 & 3 & 2 \\ 0 & 0 & 0 \\ 0 & 3 & 1 \end{bmatrix}$

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johnston-linear-matrix-algebra-F01850	180	1.000	$A=\left[\begin{array}{ccc}1 & -1 & 2 \\ 1 & 0 & 2 \\ 2 & -1 & 4\end{array}\right]$	$A = \begin{bmatrix} 1 & -1 & 2 \\ 1 & 0 & 2 \\ 2 & -1 & 4 \end{bmatrix}$	(d) $A = \begin{bmatrix} 1 & -1 & 2 \\ 1 & 0 & 2 \\ 2 & -1 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 2 & -1 \\ 1 & 2 & 0 \\ 2 & 4 & -1 \end{bmatrix}$
johnston-linear-matrix-algebra-F01851	180	1.000	$B=\left[\begin{array}{ccc}1 & 2 & -1 \\ 1 & 2 & 0 \\ 4 & -1 & \end{array}\right]$	$B = \begin{bmatrix} 1 & 2 & -1 \\ 1 & 2 & 0 \\ 2 & 4 & -1 \end{bmatrix}$	(d) $A = \begin{bmatrix} 1 & -1 & 2 \\ 1 & 0 & 2 \\ 2 & -1 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 2 & -1 \\ 1 & 2 & 0 \\ 2 & 4 & -1 \end{bmatrix}$
johnston-linear-matrix-algebra-F01852	180	0.946	$A=\left[\begin{array}{ccc}1 & -1 & 0 \\ -1 & 1 & -3 \\ 1 & -2 & -1 \\ 1 & 1 & -3\end{array}\right]$	$A = \begin{bmatrix} 1 & -1 & 0 \\ -1 & 1 & -3 \\ 1 & -2 & -1 \\ 1 & 1 & -3 \end{bmatrix}$	*(e) $A = \begin{bmatrix} 1 & -1 & 0 \\ -1 & 1 & -3 \\ 1 & -2 & -1 \\ 1 & 1 & -3 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & -1 & 3 \\ 0 & -1 & 2 \\ -1 & -1 & 1 \\ -1 & 0 & 0 \end{bmatrix}$
johnston-linear-matrix-algebra-F01853	180	0.946	$B=\left[\begin{array}{ccc}2 & -1 & 3 \\ 0 & -1 & 2 \\ -1 & -1 & 1 \\ -1 & 0 & 0\end{array}\right]$	$B = \begin{bmatrix} 2 & -1 & 3 \\ 0 & -1 & 2 \\ -1 & -1 & 1 \\ -1 & 0 & 0 \end{bmatrix}$	*(e) $A = \begin{bmatrix} 1 & -1 & 0 \\ -1 & 1 & -3 \\ 1 & -2 & -1 \\ 1 & 1 & -3 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & -1 & 3 \\ 0 & -1 & 2 \\ -1 & -1 & 1 \\ -1 & 0 & 0 \end{bmatrix}$
johnston-linear-matrix-algebra-F01854	180	1.000	$A=\left[\begin{array}{cccc}1 & 1 & 4 & 3 \\ 1 & 2 & 3 & 1 \\ 3 & 3 & 3 & 0\end{array}\right]$	$A = \begin{bmatrix} 1 & 1 & 4 & 3 \\ 1 & 2 & 3 & 1 \\ 3 & 3 & 3 & 0 \end{bmatrix}$	(f) $A = \begin{bmatrix} 1 & 1 & 4 & 3 \\ 1 & 2 & 3 & 1 \\ 3 & 3 & 3 & 0 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 2 \\ 4 & 2 & 0 & 1 \end{bmatrix}$
johnston-linear-matrix-algebra-F01855	180	1.000	$B=\left[\begin{array}{cccc}1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 2 \\ 4 & 2 & 0 & 1\end{array}\right]$	$B = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 2 \\ 4 & 2 & 0 & 1 \end{bmatrix}$	(f) $A = \begin{bmatrix} 1 & 1 & 4 & 3 \\ 1 & 2 & 3 & 1 \\ 3 & 3 & 3 & 0 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 2 \\ 4 & 2 & 0 & 1 \end{bmatrix}$
johnston-linear-matrix-algebra-F01856	180	1.000	$R \ \mathbf{x}=\mathbf{b}$	$R\mathbf{x} = \mathbf{b}$	linear systems $A\mathbf{x} = \mathbf{b}$ and $R\mathbf{x} = \mathbf{b}$ have the same
johnston-linear-matrix-algebra-F01857	180	1.000	$A \ \mathbf{v}_j=B \ \mathbf{v}_j$	$A\mathbf{v}_j = B\mathbf{v}_j$	basis of $\mathbb{R}^n$. Show that $A = B$ if and only if $A\mathbf{v}_j = B\mathbf{v}_j$ for
johnston-linear-matrix-algebra-F01858	180	0.961	4×4	4×4	2.5.4 Suppose A is a 4×4 matrix, B is a 4×3 matrix, and
johnston-linear-matrix-algebra-F01859	180	0.867	$A=B \ C$	$A = BC$	C is a 3×4 matrix such that $A = BC$. Prove that A is not
johnston-linear-matrix-algebra-F01860	180	1.000	$\operatorname{null}(A)=\operatorname{null}(A^TA \ \right)$	$\operatorname{null}(A) = \operatorname{null}(A^T A)$	(a) Show that $\operatorname{null}(A) = \operatorname{null}(A^T A)$.
johnston-linear-matrix-algebra-F01861	180	1.000	$\mathbf{x} \in \operatorname{null}(A^TA \ \right)$	$\mathbf{x} \in \operatorname{null}(A^T A)$	[Hint: If $\mathbf{x} \in \operatorname{null}(A^T A)$ then compute $\ A\mathbf{x}\ ^2$.]
johnston-linear-matrix-algebra-F01862	180	1.000	$\ A \ \mathbf{x}\ ^2$	$\ A\mathbf{x}\ ^2$	[Hint: If $\mathbf{x} \in \operatorname{null}(A^T A)$ then compute $\ A\mathbf{x}\ ^2$.]
johnston-linear-matrix-algebra-F01863	180	1.000	$\operatorname{rank}(A^TA \ \right)=\operatorname{rank}(A)$	$\operatorname{rank}(A^T A) = \operatorname{rank}(A)$	(b) Show that $\operatorname{rank}(A^T A) = \operatorname{rank}(A)$.
johnston-linear-matrix-algebra-F01864	180	0.975	$\operatorname{range}(A)=\operatorname{range}(A \ A^T \ \right)$	$\operatorname{range}(A) = \operatorname{range}(AA^T)$	(b) Show that $\operatorname{range}(A) = \operatorname{range}(AA^T)$.
johnston-linear-matrix-algebra-F01865	180	0.998	$\operatorname{rank}(A \ A^T \ \right)=\operatorname{rank}(A)$	$\operatorname{rank}(AA^T) = \operatorname{rank}(A)$	(d) Show that $\operatorname{rank}(AA^T) = \operatorname{rank}(A)$.
johnston-linear-matrix-algebra-F01866	180	0.994	$\operatorname{range}(A)=\operatorname{range}(A^TA \ \right)$	$\operatorname{range}(A) = \operatorname{range}(A^T A)$	essarily the case that $\operatorname{range}(A) = \operatorname{range}(A^T A)$ or
johnston-linear-matrix-algebra-F01867	180	1.000	$\operatorname{null}(A)=\operatorname{null}(A \ A^T \ \right)$	$\operatorname{null}(A) = \operatorname{null}(AA^T)$	$\operatorname{null}(A) = \operatorname{null}(AA^T)$.
johnston-linear-matrix-algebra-F01868	181	1.000	$a(b+c)=a \ b+a \ c$	$a(b+c) = ab+ac$	$a(b+c) = ab+ac$ for all a, b , and c in the field).
johnston-linear-matrix-algebra-F01869	181	1.000	$a+i \ b$	$a+ib$	numbers (i.e., numbers of the form $a+ib$, where $a, b \in \mathbb{R}$ and $i^2 = -1$, which
johnston-linear-matrix-algebra-F01870	181	1.000	$i^2=-1$	$i^2 = -1$	numbers (i.e., numbers of the form $a+ib$, where $a, b \in \mathbb{R}$ and $i^2 = -1$, which

Identifier	Page	Conf.	LaTeX source	Rendered	Image
johnston-linear-matrix-algebra-F01871	181	■ 0.996	$\mathbb{Z}_2$	$\mathbb{Z}_2$	For now, we work with the set $\mathbb{Z}_2$ of numbers under arithmetic mod 2. That
johnston-linear-matrix-algebra-F01872	181	■ 1.000	$\{0, 1\}$	$\{0, 1\}$	is, $\mathbb{Z}_2$ is simply the set $\{0, 1\}$, but with the understanding that addition and
johnston-linear-matrix-algebra-F01873	181	■ 0.978	$1+1=0$	$1 + 1 = 0$	tion that $1 + 1 = 0$ instead of $1 + 1 = 2$ (since there is no “2” in $\mathbb{Z}_2$). Phrased
johnston-linear-matrix-algebra-F01874	181	■ 0.978	$1+1=2$	$1 + 1 = 2$	tion that $1 + 1 = 0$ instead of $1 + 1 = 2$ (since there is no “2” in $\mathbb{Z}_2$). Phrased
johnston-linear-matrix-algebra-F01875	181	■ 1.000	$\mathbb{R}$	$\mathbb{R}$	linear systems with entries from $\mathbb{R}$.
johnston-linear-matrix-algebra-F01876	182	■ 0.996	$1=0-1$	$1 = 0 - 1$	that $1 = 0 - 1$, which
johnston-linear-matrix-algebra-F01877	182	■ 1.000	$\mathbb{Z}_2^n$	$\mathbb{Z}_2^n$	The notation $\mathbb{Z}_2^n$
johnston-linear-matrix-algebra-F01878	182	■ 1.000	$\text{left}(x_1, x_2, x_3, x_4, x_5) =$	$(x_1, x_2, x_3, x_4, x_5) =$	It follows that the unique solution of this linear system is $(x_1, x_2, x_3, x_4, x_5) =$
johnston-linear-matrix-algebra-F01879	182	■ 1.000	2^n	2^n	solutions of a linear system over $\mathbb{Z}_2$ in n variables are the 2^n vectors consisting
johnston-linear-matrix-algebra-F01880	183	■ 1.000	$0 R_j$	$0R_j$	operation, since $0R_j$
johnston-linear-matrix-algebra-F01881	183	■ 1.000	$1 R_j$	$1R_j$	operation and $1R_j$
johnston-linear-matrix-algebra-F01882	183	■ 0.980	$x_5=1, x_4=1, x_2=1-x_3$	$x_5 = 1, x_4 = 1, x_2 = 1 - x_3$	particular, $x_5 = 1, x_4 = 1, x_2 = 1 - x_3$, and $x_1 = x_3$ (recall that $x_3 = -x_3$
johnston-linear-matrix-algebra-F01883	183	■ 0.980	$x_1=x_3$	$x_1 = x_3$	particular, $x_5 = 1, x_4 = 1, x_2 = 1 - x_3$, and $x_1 = x_3$ (recall that $x_3 = -x_3$
johnston-linear-matrix-algebra-F01884	183	■ 0.980	$x_3=-x_3$	$x_3 = -x_3$	particular, $x_5 = 1, x_4 = 1, x_2 = 1 - x_3$, and $x_1 = x_3$ (recall that $x_3 = -x_3$
johnston-linear-matrix-algebra-F01885	183	■ 1.000	$x_3=0$	$x_3 = 0$	$x_3 = 0$ or $x_3 = 1$ into the equations for x_1 and x_2 that we wrote down above.
johnston-linear-matrix-algebra-F01886	183	■ 1.000	$x_3=1$	$x_3 = 1$	$x_3 = 0$ or $x_3 = 1$ into the equations for x_1 and x_2 that we wrote down above.
johnston-linear-matrix-algebra-F01887	183	■ 1.000	2^k	2^k	2^k solutions, where k is a non-negative integer (equal to the
johnston-linear-matrix-algebra-F01888	184	■ 0.999	$\mathbf{v} \in \mathbb{Z}_2^{16}$	$\mathbf{v} \in \mathbb{Z}_2^{16}$	Figure 2.27: We use a vector $\mathbf{v} \in \mathbb{Z}_2^{16}$ to encode the states of the 4×4 “Lights Out”
johnston-linear-matrix-algebra-F01889	184	■ 1.000	$\mathbf{v}_s, \mathbf{v}_e \in \mathbb{Z}_2^{16}$	$\mathbf{v}_s, \mathbf{v}_e \in \mathbb{Z}_2^{16}$	More generally, we use $\mathbf{v}_s, \mathbf{v}_e \in \mathbb{Z}_2^{16}$ to represent the vectors that encode the
johnston-linear-matrix-algebra-F01890	184	■ 0.994	$\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_{16}$	$\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_{16}$	If $\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_{16}$ denote the 16 different vectors that are potentially added to $\mathbf{v}_s$
johnston-linear-matrix-algebra-F01891	184	■ 0.994	$\mathbf{v}_s$	$\mathbf{v}_s$	If $\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_{16}$ denote the 16 different vectors that are potentially added to $\mathbf{v}_s$
johnston-linear-matrix-algebra-F01892	184	■ 1.000	$\mathbf{x} \in \mathbb{Z}_2^{16}$	$\mathbf{x} \in \mathbb{Z}_2^{16}$	when we press the 16 different buttons, and we let $\mathbf{x} \in \mathbb{Z}_2^{16}$ denote the vector
johnston-linear-matrix-algebra-F01893	184	■ 1.000	$x_j=1$	$x_j = 1$	that encodes which buttons we press (i.e., $x_j = 1$ if we press the j -th button
johnston-linear-matrix-algebra-F01894	185	■ 0.937	$x_j=0$	$x_j = 0$	and $x_j = 0$ otherwise) then our goal is to find $x_1, x_2, \dots, x_{16}$ so that
johnston-linear-matrix-algebra-F01895	185	■ 0.937	$x_1, x_2, \dots, x_{16}$	$x_1, x_2, \dots, x_{16}$	and $x_j = 0$ otherwise) then our goal is to find $x_1, x_2, \dots, x_{16}$ so that
johnston-linear-matrix-algebra-F01896	185	■ 0.919	$A \mathbf{x} = \mathbf{v}_e - \mathbf{v}_s$	$A \mathbf{x} = \mathbf{v}_e - \mathbf{v}_s$	matrix equation $A \mathbf{x} = \mathbf{v}_e - \mathbf{v}_s$, where $A = [\mathbf{a}_1 \mid \mathbf{a}_2 \mid \dots \mid \mathbf{a}_{16}]$, which we can
johnston-linear-matrix-algebra-F01897	185	■ 0.919	$A = [\mathbf{a}_1 \mid \mathbf{a}_2 \mid \dots \mid \mathbf{a}_{16}]$	$A = [\mathbf{a}_1 \mid \mathbf{a}_2 \mid \dots \mid \mathbf{a}_{16}]$	matrix equation $A \mathbf{x} = \mathbf{v}_e - \mathbf{v}_s$, where $A = [\mathbf{a}_1 \mid \mathbf{a}_2 \mid \dots \mid \mathbf{a}_{16}]$, which we can

Identifier	Page	Conf.	LaTeX source	Rendered	Image
johnston-linear-matrix-augmented F01898	185	■ 0.973	$\mathbf{v}_{\mathrm{e}} - \mathbf{v}_{\mathrm{s}} \in \mathbb{Z}_2^{16}$	$\mathbf{v}_{\mathrm{e}} - \mathbf{v}_{\mathrm{s}} \in \mathbb{Z}_2^{16}$	tion (2.A.1) and $\mathbf{v}_{\mathrm{e}} - \mathbf{v}_{\mathrm{s}} \in \mathbb{Z}_2^{16}$ is the vector with every entry equal to 1.
johnston-linear-matrix-augmented F01899	186	■ 0.999	$\left[A \mid \mathbf{v}_{\mathrm{e}} - \mathbf{v}_{\mathrm{s}} \right]$	$[A \mid \mathbf{v}_{\mathrm{e}} - \mathbf{v}_{\mathrm{s}}]$	The reduced row echelon form of the augmented matrix $[A \mid \mathbf{v}_{\mathrm{e}} - \mathbf{v}_{\mathrm{s}}]$ is
johnston-linear-matrix-augmented F01900	186	■ 1.000	$2^4 = 16$	$2^4 = 16$	which tells us that this linear system has 4 free variables and thus $2^4 = 16$
johnston-linear-matrix-augmented F01901	186	■ 1.000	x_{13}, x_{14}, x_{15}	x_{13}, x_{14}, x_{15}	the free variables x_{13}, x_{14}, x_{15} , and x_{16} . The simplest solution arises from
johnston-linear-matrix-augmented F01902	186	■ 1.000	x_{16}	x_{16}	the free variables x_{13}, x_{14}, x_{15} , and x_{16} . The simplest solution arises from
johnston-linear-matrix-augmented F01903	186	■ 0.992	$x_{13}=0, x_{14}=1, x_{15}=0, x_{16}=0$	$x_{13} = 0, x_{14} = 1, x_{15} = 0, x_{16} = 0$	choosing $x_{13} = 0, x_{14} = 1, x_{15} = 0, x_{16} = 0$, which gives
johnston-linear-matrix-augmented F01904	187	■ 0.992	$2^{16}=65536$	$2^{16} = 65536$	of that grid cannot be turned into the all-“on” configuration (of the $2^{16} = 65536$
johnston-linear-matrix-augmented F01905	187	■ 1.000	$65536 / 16=4096$	$65536/16 = 4096$	starting configurations, only $65536/16 = 4096$ are solvable). For example, of
johnston-linear-matrix-augmented F01906	188	■ 0.994	$\mathbf{v}_{\mathrm{s}}=(0,0,0,0,1,1,1,1,0)$	$\mathbf{v}_{\mathrm{s}} = (0, 0, 0, 0, 1, 1, 1, 1, 0)$	We represent the starting state via the vector $\mathbf{v}_{\mathrm{s}} = (0, 0, 0, 0, 1, 1, 1, 1, 0)$
johnston-linear-matrix-augmented F01907	188	■ 0.994	$\mathbf{v}_{\mathrm{e}}=(1,1,1,1,1,1,1,1,1)$	$\mathbf{v}_{\mathrm{e}} = (1, 1, 1, 1, 1, 1, 1, 1, 1)$	and the target state via the vector $\mathbf{v}_{\mathrm{e}} = (1, 1, 1, 1, 1, 1, 1, 1, 1)$. If we let
johnston-linear-matrix-augmented F01908	188	■ 0.916	9×9	9×9	A be the 9×9 binary matrix whose j -th column (for each $1 \leq j \leq 9$)
johnston-linear-matrix-augmented F01909	188	■ 0.916	$1 \leq j \leq 9$	$1 \leq j \leq 9$	A be the 9×9 binary matrix whose j -th column (for each $1 \leq j \leq 9$)
johnston-linear-matrix-augmented F01910	188	■ 1.000	$[I \mid \mathbf{x}]$	$[I \mid \mathbf{x}]$	echelon form of this augmented matrix has the form $[I \mid \mathbf{x}]$, where
johnston-linear-matrix-augmented F01911	188	■ 1.000	$\mathbf{x}=(1,1,0,0,0,0,1,1,0)$	$\mathbf{x} = (1, 1, 0, 0, 0, 0, 1, 1, 0)$	$\mathbf{x} = (1, 1, 0, 0, 0, 0, 1, 1, 0)$. It follows that this linear system has a unique
johnston-linear-matrix-augmented F01912	189	■ 1.000	$\mathbb{Z}_2^k$	$\mathbb{Z}_2^k$	lives in $\mathbb{Z}_2^k$, whereas
johnston-linear-matrix-augmented F01913	189	■ 1.000	$A \mathbf{x} = \mathbf{1}$	$A\mathbf{x} = \mathbf{1}$	<i>Proof.</i> Recall that the linear system we want to solve has the form $A\mathbf{x} = \mathbf{1}$,
johnston-linear-matrix-augmented F01914	189	■ 0.516	$\mathbf{1}=(1,1,\ldots,1) \in \mathbb{Z}_2^n$	$\mathbf{1} = (1, 1, \dots, 1) \in \mathbb{Z}_2^n$	where $\mathbf{1} = (1, 1, \dots, 1) \in \mathbb{Z}_2^n$ is the vector with every entry equal to 1. In other
johnston-linear-matrix-augmented F01915	189	■ 1.000	$\mathbf{1} \in \operatorname{range}(A)$	$\mathbf{1} \in \operatorname{range}(A)$	words, we want to show that $\mathbf{1} \in \operatorname{range}(A)$. Furthermore, A has the following
johnston-linear-matrix-augmented F01916	189	■ 1.000	$A^T=A$	$A^T = A$	• A is symmetric (i.e., $A^T = A$), since button i is a neighbor of button j if
johnston-linear-matrix-augmented F01917	189	■ 1.000	$\mathbf{y} \in \operatorname{null}(A)$	$\mathbf{y} \in \operatorname{null}(A)$	steps. The first step is to show that every $\mathbf{y} \in \operatorname{null}(A)$ (i.e., every $\mathbf{y} \in \mathbb{Z}_2^n$ for
johnston-linear-matrix-augmented F01918	189	■ 1.000	$\mathbf{y} \in \mathbb{Z}_2^n$	$\mathbf{y} \in \mathbb{Z}_2^n$	steps. The first step is to show that every $\mathbf{y} \in \operatorname{null}(A)$ (i.e., every $\mathbf{y} \in \mathbb{Z}_2^n$ for
johnston-linear-matrix-augmented F01919	189	■ 1.000	$A \mathbf{y} = \mathbf{0}$	$A\mathbf{y} = \mathbf{0}$	which $A\mathbf{y} = \mathbf{0}$ has an even number of entries equal to 1 (in other words, it
johnston-linear-matrix-augmented F01920	189	■ 1.000	$y_j=0$	$y_j = 0$	deleting every row and column j for which $y_j = 0$. For example, if
johnston-linear-matrix-augmented F01921	189	■ 0.808	$B^T=B$	$B^T = B$	Then $B^T = B$ as well and it also has all of its diagonal entries equal to 1.
johnston-linear-matrix-augmented F01922	189	■ 1.000	$B \mathbf{1} = \mathbf{0}$	$B\mathbf{1} = \mathbf{0}$	Furthermore, the fact that $A\mathbf{y} = \mathbf{0}$ tells us that $B\mathbf{1} = \mathbf{0}$. However, the entries of
johnston-linear-matrix-augmented F01923	189	■ 0.839	$B \mathbf{1}$	$B\mathbf{1}$	$B\mathbf{1}$ just count the number of 1s in each row of B , so every row of B contains an
johnston-linear-matrix-augmented F01924	190	■ 0.874	$\mathbf{1}$	$\mathbf{1}$	all of the rows of R results in the all-ones vector $\mathbf{1}$. It follows that $\mathbf{1} \in \operatorname{range}(A)$,

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johnston-linear-matrix-align="center" F01925	190	0.982	$\mathbb{Z}_3$	$\mathbb{Z}_3$	modulus larger than 2. For example, $\mathbb{Z}_3$ denotes the set $\{0, 1, 2\}$ under arith-																				
johnston-linear-matrix-align="center" F01926	190	0.982	$\{0, 1, 2\}$	$\{0, 1, 2\}$	modulus larger than 2. For example, $\mathbb{Z}_3$ denotes the set $\{0, 1, 2\}$ under arith-																				
johnston-linear-matrix-align="center" F01927	190	0.999	$\bmod 3$	mod3	metic mod 3. That is, we perform addition and multiplication in the usual ways,																				
johnston-linear-matrix-align="center" F01928	190	1.000	$2+2=4$	$2 + 2 = 4$	of having $2 + 2 = 4$, we have $2 + 2 = 1$ since the addition “rolls over” back to																				
johnston-linear-matrix-align="center" F01929	190	1.000	$2+2=1$	$2 + 2 = 1$	of having $2 + 2 = 4$, we have $2 + 2 = 1$ since the addition “rolls over” back to																				
johnston-linear-matrix-align="center" F01930	190	0.919	$4: 00$	$4 : 00$	it will be 4:00, since $9 + 7 = 16 = 12 + 4$, and the extra 12 is irrelevant (it																				
johnston-linear-matrix-align="center" F01931	190	0.919	$9+7=16=12+4$	$9 + 7 = 16 = 12 + 4$	it will be 4:00, since $9 + 7 = 16 = 12 + 4$, and the extra 12 is irrelevant (it																				
johnston-linear-matrix-align="center" F01932	190	0.924	$\bmod -12$	mod − 12	displayed on the clock). For example, in mod-12 arithmetic, $9 + 7 = 4$ (see																				
johnston-linear-matrix-align="center" F01933	190	0.924	$9+7=4$	$9 + 7 = 4$	displayed on the clock). For example, in mod-12 arithmetic, $9 + 7 = 4$ (see																				
johnston-linear-matrix-align="center" F01934	191	1.000	$0-1=2, 0-2=1$	$0 - 1 = 2, 0 - 2 = 1$	$0 - 1 = 2, 0 - 2 = 1$,																				
johnston-linear-matrix-align="center" F01935	191	1.000	$1-2=2$	$1 - 2 = 2$	$1 - 2 = 2$, and so on																				
johnston-linear-matrix-align="center" F01936	191	1.000	$\mathbb{Z}_p$	$\mathbb{Z}_p$	systems over $\mathbb{Z}_p$ (i.e., using just the numbers $0, 1, 2, \dots, p - 1$) when p is any																				
johnston-linear-matrix-align="center" F01937	191	1.000	$0, 1, 2, \ldots, p-1$	$0, 1, 2, \dots, p - 1$	systems over $\mathbb{Z}_p$ (i.e., using just the numbers $0, 1, 2, \dots, p - 1$) when p is any																				
johnston-linear-matrix-align="center" F01938	191	1.000	$(w, x, y, z)=$	$(w, x, y, z) =$	It follows that the unique solution of this linear system is $(w, x, y, z) =$																				
johnston-linear-matrix-align="center" F01939	192	1.000	$3^4=81$	$3^4 = 81$	striped red ovals), for a total of $3^4 = 81$ cards in the deck. Some examples																				
johnston-linear-matrix-align="center" F01940	192	1.000	$\mathbb{Z}_3^4$	$\mathbb{Z}_3^4$	the 81 cards as a vector in $\mathbb{Z}_3^4$ by encoding each of the properties as one																				
johnston-linear-matrix-align="center" F01941	192	1.000	$\mathbf{v}=(c, p, d, n)$	$\mathbf{v} = (c, p, d, n)$	Specifically, we can write the cards as vectors of the form $\mathbf{v} = (c, p, d, n)$,																				
johnston-linear-matrix-align="center" F01942	192	1.000	c, p, d	c, p, d	where c, p, d , and n encode the color, shape, shading, and number of																				
johnston-linear-matrix-align="center" F01943	193	0.516	(d)	(d)	<table><tr><th>Value</th><th>Color (c)</th><th>Shape (p)</th><th>Shading (d)</th><th>Number (n)</th></tr><tr><td>0</td><td>red</td><td>diamond</td><td>solid</td><td>one</td></tr><tr><td>1</td><td>green</td><td>squiggle</td><td>striped</td><td>two</td></tr><tr><td>2</td><td>purple</td><td>oval</td><td>hollow</td><td>three</td></tr></table>	Value	Color (c)	Shape (p)	Shading (d)	Number (n)	0	red	diamond	solid	one	1	green	squiggle	striped	two	2	purple	oval	hollow	three
Value	Color (c)	Shape (p)	Shading (d)	Number (n)																					
0	red	diamond	solid	one																					
1	green	squiggle	striped	two																					
2	purple	oval	hollow	three																					
johnston-linear-matrix-align="center" F01944	193	1.000	$\mathbf{v}=(c, p, d, n)=(0, 2, 1, 0)$	$\mathbf{v} = (c, p, d, n) = (0, 2, 1, 0)$	by the vector $\mathbf{v} = (c, p, d, n) = (0, 2, 1, 0)$.																				
johnston-linear-matrix-align="center" F01945	193	1.000	$\mathbf{v}_1, \mathbf{v}_2$	$\mathbf{v}_1, \mathbf{v}_2$	to the condition that their associated vectors $\mathbf{v}_1, \mathbf{v}_2$, and $\mathbf{v}_3$ are such that																				
johnston-linear-matrix-align="center" F01946	193	1.000	$\mathbf{v}_1+\mathbf{v}_2+\mathbf{v}_3=\mathbf{0}$	$\mathbf{v}_1 + \mathbf{v}_2 + \mathbf{v}_3 = \mathbf{0}$	$\mathbf{v}_1 + \mathbf{v}_2 + \mathbf{v}_3 = \mathbf{0}$. To see why this is the case, just notice that the only ways																				
johnston-linear-matrix-align="center" F01947	193	1.000	$\mathbf{v}_1, \mathbf{v}_2 \in \mathbb{Z}_3^4$	$\mathbf{v}_1, \mathbf{v}_2 \in \mathbb{Z}_3^4$	set with them. This is because, given $\mathbf{v}_1, \mathbf{v}_2 \in \mathbb{Z}_3^4$, there is a unique																				
johnston-linear-matrix-align="center" F01948	193	1.000	$\mathbf{v}_3 \in \mathbb{Z}_3^4$	$\mathbf{v}_3 \in \mathbb{Z}_3^4$	$\mathbf{v}_3 \in \mathbb{Z}_3^4$ such that $\mathbf{v}_1 + \mathbf{v}_2 + \mathbf{v}_3 = \mathbf{0}$ (in particular, $\mathbf{v}_3 = -\mathbf{v}_1 - \mathbf{v}_2$).																				
johnston-linear-matrix-align="center" F01949	193	1.000	$\mathbf{v}_3=-\mathbf{v}_1-\mathbf{v}_2$	$\mathbf{v}_3 = -\mathbf{v}_1 - \mathbf{v}_2$	$\mathbf{v}_3 \in \mathbb{Z}_3^4$ such that $\mathbf{v}_1 + \mathbf{v}_2 + \mathbf{v}_3 = \mathbf{0}$ (in particular, $\mathbf{v}_3 = -\mathbf{v}_1 - \mathbf{v}_2$).																				

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johnston-linear-matrix-aligned F01950	193	■ 1.000	$26 \times 3 = 78$	$26 \times 3 = 78$	If 26 sets (and thus $26 \times 3 = 78$ cards) are removed from the deck,
johnston-linear-matrix-aligned F01951	193	■ 1.000	$\mathbf{v}_1 - \mathbf{v}_2 = \mathbf{v}_2 - \mathbf{v}_3 - \mathbf{v}_3$	$\mathbf{v}_1 - \mathbf{v}_2 = \mathbf{v}_2 - \mathbf{v}_3$	Since $\mathbf{v}_1 + \mathbf{v}_2 + \mathbf{v}_3 = \mathbf{0}$ is equivalent to $\mathbf{v}_1 - \mathbf{v}_2 = \mathbf{v}_2 - \mathbf{v}_3$ in $\mathbb{Z}_3^4$,
johnston-linear-matrix-aligned F01952	193	■ 0.999	$(x, 2, y, x)$	$(x, 2, y, x)$	the form $(x, 2, y, x)$ is displayed below:
johnston-linear-matrix-aligned F01953	194	■ 1.000	$\mathbb{Z}_5$	$\mathbb{Z}_5$	exist in $\mathbb{Z}_5$ —the only
johnston-linear-matrix-aligned F01954	194	■ 1.000	$2x = 1$	$2x = 1$	in this case we run into issues like the equation $2x = 1$ not having a solution.
johnston-linear-matrix-aligned F01955	194	■ 0.982	$\{0, 1, 2, \dots, p-1\}$	$\{0, 1, 2, \dots, p-1\}$	of $\{0, 1, 2, \dots, p-1\}$, so in particular we can always “divide” by any scalar—
johnston-linear-matrix-aligned F01956	194	■ 1.000	$y \in \mathbb{Z}_p$	$y \in \mathbb{Z}_p$	given any $y \in \mathbb{Z}_p$, we can find an $x \in \mathbb{Z}_p$ such that $xy = 1$. For example, the
johnston-linear-matrix-aligned F01957	194	■ 1.000	$x \in \mathbb{Z}_p$	$x \in \mathbb{Z}_p$	given any $y \in \mathbb{Z}_p$, we can find an $x \in \mathbb{Z}_p$ such that $xy = 1$. For example, the
johnston-linear-matrix-aligned F01958	194	■ 1.000	$xy = 1$	$xy = 1$	given any $y \in \mathbb{Z}_p$, we can find an $x \in \mathbb{Z}_p$ such that $xy = 1$. For example, the
johnston-linear-matrix-aligned F01959	194	■ 1.000	$2x = 3$	$2x = 3$	For example, to solve the equation $2x = 3$ in $\mathbb{Z}_5$, instead of dividing by 2 we
johnston-linear-matrix-aligned F01960	195	■ 0.982	$2 \cdot 3 = 1$	$2 \cdot 3 = 1$	multiply by 3, since the $\mathbb{Z}_5$ multiplication table above shows us that $2 \cdot 3 = 1$.
johnston-linear-matrix-aligned F01961	195	■ 1.000	$x = 4$	$x = 4$	Multiplying the equation $2x = 3$ through by 3 then gives us $x = 4$.
johnston-linear-matrix-aligned F01962	195	■ 0.998	w, x	w, x	It follows that y is a free variable and w, x , and z are leading variables
johnston-linear-matrix-aligned F01963	195	■ 1.000	$z = 2, x = 1 - y$	$z = 2, x = 1 - y$	satisfying $z = 2, x = 1 - y$, and $w = 3 - 2y$. By letting y range over the five
johnston-linear-matrix-aligned F01964	195	■ 1.000	$w = 3 - 2y$	$w = 3 - 2y$	satisfying $z = 2, x = 1 - y$, and $w = 3 - 2y$. By letting y range over the five
johnston-linear-matrix-aligned F01965	195	■ 1.000	$p = 2$	$p = 2$	We already answered this question when $p = 2$: recall that, in this case, the
johnston-linear-matrix-aligned F01966	195	■ 1.000	p^k	p^k	solutions or exactly p^k solutions, where k is a non-negative integer (equal
johnston-linear-matrix-aligned F01967	196	■ 0.863	v, w, x, y, z	v, w, x, y, z	*2.A.7 Find integers v, w, x, y, z (or show that none exist)
johnston-linear-matrix-aligned F01968	197	■ 0.999	$k \times k$	$k \times k$	**2.A.10 Suppose that B is a symmetric $k \times k$ matrix, all
johnston-linear-matrix-aligned F01969	197	■ 0.988	$x_1, x_2, \dots, x_n \in \mathbb{R}$	$x_1, x_2, \dots, x_n \in \mathbb{R}$	linear in the variables. We typically use $x_1, x_2, \dots, x_n \in \mathbb{R}$ as the variables, so
johnston-linear-matrix-aligned F01970	198	■ 1.000	$\mathbf{x} \geq \mathbf{0}$	$\mathbf{x} \geq \mathbf{0}$	and the vector inequalities are meant entrywise (e.g., $\mathbf{x} \geq \mathbf{0}$ means that each
johnston-linear-matrix-aligned F01971	198	■ 0.781	$A \mathbf{x} \leq \mathbf{b}$	$A \mathbf{x} \leq \mathbf{b}$	entry of $\mathbf{x}$ is non-negative, and $A \mathbf{x} \leq \mathbf{b}$ means that each entry of $A \mathbf{x}$ is no larger
johnston-linear-matrix-aligned F01972	198	■ 1.000	$A \in \mathcal{M}_{m,n}, \mathbf{b} \in \mathbb{R}^m, \mathbf{c} \in \mathbb{R}^n$	$A \in \mathcal{M}_{m,n}, \mathbf{b} \in \mathbb{R}^m, \mathbf{c} \in \mathbb{R}^n$	in the following form, where $A \in \mathcal{M}_{m,n}, \mathbf{b} \in \mathbb{R}^m, \mathbf{c} \in \mathbb{R}^n$ are fixed, and
johnston-linear-matrix-aligned F01973	198	■ 0.988	$\mathbf{c} \cdot \mathbf{x}$	$\mathbf{c} \cdot \mathbf{x}$	or minimized (i.e., $\mathbf{c} \cdot \mathbf{x}$), and the optimal value is the maximal or minimal
johnston-linear-matrix-aligned F01974	198	■ 1.000	$-(\mathbf{c} \cdot \mathbf{x})$	$-(\mathbf{c} \cdot \mathbf{x})$	minimizing $\mathbf{c} \cdot \mathbf{x}$ is the same as maximizing $-(\mathbf{c} \cdot \mathbf{x})$ and then multiplying the
johnston-linear-matrix-aligned F01975	199	■ 0.699	$A, \mathbf{b}$	$A, \mathbf{b}$	If we wanted to, we could then specify exactly what $A, \mathbf{b}$, and $\mathbf{c}$ are in this
johnston-linear-matrix-aligned F01976	199	■ 0.699	$\mathbf{c}$	$\mathbf{c}$	If we wanted to, we could then specify exactly what $A, \mathbf{b}$, and $\mathbf{c}$ are in this

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johnston-linear-matrix-algebra-F01977	199	1.000	$\text{\texttt{\textbackslash left}(x_{\{1\}}, x_{\{2\}}\text{\textbackslash right})}=(1,2)$	$(x_1, x_2) = (1, 2)$	$(x_1, x_2) = (1, 2)$, but the original minimization problem has optimal value
johnston-linear-matrix-algebra-F01978	199	1.000	$\text{\texttt{-x}_{\{1\}}-2 x_{\{2\}}}=-5$	$-x_1 - 2x_2 = -5$	$-x_1 - 2x_2 = -5$, whereas its standard form has optimal value $x_1 + 2x_2 = 5$.
johnston-linear-matrix-algebra-F01979	199	1.000	$\text{\texttt{x}_{\{1\}}+2 x_{\{2\}}}=5$	$x_1 + 2x_2 = 5$	$-x_1 - 2x_2 = -5$, whereas its standard form has optimal value $x_1 + 2x_2 = 5$.
johnston-linear-matrix-algebra-F01980	199	1.000	$\text{\texttt{\textbackslash geq}}$	$\geq$	Equality constraints and “ $\geq$ ” constraints can both be converted into “ $\leq$ ” con-
johnston-linear-matrix-algebra-F01981	199	1.000	$\text{\texttt{\textbackslash leq}}$	$\leq$	Equality constraints and “ $\geq$ ” constraints can both be converted into “ $\leq$ ” con-
johnston-linear-matrix-algebra-F01982	199	1.000	$\text{\texttt{x}}=\text{\texttt{a}}$	$x = a$	equality constraints, recall that $x = a$ is equivalent to the pair of inequalities
johnston-linear-matrix-algebra-F01983	199	1.000	$\text{\texttt{x \textbackslash leq a}}$	$x \leq a$	$x \leq a$ and $x \geq a$, which in turn is equivalent to the pair of inequalities $x \leq a$ and
johnston-linear-matrix-algebra-F01984	199	1.000	$\text{\texttt{x \textbackslash geq a}}$	$x \geq a$	$x \leq a$ and $x \geq a$, which in turn is equivalent to the pair of inequalities $x \leq a$ and
johnston-linear-matrix-algebra-F01985	199	1.000	$\text{\texttt{-x \textbackslash leq -a}}$	$-x \leq -a$	$-x \leq -a$, so one equality constraint can be converted into two “ $\leq$ ” constraints.
johnston-linear-matrix-algebra-F01986	199	0.936	$\text{\texttt{3 x_{\{1\}}-4 x_{\{2\}} \textbackslash geq 1}}$	$3x_1 - 4x_2 \geq 1$	We multiply the constraint $3x_1 - 4x_2 \geq 1$ by -1 to obtain the equivalent
johnston-linear-matrix-algebra-F01987	199	1.000	$\text{\texttt{-3 x_{\{1\}}+4 x_{\{2\}} \textbackslash leq -1}}$	$-3x_1 + 4x_2 \leq -1$	constraints $-3x_1 + 4x_2 \leq -1$, and we split the equality constraint $2x_1 +$
johnston-linear-matrix-algebra-F01988	199	1.000	$\text{\texttt{2 x_{\{1\}}+}}$	$2x_1 +$	constraints $-3x_1 + 4x_2 \leq -1$, and we split the equality constraint $2x_1 +$
johnston-linear-matrix-algebra-F01989	199	1.000	$\text{\texttt{x_{\{2\}}}=5}$	$x_2 = 5$	$x_2 = 5$ into the equivalent pair of inequality constraints $2x_1 + x_2 \leq 5$ and
johnston-linear-matrix-algebra-F01990	199	1.000	$\text{\texttt{2 x_{\{1\}}+x_{\{2\}} \textbackslash leq 5}}$	$2x_1 + x_2 \leq 5$	$x_2 = 5$ into the equivalent pair of inequality constraints $2x_1 + x_2 \leq 5$ and
johnston-linear-matrix-algebra-F01991	199	1.000	$\text{\texttt{-2 x_{\{1\}}-x_{\{2\}} \textbackslash leq -5}}$	$-2x_1 - x_2 \leq -5$	$-2x_1 - x_2 \leq -5$. After making these changes, the linear program has the
johnston-linear-matrix-algebra-F01992	200	0.900	$\text{\texttt{x_{\{1\}}+3\textbackslash left(x_{\{2\}}^{\textbackslash \textasciitilde{+}}-x_{\{2\}}^{\textbackslash \textasciitilde{-}}\textbackslash right) \textbackslash leq 4}}$	$x_1 + 3(x_2^+ - x_2^-) \leq 4$	$x_1 + 3(x_2^+ - x_2^-) \leq 4$,
johnston-linear-matrix-algebra-F01993	200	1.000	$\text{\texttt{x_{\{1\}}+3 x_{\{2\}}^{\textbackslash \textasciitilde{+}}-3 x_{\{2\}}^{\textbackslash \textasciitilde{-}} \textbackslash leq 4}}$	$x_1 + 3x_2^+ - 3x_2^- \leq 4$	$x_1 + 3x_2^+ - 3x_2^- \leq 4$, so
johnston-linear-matrix-algebra-F01994	200	1.000	$\text{\texttt{x}}=\text{\texttt{x}}^{\textbackslash \textasciitilde{+}}-\text{\texttt{x}}^{\textbackslash \textasciitilde{-}}$	$x = x^+ - x^-$	since every real number x can be written in the form $x = x^+ - x^-$, where $x^+ \geq 0$
johnston-linear-matrix-algebra-F01995	200	1.000	$\text{\texttt{x}}^{\textbackslash \textasciitilde{+}} \textbackslash geq 0$	$x^+ \geq 0$	since every real number x can be written in the form $x = x^+ - x^-$, where $x^+ \geq 0$
johnston-linear-matrix-algebra-F01996	200	1.000	$\text{\texttt{x}}^{\textbackslash \textasciitilde{-}} \textbackslash geq 0$	$x^- \geq 0$	and $x^- \geq 0$. We illustrate how this fact can be used to write even more linear
johnston-linear-matrix-algebra-F01997	200	1.000	$\text{\texttt{x_{\{2\}} \textbackslash geq 0}}$	$x_2 \geq 0$	that there is no $x_2 \geq 0$ constraint. To rewrite this linear program in an
johnston-linear-matrix-algebra-F01998	200	1.000	$\text{\texttt{x_{\{2\}}=x_{\{2\}}^{\textbackslash \textasciitilde{+}}-x_{\{2\}}^{\textbackslash \textasciitilde{-}}}}$	$x_2 = x_2^+ - x_2^-$	equivalent form with all non-negative variables, we write $x_2 = x_2^+ - x_2^-$,
johnston-linear-matrix-algebra-F01999	200	1.000	$\text{\texttt{x_{\{2\}}^{\textbackslash \textasciitilde{+}}, x_{\{2\}}^{\textbackslash \textasciitilde{-}} \textbackslash geq 0}}$	$x_2^+, x_2^- \geq 0$	where $x_2^+, x_2^- \geq 0$, and then we replace all instances of x_2 in the linear
johnston-linear-matrix-algebra-F02000	200	1.000	$\text{\texttt{x_{\{2\}}^{\textbackslash \textasciitilde{+}}-x_{\{2\}}^{\textbackslash \textasciitilde{-}}}}$	$x_2^+ - x_2^-$	program with $x_2^+ - x_2^-$:
johnston-linear-matrix-algebra-F02001	201	0.989	$\text{\texttt{x_{\{1\}}^{\textbackslash \textasciitilde{+}}-x_{\{1\}}^{\textbackslash \textasciitilde{-}}}}$	$x_1^+ - x_1^-$	the unconstrained variable x_1 by $x_1^+ - x_1^-$, where $x_1^+, x_1^- \geq 0$:
johnston-linear-matrix-algebra-F02002	201	0.989	$\text{\texttt{x_{\{1\}}^{\textbackslash \textasciitilde{+}}, x_{\{1\}}^{\textbackslash \textasciitilde{-}} \textbackslash geq 0}}$	$x_1^+, x_1^- \geq 0$	the unconstrained variable x_1 by $x_1^+ - x_1^-$, where $x_1^+, x_1^- \geq 0$:
johnston-linear-matrix-algebra-F02003	201	1.000	$\text{\texttt{\textbf{x}}}=\text{\texttt{\textbackslash left}(x_{\{1\}}, x_{\{2\}}\text{\textbackslash right})}$	$\mathbf{x} = (x_1, x_2)$	region $\mathbf{x} = (x_1, x_2)$ is a subset of $\mathbb{R}^2$. The constraints $x_1 \geq 0$ and $x_2 \geq 0$ are sim-

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johnston-linear-matrix-algebra-F02004	201	1.000	$x_1 \geq 0$	$x_1 \geq 0$	region $\mathbf{x} = (x_1, x_2)$ is a subset of $\mathbb{R}^2$. The constraints $x_1 \geq 0$ and $x_2 \geq 0$ are sim-
johnston-linear-matrix-algebra-F02005	201	1.000	$x_2 \leq -x_1 + 3$	$x_2 \leq -x_1 + 3$	into the forms $x_2 \leq -x_1 + 3$ (i.e., everything on and below the line $x_2 = -x_1 + 3$)
johnston-linear-matrix-algebra-F02006	201	1.000	$x_2 = -x_1 + 3$	$x_2 = -x_1 + 3$	into the forms $x_2 \leq -x_1 + 3$ (i.e., everything on and below the line $x_2 = -x_1 + 3$)
johnston-linear-matrix-algebra-F02007	202	1.000	$f(x_1, x_2) = x_1 + 2x_2$	$f(x_1, x_2) = x_1 + 2x_2$	$f(x_1, x_2) = x_1 + 2x_2$.
johnston-linear-matrix-algebra-F02008	202	1.000	$x_2 \leq x_1 + 1$	$x_2 \leq x_1 + 1$	and $x_2 \leq x_1 + 1$ (i.e., everything on and below the line $x_2 = x_1 + 1$), respectively.
johnston-linear-matrix-algebra-F02009	202	1.000	$x_2 = x_1 + 1$	$x_2 = x_1 + 1$	and $x_2 \leq x_1 + 1$ (i.e., everything on and below the line $x_2 = x_1 + 1$), respectively.
johnston-linear-matrix-algebra-F02010	202	1.000	$z = x_1 + 2x_2$	$z = x_1 + 2x_2$	region. To this end, we define $z = x_1 + 2x_2$, so that our goal is to maximize z .
johnston-linear-matrix-algebra-F02011	202	1.000	$x_2 = -x_1 / 2 + z / 2$	$x_2 = -x_1 / 2 + z / 2$	We can rearrange this equation to get $x_2 = -x_1 / 2 + z / 2$, which is a line with
johnston-linear-matrix-algebra-F02012	202	1.000	$-1 / 2$	$-1 / 2$	slope $-1 / 2$ and y-intercept $z / 2$. We want to find the largest possible z such that
johnston-linear-matrix-algebra-F02013	202	1.000	$z / 2$	$z / 2$	slope $-1 / 2$ and y-intercept $z / 2$. We want to find the largest possible z such that
johnston-linear-matrix-algebra-F02014	202	1.000	$z = 5$	$z = 5$	in Figure 2.33(b). We see that the largest such value of z is $z = 5$ (corresponding
johnston-linear-matrix-algebra-F02015	202	1.000	$\mathbf{x} = (x_1, x_2) = (1, 2)$	$\mathbf{x} = (x_1, x_2) = (1, 2)$	it is attained at the vector $\mathbf{x} = (x_1, x_2) = (1, 2)$.
johnston-linear-matrix-algebra-F02016	203	0.999	$x_1 = x_2 = 0$	$x_1 = x_2 = 0$	$x_1 = x_2 = 0$).
johnston-linear-matrix-algebra-F02017	203	0.980	(x_1, x_2)	(x_1, x_2)	is unbounded because we can choose $x_2 = x_1 + 1$ so that (x_1, x_2) is a feasible
johnston-linear-matrix-algebra-F02018	203	1.000	$x_1 + 2x_2 = x_1 + 2(x_1 + 1) = 3x_1 + 2$	$x_1 + 2x_2 = x_1 + 2(x_1 + 1) = 3x_1 + 2$	$x_1 + 2x_2 = x_1 + 2(x_1 + 1) = 3x_1 + 2$, which can be made as large as we like
johnston-linear-matrix-algebra-F02019	203	1.000	$(x_1 - x_2) + (-x_1 + x_2) = 0 \leq -1 - 1 = -2$	$(x_1 - x_2) + (-x_1 + x_2) = 0 \leq -1 - 1 = -2$	$(x_1 - x_2) + (-x_1 + x_2) = 0 \leq -1 - 1 = -2$, which is not true no matter what
johnston-linear-matrix-algebra-F02020	203	1.000	∞	∞	optimal value is ∞ , and if it is infeasible then we might say that its optimal value
johnston-linear-matrix-algebra-F02021	203	1.000	$-\infty$	$-\infty$	is $-\infty$. Similarly, if a minimization problem is unbounded then we sometimes
johnston-linear-matrix-algebra-F02022	204	1.000	$x_2 - x_3$	$x_2 - x_3$	by $x_2 - x_3$ where $x_2, x_3 \geq 0$.
johnston-linear-matrix-algebra-F02023	204	1.000	$x_2, x_3 \geq 0$	$x_2, x_3 \geq 0$	by $x_2 - x_3$ where $x_2, x_3 \geq 0$.
johnston-linear-matrix-algebra-F02024	204	1.000	$\mathbf{b} \geq \mathbf{0}$	$\mathbf{b} \geq \mathbf{0}$	Step 1.5: Hope that $\mathbf{b} \geq \mathbf{0}$. If $\mathbf{b}$ (the vector containing the coefficients on the right-
johnston-linear-matrix-algebra-F02025	204	1.000	$\mathbf{b} = (1, 2, 3) \geq \mathbf{0}$	$\mathbf{b} = (1, 2, 3) \geq \mathbf{0}$	fix it later—for this particular example we just note that $\mathbf{b} = (1, 2, 3) \geq \mathbf{0}$,
johnston-linear-matrix-algebra-F02026	204	1.000	$x \leq 5$	$x \leq 5$	the inequality $x \leq 5$ is equivalent to the pair of constraints $s + x = 5, s \geq 0$.
johnston-linear-matrix-algebra-F02027	204	1.000	$s + x = 5, s \geq 0$	$s + x = 5, s \geq 0$	the inequality $x \leq 5$ is equivalent to the pair of constraints $s + x = 5, s \geq 0$.
johnston-linear-matrix-algebra-F02028	204	1.000	s	s	The new variable s is called a slack variable , and it just measures the
johnston-linear-matrix-algebra-F02029	204	0.997	$z = \mathbf{c} \cdot \mathbf{x}$	$z = \mathbf{c} \cdot \mathbf{x}$	$z = \mathbf{c} \cdot \mathbf{x}$), and we rephrase the linear program as maximizing over z with
johnston-linear-matrix-algebra-F02030	204	1.000	$z - \mathbf{c} \cdot \mathbf{x} = 0$	$z - \mathbf{c} \cdot \mathbf{x} = 0$	the additional constraint that $z - \mathbf{c} \cdot \mathbf{x} = 0$. Making this change to the

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johnston-linear-matrix-algebra-F02031	205	1.000	$z - \mathbf{c}^T \mathbf{x} = 0$	$z - \mathbf{c}^T \mathbf{x} = 0$	In particular, the top row simply says that $z - \mathbf{c}^T \mathbf{x} = 0$ and the bottom
johnston-linear-matrix-algebra-F02032	205	1.000	$[\mathbf{0} \mid I \mid A \mid \mathbf{b}]$	$[\mathbf{0} \mid I \mid A \mid \mathbf{b}]$	block row $[\mathbf{0} \mid I \mid A \mid \mathbf{b}]$ represents the equality constraints involving slack
johnston-linear-matrix-algebra-F02033	205	1.000	z, s_1, s_2	z, s_1, s_2	it directly. In particular, the newly-introduced variables z, s_1, s_2 , and s_3
johnston-linear-matrix-algebra-F02034	205	1.000	s_3	s_3	it directly. In particular, the newly-introduced variables z, s_1, s_2 , and s_3
johnston-linear-matrix-algebra-F02035	206	1.000	$z=0$	$z = 0$	gives us a value of $z = 0$ in the objective function, and we would like to
johnston-linear-matrix-algebra-F02036	206	1.000	$z=2x_1+x_2-x_3$	$z = 2x_1 + x_2 - x_3$	To increase the value of the objective function $z = 2x_1 + x_2 - x_3$, we
johnston-linear-matrix-algebra-F02037	206	1.000	$x_2=x_3=0$	$x_2 = x_3 = 0$	Since we are leaving $x_2 = x_3 = 0$ alone for now, the constraints simply
johnston-linear-matrix-algebra-F02038	206	1.000	$x_1=2$	$x_1 = 2$	the second one, so we wish to set $x_1 = 2$ and update the other vari-
johnston-linear-matrix-algebra-F02039	206	0.643	$\begin{array}{l} z \text{ \& } s_1 \text{ \& } s_2 \text{ \& } s_3 \text{ \& } x_1 \text{ \& } x_2 \text{ \& } x_3 \end{array}$	$z \quad s_1 \quad s_2 \quad s_3 \quad x_1 \quad x_2 \quad x_3$	Diagram illustrating the transformation of a matrix. The top matrix has columns $z, s_1, s_2, s_3, x_1, x_2, x_3$. The bottom matrix is the result of row operations: R_1+2R_3, R_2+3R_3, R_4+R_3. The new "leading" entry is highlighted in blue in the bottom matrix.
johnston-linear-matrix-algebra-F02040	206	—	$\left[\begin{array}{c ccc ccc} 1 & 0 & 0 & 0 & -2 & -1 & 1 & 0 \\ 0 & 1 & 0 & 0 & -3 & -2 & 2 & 1 \\ 0 & 0 & 1 & 0 & 1 & -1 & 1 & 2 \\ 0 & 0 & 0 & 1 & -1 & 3 & -3 & 3 \end{array} \right]$	$\left[\begin{array}{c ccc ccc} 1 & 0 & 0 & 0 & -2 & -1 & 1 & 0 \\ 0 & 1 & 0 & 0 & -3 & -2 & 2 & 1 \\ 0 & 0 & 1 & 0 & 1 & -1 & 1 & 2 \\ 0 & 0 & 0 & 1 & -1 & 3 & -3 & 3 \end{array} \right]$	—
johnston-linear-matrix-algebra-F02041	206	—	$\xrightarrow{R_1+2R_3, R_2+3R_3, R_4+R_3} \left[\begin{array}{c ccc ccc} 1 & 0 & 2 & 0 & 0 & -3 & 3 & 4 \\ 0 & 1 & 3 & 0 & 0 & -5 & 5 & 7 \\ 0 & 0 & 1 & 0 & 1 & -1 & 1 & 2 \\ 0 & 0 & 1 & 1 & 0 \uparrow & 2 & -2 & 5 \end{array} \right]$	$\xrightarrow{R_1+2R_3, R_2+3R_3, R_4+R_3} \left[\begin{array}{c ccc ccc} 1 & 0 & 2 & 0 & 0 & -3 & 3 & 4 \\ 0 & 1 & 3 & 0 & 0 & -5 & 5 & 7 \\ 0 & 0 & 1 & 0 & 1 & -1 & 1 & 2 \\ 0 & 0 & 1 & 1 & 0 \uparrow & 2 & -2 & 5 \end{array} \right]$	—
johnston-linear-matrix-algebra-F02042	206	1.000	s_1, s_3	s_1, s_3	the equality constraints directly from it. We just have to think of s_1, s_3 ,
johnston-linear-matrix-algebra-F02043	206	1.000	s_2, x_2	s_2, x_2	columns) and s_2, x_2 , and x_3 as the free variables. Setting this new set of
johnston-linear-matrix-algebra-F02044	206	1.000	$z=4$	$z = 4$	of $z = 4$ in the objective function is better than the value of $z = 0$ that we
johnston-linear-matrix-algebra-F02045	207	1.000	$z=23/2-(7/2)s_2-(3/2)s_3$	$z = 23/2 - (7/2)s_2 - (3/2)s_3$	tells us that $z = 23/2 - (7/2)s_2 - (3/2)s_3$. Since $s_2, s_3 \geq 0$ we conclude
johnston-linear-matrix-algebra-F02046	207	1.000	$s_2, s_3 \geq 0$	$s_2, s_3 \geq 0$	tells us that $z = 23/2 - (7/2)s_2 - (3/2)s_3$. Since $s_2, s_3 \geq 0$ we conclude
johnston-linear-matrix-algebra-F02047	207	1.000	$z=23/2$	$z = 23/2$	that $z = 23/2$ is the optimal value of this linear program, and it is attained
johnston-linear-matrix-algebra-F02048	207	0.636	$x_1=9/2$	$x_1 = 9/2$	when $x_1 = 9/2$ and $x_2 = 5/2$ (and $s_1 = 39/2$ and $s_2 = s_3 = x_3 = 0$).
johnston-linear-matrix-algebra-F02049	207	0.636	$x_2=5/2$	$x_2 = 5/2$	when $x_1 = 9/2$ and $x_2 = 5/2$ (and $s_1 = 39/2$ and $s_2 = s_3 = x_3 = 0$).

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johnston-linear-matrix-algebra-F02050	207	0.636	$s_{\{1\}}=39 / 2$	$s_1 = 39/2$	when $x_1 = 9/2$ and $x_2 = 5/2$ (and $s_1 = 39/2$ and $s_2 = s_3 = x_3 = 0$).
johnston-linear-matrix-algebra-F02051	207	0.636	$s_{\{2\}}=s_{\{3\}}=x_{\{3\}}=0$	$s_2 = s_3 = x_3 = 0$	when $x_1 = 9/2$ and $x_2 = 5/2$ (and $s_1 = 39/2$ and $s_2 = s_3 = x_3 = 0$).
johnston-linear-matrix-algebra-F02052	207	1.000	$\left(x_{\{1\}}, x_{\{2\}}, x_{\{3\}}\right)=(0,0,0)$	$(x_1, x_2, x_3) = (0, 0, 0)$	from $(x_1, x_2, x_3) = (0, 0, 0)$ to $(x_1, x_2, x_3) = (2, 0, 0)$ and then to $(x_1, x_2, x_3) =$
johnston-linear-matrix-algebra-F02053	207	1.000	$\left(x_{\{1\}}, x_{\{2\}}, x_{\{3\}}\right)=(2,0,0)$	$(x_1, x_2, x_3) = (2, 0, 0)$	from $(x_1, x_2, x_3) = (0, 0, 0)$ to $(x_1, x_2, x_3) = (2, 0, 0)$ and then to $(x_1, x_2, x_3) =$
johnston-linear-matrix-algebra-F02054	207	0.755	$(9 / 2, 5 / 2, 0)$	$(9/2, 5/2, 0)$	$(9/2, 5/2, 0)$, as illustrated in Figure 2.35.
johnston-linear-matrix-algebra-F02055	208	1.000	$\operatorname{binom}{m}{n}$	$\binom{m}{n}$	have up to $\binom{m}{n}$
johnston-linear-matrix-algebra-F02056	208	0.940	$\left(x_{\{1\}}, x_{\{2\}}\right)=(0,0)$	$(x_1, x_2) = (0, 0)$	with a value of $z = 0$ in the objective function at $(x_1, x_2) = (0, 0)$ and then moves
johnston-linear-matrix-algebra-F02057	208	1.000	$\left(x_{\{1\}}, x_{\{2\}}\right)=(2,0)$	$(x_1, x_2) = (2, 0)$	to $(x_1, x_2) = (2, 0)$ to get a value of $z = 4$, and finally to $(x_1, x_2) = (9/2, 5/2)$ to get the
johnston-linear-matrix-algebra-F02058	208	1.000	$\left(x_{\{1\}}, x_{\{2\}}\right)=(9 / 2, 5 / 2)$	$(x_1, x_2) = (9/2, 5/2)$	to $(x_1, x_2) = (2, 0)$ to get a value of $z = 4$, and finally to $(x_1, x_2) = (9/2, 5/2)$ to get the
johnston-linear-matrix-algebra-F02059	208	1.000	$x_{\{\mathrm{p}\}}, x_{\{\mathrm{c}\}}$	$x_{\mathrm{p}}, x_{\mathrm{c}}$	We start by formulating this problem as a linear program. Let $x_{\mathrm{p}}, x_{\mathrm{c}}$,
johnston-linear-matrix-algebra-F02060	208	1.000	$x_{\{\mathrm{pc}\}}$	x_{pc}	and x_{pc} denote the number of dozen peanut butter, chocolate chip, and
johnston-linear-matrix-algebra-F02061	209	1.000	$x_{\{\mathrm{p}\}}, x_{\{\mathrm{c}\}}, x_{\{\mathrm{pc}\}} \geq 0$	$x_{\mathrm{p}}, x_{\mathrm{c}}, x_{\mathrm{pc}} \geq 0$	and each of the variables is non-negative as well (i.e., $x_{\mathrm{p}}, x_{\mathrm{c}}, x_{\mathrm{pc}} \geq 0$) since
johnston-linear-matrix-algebra-F02062	210	0.976	$s_{\{1\}}$	s_1	$ \begin{array}{l} \begin{array}{c} z \quad s_1 \quad s_2 \quad s_3 \quad x_{\mathrm{p}} \quad x_{\mathrm{c}} \quad x_{\mathrm{pc}} \\ \left[\begin{array}{cccccc cccc} 1 & 0 & 0 & 0 & -1 & -0.8 & -1.5 & 0 \\ 0 & 1 & 0 & 0 & 0 & 1 & 1 & 100 \\ 0 & 0 & 1 & 0 & 1 & 0 & 1 & 80 \\ 0 & 0 & 0 & 1 & 1 & 1 & 1 & 150 \end{array} \right] \\ \begin{array}{l} R_1+1.5R_3 \\ R_2-R_3 \\ R_4-R_3 \end{array} \rightarrow \\ \begin{array}{c} R_1+0.8R_2 \\ R_4-R_2 \end{array} \rightarrow \\ \begin{array}{l} R_1+0.3R_4 \\ R_2+R_4 \\ R_3-R_4 \end{array} \rightarrow \end{array} \end{array} $

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johnston-linear-matrix-algebra F02066	210	■ 0.976	$\left[\begin{array}{l}1 \\ \hline 0 \\ 0 \\ 0\end{array}\right]$	$\left[\begin{array}{l}1 \\ 0 \\ 0 \\ 0\end{array}\right]$	
johnston-linear-matrix-algebra F02067	210	■ 0.976	$R_{\{1\}}+1.5 R_{\{3\}}$	$R_1+1.5 R_3$	
johnston-linear-matrix-algebra F02068	210	■ 0.976	$\left[\begin{array}{l}1 \\ 0 \\ 0 \\ 0\end{array}\right]\right.$	$\left[\begin{array}{l}1 \\ 0 \\ 0 \\ 0\end{array}\right]$	

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johnston-linear-matrix-algebra-F02069	210	■ 0.976	$\xrightarrow{R_1+0.8 R_2}$	$\xrightarrow{R_1+0.8R_2}$	
johnston-linear-matrix-algebra-F02070	210	■ 0.976	$R_1+0.3 R_4$	$R_1 + 0.3R_4$	
johnston-linear-matrix-algebra-F02071	210	■ 0.994	$\$ 151$	\$151	most money that the bakery can make per week) is \$151. Furthermore, it
johnston-linear-matrix-algebra-F02072	210	■ 1.000	$x_{\mathrm{p}}=50$	$x_p = 50$	is attained when they bake $x_p = 50$ dozen peanut butter cookies, $x_c = 70$
johnston-linear-matrix-algebra-F02073	210	■ 1.000	$x_{\mathrm{c}}=70$	$x_c = 70$	is attained when they bake $x_p = 50$ dozen peanut butter cookies, $x_c = 70$
johnston-linear-matrix-algebra-F02074	210	■ 1.000	$x_{\mathrm{pc}}=30$	$x_{pc} = 30$	dozen chocolate chip cookies, and $x_{pc} = 30$ dozen peanut butter chocolate
johnston-linear-matrix-algebra-F02075	211	■ 0.703	$\left(x_{\{1\}}, x_{\{2\}}\right)=(0.8,1.6)$	$(x_1, x_2) = (0.8, 1.6)$	is 5.6, which is attained at the vector $(x_1, x_2) = (0.8, 1.6)$. If we try to find
johnston-linear-matrix-algebra-F02076	211	■ 1.000	$\left(x_{\{1\}}, x_{\{2\}}\right)=(0,1)$	$(x_1, x_2) = (0, 1)$	these values down to $(x_1, x_2) = (0, 1)$, we run into a big problem—this
johnston-linear-matrix-algebra-F02077	211	■ 1.000	$-2 x_{\{1\}}+x_{\{2\}} \leq 0$	$-2x_1 + x_2 \leq 0$	vector is not even feasible (it violates the constraint $-2x_1 + x_2 \leq 0$)!
johnston-linear-matrix-algebra-F02078	211	■ 1.000	$x_{\{1\}}=0.8$	$x_1 = 0.8$	Of the four possible ways of rounding each of $x_1 = 0.8$ and $x_2 = 1.6$ up
johnston-linear-matrix-algebra-F02079	211	■ 1.000	$x_{\{2\}}=1.6$	$x_2 = 1.6$	Of the four possible ways of rounding each of $x_1 = 0.8$ and $x_2 = 1.6$ up

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johnston-linear-matrix-algebra-F02080	211	1.000	$\text{\texttt{\left(x_{1}, x_{2}\right)=(1,1)}}$	$(x_1, x_2) = (1, 1)$	or down, only one results in a feasible vector: $(x_1, x_2) = (1, 1)$. However,
johnston-linear-matrix-algebra-F02081	211	1.000	$\text{\texttt{x_{1}+3 x_{2}=4}}$	$x_1 + 3x_2 = 4$	this vector is not optimal—it gives a value of $x_1 + 3x_2 = 4$ in the objec-
johnston-linear-matrix-algebra-F02082	211	0.997	$\text{\texttt{\mathbf{b} \not \geq \mathbf{0}}}$	$\mathbf{b} \not\geq \mathbf{0}$	which is only feasible if $\mathbf{b} \geq \mathbf{0}$. To illustrate the problem with having $\mathbf{b} \not\geq \mathbf{0}$
johnston-linear-matrix-algebra-F02083	212	1.000	$\text{\texttt{-y}}$	$-y$	maximize $-y$.
johnston-linear-matrix-algebra-F02084	212	1.000	$\text{\texttt{y \geq 0}}$	$y \geq 0$	To construct this feasibility LP we introduce yet another variable $y \geq 0$,
johnston-linear-matrix-algebra-F02085	212	1.000	$\text{\texttt{y=2}}$	$y = 2$	$y = 2$ (in general, we set each $x_j = 0$ and then set y large enough to satisfy all
johnston-linear-matrix-algebra-F02086	212	1.000	$\text{\texttt{\left(x_{1}, x_{2}, y\right)=\left(x_{1}, x_{2}, 0\right)}}$	$(x_1, x_2, y) = (x_1, x_2, 0)$	linear program (2.B.10) is feasible, since $(x_1, x_2, y) = (x_1, x_2, 0)$ is a feasible
johnston-linear-matrix-algebra-F02087	212	1.000	$\text{\texttt{\left(x_{1}, x_{2}, y\right)=(0,0,2)}}$	$(x_1, x_2, y) = (0, 0, 2)$	“obvious” feasible vector $(x_1, x_2, y) = (0, 0, 2)$ that we noted earlier, after
johnston-linear-matrix-algebra-F02088	212	1.000	$\text{\texttt{-2 x_{1}+x_{2}-y \leq -2}}$	$-2x_1 + x_2 - y \leq -2$	corresponding to the “ $-2x_1 + x_2 - y \leq -2$ ” constraint (since this one
johnston-linear-matrix-algebra-F02089	213	0.915	$\text{\texttt{(1 / 2) R_{4}}}$	$(1/2)R_4$	$\left[\begin{array}{cccc cccc} 1 & 0 & 0 & 1 & -2 & 1 & 0 & -2 \\ 0 & 1 & 0 & 0 & 1 & 1 & 0 & 2 \\ 0 & 0 & 1 & -1 & 1 & -2 & 0 & 1 \\ 0 & 0 & 0 & -1 & 2 & -1 & 1 & 2 \end{array} \right]$ $\begin{array}{l} (1/2)R_4 \\ R_1+2R_4 \\ R_2-R_4 \\ R_3-R_4 \\ \hline \end{array} \rightarrow \left[\begin{array}{cccc cccc} 1 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1/2 & 0 & 3/2 & -1/2 & 1 \\ 0 & 0 & 1 & -1/2 & 0 & -3/2 & -1/2 & 0 \\ 0 & 0 & 0 & -1/2 & 1 & -1/2 & 1/2 & 1 \end{array} \right]$
johnston-linear-matrix-algebra-F02090	213	0.915	$\text{\texttt{R_{1}+2 R_{4}}}$	$R_1 + 2R_4$	$\left[\begin{array}{cccc cccc} 1 & 0 & 0 & 1 & -2 & 1 & 0 & -2 \\ 0 & 1 & 0 & 0 & 1 & 1 & 0 & 2 \\ 0 & 0 & 1 & -1 & 1 & -2 & 0 & 1 \\ 0 & 0 & 0 & -1 & 2 & -1 & 1 & 2 \end{array} \right]$ $\begin{array}{l} (1/2)R_4 \\ R_1+2R_4 \\ R_2-R_4 \\ R_3-R_4 \\ \hline \end{array} \rightarrow \left[\begin{array}{cccc cccc} 1 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1/2 & 0 & 3/2 & -1/2 & 1 \\ 0 & 0 & 1 & -1/2 & 0 & -3/2 & -1/2 & 0 \\ 0 & 0 & 0 & -1/2 & 1 & -1/2 & 1/2 & 1 \end{array} \right]$
johnston-linear-matrix-algebra-F02091	213	0.915	$\text{\texttt{R_{2}-R_{4}}}$	$R_2 - R_4$	$\left[\begin{array}{cccc cccc} 1 & 0 & 0 & 1 & -2 & 1 & 0 & -2 \\ 0 & 1 & 0 & 0 & 1 & 1 & 0 & 2 \\ 0 & 0 & 1 & -1 & 1 & -2 & 0 & 1 \\ 0 & 0 & 0 & -1 & 2 & -1 & 1 & 2 \end{array} \right]$ $\begin{array}{l} (1/2)R_4 \\ R_1+2R_4 \\ R_2-R_4 \\ R_3-R_4 \\ \hline \end{array} \rightarrow \left[\begin{array}{cccc cccc} 1 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1/2 & 0 & 3/2 & -1/2 & 1 \\ 0 & 0 & 1 & -1/2 & 0 & -3/2 & -1/2 & 0 \\ 0 & 0 & 0 & -1/2 & 1 & -1/2 & 1/2 & 1 \end{array} \right]$
johnston-linear-matrix-algebra-F02092	213	0.915	$\text{\texttt{R_{3}-R_{4}}}$	$R_3 - R_4$	$\left[\begin{array}{cccc cccc} 1 & 0 & 0 & 1 & -2 & 1 & 0 & -2 \\ 0 & 1 & 0 & 0 & 1 & 1 & 0 & 2 \\ 0 & 0 & 1 & -1 & 1 & -2 & 0 & 1 \\ 0 & 0 & 0 & -1 & 2 & -1 & 1 & 2 \end{array} \right]$ $\begin{array}{l} (1/2)R_4 \\ R_1+2R_4 \\ R_2-R_4 \\ R_3-R_4 \\ \hline \end{array} \rightarrow \left[\begin{array}{cccc cccc} 1 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1/2 & 0 & 3/2 & -1/2 & 1 \\ 0 & 0 & 1 & -1/2 & 0 & -3/2 & -1/2 & 0 \\ 0 & 0 & 0 & -1/2 & 1 & -1/2 & 1/2 & 1 \end{array} \right]$

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johnston-linear-matrix-algebra F02093	213	0.982	L P	LP	value of this feasibility LP is 0, so the original linear program (2.B.10) is
johnston-linear-matrix-algebra F02094	213	1.000	$x_1=1$	$x_1 = 1$	feasible. In particular, since this final tableau has $x_1 = 1$ and $x_2 = 0$, we
johnston-linear-matrix-algebra F02095	213	1.000	$x_2=0$	$x_2 = 0$	feasible. In particular, since this final tableau has $x_1 = 1$ and $x_2 = 0$, we
johnston-linear-matrix-algebra F02096	213	1.000	$\text{\texttt{\textbackslash left}(x_{\{1\}}, x_{\{2\}}\text{\textbackslash right})=(1,0)}$	$(x_1, x_2) = (1, 0)$	conclude that $(x_1, x_2) = (1, 0)$ is a feasible vector of that original LP. It is
johnston-linear-matrix-algebra F02097	213	1.000	$-y<0$	$-y < 0$	If we had instead found an optimal value of $-y < 0$ in the above example,
johnston-linear-matrix-algebra F02098	213	0.494	$\text{\texttt{\textbackslash left.0 \begin{array}{l}0 \\ 2 \\ -1 \\ -2\end{array} \text{\textbackslash right}}}$	$\begin{bmatrix} 0 \\ 2 \\ -1 \\ -2 \end{bmatrix}$	$\begin{array}{c cccc cc} z & s_1 & s_2 & s_3 & x_1 & x_2 & \\ \hline 1 & 0 & 0 & 0 & -1 & -2 & 0 \\ 0 & 1 & 0 & 0 & 1 & 1 & 2 \\ 0 & 0 & 1 & 0 & -1 & -1 & -1 \\ 0 & 0 & 0 & 1 & -2 & 1 & -2 \end{array}$ $\xrightarrow{\begin{array}{l} R_1+R_2 \\ R_3+R_2 \\ R_4+2R_2 \end{array}} \begin{array}{c cccc cc} 1 & 1 & 0 & 0 & 0 & -1 & 2 \\ \hline 0 & 1 & 0 & 0 & 1 & 1 & 2 \\ 0 & 1 & 1 & 0 & 0 & 0 & 1 \\ 0 & 2 & 0 & 1 & 0 & 3 & 2 \end{array}$
johnston-linear-matrix-algebra F02099	213	0.494	$R_{\{1\}}+R_{\{2\}}$	$R_1 + R_2$	$\begin{array}{c cccc cc} z & s_1 & s_2 & s_3 & x_1 & x_2 & \\ \hline 1 & 0 & 0 & 0 & -1 & -2 & 0 \\ 0 & 1 & 0 & 0 & 1 & 1 & 2 \\ 0 & 0 & 1 & 0 & -1 & -1 & -1 \\ 0 & 0 & 0 & 1 & -2 & 1 & -2 \end{array}$ $\xrightarrow{\begin{array}{l} R_1+R_2 \\ R_3+R_2 \\ R_4+2R_2 \end{array}} \begin{array}{c cccc cc} 1 & 1 & 0 & 0 & 0 & -1 & 2 \\ \hline 0 & 1 & 0 & 0 & 1 & 1 & 2 \\ 0 & 1 & 1 & 0 & 0 & 0 & 1 \\ 0 & 2 & 0 & 1 & 0 & 3 & 2 \end{array}$
johnston-linear-matrix-algebra F02100	214	1.000	$\text{\texttt{\textbackslash xrightarrow{\substack{(1 / 3) R_{\{4\}} \\ R_{\{1\}}+R_{\{4\}} \\ R_{\{2\}}-R_{\{4\}}\}}}}$	$\xrightarrow{\begin{array}{l} (1/3)R_4 \\ R_1+R_4 \\ R_2-R_4 \end{array}}$	$\begin{array}{c cccc cc} 1 & 1 & 0 & 0 & 0 & -1 & 2 \\ \hline 0 & 1 & 0 & 0 & 1 & 1 & 2 \\ 0 & 1 & 1 & 0 & 0 & 0 & 1 \\ 0 & 2 & 0 & 1 & 0 & 3 & 2 \end{array}$ $\xrightarrow{\begin{array}{l} (1/3)R_4 \\ R_1+R_4 \\ R_2-R_4 \end{array}} \begin{array}{c cccc cc} 1 & 5/3 & 0 & 1/3 & 0 & 0 & 8/3 \\ \hline 0 & 1/3 & 0 & -1/3 & 1 & 0 & 4/3 \\ 0 & 1 & 1 & 0 & 0 & 0 & 1 \\ 0 & 2/3 & 0 & 1/3 & 0 & 1 & 2/3 \end{array}$
johnston-linear-matrix-algebra F02101	214	—	$\text{\texttt{\textbackslash left[\begin{array}{c ccc cc} 1 & 1 & 0 & 0 & 0 & -1 & 2 \\ \hline 0 & 1 & 0 & 0 & 1 & 1 & 2 \\ 0 & 1 & 1 & 0 & 0 & 0 & 1 \\ 0 & 2 & 0 & 1 & 0 & 3 & 2 \end{array}\text{\textbackslash right}]}$	$\begin{bmatrix} 1 & 1 & 0 & 0 & 0 & -1 & 2 \\ 0 & 1 & 0 & 0 & 1 & 1 & 2 \\ 0 & 1 & 1 & 0 & 0 & 0 & 1 \\ 0 & 2 & 0 & 1 & 0 & 3 & 2 \end{bmatrix}$	—
johnston-linear-matrix-algebra F02102	214	1.000	$\text{\texttt{\textbackslash left[\begin{array}{c ccc cc} 1 & 5 / 3 & 0 & 1 / 3 & 0 & 0 & 8 / 3 \\ \hline 0 & 1 / 3 & 0 & -1 / 3 & 1 & 0 & 4 / 3 \\ 0 & 1 & 1 & 0 & 0 & 0 & 1 \\ 0 & 2 / 3 & 0 & 1 / 3 & 0 & 1 & 2 / 3 \end{array}\text{\textbackslash right}]}$	$\begin{bmatrix} 1 & 5/3 & 0 & 1/3 & 0 & 0 & 8/3 \\ 0 & 1/3 & 0 & -1/3 & 1 & 0 & 4/3 \\ 0 & 1 & 1 & 0 & 0 & 0 & 1 \\ 0 & 2/3 & 0 & 1/3 & 0 & 1 & 2/3 \end{bmatrix}$	$\begin{array}{c cccc cc} 1 & 1 & 0 & 0 & 0 & -1 & 2 \\ \hline 0 & 1 & 0 & 0 & 1 & 1 & 2 \\ 0 & 1 & 1 & 0 & 0 & 0 & 1 \\ 0 & 2 & 0 & 1 & 0 & 3 & 2 \end{array}$ $\xrightarrow{\begin{array}{l} (1/3)R_4 \\ R_1+R_4 \\ R_2-R_4 \end{array}} \begin{array}{c cccc cc} 1 & 5/3 & 0 & 1/3 & 0 & 0 & 8/3 \\ \hline 0 & 1/3 & 0 & -1/3 & 1 & 0 & 4/3 \\ 0 & 1 & 1 & 0 & 0 & 0 & 1 \\ 0 & 2/3 & 0 & 1/3 & 0 & 1 & 2/3 \end{array}$

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johnston-linear-matrix-augm F02103	214	1.000	$\left(x_1, x_2\right)=(4 / 3, 2 / 3)$	$(x_1, x_2) = (4/3, 2/3)$	attained at the vector $(x_1, x_2) = (4/3, 2/3)$. We can visualize the feasible
johnston-linear-matrix-augm F02104	215	1.000	$y_1, y_2 \geq 0$	$y_1, y_2 \geq 0$	We need $y_1, y_2 \geq 0$
johnston-linear-matrix-augm F02105	215	0.998	$\left(x_1+x_2 \leq 3\right)$	$(x_1+x_2 \leq 3)$	But if we now add the first constraint $(x_1+x_2 \leq 3)$ again, we similarly see that
johnston-linear-matrix-augm F02106	215	1.000	y_1	y_1	A bit more generally, if we add up y_1 times the first constraint and y_2 times
johnston-linear-matrix-augm F02107	215	1.000	y_2	y_2	A bit more generally, if we add up y_1 times the first constraint and y_2 times
johnston-linear-matrix-augm F02108	215	1.000	$3 y_1+y_2$	$3 y_1+y_2$	then $3 y_1+y_2$ must also be an upper bound on $x_1+2 x_2$ (i.e., the objective
johnston-linear-matrix-augm F02109	215	1.000	$x_1+2 x_2$	$x_1+2 x_2$	then $3 y_1+y_2$ must also be an upper bound on $x_1+2 x_2$ (i.e., the objective
johnston-linear-matrix-augm F02110	215	1.000	$y_1=3 / 2$	$y_1=3 / 2$	function of the linear program). It is straightforward to check that $y_1=3 / 2$,
johnston-linear-matrix-augm F02111	215	1.000	$y_2=1 / 2$	$y_2=1 / 2$	$y_2=1 / 2$ works and establishes the bound $x_1+2 x_2 \leq 3(3 / 2)+(1 / 2)=5$, just
johnston-linear-matrix-augm F02112	215	1.000	$x_1+2 x_2 \leq 3(3 / 2)+(1 / 2)=5$	$x_1+2 x_2 \leq 3(3 / 2)+(1 / 2)=5$	$y_2=1 / 2$ works and establishes the bound $x_1+2 x_2 \leq 3(3 / 2)+(1 / 2)=5$, just
johnston-linear-matrix-augm F02113	215	0.997	$A \in \mathcal{M}_{m, n}(\mathbb{R}), \mathbf{b} \in \mathbb{R}^m, \mathbf{c} \in \mathbb{R}^n$	$A \in \mathcal{M}_{m, n}(\mathbb{R}), \mathbf{b} \in \mathbb{R}^m, \mathbf{c} \in \mathbb{R}^n$	Suppose $A \in \mathcal{M}_{m, n}(\mathbb{R}), \mathbf{b} \in \mathbb{R}^m, \mathbf{c} \in \mathbb{R}^n$ are fixed, and $\mathbf{x} \in \mathbb{R}^n, \mathbf{y} \in \mathbb{R}^m$ are
johnston-linear-matrix-augm F02114	215	0.997	$\mathbf{x} \in \mathbb{R}^n, \mathbf{y} \in \mathbb{R}^m$	$\mathbf{x} \in \mathbb{R}^n, \mathbf{y} \in \mathbb{R}^m$	Suppose $A \in \mathcal{M}_{m, n}(\mathbb{R}), \mathbf{b} \in \mathbb{R}^m, \mathbf{c} \in \mathbb{R}^n$ are fixed, and $\mathbf{x} \in \mathbb{R}^n, \mathbf{y} \in \mathbb{R}^m$ are
johnston-linear-matrix-augm F02115	216	1.000	$\mathbf{y} \cdot(A \mathbf{x})=\mathbf{y}^T A \mathbf{x}=\mathbf{y} \cdot \mathbf{b}$	$\mathbf{y} \cdot(A \mathbf{x})=\mathbf{y}^T A \mathbf{x}=\mathbf{y} \cdot \mathbf{b}$	$\mathbf{y} \cdot(A \mathbf{x})=\mathbf{y}^T A \mathbf{x}=\mathbf{y} \cdot \mathbf{b}$
johnston-linear-matrix-augm F02116	216	1.000	$(A^T \mathbf{y}) \cdot \mathbf{x}$	$(A^T \mathbf{y}) \cdot \mathbf{x}$	$(A^T \mathbf{y}) \cdot \mathbf{x}$. This is a
johnston-linear-matrix-augm F02117	216	1.000	$\mathbf{y} \in \mathbb{R}^m$	$\mathbf{y} \in \mathbb{R}^m$	If $\mathbf{x} \in \mathbb{R}^n$ is a feasible vector of a (primal) linear program and $\mathbf{y} \in \mathbb{R}^m$ is a
johnston-linear-matrix-augm F02118	216	1.000	$\mathbf{y} \cdot(A \mathbf{x}) \leq \mathbf{y} \cdot \mathbf{b}$	$\mathbf{y} \cdot(A \mathbf{x}) \leq \mathbf{y} \cdot \mathbf{b}$	$\mathbf{y} \cdot(A \mathbf{x}) \leq \mathbf{y} \cdot \mathbf{b}$. However, by using the fact that $A^T \mathbf{y} \geq \mathbf{c}$, a similar argument
johnston-linear-matrix-augm F02119	216	1.000	$A^T \mathbf{y} \geq \mathbf{c}$	$A^T \mathbf{y} \geq \mathbf{c}$	$\mathbf{y} \cdot(A \mathbf{x}) \leq \mathbf{y} \cdot \mathbf{b}$. However, by using the fact that $A^T \mathbf{y} \geq \mathbf{c}$, a similar argument
johnston-linear-matrix-augm F02120	216	1.000	$\mathbf{y} \cdot(A \mathbf{x})=\left(A^T \mathbf{y}\right) \cdot \mathbf{x} \geq \mathbf{c} \cdot \mathbf{x}$	$\mathbf{y} \cdot(A \mathbf{x})=\left(A^T \mathbf{y}\right) \cdot \mathbf{x} \geq \mathbf{c} \cdot \mathbf{x}$	shows that $\mathbf{y} \cdot(A \mathbf{x})=\left(A^T \mathbf{y}\right) \cdot \mathbf{x} \geq \mathbf{c} \cdot \mathbf{x}$. Stringing these inequalities together
johnston-linear-matrix-augm F02121	216	1.000	$\mathbf{c} \cdot \mathbf{x} \leq \mathbf{y} \cdot \mathbf{b}=\mathbf{b} \cdot \mathbf{y}$	$\mathbf{c} \cdot \mathbf{x} \leq \mathbf{y} \cdot \mathbf{b}=\mathbf{b} \cdot \mathbf{y}$	shows that $\mathbf{c} \cdot \mathbf{x} \leq \mathbf{y} \cdot \mathbf{b}=\mathbf{b} \cdot \mathbf{y}$, as desired.
johnston-linear-matrix-augm F02122	216	1.000	$\mathbf{c} \cdot \mathbf{x}=\mathbf{b} \cdot \mathbf{y}$	$\mathbf{c} \cdot \mathbf{x}=\mathbf{b} \cdot \mathbf{y}$	problems, respectively, for which $\mathbf{c} \cdot \mathbf{x}=\mathbf{b} \cdot \mathbf{y}$, then they must be optimal. In
johnston-linear-matrix-augm F02123	216	0.984	$\mathbf{c} \cdot \mathbf{x}=5$	$\mathbf{c} \cdot \mathbf{x}=5$	program that gives $\mathbf{c} \cdot \mathbf{x}=5$, so 5 is a lower bound on the optimal value of
johnston-linear-matrix-augm F02124	217	1.000	$\left(y_1, y_2\right)=(3 / 2, 1 / 2)$	$(y_1, y_2) = (3/2, 1/2)$	It is straightforward to verify that $(y_1, y_2) = (3/2, 1/2)$ is a feasible vector
johnston-linear-matrix-augm F02125	217	0.679	$\mathbf{b} \cdot \mathbf{y}=5$	$\mathbf{b} \cdot \mathbf{y}=5$	of the dual program, and it gives $\mathbf{b} \cdot \mathbf{y}=5$, so 5 is an upper bound on the
johnston-linear-matrix-augm F02126	217	1.000	$\left(x_1, x_2, x_3, x_4, x_5\right)=(1, 1, 1, 1, 1)$	$(x_1, x_2, x_3, x_4, x_5) = (1, 1, 1, 1, 1)$	It does not take long to see that $(x_1, x_2, x_3, x_4, x_5) = (1, 1, 1, 1, 1)$ is a
johnston-linear-matrix-augm F02127	217	0.998	x_2+x_3	x_2+x_3	of x_2+x_3 and x_4+x_5 in order for the first and fourth constraints to still be
johnston-linear-matrix-augm F02128	217	0.998	x_4+x_5	x_4+x_5	of x_2+x_3 and x_4+x_5 in order for the first and fourth constraints to still be
johnston-linear-matrix-augm F02129	218	1.000	$\mathbf{x}_*$	$\mathbf{x}_*$	$\mathbf{x}_*$ and $\mathbf{y}_*$ are

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johnston-linear-matrix-algebra F02130	218	■ 1.000	$\mathbf{y}_*$	$\mathbf{y}_*$	$\mathbf{x}_*$ and $\mathbf{y}_*$ are
johnston-linear-matrix-algebra F02131	218	■ 1.000	$\left(y_{\{1\}}, y_{\{2\}}, y_{\{3\}}, y_{\{4\}}, y_{\{5\}}\right)=(1,1,1,1,1) / 3$	$(y_1, y_2, y_3, y_4, y_5) = (1, 1, 1, 1, 1)/3$	that the vector $(y_1, y_2, y_3, y_4, y_5) = (1, 1, 1, 1, 1)/3$ works. Since we have
johnston-linear-matrix-algebra F02132	218	■ 1.000	$\mathbf{x}_* \in \mathbb{R}^n$	$\mathbf{x}_* \in \mathbb{R}^n$	If $\mathbf{x}_* \in \mathbb{R}^n$ is a feasible vector of a (primal) linear program satisfying
johnston-linear-matrix-algebra F02133	218	■ 1.000	$\mathbf{c} \cdot \mathbf{x}_* \geq \mathbf{c} \cdot \mathbf{x}$	$\mathbf{c} \cdot \mathbf{x}_* \geq \mathbf{c} \cdot \mathbf{x}$	$\mathbf{c} \cdot \mathbf{x}_* \geq \mathbf{c} \cdot \mathbf{x}$ for all feasible vectors $\mathbf{x} \in \mathbb{R}^n$, then there exists a feasible
johnston-linear-matrix-algebra F02134	218	■ 1.000	$\mathbf{y}_* \in \mathbb{R}^m$	$\mathbf{y}_* \in \mathbb{R}^m$	vector $\mathbf{y}_* \in \mathbb{R}^m$ of its dual problem such that
johnston-linear-matrix-algebra F02135	218	■ 0.993	$\mathbf{b} \cdot \mathbf{y}$	$\mathbf{b} \cdot \mathbf{y}$	problem by some quantity $\mathbf{b} \cdot \mathbf{y}$. A similar argument shows that if a dual problem
johnston-linear-matrix-algebra F02136	219	■ 0.877	$\quad y$	y	Primal maximize: $x_1 + 2x_2$ subject to: $-x_1 + x_2 \leq 1$ $x_1, x_2 \geq 0$ Dual minimize: y subject to: $-y \geq 1$ $y \geq 2$ $y \geq 0$
johnston-linear-matrix-algebra F02137	219	■ 0.877	$-x_{\{1\}}+x_{\{2\}} \leq 1$	$-x_1 + x_2 \leq 1$	Primal maximize: $x_1 + 2x_2$ subject to: $-x_1 + x_2 \leq 1$ $x_1, x_2 \geq 0$ Dual minimize: y subject to: $-y \geq 1$ $y \geq 2$ $y \geq 0$
johnston-linear-matrix-algebra F02138	219	■ 0.877	$\quad -y \geq 1$	$-y \geq 1$	Primal maximize: $x_1 + 2x_2$ subject to: $-x_1 + x_2 \leq 1$ $x_1, x_2 \geq 0$ Dual minimize: y subject to: $-y \geq 1$ $y \geq 2$ $y \geq 0$
johnston-linear-matrix-algebra F02139	219	■ 0.877	$x_{\{1\}}, \quad x_{\{2\}} \geq 0$	$x_1, x_2 \geq 0$	Primal maximize: $x_1 + 2x_2$ subject to: $-x_1 + x_2 \leq 1$ $x_1, x_2 \geq 0$ Dual minimize: y subject to: $-y \geq 1$ $y \geq 2$ $y \geq 0$
johnston-linear-matrix-algebra F02140	219	■ 0.877	$y \geq 2$	$y \geq 2$	Primal maximize: $x_1 + 2x_2$ subject to: $-x_1 + x_2 \leq 1$ $x_1, x_2 \geq 0$ Dual minimize: y subject to: $-y \geq 1$ $y \geq 2$ $y \geq 0$
johnston-linear-matrix-algebra F02141	219	■ 1.000	$-y \geq 1$	$-y \geq 1$	possible that $-y \geq 1$ and $y \geq 0$ simultaneously).
johnston-linear-matrix-algebra F02142	219	■ 0.987	$-y_{\{1\}}-y_{\{2\}}$	$-y_1 - y_2$	Primal maximize: $x_1 + 2x_2$ subject to: $x_1 - x_2 \leq -1$ $-x_1 + x_2 \leq -1$ $x_1, x_2 \geq 0$ Dual minimize: $-y_1 - y_2$ subject to: $y_1 - y_2 \geq 1$ $-y_1 + y_2 \geq 2$ $y_1, y_2 \geq 0$

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johnston-linear-matrix-algebra F02143	219	■ 0.987	$x_{\{1\}}-x_{\{2\}} \leq -1$	$x_1 - x_2 \leq -1$	Primal maximize: $x_1 + 2x_2$ subject to: $x_1 - x_2 \leq -1$ $-x_1 + x_2 \leq -1$ $x_1, x_2 \geq 0$ Dual minimize: $-y_1 - y_2$ subject to: $y_1 - y_2 \geq 1$ $-y_1 + y_2 \geq 2$ $y_1, y_2 \geq 0$
johnston-linear-matrix-algebra F02144	219	■ 0.987	$\quad y_{\{1\}}-y_{\{2\}} \geq 1$	$y_1 - y_2 \geq 1$	Primal maximize: $x_1 + 2x_2$ subject to: $x_1 - x_2 \leq -1$ $-x_1 + x_2 \leq -1$ $x_1, x_2 \geq 0$ Dual minimize: $-y_1 - y_2$ subject to: $y_1 - y_2 \geq 1$ $-y_1 + y_2 \geq 2$ $y_1, y_2 \geq 0$
johnston-linear-matrix-algebra F02145	219	■ 0.987	$-x_{\{1\}}+x_{\{2\}} \leq -1$	$-x_1 + x_2 \leq -1$	Primal maximize: $x_1 + 2x_2$ subject to: $x_1 - x_2 \leq -1$ $-x_1 + x_2 \leq -1$ $x_1, x_2 \geq 0$ Dual minimize: $-y_1 - y_2$ subject to: $y_1 - y_2 \geq 1$ $-y_1 + y_2 \geq 2$ $y_1, y_2 \geq 0$
johnston-linear-matrix-algebra F02146	219	■ 0.987	$-y_{\{1\}}+y_{\{2\}} \geq 2$	$-y_1 + y_2 \geq 2$	Primal maximize: $x_1 + 2x_2$ subject to: $x_1 - x_2 \leq -1$ $-x_1 + x_2 \leq -1$ $x_1, x_2 \geq 0$ Dual minimize: $-y_1 - y_2$ subject to: $y_1 - y_2 \geq 1$ $-y_1 + y_2 \geq 2$ $y_1, y_2 \geq 0$
johnston-linear-matrix-algebra F02147	219	■ 0.987	$y_{\{1\}}, \quad y_{\{2\}} \geq 0$	$y_1, y_2 \geq 0$	Primal maximize: $x_1 + 2x_2$ subject to: $x_1 - x_2 \leq -1$ $-x_1 + x_2 \leq -1$ $x_1, x_2 \geq 0$ Dual minimize: $-y_1 - y_2$ subject to: $y_1 - y_2 \geq 1$ $-y_1 + y_2 \geq 2$ $y_1, y_2 \geq 0$
johnston-linear-matrix-algebra F02148	219	■ 1.000	$0 \geq 3$	$0 \geq 3$	(false) inequality $0 \geq 3$.
johnston-linear-matrix-algebra F02149	220	■ 1.000	$y_{\{1\}}-y_{\{2\}}$	$y_1 - y_2$	but rather always appear together in the form $y_1 - y_2$. We can thus replace
johnston-linear-matrix-algebra F02150	220	■ 1.000	$y_{\{*\}}$	y_*	$y_1 - y_2$ with the single (potentially negative) variable y_* , which results in the
johnston-linear-matrix-algebra F02151	220	■ 0.852	≥ 0	≥ 0	unconstrained (rather than ≥ 0). This makes a fair amount of sense intuitively—
johnston-linear-matrix-algebra F02152	221	—	$\begin{aligned} & \geq \\ & \leq \\ & \leq \end{aligned}$	$\geq$ $\leq$ $\leq$	—
johnston-linear-matrix-algebra F02153	221	—	$\begin{aligned} & \leq 0 \\ & \text{unc} \\ & \geq 0 \end{aligned}$	≤ 0 unc ≥ 0	—
johnston-linear-matrix-algebra F02154	221	—	$\begin{aligned} & \geq 0 \\ & \text{unconstrained} \\ & \leq 0 \end{aligned}$	≥ 0 unconstrained ≤ 0	—
johnston-linear-matrix-algebra F02155	221	—	$\begin{aligned} & \geq \\ & = \\ & \leq \end{aligned}$	$\geq$ $=$ $\leq$	—
johnston-linear-matrix-algebra F02156	221	■ 1.000	$\left(x_{\{1\}}, x_{\{2\}}\right)=(0,-3)$	$(x_1, x_2) = (0, -3)$	a while. It does not take long to find the vector $(x_1, x_2) = (0, -3)$, which
johnston-linear-matrix-algebra F02157	221	■ 0.999	$\mathbf{c} \cdot \mathbf{x} = -3$	$\mathbf{c} \cdot \mathbf{x} = -3$	is feasible for the primal problem and has $\mathbf{c} \cdot \mathbf{x} = -3$, and the vector

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johnston-linear-matrix-algebra-F02158	221	■ 0.965	$\left(y_{\{1\}}, y_{\{2\}}\right)=(1,0)$	$(y_1, y_2) = (1, 0)$	$(y_1, y_2) = (1, 0)$, which is feasible for the dual problem and has $\mathbf{b} \cdot \mathbf{y} = -3$.
johnston-linear-matrix-algebra-F02159	221	■ 0.965	$\mathbf{b} \cdot \mathbf{y} = -3$	$\mathbf{b} \cdot \mathbf{y} = -3$	$(y_1, y_2) = (1, 0)$, which is feasible for the dual problem and has $\mathbf{b} \cdot \mathbf{y} = -3$.
johnston-linear-matrix-algebra-F02160	222	■ 1.000	$\mathbf{x} \leq A^{-1} \mathbf{b}$	$\mathbf{x} \leq A^{-1} \mathbf{b}$	*(a) If $A\mathbf{x} \leq \mathbf{b}$ and A is invertible, then $\mathbf{x} \leq A^{-1} \mathbf{b}$.
johnston-linear-matrix-algebra-F02161	222	■ 1.000	$\mathbf{c}, \mathbf{x}, \mathbf{y} \in \mathbb{R}^n$	$\mathbf{c}, \mathbf{x}, \mathbf{y} \in \mathbb{R}^n$	(b) If $\mathbf{c}, \mathbf{x}, \mathbf{y} \in \mathbb{R}^n$ and $\mathbf{x} \geq \mathbf{y}$ then $\mathbf{c} \cdot \mathbf{x} \geq \mathbf{c} \cdot \mathbf{y}$.
johnston-linear-matrix-algebra-F02162	222	■ 1.000	$\mathbf{x} \geq \mathbf{y}$	$\mathbf{x} \geq \mathbf{y}$	(b) If $\mathbf{c}, \mathbf{x}, \mathbf{y} \in \mathbb{R}^n$ and $\mathbf{x} \geq \mathbf{y}$ then $\mathbf{c} \cdot \mathbf{x} \geq \mathbf{c} \cdot \mathbf{y}$.
johnston-linear-matrix-algebra-F02163	222	■ 1.000	$\mathbf{c} \cdot \mathbf{x} \geq \mathbf{c} \cdot \mathbf{y}$	$\mathbf{c} \cdot \mathbf{x} \geq \mathbf{c} \cdot \mathbf{y}$	(b) If $\mathbf{c}, \mathbf{x}, \mathbf{y} \in \mathbb{R}^n$ and $\mathbf{x} \geq \mathbf{y}$ then $\mathbf{c} \cdot \mathbf{x} \geq \mathbf{c} \cdot \mathbf{y}$.
johnston-linear-matrix-algebra-F02164	222	■ 1.000	$\mathbf{c}, \mathbf{x}, \mathbf{y} \in \mathbb{R}^n, \mathbf{c} \geq \mathbf{0}$	$\mathbf{c}, \mathbf{x}, \mathbf{y} \in \mathbb{R}^n, \mathbf{c} \geq \mathbf{0}$	*(c) If $\mathbf{c}, \mathbf{x}, \mathbf{y} \in \mathbb{R}^n, \mathbf{c} \geq \mathbf{0}$ and $\mathbf{x} \geq \mathbf{y}$, then $\mathbf{c} \cdot \mathbf{x} \geq \mathbf{c} \cdot \mathbf{y}$.
johnston-linear-matrix-algebra-F02165	223	■ 1.000	$0 < c \in \mathbb{R}$	$0 < c \in \mathbb{R}$	**2.B.8 Suppose $0 < c \in \mathbb{R}$ and consider the linear pro-
johnston-linear-matrix-algebra-F02166	223	■ 1.000	$\mathbf{x}_1, \dots, \mathbf{x}_k, \mathbf{y} \in \mathbb{R}^n$	$\mathbf{x}_1, \dots, \mathbf{x}_k, \mathbf{y} \in \mathbb{R}^n$	(a) Suppose $\mathbf{x}_1, \dots, \mathbf{x}_k, \mathbf{y} \in \mathbb{R}^n$. Construct a linear
johnston-linear-matrix-algebra-F02167	223	■ 1.000	$\mathbf{y} \in$	$\mathbf{y} \in$	program that determines whether or not $\mathbf{y} \in$
johnston-linear-matrix-algebra-F02168	223	■ 1.000	$\operatorname{span}(\mathbf{x}_1, \dots, \mathbf{x}_k)$	$\operatorname{span}(\mathbf{x}_1, \dots, \mathbf{x}_k)$	$\operatorname{span}(\mathbf{x}_1, \dots, \mathbf{x}_k)$.
johnston-linear-matrix-algebra-F02169	223	■ 0.827	$\mathbf{y} \in \operatorname{span}(\mathbf{x}_1, \dots, \mathbf{x}_k)$	$\mathbf{y} \in \operatorname{span}(\mathbf{x}_1, \dots, \mathbf{x}_k)$	show that $\mathbf{y} \in \operatorname{span}(\mathbf{x}_1, \dots, \mathbf{x}_k)$ if and only if $\mathbf{y}$ is
johnston-linear-matrix-algebra-F02170	223	■ 1.000	$\mathbf{x}_1, \dots, \mathbf{x}_k$	$\mathbf{x}_1, \dots, \mathbf{x}_k$	orthogonal to every vector that $\mathbf{x}_1, \dots, \mathbf{x}_k$ are orthog-
johnston-linear-matrix-algebra-F02171	223	■ 1.000	$\mathbf{x}_1 \cdot \mathbf{z} = \dots = \mathbf{x}_k \cdot \mathbf{z} = 0$	$\mathbf{x}_1 \cdot \mathbf{z} = \dots = \mathbf{x}_k \cdot \mathbf{z} = 0$	onal to (i.e., $\mathbf{x}_1 \cdot \mathbf{z} = \dots = \mathbf{x}_k \cdot \mathbf{z} = 0$ implies $\mathbf{y} \cdot \mathbf{z} = 0$).
johnston-linear-matrix-algebra-F02172	223	■ 1.000	$\mathbf{y} \cdot \mathbf{z} = 0$	$\mathbf{y} \cdot \mathbf{z} = 0$	onal to (i.e., $\mathbf{x}_1 \cdot \mathbf{z} = \dots = \mathbf{x}_k \cdot \mathbf{z} = 0$ implies $\mathbf{y} \cdot \mathbf{z} = 0$).
johnston-linear-matrix-algebra-F02173	223	■ 1.000	$\ A\mathbf{x} - \mathbf{b}\ _1$	$\ A\mathbf{x} - \mathbf{b}\ _1$	that makes $\ A\mathbf{x} - \mathbf{b}\ _1$ as small as possible.
johnston-linear-matrix-algebra-F02174	223	■ 1.000	$ a \leq b$	$ a \leq b$	[Hint: $ a \leq b$ if and only if $-b \leq a \leq b$.]
johnston-linear-matrix-algebra-F02175	223	■ 1.000	$-b \leq a \leq b$	$-b \leq a \leq b$	[Hint: $ a \leq b$ if and only if $-b \leq a \leq b$.]
johnston-linear-matrix-algebra-F02176	223	■ 1.000	$\ A\mathbf{x} - \mathbf{b}\ _\infty$	$\ A\mathbf{x} - \mathbf{b}\ _\infty$	that makes $\ A\mathbf{x} - \mathbf{b}\ _\infty$ as small as possible.
johnston-linear-matrix-algebra-F02177	223	■ 1.000	$a_{1,j} + a_{2,j} + \dots + a_{n,j} = 1$	$a_{1,j} + a_{2,j} + \dots + a_{n,j} = 1$	(i.e., $a_{1,j} + a_{2,j} + \dots + a_{n,j} = 1$ for each $1 \leq j \leq n$). Show
johnston-linear-matrix-algebra-F02178	223	■ 1.000	$A\mathbf{x} = \mathbf{x}$	$A\mathbf{x} = \mathbf{x}$	for which $A\mathbf{x} = \mathbf{x}$ and $\mathbf{x} \geq \mathbf{0}$.
johnston-linear-matrix-algebra-F02179	223	■ 1.000	$A^T \mathbf{y} = \mathbf{y} \dots$	$A^T \mathbf{y} = \mathbf{y} \dots$	“obvious” vector $\mathbf{y} \in \mathbb{R}^n$ for which $A^T \mathbf{y} = \mathbf{y}$... what is it?]
johnston-linear-matrix-algebra-F02180	223	■ 1.000	$A = CR$	$A = CR$	$A \in \mathcal{M}_{m,n}$, then we see that if we can write $A = CR$ for some $C \in \mathcal{M}_{m,r}$ and
johnston-linear-matrix-algebra-F02181	223	■ 1.000	$C \in \mathcal{M}_{m,r}$	$C \in \mathcal{M}_{m,r}$	$A \in \mathcal{M}_{m,n}$, then we see that if we can write $A = CR$ for some $C \in \mathcal{M}_{m,r}$ and
johnston-linear-matrix-algebra-F02182	223	■ 1.000	$R \in \mathcal{M}_{r,n}$	$R \in \mathcal{M}_{r,n}$	$R \in \mathcal{M}_{r,n}$, then we must have
johnston-linear-matrix-algebra-F02183	224	■ 1.000	$\operatorname{rank}(A) \leq r$	$\operatorname{rank}(A) \leq r$	<i>Proof.</i> We already showed that $\operatorname{rank}(A) \leq r$, so to complete the proof we just
johnston-linear-matrix-algebra-F02184	224	■ 1.000	$A = P\hat{R}$	$A = P\hat{R}$	recall from Theorem 2.5.3 that we can write $A = P\hat{R}$, where P is some invertible
johnston-linear-matrix-algebra-F02185	224	■ 1.000	$\hat{R}$	$\hat{R}$	matrix and $\hat{R}$ is the reduced row echelon form of A . If $r = \operatorname{rank}(A)$ then the first
johnston-linear-matrix-algebra-F02186	225	■ 1.000	C^T	C^T	We just write C^T

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johnston-linear-matrix-algebra-F02187	225	1.000	$\hat{R}=E\ A$	$\hat{R} = EA$	It follows that $\hat{R} = EA$ (and equivalently, $A = P\hat{R}$, where $P = E^{-1}$), where
johnston-linear-matrix-algebra-F02188	225	1.000	$P=E^{-1}$	$P = E^{-1}$	It follows that $\hat{R} = EA$ (and equivalently, $A = P\hat{R}$, where $P = E^{-1}$), where
johnston-linear-matrix-algebra-F02189	225	1.000	$\operatorname{rank}(A)=2$	$\operatorname{rank}(A) = 2$	Since $\hat{R}$ has two non-zero rows, it follows that $\operatorname{rank}(A) = 2$. The matrices
johnston-linear-matrix-algebra-F02190	225	1.000	$m\ r$	mr	rank- r matrix $A \in \mathcal{M}_{m,n}$, we can just keep track of the mr entries of C and rn
johnston-linear-matrix-algebra-F02191	225	1.000	$r\ n$	rn	rank- r matrix $A \in \mathcal{M}_{m,n}$, we can just keep track of the mr entries of C and rn
johnston-linear-matrix-algebra-F02192	225	0.998	$11\ \times\ 11=121$	$11 \times 11 = 121$	we could list all $11 \times 11 = 121$ of its entries. However, a more economical way
johnston-linear-matrix-algebra-F02193	225	1.000	$(11\ \times\ 2)+(2\ \times\ 11)=44$	$(11 \times 2) + (2 \times 11) = 44$	track of $(11 \times 2) + (2 \times 11) = 44$ numbers instead of the 121 numbers required
johnston-linear-matrix-algebra-F02194	225	1.000	$r=1$	$r = 1$	In the most extreme case when $r = 1$, Theorem 2.C.1 says that a non-zero
johnston-linear-matrix-algebra-F02195	226	1.000	$\left\{\mathbf{v}_j\right\}_{j=1}^r\ \subset\ \mathbb{R}^m$	$\{\mathbf{v}_j\}_{j=1}^r \subset \mathbb{R}^m$	vectors $\{\mathbf{v}_j\}_{j=1}^r \subset \mathbb{R}^m$ and $\{\mathbf{w}_j\}_{j=1}^r \subset \mathbb{R}^n$ with
johnston-linear-matrix-algebra-F02196	226	1.000	$\left\{\mathbf{w}_j\right\}_{j=1}^r\ \subset\ \mathbb{R}^n$	$\{\mathbf{w}_j\}_{j=1}^r \subset \mathbb{R}^n$	vectors $\{\mathbf{v}_j\}_{j=1}^r \subset \mathbb{R}^m$ and $\{\mathbf{w}_j\}_{j=1}^r \subset \mathbb{R}^n$ with
johnston-linear-matrix-algebra-F02197	226	1.000	$\left\{\mathbf{v}_i\right\}_{i=1}^r$	$\{\mathbf{v}_i\}_{i=1}^r$	is exactly $r = \operatorname{rank}(A)$. Furthermore, the sets $\{\mathbf{v}_i\}_{i=1}^r$ and $\{\mathbf{w}_i\}_{i=1}^r$ can be
johnston-linear-matrix-algebra-F02198	226	1.000	$\left\{\mathbf{w}_i\right\}_{i=1}^r$	$\{\mathbf{w}_i\}_{i=1}^r$	is exactly $r = \operatorname{rank}(A)$. Furthermore, the sets $\{\mathbf{v}_i\}_{i=1}^r$ and $\{\mathbf{w}_i\}_{i=1}^r$ can be
johnston-linear-matrix-algebra-F02199	226	1.000	$R\ \in$	$R \in$	<i>Proof.</i> Use the rank decomposition to write $A = CR$, where $C \in \mathcal{M}_{m,r}$ and $R \in$
johnston-linear-matrix-algebra-F02200	226	1.000	$\mathcal{M}_{r,\ n}$	$\mathcal{M}_{r,n}$	$\mathcal{M}_{r,n}$, and then write C and R in terms of their columns and rows, respectively:
johnston-linear-matrix-algebra-F02201	226	1.000	$\mathbf{v}_j\ \mathbf{w}_j^T$	$\mathbf{v}_j \mathbf{w}_j^T$	$\mathbf{v}_j \mathbf{w}_j^T$ have rank 1, so Theorem 2.4.11(a) tells us that
johnston-linear-matrix-algebra-F02202	226	1.000	$\left\{\mathbf{v}_j\right\}_{j=1}^r$	$\{\mathbf{v}_j\}_{j=1}^r$	ward way: the vectors $\{\mathbf{v}_j\}_{j=1}^r$ and $\{\mathbf{w}_j\}_{j=1}^r$ are just the columns of the matrices
johnston-linear-matrix-algebra-F02203	226	1.000	$\left\{\mathbf{w}_j\right\}_{j=1}^r$	$\{\mathbf{w}_j\}_{j=1}^r$	ward way: the vectors $\{\mathbf{v}_j\}_{j=1}^r$ and $\{\mathbf{w}_j\}_{j=1}^r$ are just the columns of the matrices
johnston-linear-matrix-algebra-F02204	228	0.998	$\operatorname{rank}(B)\ \leq\ \operatorname{rank}(A)$	$\operatorname{rank}(B) \leq \operatorname{rank}(A)$	Suppose $A \in \mathcal{M}_{m,n}$. If B is a submatrix of A then $\operatorname{rank}(B) \leq \operatorname{rank}(A)$.
johnston-linear-matrix-algebra-F02205	228	1.000	$C\ \in\ \mathcal{M}_{m,\ \ell}$	$C \in \mathcal{M}_{m,\ell}$	If a submatrix $C \in \mathcal{M}_{m,\ell}$ is created by erasing some columns of A , then
johnston-linear-matrix-algebra-F02206	228	1.000	$\operatorname{rank}(C)\ \leq\ \operatorname{rank}(A)$	$\operatorname{rank}(C) \leq \operatorname{rank}(A)$	$\operatorname{rank}(C) \leq \operatorname{rank}(A)$ since $\operatorname{range}(C)$ is the span of the columns of C , which
johnston-linear-matrix-algebra-F02207	228	1.000	$\operatorname{range}(C)$	$\operatorname{range}(C)$	$\operatorname{rank}(C) \leq \operatorname{rank}(A)$ since $\operatorname{range}(C)$ is the span of the columns of C , which
johnston-linear-matrix-algebra-F02208	228	0.972	$\operatorname{rank}(C)$	$\operatorname{rank}(C)$	On the other hand, we can also express $\operatorname{rank}(C)$ in terms of its rows
johnston-linear-matrix-algebra-F02209	229	1.000	$\operatorname{rank}(C)=\operatorname{rank}\left(C^T\right)$	$\operatorname{rank}(C) = \operatorname{rank}\left(C^T\right)$	$\operatorname{rank}(C) = \operatorname{rank}(C^T)$.
johnston-linear-matrix-algebra-F02210	229	1.000	$\operatorname{rank}(B)=3$	$\operatorname{rank}(B) = 3$	In fact, $\operatorname{rank}(B) = 3$
johnston-linear-matrix-algebra-F02211	229	0.998	$\mathbf{c}_1,\ \mathbf{c}_2,\ \ldots,\ \mathbf{c}_m:$	$\mathbf{c}_1, \mathbf{c}_2, \dots, \mathbf{c}_m:$	$\mathbf{c}_1, \mathbf{c}_2, \dots, \mathbf{c}_m:$
johnston-linear-matrix-algebra-F02212	229	1.000	$B\ \in\ \mathcal{M}_{k,\ \ell}$	$B \in \mathcal{M}_{k,\ell}$	If a submatrix $B \in \mathcal{M}_{k,\ell}$ is created by erasing some rows of C , then $\operatorname{rank}(B) \leq$
johnston-linear-matrix-algebra-F02213	229	1.000	$\operatorname{rank}(B)\ \leq$	$\operatorname{rank}(B) \leq$	If a submatrix $B \in \mathcal{M}_{k,\ell}$ is created by erasing some rows of C , then $\operatorname{rank}(B) \leq$

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johnston-linear-matrix-algebra-F02214	229	■ 0.991	$\operatorname{range}(B^T)$	$\operatorname{range}(B^T)$	$\operatorname{rank}(C)$ since $\operatorname{range}(B^T)$ is the span of the rows of B , which are a subset of
johnston-linear-matrix-algebra-F02215	229	■ 1.000	$A=\left[\begin{array}{lllll}1&1&0&0&1\\1&0&1&0&1\\1&0&0&1&1\end{array}\right]$	$A=\left[\begin{array}{ccccc}1&1&0&0&1\\1&0&1&0&1\\1&0&0&1&1\end{array}\right]$	a) $A=\left[\begin{array}{ccccc}1&1&0&0&1\\1&0&1&0&1\\1&0&0&1&1\end{array}\right]$
johnston-linear-matrix-algebra-F02216	229	■ 1.000	$B=\left[\begin{array}{ccccc}1&1&1&1&1\\1&2&4&8&16\\1&2&5&14&41\\1&3&9&27&81\end{array}\right]$	$B=\left[\begin{array}{ccccc}1&1&1&1&1\\1&2&4&8&16\\1&2&5&14&41\\1&3&9&27&81\end{array}\right]$	b) $B=\left[\begin{array}{ccccc}1&1&1&1&1\\1&2&4&8&16\\1&2&5&14&41\\1&3&9&27&81\end{array}\right]$
johnston-linear-matrix-algebra-F02217	229	■ 0.937	$\operatorname{rank}(A) \geq 3$	$\operatorname{rank}(A) \geq 3$	conclude that $\operatorname{rank}(A) \geq 3$ (in fact, since A is 3×5 , its rank cannot
johnston-linear-matrix-algebra-F02218	229	■ 0.937	3×5	3×5	conclude that $\operatorname{rank}(A) \geq 3$ (in fact, since A is 3×5 , its rank cannot
johnston-linear-matrix-algebra-F02219	229	■ 0.992	$\operatorname{rank}(B) \geq 3$	$\operatorname{rank}(B) \geq 3$	it has rank 3. It follows that $\operatorname{rank}(B) \geq 3$.
johnston-linear-matrix-algebra-F02220	229	■ 1.000	$r \times r$	$r \times r$	there exists an invertible $r \times r$ submatrix of A .
johnston-linear-matrix-algebra-F02221	229	■ 0.999	$r \leq \operatorname{rank}(A)$	$r \leq \operatorname{rank}(A)$	<i>Proof.</i> Theorem 2.C.3 tells us that $r \leq \operatorname{rank}(A)$, so we just need to show that
johnston-linear-matrix-algebra-F02222	230	■ 1.000	$\operatorname{rank}(C)=r$	$\operatorname{rank}(C)=r$	columns (in the same order that they appear in A). Then $\operatorname{rank}(C)=r$ as well,
johnston-linear-matrix-algebra-F02223	230	■ 0.999	$\operatorname{rank}(C^T)=r$	$\operatorname{rank}(C^T)=r$	so $\operatorname{rank}(C^T)=r$, so there is some set of r rows of C that span $\operatorname{range}(C^T)$. Let
johnston-linear-matrix-algebra-F02224	230	■ 0.999	$\operatorname{range}(C^T)$	$\operatorname{range}(C^T)$	so $\operatorname{rank}(C^T)=r$, so there is some set of r rows of C that span $\operatorname{range}(C^T)$. Let
johnston-linear-matrix-algebra-F02225	230	■ 1.000	$B \in \mathcal{M}_r$	$B \in \mathcal{M}_r$	$B \in \mathcal{M}_r$ be the submatrix consisting of these rows. Then $\operatorname{rank}(B)=r$ as well,
johnston-linear-matrix-algebra-F02226	230	■ 1.000	$\operatorname{rank}(B)=r$	$\operatorname{rank}(B)=r$	$B \in \mathcal{M}_r$ be the submatrix consisting of these rows. Then $\operatorname{rank}(B)=r$ as well,
johnston-linear-matrix-algebra-F02227	230	■ 1.000	$\left[\begin{array}{lll}1&2&1\\3&2&1\end{array}\right]$	$\left[\begin{array}{ccc}1&2&1\\3&2&1\end{array}\right]$	(b) $\left[\begin{array}{ccc}1&2&1\\3&2&1\end{array}\right]$
johnston-linear-matrix-algebra-F02228	230	■ 1.000	$\left[\begin{array}{ccc}1&2&3\\4&5&6\\7&8&9\\10&11&12\end{array}\right]$	$\left[\begin{array}{ccc}1&2&3\\4&5&6\\7&8&9\\10&11&12\end{array}\right]$	$\left[\begin{array}{ccc}1&2&3\\4&5&6\\7&8&9\\10&11&12\end{array}\right]$
johnston-linear-matrix-algebra-F02229	230	■ 1.000	$\left[\begin{array}{llll}1&1&0&0\\1&0&1&0\\0&0&0&1\\1&0&1&1\end{array}\right]$	$\left[\begin{array}{cccc}1&1&0&0\\1&0&1&0\\0&0&0&1\\1&0&1&1\end{array}\right]$	$\left[\begin{array}{cccc}1&1&0&0\\1&0&1&0\\0&0&0&1\\1&0&1&1\end{array}\right]$
johnston-linear-matrix-algebra-F02230	230	■ 0.644	$\operatorname{rank}(A) \leq 3$	$\operatorname{rank}(A) \leq 3$	$\operatorname{rank}(A) \leq 3$.
johnston-linear-matrix-algebra-F02231	230	■ 0.999	$\operatorname{rank}(A) \geq 4$	$\operatorname{rank}(A) \geq 4$	$\operatorname{rank}(A) \geq 4$.
johnston-linear-matrix-algebra-F02232	230	■ 0.994	$(n-1) \times (n-1)$	$(n-1) \times (n-1)$	*(c) If $A \in \mathcal{M}_n$ has an $(n-1) \times (n-1)$ submatrix equal
johnston-linear-matrix-algebra-F02233	230	■ 0.998	$\operatorname{rank}(A) \leq 1$	$\operatorname{rank}(A) \leq 1$	to the zero matrix then $\operatorname{rank}(A) \leq 1$.
johnston-linear-matrix-algebra-F02234	230	■ 0.999	$P \in \mathcal{M}_r$	$P \in \mathcal{M}_r$	and $R \in \mathcal{M}_{r,n}$. Let $P \in \mathcal{M}_r$ be any invertible matrix, and

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johnston-linear-matrix-algebra-F02235	230	■ 1.000	$\tilde{C} = C P^{-1}$	$\tilde{C} = C P^{-1}$	define $\tilde{C} = C P^{-1}$ and $\tilde{R} = P R$. Show that $A = \tilde{C} \tilde{R}$ is also a
johnston-linear-matrix-algebra-F02236	230	■ 1.000	$\tilde{R} = P R$	$\tilde{R} = P R$	define $\tilde{C} = C P^{-1}$ and $\tilde{R} = P R$. Show that $A = \tilde{C} \tilde{R}$ is also a
johnston-linear-matrix-algebra-F02237	230	■ 1.000	$A = \tilde{C} \tilde{R}$	$A = \tilde{C} \tilde{R}$	define $\tilde{C} = C P^{-1}$ and $\tilde{R} = P R$. Show that $A = \tilde{C} \tilde{R}$ is also a
johnston-linear-matrix-algebra-F02238	230	■ 0.994	$\operatorname{rank}(A) \leq z$	$\operatorname{rank}(A) \leq z$	that $\operatorname{rank}(A) \leq z$.
johnston-linear-matrix-algebra-F02239	230	■ 1.000	$A = P D Q$	$A = P D Q$	the form $A = P D Q$, where $P \in \mathcal{M}_m$ and $Q \in \mathcal{M}_n$ are
johnston-linear-matrix-algebra-F02240	230	■ 1.000	$Q \in \mathcal{M}_n$	$Q \in \mathcal{M}_n$	the form $A = P D Q$, where $P \in \mathcal{M}_m$ and $Q \in \mathcal{M}_n$ are
johnston-linear-matrix-algebra-F02241	230	■ 1.000	$A = P B Q$	$A = P B Q$	$A = P B Q$ if and only if $\operatorname{rank}(A) = \operatorname{rank}(B)$.
johnston-linear-matrix-algebra-F02242	230	■ 1.000	$\operatorname{rank}(A) = \operatorname{rank}(B)$	$\operatorname{rank}(A) = \operatorname{rank}(B)$	$A = P B Q$ if and only if $\operatorname{rank}(A) = \operatorname{rank}(B)$.
johnston-linear-matrix-algebra-F02243	230	■ 1.000	$a_{\{0\}}, a_{\{1\}}, \ldots, a_{\{m\}}$	$a_0, a_1, \ldots, a_m$	2.C.9 Suppose $a_0, a_1, \ldots, a_m$ are distinct real numbers and
johnston-linear-matrix-algebra-F02244	230	■ 1.000	$V \in \mathcal{M}_{m+1, n+1}$	$V \in \mathcal{M}_{m+1, n+1}$	let $V \in \mathcal{M}_{m+1, n+1}$ be the matrix
johnston-linear-matrix-algebra-F02245	230	■ 0.934	$\operatorname{rank}(V) = \min\{m+1, n+1\}$	$\operatorname{rank}(V) = \min\{m+1, n+1\}$	Show that $\operatorname{rank}(V) = \min\{m+1, n+1\}$ (i.e., V has full
johnston-linear-matrix-algebra-F02246	231	■ 1.000	U	U	we refer to it as U from now on. Similarly, the matrix P that we constructed
johnston-linear-matrix-algebra-F02247	231	■ 1.000	L	L	happened to be lower triangular, so we refer to it as L from now on. Being able
johnston-linear-matrix-algebra-F02248	231	■ 1.000	$\mathbf{L U}$	LU	Suppose $A \in \mathcal{M}_{m,n}$. An LU decomposition of A is any factorization of
johnston-linear-matrix-algebra-F02249	231	■ 1.000	$A = L U$	$A = L U$	the form $A = L U$, where $L \in \mathcal{M}_m$ is lower triangular and $U \in \mathcal{M}_{m,n}$ is
johnston-linear-matrix-algebra-F02250	231	■ 1.000	$L \in \mathcal{M}_m$	$L \in \mathcal{M}_m$	the form $A = L U$, where $L \in \mathcal{M}_m$ is lower triangular and $U \in \mathcal{M}_{m,n}$ is
johnston-linear-matrix-algebra-F02251	231	■ 1.000	$U \in \mathcal{M}_{m, n}$	$U \in \mathcal{M}_{m, n}$	the form $A = L U$, where $L \in \mathcal{M}_m$ is lower triangular and $U \in \mathcal{M}_{m,n}$ is
johnston-linear-matrix-algebra-F02252	231	■ 1.000	$U = R$	$U = R$	1) The matrix $U = R$ from Equation (2.D.1) was upper triangular just as a
johnston-linear-matrix-algebra-F02253	231	■ 1.000	$L = P$	$L = P$	result of being chosen to be a row echelon form of A , but why was $L = P$
johnston-linear-matrix-algebra-F02254	232	■ 1.000	$L = E^{-1}$	$L = E^{-1}$	form of A , and setting $L = E^{-1}$.
johnston-linear-matrix-algebra-F02255	232	■ 1.000	$j < i$	$j < i$	operations $R_i + c R_j$ with $j < i$ preserve lower triangularity, so E must be lower
johnston-linear-matrix-algebra-F02256	232	■ 0.873	$A = \left[\begin{array}{cccc} 2 & 1 & -2 & 1 \\ -4 & -4 & 3 & 0 \\ 2 & -5 & -2 & 8 \end{array} \right]$	$A = \left[\begin{array}{cccc} 2 & 1 & -2 & 1 \\ -4 & -4 & 3 & 0 \\ 2 & -5 & -2 & 8 \end{array} \right]$	Compute an LU decomposition of the matrix $A = \left[\begin{array}{cccc} 2 & 1 & -2 & 1 \\ -4 & -4 & 3 & 0 \\ 2 & -5 & -2 & 8 \end{array} \right]$.
johnston-linear-matrix-algebra-F02257	233	■ 1.000	$L D^{-1}$	$L D^{-1}$	entries as L then $L D^{-1}$ is a lower triangular matrix with ones on its
johnston-linear-matrix-algebra-F02258	233	■ 1.000	$D U$	$D U$	diagonal and $D U$ is an upper triangular matrix, so $A = (L D^{-1})(D U)$ is a
johnston-linear-matrix-algebra-F02259	233	■ 1.000	$A = (L D^{-1})(D U)$	$A = (L D^{-1})(D U)$	diagonal and $D U$ is an upper triangular matrix, so $A = (L D^{-1})(D U)$ is a
johnston-linear-matrix-algebra-F02260	234	■ 0.976	$R_{\{2\}} - \mathbf{2} R_{\{1\}}, R_{\{3\}} - \mathbf{3} R_{\{1\}}$	$R_2 - \mathbf{2} R_1, R_3 - \mathbf{3} R_1$	$R_2 - \mathbf{2} R_1, R_3 - \mathbf{3} R_1$,
johnston-linear-matrix-algebra-F02261	234	■ 0.871	$R_{\{3\}} - \mathbf{4} R_{\{2\}}$	$R_3 - \mathbf{4} R_2$	and $R_3 - \mathbf{4} R_2$.

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johnston-linear-matrix-algebra-F02262	234	■ 1.000	$\ell_{i,j}$	$\ell_{i,j}$	Here, $\ell_{i,j}$ refers to
johnston-linear-matrix-algebra-F02263	234	■ 1.000	$j=1$	$j = 1$	If $j = 1$ then the sum
johnston-linear-matrix-algebra-F02264	234	■ 0.882	$\mathbf{a}_1^T = \mathbf{u}_1^T$	$\mathbf{a}_1^T = \mathbf{u}_1^T$	(so $\mathbf{a}_1^T = \mathbf{u}_1^T$).
johnston-linear-matrix-algebra-F02265	234	■ 1.000	$n(n+1) / 2$	$n(n+1)/2$	specified by the $n(n+1)/2$ potentially non-zero entries in each of L and
johnston-linear-matrix-algebra-F02266	234	■ 1.000	$n(n+1)=n^2+n$	$n(n+1) = n^2 + n$	U , for a total of $n(n+1) = n^2 + n$ numbers, so it seems reasonable that we
johnston-linear-matrix-algebra-F02267	234	■ 1.000	n^2	n^2	can freely choose n of those numbers while still recovering the n^2 entries
johnston-linear-matrix-algebra-F02268	234	■ 1.000	$[U \mid E] \left(R_2 + 2R_1, R_3 - R_1 \right)$	$[U \mid E] (R_2 + 2R_1, R_3 - R_1$	that we used to put $[A \mid I]$ into row echelon form $[U \mid E] (R_2 + 2R_1, R_3 - R_1,$
johnston-linear-matrix-algebra-F02269	234	■ 1.000	$R_3 - 3R_2$	$R_3 - 3R_2$	and $R_3 - 3R_2$). This is not a coincidence—roughly speaking, it is a result of
johnston-linear-matrix-algebra-F02270	234	■ 1.000	$A, U \in \mathcal{M}_{m,n}$	$A, U \in \mathcal{M}_{m,n}$	Suppose $A, U \in \mathcal{M}_{m,n}$. A unit lower triangular matrix $L \in \mathcal{M}_m$ is such that
johnston-linear-matrix-algebra-F02271	234	■ 1.000	$\mathbf{a}_i^T$	$\mathbf{a}_i^T$	In particular, if we let $\mathbf{a}_i^T$ and $\mathbf{u}_i^T$ denote the i -th rows of A and U , respectively,
johnston-linear-matrix-algebra-F02272	234	■ 1.000	$\mathbf{u}_i^T$	$\mathbf{u}_i^T$	In particular, if we let $\mathbf{a}_i^T$ and $\mathbf{u}_i^T$ denote the i -th rows of A and U , respectively,
johnston-linear-matrix-algebra-F02273	234	■ 1.000	$R_i + \ell_{i,j} R_j$	$R_i + \ell_{i,j} R_j$	tions of the form $R_i + \ell_{i,j} R_j$ (with $j < i$) that converts U into A . For example,
johnston-linear-matrix-algebra-F02274	235	■ 1.000	$R_i - \ell_{i,j} R_j$	$R_i - \ell_{i,j} R_j$	is $R_i - \ell_{i,j} R_j$.
johnston-linear-matrix-algebra-F02275	235	■ 1.000	$R_2 - 2R_1, R_3 + 3R_1$	$R_2 - 2R_1, R_3 + 3R_1$	$R_2 - 2R_1, R_3 + 3R_1$,
johnston-linear-matrix-algebra-F02276	235	■ 1.000	$R_4 + R_1$	$R_4 + R_1$	and $R_4 + R_1$, so the
johnston-linear-matrix-algebra-F02277	235	■ 1.000	$\mathbf{a}_m^T$	$\mathbf{a}_m^T$	which is exactly $\mathbf{a}_m^T$, via Equation (2.D.2). Similar sequences of row operations
johnston-linear-matrix-algebra-F02278	235	■ 1.000	R_{m-1}	R_{m-1}	that involves adding a multiple of R_{m-1} , then all row operations that involve
johnston-linear-matrix-algebra-F02279	235	■ 1.000	R_{m-2}	R_{m-2}	adding a multiple of R_{m-2} , and so on. In particular, U is transformed into A via
johnston-linear-matrix-algebra-F02280	236	■ 1.000	$i > j$	$i > j$	operations of the form $R_i + cR_j$ with $i > j$.
johnston-linear-matrix-algebra-F02281	236	■ 1.000	$L=I$	$L = I$	just choose $L = I$ and $U = A$) but requires a swap row operation (or an operation
johnston-linear-matrix-algebra-F02282	236	■ 1.000	$U=A$	$U = A$	just choose $L = I$ and $U = A$) but requires a swap row operation (or an operation
johnston-linear-matrix-algebra-F02283	236	■ 1.000	$i < j$	$i < j$	of the form $R_i + cR_j$ for some $i < j$ instead of $i > j$) to put it in row echelon
johnston-linear-matrix-algebra-F02284	236	■ 1.000	$LU\mathbf{x} = \mathbf{b}$	$LU\mathbf{x} = \mathbf{b}$	the coefficient matrix of the linear system $A\mathbf{x} = \mathbf{b}$. Then $LU\mathbf{x} = \mathbf{b}$, which is a
johnston-linear-matrix-algebra-F02285	236	■ 1.000	$L\mathbf{y} = \mathbf{b}$	$L\mathbf{y} = \mathbf{b}$	• First, solve the linear system $L\mathbf{y} = \mathbf{b}$ for the vector $\mathbf{y}$. This linear system
johnston-linear-matrix-algebra-F02286	236	■ 1.000	$U\mathbf{x} = \mathbf{y}$	$U\mathbf{x} = \mathbf{y}$	• Next, solve the linear system $U\mathbf{x} = \mathbf{y}$ for the vector $\mathbf{x}$. This linear system
johnston-linear-matrix-algebra-F02287	237	■ 1.000	$A \in$	$A \in$	shape of the linear systems saves us a lot of work in the end. In particular, if $A \in$
johnston-linear-matrix-algebra-F02288	237	■ 1.000	$2n^3 / 3$	$2n^3/3$	elimination requires roughly $2n^3/3$ operations (i.e., additions, subtractions,

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johnston-linear-matrix-algebra-F02289	237	1.000	$L \mathbf{y} =$	$L\mathbf{y} =$	multiplications, and/or divisions), whereas each of the triangular systems $L\mathbf{y} =$
johnston-linear-matrix-algebra-F02290	237	1.000	$2n^2$	$2n^2$	these two triangular systems via roughly $2n^2$ operations is considerably quicker
johnston-linear-matrix-algebra-F02291	237	1.000	$\mathbf{y} = (0, 2, -6, -15)$	$\mathbf{y} = (0, 2, -6, -15)$	so $\mathbf{y} = (0, 2, -6, -15)$. Next, we solve the upper triangular system $U\mathbf{x} = \mathbf{y}$:
johnston-linear-matrix-algebra-F02292	238	0.951	$\mathbf{x} = (w, x, y, z) = (22, -9, 3, 5)$	$\mathbf{x} = (w, x, y, z) = (22, -9, 3, 5)$	$\mathbf{x} = (w, x, y, z) = (22, -9, 3, 5)$.
johnston-linear-matrix-algebra-F02293	238	0.593	$\{(0, 3), (1, -1), (2, -3), (3, 9)\}$	$\{(0, 3), (1, -1), (2, -3), (3, 9)\}$	a) $\{(0, 3), (1, -1), (2, -3), (3, 9)\}$
johnston-linear-matrix-algebra-F02294	238	1.000	$\{(0, -1), (1, -1), (2, -1), (3, 5)\}$	$\{(0, -1), (1, -1), (2, -1), (3, 5)\}$	b) $\{(0, -1), (1, -1), (2, -1), (3, 5)\}$
johnston-linear-matrix-algebra-F02295	238	1.000	$p(x) = c_3x^3 + c_2x^2 + c_1x + c_0$	$p(x) = c_3x^3 + c_2x^2 + c_1x + c_0$	a) If the polynomial that we seek is $p(x) = c_3x^3 + c_2x^2 + c_1x + c_0$, then
johnston-linear-matrix-algebra-F02296	238	1.000	$p(0)=3, p(1)=0, p(2)=1$	$p(0) = 3, p(1) = 0, p(2) = 1$	the equations $p(0) = 3, p(1) = 0, p(2) = 1$, and $p(3) = 18$ can be
johnston-linear-matrix-algebra-F02297	238	1.000	$p(3)=18$	$p(3) = 18$	the equations $p(0) = 3, p(1) = 0, p(2) = 1$, and $p(3) = 18$ can be
johnston-linear-matrix-algebra-F02298	239	1.000	$L \mathbf{y} = \mathbf{b} = (3, -1, 3, 9)$	$L\mathbf{y} = \mathbf{b} = (3, -1, 3, 9)$	Then solving the lower triangular linear system $L\mathbf{y} = \mathbf{b} = (3, -1, 3, 9)$
johnston-linear-matrix-algebra-F02299	239	1.000	$\mathbf{y} = (3, -4, 2, 12)$	$\mathbf{y} = (3, -4, 2, 12)$	gives $\mathbf{y} = (3, -4, 2, 12)$, and solving the upper triangular linear sys-
johnston-linear-matrix-algebra-F02300	239	1.000	$U \mathbf{x} = \mathbf{y} = (3, -4, 2, 12)$	$U\mathbf{x} = \mathbf{y} = (3, -4, 2, 12)$	tem $U\mathbf{x} = \mathbf{y} = (3, -4, 2, 12)$ gives $\mathbf{x} = (c_0, c_1, c_2, c_3) = (3, -1, -5, 2)$.
johnston-linear-matrix-algebra-F02301	239	1.000	$\mathbf{x} = (c_0, c_1, c_2, c_3) = (3, -1, -5, 2)$	$\mathbf{x} = (c_0, c_1, c_2, c_3) = (3, -1, -5, 2)$	tem $U\mathbf{x} = \mathbf{y} = (3, -4, 2, 12)$ gives $\mathbf{x} = (c_0, c_1, c_2, c_3) = (3, -1, -5, 2)$.
johnston-linear-matrix-algebra-F02302	239	1.000	$x = 0, 1, 2$	$x = 0, 1, 2$	straightforward to verify by plugging $x = 0, 1, 2$ and 3 into it, and it
johnston-linear-matrix-algebra-F02303	239	1.000	$L \mathbf{y} = \mathbf{b} = (-1, -1, -1, 5)$	$L\mathbf{y} = \mathbf{b} = (-1, -1, -1, 5)$	ear system $L\mathbf{y} = \mathbf{b} = (-1, -1, -1, 5)$, which gives $\mathbf{y} = (-1, 0, 0, 6)$.
johnston-linear-matrix-algebra-F02304	239	1.000	$\mathbf{y} = (-1, 0, 0, 6)$	$\mathbf{y} = (-1, 0, 0, 6)$	ear system $L\mathbf{y} = \mathbf{b} = (-1, -1, -1, 5)$, which gives $\mathbf{y} = (-1, 0, 0, 6)$.
johnston-linear-matrix-algebra-F02305	239	1.000	$U \mathbf{x} = \mathbf{y} = (-1, 0, 0, 6)$	$U\mathbf{x} = \mathbf{y} = (-1, 0, 0, 6)$	Solving the upper triangular linear system $U\mathbf{x} = \mathbf{y} = (-1, 0, 0, 6)$
johnston-linear-matrix-algebra-F02306	239	1.000	$\mathbf{x} = (c_0, c_1, c_2, c_3) = (-1, 2, -3, 1)$	$\mathbf{x} = (c_0, c_1, c_2, c_3) = (-1, 2, -3, 1)$	then gives $\mathbf{x} = (c_0, c_1, c_2, c_3) = (-1, 2, -3, 1)$, so the (unique) degree-
johnston-linear-matrix-algebra-F02307	239	0.817	$(0, -1), (1, -1), (2, -1)$	$(0, -1), (1, -1), (2, -1)$	goes through the four points $(0, -1), (1, -1), (2, -1)$, and $(3, 5)$:
johnston-linear-matrix-algebra-F02308	239	0.817	$(3, 5)$	$(3, 5)$	goes through the four points $(0, -1), (1, -1), (2, -1)$, and $(3, 5)$:
johnston-linear-matrix-algebra-F02309	240	1.000	$\mathbf{x} = A^{-1}\mathbf{b}, \mathbf{y} = A^{-1}\mathbf{c}$	$\mathbf{x} = A^{-1}\mathbf{b}, \mathbf{y} = A^{-1}\mathbf{c}$	$\mathbf{x} = A^{-1}\mathbf{b}, \mathbf{y} = A^{-1}\mathbf{c}$, and $\mathbf{z} = A^{-1}\mathbf{d}$. Both this approach and the one based
johnston-linear-matrix-algebra-F02310	240	1.000	$\mathbf{z} = A^{-1}\mathbf{d}$	$\mathbf{z} = A^{-1}\mathbf{d}$	$\mathbf{x} = A^{-1}\mathbf{b}, \mathbf{y} = A^{-1}\mathbf{c}$, and $\mathbf{z} = A^{-1}\mathbf{d}$. Both this approach and the one based
johnston-linear-matrix-algebra-F02311	240	1.000	$2n^3$	$2n^3$	quires roughly $2n^3$ operations (additions, subtractions, multiplica-
johnston-linear-matrix-algebra-F02312	241	1.000	$A = E^{-1}R$	$A = E^{-1}R$	until it is in row echelon form $[R \mid E]$ and noting that $A = E^{-1}R$:
johnston-linear-matrix-algebra-F02313	241	1.000	E^{-1}	E^{-1}	decomposition). On the other hand, E^{-1} is not lower triangular, but its rows
johnston-linear-matrix-algebra-F02314	242	1.000	$P = I$	$P = I$	$P = I$ then we get
johnston-linear-matrix-algebra-F02315	242	0.793	$0)$	$0)$	$0)$. In this case, we can write

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johnston-linear-matrix-algebra F02316	242	■ 0.999	$A=E^{-1} \ U=P \ L \ U$	$A = E^{-1}U = PLU$	It follows that $A = E^{-1}U = PLU$, where P is a permutation matrix, L is
johnston-linear-matrix-algebra F02317	242	■ 1.000	$A=P \ L \ U$	$A = PLU$	the form $A = PLU$, where $P \in \mathcal{M}_m$ is a permutation matrix, $L \in \mathcal{M}_m$ is
johnston-linear-matrix-algebra F02318	242	■ 1.000	$P \in \mathcal{M}_n$	$P \in \mathcal{M}_n$	matrices $P \in \mathcal{M}_n$ are special for the fact that are not only invertible, but in
johnston-linear-matrix-algebra F02319	242	■ 1.000	$P^{-1}=P^T$	$P^{-1} = P^T$	fact their inverse is simply their transpose: $P^{-1} = P^T$. We prove this fact in
johnston-linear-matrix-algebra F02320	242	■ 1.000	$[U \mid E]$	$[U \mid E]$	If Gaussian elimination row-reduces $[A \mid I]$ to $[U \mid E]$, where U is a row
johnston-linear-matrix-algebra F02321	242	■ 1.000	$E^{-1}=P \ L$	$E^{-1} = PL$	triangular matrix L such that $E^{-1} = PL$. In particular,
johnston-linear-matrix-algebra F02322	243	■ 1.000	$P_* \in \mathcal{M}_m$	$P_* \in \mathcal{M}_m$	into lower triangular form. That is, we can find a permutation matrix $P_* \in \mathcal{M}_m$
johnston-linear-matrix-algebra F02323	243	■ 1.000	$L_* \in \mathcal{M}_m$	$L_* \in \mathcal{M}_m$	and a lower triangular matrix $L_* \in \mathcal{M}_m$ such that $E = L_*P_*$. Taking the inverse
johnston-linear-matrix-algebra F02324	243	■ 1.000	$E=L_* \ P_*$	$E = L_*P_*$	and a lower triangular matrix $L_* \in \mathcal{M}_m$ such that $E = L_*P_*$. Taking the inverse
johnston-linear-matrix-algebra F02325	243	■ 1.000	$P_*^{-1}=P_*^T$	$P_*^{-1} = P_*^T$	Since $P_*^{-1} = P_*^T$ is also a permutation matrix, and L_*^{-1} is also lower triangular,
johnston-linear-matrix-algebra F02326	243	■ 1.000	L_*^{-1}	L_*^{-1}	Since $P_*^{-1} = P_*^T$ is also a permutation matrix, and L_*^{-1} is also lower triangular,
johnston-linear-matrix-algebra F02327	243	■ 1.000	$P=P_*^{-1}$	$P = P_*^{-1}$	it follows that we can set $P = P_*^{-1}$ and $L = L_*^{-1}$ to complete the proof.
johnston-linear-matrix-algebra F02328	243	■ 1.000	$L=L_*^{-1}$	$L = L_*^{-1}$	it follows that we can set $P = P_*^{-1}$ and $L = L_*^{-1}$ to complete the proof.
johnston-linear-matrix-algebra F02329	243	■ 1.000	$E \ P$	EP	The reason this that works is simply that EP is lower triangular (which is
johnston-linear-matrix-algebra F02330	243	■ 1.000	$P^{-1} \ E^{-1}$	$P^{-1}E^{-1}$	tion), so its inverse $P^{-1}E^{-1}$ is lower triangular too. If we define $L = P^{-1}E^{-1}$
johnston-linear-matrix-algebra F02331	243	■ 1.000	$L=P^{-1} \ E^{-1}$	$L = P^{-1}E^{-1}$	tion), so its inverse $P^{-1}E^{-1}$ is lower triangular too. If we define $L = P^{-1}E^{-1}$
johnston-linear-matrix-algebra F02332	243	■ 0.999	$A=\left[\begin{array}{cccc}1 & 1 & 2 & 1 \\ 1 & 1 & 1 & 2 \\ 1 & 2 & 1 & 1 \\ 0 & 1 & 1 & 1\end{array}\right]$	$A = \begin{bmatrix} 1 & 1 & 2 & 1 \\ 1 & 1 & 1 & 2 \\ 1 & 2 & 1 & 1 \\ 0 & 1 & 1 & 1 \end{bmatrix}$	Construct a PLU decomposition of the matrix $A = \begin{bmatrix} 1 & 1 & 2 & 1 \\ 1 & 1 & 1 & 2 \\ 1 & 2 & 1 & 1 \\ 0 & 1 & 1 & 1 \end{bmatrix}$.
johnston-linear-matrix-algebra F02333	243	■ 1.000	P^T	P^T	We then choose P^T to have ones in its entries corresponding to the topmost
johnston-linear-matrix-algebra F02334	244	■ 1.000	$\mathbf{c}=P^T \ \mathbf{b}$	$\mathbf{c} = P^T \mathbf{b}$	so $\mathbf{c} = P^T \mathbf{b}$ means
johnston-linear-matrix-algebra F02335	244	■ 1.000	$P \ \mathbf{c}=\mathbf{b}$	$P\mathbf{c} = \mathbf{b}$	$P\mathbf{c} = \mathbf{b}$.
johnston-linear-matrix-algebra F02336	244	■ 1.000	$P \ L \ U \ \mathbf{x}=\mathbf{b}$	$PLU\mathbf{x} = \mathbf{b}$	decomposition $A = PLU$, then $PLU\mathbf{x} = \mathbf{b}$ is a linear system that we can solve
johnston-linear-matrix-algebra F02337	244	■ 1.000	$L \ \mathbf{y}=\mathbf{c}$	$L\mathbf{y} = \mathbf{c}$	• Solve the linear system $L\mathbf{y} = \mathbf{c}$ for $\mathbf{y}$.
johnston-linear-matrix-algebra F02338	245	■ 1.000	$\mathbf{b}=(0,1,2,3)$	$\mathbf{b} = (0, 1, 2, 3)$	to solve the linear system $A\mathbf{x} = \mathbf{b}$, where $\mathbf{b} = (0, 1, 2, 3)$.
johnston-linear-matrix-algebra F02339	245	■ 1.000	$P, \ L$	P, L	We let P, L , and U be the matrices that we computed in Example 2.D.5
johnston-linear-matrix-algebra F02340	245	■ 1.000	$\mathbf{y}=(0,2,1,3 \ / \ 2)$	$\mathbf{y} = (0, 2, 1, 3/2)$	so $\mathbf{y} = (0, 2, 1, 3/2)$.

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johnston-linear-matrix-algebra-F02341	245	1.000	$\mathbf{x}=(-3,2,0,1)$	$\mathbf{x} = (-3, 2, 0, 1)$	so the unique solution of this linear system is $\mathbf{x} = (-3, 2, 0, 1)$.
johnston-linear-matrix-algebra-F02342	245	0.840	$1 \leq k \leq r$	$1 \leq k \leq r$	invertible for all $1 \leq k \leq r$ then A has a unit LU decomposition.
johnston-linear-matrix-algebra-F02343	245	0.835	$a_{1,1}$	$a_{1,1}$	result is trivial, since any 1×1 matrix $A = [a_{1,1}]$ can be written in the form
johnston-linear-matrix-algebra-F02344	245	1.000	$L=[1]$	$L = [1]$	$A = LU$ with $L = [1]$ and $U = [a_{1,1}]$.
johnston-linear-matrix-algebra-F02345	245	1.000	$U=\left[a_{1,1}\right]$	$U = [a_{1,1}]$	$A = LU$ with $L = [1]$ and $U = [a_{1,1}]$.
johnston-linear-matrix-algebra-F02346	246	1.000	$L\ U=\left(L\ D^{-1}\right)(D\ U)$	$LU = (LD^{-1})(DU)$	$LU = (LD^{-1})(DU)$
johnston-linear-matrix-algebra-F02347	246	0.743	$B \in \mathcal{M}_{n-1}$	$B \in \mathcal{M}_{n-1}$	Since $B \in \mathcal{M}_{n-1}$, we know by the inductive hypothesis that it has a unit LU
johnston-linear-matrix-algebra-F02348	246	1.000	$B=L\ U$	$B = LU$	decomposition $B = LU$. Our goal is to extend this to a unit LU decomposition
johnston-linear-matrix-algebra-F02349	246	1.000	$\mathbf{x}, \mathbf{y} \in \mathbb{R}^{n-1}$	$\mathbf{x}, \mathbf{y} \in \mathbb{R}^{n-1}$	of A . To this end, we want to find column vectors $\mathbf{x}, \mathbf{y} \in \mathbb{R}^{n-1}$ and a scalar
johnston-linear-matrix-algebra-F02350	246	1.000	$d \in \mathbb{R}$	$d \in \mathbb{R}$	$d \in \mathbb{R}$ such that
johnston-linear-matrix-algebra-F02351	246	1.000	$c=d+\mathbf{x}^T\ \mathbf{y}$	$c = d + \mathbf{x}^T \mathbf{y}$	chose L and U in the first place. The bottom-right block requires $c = d + \mathbf{x}^T \mathbf{y}$,
johnston-linear-matrix-algebra-F02352	246	1.000	$d=c-\mathbf{x}^T\ \mathbf{y}$	$d = c - \mathbf{x}^T \mathbf{y}$	which can be satisfied just by choosing $d = c - \mathbf{x}^T \mathbf{y}$. The top-right block
johnston-linear-matrix-algebra-F02353	246	0.989	$L\ \mathbf{y}=\mathbf{v}$	$L\mathbf{y} = \mathbf{v}$	requires us to find a $\mathbf{y}$ such that $L\mathbf{y} = \mathbf{v}$, which can be be done straightforwardly
johnston-linear-matrix-algebra-F02354	246	1.000	$\mathbf{y}=L^{-1}\ \mathbf{v}$	$\mathbf{y} = L^{-1} \mathbf{v}$	$\mathbf{y} = L^{-1} \mathbf{v}$.
johnston-linear-matrix-algebra-F02355	246	1.000	$\mathbf{w}^T=\mathbf{x}^T\ U$	$\mathbf{w}^T = \mathbf{x}^T U$	block is correct: $\mathbf{w}^T = \mathbf{x}^T U$. To this end, we split into two separate cases:
johnston-linear-matrix-algebra-F02356	246	0.999	$U=B\ L^{-1}$	$U = BL^{-1}$	Case 1: B is invertible. In this case, $B = LU$, so $U = BL^{-1}$ is also invertible, so
johnston-linear-matrix-algebra-F02357	246	1.000	$\mathbf{x}=\left(U^{-1}\right)^T\ \mathbf{w}$	$\mathbf{x} = (U^{-1})^T \mathbf{w}$	we can simply choose $\mathbf{x} = (U^{-1})^T \mathbf{w}$.
johnston-linear-matrix-algebra-F02358	246	1.000	$\operatorname{rank}(A)=$	$\operatorname{rank}(A) =$	(by one of the hypotheses of the theorem), we conclude that $\operatorname{rank}(A) =$
johnston-linear-matrix-algebra-F02359	246	0.987	$\mathbf{w}^T$	$\mathbf{w}^T$	$\operatorname{rank}(B) = r$. It follows that $\mathbf{w}^T$ must be a linear combination of the rows
johnston-linear-matrix-algebra-F02360	246	1.000	$\operatorname{rank}(A)>r$	$\operatorname{rank}(A) > r$	which in turn would imply $\operatorname{rank}(A) > r$.
johnston-linear-matrix-algebra-F02361	246	1.000	$\mathbf{x}^T$	$\mathbf{x}^T$	equation $\mathbf{w}^T = \mathbf{x}^T U$ says—in particular, we choose the entries of $\mathbf{x}^T$ to
johnston-linear-matrix-algebra-F02362	246	0.997	$d,\ \mathbf{x}$	$d, \mathbf{x}$	Now that we know how to construct each of d , $\mathbf{x}$, and $\mathbf{y}$, the inductive step and
johnston-linear-matrix-algebra-F02363	246	0.523	$A=\left(\begin{array}{ccc}3 & -2 & -1 \\ 3 & -1 & 4 \\ 1 & 5 & 2\end{array}\right)$	$A = \begin{bmatrix} 3 & -2 & -1 \\ 3 & -1 & 4 \\ 1 & 5 & 2 \end{bmatrix}$	Show that the matrix $A = \begin{bmatrix} 3 & -2 & -1 \\ 3 & -1 & 4 \\ 1 & 5 & 2 \end{bmatrix}$ has a unit LU decomposition.
johnston-linear-matrix-algebra-F02364	247	0.852	$\left(\begin{array}{ll}1 & 2 \\ 2 & 5\end{array}\right)$	$\begin{bmatrix} 1 & 2 \\ 2 & 5 \end{bmatrix}$	$\ast(\mathbf{a}) \begin{bmatrix} 1 & 2 \\ 2 & 5 \end{bmatrix}$

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johnston-linear-matrix-algebra F02365	247	1.000	$\left[\begin{array}{cc}2 & -1 \\ 4 & 1\end{array}\right]$	$\begin{bmatrix} 2 & -1 \\ 4 & 1 \end{bmatrix}$	(b) $\begin{bmatrix} 2 & -1 \\ 4 & 1 \end{bmatrix}$
johnston-linear-matrix-algebra F02366	247	0.995	$\left[\begin{array}{ccc}3 & 1 & 2 \\ -3 & -3 & -1\end{array}\right]$	$\begin{bmatrix} 3 & 1 & 2 \\ -3 & -3 & -1 \end{bmatrix}$	*(c) $\begin{bmatrix} 3 & 1 & 2 \\ -3 & -3 & -1 \end{bmatrix}$
johnston-linear-matrix-algebra F02367	247	1.000	$\left[\begin{array}{ccc}1 & 2 & -1 \\ -1 & -3 & -2 \\ 5 & -8\end{array}\right]$	$\begin{bmatrix} 1 & 2 & -1 \\ -1 & -3 & -2 \\ 3 & 5 & -8 \end{bmatrix}$	(f) $\begin{bmatrix} 1 & 2 & -1 \\ -1 & -3 & -2 \\ 3 & 5 & -8 \end{bmatrix}$
johnston-linear-matrix-algebra F02368	247	0.546	$\left[\begin{array}{ccc}1 & -4 & 5 \\ 3 & -9 & 8 \\ 5 & -2\end{array}\right]$	$\begin{bmatrix} 1 & -4 & 5 \\ 3 & -9 & 8 \\ -2 & 5 & -2 \end{bmatrix}$	*(g) $\begin{bmatrix} 1 & -4 & 5 \\ 3 & -9 & 8 \\ -2 & 5 & -2 \end{bmatrix}$
johnston-linear-matrix-algebra F02369	247	1.000	$\left[\begin{array}{cccc}2 & -1 & 4 & 3 \\ -4 & 4 & -7 & -6 \\ 6 & -7 & 12 & 10\end{array}\right]$	$\begin{bmatrix} 2 & -1 & 4 & 3 \\ -4 & 4 & -7 & -6 \\ 6 & -7 & 12 & 10 \end{bmatrix}$	(h) $\begin{bmatrix} 2 & -1 & 4 & 3 \\ -4 & 4 & -7 & -6 \\ 6 & -7 & 12 & 10 \end{bmatrix}$
johnston-linear-matrix-algebra F02370	247	0.947	$\left[\begin{array}{ll}1 & 0 \\ 2 & 1\end{array}\right]\left[\begin{array}{ll}2 & 3 \\ 0 & 1\end{array}\right]\left[\begin{array}{l}x \\ y\end{array}\right]=\left[\begin{array}{l}1 \\ -1\end{array}\right]$	$\begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}\begin{bmatrix} 2 & 3 \\ 0 & 1 \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix}=\begin{bmatrix} 1 \\ -1 \end{bmatrix}$	*(a) $\begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}\begin{bmatrix} 2 & 3 \\ 0 & 1 \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix}=\begin{bmatrix} 1 \\ -1 \end{bmatrix}$
johnston-linear-matrix-algebra F02371	247	0.518	$\left[\begin{array}{ccc}1 & 0 & 0 \\ 2 & 1 & 0 \\ -1 & 1 & 1\end{array}\right]\left[\begin{array}{ccc}2 & 1 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 2\end{array}\right]\left[\begin{array}{l}x \\ y \\ z\end{array}\right]=\left[\begin{array}{l}3 \\ 1 \\ 2\end{array}\right]$	$\begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -1 & 1 & 1 \end{bmatrix}\begin{bmatrix} 2 & 1 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 2 \end{bmatrix}\begin{bmatrix} x \\ y \\ z \end{bmatrix}=\begin{bmatrix} 3 \\ 1 \\ 2 \end{bmatrix}$	(b) $\begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -1 & 1 & 1 \end{bmatrix}\begin{bmatrix} 2 & 1 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 2 \end{bmatrix}\begin{bmatrix} x \\ y \\ z \end{bmatrix}=\begin{bmatrix} 3 \\ 1 \\ 2 \end{bmatrix}$
johnston-linear-matrix-algebra F02372	247	0.556	$\left[\begin{array}{lll}0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0\end{array}\right]\left[\begin{array}{lll}1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & 1\end{array}\right]\left[\begin{array}{l}x \\ y \\ z\end{array}\right]=\left[\begin{array}{l}1 \\ 2 \\ 3\end{array}\right]$	$\begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}\begin{bmatrix} x \\ y \\ z \end{bmatrix}=\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$	*(c) $\begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}\begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix}\begin{bmatrix} x \\ y \\ z \end{bmatrix}=\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$
johnston-linear-matrix-algebra F02373	247	0.917	$\left[\begin{array}{llll}1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 2 & 1 & 1 & 0 \\ 1 & 1 & 3 & 1\end{array}\right]\left[\begin{array}{llll}2 & 1 & 0 & 1 \\ 0 & 1 & 2 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 3\end{array}\right]\left[\begin{array}{l}w \\ x \\ y \\ z\end{array}\right]=\left[\begin{array}{l}1 \\ 1 \\ 1 \\ 1\end{array}\right]$	$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 2 & 1 & 1 & 0 \\ 1 & 1 & 3 & 1 \end{bmatrix}\begin{bmatrix} 2 & 1 & 0 & 1 \\ 0 & 1 & 2 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 3 \end{bmatrix}\begin{bmatrix} w \\ x \\ y \\ z \end{bmatrix}=\begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$	(d) $\begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 2 & 1 & 1 & 0 \\ 1 & 1 & 3 & 1 \end{bmatrix}\begin{bmatrix} 2 & 1 & 0 & 1 \\ 0 & 1 & 2 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 3 \end{bmatrix}\begin{bmatrix} w \\ x \\ y \\ z \end{bmatrix}=\begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$
johnston-linear-matrix-algebra F02374	247	0.661	$\left[\begin{array}{ll}0 & 2 \\ 1 & 3\end{array}\right]$	$\begin{bmatrix} 0 & 2 \\ 1 & 3 \end{bmatrix}$	*(a) $\begin{bmatrix} 0 & 2 \\ 1 & 3 \end{bmatrix}$
johnston-linear-matrix-algebra F02375	247	0.950	$\left[\begin{array}{lll}1 & 1 & 1 \\ 1 & 1 & 2 \\ 1 & 2 & 3\end{array}\right]$	$\begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 2 \\ 1 & 2 & 3 \end{bmatrix}$	*(c) $\begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 2 \\ 1 & 2 & 3 \end{bmatrix}$

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johnston-linear-matrix-algebra-F02376	247	■ 0.999	$\left[\begin{array}{llll}0 & 2 & 3 & 3 \\ 1 & 2 & 1 & 2 \\ 1 & 2 & 1 & 4\end{array}\right]$	$\begin{bmatrix} 0 & 2 & 3 & 3 \\ 1 & 2 & 1 & 2 \\ 1 & 2 & 1 & 4 \end{bmatrix}$	(d) $\begin{bmatrix} 0 & 2 & 3 & 3 \\ 1 & 2 & 1 & 2 \\ 1 & 2 & 1 & 4 \end{bmatrix}$
johnston-linear-matrix-algebra-F02377	248	■ 0.584	$\left[\begin{array}{ll}1 & 3 \\ 3 & 2\end{array}\right]$	$\begin{bmatrix} 1 & 3 \\ 3 & 2 \end{bmatrix}$	*(a) $\begin{bmatrix} 1 & 3 \\ 3 & 2 \end{bmatrix}$
johnston-linear-matrix-algebra-F02378	248	■ 1.000	$\left[\begin{array}{cc}0 & 1 \\ 1 & -2\end{array}\right]$	$\begin{bmatrix} 0 & 1 \\ 1 & -2 \end{bmatrix}$	(b) $\begin{bmatrix} 0 & 1 \\ 1 & -2 \end{bmatrix}$
johnston-linear-matrix-algebra-F02379	248	■ 0.997	$\left[\begin{array}{llll}4 & 5 & 5 & 0 \\ 3 & 4 & 3 & 3 \\ 1 & 1 & 2 & 3 \\ 3 & 3 & 4 & 2\end{array}\right]$	$\begin{bmatrix} 4 & 5 & 5 & 0 \\ 3 & 4 & 3 & 3 \\ 1 & 1 & 2 & 3 \\ 3 & 3 & 4 & 2 \end{bmatrix}$	(d) $\begin{bmatrix} 4 & 5 & 5 & 0 \\ 3 & 4 & 3 & 3 \\ 1 & 1 & 2 & 3 \\ 3 & 3 & 4 & 2 \end{bmatrix}$
johnston-linear-matrix-algebra-F02380	248	■ 1.000	$A=L \ D \ U$	$A = LDU$	is any factorization of the form $A = LDU$, where L and U
johnston-linear-matrix-algebra-F02381	248	■ 1.000	$A=L \ D \ L^{\mathrm{T}}$	$A = LDL^T$	then it is unique and has the form $A = LDL^T$.
johnston-linear-matrix-algebra-F02382	248	■ 0.994	$A=\left[\begin{array}{ll}0 & 0 \\ 1 & 1\end{array}\right]$	$A = \begin{bmatrix} 0 & 0 \\ 1 & 1 \end{bmatrix}$	**2.D.13 Consider the matrix $A = \begin{bmatrix} 0 & 0 \\ 1 & 1 \end{bmatrix}$.
johnston-linear-matrix-algebra-F02383	249	■ 1.000	$A \ \mathbf{e}_1=(2,1)$	$A\mathbf{e}_1 = (2,1)$	$A\mathbf{e}_1 = (2,1)$, which is
johnston-linear-matrix-algebra-F02384	249	■ 1.000	$A \ \mathbf{e}_2=(1,2)$	$A\mathbf{e}_2 = (1,2)$	$A\mathbf{e}_2 = (1,2)$, which is
johnston-linear-matrix-algebra-F02385	249	■ 1.000	$A=\left[\begin{array}{ll}2 & 1 \\ 1 & 2\end{array}\right]$	$A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$	Figure 3.1: The matrix $A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$ acting as a linear transformation on $\mathbb{R}^2$.
johnston-linear-matrix-algebra-F02386	249	■ 1.000	$\mathbf{v}_1=(1,1)$	$\mathbf{v}_1 = (1,1)$	and instead consider how it acts on the input grid defined by $\mathbf{v}_1 = (1,1)$ and
johnston-linear-matrix-algebra-F02387	249	■ 1.000	$\mathbf{v}_2=(-1,1)$	$\mathbf{v}_2 = (-1,1)$	$\mathbf{v}_2 = (-1,1)$, then this matrix transforms $\mathbb{R}^2$ in the manner shown in Figure 3.2,
johnston-linear-matrix-algebra-F02388	250	■ 1.000	$(1,0)$	$(1,0)$	away either of $(1,0)$
johnston-linear-matrix-algebra-F02389	250	■ 0.993	$\left\{\mathbf{v}_1, \mathbf{v}_2\right\}$	$\{\mathbf{v}_1, \mathbf{v}_2\}$	different input grid), we investigate how to find a particular basis ($\{\mathbf{v}_1, \mathbf{v}_2\}$ in
johnston-linear-matrix-algebra-F02390	250	■ 1.000	$\mathbb{R}^2=\operatorname{span}((1,0),(0,1),(1,1))$	$\mathbb{R}^2 = \operatorname{span}((1,0), (0,1), (1,1))$	For example, it is the case that $\mathbb{R}^2 = \operatorname{span}((1,0), (0,1), (1,1))$, so every
johnston-linear-matrix-algebra-F02391	250	■ 0.985	$(1,0),(0,1)$	$(1,0), (0,1)$	vector $(x,y) \in \mathbb{R}^2$ can be written as a linear combination of $(1,0), (0,1)$, and
johnston-linear-matrix-algebra-F02392	250	■ 1.000	$\{(1,0),(0,1),(1,1)\}$	$\{(1,0), (0,1), (1,1)\}$	that the set $\{(1,0), (0,1), (1,1)\}$ is not linearly independent (and thus not a
johnston-linear-matrix-algebra-F02393	250	■ 1.000	$(1,1)$	$(1,1)$	basis). If we throw away the vector $(1,1)$ then we arrive at the standard basis
johnston-linear-matrix-algebra-F02394	250	■ 0.975	$\{(1,0),(0,1)\}$	$\{(1,0), (0,1)\}$	$\{(1,0), (0,1)\}$, and linear combinations of the standard basis vectors <i>are</i> unique
johnston-linear-matrix-algebra-F02395	251	■ 0.998	$(2,1)=2(1,0)+(0,1)$	$(2,1) = 2(1,0) + (0,1)$	standard basis vectors is simply $(2,1) = 2(1,0) + (0,1)$.
johnston-linear-matrix-algebra-F02396	252	■ 1.000	$\mathbb{R}^n, B=\left\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_k, \right.$ $\left. \ldots, \mathbf{v}_k\right\}$	$\mathbb{R}^n, B = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}$	Suppose $\mathcal{S}$ is a subspace of $\mathbb{R}^n$, $B = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}$ is a basis of $\mathcal{S}$, and
johnston-linear-matrix-algebra-F02397	252	■ 1.000	$\left\{\mathbf{e}_1, \mathbf{e}_2\right\}$	$\{\mathbf{e}_1, \mathbf{e}_2\}$	that order does not matter for bases (e.g., $\{\mathbf{e}_1, \mathbf{e}_2\}$ and $\{\mathbf{e}_2, \mathbf{e}_1\}$ are the
johnston-linear-matrix-algebra-F02398	252	■ 1.000	$\left\{\mathbf{e}_2, \mathbf{e}_1\right\}$	$\{\mathbf{e}_2, \mathbf{e}_1\}$	that order does not matter for bases (e.g., $\{\mathbf{e}_1, \mathbf{e}_2\}$ and $\{\mathbf{e}_2, \mathbf{e}_1\}$ are the

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johnston-linear-matrix-augmented-252 F02399	252	■ 0.992	$\left(c_{\{1\}}, c_{\{2\}}\right)$	(c_1, c_2)	(c_1, c_2) and (c_2, c_1) are different vectors). In other words, we want to
johnston-linear-matrix-augmented-252 F02400	252	■ 0.992	$\left(c_{\{2\}}, c_{\{1\}}\right)$	(c_2, c_1)	(c_1, c_2) and (c_2, c_1) are different vectors). In other words, we want to
johnston-linear-matrix-augmented-252 F02401	252	■ 1.000	$\mathcal{S}=\mathbb{R}^n$	$\mathcal{S} = \mathbb{R}^n$	When working with the standard basis of $\mathbb{R}^n$ (i.e., when $\mathcal{S} = \mathbb{R}^n$ and $B =$
johnston-linear-matrix-augmented-252 F02402	252	■ 0.999	$c_{\{1\}}, c_{\{2\}}, \ldots, c_{\{n\}}$	$c_1, c_2, \dots, c_n$	to the standard basis. This is because the unique scalars $c_1, c_2, \dots, c_n$ for which
johnston-linear-matrix-augmented-252 F02403	252	■ 1.000	$\mathbf{v}=\left(c_{\{1\}}, c_{\{2\}}, \ldots, c_{\{n\}}\right)=[\mathbf{v}]_B$	$\mathbf{v} = (c_1, c_2, \dots, c_n) = [\mathbf{v}]_B$	$\mathbf{v} = (c_1, c_2, \dots, c_n) = [\mathbf{v}]_B$. The true utility of coordinates and coordinate vectors
johnston-linear-matrix-augmented-252 F02404	252	■ 1.000	$\mathbf{v}=(5,1), B=\{(2,1),(1,2)\}, \mathcal{S}=\mathbb{R}^2$	$\mathbf{v} = (5, 1), B = \{(2, 1), (1, 2)\}, \mathcal{S} = \mathbb{R}^2$	a) $\mathbf{v} = (5, 1), B = \{(2, 1), (1, 2)\}, \mathcal{S} = \mathbb{R}^2$
johnston-linear-matrix-augmented-252 F02405	252	■ 1.000	$\mathbf{v}=(5,4,3), B=\{(1,2,1),(2,1,1)\}, \mathcal{S}=\operatorname{span}(B)$	$\mathbf{v} = (5, 4, 3), B = \{(1, 2, 1), (2, 1, 1)\}, \mathcal{S} = \operatorname{span}(B)$	b) $\mathbf{v} = (5, 4, 3), B = \{(1, 2, 1), (2, 1, 1)\}, \mathcal{S} = \operatorname{span}(B)$
johnston-linear-matrix-augmented-253 F02406	253	■ 1.000	$(5,1)$	$(5, 1)$	columns in a matrix with $(5, 1)$ as the augmented right-hand side:
johnston-linear-matrix-augmented-253 F02407	253	■ 1.000	$c_{\{1\}}=3, c_{\{2\}}=-1$	$c_1 = 3, c_2 = -1$	see that the unique solution is $c_1 = 3, c_2 = -1$, so $[\mathbf{v}]_B = (3, -1)$:
johnston-linear-matrix-augmented-253 F02408	253	■ 1.000	$[\mathbf{v}]_B=(3,-1)$	$[\mathbf{v}]_B = (3, -1)$	see that the unique solution is $c_1 = 3, c_2 = -1$, so $[\mathbf{v}]_B = (3, -1)$:
johnston-linear-matrix-augmented-253 F02409	253	■ 0.997	$c_{\{1\}}=1, c_{\{2\}}=2$	$c_1 = 1, c_2 = 2$	see that the unique solution is $c_1 = 1, c_2 = 2$, so $[\mathbf{v}]_B = (1, 2)$:
johnston-linear-matrix-augmented-253 F02410	253	■ 0.997	$[\mathbf{v}]_B=(1,2)$	$[\mathbf{v}]_B = (1, 2)$	see that the unique solution is $c_1 = 1, c_2 = 2$, so $[\mathbf{v}]_B = (1, 2)$:
johnston-linear-matrix-augmented-253 F02411	253	■ 1.000	$\left[\mathbf{v}\right]_B=\mathbf{e}_j$	$[\mathbf{v}]_B = \mathbf{e}_j$	It is worth noting that if $B = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}$ is a basis then $[\mathbf{v}]_B = \mathbf{e}_j$ (the
johnston-linear-matrix-augmented-253 F02412	253	■ 1.000	$1 \leq j \leq k$	$1 \leq j \leq k$	j -th standard basis vector) for all $1 \leq j \leq k$, since the unique linear combination
johnston-linear-matrix-augmented-253 F02413	253	■ 1.000	$\mathbf{v}_j$	$\mathbf{v}_j$	of $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}$ resulting in $\mathbf{v}_j$ is quite trivial:
johnston-linear-matrix-augmented-253 F02414	253	■ 1.000	$\mathbf{e}_1, \mathbf{v}_2$	$\mathbf{e}_1, \mathbf{v}_2$	so that $\mathbf{v}_1$ looks like $\mathbf{e}_1$, $\mathbf{v}_2$ looks like $\mathbf{e}_2$, and so on (see Figure 3.4). It is worth
johnston-linear-matrix-augmented-254 F02415	254	■ 0.844	$\mathcal{S}=\operatorname{span}((1,2,1),(2,1,1))$	$\mathcal{S} = \operatorname{span}((1, 2, 1), (2, 1, 1))$	$\mathbb{R}^3$, it also lives in the 2-dimensional plane $\mathcal{S} = \operatorname{span}((1, 2, 1), (2, 1, 1))$ and
johnston-linear-matrix-augmented-254 F02416	254	■ 1.000	$\mathcal{S}=\mathbb{R}^2$	$\mathcal{S} = \mathbb{R}^2$	$\mathcal{S} = \mathbb{R}^2$ and the basis is $B = \{(2, 1), (1, 2)\}$, as in Example 3.1.1(a).
johnston-linear-matrix-augmented-254 F02417	254	■ 1.000	$B=\{(2,1),(1,2)\}$	$B = \{(2, 1), (1, 2)\}$	$\mathcal{S} = \mathbb{R}^2$ and the basis is $B = \{(2, 1), (1, 2)\}$, as in Example 3.1.1(a).
johnston-linear-matrix-augmented-254 F02418	254	■ 1.000	$\mathbb{R}^{85}$	$\mathbb{R}^{85}$	of $\mathbb{R}^{85}$ (for example), then we can use coordinate vectors to describe vectors in
johnston-linear-matrix-augmented-254 F02419	254	■ 1.000	$[\mathbf{v}]_B$	$[\mathbf{v}]_B$	the coordinate vector $[\mathbf{v}]_B$ of the vector $\mathbf{v} = (2, 1, -3, 1, 2) \in \operatorname{range}(A)$:
johnston-linear-matrix-augmented-254 F02420	254	■ 1.000	$\mathbf{v}=(2,1,-3,1,2) \in \operatorname{range}(A)$	$\mathbf{v} = (2, 1, -3, 1, 2) \in \operatorname{range}(A)$	the coordinate vector $[\mathbf{v}]_B$ of the vector $\mathbf{v} = (2, 1, -3, 1, 2) \in \operatorname{range}(A)$:
johnston-linear-matrix-augmented-255 F02421	255	■ 1.000	$c_{\{1\}}=2$	$c_1 = 2$	It follows that $c_1 = 2$ and $c_2 = -3$, so $[\mathbf{v}]_B = (2, -3)$.
johnston-linear-matrix-augmented-255 F02422	255	■ 1.000	$c_{\{2\}}=-3$	$c_2 = -3$	It follows that $c_1 = 2$ and $c_2 = -3$, so $[\mathbf{v}]_B = (2, -3)$.
johnston-linear-matrix-augmented-255 F02423	255	■ 1.000	$[\mathbf{v}]_B=(2,-3)$	$[\mathbf{v}]_B = (2, -3)$	It follows that $c_1 = 2$ and $c_2 = -3$, so $[\mathbf{v}]_B = (2, -3)$.
johnston-linear-matrix-augmented-255 F02424	255	■ 0.991	$\mathbf{v}=(2,1,-3,1,2)$	$\mathbf{v} = (2, 1, -3, 1, 2)$	2-dimensional (i.e., its rank is 2), so representing the vector $\mathbf{v} = (2, 1, -3, 1, 2)$
johnston-linear-matrix-augmented-255 F02425	255	■ 1.000	$[\mathbf{v}+\mathbf{w}]_B=[\mathbf{v}]_B+[\mathbf{w}]_B$	$[\mathbf{v} + \mathbf{w}]_B = [\mathbf{v}]_B + [\mathbf{w}]_B$	a) $[\mathbf{v} + \mathbf{w}]_B = [\mathbf{v}]_B + [\mathbf{w}]_B$, and

Identifier	Page	Conf.	LaTeX source	Rendered	Image
johnston-linear-matrix-algebra-F02426	255	■ 0.997	$[c \, \mathbf{v}]_B = c[\mathbf{v}]_B$	$[c\mathbf{v}]_B = c[\mathbf{v}]_B$	b) $[c\mathbf{v}]_B = c[\mathbf{v}]_B$.
johnston-linear-matrix-algebra-F02427	255	■ 1.000	$B = \{(1, 2, 0, 2, 1), (0, 1, 1, 1, 0)\}$	$B = \{(1, 2, 0, 2, 1), (0, 1, 1, 1, 0)\}$	Let $B = \{(1, 2, 0, 2, 1), (0, 1, 1, 1, 0)\}$ be a basis of the subspace
johnston-linear-matrix-algebra-F02428	255	■ 1.000	$\mathbf{w} = (1, 0, -2, 0, 1)$	$\mathbf{w} = (1, 0, -2, 0, 1)$	and suppose $\mathbf{v} = (2, 1, -3, 1, 2)$ and $\mathbf{w} = (1, 0, -2, 0, 1)$. Compute the fol-
johnston-linear-matrix-algebra-F02429	255	■ 1.000	$[2 \, \mathbf{v} - 5 \, \mathbf{w}]_B$	$[2\mathbf{v} - 5\mathbf{w}]_B$	a) $[2\mathbf{v} - 5\mathbf{w}]_B$
johnston-linear-matrix-algebra-F02430	255	■ 1.000	$2[\mathbf{v}]_B - 5[\mathbf{w}]_B$	$2[\mathbf{v}]_B - 5[\mathbf{w}]_B$	b) $2[\mathbf{v}]_B - 5[\mathbf{w}]_B$
johnston-linear-matrix-algebra-F02431	255	■ 1.000	$2 \, \mathbf{v} - 5 \, \mathbf{w}$	$2\mathbf{v} - 5\mathbf{w}$	Next, we can compute the coordinate vector of $2\mathbf{v} - 5\mathbf{w}$ in the same
johnston-linear-matrix-algebra-F02432	256	■ 1.000	$\mathbf{v}_1 = (1, 0.002)$	$\mathbf{v}_1 = (1, 0.002)$	if $\mathbf{v}_1 = (1, 0.002)$ and
johnston-linear-matrix-algebra-F02433	256	■ 1.000	$\mathbf{v}_2 = (1, 0.001)$	$\mathbf{v}_2 = (1, 0.001)$	$\mathbf{v}_2 = (1, 0.001)$ then
johnston-linear-matrix-algebra-F02434	256	■ 1.000	$[(-1, 1)]_B =$	$[(-1, 1)]_B =$	$[(-1, 1)]_B =$
johnston-linear-matrix-algebra-F02435	256	■ 1.000	$[2 \, \mathbf{v} - 5 \, \mathbf{w}]_B = (-1, 4)$	$[2\mathbf{v} - 5\mathbf{w}]_B = (-1, 4)$	so $[2\mathbf{v} - 5\mathbf{w}]_B = (-1, 4)$.
johnston-linear-matrix-algebra-F02436	256	■ 1.000	$[\mathbf{w}]_B$	$[\mathbf{w}]_B$	just need to compute $[\mathbf{w}]_B$:
johnston-linear-matrix-algebra-F02437	256	■ 1.000	$[\mathbf{w}]_B = (1, -2)$	$[\mathbf{w}]_B = (1, -2)$	It follows that $[\mathbf{w}]_B = (1, -2)$, so we see that
johnston-linear-matrix-algebra-F02438	256	■ 1.000	$\mathbf{v}_1 = (1, 0.2)$	$\mathbf{v}_1 = (1, 0.2)$	this, suppose $\mathbf{v}_1 = (1, 0.2)$ and $\mathbf{v}_2 = (1, 0.1)$, and consider the basis $B =$
johnston-linear-matrix-algebra-F02439	256	■ 1.000	$\mathbf{v}_2 = (1, 0.1)$	$\mathbf{v}_2 = (1, 0.1)$	this, suppose $\mathbf{v}_1 = (1, 0.2)$ and $\mathbf{v}_2 = (1, 0.1)$, and consider the basis $B =$
johnston-linear-matrix-algebra-F02440	257	■ 1.000	$\mathbf{v} \neq \mathbf{w} \in B$	$\mathbf{v} \neq \mathbf{w} \in B$	a) $\mathbf{v} \cdot \mathbf{w} = 0$ for all $\mathbf{v} \neq \mathbf{w} \in B$, and
johnston-linear-matrix-algebra-F02441	257	■ 1.000	$\mathbf{v} \in B$	$\mathbf{v} \in B$	b) $\ \mathbf{v}\ = 1$ for all $\mathbf{v} \in B$.
johnston-linear-matrix-algebra-F02442	257	■ 0.998	$\ \mathbf{v}\ = \ [\mathbf{v}]_B \ $	$\ \mathbf{v}\ = \ [\mathbf{v}]_B \ $	If B is an orthonormal basis then $\ \mathbf{v}\ = \ [\mathbf{v}]_B \ $, so $[\mathbf{v}]_B$ cannot be large
johnston-linear-matrix-algebra-F02443	257	■ 1.000	$B = \{(1, 1), (1, -1)\}$	$B = \{(1, 1), (1, -1)\}$	Let $B = \{(1, 1), (1, -1)\}$ and $C = \{(3, 1), (0, 1)\}$ be bases of $\mathbb{R}^2$, and
johnston-linear-matrix-algebra-F02444	257	■ 1.000	$C = \{(3, 1), (0, 1)\}$	$C = \{(3, 1), (0, 1)\}$	Let $B = \{(1, 1), (1, -1)\}$ and $C = \{(3, 1), (0, 1)\}$ be bases of $\mathbb{R}^2$, and
johnston-linear-matrix-algebra-F02445	257	■ 1.000	$[\mathbf{v}]_B = (5, 1)$	$[\mathbf{v}]_B = (5, 1)$	suppose that $[\mathbf{v}]_B = (5, 1)$. Compute $[\mathbf{v}]_C$.
johnston-linear-matrix-algebra-F02446	257	■ 1.000	$[\mathbf{v}]_C$	$[\mathbf{v}]_C$	suppose that $[\mathbf{v}]_B = (5, 1)$. Compute $[\mathbf{v}]_C$.
johnston-linear-matrix-algebra-F02447	257	■ 1.000	$\mathbf{v} = 5(1, 1) + (1, -1) = (6, 4)$	$\mathbf{v} = 5(1, 1) + (1, -1) = (6, 4)$	Since $[\mathbf{v}]_B = (5, 1)$ we know that $\mathbf{v} = 5(1, 1) + (1, -1) = (6, 4)$. Next,
johnston-linear-matrix-algebra-F02448	257	■ 1.000	$\mathbf{v} = (6, 4)$	$\mathbf{v} = (6, 4)$	we simply represent $\mathbf{v} = (6, 4)$ in the basis C by finding scalars c_1 and c_2
johnston-linear-matrix-algebra-F02449	257	■ 1.000	$(6, 4) = c_1(3, 1) + c_2(0, 1)$	$(6, 4) = c_1(3, 1) + c_2(0, 1)$	so that $(6, 4) = c_1(3, 1) + c_2(0, 1)$:
johnston-linear-matrix-algebra-F02450	257	■ 1.000	$[\mathbf{v}]_C = (2, 2)$	$[\mathbf{v}]_C = (2, 2)$	It follows that $c_1 = 2$ and $c_2 = 2$, so we conclude that $[\mathbf{v}]_C = (2, 2)$, as
johnston-linear-matrix-algebra-F02451	258	■ 1.000	$P_{C \leftarrow B}$	$P_{C \leftarrow B}$	notation $P_{C \leftarrow B}$
johnston-linear-matrix-algebra-F02452	258	■ 1.000	$P_{B \rightarrow C}$	$P_{B \rightarrow C}$	instead of $P_{B \rightarrow C}$, but

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johnston-linear-matrix-algebra-F02453	258	1.000	$\left[\mathbf{v}_1\right]_C, \left[\mathbf{v}_2\right]_C, \dots, \left[\mathbf{v}_k\right]_C$	$[\mathbf{v}_1]_C, [\mathbf{v}_2]_C, \dots, [\mathbf{v}_k]_C$	whose columns are the coordinate vectors $[\mathbf{v}_1]_C, [\mathbf{v}_2]_C, \dots, [\mathbf{v}_k]_C$:
johnston-linear-matrix-algebra-F02454	259	1.000	$\mathcal{S}=\operatorname{span}(C)$	$\mathcal{S} = \operatorname{span}(C)$	that $\mathcal{S} = \operatorname{span}(C)$.
johnston-linear-matrix-algebra-F02455	259	1.000	$P_{\{C \rightarrow B\}}[\mathbf{v}]_B = [\mathbf{v}]_C$	$P_{C \leftarrow B}[\mathbf{v}]_B = [\mathbf{v}]_C$	a) $P_{C \leftarrow B}[\mathbf{v}]_B = [\mathbf{v}]_C$ for all $\mathbf{v} \in \mathcal{S}$, and
johnston-linear-matrix-algebra-F02456	259	1.000	$\left(P_{\{C \rightarrow B\}}\right)^{-1}=P_{\{B \rightarrow C\}}$	$(P_{C \leftarrow B})^{-1} = P_{B \leftarrow C}$	b) $P_{C \leftarrow B}$ is invertible and $(P_{C \leftarrow B})^{-1} = P_{B \leftarrow C}$.
johnston-linear-matrix-algebra-F02457	259	1.000	$\mathbf{v}=c_1 \mathbf{v}_1+c_2 \mathbf{v}_2+\cdots+c_k \mathbf{v}_k$	$\mathbf{v} = c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 + \cdots + c_k \mathbf{v}_k$	see that property (a) holds, suppose $\mathbf{v} \in \mathcal{S}$ and write $\mathbf{v} = c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 + \cdots + c_k \mathbf{v}_k$,
johnston-linear-matrix-algebra-F02458	259	1.000	$[\mathbf{v}]_B=\left(c_1, c_2, \dots, c_k\right)$	$[\mathbf{v}]_B = (c_1, c_2, \dots, c_k)$	so that $[\mathbf{v}]_B = (c_1, c_2, \dots, c_k)$. We can then directly compute
johnston-linear-matrix-algebra-F02459	259	1.000	$P \in \mathcal{M}_k$	$P \in \mathcal{M}_k$	suppose $P \in \mathcal{M}_k$ is any matrix for which $P[\mathbf{v}]_B = [\mathbf{v}]_C$ for all $\mathbf{v} \in \mathcal{S}$. For every
johnston-linear-matrix-algebra-F02460	259	1.000	$P[\mathbf{v}]_B = [\mathbf{v}]_C$	$P[\mathbf{v}]_B = [\mathbf{v}]_C$	suppose $P \in \mathcal{M}_k$ is any matrix for which $P[\mathbf{v}]_B = [\mathbf{v}]_C$ for all $\mathbf{v} \in \mathcal{S}$. For every
johnston-linear-matrix-algebra-F02461	259	0.999	$\mathbf{v}=\mathbf{v}_j$	$\mathbf{v} = \mathbf{v}_j$	$1 \leq j \leq k$, if $\mathbf{v} = \mathbf{v}_j$ then we see that $[\mathbf{v}]_B = [\mathbf{v}_j]_B = \mathbf{e}_j$ (the j -th standard basis
johnston-linear-matrix-algebra-F02462	259	0.999	$[\mathbf{v}]_B=\left[\mathbf{v}_j\right]_B=\mathbf{e}_j$	$[\mathbf{v}]_B = [\mathbf{v}_j]_B = \mathbf{e}_j$	$1 \leq j \leq k$, if $\mathbf{v} = \mathbf{v}_j$ then we see that $[\mathbf{v}]_B = [\mathbf{v}_j]_B = \mathbf{e}_j$ (the j -th standard basis
johnston-linear-matrix-algebra-F02463	259	1.000	$P[\mathbf{v}]_B=P \mathbf{e}_j$	$P[\mathbf{v}]_B = P \mathbf{e}_j$	vector), so $P[\mathbf{v}]_B = P \mathbf{e}_j$ is the j -th column of P . On the other hand, it is also the
johnston-linear-matrix-algebra-F02464	259	1.000	$P[\mathbf{v}]_B = [\mathbf{v}]_C = [\mathbf{v}_j]_C$	$P[\mathbf{v}]_B = [\mathbf{v}]_C = [\mathbf{v}_j]_C$	case that $P[\mathbf{v}]_B = [\mathbf{v}]_C = [\mathbf{v}_j]_C$. The j -th column of P thus equals $[\mathbf{v}_j]_C$ for each
johnston-linear-matrix-algebra-F02465	259	1.000	$\left[\mathbf{v}_j\right]_C$	$[\mathbf{v}_j]_C$	case that $P[\mathbf{v}]_B = [\mathbf{v}]_C = [\mathbf{v}_j]_C$. The j -th column of P thus equals $[\mathbf{v}_j]_C$ for each
johnston-linear-matrix-algebra-F02466	259	1.000	$P=P_{\{C \rightarrow B\}}$	$P = P_{C \leftarrow B}$	$1 \leq j \leq k$, so $P = P_{C \leftarrow B}$.
johnston-linear-matrix-algebra-F02467	259	1.000	$P_{B \leftarrow C} P_{C \leftarrow B}$	$P_{B \leftarrow C} P_{C \leftarrow B}$	It follows that the j -th column of $P_{B \leftarrow C} P_{C \leftarrow B}$ equals $\mathbf{e}_j$ for each $1 \leq j \leq k$,
johnston-linear-matrix-algebra-F02468	259	1.000	$P_{B \leftarrow C} P_{C \leftarrow B}=I$	$P_{B \leftarrow C} P_{C \leftarrow B} = I$	so $P_{B \leftarrow C} P_{C \leftarrow B} = I$. Theorem 2.2.7 then tells us that $P_{C \leftarrow B}$ is invertible with
johnston-linear-matrix-algebra-F02469	259	0.999	$[\mathbf{v}]_B =$	$[\mathbf{v}]_B =$	Equation (3.1.1) we used the change-of-basis matrix $P_{C \leftarrow B}$ to change $[\mathbf{v}]_B =$
johnston-linear-matrix-algebra-F02470	259	1.000	$[\mathbf{w}]_B=(-1,4)$	$[\mathbf{w}]_B = (-1, 4)$	have $[\mathbf{w}]_B = (-1, 4)$ then we can re-use that same change-of-basis matrix to
johnston-linear-matrix-algebra-F02471	259	1.000	$B=\left\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\right\}$	$B = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$	Then $B = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ and $C = \{\mathbf{w}_1, \mathbf{w}_2, \mathbf{w}_3\}$ are bases of $\mathcal{S} = \operatorname{span}(B)$.
johnston-linear-matrix-algebra-F02472	259	1.000	$C=\left\{\mathbf{w}_1, \mathbf{w}_2, \mathbf{w}_3\right\}$	$C = \{\mathbf{w}_1, \mathbf{w}_2, \mathbf{w}_3\}$	Then $B = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ and $C = \{\mathbf{w}_1, \mathbf{w}_2, \mathbf{w}_3\}$ are bases of $\mathcal{S} = \operatorname{span}(B)$.
johnston-linear-matrix-algebra-F02473	259	1.000	$[\mathbf{v}]_B=(1,2,3)$	$[\mathbf{v}]_B = (1, 2, 3)$	b) Compute $[\mathbf{v}]_C$ if $[\mathbf{v}]_B = (1, 2, 3)$.
johnston-linear-matrix-algebra-F02474	260	1.000	$\mathbf{v}_1 =$	$\mathbf{v}_1 =$	$\mathbf{v}_1 =$
johnston-linear-matrix-algebra-F02475	260	0.604	$c_1 \mathbf{w}_1+c_2 \mathbf{w}_2+c_3 \mathbf{w}_3$	$c_1 \mathbf{w}_1 + c_2 \mathbf{w}_2 + c_3 \mathbf{w}_3$	$c_1 \mathbf{w}_1 + c_2 \mathbf{w}_2 + c_3 \mathbf{w}_3$,
johnston-linear-matrix-algebra-F02476	260	1.000	$[\mathbf{w}]_C$	$[\mathbf{w}]_C$	c) Compute $[\mathbf{w}]_C$ if $[\mathbf{w}]_B = (2, 0, -1)$.
johnston-linear-matrix-algebra-F02477	260	1.000	$[\mathbf{w}]_B=(2,0,-1)$	$[\mathbf{w}]_B = (2, 0, -1)$	c) Compute $[\mathbf{w}]_C$ if $[\mathbf{w}]_B = (2, 0, -1)$.
johnston-linear-matrix-algebra-F02478	260	1.000	$\left[\mathbf{v}_1\right]_C, \left[\mathbf{v}_2\right]_C$	$[\mathbf{v}_1]_C, [\mathbf{v}_2]_C$	a) We must first compute $[\mathbf{v}_1]_C, [\mathbf{v}_2]_C$, and $[\mathbf{v}_3]_C$ (i.e., we must write
johnston-linear-matrix-algebra-F02479	260	1.000	$\left[\mathbf{v}_3\right]_C$	$[\mathbf{v}_3]_C$	a) We must first compute $[\mathbf{v}_1]_C, [\mathbf{v}_2]_C$, and $[\mathbf{v}_3]_C$ (i.e., we must write

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johnston-linear-matrix-a460a-F02480	260	1.000	$\mathbf{w}_1, \mathbf{w}_2$	$\mathbf{w}_1, \mathbf{w}_2$	$\mathbf{v}_1, \mathbf{v}_2$, and $\mathbf{v}_3$ each as a linear combination of $\mathbf{w}_1, \mathbf{w}_2$, and $\mathbf{w}_3$). For
johnston-linear-matrix-a460a-F02481	260	1.000	$\mathbf{w}_3$	$\mathbf{w}_3$	$\mathbf{v}_1, \mathbf{v}_2$, and $\mathbf{v}_3$ each as a linear combination of $\mathbf{w}_1, \mathbf{w}_2$, and $\mathbf{w}_3$). For
johnston-linear-matrix-a460a-F02482	260	1.000	$c_1=1, c_2=-1, c_3=1$	$c_1 = 1, c_2 = -1, c_3 = 1$	that $c_1 = 1, c_2 = -1, c_3 = 1$, so $[\mathbf{v}_1]_C = (1, -1, 1)$. Similar compu-
johnston-linear-matrix-a460a-F02483	260	1.000	$\left[\mathbf{v}_1\right]_C=(1,-1,1)$	$[\mathbf{v}_1]_C = (1, -1, 1)$	that $c_1 = 1, c_2 = -1, c_3 = 1$, so $[\mathbf{v}_1]_C = (1, -1, 1)$. Similar compu-
johnston-linear-matrix-a460a-F02484	260	1.000	$\left[\mathbf{v}_2\right]_C=(0,-1,1)$	$[\mathbf{v}_2]_C = (0, -1, 1)$	tations show that $[\mathbf{v}_2]_C = (0, -1, 1)$ and $[\mathbf{v}_3]_C = (0, -2, 1)$, so
johnston-linear-matrix-a460a-F02485	260	1.000	$\left[\mathbf{v}_3\right]_C=(0,-2,1)$	$[\mathbf{v}_3]_C = (0, -2, 1)$	tations show that $[\mathbf{v}_2]_C = (0, -1, 1)$ and $[\mathbf{v}_3]_C = (0, -2, 1)$, so
johnston-linear-matrix-a460a-F02486	260	1.000	$\mathbb{R}^k$	$\mathbb{R}^k$	different ways of making the subspace S look like $\mathbb{R}^k$, and the change-of-basis
johnston-linear-matrix-a460a-F02487	260	1.000	$P_{B \leftarrow C}$	$P_{B \leftarrow C}$	matrices $P_{C \leftarrow B}$ and $P_{B \leftarrow C}$ are linear transformations that convert these two
johnston-linear-matrix-a460a-F02488	261	1.000	$T: \mathbb{R}^k \rightarrow \mathbb{R}^k$	$T: \mathbb{R}^k \rightarrow \mathbb{R}^k$	transformation $T: \mathbb{R}^k \rightarrow \mathbb{R}^k$ (Theorem 1.4.1):
johnston-linear-matrix-a460a-F02489	261	1.000	$\left[\mathbf{v}_j\right]_B$	$[\mathbf{v}_j]_B$	transformation that sends $[\mathbf{v}_j]_B$ to $[\mathbf{v}_j]_C$ for all $1 \leq j \leq k$.
johnston-linear-matrix-a460a-F02490	261	1.000	$E=\left\{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\right\}$	$E = \{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\}$	Let $B = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ and $E = \{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\}$ be bases of $\mathbb{R}^3$, where
johnston-linear-matrix-a460a-F02491	261	1.000	$P_{E \leftarrow B}$	$P_{E \leftarrow B}$	Compute $P_{E \leftarrow B}$ and $P_{B \leftarrow E}$.
johnston-linear-matrix-a460a-F02492	261	1.000	$P_{B \leftarrow E}$	$P_{B \leftarrow E}$	Compute $P_{E \leftarrow B}$ and $P_{B \leftarrow E}$.
johnston-linear-matrix-a460a-F02493	261	0.998	$[\mathbf{v}]_E=\mathbf{v}$	$[\mathbf{v}]_E = \mathbf{v}$	already been done for us (i.e., recall that $[\mathbf{v}]_E = \mathbf{v}$ for all vectors $\mathbf{v}$):
johnston-linear-matrix-a460a-F02494	262	0.950	$m \neq n$	$m \neq n$	$m \neq n$), even with
johnston-linear-matrix-a460a-F02495	262	1.000	$\left[\mathbf{e}_1\right]_B, \left[\mathbf{e}_2\right]_B$	$[\mathbf{e}_1]_B, [\mathbf{e}_2]_B$	To construct $P_{B \leftarrow E}$, we could compute each of $[\mathbf{e}_1]_B$, $[\mathbf{e}_2]_B$, and $[\mathbf{e}_3]_B$
johnston-linear-matrix-a460a-F02496	262	1.000	$\left[\mathbf{e}_3\right]_B$	$[\mathbf{e}_3]_B$	To construct $P_{B \leftarrow E}$, we could compute each of $[\mathbf{e}_1]_B$, $[\mathbf{e}_2]_B$, and $[\mathbf{e}_3]_B$
johnston-linear-matrix-a460a-F02497	262	1.000	$P_{B \leftarrow E}=P_{E \leftarrow B}^{-1}$	$P_{B \leftarrow E} = P_{E \leftarrow B}^{-1}$	$P_{B \leftarrow E} = P_{E \leftarrow B}^{-1}$, so we can just invert the change of basis matrix that we
johnston-linear-matrix-a460a-F02498	262	1.000	$B=\left\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n\right\}$	$B = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$	Suppose $B = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ is a basis of $\mathbb{R}^n$ and $T: \mathbb{R}^n \rightarrow \mathbb{R}^n$ is a linear
johnston-linear-matrix-a460a-F02499	262	1.000	$[T]_B \in \mathcal{M}_n$	$[T]_B \in \mathcal{M}_n$	transformation. Then there exists a unique matrix $[T]_B \in \mathcal{M}_n$ for which
johnston-linear-matrix-a460a-F02500	262	1.000	$[T]_B$	$[T]_B$	the coordinate vector of the input and then multiply by the matrix $[T]_B$ (see
johnston-linear-matrix-a460a-F02501	262	1.000	$B=\left\{\mathbf{e}_1, \mathbf{e}_2, \ldots, \mathbf{e}_n\right\}$	$B = \{\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n\}$	we get if we use the standard basis $B = \{\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n\}$.
johnston-linear-matrix-a460a-F02502	262	1.000	$\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n$	$\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$	rem 1.4.1 (just replace the standard basis vectors $\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n$ by $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$
johnston-linear-matrix-a460a-F02503	262	1.000	$S, T: \mathbb{R}^n \rightarrow \mathbb{R}^n$	$S, T: \mathbb{R}^n \rightarrow \mathbb{R}^n$	$S, T: \mathbb{R}^n \rightarrow \mathbb{R}^n$ are linear transformations then $[S \circ T]_B = [S]_B[T]_B$, which is
johnston-linear-matrix-a460a-F02504	262	1.000	$[S \circ T]_B=[S]_B[T]_B$	$[S \circ T]_B = [S]_B[T]_B$	$S, T: \mathbb{R}^n \rightarrow \mathbb{R}^n$ are linear transformations then $[S \circ T]_B = [S]_B[T]_B$, which is
johnston-linear-matrix-a460a-F02505	263	1.000	$[T(\mathbf{v})]_B$	$[T(\mathbf{v})]_B$	$[\mathbf{v}]_B$ to $[T(\mathbf{v})]_B$. In other words, Theorem 3.1.4 says that we can first compute $T(\mathbf{v})$
johnston-linear-matrix-a460a-F02506	263	0.995	$T(x, y)=(2 x+y, x+2 y)$	$T(x, y) = (2x + y, x + 2y)$	Let $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be defined by $T(x, y) = (2x + y, x + 2y)$. Compute $[T]_B$

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johnston-linear-matrix-algebra F02507	263	1.000	$B=\left\{\mathbf{e}_1, \mathbf{e}_2\right\}$	$B=\{\mathbf{e}_1, \mathbf{e}_2\}$	a) $B=\{\mathbf{e}_1, \mathbf{e}_2\}$
johnston-linear-matrix-algebra F02508	263	1.000	$T(1,1)$	$T(1,1)$	b) First, we need to compute $T(1,1)$ and $T(1,-1)$:
johnston-linear-matrix-algebra F02509	263	1.000	$T(1,-1)$	$T(1,-1)$	b) First, we need to compute $T(1,1)$ and $T(1,-1)$:
johnston-linear-matrix-algebra F02510	265	1.000	$P_{C \leftarrow B}[T]_B P_{B \leftarrow C}$	$P_{C \leftarrow B}[T]_B P_{B \leftarrow C}$	<i>Proof.</i> We simply multiply the matrix $P_{C \leftarrow B}[T]_B P_{B \leftarrow C}$ on the right by an arbi-
johnston-linear-matrix-algebra F02511	265	1.000	$[T]_C$	$[T]_C$	However, we know from Theorem 3.1.4 that $[T]_C$ is the <i>unique</i> matrix for which
johnston-linear-matrix-algebra F02512	265	1.000	$[T]_C[\mathbf{v}]_C=[T(\mathbf{v})]_C$	$[T]_C[\mathbf{v}]_C=[T(\mathbf{v})]_C$	$[T]_C[\mathbf{v}]_C=[T(\mathbf{v})]_C$ for all $\mathbf{v} \in \mathbb{R}^n$, so it follows that $P_{C \leftarrow B}[T]_B P_{B \leftarrow C}=[T]_C$, as
johnston-linear-matrix-algebra F02513	265	1.000	$P_{C \leftarrow B}[T]_B P_{B \leftarrow C}=[T]_C$	$P_{C \leftarrow B}[T]_B P_{B \leftarrow C}=[T]_C$	$[T]_C[\mathbf{v}]_C=[T(\mathbf{v})]_C$ for all $\mathbf{v} \in \mathbb{R}^n$, so it follows that $P_{C \leftarrow B}[T]_B P_{B \leftarrow C}=[T]_C$, as
johnston-linear-matrix-algebra F02514	265	1.000	$[T(\mathbf{v})]_C$	$[T(\mathbf{v})]_C$	are numerous different ways of computing $[T(\mathbf{v})]_C$ from $\mathbf{v}$, such as:
johnston-linear-matrix-algebra F02515	265	1.000	$B=\{(1,2,3),(2,-1,0),(-1,1,2)\}$	$B=\{(1,2,3),(2,-1,0),(-1,1,2)\}$	Let $B=\{(1,2,3),(2,-1,0),(-1,1,2)\}$ be a basis of $\mathbb{R}^3$ and let $T: \mathbb{R}^3 \rightarrow$
johnston-linear-matrix-algebra F02516	265	1.000	$T: \mathbb{R}^3 \rightarrow$	$T: \mathbb{R}^3 \rightarrow$	Let $B=\{(1,2,3),(2,-1,0),(-1,1,2)\}$ be a basis of $\mathbb{R}^3$ and let $T: \mathbb{R}^3 \rightarrow$
johnston-linear-matrix-algebra F02517	266	1.000	$[T]_E$	$[T]_E$	written $[T]$ as $[T]_E$.
johnston-linear-matrix-algebra F02518	266	1.000	$[T]_B=P_{B \leftarrow E}[T] P_{E \leftarrow B}$	$[T]_B=P_{B \leftarrow E}[T] P_{E \leftarrow B}$	Theorem 3.1.5 tells us that $[T]_B=P_{B \leftarrow E}[T] P_{E \leftarrow B}$. Well, we recall from
johnston-linear-matrix-algebra F02519	266	1.000	$A=[T]_C$	$A=[T]_C$	<i>common</i> linear transformation T such that $A=[T]_C$ and $B=[T]_D$?
johnston-linear-matrix-algebra F02520	266	1.000	$B=[T]_D$	$B=[T]_D$	<i>common</i> linear transformation T such that $A=[T]_C$ and $B=[T]_D$?
johnston-linear-matrix-algebra F02521	266	1.000	$B=P_{D \leftarrow C} A P_{C \leftarrow D}$	$B=P_{D \leftarrow C} A P_{C \leftarrow D}$	we know that $A=[T]_C$ and $B=[T]_D$ if and only if $B=P_{D \leftarrow C} A P_{C \leftarrow D}$. However,
johnston-linear-matrix-algebra F02522	266	1.000	$P_{C \leftarrow D}=P_{D \leftarrow C}^{-1}$	$P_{C \leftarrow D}=P_{D \leftarrow C}^{-1}$	we also know from Theorem 3.1.3(b) that $P_{C \leftarrow D}=P_{D \leftarrow C}^{-1}$, so we have
johnston-linear-matrix-algebra F02523	267	1.000	$A=P B P^{-1}$	$A=P B P^{-1}$	$A=P B P^{-1}$. However,
johnston-linear-matrix-algebra F02524	267	1.000	$B=P^{-1} A P$	$B=P^{-1} A P$	since $B=P^{-1} A P$,
johnston-linear-matrix-algebra F02525	267	1.000	$B=P A P^{-1}$	$B=P A P^{-1}$	is an invertible matrix $P \in \mathcal{M}_n$ such that $B=P A P^{-1}$. This property is important
johnston-linear-matrix-algebra F02526	267	1.000	$P=\left[\begin{array}{cc} 1 & 1 \\ 1 & -1 \end{array}\right]$	$P=\left[\begin{array}{cc} 1 & 1 \\ 1 & -1 \end{array}\right]$	are similar, since straightforward calculation shows that if $P=\left[\begin{array}{cc} 1 & 1 \\ 1 & -1 \end{array}\right]$ then
johnston-linear-matrix-algebra F02527	267	1.000	$A=\left[\begin{array}{ll} 1 & 2 \\ 2 & 4 \end{array}\right]$	$A=\left[\begin{array}{cc} 1 & 2 \\ 2 & 4 \end{array}\right]$	Show that the matrices $A=\left[\begin{array}{cc} 1 & 2 \\ 2 & 4 \end{array}\right]$ and $B=\left[\begin{array}{cc} 1 & 2 \\ 3 & 4 \end{array}\right]$ are not similar.
johnston-linear-matrix-algebra F02528	267	1.000	$B=\left[\begin{array}{ll} 1 & 2 \\ 3 & 4 \end{array}\right]$	$B=\left[\begin{array}{cc} 1 & 2 \\ 3 & 4 \end{array}\right]$	Show that the matrices $A=\left[\begin{array}{cc} 1 & 2 \\ 2 & 4 \end{array}\right]$ and $B=\left[\begin{array}{cc} 1 & 2 \\ 3 & 4 \end{array}\right]$ are not similar.
johnston-linear-matrix-algebra F02529	268	1.000	$\operatorname{tr}(A)$	$\operatorname{tr}(A)$	Suppose $A \in \mathcal{M}_n$ is a square matrix. Then the trace of A , denoted by $\operatorname{tr}(A)$,
johnston-linear-matrix-algebra F02530	268	1.000	$\left[\begin{array}{ccc} 3 & 6 & -3 \\ 0 & -2 & 3 \\ 2 & 4 & -1 \end{array}\right]$	$\left[\begin{array}{ccc} 3 & 6 & -3 \\ 0 & -2 & 3 \\ 2 & 4 & -1 \end{array}\right]$	b) $\left[\begin{array}{ccc} 3 & 6 & -3 \\ 0 & -2 & 3 \\ 2 & 4 & -1 \end{array}\right]$

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johnston-linear-matrix-algebra-F02531	268	0.884	$\operatorname{tr}\left(\begin{bmatrix} 2 & 2 \\ 4 & 5 \end{bmatrix}\right)=2+5=7$	$\operatorname{tr}\left(\begin{bmatrix} 2 & 2 \\ 4 & 5 \end{bmatrix}\right)=2+5=7$	a) $\operatorname{tr}\left(\begin{bmatrix} 2 & 2 \\ 4 & 5 \end{bmatrix}\right)=2+5=7.$
johnston-linear-matrix-algebra-F02532	268	1.000	$\operatorname{tr}\left(\begin{bmatrix} 3 & 6 & -3 \\ 0 & -2 & 3 \\ 2 & 4 & -1 \end{bmatrix}\right)=3-2-1=0$	$\operatorname{tr}\left(\begin{bmatrix} 3 & 6 & -3 \\ 0 & -2 & 3 \\ 2 & 4 & -1 \end{bmatrix}\right)=3-2-1=0$	b) $\operatorname{tr}\left(\begin{bmatrix} 3 & 6 & -3 \\ 0 & -2 & 3 \\ 2 & 4 & -1 \end{bmatrix}\right)=3-2-1=0.$
johnston-linear-matrix-algebra-F02533	268	1.000	$\operatorname{tr}(A+B)=\operatorname{tr}(A)+\operatorname{tr}(B)$	$\operatorname{tr}(A+B)=\operatorname{tr}(A)+\operatorname{tr}(B)$	a) $\operatorname{tr}(A+B)=\operatorname{tr}(A)+\operatorname{tr}(B)$
johnston-linear-matrix-algebra-F02534	268	1.000	$\operatorname{tr}(cA)=c\operatorname{tr}(A)$	$\operatorname{tr}(cA)=c\operatorname{tr}(A)$	b) $\operatorname{tr}(cA)=c\operatorname{tr}(A)$
johnston-linear-matrix-algebra-F02535	268	1.000	$\operatorname{tr}(AB)=\operatorname{tr}(BA)$	$\operatorname{tr}(AB)=\operatorname{tr}(BA)$	c) $\operatorname{tr}(AB)=\operatorname{tr}(BA)$
johnston-linear-matrix-algebra-F02536	269	1.000	$[AB]_{i,i}$	$[AB]_{i,i}$	notation $[AB]_{i,i}$
johnston-linear-matrix-algebra-F02537	269	1.000	(i,i)	(i,i)	means the (i,i) -entry
johnston-linear-matrix-algebra-F02538	269	1.000	$a_{i,j}b_{j,i}=b_{j,i}a_{i,j}$	$a_{i,j}b_{j,i}=b_{j,i}a_{i,j}$	where the central step is accomplished by noting that $a_{i,j}b_{j,i}=b_{j,i}a_{i,j}$ and the
johnston-linear-matrix-algebra-F02539	269	1.000	$B=\begin{bmatrix} 1 & -1 \\ -2 & 1 \end{bmatrix}$	$B=\begin{bmatrix} 1 & -1 \\ -2 & 1 \end{bmatrix}$	Suppose $A=\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ and $B=\begin{bmatrix} 1 & -1 \\ -2 & 1 \end{bmatrix}$. Compute $\operatorname{tr}(AB)$ and $\operatorname{tr}(BA)$.
johnston-linear-matrix-algebra-F02540	269	1.000	$\operatorname{tr}(AB)$	$\operatorname{tr}(AB)$	Suppose $A=\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ and $B=\begin{bmatrix} 1 & -1 \\ -2 & 1 \end{bmatrix}$. Compute $\operatorname{tr}(AB)$ and $\operatorname{tr}(BA)$.
johnston-linear-matrix-algebra-F02541	269	1.000	$\operatorname{tr}(BA)$	$\operatorname{tr}(BA)$	Suppose $A=\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ and $B=\begin{bmatrix} 1 & -1 \\ -2 & 1 \end{bmatrix}$. Compute $\operatorname{tr}(AB)$ and $\operatorname{tr}(BA)$.
johnston-linear-matrix-algebra-F02542	269	0.997	$\operatorname{tr}(A)=\operatorname{tr}(B)$	$\operatorname{tr}(A)=\operatorname{tr}(B)$	 If $A \in \mathcal{M}_n$ and $B \in \mathcal{M}_n$ are similar then $\operatorname{tr}(A)=\operatorname{tr}(B)$.
johnston-linear-matrix-algebra-F02543	270	1.000	$B=\begin{bmatrix} 1 & 2 \\ 3 & 5 \end{bmatrix}$	$B=\begin{bmatrix} 1 & 2 \\ 3 & 5 \end{bmatrix}$	Show that the matrices $A=\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ and $B=\begin{bmatrix} 1 & 2 \\ 3 & 5 \end{bmatrix}$ are not similar.
johnston-linear-matrix-algebra-F02544	270	0.991	$\operatorname{rank}(A)=\operatorname{rank}(B)=2$	$\operatorname{rank}(A)=\operatorname{rank}(B)=2$	The rank does not help us here, since $\operatorname{rank}(A)=\operatorname{rank}(B)=2$. On the
johnston-linear-matrix-algebra-F02545	270	1.000	$\operatorname{tr}(A)=5$	$\operatorname{tr}(A)=5$	other hand, $\operatorname{tr}(A)=5$ and $\operatorname{tr}(B)=6$, so we know that A and B are not
johnston-linear-matrix-algebra-F02546	270	1.000	$\operatorname{tr}(B)=6$	$\operatorname{tr}(B)=6$	other hand, $\operatorname{tr}(A)=5$ and $\operatorname{tr}(B)=6$, so we know that A and B are not
johnston-linear-matrix-algebra-F02547	270	0.995	$\operatorname{tr}(ABC)=\operatorname{tr}(CAB)$	$\operatorname{tr}(ABC)=\operatorname{tr}(CAB)$	and similarly that $\operatorname{tr}(ABC)=\operatorname{tr}(CAB)$. However, the other similar-looking
johnston-linear-matrix-algebra-F02548	270	0.856	$\operatorname{tr}(ABC)=\operatorname{tr}(ACB)$	$\operatorname{tr}(ABC)=\operatorname{tr}(ACB)$	equalities like $\operatorname{tr}(ABC)=\operatorname{tr}(ACB)$ do <i>not</i> hold in general. For example, if
johnston-linear-matrix-algebra-F02549	270	1.000	$\operatorname{tr}(ABC)=1$	$\operatorname{tr}(ABC)=1$	then direct calculation reveals that $\operatorname{tr}(ABC)=1$, but $\operatorname{tr}(ACB)=0$.
johnston-linear-matrix-algebra-F02550	270	1.000	$\operatorname{tr}(ACB)=0$	$\operatorname{tr}(ACB)=0$	then direct calculation reveals that $\operatorname{tr}(ABC)=1$, but $\operatorname{tr}(ACB)=0$.
johnston-linear-matrix-algebra-F02551	270	1.000	$A, B, C, D \in \mathcal{M}_n$	$A, B, C, D \in \mathcal{M}_n$	if $A, B, C, D \in \mathcal{M}_n$ then
johnston-linear-matrix-algebra-F02552	270	1.000	$\operatorname{tr}(ABCD)$	$\operatorname{tr}(ABCD)$	However, $\operatorname{tr}(ABCD)$ does not necessarily equal the trace of any of the other 20

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johnston-linear-matrix-algebra-F02553	270	0.848	A, B, C	A, B, C	possible products of A, B, C , and D .
johnston-linear-matrix-algebra-F02554	270	0.999	$\mathbf{v}=(6,1)$, $B=\{(3,0),(0,2)\}$, $\mathcal{S}=\mathbb{R}^2$	$\mathbf{v}=(6,1), B=\{(3,0),(0,2)\}, \mathcal{S}=\mathbb{R}^2$	*(a) $\mathbf{v}=(6,1), B=\{(3,0),(0,2)\}, \mathcal{S}=\mathbb{R}^2$
johnston-linear-matrix-algebra-F02555	270	0.991	$\mathbf{v}=(4,2)$, $B=\{(1,1),(1,-1)\}$, $\mathcal{S}=\mathbb{R}^2$	$\mathbf{v}=(4,2), B=\{(1,1),(1,-1)\}, \mathcal{S}=\mathbb{R}^2$	(b) $\mathbf{v}=(4,2), B=\{(1,1),(1,-1)\}, \mathcal{S}=\mathbb{R}^2$
johnston-linear-matrix-algebra-F02556	270	0.987	$\mathbf{v}=(2,3,1)$, $B=\{(1,0,-1),(1,2,1)\}$, $\mathcal{S}$	$\mathbf{v}=(2,3,1), B=\{(1,0,-1),(1,2,1)\}, \mathcal{S}$	*(c) $\mathbf{v}=(2,3,1), B=\{(1,0,-1),(1,2,1)\}, \mathcal{S}$ is the
johnston-linear-matrix-algebra-F02557	270	1.000	$x+y+z=0$	$x-y+z=0$	plane in $\mathbb{R}^3$ with equation $x-y+z=0$.
johnston-linear-matrix-algebra-F02558	270	1.000	$\mathbf{v}=(1,1,1)$, $B=\{(1,2,3),(-1,0,1),(2,2,1)\}$, $\mathcal{S}$	$\mathbf{v}=(1,1,1), B=\{(1,2,3),(-1,0,1),(2,2,1)\}, \mathcal{S} =$	(d) $\mathbf{v}=(1,1,1), B=\{(1,2,3),(-1,0,1),(2,2,1)\}, \mathcal{S} =$
johnston-linear-matrix-algebra-F02559	270	1.000	$\mathbf{v}=(2,3)$	$\mathbf{v}=(2,3)$	3.1.2 Suppose $\mathbf{v}=(2,3)$ and $\mathbf{w}=(2,-1)$.
johnston-linear-matrix-algebra-F02560	270	1.000	$\mathbf{w}=(2,-1)$	$\mathbf{w}=(2,-1)$	3.1.2 Suppose $\mathbf{v}=(2,3)$ and $\mathbf{w}=(2,-1)$.
johnston-linear-matrix-algebra-F02561	270	0.999	$[\mathbf{v}]_B=(1,0)$	$[\mathbf{v}]_B=(1,0)$	*(a) Find a basis B of $\mathbb{R}^2$ such that $[\mathbf{v}]_B=(1,0)$ and
johnston-linear-matrix-algebra-F02562	270	0.992	$[\mathbf{w}]_B=(0,1)$	$[\mathbf{w}]_B=(0,1)$	$[\mathbf{w}]_B=(0,1)$.
johnston-linear-matrix-algebra-F02563	270	0.998	$[\mathbf{w}]_B=(-1,2)$	$[\mathbf{w}]_B=(-1,2)$	$[\mathbf{w}]_B=(-1,2)$.
johnston-linear-matrix-algebra-F02564	270	1.000	$\mathbf{v}=(1,5,6,8,-9)$	$\mathbf{v}=(1,5,6,8,-9)$	and $\mathbf{v}=(1,5,6,8,-9)$, then $\mathbf{v} \in \text{span}(B)$, and compute $[\mathbf{v}]_B$.
johnston-linear-matrix-algebra-F02565	270	1.000	$\mathbf{v} \in \text{span}(B)$	$\mathbf{v} \in \text{span}(B)$	and $\mathbf{v}=(1,5,6,8,-9)$, then $\mathbf{v} \in \text{span}(B)$, and compute $[\mathbf{v}]_B$.
johnston-linear-matrix-algebra-F02566	270	0.995	$B=\{(1,2),(3,4)\}$, $C=\{(1,0),(0,1)\}$	$B=\{(1,2),(3,4)\}, C=\{(1,0),(0,1)\}$	*(a) $B=\{(1,2),(3,4)\}, C=\{(1,0),(0,1)\}$
johnston-linear-matrix-algebra-F02567	270	1.000	$B=\{(1,0),(0,1)\}$, $C=\{(2,1),(-4,3)\}$	$B=\{(1,0),(0,1)\}, C=\{(2,1),(-4,3)\}$	(b) $B=\{(1,0),(0,1)\}, C=\{(2,1),(-4,3)\}$
johnston-linear-matrix-algebra-F02568	270	0.964	$B=\{(1,2),(3,4)\}$, $C=\{(2,1),(-4,3)\}$	$B=\{(1,2),(3,4)\}, C=\{(2,1),(-4,3)\}$	*(c) $B=\{(1,2),(3,4)\}, C=\{(2,1),(-4,3)\}$
johnston-linear-matrix-algebra-F02569	270	1.000	$B=\{(4,0,0),(0,4,0),(1,4,1)\}$	$B=\{(4,0,0),(0,4,0),(1,4,1)\}$	(d) $B=\{(4,0,0),(0,4,0),(1,4,1)\},$
johnston-linear-matrix-algebra-F02570	270	1.000	$C=\{(1,2,3),(1,0,1),(-1,1,2)\}$	$C=\{(1,2,3),(1,0,1),(-1,1,2)\}$	$C=\{(1,2,3),(1,0,1),(-1,1,2)\}$
johnston-linear-matrix-algebra-F02571	270	0.925	$\{(2,3),(1,-1)\}$	$\{(2,3),(1,-1)\}$	$\{(2,3),(1,-1)\}$:
johnston-linear-matrix-algebra-F02572	270	0.996	$T(\mathbf{v}_1, \mathbf{v}_2)=\left(\mathbf{v}_1-\mathbf{v}_2, \mathbf{v}_1+3\mathbf{v}_2\right)$	$T(v_1, v_2)=(v_1-v_2, v_1+3v_2)$	*(a) $T(v_1, v_2)=(v_1-v_2, v_1+3v_2)$.
johnston-linear-matrix-algebra-F02573	270	1.000	$T(\mathbf{v}_1, \mathbf{v}_2)=\left(\mathbf{v}_1+2\mathbf{v}_2, \mathbf{v}_1+2\mathbf{v}_2\right)$	$T(v_1, v_2)=(v_1+2v_2, v_1+2v_2)$	(b) $T(v_1, v_2)=(v_1+2v_2, v_1+2v_2)$.
johnston-linear-matrix-algebra-F02574	270	1.000	$T(\mathbf{v}_1, \mathbf{v}_2)=\left(-\mathbf{v}_1+2\mathbf{v}_2, 3\mathbf{v}_1\right)$	$T(v_1, v_2)=(-v_1+2v_2, 3v_1)$	*(c) $T(v_1, v_2)=(-v_1+2v_2, 3v_1)$.
johnston-linear-matrix-algebra-F02575	270	1.000	$T(\mathbf{v}_1, \mathbf{v}_2)=\left(4\mathbf{v}_1+3\mathbf{v}_2, 5\mathbf{v}_1+\mathbf{v}_2\right)$	$T(v_1, v_2)=(4v_1+3v_2, 5v_1+v_2)$	(d) $T(v_1, v_2)=(4v_1+3v_2, 5v_1+v_2)$.
johnston-linear-matrix-algebra-F02576	271	1.000	$B=\{(3,1,2),(0,-1,-1),(2,-1,0)\}$	$B=\{(3,1,2),(0,-1,-1),(2,-1,0)\}$	to the basis $B=\{(3,1,2),(0,-1,-1),(2,-1,0)\}$:
johnston-linear-matrix-algebra-F02577	271	0.595	$T(\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3)=$	$T(v_1, v_2, v_3) =$	*(a) $T(v_1, v_2, v_3) =$

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johnston-linear-matrix-algebra-F02578	271	0.802	$\begin{aligned} & \backslash\begin{array}{cc} 1 & -1 \\ -1 & 1 \end{array} \backslash\text{ and } B=\backslash\begin{array}{cc} 2 & -1 \\ -1 & 0 \end{array} \\ & \backslash\text{ (b) } A=\backslash\begin{array}{cc} 1 & -1 \\ -1 & 1 \end{array} \backslash\text{ and } B=\backslash\begin{array}{cc} 1 & -2 \\ -2 & 4 \end{array} \end{aligned}$	$A = \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} \text{ and } B = \begin{bmatrix} 2 & -1 \\ -1 & 0 \end{bmatrix}$ $(b) A = \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} \text{ and } B = \begin{bmatrix} 1 & -2 \\ -2 & 4 \end{bmatrix}$	$\ast(a) A = \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} \text{ and } B = \begin{bmatrix} 2 & -1 \\ -1 & 0 \end{bmatrix}$ $(b) A = \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} \text{ and } B = \begin{bmatrix} 1 & -2 \\ -2 & 4 \end{bmatrix}$
johnston-linear-matrix-algebra-F02579	271	0.816	$A=\backslash\begin{array}{lll} 0 & 1 & 2 \\ 1 & 2 & 3 \\ 2 & 3 & 4 \end{array} \backslash\text{right}]$	$A = \begin{bmatrix} 0 & 1 & 2 \\ 1 & 2 & 3 \\ 2 & 3 & 4 \end{bmatrix}$	$\ast(c) A = \begin{bmatrix} 0 & 1 & 2 \\ 1 & 2 & 3 \\ 2 & 3 & 4 \end{bmatrix} \text{ and } B = \begin{bmatrix} 1 & -1 & 0 \\ 3 & 1 & 2 \\ 1 & 1 & 1 \end{bmatrix}$
johnston-linear-matrix-algebra-F02580	271	0.816	$B=\backslash\begin{array}{ccc} 1 & -1 & 0 \\ 3 & 1 & 2 \\ 1 & 1 & 1 \end{array} \backslash\text{right}]$	$B = \begin{bmatrix} 1 & -1 & 0 \\ 3 & 1 & 2 \\ 1 & 1 & 1 \end{bmatrix}$	$\ast(c) A = \begin{bmatrix} 0 & 1 & 2 \\ 1 & 2 & 3 \\ 2 & 3 & 4 \end{bmatrix} \text{ and } B = \begin{bmatrix} 1 & -1 & 0 \\ 3 & 1 & 2 \\ 1 & 1 & 1 \end{bmatrix}$
johnston-linear-matrix-algebra-F02581	271	1.000	$A=\backslash\begin{array}{ccc} 2 & 0 & 1 \\ 1 & 1 & 3 \\ -1 & 2 & 1 \end{array} \backslash\text{right}]$	$A = \begin{bmatrix} 2 & 0 & 1 \\ 1 & 1 & 3 \\ -1 & 2 & 1 \end{bmatrix}$	$(d) A = \begin{bmatrix} 2 & 0 & 1 \\ 1 & 1 & 3 \\ -1 & 2 & 1 \end{bmatrix} \text{ and } B = \begin{bmatrix} 1 & -1 & 1 \\ -1 & 1 & 1 \\ 0 & 0 & 2 \end{bmatrix}$
johnston-linear-matrix-algebra-F02582	271	1.000	$B=\backslash\begin{array}{ccc} 1 & -1 & 1 \\ -1 & 1 & 1 \\ 0 & 0 & 2 \end{array} \backslash\text{right}]$	$B = \begin{bmatrix} 1 & -1 & 1 \\ -1 & 1 & 1 \\ 0 & 0 & 2 \end{bmatrix}$	$(d) A = \begin{bmatrix} 2 & 0 & 1 \\ 1 & 1 & 3 \\ -1 & 2 & 1 \end{bmatrix} \text{ and } B = \begin{bmatrix} 1 & -1 & 1 \\ -1 & 1 & 1 \\ 0 & 0 & 2 \end{bmatrix}$
johnston-linear-matrix-algebra-F02583	271	0.997	$A=\backslash\begin{array}{ll} 1 & 0 \\ 0 & 2 \end{array} \backslash\text{right}]$	$A = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$	$\ast(a) A = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix} \text{ and } B = \begin{bmatrix} 3 & -2 \\ 1 & 0 \end{bmatrix}$
johnston-linear-matrix-algebra-F02584	271	0.997	$B=\backslash\begin{array}{cc} 3 & -2 \\ 1 & 0 \end{array} \backslash\text{right}]$	$B = \begin{bmatrix} 3 & -2 \\ 1 & 0 \end{bmatrix}$	$\ast(a) A = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix} \text{ and } B = \begin{bmatrix} 3 & -2 \\ 1 & 0 \end{bmatrix}$
johnston-linear-matrix-algebra-F02585	271	1.000	$A=\backslash\begin{array}{ll} 1 & 2 \\ 0 & 3 \end{array} \backslash\text{right}]$	$A = \begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix}$	$(b) A = \begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix} \text{ and } B = \begin{bmatrix} 2 & 0 \\ 1 & 2 \end{bmatrix}$
johnston-linear-matrix-algebra-F02586	271	1.000	$B=\backslash\begin{array}{ll} 2 & 0 \\ 1 & 2 \end{array} \backslash\text{right}]$	$B = \begin{bmatrix} 2 & 0 \\ 1 & 2 \end{bmatrix}$	$(b) A = \begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix} \text{ and } B = \begin{bmatrix} 2 & 0 \\ 1 & 2 \end{bmatrix}$
johnston-linear-matrix-algebra-F02587	271	0.995	$P_{\{B \rightarrow B\}}=I$	$P_{B \leftarrow B} = I$	$\ast(a) \text{ If } B \text{ is a basis of a subspace } S \text{ of } \mathbb{R}^n \text{ then } P_{B \leftarrow B} = I.$
johnston-linear-matrix-algebra-F02588	271	1.000	$\operatorname{tr}(A B)=\operatorname{tr}(A) \operatorname{tr}(B)$	$\operatorname{tr}(AB) = \operatorname{tr}(A) \operatorname{tr}(B)$	$\operatorname{tr}(AB) = \operatorname{tr}(A) \operatorname{tr}(B).$
johnston-linear-matrix-algebra-F02589	271	1.000	$\operatorname{tr}\left(A^T\right)=\operatorname{tr}(A)$	$\operatorname{tr}\left(A^T\right)=\operatorname{tr}(A)$	$(h) \text{ If } A \text{ is a square matrix then } \operatorname{tr}\left(A^T\right)=\operatorname{tr}(A).$
johnston-linear-matrix-algebra-F02590	271	1.000	$C=\backslash\begin{array}{l} \mathbf{w}_1, \ldots, \mathbf{w}_m \end{array} \backslash\text{right} \backslash \backslash \subset \backslash\mathcal{S}$	$C = \{\mathbf{w}_1, \dots, \mathbf{w}_m\} \subset \mathcal{S}$	$\text{basis } B. \text{ Let } C = \{\mathbf{w}_1, \dots, \mathbf{w}_m\} \subset \mathcal{S} \text{ be a set of vectors and}$
johnston-linear-matrix-algebra-F02591	271	1.000	$D=\backslash\begin{array}{l} \left[\mathbf{w}_1\right]_B, \ldots, \left[\mathbf{w}_m\right]_B \end{array} \backslash\text{right} \backslash \backslash \left[\mathbf{w}_m\right]_B \backslash\text{right} \backslash \backslash \left[\mathbf{w}_m\right]_B \backslash\text{right} \backslash \backslash$	$D = \{[\mathbf{w}_1]_B, \dots, [\mathbf{w}_m]_B\}$	$\text{let } D = \{[\mathbf{w}_1]_B, \dots, [\mathbf{w}_m]_B\} \text{ be the set of corresponding co-}$
johnston-linear-matrix-algebra-F02592	271	1.000	$P=P_{\{B \rightarrow C\}}$	$P = P_{B \leftarrow C}$	$\text{a basis } C \text{ of } \mathcal{S} \text{ such that } P = P_{B \leftarrow C}.$
johnston-linear-matrix-algebra-F02593	271	0.997	B, C	B, C	$\ast\ast\textbf{3.1.14} \text{ Let } S \text{ be a subspace of } \mathbb{R}^n \text{ with bases } B, C, \text{ and } E.$
johnston-linear-matrix-algebra-F02594	271	1.000	$P_{\{C \rightarrow B\}}=P_{\{C \rightarrow E\}} P_{\{E \rightarrow B\}}$	$P_{C \leftarrow B} = P_{C \leftarrow E} P_{E \leftarrow B}$	$(a) \text{ Show that } P_{C \leftarrow B} = P_{C \leftarrow E} P_{E \leftarrow B}.$
johnston-linear-matrix-algebra-F02595	271	1.000	$\left[P_{\{E \rightarrow C\}} \mid P_{\{E \rightarrow B\}}\right]$	$[P_{E \leftarrow C} \mid P_{E \leftarrow B}]$	$[P_{E \leftarrow C} \mid P_{E \leftarrow B}] \text{ is } [I \mid P_{C \leftarrow B}].$

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johnston-linear-matrix-algebra-F02596	271	■ 1.000	$\left[I \mid P_{C \leftarrow B}\right]$	$[I \mid P_{C \leftarrow B}]$	$[P_{E \leftarrow C} \mid P_{E \leftarrow B}]$ is $[I \mid P_{C \leftarrow B}]$.
johnston-linear-matrix-algebra-F02597	271	■ 0.998	$\operatorname{nullity}(A)=\operatorname{nullity}(B)$	$\operatorname{nullity}(A) = \operatorname{nullity}(B)$	similar, then $\operatorname{nullity}(A) = \operatorname{nullity}(B)$.
johnston-linear-matrix-algebra-F02598	272	■ 0.999	$f: \mathcal{M}_n \rightarrow \mathbb{R}$	$f: \mathcal{M}_n \rightarrow \mathbb{R}$	**3.1.20 Show that the only linear function $f: \mathcal{M}_n \rightarrow \mathbb{R}$
johnston-linear-matrix-algebra-F02599	272	■ 1.000	$f(A \ B \ C)=f(A \ C \ B)$	$f(ABC) = f(ACB)$	with the property that $f(ABC) = f(ACB)$ for all $A, B, C \in$
johnston-linear-matrix-algebra-F02600	272	■ 1.000	$A, B, C \in$	$A, B, C \in$	with the property that $f(ABC) = f(ACB)$ for all $A, B, C \in$
johnston-linear-matrix-algebra-F02601	272	■ 1.000	$f(A)=0$	$f(A) = 0$	$f(A) = 0$ for all $A \in \mathcal{M}_n$.
johnston-linear-matrix-algebra-F02602	272	■ 1.000	$f(\left(\mathbf{e}_i \ \mathbf{e}_j^T\right))$	$f(\mathbf{e}_i \mathbf{e}_j^T)$	[Hint: Compute $f(\mathbf{e}_i \mathbf{e}_j^T)$ in multiple different ways.]
johnston-linear-matrix-algebra-F02603	272	■ 1.000	$\mathbf{v} \cdot \mathbf{w} = [\mathbf{v}]_B \cdot [\mathbf{w}]_B$	$\mathbf{v} \cdot \mathbf{w} = [\mathbf{v}]_B \cdot [\mathbf{w}]_B$	Remark 3.1.2) then $\mathbf{v} \cdot \mathbf{w} = [\mathbf{v}]_B \cdot [\mathbf{w}]_B$.
johnston-linear-matrix-algebra-F02604	272	■ 1.000	$B \subset \mathbb{R}^n$	$B \subset \mathbb{R}^n$	**3.1.22 Suppose $B \subset \mathbb{R}^n$ is a set consisting of non-zero
johnston-linear-matrix-algebra-F02605	272	■ 1.000	$\left\{R^{\theta}(\mathbf{e}_1), R^{\theta}(\mathbf{e}_2)\right\}$	$\{R^{\theta}(\mathbf{e}_1), R^{\theta}(\mathbf{e}_2)\}$	(a) Show that $\{R^{\theta}(\mathbf{e}_1), R^{\theta}(\mathbf{e}_2)\}$ is an orthonormal basis
johnston-linear-matrix-algebra-F02606	272	■ 1.000	$A \mathbf{x} \in \mathbb{R}^n$	$A \mathbf{x} \in \mathbb{R}^n$	linear transformation that sends $\mathbf{x} \in \mathbb{R}^n$ to $A \mathbf{x} \in \mathbb{R}^n$. That linear transformation
johnston-linear-matrix-algebra-F02607	272	■ 1.000	$A: A \mathbf{e}_1, A \mathbf{e}_2, \dots, A \mathbf{e}_n$	$A: A \mathbf{e}_1, A \mathbf{e}_2, \dots, A \mathbf{e}_n$	equal to the columns of A : $A \mathbf{e}_1, A \mathbf{e}_2, \dots, A \mathbf{e}_n$ (see Figure 3.10).
johnston-linear-matrix-algebra-F02608	272	■ 1.000	$\operatorname{det}(A)$	$\det(A)$	The determinant of A , which we denote by $\det(A)$, is the area (or volume,
johnston-linear-matrix-algebra-F02609	273	■ 1.000	$\operatorname{det}(A)=2$	$\det(A) = 2$	area of the unit square (i.e., $\det(A) = 2$), then it also doubles the area of any
johnston-linear-matrix-algebra-F02610	273	■ 0.534	$A(\operatorname{det}(A))=2.46$	$A(\det(A) = 2.46$	amount by the matrix A ($\det(A) = 2.46$ in this case).
johnston-linear-matrix-algebra-F02611	273	■ 1.000	$\operatorname{det}(I)=1$	$\det(I) = 1$	$\det(I) = 1$. Secondly, since every $A \in \mathcal{M}_n$ expands space by a factor of $\det(A)$,
johnston-linear-matrix-algebra-F02612	273	■ 1.000	$\operatorname{det}(B)$	$\det(B)$	and similarly each $B \in \mathcal{M}_n$ expands space by a factor of $\det(B)$, it must be the
johnston-linear-matrix-algebra-F02613	273	■ 1.000	$n=2$	$n = 2$	when $n = 2$ in Figure 3.13.
johnston-linear-matrix-algebra-F02614	274	■ 1.000	$\mathbf{a}_1$	$\mathbf{a}_1$	parallel to the side $\mathbf{a}_1$.
johnston-linear-matrix-algebra-F02615	274	■ 1.000	$A=\left[\mathbf{a}_1 \ \mathbf{a}_2\right]$	$A = [\mathbf{a}_1 \mid \mathbf{a}_2]$	Figure 3.12: Scaling one of the columns of $A = [\mathbf{a}_1 \mid \mathbf{a}_2]$ scales the area of the
johnston-linear-matrix-algebra-F02616	274	■ 1.000	$2 \ \mathbf{a}_2$	$2 \mathbf{a}_2$	amount. Here, the parallelogram with sides $\mathbf{a}_1$ and $2 \mathbf{a}_2$ has twice the area of the
johnston-linear-matrix-algebra-F02617	274	■ 1.000	$\mathbf{a}_2$	$\mathbf{a}_2$	parallelogram with sides $\mathbf{a}_1$ and $\mathbf{a}_2$.
johnston-linear-matrix-algebra-F02618	274	■ 0.998	$\operatorname{det}: \mathcal{M}_n \rightarrow \mathbb{R}$	$\det: \mathcal{M}_n \rightarrow \mathbb{R}$	The determinant is the (unique!) function $\det: \mathcal{M}_n \rightarrow \mathbb{R}$ that satisfies the
johnston-linear-matrix-algebra-F02619	274	■ 1.000	$\operatorname{det}(A \ B)=\operatorname{det}(A) \operatorname{det}(B)$	$\det(AB) = \det(A) \det(B)$	b) $\det(AB) = \det(A) \det(B)$ for all $A, B \in \mathcal{M}_n$, and
johnston-linear-matrix-algebra-F02620	274	■ 1.000	$\mathbf{v}, \mathbf{w}, \mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_n \in \mathbb{R}^n$	$\mathbf{v}, \mathbf{w}, \mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_n \in \mathbb{R}^n$	c) for all $c \in \mathbb{R}$ and all $\mathbf{v}, \mathbf{w}, \mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_n \in \mathbb{R}^n$, it is the case that
johnston-linear-matrix-algebra-F02621	275	■ 1.000	$\operatorname{det}(A) \neq 0$	$\det(A) \neq 0$	It follows that $\det(A) \neq 0$ and $\det(A^{-1}) = 1/\det(A)$. This makes sense geomet-
johnston-linear-matrix-algebra-F02622	275	■ 1.000	$\operatorname{det}\left(A^{-1}\right)=1 / \operatorname{det}(A)$	$\det\left(A^{-1}\right) = 1/\det(A)$	It follows that $\det(A) \neq 0$ and $\det(A^{-1}) = 1/\det(A)$. This makes sense geomet-
johnston-linear-matrix-algebra-F02623	275	■ 1.000	$\operatorname{det}(A)=0$	$\det(A) = 0$	A is not invertible then $\det(A) = 0$ (see Figure 3.14).

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johnston-linear-matrix-algebra F02624	275	1.000	$\operatorname{det}(P^{-1} A P)=0$	$\det(P^{-1}AP) = 0$	so our goal now is to show that $\det(P^{-1}AP) = 0$. By recalling that the first
johnston-linear-matrix-algebra F02625	275	0.817	$P \, \mathbf{e}_1=\mathbf{x}$	$P\mathbf{e}_1 = \mathbf{x}$	column of P is $\mathbf{x}$ (i.e., $P\mathbf{e}_1 = \mathbf{x}$), we see that
johnston-linear-matrix-algebra F02626	275	1.000	$P^{-1} A P$	$P^{-1}AP$	In other words, the first column of $P^{-1}AP$ is the zero vector $\mathbf{0}$, so our goal
johnston-linear-matrix-algebra F02627	276	1.000	$\operatorname{det}(P^{-1} A P)=\operatorname{det}(A)$	$\det(P^{-1}AP) = \det(A)$	that $\det(P^{-1}AP) = \det(A)$ for every invertible matrix P . In other words, the
johnston-linear-matrix-algebra F02628	276	0.965	$\operatorname{det}(A)=\operatorname{det}(B)$	$\det(A) = \det(B)$	 If $A \in \mathcal{M}_n$ and $B \in \mathcal{M}_n$ are similar then $\det(A) = \det(B)$.
johnston-linear-matrix-algebra F02629	276	1.000	$\operatorname{det}(c A)=c^n \operatorname{det}(A)$	$\det(cA) = c^n \det(A)$	a) $\det(cA) = c^n \det(A)$, and
johnston-linear-matrix-algebra F02630	276	1.000	$\operatorname{det}(A^T)=\operatorname{det}(A)$	$\det(A^T) = \det(A)$	b) $\det(A^T) = \det(A)$.
johnston-linear-matrix-algebra F02631	276	1.000	c^n	c^n	resulting volume by c^n (rather than just by c , which is what we might mistakenly
johnston-linear-matrix-algebra F02632	276	1.000	$2^2=4$	$2^2 = 4$	determinant) by a factor of $2^2 = 4$.
johnston-linear-matrix-algebra F02633	277	0.998	$A, B, C \in \mathcal{M}_3$	$A, B, C \in \mathcal{M}_3$	Suppose that $A, B, C \in \mathcal{M}_3$ are matrices with $\det(A) = 2$, $\det(B) = 3$, and
johnston-linear-matrix-algebra F02634	277	0.998	$\operatorname{det}(A)=2, \operatorname{det}(B)=3$	$\det(A) = 2, \det(B) = 3$	Suppose that $A, B, C \in \mathcal{M}_3$ are matrices with $\det(A) = 2$, $\det(B) = 3$, and
johnston-linear-matrix-algebra F02635	277	1.000	$\operatorname{det}(C)=5$	$\det(C) = 5$	$\det(C) = 5$. Compute each of the following determinants:
johnston-linear-matrix-algebra F02636	277	1.000	$\operatorname{det}(A B)$	$\det(AB)$	a) $\det(AB)$
johnston-linear-matrix-algebra F02637	277	1.000	$\operatorname{det}(A^2 C^T B^{-1})$	$\det(A^2C^TB^{-1})$	b) $\det(A^2C^TB^{-1})$
johnston-linear-matrix-algebra F02638	277	1.000	$\operatorname{det}(3 A B^{-2} C^2)$	$\det(3AB^{-2}C^2)$	c) $\det(3AB^{-2}C^2)$
johnston-linear-matrix-algebra F02639	277	1.000	$\operatorname{det}(2 A^{-3} B^{-2} C^T B \right)^4(C / 5)^2)$	$\det\left(2A^{-3}B^{-2}(C^TB)^4(C/5)^2\right)$	d) $\det\left(2A^{-3}B^{-2}(C^TB)^4(C/5)^2\right)$
johnston-linear-matrix-algebra F02640	277	1.000	$\operatorname{det}(A B)=\operatorname{det}(A) \operatorname{det}(B)=2 \cdot 3=6$	$\det(AB) = \det(A) \det(B) = 2 \cdot 3 = 6$	a) $\det(AB) = \det(A) \det(B) = 2 \cdot 3 = 6$.
johnston-linear-matrix-algebra F02641	277	1.000	$\operatorname{det}(A^2 C^T B^{-1})=\operatorname{det}(A)^2 \operatorname{det}(C^T) \operatorname{det}(B^{-1})=2^2 \cdot 5 \cdot(1 / 3)=20 / 3$	$\det(A^2C^TB^{-1}) = \det(A)^2 \det(C^T) \det(B^{-1}) = 2^2 \cdot 5 \cdot (1/3) = 20/3$	b) $\det(A^2C^TB^{-1}) = \det(A)^2 \det(C^T) \det(B^{-1}) = 2^2 \cdot 5 \cdot (1/3) = 20/3$.
johnston-linear-matrix-algebra F02642	277	1.000	$\operatorname{det}(3 A B^{-2} C^2)=3^3 \operatorname{det}(A) \operatorname{det}(B)^{-2} \operatorname{det}(C)^2=27 \cdot 2 \cdot(1 / 9) \cdot 25=25$	$\det(3AB^{-2}C^2) = 3^3 \det(A) \det(B)^{-2} \det(C)^2 = 27 \cdot 2 \cdot (1/9) \cdot 25 =$	c) $\det(3AB^{-2}C^2) = 3^3 \det(A) \det(B)^{-2} \det(C)^2 = 27 \cdot 2 \cdot (1/9) \cdot 25 =$
johnston-linear-matrix-algebra F02643	278	1.000	$c \, \mathbf{e}_i$	$c\mathbf{e}_i$	by adding $c\mathbf{e}_i$ to the j -th column of I , so we can use multilinearity (defining
johnston-linear-matrix-algebra F02644	278	0.523	$1+c \cdot 0=1$	$1 + c \cdot 0 = 1$	determinant is 0. This elementary matrix thus has determinant $1 + c \cdot 0 = 1$.
johnston-linear-matrix-algebra F02645	278	1.000	$R_2+\frac{1}{2} R_1$	$R_2 + \frac{1}{2}R_1$	Figure 3.17: The elementary matrix corresponding to the row operation $R_2 + \frac{1}{2}R_1$
johnston-linear-matrix-algebra F02646	279	1.000	$\mathbf{e}_i+\mathbf{e}_j$	$\mathbf{e}_i + \mathbf{e}_j$	equal $\mathbf{e}_i + \mathbf{e}_j$. Since this matrix has two equal columns, it is not invertible, so
johnston-linear-matrix-algebra F02647	280	1.000	$f(x)$	$f(x)$	is defined as the area under the graph of $f(x)$ from $x = a$ to $x = b$:
johnston-linear-matrix-algebra F02648	280	1.000	$x=b$	$x = b$	is defined as the area under the graph of $f(x)$ from $x = a$ to $x = b$:

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johnston-linear-matrix-algebra-F02649	280	■ 1.000	$f(x) \geq 0$	$f(x) \geq 0$	However, this geometric definition is only accurate if $f(x) \geq 0$ for all
johnston-linear-matrix-algebra-F02650	281	■ 1.000	$c_{\{1\}} R_{\{i_{\{1\}}\}}, c_{\{2\}} R_{\{i_{\{2\}}\}}, \ldots, c_{\{k\}} R_{\{i_{\{k\}}\}}$	$c_1 R_{i_1}, c_2 R_{i_2}, \dots, c_k R_{i_k}$	via k “multiplication” row operations $c_1 R_{i_1}, c_2 R_{i_2}, \dots, c_k R_{i_k}$ as well as s
johnston-linear-matrix-algebra-F02651	281	■ 1.000	$E_{\{m\}} \cdots E_{\{2\}} E_{\{1\}} A = I$	$E_m \cdots E_2 E_1 A = I$	<i>Proof.</i> Recall that if A is invertible then we can write $E_m \cdots E_2 E_1 A = I$, where
johnston-linear-matrix-algebra-F02652	281	■ 0.999	$E_{\{1\}}, E_{\{2\}}, \ldots, E_{\{m\}}$	$E_1, E_2, \dots, E_m$	$E_1, E_2, \dots, E_m$ are the elementary matrices corresponding to the row operations
johnston-linear-matrix-algebra-F02653	282	■ 1.000	$1 / (-1)^s = (-1)^s$	$1/(-1)^s = (-1)^s$	since $1/(-1)^s = (-1)^s$.
johnston-linear-matrix-algebra-F02654	282	■ 1.000	$\left[\begin{array}{ll} 0 & 2 \\ 3 & 4 \end{array} \right]$	$\begin{bmatrix} 0 & 2 \\ 3 & 4 \end{bmatrix}$	b) $\begin{bmatrix} 0 & 2 \\ 3 & 4 \end{bmatrix}$
johnston-linear-matrix-algebra-F02655	282	■ 1.000	$\frac{1}{2} R_1$	$\frac{1}{2} R_1$	the only “multiplication” row operation that we performed was $\frac{1}{2} R_1$,
johnston-linear-matrix-algebra-F02656	282	■ 1.000	$1 / (1 / 2) = 2$	$1/(1/2) = 2$	we conclude that the determinant of this matrix is $1/(1/2) = 2$.
johnston-linear-matrix-algebra-F02657	282	■ 1.000	$s=1$	$s = 1$	Since we performed $s = 1$ “swap” row operation, as well as the
johnston-linear-matrix-algebra-F02658	282	■ 0.999	$\frac{1}{3} R_1$	$\frac{1}{3} R_1$	two “multiplication” row operations $\frac{1}{3} R_1$ and $\frac{1}{2} R_2$, it follows that
johnston-linear-matrix-algebra-F02659	282	■ 0.997	$(-1) / ((1 / 2)(1 / 3)) = -6$	$(-1)/((1/2)(1/3)) = -6$	the determinant of this matrix is $(-1)/((1/2)(1/3)) = -6$.
johnston-linear-matrix-algebra-F02660	283	■ 1.000	$\frac{1}{2} R_3$	$\frac{1}{2} R_3$	“multiplication” row operation that we performed was $\frac{1}{2} R_3$, we con-
johnston-linear-matrix-algebra-F02661	283	■ 1.000	$a_{\{1,1\}}, a_{\{2,2\}}, \ldots, a_{\{n,n\}} \neq 0$	$a_{1,1}, a_{2,2}, \dots, a_{n,n} \neq 0$	$a_{1,1}, a_{2,2}, \dots, a_{n,n} \neq 0$. Then we can construct a sequence of row operations that
johnston-linear-matrix-algebra-F02662	283	■ 1.000	$a_{\{n,n\}}$	$a_{n,n}$	row-reduces A to I . First, divide the n -th row of A by $a_{n,n}$ and then, for each
johnston-linear-matrix-algebra-F02663	283	■ 1.000	$1 \leq i \leq n-1$	$1 \leq i \leq n-1$	$1 \leq i \leq n-1$, subtract $a_{i,n} R_n$ from R_i :
johnston-linear-matrix-algebra-F02664	283	■ 1.000	$a_{\{i,n\}} R_{\{n\}}$	$a_{i,n} R_n$	$1 \leq i \leq n-1$, subtract $a_{i,n} R_n$ from R_i :
johnston-linear-matrix-algebra-F02665	283	■ 1.000	$R_{\{i\}}$	R_i	$1 \leq i \leq n-1$, subtract $a_{i,n} R_n$ from R_i :
johnston-linear-matrix-algebra-F02666	283	■ 1.000	$a_{\{n-1,n-1\}}$	$a_{n-1,n-1}$	$(n-1)$ -th row by $a_{n-1,n-1}$ and then, for each $1 \leq i \leq n-2$, subtract $a_{i,n-1} R_{n-1}$
johnston-linear-matrix-algebra-F02667	283	■ 1.000	$1 \leq i \leq n-2$	$1 \leq i \leq n-2$	$(n-1)$ -th row by $a_{n-1,n-1}$ and then, for each $1 \leq i \leq n-2$, subtract $a_{i,n-1} R_{n-1}$
johnston-linear-matrix-algebra-F02668	283	■ 1.000	$a_{\{i,n-1\}} R_{\{n-1\}}$	$a_{i,n-1} R_{n-1}$	$(n-1)$ -th row by $a_{n-1,n-1}$ and then, for each $1 \leq i \leq n-2$, subtract $a_{i,n-1} R_{n-1}$
johnston-linear-matrix-algebra-F02669	284	■ 0.894	$a_{\{1,1\}} a_{\{2,2\}} \cdots a_{\{n,n\}} =$	$a_{1,1} a_{2,2} \cdots a_{n,n} =$	On the other hand, if A has a diagonal entry equal to 0, then $a_{1,1} a_{2,2} \cdots a_{n,n} =$
johnston-linear-matrix-algebra-F02670	284	■ 1.000	$a_{\{i,i\}} = 0$	$a_{i,i} = 0$	is not invertible). To see this, suppose that the $a_{i,i} = 0$ (and furthermore, i is
johnston-linear-matrix-algebra-F02671	284	■ 0.958	$r_{\{1,1\}}, r_{\{2,2\}}, \ldots, r_{\{n,n\}}$	$r_{1,1}, r_{2,2}, \dots, r_{n,n}$	where $r_{1,1}, r_{2,2}, \dots, r_{n,n}$ are the diagonal entries of the row echelon form R .
johnston-linear-matrix-algebra-F02672	284	■ 1.000	$\left[\begin{array}{cccc} 0 & 2 & 1 & 2 \\ 1 & -1 & 1 & 0 \\ 2 & 1 & 0 & 1 \\ -2 & 0 & 1 & 1 \end{array} \right]$	$\begin{bmatrix} 0 & 2 & 1 & 2 \\ 1 & -1 & 1 & 0 \\ 2 & 1 & 0 & 1 \\ -2 & 0 & 1 & 1 \end{bmatrix}$	d) $\begin{bmatrix} 0 & 2 & 1 & 2 \\ 1 & -1 & 1 & 0 \\ 2 & 1 & 0 & 1 \\ -2 & 0 & 1 & 1 \end{bmatrix}$

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johnston-linear-matrix-algebra-F02673	284	■ 1.000	$2 \cdot 1 = 2$	$2 \cdot 1 = 2$	$2 \cdot 1 = 2.$
johnston-linear-matrix-algebra-F02674	285	■ 1.000	$(1 / 3) R_{\{1\}}$	$(1/3)R_1$	that we applied was $(1/3)R_1$, we conclude that the determinant of
johnston-linear-matrix-algebra-F02675	285	■ 0.999	$(1 \cdot (-2) \cdot 1) / (1 / 3) = -6$	$(1 \cdot (-2) \cdot 1) / (1/3) = -6$	the original matrix is $(1 \cdot (-2) \cdot 1) / (1/3) = -6.$
johnston-linear-matrix-algebra-F02676	285	■ 1.000	$(-1) \cdot 1 \cdot 2 \cdot (-7 / 2) \cdot (5 / 7) = 5$	$(-1) \cdot 1 \cdot 2 \cdot (-7/2) \cdot (5/7) = 5$	original matrix is $(-1) \cdot 1 \cdot 2 \cdot (-7/2) \cdot (5/7) = 5.$
johnston-linear-matrix-algebra-F02677	285	■ 0.435	$[a]$	$[a]$	of a 1×1 matrix $[a]$
johnston-linear-matrix-algebra-F02678	286	■ 1.000	$B = \begin{bmatrix} 0 & 2 \\ 3 & 5 \end{bmatrix}$	$B = \begin{bmatrix} 0 & 2 \\ 3 & 5 \end{bmatrix}$	Show that the matrices $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} 0 & 2 \\ 3 & 5 \end{bmatrix}$ are not similar.
johnston-linear-matrix-algebra-F02679	286	■ 0.966	$\text{tr}(A) = \text{tr}(B) = 5$	$\text{tr}(A) = \text{tr}(B) = 5$	similar, since $\text{rank}(A) = \text{rank}(B) = 2$ and $\text{tr}(A) = \text{tr}(B) = 5$. However,
johnston-linear-matrix-algebra-F02680	286	■ 1.000	$\text{det}(A) \neq \text{det}(B)$	$\text{det}(A) \neq \text{det}(B)$	Since $\text{det}(A) \neq \text{det}(B)$, we conclude that A and B are not similar.
johnston-linear-matrix-algebra-F02681	288	■ 1.000	$a e i + b f g + c d h - a f h - b d i - c e g$	$aei + bfg + cdh - afh - bdi - ceg$	Determinant: $aei + bfg + cdh - afh - bdi - ceg$
johnston-linear-matrix-algebra-F02682	288	■ 1.000	$\begin{bmatrix} 0 & 2 & 1 \\ 1 & -1 & 1 \\ 2 & 1 & 0 \end{bmatrix}$	$\begin{bmatrix} 0 & 2 & 1 \\ 1 & -1 & 1 \\ 2 & 1 & 0 \end{bmatrix}$	d) $\begin{bmatrix} 0 & 2 & 1 \\ 1 & -1 & 1 \\ 2 & 1 & 0 \end{bmatrix}$
johnston-linear-matrix-algebra-F02683	289	■ 1.000	$m_{\{i, j\}}$	$m_{i,j}$	We typically use $m_{i,j}$
johnston-linear-matrix-algebra-F02684	289	■ 1.000	$c_{\{i, j\}}$	$c_{i,j}$	$c_{i,j}$ to denote the
johnston-linear-matrix-algebra-F02685	289	■ 0.898	$(\mathbf{i}, \mathbf{j})$	$(\mathbf{i}, \mathbf{j})$	a) the $(\mathbf{i}, \mathbf{j})$-minor of A is the determinant of the $(n - 1) \times (n - 1)$
johnston-linear-matrix-algebra-F02686	289	■ 0.843	$\mathbf{i}, \mathbf{j}$	$\mathbf{i}, \mathbf{j}$	b) the $(\mathbf{i}, \mathbf{j})$-cofactor of A is the quantity $(-1)^{i+j} m_{i,j}$, where $m_{i,j}$ is the
johnston-linear-matrix-algebra-F02687	289	■ 0.843	$(-1)^{i+j} m_{\{i, j\}}$	$(-1)^{i+j} m_{i,j}$	b) the $(\mathbf{i}, \mathbf{j})$-cofactor of A is the quantity $(-1)^{i+j} m_{i,j}$, where $m_{i,j}$ is the
johnston-linear-matrix-algebra-F02688	289	■ 0.860	$(-1)^{i+j}$	$(-1)^{i+j}$	minor $m_{i,j}$ by $(-1)^{i+j}$. In this case, the $(1, 2)$ -cofactor of A is
johnston-linear-matrix-algebra-F02689	289	■ 1.000	$c_{\{1, 2\}} = -m_{\{1, 2\}} = -(-6) = 6$	$c_{1,2} = -m_{1,2} = -(-6) = 6$	$c_{1,2} = -m_{1,2} = -(-6) = 6$ that we did earlier).
johnston-linear-matrix-algebra-F02690	290	■ 0.793	$(3, 2)$	$(3, 2)$	$(3, 2)$ -cofactors have their signs switched (we have highlighted these
johnston-linear-matrix-algebra-F02691	291	■ 1.000	$\{c_{\{i, j\}}\}_{i,j=1}^n$	$\{c_{i,j}\}_{i,j=1}^n$	Suppose $A \in \mathcal{M}_n$ has cofactors $\{c_{i,j}\}_{i,j=1}^n$. Then
johnston-linear-matrix-algebra-F02692	293	■ 1.000	$c_{\{1, 1\}}$	$c_{1,1}$	$c_{1,1}$. To see why the first equality holds, notice that if the matrix
johnston-linear-matrix-algebra-F02693	294	■ 1.000	$i - 1$	$i - 1$	for the i -th term in the sum, we perform $i - 1$ swap operations: first we swap
johnston-linear-matrix-algebra-F02694	294	■ 1.000	$i - 2$	$i - 2$	rows i and $i - 1$, then $i - 1$ and $i - 2$, then $i - 2$ and $i - 3$, and so on until rows
johnston-linear-matrix-algebra-F02695	294	■ 1.000	$i - 3$	$i - 3$	rows i and $i - 1$, then $i - 1$ and $i - 2$, then $i - 2$ and $i - 3$, and so on until rows
johnston-linear-matrix-algebra-F02696	294	■ 0.992	$(-1)^{i-1}$	$(-1)^{i-1}$	2 and 1. These swaps introduce a sign of $(-1)^{i-1}$, which explains why the $c_{i,1}$
johnston-linear-matrix-algebra-F02697	294	■ 0.992	$c_{\{i, 1\}}$	$c_{i,1}$	2 and 1. These swaps introduce a sign of $(-1)^{i-1}$, which explains why the $c_{i,1}$

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johnston-linear-matrix-algebra-F02698	294	1.000	$\operatorname{det}(A)=\operatorname{det}(\left(A^T\right)$	$\det(A) = \det(A^T)$	columns), recall from Theorem 3.2.2(b) that $\det(A) = \det(A^T)$, and a cofactor
johnston-linear-matrix-algebra-F02699	296	1.000	$f(I)=2$	$f(I) = 2$	require that $f(I) = 2$ instead of $f(I) = 1$:
johnston-linear-matrix-algebra-F02700	296	1.000	$f(I)=1$	$f(I) = 1$	require that $f(I) = 2$ instead of $f(I) = 1$:
johnston-linear-matrix-algebra-F02701	296	1.000	$f(A\ B)=f(A)\ f(B)$	$f(AB) = f(A)f(B)$	b) $f(AB) = f(A)f(B)$ for all $A, B \in \mathcal{M}_n$.
johnston-linear-matrix-algebra-F02702	296	1.000	$A=B=I$	$A = B = I$	properties (a) and (b) described above. To see why, simply let $A = B = I$
johnston-linear-matrix-algebra-F02703	296	0.998	$\left[\begin{array}{ll}1 & 2 \\ 3 & 5\end{array}\right]$	$\left[\begin{array}{cc}1 & 2 \\ 3 & 5\end{array}\right]$	(b) $\left[\begin{array}{cc}1 & 2 \\ 3 & 5\end{array}\right]$
johnston-linear-matrix-algebra-F02704	296	1.000	$\left[\begin{array}{ccc}1 & 3 & 0 \\ 0 & -2 & 2 \\ -1 & 0 & 1\end{array}\right]$	$\left[\begin{array}{ccc}1 & 3 & 0 \\ 0 & -2 & 2 \\ -1 & 0 & 1\end{array}\right]$	(d) $\left[\begin{array}{ccc}1 & 3 & 0 \\ 0 & -2 & 2 \\ -1 & 0 & 1\end{array}\right]$
johnston-linear-matrix-algebra-F02705	296	0.832	$\left[\begin{array}{ccc}2 & 0 & 1 \\ 5 & 2 & 8 \\ 3 & -2 & 7\end{array}\right]$	$\left[\begin{array}{ccc}2 & 0 & 1 \\ 5 & 2 & 8 \\ 3 & -2 & 7\end{array}\right]$	*(e) $\left[\begin{array}{ccc}2 & 0 & 1 \\ 5 & 2 & 8 \\ 3 & -2 & 7\end{array}\right]$
johnston-linear-matrix-algebra-F02706	296	1.000	$\left[\begin{array}{ccc}3 & 2 & -6 \\ 0 & 2 & 2 \\ 0 & 0 & 3\end{array}\right]$	$\left[\begin{array}{ccc}3 & 2 & -6 \\ 0 & 2 & 2 \\ 0 & 0 & 3\end{array}\right]$	$\left[\begin{array}{ccc}3 & 2 & -6 \\ 0 & 2 & 2 \\ 0 & 0 & 3\end{array}\right]$
johnston-linear-matrix-algebra-F02707	296	1.000	$\left[\begin{array}{cccc}1 & 2 & 1 & 0 \\ 2 & 1 & 2 & -3 \\ 4 & 3 & 1 & 2 \\ 0 & 0 & 2 & -2\end{array}\right]$	$\left[\begin{array}{cccc}1 & 2 & 1 & 0 \\ 2 & 1 & 2 & -3 \\ 4 & 3 & 1 & 2 \\ 0 & 0 & 2 & -2\end{array}\right]$	$\left[\begin{array}{cccc}1 & 2 & 1 & 0 \\ 2 & 1 & 2 & -3 \\ 4 & 3 & 1 & 2 \\ 0 & 0 & 2 & -2\end{array}\right]$
johnston-linear-matrix-algebra-F02708	297	0.994	$\left[\begin{array}{cccc}3 & 1 & 6 & 4 \\ -2 & 5 & 3 & 3 \\ 6 & -2 & 4 & 1 \\ 3 & 4 & -2 & 6\end{array}\right]$	$\left[\begin{array}{cccc}3 & 1 & 6 & 4 \\ -2 & 5 & 3 & 3 \\ 6 & -2 & 4 & 1 \\ 3 & 4 & -2 & 6\end{array}\right]$	*(a) $\left[\begin{array}{cccc}3 & 1 & 6 & 4 \\ -2 & 5 & 3 & 3 \\ 6 & -2 & 4 & 1 \\ 3 & 4 & -2 & 6\end{array}\right]$
johnston-linear-matrix-algebra-F02709	297	1.000	$\left[\begin{array}{ccccc}-2 & 6 & -1 & 6 & -1 \\ 5 & 4 & -1 & 3 & -1 \\ 4 & 3 & 4 & -2 & 1 \\ 6 & 6 & 5 & 4 & 2 \\ -1 & 2 & -2 & -1 & -2\end{array}\right]$	$\left[\begin{array}{ccccc}-2 & 6 & -1 & 6 & -1 \\ 5 & 4 & -1 & 3 & -1 \\ 4 & 3 & 4 & -2 & 1 \\ 6 & 6 & 5 & 4 & 2 \\ -1 & 2 & -2 & -1 & -2\end{array}\right]$	(b) $\left[\begin{array}{ccccc}-2 & 6 & -1 & 6 & -1 \\ 5 & 4 & -1 & 3 & -1 \\ 4 & 3 & 4 & -2 & 1 \\ 6 & 6 & 5 & 4 & 2 \\ -1 & 2 & -2 & -1 & -2\end{array}\right]$
johnston-linear-matrix-algebra-F02710	297	0.999	$\left[\begin{array}{cccccc}2 & 2 & 5 & -1 & 1 & -1 \\ 3 & 0 & 0 & 1 & 1 & 1 \\ 1 & 2 & 5 & -1 & 0 & 3 \\ 4 & 0 & 5 & 5 & -1 & 3 \\ 5 & -1 & 5 & 2 & 0 & -1 \\ 3 & 0 & 3 & 3 & 4 & 3\end{array}\right]$	$\left[\begin{array}{cccccc}2 & 2 & 5 & -1 & 1 & -1 \\ 1 & 3 & 0 & 0 & 1 & 1 \\ 1 & 2 & 5 & -1 & 0 & 3 \\ 4 & 0 & 5 & 5 & -1 & 3 \\ 5 & -1 & 5 & 2 & 0 & -1 \\ 3 & 0 & 3 & 3 & 4 & 3\end{array}\right]$	*(c) $\left[\begin{array}{cccccc}2 & 2 & 5 & -1 & 1 & -1 \\ 1 & 3 & 0 & 0 & 1 & 1 \\ 1 & 2 & 5 & -1 & 0 & 3 \\ 4 & 0 & 5 & 5 & -1 & 3 \\ 5 & -1 & 5 & 2 & 0 & -1 \\ 3 & 0 & 3 & 3 & 4 & 3\end{array}\right]$

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johnston-linear-matrix-algebra-F02711	297	1.000	$\left[\begin{array}{cccccc}-2 & 3 & 2 & 5 & 2 & 4 \\ -1 & 2 & -1 & -3 & 2 & 0 \\ 6 & -2 & 3 & 5 & 4 & -1 \\ -2 & 2 & -2 & -2 & -3 & 3 \\ 3 & 3 & 2 & 2 & -3 & 1 \\ 1 & 6 & 0 & -1 & -3 & -2\end{array}\right]$		
johnston-linear-matrix-algebra-F02712	297	1.000	$A, B, C \in \mathcal{M}_5$	$A, B, C \in \mathcal{M}_5$	3.2.3 Suppose that $A, B, C \in \mathcal{M}_5$ are such that $\det(A) = 2$,
johnston-linear-matrix-algebra-F02713	297	1.000	$\operatorname{det}(B)=3$	$\det(B) = 3$	$\det(B) = 3$, and $\det(C) = 0$. Compute the determinant of
johnston-linear-matrix-algebra-F02714	297	1.000	$\operatorname{det}(C)=0$	$\det(C) = 0$	$\det(B) = 3$, and $\det(C) = 0$. Compute the determinant of
johnston-linear-matrix-algebra-F02715	297	0.904	$A B^T$	AB^T	*(a) AB^T
johnston-linear-matrix-algebra-F02716	297	1.000	$A^2 B$	A^2B	*(c) A^2B
johnston-linear-matrix-algebra-F02717	297	1.000	$2 A^2$	$2A^2$	(d) $2A^2$
johnston-linear-matrix-algebra-F02718	297	1.000	$A^{-1} B^3$	$A^{-1}B^3$	*(e) $A^{-1}B^3$
johnston-linear-matrix-algebra-F02719	297	1.000	$A^3 B^{-2} A^{-4} B^2 A$	$A^3B^{-2}A^{-4}B^2A$	(f) $A^3B^{-2}A^{-4}B^2A$
johnston-linear-matrix-algebra-F02720	297	1.000	$3 A^2 \left(B^T B / 2\right)^{-1}$	$3A^2 \left(B^TB/2\right)^{-1}$	*(g) $3A^2(B^TB/2)^{-1}$
johnston-linear-matrix-algebra-F02721	297	1.000	$A^{64} B^{-14} C^9 A^{-6} B^{13}$	$A^{64}B^{-14}C^9A^{-6}B^{13}$	(h) $A^{64}B^{-14}C^9A^{-6}B^{13}$
johnston-linear-matrix-algebra-F02722	297	1.000	$\operatorname{det}(A)=6$	$\det(A) = 6$	3.2.4 Suppose that $A \in \mathcal{M}_4$ is a matrix with $\det(A) = 6$.
johnston-linear-matrix-algebra-F02723	297	1.000	$3 R_3$	$3R_3$	(b) $3R_3$
johnston-linear-matrix-algebra-F02724	297	0.961	$R_1 \leftrightarrow R_4$	$R_1 \leftrightarrow R_4$	*(c) $R_1 \leftrightarrow R_4$
johnston-linear-matrix-algebra-F02725	297	1.000	$R_2-2 R_1, R_3-4 R_1$	R_2-2R_1, R_3-4R_1	(d) R_2-2R_1, R_3-4R_1
johnston-linear-matrix-algebra-F02726	297	0.991	$2 R_1, R_2 \leftrightarrow R_4$	$2R_1, R_2 \leftrightarrow R_4$	*(e) $2R_1, R_2 \leftrightarrow R_4$
johnston-linear-matrix-algebra-F02727	297	1.000	$R_1 \leftrightarrow R_2, R_2 \leftrightarrow R_3$	$R_1 \leftrightarrow R_2, R_2 \leftrightarrow R_3$	(f) $R_1 \leftrightarrow R_2, R_2 \leftrightarrow R_3$
johnston-linear-matrix-algebra-F02728	297	1.000	$2 R_2, 3 R_3, 4 R_4$	$2R_2, 3R_3, 4R_4$	*(g) $2R_2, 3R_3, 4R_4$
johnston-linear-matrix-algebra-F02729	297	1.000	$R_1 \leftrightarrow R_3, 2 R_2, R_3-3 R_1, R_4-7 R_2, 3 R_4$	$R_1 \leftrightarrow R_3, 2R_2, R_3-3R_1, R_4-7R_2, 3R_4$	(h) $R_1 \leftrightarrow R_3, 2R_2, R_3-3R_1, R_4-7R_2, 3R_4$
johnston-linear-matrix-algebra-F02730	297	0.979	$\left[\begin{array}{lll} g & h & i \\ d & e & f \\ a & b & c \end{array}\right]$		
johnston-linear-matrix-algebra-F02731	297	1.000	$\left[\begin{array}{ccc} a & b & c \\ 2 d & 2 e & 2 f \\ 3 g & 3 h & 3 i \end{array}\right]$		
johnston-linear-matrix-algebra-F02732	297	0.566	$\left[\begin{array}{ccc} a & b+a & 2 c \\ d & e+d & 2 f \\ g & h+g & 2 i \end{array}\right]$		

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johnston-linear-matrix-algebra-F02733	297	■ 1.000	$\left[\begin{array}{ccc}a & 2b & 3c \\ 2d & 4e & 6f \\ 3g & 6h & 9i\end{array}\right]$	$\begin{bmatrix} a & 2b & 3c \\ 2d & 4e & 6f \\ 3g & 6h & 9i \end{bmatrix}$	(d) $\begin{bmatrix} a & 2b & 3c \\ 2d & 4e & 6f \\ 3g & 6h & 9i \end{bmatrix}$
johnston-linear-matrix-algebra-F02734	297	■ 0.997	$\operatorname{det}(-A)=$	$\det(-A) =$	*(a) For all square matrices A , it is true that $\det(-A) =$
johnston-linear-matrix-algebra-F02735	297	■ 1.000	$-\operatorname{det}(A)$	$-\det(A)$	$-\det(A)$.
johnston-linear-matrix-algebra-F02736	297	■ 1.000	$\operatorname{det}(A\ B)=\operatorname{det}(B\ A)$	$\det(AB) = \det(BA)$	$\det(AB) = \det(BA)$.
johnston-linear-matrix-algebra-F02737	297	■ 1.000	$\operatorname{det}(A+B)=\operatorname{det}(A)+\operatorname{det}(B)$	$\det(A+B) = \det(A) + \det(B)$	$\det(A+B) = \det(A) + \det(B)$.
johnston-linear-matrix-algebra-F02738	297	■ 0.952	$\operatorname{det}(A)=3$	$\det(A) = 3$	(d) If A is a 3×3 matrix with $\det(A) = 3$, then $\operatorname{rank}(A) =$
johnston-linear-matrix-algebra-F02739	297	■ 1.000	$\operatorname{det}(A)=1$	$\det(A) = 1$	form is I , then $\det(A) = 1$.
johnston-linear-matrix-algebra-F02740	297	■ 1.000	$B_{\{n\}}$	B_n	3.2.7 Let B_n be the $n \times n$ “backwards identity” matrix
johnston-linear-matrix-algebra-F02741	297	■ 1.000	$\operatorname{det}\left(B_{\{n\}}\right)$	$\det(B_n)$	Find a formula for $\det(B_n)$ in terms of n .
johnston-linear-matrix-algebra-F02742	297	■ 0.977	$\left[P_{\{\mathbf{u}\}}\right]$	$[P_{\mathbf{u}}]$	*3.2.10 Let $[P_{\mathbf{u}}]$ be the standard matrix of a projection
johnston-linear-matrix-algebra-F02743	297	■ 1.000	$\operatorname{det}\left(\left[P_{\{\mathbf{u}\}}\right]\right)$	$\det([P_{\mathbf{u}}])$	onto a line (see Section 1.4.2). Compute $\det([P_{\mathbf{u}}])$.
johnston-linear-matrix-algebra-F02744	297	■ 1.000	$\operatorname{det}\left(A^{\{r\}}\right)=\left(\operatorname{det}(A)\right)^{\{r\}}$	$\det(A^r) = (\det(A))^r$	$\det(A^r) = (\det(A))^r$.
johnston-linear-matrix-algebra-F02745	297	■ 1.000	$A^{\{T\}}=-A$	$A^T = -A$	$A^T = -A$. Show that skew-symmetric matrices are not in-
johnston-linear-matrix-algebra-F02746	298	■ 1.000	$A^{\{R\}}$	A^R	3.2.14 Given a matrix $A \in \mathcal{M}_n$, let A^R denote the matrix
johnston-linear-matrix-algebra-F02747	298	■ 1.000	$\operatorname{det}\left(A^{\{R\}}\right)$	$\det(A^R)$	Find a formula for $\det(A^R)$ in terms of $\det(A)$ and n . [Hint:
johnston-linear-matrix-algebra-F02748	298	■ 1.000	$\operatorname{det}\left(A_{\{n\}}\right)$	$\det(A_n)$	(a) Use computer software to compute $\det(A_n)$ when
johnston-linear-matrix-algebra-F02749	298	■ 1.000	$n=2,3,4,5$	$n = 2, 3, 4, 5$	$n = 2, 3, 4, 5$.
johnston-linear-matrix-algebra-F02750	298	■ 0.971	$(*)$	$(*)$	where asterisks $(*)$ denote blocks whose values are
johnston-linear-matrix-algebra-F02751	298	■ 1.000	$\prod_{j=1}^{\{n\}}$	$\prod_{j=1}^n$	[Side note: The notation $\prod_{j=1}^n$ refers to a product
johnston-linear-matrix-algebra-F02752	298	■ 1.000	$A \in \mathcal{M}_{\{m\}}, B \in \mathcal{M}_{\{m, n\}}, C \in \mathcal{M}_{\{n, m\}}$	$A \in \mathcal{M}_m, B \in \mathcal{M}_{m,n}, C \in \mathcal{M}_{n,m}$	3.2.17 Suppose $A \in \mathcal{M}_m, B \in \mathcal{M}_{m,n}, C \in \mathcal{M}_{n,m}$, and
johnston-linear-matrix-algebra-F02753	298	■ 0.922	$\operatorname{det}\left(I_{\{n\}}+\mathbf{v} \ \mathbf{w}^{\{T\}}\right)=1+\mathbf{w}^{\{T\}} \ \mathbf{v}$	$\det(I_n + \mathbf{vw}^T) = 1 + \mathbf{w}^T \mathbf{v}$	(b) Use part (a) to show that $\det(I_n + \mathbf{vw}^T) = 1 + \mathbf{w}^T \mathbf{v}$.
johnston-linear-matrix-algebra-F02754	298	■ 0.827	$\operatorname{det}\left(A+\mathbf{v} \ \mathbf{w}^{\{T\}}\right)=\left(1+\mathbf{w}^{\{T\}} A^{-1} \ \mathbf{v}\right) \operatorname{det}(A)$	$\det(A + \mathbf{vw}^T) = (1 + \mathbf{w}^T A^{-1} \mathbf{v}) \det(A)$	(c) Show that $\det(A + \mathbf{vw}^T) = (1 + \mathbf{w}^T A^{-1} \mathbf{v}) \det(A)$.
johnston-linear-matrix-algebra-F02755	298	■ 0.971	$A+\mathbf{v} \ \mathbf{w}^{\{T\}}=A\left(I_{\{n\}}+A^{-1} \ \mathbf{v} \ \mathbf{w}^{\{T\}}\right)$	$A + \mathbf{vw}^T = A(I_n + A^{-1} \mathbf{vw}^T)$	[Hint: Write $A + \mathbf{vw}^T = A(I_n + A^{-1} \mathbf{vw}^T)$.]
johnston-linear-matrix-algebra-F02756	298	■ 1.000	$(-1)^{\{n+1\}}$	$(-1)^{n+1}$	equal to $(-1)^{n+1}$.
johnston-linear-matrix-algebra-F02757	298	■ 1.000	$f: \mathcal{M}_{\{n\}} \rightarrow$	$f: \mathcal{M}_n \rightarrow$	3.2.22 Show that there does not exist a function $f: \mathcal{M}_n \rightarrow$
johnston-linear-matrix-algebra-F02758	298	■ 1.000	$f(A+c\ B)=$	$f(A + cB) =$	c) for all $c \in \mathbb{R}$ and $A, B \in \mathcal{M}_n$, we have $f(A + cB) =$
johnston-linear-matrix-algebra-F02759	298	■ 1.000	$f(A)+c\ f(B)$	$f(A) + cf(B)$	$f(A) + cf(B)$.

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johnston-linear-matrix-algebra-F02760	299	1.000	λ	λ	to λ . It does not
johnston-linear-matrix-algebra-F02761	299	1.000	$A \mathbf{e}_j = a_{j,j} \mathbf{e}_j$	$A \mathbf{e}_j = a_{j,j} \mathbf{e}_j$	diagonal then $A \mathbf{e}_j = a_{j,j} \mathbf{e}_j$ for all $1 \leq j \leq n$.
johnston-linear-matrix-algebra-F02762	299	1.000	$a_{1,1}, a_{2,2}, \ldots, a_{n,n}$	$a_{1,1}, a_{2,2}, \ldots, a_{n,n}$	A 's diagonal entries $a_{1,1}, a_{2,2}, \ldots, a_{n,n}$.
johnston-linear-matrix-algebra-F02763	300	1.000	$A \mathbf{0} = \lambda \mathbf{0}$	$A \mathbf{0} = \lambda \mathbf{0}$	that $A \mathbf{0} = \lambda \mathbf{0}$ for all
johnston-linear-matrix-algebra-F02764	300	1.000	$A \mathbf{v} = 3 \mathbf{v}$	$A \mathbf{v} = 3 \mathbf{v}$	Since $A \mathbf{v} = 3 \mathbf{v}$, the eigenvalue corresponding to $\mathbf{v}$ is $\lambda = 3$. The effect of
johnston-linear-matrix-algebra-F02765	300	1.000	$\lambda = 3$	$\lambda = 3$	Since $A \mathbf{v} = 3 \mathbf{v}$, the eigenvalue corresponding to $\mathbf{v}$ is $\lambda = 3$. The effect of
johnston-linear-matrix-algebra-F02766	300	1.000	$\lambda = 1$	$\lambda = 1$	$\lambda = 1$ is an eigenvalue of A , and find an eigenvector that it corresponds to.
johnston-linear-matrix-algebra-F02767	300	1.000	$A \mathbf{v} = 1 \mathbf{v}$	$A \mathbf{v} = 1 \mathbf{v}$	We want to find a non-zero vector $\mathbf{v} \in \mathbb{R}^2$ such that $A \mathbf{v} = 1 \mathbf{v}$. This is a
johnston-linear-matrix-algebra-F02768	300	1.000	$\mathbf{v}_1 = -\mathbf{v}_2$	$v_1 = -v_2$	$v_1 = -v_2$. In other words, the eigenvectors are the (non-zero) vectors of
johnston-linear-matrix-algebra-F02769	300	1.000	$\mathbf{v} = (-v_2, v_2)$	$\mathbf{v} = (-v_2, v_2)$	the form $\mathbf{v} = (-v_2, v_2)$. We (arbitrarily) choose $v_2 = 1$ so that we get
johnston-linear-matrix-algebra-F02770	300	1.000	$v_2 = 1$	$v_2 = 1$	the form $\mathbf{v} = (-v_2, v_2)$. We (arbitrarily) choose $v_2 = 1$ so that we get
johnston-linear-matrix-algebra-F02771	300	1.000	$\mathbf{v} = (-1, 1)$	$\mathbf{v} = (-1, 1)$	$\mathbf{v} = (-1, 1)$ as an eigenvector of A with corresponding eigenvalue $\lambda = 1$.
johnston-linear-matrix-algebra-F02772	301	1.000	$A \mathbf{v} - \lambda \mathbf{v}$	$A \mathbf{v} - \lambda \mathbf{v}$	factoring $A \mathbf{v} - \lambda \mathbf{v}$. It is
johnston-linear-matrix-algebra-F02773	301	1.000	$(A - \lambda) \mathbf{v} = \mathbf{0}$	$(A - \lambda) \mathbf{v} = \mathbf{0}$	$(A - \lambda) \mathbf{v} = \mathbf{0}$, but this
johnston-linear-matrix-algebra-F02774	301	0.955	$A \mathbf{v} = \lambda \mathbf{v}$	$A \mathbf{v} = \lambda \mathbf{v}$	only if there is a non-zero vector $\mathbf{v}$ such that $A \mathbf{v} = \lambda \mathbf{v}$. By moving both
johnston-linear-matrix-algebra-F02775	301	1.000	$(A - \lambda I) \mathbf{v} = \mathbf{0}$	$(A - \lambda I) \mathbf{v} = \mathbf{0}$	$(A - \lambda I) \mathbf{v} = \mathbf{0}$ has a non-zero solution. By Theorem 2.2.4, this is equiva-
johnston-linear-matrix-algebra-F02776	301	1.000	$A - \lambda I$	$A - \lambda I$	lent to $A - \lambda I$ not being invertible. We have thus arrived at the following
johnston-linear-matrix-algebra-F02777	301	1.000	$\det(A - \lambda I)$	$\det(A - \lambda I)$	is to compute $\det(A - \lambda I)$ via one of the explicit formulas for the
johnston-linear-matrix-algebra-F02778	301	1.000	$\det(A - \lambda I) = 0$	$\det(A - \lambda I) = 0$	is not invertible if and only if $\det(A - \lambda I) = 0$, by Theorem 3.2.1). We
johnston-linear-matrix-algebra-F02779	301	0.998	$A = \begin{bmatrix} 1 & 2 \\ 5 & 4 \end{bmatrix}$	$A = \begin{bmatrix} 1 & 2 \\ 5 & 4 \end{bmatrix}$	Compute all of the eigenvalues of the matrix $A = \begin{bmatrix} 1 & 2 \\ 5 & 4 \end{bmatrix}$.
johnston-linear-matrix-algebra-F02780	302	1.000	$\lambda = -1$	$\lambda = -1$	so the eigenvalues of A are $\lambda = -1$ and $\lambda = 6$.
johnston-linear-matrix-algebra-F02781	302	1.000	$\lambda = 6$	$\lambda = 6$	so the eigenvalues of A are $\lambda = -1$ and $\lambda = 6$.
johnston-linear-matrix-algebra-F02782	302	1.000	$\lambda^2 - 5\lambda - 6$	$\lambda^2 - 5\lambda - 6$	As an alternative to factoring the polynomial $\lambda^2 - 5\lambda - 6$, we could have
johnston-linear-matrix-algebra-F02783	302	1.000	$a\lambda^2 + b\lambda + c = 0$	$a\lambda^2 + b\lambda + c = 0$	solutions of $a\lambda^2 + b\lambda + c = 0$ are
johnston-linear-matrix-algebra-F02784	302	1.000	$\lambda = (5 \pm 7)/2$	$\lambda = (5 \pm 7)/2$	In this case, the quadratic formula gives $\lambda = (5 \pm 7)/2$, which of course
johnston-linear-matrix-algebra-F02785	302	1.000	$\mathbf{v} \in \text{null}(A - \lambda I)$	$\mathbf{v} \in \text{null}(A - \lambda I)$	sponding eigenvalue λ if and only if $\mathbf{v} \in \text{null}(A - \lambda I)$.
johnston-linear-matrix-algebra-F02786	302	1.000	$(A + I) \mathbf{v} = \mathbf{0}$	$(A + I) \mathbf{v} = \mathbf{0}$	systems $(A + I) \mathbf{v} = \mathbf{0}$ and $(A - 6I) \mathbf{v} = \mathbf{0}$, respectively:

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johnston-linear-matrix-algebra-F02787	302	1.000	(A-6 I) \mathbf{v}=\mathbf{0}	$(A - 6I)\mathbf{v} = \mathbf{0}$	systems $(A + I)\mathbf{v} = \mathbf{0}$ and $(A - 6I)\mathbf{v} = \mathbf{0}$, respectively:
johnston-linear-matrix-algebra-F02788	302	0.998	(A-\lambda I) \mathbf{v}=	$(A - \lambda I)\mathbf{v} =$	$\lambda = -1$: In this case, we want to solve the linear system $(A - \lambda I)\mathbf{v} =$
johnston-linear-matrix-algebra-F02789	302	1.000	\left(-v_2, v_2\right)=v_2(-1,1)	$(-v_2, v_2) = v_2(-1, 1)$	$(-v_2, v_2) = v_2(-1, 1)$.
johnston-linear-matrix-algebra-F02790	303	0.999	(2 / 5,1)	$(2/5, 1)$	By multiplying $(2/5, 1)$
johnston-linear-matrix-algebra-F02791	303	1.000	v_1=2 v_2 / 5	$v_1 = 2v_2/5$	ing variable with $v_1 = 2v_2/5$, so the eigenvectors with corre-
johnston-linear-matrix-algebra-F02792	303	0.999	\mathbf{v}=\left(2 v_2 / 5, v_2\right)=v_2(2 / 5,1)	$\mathbf{v} = (2v_2/5, v_2) = v_2(2/5, 1)$	$\mathbf{v} = (2v_2/5, v_2) = v_2(2/5, 1)$.
johnston-linear-matrix-algebra-F02793	304	1.000	(2-\lambda)	$(2 - \lambda)$	$(2 - \lambda)$, since this
johnston-linear-matrix-algebra-F02794	304	0.816	\mathbf{v}_1=(-1,1)	$\mathbf{v}_1 = (-1, 1)$	$\mathbf{v}_1 = (-1, 1)$ and $\mathbf{v}_2 = (2/5, 1)$, which have corresponding eigenvalues -1 and 6 ,
johnston-linear-matrix-algebra-F02795	304	0.816	\mathbf{v}_2=(2 / 5,1)	$\mathbf{v}_2 = (2/5, 1)$	$\mathbf{v}_1 = (-1, 1)$ and $\mathbf{v}_2 = (2/5, 1)$, which have corresponding eigenvalues -1 and 6 ,
johnston-linear-matrix-algebra-F02796	304	1.000	\lambda=-2, \lambda=2	$\lambda = -2, \lambda = 2$	gives us $\lambda = -2$, $\lambda = 2$, and $\lambda = 4$. To find eigenvectors corresponding
johnston-linear-matrix-algebra-F02797	304	1.000	\lambda=4	$\lambda = 4$	gives us $\lambda = -2$, $\lambda = 2$, and $\lambda = 4$. To find eigenvectors corresponding
johnston-linear-matrix-algebra-F02798	304	1.000	\lambda=-2	$\lambda = -2$	$\lambda = -2$: In this case, the linear system is $(A + 2I)\mathbf{v} = \mathbf{0}$, which we can
johnston-linear-matrix-algebra-F02799	304	1.000	(A+2 I) \mathbf{v}=\mathbf{0}	$(A + 2I)\mathbf{v} = \mathbf{0}$	$\lambda = -2$: In this case, the linear system is $(A + 2I)\mathbf{v} = \mathbf{0}$, which we can
johnston-linear-matrix-algebra-F02800	304	1.000	v_3=0	$v_3 = 0$	ing variables with $v_3 = 0$ and $v_1 = -v_2$. The eigenvectors cor-
johnston-linear-matrix-algebra-F02801	304	0.954	(-1,1,0)	$(-1, 1, 0)$	multiples of $(-1, 1, 0)$.
johnston-linear-matrix-algebra-F02802	304	1.000	\lambda=2	$\lambda = 2$	$\lambda = 2$: We work through this eigenvector calculation a bit more quickly.
johnston-linear-matrix-algebra-F02803	305	0.706	\mathbf{3} \times \mathbf{3}	$\mathbf{3} \times \mathbf{3}$	Ugly 3×3 Matrix
johnston-linear-matrix-algebra-F02804	305	1.000	v_1=0	$v_1 = 0$	ables with $v_1 = 0$ and $v_2 = -v_3$. The eigenvectors correspond-
johnston-linear-matrix-algebra-F02805	305	1.000	v_2=-v_3	$v_2 = -v_3$	ables with $v_1 = 0$ and $v_2 = -v_3$. The eigenvectors correspond-
johnston-linear-matrix-algebra-F02806	305	1.000	(A-4 I) \mathbf{v}=\mathbf{0}	$(A - 4I)\mathbf{v} = \mathbf{0}$	$\lambda = 4$: In this case, the linear system is $(A - 4I)\mathbf{v} = \mathbf{0}$. Since the reduced
johnston-linear-matrix-algebra-F02807	305	1.000	A-4 I	$A - 4I$	row echelon form of $A - 4I$ is
johnston-linear-matrix-algebra-F02808	305	1.000	v_1=v_2	$v_1 = v_2$	ables. Furthermore, $v_1 = v_2$ and $v_3 = 0$, so the eigenvectors
johnston-linear-matrix-algebra-F02809	305	1.000	A=\left[\begin{array}{ccc}1 & 2 & 3 \\ 1 & -2 & 1 \\ 3 & 2 & 1\end{array}\right]	$A = \begin{bmatrix} 1 & 2 & 3 \\ 1 & -2 & 1 \\ 3 & 2 & 1 \end{bmatrix}$	Compute the eigenvalues of the matrix $A = \begin{bmatrix} 1 & 2 & 3 \\ 1 & -2 & 1 \\ 3 & 2 & 1 \end{bmatrix}$.
johnston-linear-matrix-algebra-F02810	305	1.000	-\lambda^3+16 \lambda+24=0	$-\lambda^3 + 16\lambda + 24 = 0$	degree polynomial equations) like $-\lambda^3 + 16\lambda + 24 = 0$ are covered in
johnston-linear-matrix-algebra-F02811	306	0.996	-(-2)^3+16(-2)+	$-(-2)^3 + 16(-2) +$	only one of them that is actually a solution (indeed, $-(-2)^3 + 16(-2) +$
johnston-linear-matrix-algebra-F02812	306	1.000	24=8-32+24=0	$24 = 8 - 32 + 24 = 0$	$24 = 8 - 32 + 24 = 0$).

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johnston-linear-matrix-algebra-F02813	306	■ 1.000	$-\lambda^3+16\lambda+$	$-\lambda^3+16\lambda+$	use polynomial long division to obtain the result of dividing $-\lambda^3+16\lambda+$
johnston-linear-matrix-algebra-F02814	306	■ 1.000	$\lambda+2$	$\lambda+2$	24 by $\lambda+2$ (which we know is a factor since $\lambda=-2$ is a root of the cubic
johnston-linear-matrix-algebra-F02815	306	■ 0.990	$-\lambda^3+16\lambda+24=(\lambda+2)(-\lambda^2+2\lambda+12)$	$-\lambda^3+16\lambda+24=(\lambda+2)(-\lambda^2+2\lambda+12)$	It follows that $-\lambda^3+16\lambda+24=(\lambda+2)(-\lambda^2+2\lambda+12)$, so the roots of
johnston-linear-matrix-algebra-F02816	306	■ 1.000	$-\lambda^3+16\lambda+24$	$-\lambda^3+16\lambda+24$	$-\lambda^3+16\lambda+24$ (i.e., the eigenvalues of A) are $\lambda=-2$ as well as the two
johnston-linear-matrix-algebra-F02817	306	■ 1.000	$-\lambda^2+2\lambda+12$	$-\lambda^2+2\lambda+12$	roots of $-\lambda^2+2\lambda+12$, which we can compute via the quadratic formula:
johnston-linear-matrix-algebra-F02818	306	■ 0.998	$\lambda=-2, \lambda=1+\sqrt{13}$	$\lambda=-2, \lambda=1+\sqrt{13}$	The eigenvalues of A are thus $\lambda=-2, \lambda=1+\sqrt{13}$, and $\lambda=1-$
johnston-linear-matrix-algebra-F02819	306	■ 0.998	$\lambda=1-$	$\lambda=1-$	The eigenvalues of A are thus $\lambda=-2, \lambda=1+\sqrt{13}$, and $\lambda=1-$
johnston-linear-matrix-algebra-F02820	306	■ 1.000	$\sqrt{13}$	$\sqrt{13}$	$\sqrt{13}$. If we were feeling adventurous, we could also find the eigenvectors
johnston-linear-matrix-algebra-F02821	306	■ 1.000	$A-$	$A-$	corresponding to these eigenvalues by solving the linear system $(A-$
johnston-linear-matrix-algebra-F02822	306	■ 1.000	$(\lambda I-\mathbf{A})\mathbf{v}=\mathbf{0}$	$\lambda I)\mathbf{v}=\mathbf{0}$	$\lambda I)\mathbf{v}=\mathbf{0}$ for each of these three choices of λ . However, it is perhaps a
johnston-linear-matrix-algebra-F02823	306	■ 0.921	5×5	5×5	be <i>impossible</i> to do so if the matrix is 5×5 or larger (see Remark A.2.1). For
johnston-linear-matrix-algebra-F02824	307	■ 1.000	$\operatorname{det}(\lambda I-A)$	$\det(\lambda I-A)$	$\det(\lambda I-A)$. These two
johnston-linear-matrix-algebra-F02825	307	■ 1.000	$p_A:\mathbb{R}\rightarrow\mathbb{R}$	$p_A:\mathbb{R}\rightarrow\mathbb{R}$	Suppose A is a square matrix. The function $p_A:\mathbb{R}\rightarrow\mathbb{R}$ defined by
johnston-linear-matrix-algebra-F02826	307	■ 1.000	$(a_{1,1}-\lambda)(a_{2,2}-\lambda)\cdots(a_{n,n}-\lambda)$	$(a_{1,1}-\lambda)(a_{2,2}-\lambda)\cdots(a_{n,n}-\lambda)$	product of the diagonal entries, $(a_{1,1}-\lambda)(a_{2,2}-\lambda)\cdots(a_{n,n}-\lambda)$, which has
johnston-linear-matrix-algebra-F02827	307	■ 1.000	p_A	p_A	A bit more explicitly, if the characteristic polynomial of A is p_A and λ_0 is a
johnston-linear-matrix-algebra-F02828	307	■ 1.000	λ_0	λ_0	A bit more explicitly, if the characteristic polynomial of A is p_A and λ_0 is a
johnston-linear-matrix-algebra-F02829	307	■ 1.000	$(\lambda_0-\lambda)$	$(\lambda_0-\lambda)$	of the term $(\lambda_0-\lambda)$ in the factored form of $p_A(\lambda)$. We now work through an
johnston-linear-matrix-algebra-F02830	307	■ 1.000	$p_A(\lambda)$	$p_A(\lambda)$	of the term $(\lambda_0-\lambda)$ in the factored form of $p_A(\lambda)$. We now work through an
johnston-linear-matrix-algebra-F02831	308	■ 1.000	$\sqrt{-1}=i$	$\sqrt{-1}=i$	$\sqrt{-1}=i$. See
johnston-linear-matrix-algebra-F02832	309	■ 1.000	$a+bi$	$a+bi$	conjugate of $a+bi$
johnston-linear-matrix-algebra-F02833	309	■ 1.000	$\overline{a+bi}=a-bi$	$\overline{a+bi}=a-bi$	$\overline{a+bi}=a-bi$.
johnston-linear-matrix-algebra-F02834	309	■ 1.000	$\mathcal{M}_{m,n}(X)$	$\mathcal{M}_{m,n}(X)$	$\mathcal{M}_{m,n}(X)$ is the set of
johnston-linear-matrix-algebra-F02835	309	■ 1.000	$\mathcal{M}_n(X)$	$\mathcal{M}_n(X)$	use $\mathcal{M}_n(X)$ to
johnston-linear-matrix-algebra-F02836	309	■ 0.999	$\overline{-1+2i}=-1-2i$	$\overline{-1+2i}=-1-2i$	complex conjugates of each other: $\overline{-1+2i}=-1-2i$. This always happens for
johnston-linear-matrix-algebra-F02837	309	■ 1.000	$p_A(\lambda)=0$	$p_A(\lambda)=0$	coefficients, so if $p_A(\lambda)=0$ then $p_A(\bar{\lambda})=\overline{p_A(\lambda)}=0$ as well.
johnston-linear-matrix-algebra-F02838	309	■ 1.000	$p_A(\bar{\lambda})=\overline{p_A(\lambda)}=0$	$p_A(\bar{\lambda})=\overline{p_A(\lambda)}=0$	coefficients, so if $p_A(\lambda)=0$ then $p_A(\bar{\lambda})=\overline{p_A(\lambda)}=0$ as well.
johnston-linear-matrix-algebra-F02839	309	■ 1.000	$\mathcal{M}_{m,n}(\mathbb{R})$	$\mathcal{M}_{m,n}(\mathbb{R})$	matrix can have, we use the notation $\mathcal{M}_{m,n}(\mathbb{R})$ to denote the set of $m\times n$

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johnston-linear-matrix-algebra-F02840	309	1.000	$\mathcal{M}_{m,n}(\mathbb{C})$	$\mathcal{M}_{m,n}(\mathbb{C})$	matrices with real entries and $\mathcal{M}_{m,n}(\mathbb{C})$ to denote the set of $m \times n$ matrices
johnston-linear-matrix-algebra-F02841	309	1.000	$p_A(\lambda)=p_B(\lambda)$	$p_A(\lambda) = p_B(\lambda)$	polynomials are the same: $p_A(\lambda) = p_B(\lambda)$.
johnston-linear-matrix-algebra-F02842	310	1.000	λ^n	λ^n	The coefficient of λ^n
johnston-linear-matrix-algebra-F02843	310	0.990	$(-1)^n$	$(-1)^n$	$(-1)^n$.
johnston-linear-matrix-algebra-F02844	310	1.000	$A=\left[\begin{array}{ccc}4 & 0 & 2 \\ -2 & 6 & 2 \\ 0 & 4 & 2\end{array}\right]$	$A = \begin{bmatrix} 4 & 0 & 2 \\ -2 & 6 & 2 \\ 2 & 0 & 4 \end{bmatrix}$	Show that $A = \begin{bmatrix} 4 & 0 & 2 \\ -2 & 6 & 2 \\ 2 & 0 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} 8 & -5 & -5 \\ 5 & -2 & -5 \\ -5 & 5 & 8 \end{bmatrix}$ are not similar.
johnston-linear-matrix-algebra-F02845	310	1.000	$B=\left[\begin{array}{ccc}8 & -5 & -5 \\ 5 & -2 & -5 \\ -5 & 5 & 8\end{array}\right]$	$B = \begin{bmatrix} 8 & -5 & -5 \\ 5 & -2 & -5 \\ -5 & 5 & 8 \end{bmatrix}$	Show that $A = \begin{bmatrix} 4 & 0 & 2 \\ -2 & 6 & 2 \\ 2 & 0 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} 8 & -5 & -5 \\ 5 & -2 & -5 \\ -5 & 5 & 8 \end{bmatrix}$ are not similar.
johnston-linear-matrix-algebra-F02846	310	0.975	$\operatorname{rank}(A)=\operatorname{rank}(B)=3,$ $\operatorname{tr}(A)=\operatorname{tr}(B)=14$	$\operatorname{rank}(A) = \operatorname{rank}(B) = 3, \operatorname{tr}(A) = \operatorname{tr}(B) = 14$	$\operatorname{rank}(A) = \operatorname{rank}(B) = 3, \operatorname{tr}(A) = \operatorname{tr}(B) = 14,$ and $\det(A) = \det(B) = 72$.
johnston-linear-matrix-algebra-F02847	310	0.975	$\operatorname{det}(A)=\operatorname{det}(B)=72$	$\det(A) = \det(B) = 72$	$\operatorname{rank}(A) = \operatorname{rank}(B) = 3, \operatorname{tr}(A) = \operatorname{tr}(B) = 14,$ and $\det(A) = \det(B) = 72$.
johnston-linear-matrix-algebra-F02848	310	1.000	$p_A(\lambda) \neq p_B(\lambda)$	$p_A(\lambda) \neq p_B(\lambda)$	Since $p_A(\lambda) \neq p_B(\lambda)$, we conclude that A and B are not similar.
johnston-linear-matrix-algebra-F02849	310	1.000	λ^2	λ^2	Notice in the above example that the coefficient of λ^2 in the characteristic
johnston-linear-matrix-algebra-F02850	310	1.000	$A \in \mathcal{M}_n(\mathbb{C})$	$A \in \mathcal{M}_n(\mathbb{C})$	Denote the n eigenvalues of $A \in \mathcal{M}_n(\mathbb{C})$ (listed according to algebraic
johnston-linear-matrix-algebra-F02851	310	1.000	$\lambda_1, \lambda_2, \ldots, \lambda_n$	$\lambda_1, \lambda_2, \ldots, \lambda_n$	multiplicity) by $\lambda_1, \lambda_2, \ldots, \lambda_n$, and its characteristic polynomial by
johnston-linear-matrix-algebra-F02852	310	1.000	$c_0=\operatorname{det}(A)=\lambda_1 \lambda_2 \cdots \lambda_n$	$c_0 = \det(A) = \lambda_1 \lambda_2 \cdots \lambda_n$	<i>Proof.</i> We start by proving the claim that $c_0 = \det(A) = \lambda_1 \lambda_2 \cdots \lambda_n$. To see
johnston-linear-matrix-algebra-F02853	311	1.000	$\lambda=0$	$\lambda = 0$	$\lambda = 0$ into the above formulas shows that
johnston-linear-matrix-algebra-F02854	311	1.000	$(-1)^{n-1} c_{n-1}=\lambda_1+\lambda_2+\cdots+\lambda_n$	$(-1)^{n-1} c_{n-1} = \lambda_1 + \lambda_2 + \cdots + \lambda_n$	To prove the claim that $(-1)^{n-1} c_{n-1} = \lambda_1 + \lambda_2 + \cdots + \lambda_n$, we simply mul-
johnston-linear-matrix-algebra-F02855	311	1.000	λ^{n-1}	λ^{n-1}	and observe that the coefficient of λ^{n-1} on the right-hand side (i.e., c_{n-1}) is
johnston-linear-matrix-algebra-F02856	311	1.000	c_{n-1}	c_{n-1}	and observe that the coefficient of λ^{n-1} on the right-hand side (i.e., c_{n-1}) is
johnston-linear-matrix-algebra-F02857	311	1.000	$(-1)^{n-1}\left(\lambda_1+\lambda_2+\cdots+\lambda_n\right)$	$(-1)^{n-1}(\lambda_1 + \lambda_2 + \cdots + \lambda_n)$	$(-1)^{n-1}(\lambda_1 + \lambda_2 + \cdots + \lambda_n)$.
johnston-linear-matrix-algebra-F02858	311	1.000	$(-1)^{n-1} c_{n-1}=\operatorname{tr}(A)$	$(-1)^{n-1} c_{n-1} = \operatorname{tr}(A)$	On the other hand, we can show that $(-1)^{n-1} c_{n-1} = \operatorname{tr}(A)$ by considering
johnston-linear-matrix-algebra-F02859	311	0.997	$n-1$	$n-1$	term of the form λ^{n-1} is the first one, since it is the only one containing $n-1$
johnston-linear-matrix-algebra-F02860	312	1.000	$2 \times 3=6$	$2 \times 3 = 6$	$2 \times 3 = 6$.
johnston-linear-matrix-algebra-F02861	313	1.000	$A \in \mathcal{M}_{m,n}(\mathbb{C})$	$A \in \mathcal{M}_{m,n}(\mathbb{C})$	$A \in \mathcal{M}_{m,n}(\mathbb{C})$ means
johnston-linear-matrix-algebra-F02862	313	0.742	$\lambda=1,1$	$\lambda = 1, 1$	It follows that A has eigenvalues $\lambda = 1, 1$, and 2 , so its trace is $\operatorname{tr}(A) =$
johnston-linear-matrix-algebra-F02863	313	0.742	$\operatorname{tr}(A)=$	$\operatorname{tr}(A) =$	It follows that A has eigenvalues $\lambda = 1, 1$, and 2 , so its trace is $\operatorname{tr}(A) =$

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johnston-linear-matrix-algebra F02864	313	1.000	1+1+2=4	$1 + 1 + 2 = 4$	$1 + 1 + 2 = 4$ and its determinant is $\det(A) = 1 \times 1 \times 2 = 2$, which agree
johnston-linear-matrix-algebra F02865	313	1.000	$\operatorname{det}(A)=1 \times 1 \times 2=2$	$\det(A) = 1 \times 1 \times 2 = 2$	$1 + 1 + 2 = 4$ and its determinant is $\det(A) = 1 \times 1 \times 2 = 2$, which agree
johnston-linear-matrix-algebra F02866	313	1.000	$A^{\{*\}}$	A^*	construct its conjugate transpose A^* (instead of its standard transpose A^T),
johnston-linear-matrix-algebra F02867	313	1.000	$\left[\begin{array}{cc}2 i & 1-3 i \\-3 & 5+2 i\end{array}\right]$	$\left[\begin{array}{cc}2 i & 1-3 i \\-3 & 5+2 i\end{array}\right]$	a) $\left[\begin{array}{cc}2 i & 1-3 i \\-3 & 5+2 i\end{array}\right]$
johnston-linear-matrix-algebra F02868	313	1.000	$\left[\begin{array}{cc}0 & 2+3 i \\2-3 i & 4\end{array}\right]$	$\left[\begin{array}{cc}0 & 2+3 i \\2-3 i & 4\end{array}\right]$	b) $\left[\begin{array}{cc}0 & 2+3 i \\2-3 i & 4\end{array}\right]$
johnston-linear-matrix-algebra F02869	313	1.000	$\left[\begin{array}{cc}i & 1-i \\-2 & -2-i \\1+i & 2+3 i\end{array}\right]$	$\left[\begin{array}{cc}i & 1-i \\-2 & -2-i \\1+i & 2+3 i\end{array}\right]$	d) $\left[\begin{array}{cc}i & 1-i \\-2 & -2-i \\1+i & 2+3 i\end{array}\right]$
johnston-linear-matrix-algebra F02870	314	1.000	$\mathbf{v} \cdot(A \mathbf{w})=\left(A^{\{*\}} \mathbf{v}\right) \cdot \mathbf{w}$	$\mathbf{v} \cdot(A \mathbf{w})=(A^* \mathbf{v}) \cdot \mathbf{w}$	$\mathbf{v} \cdot(A \mathbf{w})=(A^* \mathbf{v}) \cdot \mathbf{w}$
johnston-linear-matrix-algebra F02871	314	1.000	$\mathbf{v}^{\{*\}} A \mathbf{v}$	$\mathbf{v}^* A \mathbf{v}$	Since $\mathbf{v}^* A \mathbf{v}$ is a scalar,
johnston-linear-matrix-algebra F02872	314	1.000	$\bar{z} z= z ^2$	$\bar{z} z= z ^2$	$\bar{z} z= z ^2$ for all $z \in \mathbb{C}$.
johnston-linear-matrix-algebra F02873	314	1.000	$z \in \mathbb{C}$	$z \in \mathbb{C}$	$\bar{z} z= z ^2$ for all $z \in \mathbb{C}$.
johnston-linear-matrix-algebra F02874	314	1.000	$A^{\{*\}}=A$	$A^*=A$	In the special case when a matrix $A \in \mathcal{M}_n(\mathbb{C})$ is such that $A^*=A$, like in
johnston-linear-matrix-algebra F02875	314	1.000	$\mathbf{v} \in \mathbb{C}^n$	$\mathbf{v} \in \mathbb{C}^n$	<i>Proof.</i> Suppose that $\mathbf{v} \in \mathbb{C}^n$ is an eigenvector of A with corresponding eigen-
johnston-linear-matrix-algebra F02876	314	1.000	$\lambda \in \mathbb{C}$	$\lambda \in \mathbb{C}$	value $\lambda \in \mathbb{C}$. Then computing $\mathbf{v}^* A \mathbf{v}$ shows that
johnston-linear-matrix-algebra F02877	315	1.000	$\lambda \mathbf{v}^{\{*\}} \mathbf{v}=\bar{\lambda} \mathbf{v}^{\{*\}} \mathbf{v}$	$\lambda \mathbf{v}^* \mathbf{v}=\bar{\lambda} \mathbf{v}^* \mathbf{v}$	$\lambda \mathbf{v}^* \mathbf{v}=\bar{\lambda} \mathbf{v}^* \mathbf{v}$ by $\mathbf{v}^* \mathbf{v}$ to see that $\lambda=\bar{\lambda}$, which shows that $\lambda \in \mathbb{R}$.
johnston-linear-matrix-algebra F02878	315	1.000	$\mathbf{v}^{\{*\}} \mathbf{v}$	$\mathbf{v}^* \mathbf{v}$	$\lambda \mathbf{v}^* \mathbf{v}=\bar{\lambda} \mathbf{v}^* \mathbf{v}$ by $\mathbf{v}^* \mathbf{v}$ to see that $\lambda=\bar{\lambda}$, which shows that $\lambda \in \mathbb{R}$.
johnston-linear-matrix-algebra F02879	315	1.000	$\lambda=\bar{\lambda}$	$\lambda=\bar{\lambda}$	$\lambda \mathbf{v}^* \mathbf{v}=\bar{\lambda} \mathbf{v}^* \mathbf{v}$ by $\mathbf{v}^* \mathbf{v}$ to see that $\lambda=\bar{\lambda}$, which shows that $\lambda \in \mathbb{R}$.
johnston-linear-matrix-algebra F02880	315	1.000	$\lambda \in \mathbb{R}$	$\lambda \in \mathbb{R}$	$\lambda \mathbf{v}^* \mathbf{v}=\bar{\lambda} \mathbf{v}^* \mathbf{v}$ by $\mathbf{v}^* \mathbf{v}$ to see that $\lambda=\bar{\lambda}$, which shows that $\lambda \in \mathbb{R}$.
johnston-linear-matrix-algebra F02881	315	1.000	$\left[\begin{array}{cc}2 & 1+i \\1-i & 3\end{array}\right]$	$\left[\begin{array}{cc}2 & 1+i \\1-i & 3\end{array}\right]$	a) $\left[\begin{array}{cc}2 & 1+i \\1-i & 3\end{array}\right]$
johnston-linear-matrix-algebra F02882	315	1.000	$\left[\begin{array}{ccc}0 & 1 & -1 \\1 & -3 & 2 \\-1 & 2 & 1\end{array}\right]$	$\left[\begin{array}{ccc}0 & 1 & -1 \\1 & -3 & 2 \\-1 & 2 & 1\end{array}\right]$	b) $\left[\begin{array}{ccc}0 & 1 & -1 \\1 & -3 & 2 \\-1 & 2 & 1\end{array}\right]$
johnston-linear-matrix-algebra F02883	315	1.000	$-\lambda^3-2 \lambda^2+9 \lambda-2$	$-\lambda^3-2 \lambda^2+9 \lambda-2$	polynomial long division to divide $-\lambda^3-2 \lambda^2+9 \lambda-2$ by $\lambda-2$:
johnston-linear-matrix-algebra F02884	315	1.000	$\lambda-2$	$\lambda-2$	polynomial long division to divide $-\lambda^3-2 \lambda^2+9 \lambda-2$ by $\lambda-2$:
johnston-linear-matrix-algebra F02885	315	0.972	$-\lambda^3-2 \lambda^2+9 \lambda-2=(\lambda-2)\left(-\lambda^2-4 \lambda+1\right)$	$-\lambda^3-2 \lambda^2+9 \lambda-2=(\lambda-2)\left(-\lambda^2-4 \lambda+1\right)$	It follows that $-\lambda^3-2 \lambda^2+9 \lambda-2=(\lambda-2)\left(-\lambda^2-4 \lambda+1\right)$, so
johnston-linear-matrix-algebra F02886	315	1.000	$-\lambda^2-4 \lambda+1$	$-\lambda^2-4 \lambda+1$	are $\lambda=2$ as well as the two roots of $-\lambda^2-4 \lambda+1$, which can be

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johnston-linear-matrix-algebra-F02887	316	1.000	$\operatorname{null}(A-\lambda I)$	$\operatorname{null}(A-\lambda I)$	$\operatorname{null}(A-\lambda I)$.
johnston-linear-matrix-algebra-F02888	316	0.943	$(-1, 1, 0), (0, 1, -1)$	$(-1, 1, 0), (0, 1, -1)$	the non-zero multiples of $(-1, 1, 0)$, $(0, 1, -1)$, and $(1, 1, 0)$, respectively. It fol-
johnston-linear-matrix-algebra-F02889	316	0.943	$(1, 1, 0)$	$(1, 1, 0)$	the non-zero multiples of $(-1, 1, 0)$, $(0, 1, -1)$, and $(1, 1, 0)$, respectively. It fol-
johnston-linear-matrix-algebra-F02890	316	1.000	$\{(-1, 1, 0)\}, \{(0, 1, -1)\}$	$\{(-1, 1, 0)\}, \{(0, 1, -1)\}$	lows that $\{(-1, 1, 0)\}, \{(0, 1, -1)\}$, and $\{(1, 1, 0)\}$ are bases of the eigenspaces
johnston-linear-matrix-algebra-F02891	316	1.000	$\{(1, 1, 0)\}$	$\{(1, 1, 0)\}$	lows that $\{(-1, 1, 0)\}, \{(0, 1, -1)\}$, and $\{(1, 1, 0)\}$ are bases of the eigenspaces
johnston-linear-matrix-algebra-F02892	316	1.000	$A=\left[\begin{array}{ccc}2 & 0 & -3 \\ 1 & -1 & -1 \\ 0 & 0 & -1\end{array}\right]$	$A=\left[\begin{array}{ccc}2 & 0 & -3 \\ 1 & -1 & -1 \\ 0 & 0 & -1\end{array}\right]$	Compute bases of the eigenspaces of the matrix $A=\left[\begin{array}{ccc}2 & 0 & -3 \\ 1 & -1 & -1 \\ 0 & 0 & -1\end{array}\right]$.
johnston-linear-matrix-algebra-F02893	317	0.925	$(A-\lambda I)$	$(A-\lambda I)$	$\operatorname{nullity}(A-\lambda I)$.
johnston-linear-matrix-algebra-F02894	317	0.999	$(A-2 I) \mathbf{v}=\mathbf{0}$	$(A-2 I)\mathbf{v}=\mathbf{0}$	by solving the linear system $(A-2 I)\mathbf{v}=\mathbf{0}$ as follows:
johnston-linear-matrix-algebra-F02895	317	1.000	$v_1=3 v_2$	$v_1=3 v_2$	variables with $v_3=0$ and $v_1=3 v_2$. The eigenvectors corre-
johnston-linear-matrix-algebra-F02896	317	1.000	$v_2(3, 1, 0)$	$v_2(3, 1, 0)$	sponding to the eigenvalue $\lambda=2$ thus have the form $v_2(3, 1, 0)$,
johnston-linear-matrix-algebra-F02897	317	1.000	$\{(3, 1, 0)\}$	$\{(3, 1, 0)\}$	so $\{(3, 1, 0)\}$ is a basis of this eigenspace.
johnston-linear-matrix-algebra-F02898	317	1.000	$v_1=v_3$	$v_1=v_3$	variable with $v_1=v_3$. The eigenvectors corresponding to the
johnston-linear-matrix-algebra-F02899	317	1.000	$v_2(0, 1, 0)+v_3(1, 0, 1)$	$v_2(0, 1, 0)+v_3(1, 0, 1)$	eigenvalue $\lambda=-1$ thus have the form $v_2(0, 1, 0)+v_3(1, 0, 1)$,
johnston-linear-matrix-algebra-F02900	317	1.000	$\{(0, 1, 0), (1, 0, 1)\}$	$\{(0, 1, 0), (1, 0, 1)\}$	so $\{(0, 1, 0), (1, 0, 1)\}$ is a basis of this eigenspace.
johnston-linear-matrix-algebra-F02901	318	1.000	$(B-\lambda I)$	$(B-\lambda I)$	multiplicity of λ as an eigenvalue of A (and $\operatorname{nullity}(B-\lambda I)$ is its geometric
johnston-linear-matrix-algebra-F02902	318	1.000	$B=\left[\begin{array}{ccc}2 & 2 & 3 \\ 0 & -1 & 0 \\ 0 & 1 & -1\end{array}\right]$	$B=\left[\begin{array}{ccc}2 & 2 & 3 \\ 0 & -1 & 0 \\ 0 & 1 & -1\end{array}\right]$	Show that $A=\left[\begin{array}{ccc}2 & 0 & -3 \\ 1 & -1 & -1 \\ 0 & 0 & -1\end{array}\right]$ and $B=\left[\begin{array}{ccc}2 & 2 & 3 \\ 0 & -1 & 0 \\ 0 & 1 & -1\end{array}\right]$ are not similar.
johnston-linear-matrix-algebra-F02903	318	1.000	$(B+I) \mathbf{v}=\mathbf{0}$	$(B+I)\mathbf{v}=\mathbf{0}$	multiplicity as an eigenvalue of B , we solve the linear system $(B+I)\mathbf{v}=\mathbf{0}$
johnston-linear-matrix-algebra-F02904	318	1.000	$v_1=-v_3$	$v_1=-v_3$	with $v_1=-v_3$ and $v_2=0$. The eigenvectors corresponding to $\lambda=-1$
johnston-linear-matrix-algebra-F02905	318	1.000	$v_2=0$	$v_2=0$	with $v_1=-v_3$ and $v_2=0$. The eigenvectors corresponding to $\lambda=-1$
johnston-linear-matrix-algebra-F02906	318	1.000	$v_3(-1, 0, 1)$	$v_3(-1, 0, 1)$	as an eigenvalue of B are thus of the form $v_3(-1, 0, 1)$, so $\{(-1, 0, 1)\}$ is
johnston-linear-matrix-algebra-F02907	318	1.000	$\{(-1, 0, 1)\}$	$\{(-1, 0, 1)\}$	as an eigenvalue of B are thus of the form $v_3(-1, 0, 1)$, so $\{(-1, 0, 1)\}$ is
johnston-linear-matrix-algebra-F02908	319	1.000	λ_1	λ_1	eigenvalue λ_1
johnston-linear-matrix-algebra-F02909	319	0.999	$\left\{\mathbf{v}_1, \ldots, \mathbf{v}_k\right\}$	$\{\mathbf{v}_1, \ldots, \mathbf{v}_k\}$	k , so that there exists a basis $\{\mathbf{v}_1, \ldots, \mathbf{v}_k\}$ of the eigenspace $\operatorname{nullity}(A-\lambda_1 I)$.
johnston-linear-matrix-algebra-F02910	319	0.999	$\operatorname{nullity}\left(A-\lambda_1 I\right)$	$\operatorname{nullity}(A-\lambda_1 I)$	k , so that there exists a basis $\{\mathbf{v}_1, \ldots, \mathbf{v}_k\}$ of the eigenspace $\operatorname{nullity}(A-\lambda_1 I)$.
johnston-linear-matrix-algebra-F02911	319	1.000	$V \in \mathcal{M}_{n, n-k}$	$V \in \mathcal{M}_{n, n-k}$	$V \in \mathcal{M}_{n, n-k}$; they are just chosen to make P invertible).

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johnston-linear-matrix-algebra-F02912	319	1.000	$B=\mathcal{M}_{k, n-k}$	$B = \mathcal{M}_{k,n-k}$	where $B = \mathcal{M}_{k,n-k}$ and $C \in \mathcal{M}_{n-k,n-k}$ are ugly matrices whose entries we do
johnston-linear-matrix-algebra-F02913	319	1.000	$C \in \mathcal{M}_{n-k, n-k}$	$C \in \mathcal{M}_{n-k,n-k}$	where $B = \mathcal{M}_{k,n-k}$ and $C \in \mathcal{M}_{n-k,n-k}$ are ugly matrices whose entries we do
johnston-linear-matrix-algebra-F02914	319	1.000	$\left(\lambda_1-\lambda\right)^k$	$(\lambda_1 - \lambda)^k$	The characteristic polynomial of A thus has $(\lambda_1 - \lambda)^k$ as a factor, so the alge-
johnston-linear-matrix-algebra-F02915	320	0.614	$\{(0,0,1,2)\}$	$\{(0,0,1,2)\}$	has a 1-dimensional eigenspace with $\{(0,0,1,2)\}$ as a basis, but this is
johnston-linear-matrix-algebra-F02916	320	1.000	$\lambda=-3$	$\lambda = -3$	For $\lambda = -3$, all we know is that its geometric multiplicity is between
johnston-linear-matrix-algebra-F02917	320	1.000	$(A+3\ I)\ \mathbf{v}=\mathbf{0}$	$(A + 3I)\mathbf{v} = \mathbf{0}$	this end, we solve the linear system $(A + 3I)\mathbf{v} = \mathbf{0}$ to find a basis of this
johnston-linear-matrix-algebra-F02918	320	1.000	$\{(0,1,0,0)\}$	$\{(0,1,0,0)\}$	ear system and see that $\{(0,1,0,0)\}$ is a basis of the eigenspace, so the
johnston-linear-matrix-algebra-F02919	321	0.997	$\left(a_{2,2},\ \ldots,\ a_{n,\ n}\right)$	$a_{2,2}, \dots, a_{n,n})$	$a_{2,2}, \dots, a_{n,n})$.
johnston-linear-matrix-algebra-F02920	321	0.996	$\left[\begin{array}{lll}1&2&3\\0&4&5\\0&0&6\end{array}\right]$	$\left[\begin{array}{ccc}1 & 2 & 3 \\ 0 & 4 & 5 \\ 0 & 0 & 6\end{array}\right]$	a) $\left[\begin{array}{ccc}1 & 2 & 3 \\ 0 & 4 & 5 \\ 0 & 0 & 6\end{array}\right]$
johnston-linear-matrix-algebra-F02921	321	1.000	$\left[\begin{array}{ccc}7&0&0\\0&2&0\\0&0&-3\end{array}\right]$	$\left[\begin{array}{ccc}7 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & -3\end{array}\right]$	b) $\left[\begin{array}{ccc}7 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & -3\end{array}\right]$
johnston-linear-matrix-algebra-F02922	321	1.000	$\left[\begin{array}{lll}2&1&0\\0&2&1\\0&0&2\end{array}\right]$	$\left[\begin{array}{ccc}2 & 1 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 2\end{array}\right]$	c) $\left[\begin{array}{ccc}2 & 1 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 2\end{array}\right]$
johnston-linear-matrix-algebra-F02923	322	0.852	$\{(1,0,0)\}$	$\{(1,0,0)\}$	One basis of this null space is thus $\{(1,0,0)\}$.
johnston-linear-matrix-algebra-F02924	322	1.000	$A-\lambda I=$	$A - \lambda I =$	$\lambda = 4$: We want to find the null space of the matrix $A - \lambda I =$
johnston-linear-matrix-algebra-F02925	322	0.954	$\{(2,3,0)\}$	$\{(2,3,0)\}$	One basis of this null space is thus $\{(2,3,0)\}$.
johnston-linear-matrix-algebra-F02926	322	1.000	$A-6\ I$	$A - 6I$	$A - 6I$:
johnston-linear-matrix-algebra-F02927	322	0.813	$\{(16,25,10)\}$	$\{(16,25,10)\}$	One basis of this null space is thus $\{(16,25,10)\}$.
johnston-linear-matrix-algebra-F02928	322	0.938	$[\mathrm{v}],\left\{\mathbf{e}_1\right\},\left\{\mathbf{e}_2\right\}$	$[\mathbf{v}], \{\mathbf{e}_1\}, \{\mathbf{e}_2\}$	that bases of the corresponding eigenspaces are $[\mathbf{v}], \{\mathbf{e}_1\}, \{\mathbf{e}_2\}$, and
johnston-linear-matrix-algebra-F02929	322	1.000	$\left\{\mathbf{e}_3\right\}$	$\{\mathbf{e}_3\}$	$\{\mathbf{e}_3\}$, respectively.
johnston-linear-matrix-algebra-F02930	322	0.957	$A=\left[\begin{array}{ll}1&5\\4&2\end{array}\right],\ \mathbf{v}=\left[\begin{array}{l}1\\1\end{array}\right]$	$A = \left[\begin{array}{cc}1 & 5 \\ 4 & 2\end{array}\right], \mathbf{v} = \left[\begin{array}{c}1 \\ 1\end{array}\right]$	*(a) $A = \left[\begin{array}{cc}1 & 5 \\ 4 & 2\end{array}\right], \mathbf{v} = \left[\begin{array}{c}1 \\ 1\end{array}\right]$.
johnston-linear-matrix-algebra-F02931	322	1.000	$A=\left[\begin{array}{cc}1&-1\\3&6\end{array}\right],\ \mathbf{v}=\left[\begin{array}{c}-2\\5+\sqrt{13}\end{array}\right]$	$A = \left[\begin{array}{cc}1 & -1 \\ 3 & 6\end{array}\right], \mathbf{v} = \left[\begin{array}{c}-2 \\ 5 + \sqrt{13}\end{array}\right]$	(b) $A = \left[\begin{array}{cc}1 & -1 \\ 3 & 6\end{array}\right], \mathbf{v} = \left[\begin{array}{c}-2 \\ 5 + \sqrt{13}\end{array}\right]$.

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johnston-linear-matrix-algebra F02932	322	0.921	$A=\left[\begin{array}{ccc}4 & 1 & 1 \\ 3 & 3 & -1 \\ 4 & 1 & 1\end{array}\right], \mathbf{v}=\left[\begin{array}{c}-4 \\ 7 \\ 9\end{array}\right]$	$A = \begin{bmatrix} 4 & 1 & 1 \\ 3 & 3 & -1 \\ 4 & 1 & 1 \end{bmatrix}, \mathbf{v} = \begin{bmatrix} -4 \\ 7 \\ 9 \end{bmatrix}$	*(c) $A = \begin{bmatrix} 4 & 1 & 1 \\ 3 & 3 & -1 \\ 4 & 1 & 1 \end{bmatrix}, \mathbf{v} = \begin{bmatrix} -4 \\ 7 \\ 9 \end{bmatrix}.$
johnston-linear-matrix-algebra F02933	322	1.000	$A=\left[\begin{array}{ccc}2 & -1 & -1 \\ 3 & 2 & 4\end{array}\right], \mathbf{v}=\left[\begin{array}{c}2 \\ -1+i \\ -1-3i\end{array}\right]$	$A = \begin{bmatrix} 2 & -1 & -1 \\ -1 & 2 & 0 \\ 3 & 2 & 4 \end{bmatrix}, \mathbf{v} = \begin{bmatrix} 2 \\ -1+i \\ -1-3i \end{bmatrix}$	(d) $A = \begin{bmatrix} 2 & -1 & -1 \\ -1 & 2 & 0 \\ 3 & 2 & 4 \end{bmatrix}, \mathbf{v} = \begin{bmatrix} 2 \\ -1+i \\ -1-3i \end{bmatrix}.$
johnston-linear-matrix-algebra F02934	323	0.884	$A=\left[\begin{array}{llll}3 & 1 & 5 & 5 \\ 3 & 2 & 2 & 5 \\ 3 & 3 & 5 & 5 \\ 2 & 5 & 2 & 0\end{array}\right], \mathbf{v}=\left[\begin{array}{c}13 \\ 14 \\ 6 \\ -27\end{array}\right]$	$A = \begin{bmatrix} 3 & 1 & 5 & 5 \\ 3 & 2 & 2 & 5 \\ 3 & 3 & 5 & 5 \\ 2 & 5 & 2 & 0 \end{bmatrix}, \mathbf{v} = \begin{bmatrix} 13 \\ 14 \\ 6 \\ -27 \end{bmatrix}$	*(e) $A = \begin{bmatrix} 3 & 1 & 5 & 5 \\ 3 & 2 & 2 & 5 \\ 3 & 3 & 5 & 5 \\ 2 & 5 & 2 & 0 \end{bmatrix}, \mathbf{v} = \begin{bmatrix} 13 \\ 14 \\ 6 \\ -27 \end{bmatrix}.$
johnston-linear-matrix-algebra F02935	323	0.754	$\lambda=-3, A=\left[\begin{array}{ll}1 & 5 \\ 4 & 2\end{array}\right]$	$\lambda = -3, A = \begin{bmatrix} 1 & 5 \\ 4 & 2 \end{bmatrix}$	*(a) $\lambda = -3, A = \begin{bmatrix} 1 & 5 \\ 4 & 2 \end{bmatrix}.$
johnston-linear-matrix-algebra F02936	323	1.000	$\lambda=\sqrt{2}, A=\left[\begin{array}{ll}0 & 1 \\ 2 & 0\end{array}\right]$	$\lambda = \sqrt{2}, A = \begin{bmatrix} 0 & 1 \\ 2 & 0 \end{bmatrix}$	(b) $\lambda = \sqrt{2}, A = \begin{bmatrix} 0 & 1 \\ 2 & 0 \end{bmatrix}.$
johnston-linear-matrix-algebra F02937	323	0.985	$\lambda=2, A=\left[\begin{array}{ccc}0 & 3 & -1 \\ 2 & -1 & -1 \\ -2 & 3 & 0\end{array}\right]$	$\lambda = 2, A = \begin{bmatrix} 0 & 3 & -1 \\ 2 & -1 & -1 \\ -2 & 3 & 0 \end{bmatrix}$	*(c) $\lambda = 2, A = \begin{bmatrix} 0 & 3 & -1 \\ 2 & -1 & -1 \\ -2 & 3 & 0 \end{bmatrix}.$
johnston-linear-matrix-algebra F02938	323	1.000	$\lambda=2+i, A=\left[\begin{array}{ccc}-1 & 0 & -1 \\ 2 & 1 & 2 \\ 2 & -2 & 3\end{array}\right]$	$\lambda = 2+i, A = \begin{bmatrix} -1 & 0 & -1 \\ 2 & 2 & 1 \\ 2 & -2 & 3 \end{bmatrix}$	(d) $\lambda = 2+i, A = \begin{bmatrix} -1 & 0 & -1 \\ 2 & 2 & 1 \\ 2 & -2 & 3 \end{bmatrix}.$
johnston-linear-matrix-algebra F02939	323	0.969	$\lambda=-1, A=\left[\begin{array}{cccc}3 & 2 & 3 & 2 \\ 0 & 0 & 3 & -1 \\ 4 & 1 & 3 & 3 \\ 2 & 1 & 1 & 0\end{array}\right]$	$\lambda = -1, A = \begin{bmatrix} 3 & 2 & 3 & 2 \\ 0 & 0 & 3 & -1 \\ 4 & 1 & 3 & 3 \\ 2 & 1 & 1 & 0 \end{bmatrix}$	*(e) $\lambda = -1, A = \begin{bmatrix} 3 & 2 & 3 & 2 \\ 0 & 0 & 3 & -1 \\ 4 & 1 & 3 & 3 \\ 2 & 1 & 1 & 0 \end{bmatrix}.$
johnston-linear-matrix-algebra F02940	323	0.976	$\left[\begin{array}{cc}1 & 2 \\ -1 & -2\end{array}\right]$	$\begin{bmatrix} 1 & 2 \\ -1 & -2 \end{bmatrix}$	*(a) $\begin{bmatrix} 1 & 2 \\ -1 & -2 \end{bmatrix}$
johnston-linear-matrix-algebra F02941	323	0.916	$\left[\begin{array}{ll}0 & 1 \\ 0 & 0\end{array}\right]$	$\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$	*(c) $\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$
johnston-linear-matrix-algebra F02942	323	1.000	$\left[\begin{array}{cc}2 & 1 \\ 3 & -1\end{array}\right]$	$\begin{bmatrix} 2 & 1 \\ 3 & -1 \end{bmatrix}$	(d) $\begin{bmatrix} 2 & 1 \\ 3 & -1 \end{bmatrix}$
johnston-linear-matrix-algebra F02943	323	0.599	$\left[\begin{array}{ccc}3 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & 7\end{array}\right]$	$\begin{bmatrix} 3 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & 7 \end{bmatrix}$	*(e) $\begin{bmatrix} 3 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & 7 \end{bmatrix}$
johnston-linear-matrix-algebra F02944	323	0.992	$\left[\begin{array}{lll}2 & 3 & 0 \\ 3 & 0 & 1 \\ 0 & 1 & 2\end{array}\right]$	$\begin{bmatrix} 2 & 3 & 0 \\ 3 & 0 & 1 \\ 0 & 1 & 2 \end{bmatrix}$	(f) $\begin{bmatrix} 2 & 3 & 0 \\ 3 & 0 & 1 \\ 0 & 1 & 2 \end{bmatrix}$

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johnston-linear-matrix-algebra-F02945	323	0.981	$\left[\begin{array}{cccc}2 & 1 & 0 & 0 \\ 0 & -3 & 2 & 0 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 2\end{array}\right]$	$\begin{bmatrix} 2 & 1 & 0 & 0 \\ 0 & -3 & 2 & 0 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 2 \end{bmatrix}$	*(g)
johnston-linear-matrix-algebra-F02946	323	1.000	$\left[\begin{array}{cccc}2 & 1 & -1 & -1 \\ 0 & 3 & 1 & 1 \\ 0 & 1 & 3 & 1 \\ 0 & 0 & 0 & 2\end{array}\right]$	$\begin{bmatrix} 2 & 1 & -1 & -1 \\ 0 & 3 & 1 & 1 \\ 0 & 1 & 3 & 1 \\ 0 & 0 & 0 & 2 \end{bmatrix}$	(h)
johnston-linear-matrix-algebra-F02947	323	0.999	$\left[\begin{array}{cccc}19 & 10 & 5 & 22 \\ 9 & 17 & 5 & 19 \\ -8 & -10 & 2 & -18 \\ -11 & -10 & -5 & -14\end{array}\right]$	$\begin{bmatrix} 19 & 10 & 5 & 22 \\ 9 & 17 & 5 & 19 \\ -8 & -10 & 2 & -18 \\ -11 & -10 & -5 & -14 \end{bmatrix}$	*(a)
johnston-linear-matrix-algebra-F02948	323	1.000	$\left[\begin{array}{cccc}12 & -3 & -6 & -3 \\ 1 & 16 & 2 & 1 \\ -3 & -3 & 9 & -3 \\ -1 & -1 & -2 & 14\end{array}\right]$	$\begin{bmatrix} 12 & -3 & -6 & -3 \\ 1 & 16 & 2 & 1 \\ -3 & -3 & 9 & -3 \\ -1 & -1 & -2 & 14 \end{bmatrix}$	(b)
johnston-linear-matrix-algebra-F02949	323	1.000	$\left[\begin{array}{ccccc}6 & -4 & 4 & 0 & 4 \\ 0 & 10 & -6 & 0 & -6 \\ 16 & 20 & -20 & -8 & -26 \\ -8 & -8 & 11 & 10 & 11 \\ -16 & -20 & 24 & 8 & 30\end{array}\right]$	$\begin{bmatrix} 6 & -4 & 4 & 0 & 4 \\ 0 & 10 & -6 & 0 & -6 \\ 16 & 20 & -20 & -8 & -26 \\ -8 & -8 & 11 & 10 & 11 \\ -16 & -20 & 24 & 8 & 30 \end{bmatrix}$	*(c)
johnston-linear-matrix-algebra-F02950	323	1.000	$\operatorname{det}(A - \lambda I) = \operatorname{det}(\lambda I - A)$	$\det(A - \lambda I) = \det(\lambda I - A)$	*(c) If $A \in \mathcal{M}_n$ then $\det(A - \lambda I) = \det(\lambda I - A)$.
johnston-linear-matrix-algebra-F02951	323	0.991	$A \in \mathcal{M}_n(\mathbb{R})$	$A \in \mathcal{M}_n(\mathbb{R})$	*(g) Every diagonal matrix $A \in \mathcal{M}_n(\mathbb{R})$ is symmetric.
johnston-linear-matrix-algebra-F02952	323	1.000	$k \in \mathbb{R}$	$k \in \mathbb{R}$	*3.3.6 Let $k \in \mathbb{R}$ and consider the following matrix:
johnston-linear-matrix-algebra-F02953	323	0.999	$A = \begin{bmatrix} 1 & 0 & 0 \\ -8 & 4 & -5 \\ 8 & 0 & 9 \end{bmatrix}$	$A = \begin{bmatrix} 1 & 0 & 0 \\ -8 & 4 & -5 \\ 8 & 0 & 9 \end{bmatrix}$	3.3.7 Let $A = \begin{bmatrix} 1 & 0 & 0 \\ -8 & 4 & -5 \\ 8 & 0 & 9 \end{bmatrix}$.
johnston-linear-matrix-algebra-F02954	323	1.000	$A^2 - 3A + I$	$A^2 - 3A + I$	(c) Find all eigenvalues of $A^2 - 3A + I$.
johnston-linear-matrix-algebra-F02955	323	0.998	$A^2 -$	$A^2 -$	[Hint: You can explicitly calculate the matrix $A^2 -$
johnston-linear-matrix-algebra-F02956	323	1.000	$3A + I$	$3A + I$	$3A + I$ and find its eigenvalues, but there is a much
johnston-linear-matrix-algebra-F02957	323	1.000	$\overline{\mathbf{v}}$	$\bar{\mathbf{v}}$	value λ then $\bar{\mathbf{v}}$ is an eigenvector of A corresponding to the
johnston-linear-matrix-algebra-F02958	323	1.000	$\bar{\lambda}$	$\bar{\lambda}$	eigenvalue $\bar{\lambda}$.
johnston-linear-matrix-algebra-F02959	324	0.928	$A \in \mathcal{M}_2(\mathbb{C})$	$A \in \mathcal{M}_2(\mathbb{C})$	**3.3.10 Suppose $A \in \mathcal{M}_2(\mathbb{C})$. Find a formula for the
johnston-linear-matrix-algebra-F02960	324	1.000	$A^4 = I$	$A^4 = I$	$A^4 = I$. Find all possible eigenvalues of A .
johnston-linear-matrix-algebra-F02961	324	1.000	$A^* = -A$	$A^* = -A$	if $A^* = -A$. Show that the eigenvalues of skew-Hermitian
johnston-linear-matrix-algebra-F02962	324	0.991	$b \in \mathbb{R}$	$b \in \mathbb{R}$	matrices are imaginary (i.e., of the form bi for some $b \in \mathbb{R}$).
johnston-linear-matrix-algebra-F02963	324	0.955	$\mathbf{v}^* \mathbf{w}$	$\mathbf{v}^* \mathbf{w}$	which equals $\mathbf{v}^* \mathbf{w}$ if $\mathbf{v}$ and $\mathbf{w}$ are column vectors.

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johnston-linear-matrix-algebra-F02964	324	1.000	$\mathbf{v} \in \mathbb{C}^m$	$\mathbf{v} \in \mathbb{C}^m$	(a) Show that $\mathbf{v} \cdot (A\mathbf{w}) = (A^*\mathbf{v}) \cdot \mathbf{w}$ for all $\mathbf{v} \in \mathbb{C}^m$ and
johnston-linear-matrix-algebra-F02965	324	1.000	$\mathbf{w} \in \mathbb{C}^n$	$\mathbf{w} \in \mathbb{C}^n$	$\mathbf{w} \in \mathbb{C}^n$.
johnston-linear-matrix-algebra-F02966	324	1.000	$B \in \mathcal{M}_{n,m}(\mathbb{C})$	$B \in \mathcal{M}_{n,m}(\mathbb{C})$	(b) Show that if $B \in \mathcal{M}_{n,m}(\mathbb{C})$ is such that $\mathbf{v} \cdot (A\mathbf{w}) =$
johnston-linear-matrix-algebra-F02967	324	1.000	$\mathbf{v} \cdot (A\mathbf{w}) =$	$\mathbf{v} \cdot (A\mathbf{w}) =$	(b) Show that if $B \in \mathcal{M}_{n,m}(\mathbb{C})$ is such that $\mathbf{v} \cdot (A\mathbf{w}) =$
johnston-linear-matrix-algebra-F02968	324	1.000	$(B\mathbf{v}) \cdot \mathbf{w}$	$(B\mathbf{v}) \cdot \mathbf{w}$	$(B\mathbf{v}) \cdot \mathbf{w}$ for all $\mathbf{v} \in \mathbb{C}^m$ and $\mathbf{w} \in \mathbb{C}^n$, then $B = A^*$.
johnston-linear-matrix-algebra-F02969	324	1.000	$B = A^*$	$B = A^*$	$(B\mathbf{v}) \cdot \mathbf{w}$ for all $\mathbf{v} \in \mathbb{C}^m$ and $\mathbf{w} \in \mathbb{C}^n$, then $B = A^*$.
johnston-linear-matrix-algebra-F02970	324	1.000	$P\mathbf{v}$	$P\mathbf{v}$	eigenvector of B if and only if $P\mathbf{v}$ is an eigenvector of A .
johnston-linear-matrix-algebra-F02971	324	1.000	$\lambda \neq 0$	$\lambda \neq 0$	(a) Show that if $\lambda \neq 0$ is an eigenvalue of AB then it is
johnston-linear-matrix-algebra-F02972	324	0.999	$BAB\mathbf{v}$	$BAB\mathbf{v}$	tor of AB corresponding to λ and simplify $BAB\mathbf{v}$.]
johnston-linear-matrix-algebra-F02973	324	1.000	$\mu \neq \bar{\lambda}$	$\mu \neq \bar{\lambda}$	to an eigenvalue $\mu \neq \bar{\lambda}$ of A^* . Show that $\mathbf{v} \cdot \mathbf{w} = 0$.
johnston-linear-matrix-algebra-F02974	324	0.997	$\lambda_1, \lambda_2, \dots$	$\lambda_1, \lambda_2, \dots$	**3.3.22 Suppose $A \in \mathcal{M}_n(\mathbb{C})$ has eigenvalues $\lambda_1, \lambda_2, \dots$,
johnston-linear-matrix-algebra-F02975	324	0.992	λ_n	λ_n	λ_n with corresponding unit (i.e., length 1) eigenvectors $\mathbf{v}_1$,
johnston-linear-matrix-algebra-F02976	324	1.000	$\mathbf{v}_2, \dots, \mathbf{v}_n$	$\mathbf{v}_2, \dots, \mathbf{v}_n$	$\mathbf{v}_2, \dots, \mathbf{v}_n$. Let $B = A - \lambda_1 \mathbf{v}_1 \mathbf{v}_1^*$.
johnston-linear-matrix-algebra-F02977	324	1.000	$B = A - \lambda_1 \mathbf{v}_1 \mathbf{v}_1^*$	$B = A - \lambda_1 \mathbf{v}_1 \mathbf{v}_1^*$	$\mathbf{v}_2, \dots, \mathbf{v}_n$. Let $B = A - \lambda_1 \mathbf{v}_1 \mathbf{v}_1^*$.
johnston-linear-matrix-algebra-F02978	324	1.000	$0, \lambda_2, \lambda_3, \dots, \lambda_n$	$0, \lambda_2, \lambda_3, \dots, \lambda_n$	(b) Show that B has eigenvalues $0, \lambda_2, \lambda_3, \dots, \lambda_n$. [Side
johnston-linear-matrix-algebra-F02979	324	1.000	$\lambda_1 \neq \lambda_2$	$\lambda_1 \neq \lambda_2$	note: If it helps, you can assume that $\lambda_1 \neq \lambda_2$,
johnston-linear-matrix-algebra-F02980	324	0.999	$\lambda_1 \neq \lambda_3, \dots, \lambda_1 \neq \lambda_n$	$\lambda_1 \neq \lambda_3, \dots, \lambda_1 \neq \lambda_n$	$\lambda_1 \neq \lambda_3, \dots, \lambda_1 \neq \lambda_n$.] [Hint: Use Exercise 3.3.21.]
johnston-linear-matrix-algebra-F02981	324	1.000	$(-1)^n \lambda^n$	$(-1)^n \lambda^n$	nomial (with leading term $(-1)^n \lambda^n$) of our choosing.
johnston-linear-matrix-algebra-F02982	324	1.000	$p_C(\lambda) = -(\lambda^3 + a_2 \lambda^2 + a_1 \lambda + a_0)$	$p_C(\lambda) = -(\lambda^3 + a_2 \lambda^2 + a_1 \lambda + a_0)$	is $p_C(\lambda) = -(\lambda^3 + a_2 \lambda^2 + a_1 \lambda + a_0)$.
johnston-linear-matrix-algebra-F02983	324	1.000	$p_C(\lambda) = (-1)^n (\lambda^n + a_{n-1} \lambda^{n-1} + \dots + a_1 \lambda + a_0)$	$p_C(\lambda) = (-1)^n (\lambda^n + a_{n-1} \lambda^{n-1} + \dots + a_1 \lambda + a_0)$	is $p_C(\lambda) = (-1)^n (\lambda^n + a_{n-1} \lambda^{n-1} + \dots + a_1 \lambda + a_0)$.
johnston-linear-matrix-algebra-F02984	324	1.000	p_C	p_C	of the polynomial p_C . Although we have used polynomials
johnston-linear-matrix-algebra-F02985	325	0.987	v_i	v_i	that its largest entry v_i has $v_i = 1$.]
johnston-linear-matrix-algebra-F02986	325	0.987	$v_i = 1$	$v_i = 1$	that its largest entry v_i has $v_i = 1$.]
johnston-linear-matrix-algebra-F02987	325	1.000	$A = P D P^{-1}$	$A = P D P^{-1}$	$D \in \mathcal{M}_n$ and an invertible matrix $P \in \mathcal{M}_n$ such that $A = P D P^{-1}$.
johnston-linear-matrix-algebra-F02988	325	1.000	D^k	D^k	Since D is diagonal, D^k can be computed entry-wise, so we can compute
johnston-linear-matrix-algebra-F02989	325	1.000	P^{-1}	P^{-1}	right by P^{-1} .
johnston-linear-matrix-algebra-F02990	326	0.999	$P D$	$P D$	$P D$ in terms of scaled
johnston-linear-matrix-algebra-F02991	326	1.000	$P, D \in \mathcal{M}_n$	$P, D \in \mathcal{M}_n$	Let $A \in \mathcal{M}_n$ and suppose $P, D \in \mathcal{M}_n$ are such that P is invertible and D is
johnston-linear-matrix-algebra-F02992	326	1.000	$A P = P D$	$A P = P D$	it is equivalent to $A P = P D$. If we write P in terms of its columns $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$

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johnston-linear-matrix-algebra-F02993	326	1.000	A P	AP	Since $AP = PD$ if and only if the columns of AP equal the columns of PD
johnston-linear-matrix-algebra-F02994	326	1.000	$A \mathbf{v}_j = d_{j,j} \mathbf{v}_j$	$A\mathbf{v}_j = d_{j,j}\mathbf{v}_j$	in the same order, we conclude that $AP = PD$ if and only if $A\mathbf{v}_j = d_{j,j}\mathbf{v}_j$ for all
johnston-linear-matrix-algebra-F02995	326	1.000	$d_{1,1}, d_{2,2}, \ldots, d_{n,n}$	$d_{1,1}, d_{2,2}, \ldots, d_{n,n}$	of P) are eigenvectors of A with corresponding eigenvalues $d_{1,1}, d_{2,2}, \ldots, d_{n,n}$,
johnston-linear-matrix-algebra-F02996	326	1.000	$\lambda_1 = -1$	$\lambda_1 = -1$	ues $\lambda_1 = -1$ and $\lambda_2 = 6$ corresponding to the eigenvectors $\mathbf{v}_1 = (-1, 1)$
johnston-linear-matrix-algebra-F02997	326	1.000	$\lambda_2 = 6$	$\lambda_2 = 6$	ues $\lambda_1 = -1$ and $\lambda_2 = 6$ corresponding to the eigenvectors $\mathbf{v}_1 = (-1, 1)$
johnston-linear-matrix-algebra-F02998	326	0.982	$\mathbf{v}_2 = (2, 5)$	$\mathbf{v}_2 = (2, 5)$	and $\mathbf{v}_2 = (2, 5)$, , respectively. We thus stick these eigenvalues along the
johnston-linear-matrix-algebra-F02999	327	1.000	$\mathbf{v}_2 = (0, 1, 0)$	$\mathbf{v}_2 = (0, 1, 0)$	$\mathbf{v}_2 = (0, 1, 0)$ and
johnston-linear-matrix-algebra-F03000	327	0.998	$\mathbf{v}_3 = (1, 0, 1)$	$\mathbf{v}_3 = (1, 0, 1)$	$\mathbf{v}_3 = (1, 0, 1)$, or
johnston-linear-matrix-algebra-F03001	327	1.000	$\mathbf{v}_2 = (1, 0, 1)$	$\mathbf{v}_2 = (1, 0, 1)$	$\mathbf{v}_2 = (1, 0, 1)$ and
johnston-linear-matrix-algebra-F03002	327	1.000	$\mathbf{v}_3 = (0, 1, 0)$	$\mathbf{v}_3 = (0, 1, 0)$	$\mathbf{v}_3 = (0, 1, 0)$. Either
johnston-linear-matrix-algebra-F03003	327	1.000	$P D P^{-1}$	PDP^{-1}	let's multiply out PDP^{-1} and see that it does indeed equal A :
johnston-linear-matrix-algebra-F03004	327	1.000	$B = \{\mathbf{v}_1, \mathbf{v}_2\} = \{(-1, 1), (2, 5)\}$	$B = \{\mathbf{v}_1, \mathbf{v}_2\} = \{(-1, 1), (2, 5)\}$	transformation in the basis $B = \{\mathbf{v}_1, \mathbf{v}_2\} = \{(-1, 1), (2, 5)\}$ consisting of
johnston-linear-matrix-algebra-F03005	327	1.000	$[T]_B = D$	$[T]_B = D$	eigenvectors of A then $[T]_B = D$ is diagonal:
johnston-linear-matrix-algebra-F03006	327	1.000	$\lambda_1 = 2$	$\lambda_1 = 2$	according to algebraic multiplicity) $\lambda_1 = 2$ and $\lambda_2 = \lambda_3 = -1$, and we
johnston-linear-matrix-algebra-F03007	327	1.000	$\lambda_2 = \lambda_3 = -1$	$\lambda_2 = \lambda_3 = -1$	according to algebraic multiplicity) $\lambda_1 = 2$ and $\lambda_2 = \lambda_3 = -1$, and we
johnston-linear-matrix-algebra-F03008	327	1.000	$\mathbf{v}_1 = (3, 1, 0), \mathbf{v}_2 = (0, 1, 0)$	$\mathbf{v}_1 = (3, 1, 0), \mathbf{v}_2 = (0, 1, 0)$	from the corresponding eigenvectors $\mathbf{v}_1 = (3, 1, 0)$, $\mathbf{v}_2 = (0, 1, 0)$, and
johnston-linear-matrix-algebra-F03009	328	1.000	$B_1, B_2, \ldots, B_m$	$B_1, B_2, \ldots, B_m$	Suppose A is a square matrix. If $B_1, B_2, \ldots, B_m$ are bases of eigenspaces
johnston-linear-matrix-algebra-F03010	328	1.000	$\mathbf{v}_1 \in \operatorname{span}(B_1), \mathbf{v}_2 \in \operatorname{span}(B_2), \ldots, \mathbf{v}_m \in \operatorname{span}(B_m)$	$\mathbf{v}_1 \in \operatorname{span}(B_1), \mathbf{v}_2 \in \operatorname{span}(B_2), \ldots, \mathbf{v}_m \in \operatorname{span}(B_m)$	<i>Proof.</i> First, let $\mathbf{v}_1 \in \operatorname{span}(B_1), \mathbf{v}_2 \in \operatorname{span}(B_2), \ldots, \mathbf{v}_m \in \operatorname{span}(B_m)$ be arbitrary
johnston-linear-matrix-algebra-F03011	329	1.000	γ_j	γ_j	Here, γ_j is the
johnston-linear-matrix-algebra-F03012	329	1.000	B_j	B_j	basis B_j .
johnston-linear-matrix-algebra-F03013	329	1.000	$\lambda_1, \lambda_2, \ldots, \lambda_m$	$\lambda_1, \lambda_2, \ldots, \lambda_m$	of A corresponding to eigenvalues $\lambda_1, \lambda_2, \ldots, \lambda_m$, respectively). We first show
johnston-linear-matrix-algebra-F03014	329	1.000	$1 \leq \ell \leq m$	$1 \leq \ell \leq m$	a) If $1 \leq \ell \leq m$ is such that $\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_\ell\}$ is linearly dependent, then at
johnston-linear-matrix-algebra-F03015	329	1.000	$\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_\ell\}$	$\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_\ell\}$	a) If $1 \leq \ell \leq m$ is such that $\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_\ell\}$ is linearly dependent, then at
johnston-linear-matrix-algebra-F03016	329	1.000	$\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_\ell$	$\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_\ell$	least one of $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_\ell$ is the zero vector.
johnston-linear-matrix-algebra-F03017	329	1.000	$\mathbf{v}_1 + \mathbf{v}_2 + \cdots + \mathbf{v}_m = \mathbf{0}$	$\mathbf{v}_1 + \mathbf{v}_2 + \cdots + \mathbf{v}_m = \mathbf{0}$	b) If $\mathbf{v}_1 + \mathbf{v}_2 + \cdots + \mathbf{v}_m = \mathbf{0}$ then $\mathbf{v}_1 = \mathbf{v}_2 = \cdots = \mathbf{v}_m = \mathbf{0}$.
johnston-linear-matrix-algebra-F03018	329	1.000	$\mathbf{v}_1 = \mathbf{v}_2 = \cdots = \mathbf{v}_m = \mathbf{0}$	$\mathbf{v}_1 = \mathbf{v}_2 = \cdots = \mathbf{v}_m = \mathbf{0}$	b) If $\mathbf{v}_1 + \mathbf{v}_2 + \cdots + \mathbf{v}_m = \mathbf{0}$ then $\mathbf{v}_1 = \mathbf{v}_2 = \cdots = \mathbf{v}_m = \mathbf{0}$.
johnston-linear-matrix-algebra-F03019	329	1.000	$k \leq \ell$	$k \leq \ell$	$k \leq \ell$ be the smallest integer such that $\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_k\}$ is linearly dependent

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johnston-linear-matrix-algebra-F03020	329	■ 0.934	$c_{\{1\}}, c_{\{2\}}, \ldots, c_{\{k-1\}}$	$c_1, c_2, \dots, c_{k-1}$	then there exist scalars $c_1, c_2, \dots, c_{k-1}$ such that
johnston-linear-matrix-algebra-F03021	329	■ 1.000	$\lambda_{\{k\}}$	λ_k	Also, multiplying both sides of Equation (3.4.1) by λ_k gives
johnston-linear-matrix-algebra-F03022	329	■ 1.000	$\left\{ \mathbf{v}_{\{1\}}, \mathbf{v}_{\{2\}}, \ldots, \mathbf{v}_{\{k-1\}} \right\}$	$\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_{k-1}\}$	linearly dependent, $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_{k-1}\}$ is linearly independent, which implies
johnston-linear-matrix-algebra-F03023	329	■ 1.000	$c_{\{j\}} \left(\lambda_{\{j\}} - \lambda_{\{k\}} \right) = 0$	$c_j (\lambda_j - \lambda_k) = 0$	$c_j (\lambda_j - \lambda_k) = 0$ for all $1 \leq j \leq k-1$. Since $\lambda_1, \lambda_2, \dots, \lambda_m$ are distinct, this is
johnston-linear-matrix-algebra-F03024	329	■ 1.000	$1 \leq j \leq k-1$	$1 \leq j \leq k-1$	$c_j (\lambda_j - \lambda_k) = 0$ for all $1 \leq j \leq k-1$. Since $\lambda_1, \lambda_2, \dots, \lambda_m$ are distinct, this is
johnston-linear-matrix-algebra-F03025	329	■ 1.000	$c_{\{1\}} = c_{\{2\}} = \cdots = c_{\{k-1\}} = 0$	$c_1 = c_2 = \cdots = c_{k-1} = 0$	only possible if $c_1 = c_2 = \cdots = c_{k-1} = 0$, so $\mathbf{v}_k = \mathbf{0}$, which proves fact (a).
johnston-linear-matrix-algebra-F03026	329	■ 1.000	$\mathbf{v}_{\{k\}} = \mathbf{0}$	$\mathbf{v}_k = \mathbf{0}$	only possible if $c_1 = c_2 = \cdots = c_{k-1} = 0$, so $\mathbf{v}_k = \mathbf{0}$, which proves fact (a).
johnston-linear-matrix-algebra-F03027	329	■ 1.000	$\mathbf{v}_{\{1\}}, \mathbf{v}_{\{2\}}, \ldots, \mathbf{v}_{\{m\}}$	$\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_m$	then at least one of $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_m$ is the zero vector. Suppose that $\mathbf{v}_m = \mathbf{0}$ (but a
johnston-linear-matrix-algebra-F03028	329	■ 1.000	$\mathbf{v}_{\{m\}} = \mathbf{0}$	$\mathbf{v}_m = \mathbf{0}$	then at least one of $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_m$ is the zero vector. Suppose that $\mathbf{v}_m = \mathbf{0}$ (but a
johnston-linear-matrix-algebra-F03029	329	■ 0.958	$\mathbf{v}_{\{1\}}, \mathbf{v}_{\{2\}}, \ldots, \mathbf{v}_{\{m-1\}}$	$\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_{m-1}$	similar argument works if it's one of $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_{m-1}$ that equals $\mathbf{0}$). Then
johnston-linear-matrix-algebra-F03030	329	■ 1.000	$B_{\{1\}} \cup B_{\{2\}} \cup \cdots \cup B_{\{m\}}$	$B_1 \cup B_2 \cup \cdots \cup B_m$	To now show that $B_1 \cup B_2 \cup \cdots \cup B_m$ is linearly independent, we give names
johnston-linear-matrix-algebra-F03031	329	■ 1.000	$B_{\{j\}} = \left\{ \mathbf{v}_{\{j, 1\}}, \mathbf{v}_{\{j, 2\}}, \ldots, \mathbf{v}_{\{j, \gamma_j\}} \right\}$	$B_j = \{\mathbf{v}_{j,1}, \mathbf{v}_{j,2}, \dots, \mathbf{v}_{j,\gamma_j}\}$	to the vectors in each of these bases via $B_j = \{\mathbf{v}_{j,1}, \mathbf{v}_{j,2}, \dots, \mathbf{v}_{j,\gamma_j}\}$ for each
johnston-linear-matrix-algebra-F03032	329	■ 1.000	$1 \leq j \leq m$	$1 \leq j \leq m$	$1 \leq j \leq m$. Now suppose that some linear combination of these vectors equals
johnston-linear-matrix-algebra-F03033	329	■ 1.000	$c_{\{j, 1\}} = c_{\{j, 2\}} = \cdots = c_{\{j, \gamma_j\}} = 0$	$c_{j,1} = c_{j,2} = \cdots = c_{j,\gamma_j} = 0$	tells us that $c_{j,1} = c_{j,2} = \cdots = c_{j,\gamma_j} = 0$ for all $1 \leq j \leq m$. It follows that every
johnston-linear-matrix-algebra-F03034	330	■ 1.000	$A = \begin{bmatrix} 3 & -1 \\ 1 & 1 \end{bmatrix}$	$A = \begin{bmatrix} 3 & -1 \\ 1 & 1 \end{bmatrix}$	Show that the matrix $A = \begin{bmatrix} 3 & -1 \\ 1 & 1 \end{bmatrix}$ cannot be diagonalized.
johnston-linear-matrix-algebra-F03035	330	■ 1.000	$\mathbf{v} = \mathbf{v}_{\{2\}}(1, 1)$	$\mathbf{v} = v_2(1, 1)$	v_2 , so all eigenvectors corresponding to $\lambda = 2$ are of the form $\mathbf{v} = v_2(1, 1)$.
johnston-linear-matrix-algebra-F03036	331	■ 1.000	$D, P \in \mathcal{M}_n(\mathbb{R})$	$D, P \in \mathcal{M}_n(\mathbb{R})$	onalized as $A = PDP^{-1}$ via <i>real</i> matrices $D, P \in \mathcal{M}_n(\mathbb{R})$ then we say that A
johnston-linear-matrix-algebra-F03037	331	■ 1.000	$D, P \in \mathcal{M}_n(\mathbb{C})$	$D, P \in \mathcal{M}_n(\mathbb{C})$	$D, P \in \mathcal{M}_n(\mathbb{C})$ then we say that A is diagonalizable over $\mathbb{C}$. If we just say
johnston-linear-matrix-algebra-F03038	331	■ 1.000	$\mathbb{C}$	$\mathbb{C}$	$D, P \in \mathcal{M}_n(\mathbb{C})$ then we say that A is diagonalizable over $\mathbb{C}$. If we just say
johnston-linear-matrix-algebra-F03039	331	■ 1.000	$B_{\{1\}}, B_{\{2\}}, \ldots, B_{\{m\}} \subseteq B$	$B_1, B_2, \dots, B_m \subseteq B$	to their corresponding eigenvalues. Specifically, let $B_1, B_2, \dots, B_m \subseteq B$ be
johnston-linear-matrix-algebra-F03040	331	■ 0.995	$B_{\{1\}}, B_{\{2\}}, \ldots$	$B_1, B_2, \dots$	and note that the fact that $\lambda_1, \lambda_2, \dots, \lambda_m$ are distinct ensures that $B_1, B_2, \dots,$
johnston-linear-matrix-algebra-F03041	331	■ 1.000	$B_{\{m\}}$	B_m	B_m are disjoint (after all, no non-zero vector can be scaled by two different
johnston-linear-matrix-algebra-F03042	331	■ 0.999	$\lambda_{\{j\}}$	λ_j	of λ_j is at least $ B_j $. Then the fact that $n = B = B_1 + \cdots + B_m $ implies that
johnston-linear-matrix-algebra-F03043	331	■ 0.999	$ B_{\{j\}} $	$ B_j $	of λ_j is at least $ B_j $. Then the fact that $n = B = B_1 + \cdots + B_m $ implies that
johnston-linear-matrix-algebra-F03044	331	■ 0.999	$n = B = B_{\{1\}} + \cdots + B_{\{m\}} $	$n = B = B_1 + \cdots + B_m $	of λ_j is at least $ B_j $. Then the fact that $n = B = B_1 + \cdots + B_m $ implies that
johnston-linear-matrix-algebra-F03045	332	■ 1.000	$ B_{\{1\}} , B_{\{2\}} , \ldots, B_{\{m\}} $	$ B_1 , B_2 , \dots, B_m $	the eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_m$, respectively. Since $ B_1 , B_2 , \dots, B_m $ are

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johnston-linear-matrix-algebra-F03046	332	0.998	$\left B_{\{1\}}\right +\left B_{\{2\}}\right +\cdots+\left B_{\{m\}}\right =n$	$ B_1 + B_2 + \cdots + B_m = n$	$ B_1 + B_2 + \cdots + B_m = n$. If we define $B = B_1 \cup B_2 \cup \cdots \cup B_m$, then $ B = n$
johnston-linear-matrix-algebra-F03047	332	0.998	$B=B_{\{1\}} \cup B_{\{2\}} \cup \cdots \cup B_{\{m\}}$	$B = B_1 \cup B_2 \cup \cdots \cup B_m$	$ B_1 + B_2 + \cdots + B_m = n$. If we define $B = B_1 \cup B_2 \cup \cdots \cup B_m$, then $ B = n$
johnston-linear-matrix-algebra-F03048	332	0.998	$ B =n$	$ B = n$	$ B_1 + B_2 + \cdots + B_m = n$. If we define $B = B_1 \cup B_2 \cup \cdots \cup B_m$, then $ B = n$
johnston-linear-matrix-algebra-F03049	332	0.979	$1+2=3=n$	$1 + 2 = 3 = n$	plicities of its eigenvalues were 1 and 2, and $1 + 2 = 3 = n$.
johnston-linear-matrix-algebra-F03050	332	0.793	$\operatorname{not} A=\left[\begin{array}{ccc}5 & 1 & -1 \\ 1 & 3 & -1 \\ 2 & 0 & 2\end{array}\right]$	$\operatorname{not} A = \begin{bmatrix} 5 & 1 & -1 \\ 1 & 3 & -1 \\ 2 & 0 & 2 \end{bmatrix}$	Determine whether or not $A = \begin{bmatrix} 5 & 1 & -1 \\ 1 & 3 & -1 \\ 2 & 0 & 2 \end{bmatrix}$ can be diagonalized.
johnston-linear-matrix-algebra-F03051	333	1.000	$a \lambda^2+b \lambda+c$	$a\lambda^2 + b\lambda + c$	$a\lambda^2 + b\lambda + c$ are
johnston-linear-matrix-algebra-F03052	333	0.997	$1+$	$1+$	Since there are n distinct eigenvalues, their geometric multiplies thus sum to $1 +$
johnston-linear-matrix-algebra-F03053	333	0.997	$1+\cdots+1=n$	$1 + \cdots + 1 = n$	$1 + \cdots + 1 = n$. It then follows from Theorem 3.4.3 that A is diagonalizable.
johnston-linear-matrix-algebra-F03054	333	0.972	$\operatorname{not} A=\left[\begin{array}{ll}1 & 1 \\ 1 & 0\end{array}\right]$	$\operatorname{not} A = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}$	Determine whether or not $A = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}$ is diagonalizable.
johnston-linear-matrix-algebra-F03055	334	1.000	$1+\varepsilon$	$1 + \varepsilon$	$1 + \varepsilon$ results in its eigenvalues being
johnston-linear-matrix-algebra-F03056	334	1.000	$\varepsilon \notin\{0,4\}$	$\varepsilon \notin \{0,4\}$	which are distinct as long as $\varepsilon \notin \{0,4\}$. It follows that nearby matrices
johnston-linear-matrix-algebra-F03057	335	1.000	$Q^{\ast}$	Q^*	Recall that Q^* is the
johnston-linear-matrix-algebra-F03058	335	1.000	Q	Q	of Q , which was
johnston-linear-matrix-algebra-F03059	335	1.000	$A^{\{500\}}$	A^{500}	it is much simpler to compute A^{500} by performing this diagonalization and then
johnston-linear-matrix-algebra-F03060	335	1.000	$k=500$	$k = 500$	plugging $k = 500$ into that formula, rather than multiplying A by itself 499
johnston-linear-matrix-algebra-F03061	335	1.000	$P, Q, D \in \mathcal{M}_n$	$P, Q, D \in \mathcal{M}_n$	Suppose $P, Q, D \in \mathcal{M}_n$ are such that D is diagonal with diagonal entries
johnston-linear-matrix-algebra-F03062	335	0.999	$d_{\{1\}}, d_{\{2\}}, \ldots, d_{\{n\}}$	$d_1, d_2, \dots, d_n$	$d_1, d_2, \dots, d_n$, in that order. Write P and Q in terms of their columns:
johnston-linear-matrix-algebra-F03063	336	1.000	$\mathbf{p}_{\{1\}} \mathbf{q}_{\{1\}}^{\ast}$	$\mathbf{p}_1 \mathbf{q}_1^*$	like $\mathbf{p}_1 \mathbf{q}_1^*$. A column
johnston-linear-matrix-algebra-F03064	336	1.000	$(-1)^{\{k\}}$	$(-1)^k$	by $(-1)^k$ here if we
johnston-linear-matrix-algebra-F03065	336	1.000	$\mathbf{p}_{\{j\}}$	$\mathbf{p}_j$	where $\mathbf{p}_j$ is the j -th column of P and $\mathbf{q}_j^*$ is the j -th row of P^{-1} . Furthermore, in
johnston-linear-matrix-algebra-F03066	336	1.000	$\mathbf{q}_{\{j\}}^{\ast}$	$\mathbf{q}_j^*$	where $\mathbf{p}_j$ is the j -th column of P and $\mathbf{q}_j^*$ is the j -th row of P^{-1} . Furthermore, in
johnston-linear-matrix-algebra-F03067	336	1.000	$A^{\{k\}}=P D^{\{k\}} P^{-1}$	$A^k = PD^k P^{-1}$	this case we have $A^k = PD^k P^{-1}$, from which the above theorem implies
johnston-linear-matrix-algebra-F03068	336	0.996	$\mathbf{p}_{\{j\}} \mathbf{q}_{\{j\}}^{\ast}$	$\mathbf{p}_j \mathbf{q}_j^*$	$\mathbf{p}_j$ is a column vector and each $\mathbf{q}_j^*$ is a row vector, the products $\mathbf{p}_j \mathbf{q}_j^*$ are rank-1
johnston-linear-matrix-algebra-F03069	337	1.000	$F_{\{3\}}=F_{\{2\}}+F_{\{1\}}=1+1=2$	$F_3 = F_2 + F_1 = 1 + 1 = 2$	$F_3 = F_2 + F_1 = 1 + 1 = 2$,
johnston-linear-matrix-algebra-F03070	337	0.999	$F_{\{4\}}=F_{\{3\}}+F_{\{2\}}=2+1=3$	$F_4 = F_3 + F_2 = 2 + 1 = 3$	$F_4 = F_3 + F_2 = 2 + 1 = 3$,

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johnston-linear-matrix-algebra F03071	337	■ 1.000	$\phi=1.61803 \ \ldots$	$\phi = 1.61803 \dots$	starts $\phi = 1.61803 \dots$
johnston-linear-matrix-algebra F03072	337	■ 1.000	$\mathbf{p}_1 \mathbf{q}_1^*, \mathbf{p}_2 \mathbf{q}_2^*, \dots, \mathbf{p}_n \mathbf{q}_n^*$ $\mathbf{q}_2^*, \ \ldots, \ \mathbf{p}_n \mathbf{q}_n^*$ $\mathbf{q}_n^*$	$\mathbf{p}_1 \mathbf{q}_1^*, \mathbf{p}_2 \mathbf{q}_2^*, \dots, \mathbf{p}_n \mathbf{q}_n^*$	$\mathbf{p}_1 \mathbf{q}_1^*, \mathbf{p}_2 \mathbf{q}_2^*, \dots, \mathbf{p}_n \mathbf{q}_n^*$ (we have not yet discussed linear combinations of ma-
johnston-linear-matrix-algebra F03073	337	■ 0.988	$0, 1, 1, 2, 3, 5, 8, 13, 21, 34, \ \ldots$	$0, 1, 1, 2, 3, 5, 8, 13, 21, 34, \dots$	The first few terms of this sequence are $0, 1, 1, 2, 3, 5, 8, 13, 21, 34, \dots$. Find
johnston-linear-matrix-algebra F03074	337	■ 1.000	$F_{\{n\}}$	F_n	an explicit formula for F_n that does not depend on the previous terms in
johnston-linear-matrix-algebra F03075	337	■ 1.000	$F_{\{n+1\}}=F_{\{n\}}+F_{\{n-1\}}$	$F_{n+1} = F_n + F_{n-1}$	terms of matrix multiplication—the fact that $F_{n+1} = F_n + F_{n-1}$ is equivalent
johnston-linear-matrix-algebra F03076	337	■ 1.000	$\lambda^2-\lambda-1$	$\lambda^2 - \lambda - 1$	recall from Example 3.4.5 that its characteristic polynomial is $\lambda^2 - \lambda - 1$,
johnston-linear-matrix-algebra F03077	337	■ 1.000	$\phi=(1+\sqrt{5}) \ / \ 2$	$\phi = (1 + \sqrt{5})/2$	For convenience, we define $\phi = (1 + \sqrt{5})/2$, so that the eigenvalues of A
johnston-linear-matrix-algebra F03078	337	■ 0.999	$1-\phi$	$1 - \phi$	are ϕ and $1 - \phi$.
johnston-linear-matrix-algebra F03079	337	■ 1.000	$\lambda=\phi$	$\lambda = \phi$	Next, we find the corresponding eigenvectors. For $\lambda = \phi$, we find a
johnston-linear-matrix-algebra F03080	338	■ 1.000	$1-(1-\phi)(-\phi)=$	$1 - (1 - \phi)(-\phi) =$	$1 - (1 - \phi)(-\phi) =$
johnston-linear-matrix-algebra F03081	338	■ 0.628	$\phi^2+\phi+1=0$	$\phi^2 + \phi + 1 = 0$	$-\phi^2 + \phi + 1 = 0$, since
johnston-linear-matrix-algebra F03082	338	■ 1.000	$A^{\{n\}}=P \ D^{\{n\}} \ P^{\{-1\}}$	$A^n = PD^n P^{-1}$	then $A^n = PD^n P^{-1}$.
johnston-linear-matrix-algebra F03083	338	■ 0.935	$\left(F_{\{n\}}\right)$	(F_n)	an integer (F_n) for all
johnston-linear-matrix-algebra F03084	338	■ 0.830	$\operatorname{null}(A-\phi \ I)$	$\operatorname{null}(A - \phi I)$	basis of $\operatorname{null}(A - \phi I)$ as follows:
johnston-linear-matrix-algebra F03085	338	■ 1.000	$\{(\phi, \ 1)\}$	$\{(\phi, 1)\}$	One basis of this null space is thus $\{(\phi, 1)\}$. On the other hand, for $\lambda =$
johnston-linear-matrix-algebra F03086	338	■ 1.000	$\lambda=$	$\lambda =$	One basis of this null space is thus $\{(\phi, 1)\}$. On the other hand, for $\lambda =$
johnston-linear-matrix-algebra F03087	338	■ 0.950	$\operatorname{null}(A-(1-\phi) \ I)$	$\operatorname{null}(A - (1 - \phi)I)$	$1 - \phi$, we find a basis of $\operatorname{null}(A - (1 - \phi)I)$:
johnston-linear-matrix-algebra F03088	338	■ 0.997	$\{(1-\phi, \ 1)\}$	$\{(1 - \phi, 1)\}$	It follows that one basis of this null space is $\{(1 - \phi, 1)\}$.
johnston-linear-matrix-algebra F03089	339	■ 0.986	$\lambda_3=\lambda_4=0$	$\lambda_3 = \lambda_4 = 0$	$\lambda_3 = \lambda_4 = 0$ might
johnston-linear-matrix-algebra F03090	339	■ 1.000	$\left[P \ D^{\{k\}} \ P^{\{-1\}}\right]_{1,4}$	$[PD^k P^{-1}]_{1,4}$	$[PD^k P^{-1}]_{1,4}$ is the dot
johnston-linear-matrix-algebra F03091	339	■ 1.000	$D^{\{k\}} \ P^{\{-1\}}$	$D^k P^{-1}$	row of $D^k P^{-1}$.
johnston-linear-matrix-algebra F03092	339	■ 0.978	$A \ \text{in} \ \mathcal{M}_{\{5\}}$	$A \in \mathcal{M}_5$	$A \in \mathcal{M}_5$ of this graph—the matrix whose (i, j) -entry equals 1 if there is
johnston-linear-matrix-algebra F03093	339	■ 1.000	$\left[A^{\{k\}}\right]_{1,4}$	$[A^k]_{1,4}$	equal to $[A^k]_{1,4}$.
johnston-linear-matrix-algebra F03094	339	■ 1.000	$A^{\{k\}}=$	$A^k =$	so we now have a complete diagonalization of A . It follows that $A^k =$
johnston-linear-matrix-algebra F03095	339	■ 1.000	$P \ D^{\{k\}} \ P^{\{-1\}}$	$PD^k P^{-1}$	$PD^k P^{-1}$ for all integers $k \geq 0$, so the formula that we desire is simply the
johnston-linear-matrix-algebra F03096	339	■ 1.000	$\left[A^{\{k\}}\right]_{1,4}=\left[P \ D^{\{k\}} \ P^{\{-1\}}\right]_{1,4}$	$[A^k]_{1,4} = [PD^k P^{-1}]_{1,4}$	save some effort by recalling that $[A^k]_{1,4} = [PD^k P^{-1}]_{1,4}$ is the dot product
johnston-linear-matrix-algebra F03097	340	■ 1.000	$r \ \text{in} \ \mathbb{C}$	$r \in \mathbb{C}$	if $r \in \mathbb{C}$ too.

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johnston-linear-matrix-algebra-F03098	340	1.000	$P^{-1}D^k$	PD^k	of the 1st row of PD^k with the 4th column of P^{-1} . That is, the number of
johnston-linear-matrix-algebra-F03099	340	0.991	D^r	D^r	where D^r is obtained by raising each of its diagonal entries to the r -th
johnston-linear-matrix-algebra-F03100	340	1.000	A^r	A^r	we know that A^r does not change if we choose a different diagonalization
johnston-linear-matrix-algebra-F03101	340	1.000	$(-1)^{1/2} = i$	$(-1)^{1/2} = i$	an exponentiation will be complex in general, since we have $(-1)^{1/2} = i$,
johnston-linear-matrix-algebra-F03102	340	1.000	$A = \begin{bmatrix} 0 & 4 \\ -1 & 5 \end{bmatrix}$	$A = \begin{bmatrix} 0 & 4 \\ -1 & 5 \end{bmatrix}$	Use Definition 3.4.2 to compute the following powers of $A = \begin{bmatrix} 0 & 4 \\ -1 & 5 \end{bmatrix}$.
johnston-linear-matrix-algebra-F03103	340	1.000	$A^{1/2}$	$A^{1/2}$	a) $A^{1/2}$
johnston-linear-matrix-algebra-F03104	340	1.000	A^π	A^π	c) A^π
johnston-linear-matrix-algebra-F03105	340	0.710	$(4, 1)$	$(4, 1)$	vectors $(4, 1)$ and $(1, 1)$, respectively. We can thus diagonalize A as
johnston-linear-matrix-algebra-F03106	341	1.000	$r = 1/2$	$r = 1/2$	If we plug $r = 1/2$
johnston-linear-matrix-algebra-F03107	341	1.000	$r = -1$	$r = -1$	instead find an explicit formula for A^r and then plug $r = -1$ into
johnston-linear-matrix-algebra-F03108	341	1.000	$r = \pi$	$r = \pi$	c) Plugging $r = \pi$ into the formula that we found in part (b) gives
johnston-linear-matrix-algebra-F03109	342	1.000	λ_j^{-1}	λ_j^{-1}	only if each λ_j^{-1}
johnston-linear-matrix-algebra-F03110	342	1.000	$B^2 = A$	$B^2 = A$	a matrix $B \in \mathcal{M}_n$ such that $B^2 = A$. Well, if $A = PDP^{-1}$ is a diagonalization of
johnston-linear-matrix-algebra-F03111	342	1.000	$B^k = A$	$B^k = A$	$A \in \mathcal{M}_n$ if $B^k = A$, and the same argument as above shows that $B = A^{1/k}$ is one
johnston-linear-matrix-algebra-F03112	342	1.000	$B = A^{1/k}$	$B = A^{1/k}$	$A \in \mathcal{M}_n$ if $B^k = A$, and the same argument as above shows that $B = A^{1/k}$ is one
johnston-linear-matrix-algebra-F03113	342	1.000	$A = \begin{bmatrix} -2 & 6 \\ 3 & -5 \end{bmatrix}$	$A = \begin{bmatrix} -2 & 6 \\ 3 & -5 \end{bmatrix}$	Find a cube root of the matrix $A = \begin{bmatrix} -2 & 6 \\ 3 & -5 \end{bmatrix}$.
johnston-linear-matrix-algebra-F03114	342	0.719	$(1, -1)$	$(1, -1)$	$(2, 1)$ and $(1, -1)$, respectively. We can thus diagonalize A as follows:
johnston-linear-matrix-algebra-F03115	343	1.000	$A^{1/2}, B$	$A^{1/2}, B$	like $A^{1/2}$, B really is $\tilde{a}$
johnston-linear-matrix-algebra-F03116	343	1.000	$B^3 = A$	$B^3 = A$	$B^3 = A$:
johnston-linear-matrix-algebra-F03117	343	1.000	$-A^{1/2}$	$-A^{1/2}$	is one, and $-A^{1/2}$ is another one. To find two more square roots of A ,
johnston-linear-matrix-algebra-F03118	343	0.978	$-B$	$-B$	as well as $-B$.
johnston-linear-matrix-algebra-F03119	344	0.531	$F_{\mathbf{u}}^2 = I$	$F_{\mathbf{u}}^2 = I$	unit vector $\mathbf{u}$, then $F_{\mathbf{u}}^2 = I$:
johnston-linear-matrix-algebra-F03120	344	0.990	$[F_{\mathbf{u}}] = 2\mathbf{u}\mathbf{u}^T - I$	$[F_{\mathbf{u}}] = 2\mathbf{u}\mathbf{u}^T - I$	Since the standard matrix of $F_{\mathbf{u}}$ is $[F_{\mathbf{u}}] = 2\mathbf{u}\mathbf{u}^T - I$, every matrix of the form
johnston-linear-matrix-algebra-F03121	344	0.446	$2\mathbf{u}\mathbf{u}^T - I$	$2\mathbf{u}\mathbf{u}^T - I$	$2\mathbf{u}\mathbf{u}^T - I$ is a square root of I (i.e., $(2\mathbf{u}\mathbf{u}^T - I)^2 = I$ for all unit vectors $\mathbf{u}$).
johnston-linear-matrix-algebra-F03122	344	0.446	$(2\mathbf{u}\mathbf{u}^T - I)^2 = I$	$(2\mathbf{u}\mathbf{u}^T - I)^2 = I$	$2\mathbf{u}\mathbf{u}^T - I$ is a square root of I (i.e., $(2\mathbf{u}\mathbf{u}^T - I)^2 = I$ for all unit vectors $\mathbf{u}$).

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johnston-linear-matrix-algebra-F03123	344	1.000	$k > 1$	$k > 1$	Suppose $A \in \mathcal{M}_n(\mathbb{C})$ is diagonalizable and let $k > 1$ be an integer.
johnston-linear-matrix-algebra-F03124	344	1.000	k^n	k^n	a) If A 's eigenvalues are distinct, it has exactly k^n different k -th roots.
johnston-linear-matrix-algebra-F03125	344	1.000	$A^{r+s} = A^r A^s$	$A^{r+s} = A^r A^s$	matrices with non-negative real eigenvalues. That is, $A^{r+s} = A^r A^s$ and $(A^r)^s =$
johnston-linear-matrix-algebra-F03126	344	1.000	$\left(A^r\right)^s =$	$(A^r)^s =$	matrices with non-negative real eigenvalues. That is, $A^{r+s} = A^r A^s$ and $(A^r)^s =$
johnston-linear-matrix-algebra-F03127	344	1.000	A^{rs}	A^{rs}	A^{rs} for all real numbers r and s , as long as A has non-negative real eigenvalues
johnston-linear-matrix-algebra-F03128	345	1.000	$p(D)$	$p(D)$	where $p(D)$ is obtained by applying p to each diagonal entry of D .
johnston-linear-matrix-algebra-F03129	345	1.000	$p(x) = c_k x^k + \cdots + c_2 x^2 + c_1 x + c_0$	$p(x) = c_k x^k + \cdots + c_2 x^2 + c_1 x + c_0$	<i>Proof.</i> We just write $p(x) = c_k x^k + \cdots + c_2 x^2 + c_1 x + c_0$, and then make use of
johnston-linear-matrix-algebra-F03130	345	1.000	$\left(P D P^{-1}\right)^r = P D^r P^{-1}$	$(P D P^{-1})^r = P D^r P^{-1}$	the fact that $(P D P^{-1})^r = P D^r P^{-1}$ for all r :
johnston-linear-matrix-algebra-F03131	345	1.000	$p(\lambda)$	$p(\lambda)$	$\mathcal{M}_n$ has an eigenvalue λ with corresponding eigenvector $\mathbf{v}$, then $p(\lambda)$ is an
johnston-linear-matrix-algebra-F03132	345	0.979	$p(A)$	$p(A)$	eigenvalue of $p(A)$ corresponding to the same eigenvector $\mathbf{v}$. In fact, this is true
johnston-linear-matrix-algebra-F03133	345	1.000	$p(x) = x^3 - 3x^2 + 2x - 4$	$p(x) = x^3 - 3x^2 + 2x - 4$	$p(x) = x^3 - 3x^2 + 2x - 4$ and $A = \begin{bmatrix} 0 & 4 \\ -1 & 5 \end{bmatrix}$.
johnston-linear-matrix-algebra-F03134	346	0.991	$f(A)$	$f(A)$	Don't worry— $f(A)$ is
johnston-linear-matrix-algebra-F03135	346	1.000	$f(D)$	$f(D)$	where $f(D)$ is obtained by applying f to each of the diagonal entries of D .
johnston-linear-matrix-algebra-F03136	346	1.000	e^A	e^A	a) e^A
johnston-linear-matrix-algebra-F03137	346	1.000	$\sin(A)$	$\sin(A)$	b) $\sin(A)$
johnston-linear-matrix-algebra-F03138	347	1.000	$\sin(x)$	$\sin(x)$	function $\sin(x)$ instead of e^x to the diagonal entries of D , gives
johnston-linear-matrix-algebra-F03139	347	1.000	e^x	e^x	function $\sin(x)$ instead of e^x to the diagonal entries of D , gives
johnston-linear-matrix-algebra-F03140	347	0.996	$f(\lambda)$	$f(\lambda)$	$f(\lambda)$ must be an eigenvalue of $f(A)$ corresponding to the same eigenvector $\mathbf{v}$.
johnston-linear-matrix-algebra-F03141	347	1.000	(a, b)	(a, b)	variable—on some open interval (a, b) :
johnston-linear-matrix-algebra-F03142	348	1.000	$f(A) = e^A$	$f(A) = e^A$	and polynomials) is the matrix exponential $f(A) = e^A$. The following theorem
johnston-linear-matrix-algebra-F03143	348	1.000	$e^0 = 1$	$e^0 = 1$	analogous to the facts that, for real numbers, $e^0 = 1$ and $1/e^x = e^{-x}$.
johnston-linear-matrix-algebra-F03144	348	1.000	$1/e^x = e^{-x}$	$1/e^x = e^{-x}$	analogous to the facts that, for real numbers, $e^0 = 1$ and $1/e^x = e^{-x}$.
johnston-linear-matrix-algebra-F03145	348	1.000	$e^0 = I$	$e^0 = I$	a) $e^0 = I$, and
johnston-linear-matrix-algebra-F03146	348	1.000	$\left(e^A\right)^{-1} = e^{-A}$	$(e^A)^{-1} = e^{-A}$	b) e^A is invertible and $(e^A)^{-1} = e^{-A}$.
johnston-linear-matrix-algebra-F03147	348	1.000	e^{-A}	e^{-A}	Similarly, for part (b) we just multiply e^A by e^{-A} and work with their diagonal
johnston-linear-matrix-algebra-F03148	349	1.000	$e^{A+B} = e^A e^B$	$e^{A+B} = e^A e^B$	Note, however, that it is typically not the case that $e^{A+B} = e^A e^B$ (even

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johnston-linear-matrix-algebra-F03149	349	1.000	$e^{x+y}=e^x e^y$	$e^{x+y} = e^x e^y$	though $e^{x+y} = e^x e^y$ for all real numbers x and y). The reason that this property
johnston-linear-matrix-algebra-F03150	349	1.000	$\operatorname{det}(e^A)=e^{\operatorname{tr}(A)}$	$\det(e^A) = e^{\operatorname{tr}(A)}$	If $A \in \mathcal{M}_n$ is diagonalizable then $\det(e^A) = e^{\operatorname{tr}(A)}$.
johnston-linear-matrix-algebra-F03151	349	1.000	$e^{\lambda_1}, e^{\lambda_2}, \dots, e^{\lambda_n}$	$e^{\lambda_1}, e^{\lambda_2}, \dots, e^{\lambda_n}$	that the eigenvalues of e^A are $e^{\lambda_1}, e^{\lambda_2}, \dots, e^{\lambda_n}$:
johnston-linear-matrix-algebra-F03152	350	0.214	$(\mathbf{c}) \left[\begin{array}{l} 1 \\ 1 \\ 0 \end{array} \right]$	$(\mathbf{c}) \left[\begin{array}{cc} 1 & \\ 1 & 1 \\ 0 & \end{array} \right]$	$(\mathbf{a}) \left[\begin{array}{cc} 1 & 1 \\ 1 & 1 \end{array} \right]$ $(\mathbf{c}) \left[\begin{array}{cc} 1 & 1 \\ 0 & 1 \end{array} \right]$
johnston-linear-matrix-algebra-F03153	350	0.643	$\left[\begin{array}{cc} 2 & 1 \\ -1 & 2 \end{array} \right] \text{ (d) } \left[\begin{array}{cc} -2 & 6 \\ -3 & 7 \end{array} \right]$	$\left[\begin{array}{cc} 2 & 1 \\ -1 & 2 \end{array} \right]$ (d) $\left[\begin{array}{cc} -2 & 6 \\ -3 & 7 \end{array} \right]$	(b) $\left[\begin{array}{cc} 2 & 1 \\ -1 & 2 \end{array} \right]$ (d) $\left[\begin{array}{cc} -2 & 6 \\ -3 & 7 \end{array} \right]$
johnston-linear-matrix-algebra-F03154	350	0.395	$\left[\begin{array}{ccc} 2 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{array} \right]$	$\left[\begin{array}{ccc} 2 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{array} \right]$	$(\mathbf{e}) \left[\begin{array}{ccc} 2 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{array} \right]$
johnston-linear-matrix-algebra-F03155	350	1.000	$\left[\begin{array}{ccc} 5 & 0 & -3 \\ -3 & 2 & 3 \\ 6 & 0 & -4 \end{array} \right]$	$\left[\begin{array}{ccc} 5 & 0 & -3 \\ -3 & 2 & 3 \\ 6 & 0 & -4 \end{array} \right]$	(f) $\left[\begin{array}{ccc} 5 & 0 & -3 \\ -3 & 2 & 3 \\ 6 & 0 & -4 \end{array} \right]$
johnston-linear-matrix-algebra-F03156	350	0.705	$\left[\begin{array}{ccc} 3 & 0 & 1 \\ 0 & -1 & 2 \\ 0 & 0 & 2 \end{array} \right]$	$\left[\begin{array}{ccc} 3 & 0 & 1 \\ 0 & -1 & 2 \\ 0 & 0 & 2 \end{array} \right]$	$(\mathbf{g}) \left[\begin{array}{ccc} 3 & 0 & 1 \\ 0 & -1 & 2 \\ 0 & 0 & 2 \end{array} \right]$
johnston-linear-matrix-algebra-F03157	350	1.000	$\left[\begin{array}{ccc} -3 & 4 & 5 \\ -3 & 5 & 3 \\ 1 & -2 & 1 \end{array} \right]$	$\left[\begin{array}{ccc} -3 & 4 & 5 \\ -3 & 5 & 3 \\ 1 & -2 & 1 \end{array} \right]$	(h) $\left[\begin{array}{ccc} -3 & 4 & 5 \\ -3 & 5 & 3 \\ 1 & -2 & 1 \end{array} \right]$
johnston-linear-matrix-algebra-F03158	350	0.262	$\left[\begin{array}{cc} 1 & 1 \\ 2 & 1 \end{array} \right] \left[\begin{array}{cc} 2 & 1 \\ -1 & 2 \end{array} \right]$	$(\mathbf{a}) \left[\begin{array}{cc} 1 & 1 \\ 1 & 1 \end{array} \right] \left[\begin{array}{cc} 2 & 1 \\ -1 & 2 \end{array} \right]$ $(\mathbf{c}) \left[\begin{array}{cc} 1 & 1 \\ 1 & 1 \end{array} \right] \left[\begin{array}{cc} 2 & 1 \\ -1 & 2 \end{array} \right]$	$(\mathbf{a}) \left[\begin{array}{cc} 1 & 1 \\ 1 & 1 \end{array} \right]$ $(\mathbf{c}) \left[\begin{array}{cc} 2 & 1 \\ -1 & 2 \end{array} \right]$
johnston-linear-matrix-algebra-F03159	350	1.000	$\left[\begin{array}{cc} 2+2i & 1-i \\ 2 & 1-i \end{array} \right]$	$\left[\begin{array}{cc} 2+2i & 1-i \\ 2 & 1-i \end{array} \right]$	(b) $\left[\begin{array}{cc} 2+2i & 1-i \\ 2 & 1-i \end{array} \right]$
johnston-linear-matrix-algebra-F03160	350	1.000	$\left[\begin{array}{cc} 1 & 1 \\ -9 & -5 \end{array} \right]$	$\left[\begin{array}{cc} 1 & 1 \\ -9 & -5 \end{array} \right]$	(d) $\left[\begin{array}{cc} 1 & 1 \\ -9 & -5 \end{array} \right]$
johnston-linear-matrix-algebra-F03161	350	0.741	$\left[\begin{array}{ccc} 2 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{array} \right]$	$\left[\begin{array}{ccc} 2 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{array} \right]$	$(\mathbf{e}) \left[\begin{array}{ccc} 2 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{array} \right]$
johnston-linear-matrix-algebra-F03162	350	1.000	$\left[\begin{array}{ccc} -2 & -3 & -2 \\ 4 & 6 & 4 \\ -4 & -5 & -4 \end{array} \right]$	$\left[\begin{array}{ccc} -2 & -3 & -2 \\ 4 & 6 & 4 \\ -4 & -5 & -4 \end{array} \right]$	(f) $\left[\begin{array}{ccc} -2 & -3 & -2 \\ 4 & 6 & 4 \\ -4 & -5 & -4 \end{array} \right]$

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johnston-linear-matrix-algebra-F03163	350	0.932	$\left[\begin{array}{ccc}1+i & -1 & 0 \\ 1 & -1+i & 0 \\ 2 & -2 & i\end{array}\right]$	$\begin{bmatrix} 1+i & -1 & 0 \\ 1 & -1+i & 0 \\ 2 & -2 & i \end{bmatrix}$	$\ast(\mathbf{g}) \begin{bmatrix} 1+i & -1 & 0 \\ 1 & -1+i & 0 \\ 2 & -2 & i \end{bmatrix}$
johnston-linear-matrix-algebra-F03164	350	1.000	$\left[\begin{array}{ccc}2 & 4 & -2 \\ 1 & 3 & -1 \\ 3 & -1 & 1\end{array}\right]$	$\begin{bmatrix} 2 & 4 & -2 \\ 1 & 3 & -1 \\ 3 & -1 & 1 \end{bmatrix}$	$(\mathbf{h}) \begin{bmatrix} 2 & 4 & -2 \\ 1 & 3 & -1 \\ 3 & -1 & 1 \end{bmatrix}$
johnston-linear-matrix-algebra-F03165	350	1.000	$\sqrt{A}$	$\sqrt{A}$	$(\mathbf{c}) \sqrt{A}$, Exercise 3.4.1(d).
johnston-linear-matrix-algebra-F03166	351	1.000	L_n	L_n	Find an explicit formula for L_n . [Hint: This sequence satis-
johnston-linear-matrix-algebra-F03167	351	1.000	P_n	P_n	Find an explicit formula for P_n .
johnston-linear-matrix-algebra-F03168	351	1.000	$F_{n+1} F_{n-1} - F_n^2 = (-1)^n$	$F_{n+1}F_{n-1} - F_n^2 = (-1)^n$	$(\mathbf{b})$ Show that $F_{n+1}F_{n-1} - F_n^2 = (-1)^n$ for all $n \geq 1$.
johnston-linear-matrix-algebra-F03169	351	1.000	$F_{m+n} = F_{m+1}F_n + F_mF_{n-1}$	$F_{m+n} = F_{m+1}F_n + F_mF_{n-1}$	$(\mathbf{c})$ Show that $F_{m+n} = F_{m+1}F_n + F_mF_{n-1}$ for all $m, n \geq 1$.
johnston-linear-matrix-algebra-F03170	351	1.000	$m, n \geq 1$	$m, n \geq 1$	$(\mathbf{c})$ Show that $F_{m+n} = F_{m+1}F_n + F_mF_{n-1}$ for all $m, n \geq 1$.
johnston-linear-matrix-algebra-F03171	351	0.996	$A^m A^n = A^{m+n}$	$A^m A^n = A^{m+n}$	[Hint: $A^m A^n = A^{m+n}$ for all square matrices A .]
johnston-linear-matrix-algebra-F03172	351	1.000	$A = \mathbf{x} \mathbf{y}^T$	$A = \mathbf{xy}^T$	3.4.13 Suppose $A = \mathbf{xy}^T$ for some non-zero column vec-
johnston-linear-matrix-algebra-F03173	351	0.954	$\mathbf{x} \cdot \mathbf{y}$	$\mathbf{x} \cdot \mathbf{y}$	equal to $\mathbf{x} \cdot \mathbf{y}$, with corresponding eigenvector $\mathbf{x}$.
johnston-linear-matrix-algebra-F03174	351	1.000	$\mathbf{x} \cdot \mathbf{y} \neq 0$	$\mathbf{x} \cdot \mathbf{y} \neq 0$	$(\mathbf{b})$ Show that A is diagonalizable if and only if $\mathbf{x} \cdot \mathbf{y} \neq 0$.
johnston-linear-matrix-algebra-F03175	351	1.000	$A \neq B$	$A \neq B$	able then it might be the case that $A \neq B$.
johnston-linear-matrix-algebra-F03176	351	1.000	$P_1 D_1 P_1^{-1} = P_2 D_2 P_2^{-1}$	$P_1 D_1 P_1^{-1} = P_2 D_2 P_2^{-1}$	inition 3.4.2. That is, show that if $P_1 D_1 P_1^{-1} = P_2 D_2 P_2^{-1}$ are
johnston-linear-matrix-algebra-F03177	351	1.000	$P_1 D_1^r P_1^{-1} = P_2 D_2^r P_2^{-1}$	$P_1 D_1^r P_1^{-1} = P_2 D_2^r P_2^{-1}$	two diagonalizations of a matrix then $P_1 D_1^r P_1^{-1} = P_2 D_2^r P_2^{-1}$
johnston-linear-matrix-algebra-F03178	351	1.000	$P_1 f(D_1) P_1^{-1} =$	$P_1 f(D_1) P_1^{-1} =$	are two diagonalizations of a matrix then $P_1 f(D_1) P_1^{-1} =$
johnston-linear-matrix-algebra-F03179	351	1.000	$P_2 f(D_2) P_2^{-1}$	$P_2 f(D_2) P_2^{-1}$	$P_2 f(D_2) P_2^{-1}$ for all functions f .
johnston-linear-matrix-algebra-F03180	351	1.000	λ^k	λ^k	tor of A^k with eigenvalue $\lambda^{\tilde{k}}$.
johnston-linear-matrix-algebra-F03181	351	1.000	$(A^r)^s = A^{rs}$	$(A^r)^s = A^{rs}$	$(\mathbf{b}) (A^r)^s = A^{rs}$.
johnston-linear-matrix-algebra-F03182	351	0.999	$e^{\left(A^T\right)} = \left(e^A\right)^T$	$e^{(A^T)} = (e^A)^T$	$e^{(A^T)} = (e^A)^T$.
johnston-linear-matrix-algebra-F03183	351	1.000	$e^{A+B} =$	$e^{A+B} =$	[Be careful: remember that it is <i>not</i> the case that $e^{A+B} =$
johnston-linear-matrix-algebra-F03184	351	1.000	$e^A e^B$	$e^A e^B$	$e^A e^B$ in general.]
johnston-linear-matrix-algebra-F03185	351	1.000	$\sin^2(A) + \cos^2(A) = I$	$\sin^2(A) + \cos^2(A) = I$	$\sin^2(A) + \cos^2(A) = I$.
johnston-linear-matrix-algebra-F03186	351	1.000	$p(\lambda) = \lambda^n + a_{n-1} \lambda^{n-1} + \cdots +$	$p(\lambda) = \lambda^n + a_{n-1} \lambda^{n-1} + \cdots +$	matrix of the polynomial $p(\lambda) = \lambda^n + a_{n-1} \lambda^{n-1} + \cdots +$
johnston-linear-matrix-algebra-F03187	351	1.000	$a_1 \lambda + a_0$	$a_1 \lambda + a_0$	$a_1 \lambda + a_0$ is the matrix
johnston-linear-matrix-algebra-F03188	351	1.000	$C = V D V^{-1}$	$C = V D V^{-1}$	That is, show that $C = V D V^{-1}$, where D is the diag-
johnston-linear-matrix-algebra-F03189	352	1.000	$A = P D P^{-1} \in \mathcal{M}_2$	$A = P D P^{-1} \in \mathcal{M}_2$	Figure 3.25: If $A = P D P^{-1} \in \mathcal{M}_2$ has eigenvectors $\mathbf{v}_1$ and $\mathbf{v}_2$ with corresponding

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johnston-linear-matrix-algebra- F03190	352	1.000	λ_2	λ_2	eigenvalues λ_1 and λ_2 , respectively, then its diagonalization skews the basis $\{v_1, v_2\}$
johnston-linear-matrix-algebra- F03191	353	1.000	N	N	does not hold for non-diagonalizable matrices, since N has no non-zero
johnston-linear-matrix-algebra- F03192	354	1.000	$A, B \in \mathcal{M}_n(\mathbb{C})$	$A, B \in \mathcal{M}_n(\mathbb{C})$	Suppose $A, B \in \mathcal{M}_n(\mathbb{C})$ are diagonalizable. Then A and B are similar if
johnston-linear-matrix-algebra- F03193	354	1.000	$B = Q D Q^{-1}$	$B = Q D Q^{-1}$	Multiplying the equation $B = Q D Q^{-1}$ on the left by Q^{-1} and on the right by Q
johnston-linear-matrix-algebra- F03194	354	1.000	Q^{-1}	Q^{-1}	Multiplying the equation $B = Q D Q^{-1}$ on the left by Q^{-1} and on the right by Q
johnston-linear-matrix-algebra- F03195	354	1.000	$Q^{-1} B Q = D$	$Q^{-1} B Q = D$	shows that $Q^{-1} B Q = D$. Substituting this into the equation $A = P D P^{-1}$ shows
johnston-linear-matrix-algebra- F03196	354	0.983	$(P Q^{-1})^{-1} = Q P^{-1}$	$(P Q^{-1})^{-1} = Q P^{-1}$	$(P Q^{-1})^{-1} = Q P^{-1}$
johnston-linear-matrix-algebra- F03197	354	0.999	$B = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$	$B = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$	*(a) $A = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$
johnston-linear-matrix-algebra- F03198	354	1.000	$B = \begin{bmatrix} 2 & 0 \\ 3 & 0 \end{bmatrix}$	$B = \begin{bmatrix} 2 & 0 \\ 3 & 0 \end{bmatrix}$	(b) $A = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & 0 \\ 3 & 0 \end{bmatrix}$
johnston-linear-matrix-algebra- F03199	354	0.996	$A = \begin{bmatrix} 3 & 1 \\ 2 & 3 \end{bmatrix}$	$A = \begin{bmatrix} 3 & 1 \\ 2 & 3 \end{bmatrix}$	*(c) $A = \begin{bmatrix} 3 & 1 \\ 2 & 3 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & -1 \\ 3 & 5 \end{bmatrix}$
johnston-linear-matrix-algebra- F03200	354	0.996	$B = \begin{bmatrix} 2 & -1 \\ 3 & 5 \end{bmatrix}$	$B = \begin{bmatrix} 2 & -1 \\ 3 & 5 \end{bmatrix}$	*(c) $A = \begin{bmatrix} 3 & 1 \\ 2 & 3 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & -1 \\ 3 & 5 \end{bmatrix}$
johnston-linear-matrix-algebra- F03201	354	1.000	$A = \begin{bmatrix} 2 & 1 \\ -3 & 3 \end{bmatrix}$	$A = \begin{bmatrix} 2 & 1 \\ -3 & 3 \end{bmatrix}$	(d) $A = \begin{bmatrix} 2 & 1 \\ -3 & 3 \end{bmatrix}$ and $B = \begin{bmatrix} 4 & 2 \\ -1 & 1 \end{bmatrix}$
johnston-linear-matrix-algebra- F03202	354	1.000	$B = \begin{bmatrix} 4 & 2 \\ -1 & 1 \end{bmatrix}$	$B = \begin{bmatrix} 4 & 2 \\ -1 & 1 \end{bmatrix}$	(d) $A = \begin{bmatrix} 2 & 1 \\ -3 & 3 \end{bmatrix}$ and $B = \begin{bmatrix} 4 & 2 \\ -1 & 1 \end{bmatrix}$
johnston-linear-matrix-algebra- F03203	354	0.980	$A = \begin{bmatrix} 3 & 3 \\ 0 & 2 \end{bmatrix}$	$A = \begin{bmatrix} 3 & 3 \\ 0 & 2 \end{bmatrix}$	*(e) $A = \begin{bmatrix} 3 & 3 \\ 0 & 2 \end{bmatrix}$ and $B = \begin{bmatrix} 4 & 2 \\ -1 & 1 \end{bmatrix}$
johnston-linear-matrix-algebra- F03204	354	0.999	$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$	$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$	(f) $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 2 & 3 \\ 8 & 9 & 4 \\ 7 & 6 & 5 \end{bmatrix}$
johnston-linear-matrix-algebra- F03205	354	0.999	$B = \begin{bmatrix} 1 & 2 & 3 \\ 8 & 9 & 4 \\ 7 & 6 & 5 \end{bmatrix}$	$B = \begin{bmatrix} 1 & 2 & 3 \\ 8 & 9 & 4 \\ 7 & 6 & 5 \end{bmatrix}$	(f) $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 2 & 3 \\ 8 & 9 & 4 \\ 7 & 6 & 5 \end{bmatrix}$
johnston-linear-matrix-algebra- F03206	354	0.962	$A = \begin{bmatrix} 3 & 1 & -1 \\ 0 & 2 & 1 \\ -1 & 0 & 2 \end{bmatrix}$	$A = \begin{bmatrix} 3 & 1 & -1 \\ 0 & 2 & 1 \\ -1 & 0 & 2 \end{bmatrix}$	*(g) $A = \begin{bmatrix} 3 & 1 & -1 \\ 0 & 2 & 1 \\ -1 & 0 & 2 \end{bmatrix}$ and $B = \begin{bmatrix} 3 & 1 & 2 \\ -2 & 3 & 0 \\ 1 & -1 & 1 \end{bmatrix}$
johnston-linear-matrix-algebra- F03207	354	0.962	$B = \begin{bmatrix} 3 & 1 & 2 \\ -2 & 3 & 0 \\ 1 & -1 & 1 \end{bmatrix}$	$B = \begin{bmatrix} 3 & 1 & 2 \\ -2 & 3 & 0 \\ 1 & -1 & 1 \end{bmatrix}$	*(g) $A = \begin{bmatrix} 3 & 1 & -1 \\ 0 & 2 & 1 \\ -1 & 0 & 2 \end{bmatrix}$ and $B = \begin{bmatrix} 3 & 1 & 2 \\ -2 & 3 & 0 \\ 1 & -1 & 1 \end{bmatrix}$
johnston-linear-matrix-algebra- F03208	354	0.736	$B \in \mathcal{M}_3$	$B \in \mathcal{M}_3$	(f) If $A \in \mathcal{M}_3$ and $B \in \mathcal{M}_3$ each have eigenvalues 1, 2,

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johnston-linear-matrix-algebra-F03209	355	1.000	$A = P B$	$AP = PB$	[Hint: Rewrite $A = PBP^{-1}$ as $AP = PB$ and solve a
johnston-linear-matrix-algebra-F03210	355	1.000	$A^2 = N$	$A^2 = N$	not exist a matrix $A \in \mathcal{M}_2(\mathbb{C})$ such that $A^2 = N$.
johnston-linear-matrix-algebra-F03211	355	1.000	$c_{i,j} = (-1)^{i+j} m_{i,j}$	$c_{i,j} = (-1)^{i+j} m_{i,j}$	matrix $A \in \mathcal{M}_n$ is the quantity $c_{i,j} = (-1)^{i+j} m_{i,j}$, where $m_{i,j}$ is the determinant
johnston-linear-matrix-algebra-F03212	355	0.993	$\operatorname{cof}(A)$	$\operatorname{cof}(A)$	The cofactor matrix of $A \in \mathcal{M}_n$, denoted by $\operatorname{cof}(A)$, is the matrix whose
johnston-linear-matrix-algebra-F03213	355	1.000	$\left[\begin{array}{cc} 2 & 3 \\ -1 & 4 \end{array} \right]$	$\begin{bmatrix} 2 & 3 \\ -1 & 4 \end{bmatrix}$	a) $\begin{bmatrix} 2 & 3 \\ -1 & 4 \end{bmatrix}$
johnston-linear-matrix-algebra-F03214	356	1.000	$\operatorname{cof}(A)^T$	$\operatorname{cof}(A)^T$	The matrix $\operatorname{cof}(A)^T$ is
johnston-linear-matrix-algebra-F03215	356	0.998	$\operatorname{adj}(A)$	$\operatorname{adj}(A)$	$\operatorname{adj}(A)$.
johnston-linear-matrix-algebra-F03216	356	0.980	$A \operatorname{cof}(A)^T$	$A \operatorname{cof}(A)^T$	the product $A \operatorname{cof}(A)^T$ straight from the definition of matrix multiplication, we
johnston-linear-matrix-algebra-F03217	356	0.660	$\operatorname{product} A \operatorname{cof}(A)^T$	$\operatorname{product} A \operatorname{cof}(A)^T$	On the other hand, if $i \neq j$ then the (i, j) -entry of the product $A \operatorname{cof}(A)^T$ is
johnston-linear-matrix-algebra-F03218	357	1.000	$\left[A \operatorname{cof}(A)^T \right]_{i,j}$	$[A \operatorname{cof}(A)^T]_{i,j}$	which is exactly the quantity $[A \operatorname{cof}(A)^T]_{i,j}$ from Equation (3.A.1). Since B
johnston-linear-matrix-algebra-F03219	357	1.000	$0 = \operatorname{det}(B) =$	$0 = \det(B) =$	has two identical rows, it is not invertible, so we conclude that $0 = \det(B) =$
johnston-linear-matrix-algebra-F03220	357	1.000	$A \operatorname{cof}(A)^T = \operatorname{det}(A) I$	$A \operatorname{cof}(A)^T = \det(A) I$	$A \operatorname{cof}(A)^T = \det(A) I$. By dividing both sides by $\det(A)$, we see that
johnston-linear-matrix-algebra-F03221	357	1.000	$\operatorname{det}(A) = ad - bc$	$\det(A) = ad - bc$	Since $\det(A) = ad - bc$, it follows that
johnston-linear-matrix-algebra-F03222	358	1.000	$A_{\{j\}}$	A_j	Suppose $A \in \mathcal{M}_n$ is invertible and $\mathbf{b} \in \mathbb{R}^n$. Define A_j to be the matrix that
johnston-linear-matrix-algebra-F03223	359	1.000	$A^{-1} \mathbf{b}$	$A^{-1} \mathbf{b}$	the entries of $A^{-1} \mathbf{b}$, which we do via Theorem 3.A.1:
johnston-linear-matrix-algebra-F03224	359	1.000	$c_{1,j}, c_{2,j}, \dots, c_{n,j}$	$c_{1,j}, c_{2,j}, \dots, c_{n,j}$	where $c_{1,j}, c_{2,j}, \dots, c_{n,j}$ are cofactors of A and the final equality above follows
johnston-linear-matrix-algebra-F03225	359	1.000	$c_{1,j} b_1 + c_{2,j} b_2 +$	$c_{1,j} b_1 + c_{2,j} b_2 +$	from the definition of matrix multiplication. Well, the quantity $c_{1,j} b_1 + c_{2,j} b_2 +$
johnston-linear-matrix-algebra-F03226	359	1.000	$\dots + c_{n,j} b_n$	$\dots + c_{n,j} b_n$	$\dots + c_{n,j} b_n$ above on the right is exactly the cofactor expansion of A_j along its
johnston-linear-matrix-algebra-F03227	359	1.000	$c_{1,j} b_1 +$	$c_{1,j} b_1 +$	It follows from the cofactor expansion theorem (Theorem 3.2.8) that $c_{1,j} b_1 +$
johnston-linear-matrix-algebra-F03228	359	1.000	$c_{2,j} b_2 + \dots + c_{n,j} b_n = \operatorname{det}(A_j)$	$c_{2,j} b_2 + \dots + c_{n,j} b_n = \det(A_j)$	$c_{2,j} b_2 + \dots + c_{n,j} b_n = \det(A_j)$, so $x_j = \det(A_j) / \det(A)$, as desired.
johnston-linear-matrix-algebra-F03229	359	1.000	$x_j = \operatorname{det}(A_j) / \operatorname{det}(A)$	$x_j = \det(A_j) / \det(A)$	$c_{2,j} b_2 + \dots + c_{n,j} b_n = \det(A_j)$, so $x_j = \det(A_j) / \det(A)$, as desired.
johnston-linear-matrix-algebra-F03230	359	1.000	$A, A_{\{1\}}, A_{\{2\}}$	A, A_1, A_2	We start by writing out the four matrices A, A_1, A_2 , and A_3 whose
johnston-linear-matrix-algebra-F03231	359	1.000	$A_{\{3\}}$	A_3	We start by writing out the four matrices A, A_1, A_2 , and A_3 whose
johnston-linear-matrix-algebra-F03232	361	1.000	$\sigma: \{1, 2, \dots, n\} \rightarrow$	$\sigma: \{1, 2, \dots, n\} \rightarrow$	$\sigma: \{1, 2, \dots, n\} \rightarrow$
johnston-linear-matrix-algebra-F03233	361	0.999	$\{1, 2, \dots, n\}$	$\{1, 2, \dots, n\}$	$\{1, 2, \dots, n\}$ means
johnston-linear-matrix-algebra-F03234	361	1.000	σ	σ	that σ is a function
johnston-linear-matrix-algebra-F03235	361	1.000	$1, 2, \dots, n$	$1, 2, \dots, n$	that takes $1, 2, \dots, n$

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johnston-linear-matrix-algebra-F03236	361	1.000	$\sigma:\{1,2,\ldots,n\}\rightarrow\{1,2,\ldots,n\}$	$\sigma:\{1,2,\ldots,n\}\rightarrow\{1,2,\ldots,n\}$	A function $\sigma:\{1,2,\ldots,n\}\rightarrow\{1,2,\ldots,n\}$ is called a permutation if
johnston-linear-matrix-algebra-F03237	361	0.996	$\sigma(i)\neq\sigma(j)$	$\sigma(i)\neq\sigma(j)$	$\sigma(i)\neq\sigma(j)$ whenever $i\neq j$.
johnston-linear-matrix-algebra-F03238	361	1.000	$\sigma:\{1,2,3\}\rightarrow\{1,2,3\}$	$\sigma:\{1,2,3\}\rightarrow\{1,2,3\}$	For example, the function $\sigma:\{1,2,3\}\rightarrow\{1,2,3\}$ defined by $\sigma(1)=2$,
johnston-linear-matrix-algebra-F03239	361	1.000	$\sigma(1)=2$	$\sigma(1)=2$	For example, the function $\sigma:\{1,2,3\}\rightarrow\{1,2,3\}$ defined by $\sigma(1)=2$,
johnston-linear-matrix-algebra-F03240	361	0.973	$\sigma(2)=3$	$\sigma(2)=3$	$\sigma(2)=3$, and $\sigma(3)=1$ is a permutation (it just shuffles 1, 2, and 3 around),
johnston-linear-matrix-algebra-F03241	361	0.973	$\sigma(3)=1$	$\sigma(3)=1$	$\sigma(2)=3$, and $\sigma(3)=1$ is a permutation (it just shuffles 1, 2, and 3 around),
johnston-linear-matrix-algebra-F03242	361	1.000	$f:\{1,2,3\}\rightarrow\{1,2,3\}$	$f:\{1,2,3\}\rightarrow\{1,2,3\}$	but the function $f:\{1,2,3\}\rightarrow\{1,2,3\}$ defined by $f(1)=2$, $f(2)=2$, and
johnston-linear-matrix-algebra-F03243	361	1.000	$f(1)=2$, $f(2)=2$	$f(1)=2$, $f(2)=2$	but the function $f:\{1,2,3\}\rightarrow\{1,2,3\}$ defined by $f(1)=2$, $f(2)=2$, and
johnston-linear-matrix-algebra-F03244	361	1.000	$f(3)=1$	$f(3)=1$	$f(3)=1$ is not (since $f(1)=f(2)$).
johnston-linear-matrix-algebra-F03245	361	1.000	$f(1)=f(2)$	$f(1)=f(2)$	$f(3)=1$ is not (since $f(1)=f(2)$).
johnston-linear-matrix-algebra-F03246	361	1.000	$(\sigma(1)\sigma(2)\cdots\sigma(n))$	$(\sigma(1)\sigma(2)\cdots\sigma(n))$	parentheses: $(\sigma(1)\sigma(2)\cdots\sigma(n))$. For example, the permutation discussed
johnston-linear-matrix-algebra-F03247	361	1.000	$\sigma(1)=2$, $\sigma(2)=3$	$\sigma(1)=2$, $\sigma(2)=3$	earlier with $\sigma(1)=2$, $\sigma(2)=3$, and $\sigma(3)=1$ is described succinctly by its
johnston-linear-matrix-algebra-F03248	361	0.699	$\sigma=(4132)$	$\sigma=(4132)$	via its one-line notation $\sigma=(4\ 1\ 3\ 2)$ then we know that it acts on $\{1,2,3,4\}$
johnston-linear-matrix-algebra-F03249	361	1.000	$\sigma(1)=4$, $\sigma(2)=1$, $\sigma(3)=3$	$\sigma(1)=4$, $\sigma(2)=1$, $\sigma(3)=3$	via $\sigma(1)=4$, $\sigma(2)=1$, $\sigma(3)=3$, and $\sigma(4)=2$.
johnston-linear-matrix-algebra-F03250	361	1.000	$\sigma(4)=2$	$\sigma(4)=2$	via $\sigma(1)=4$, $\sigma(2)=1$, $\sigma(3)=3$, and $\sigma(4)=2$.
johnston-linear-matrix-algebra-F03251	361	1.000	$n!=n\cdot(n-1)\cdot(n-2)\cdots2\cdot1$	$n!=n\cdot(n-1)\cdot(n-2)\cdots2\cdot1$	There are exactly $n!=n\cdot(n-1)\cdot(n-2)\cdots2\cdot1$ permutations σ that act
johnston-linear-matrix-algebra-F03252	361	1.000	$\sigma(1)$	$\sigma(1)$	on the set $\{1,2,\ldots,n\}$. To see this, notice that $\sigma(1)$ can be any of n different
johnston-linear-matrix-algebra-F03253	361	1.000	$\sigma(2)$	$\sigma(2)$	values, $\sigma(2)$ can be any of the $n-1$ remaining values (since it cannot be the
johnston-linear-matrix-algebra-F03254	361	0.970	$\sigma(3)$	$\sigma(3)$	same as $\sigma(1)$), $\sigma(3)$ can be any of the $n-2$ remaining values not equal to $\sigma(1)$
johnston-linear-matrix-algebra-F03255	361	0.970	$n-2$	$n-2$	same as $\sigma(1)$), $\sigma(3)$ can be any of the $n-2$ remaining values not equal to $\sigma(1)$
johnston-linear-matrix-algebra-F03256	361	1.000	S_n	S_n	is denoted by S_n .
johnston-linear-matrix-algebra-F03257	361	1.000	S_2	S_2	a) S_2
johnston-linear-matrix-algebra-F03258	361	1.000	S_3	S_3	b) S_3
johnston-linear-matrix-algebra-F03259	361	0.772	$2!=2$	$2!=2$	a) There are only $2!=2$ permutations that act on the set $\{1,2\}$, and
johnston-linear-matrix-algebra-F03260	361	0.772	$\{1,2\}$	$\{1,2\}$	a) There are only $2!=2$ permutations that act on the set $\{1,2\}$, and
johnston-linear-matrix-algebra-F03261	361	0.984	$3!=6$	$3!=6$	b) There are $3!=6$ permutations that act on the set $\{1,2,3\}$, and they
johnston-linear-matrix-algebra-F03262	361	0.984	$\{1,2,3\}$	$\{1,2,3\}$	b) There are $3!=6$ permutations that act on the set $\{1,2,3\}$, and they

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johnston-linear-matrix-algebra-F03263	361	1.000	$\sigma, \tau \in S_n$	$\sigma, \tau \in S_n$	just like we do with linear transformations. Specifically, if $\sigma, \tau \in S_n$ then we
johnston-linear-matrix-algebra-F03264	361	1.000	$\sigma \circ \tau$	$\sigma \circ \tau$	define their composition $\sigma \circ \tau$ by
johnston-linear-matrix-algebra-F03265	361	0.600	ι	ι	Similarly, the identity permutation , which we denote by ι , is the one that does
johnston-linear-matrix-algebra-F03266	361	0.644	$\iota = (1\ 2\ \cdots\ n)$	$\iota = (1\ 2\ \cdots\ n)$	nothing at all: $\iota = (1\ 2\ \cdots\ n)$, so $\iota(j) = j$ for all $1 \leq j \leq n$, and the inverse
johnston-linear-matrix-algebra-F03267	361	0.644	$\iota(j) = j$	$\iota(j) = j$	nothing at all: $\iota = (1\ 2\ \cdots\ n)$, so $\iota(j) = j$ for all $1 \leq j \leq n$, and the inverse
johnston-linear-matrix-algebra-F03268	361	1.000	σ^{-1}	σ^{-1}	σ^{-1} of a permutation σ is the one that “undoes” it:
johnston-linear-matrix-algebra-F03269	362	1.000	$\sigma(\tau(1))$	$\sigma(\tau(1))$	by writing $\sigma(\tau(1))$.
johnston-linear-matrix-algebra-F03270	362	1.000	$\sigma(\tau(2)), \dots, \sigma(\tau(5))$	$\sigma(\tau(2)), \dots, \sigma(\tau(5))$	$\sigma(\tau(2)), \dots, \sigma(\tau(5))$ in
johnston-linear-matrix-algebra-F03271	362	1.000	$\sigma(j) = k$	$\sigma(j) = k$	If $\sigma(j) = k$ then
johnston-linear-matrix-algebra-F03272	362	1.000	$\sigma^{-1}(k) = j$	$\sigma^{-1}(k) = j$	$\sigma^{-1}(k) = j$.
johnston-linear-matrix-algebra-F03273	362	0.306	$\sigma = (3\ 2\ 5\ 1\ 4)$	$\sigma = (3\ 2\ 5\ 1\ 4)$	Suppose $\sigma = (3\ 2\ 5\ 1\ 4)$ and $\tau = (5\ 1\ 4\ 2\ 3)$. Compute the one-line
johnston-linear-matrix-algebra-F03274	362	0.306	$\tau = (5\ 1\ 4\ 2\ 3)$	$\tau = (5\ 1\ 4\ 2\ 3)$	Suppose $\sigma = (3\ 2\ 5\ 1\ 4)$ and $\tau = (5\ 1\ 4\ 2\ 3)$. Compute the one-line
johnston-linear-matrix-algebra-F03275	362	1.000	$\tau \circ \sigma$	$\tau \circ \sigma$	b) $\tau \circ \sigma$
johnston-linear-matrix-algebra-F03276	362	1.000	τ^{-1}	τ^{-1}	d) τ^{-1}
johnston-linear-matrix-algebra-F03277	362	1.000	$\sigma(\tau(j))$	$\sigma(\tau(j))$	a) We just directly compute $\sigma(\tau(j))$ for each $1 \leq j \leq 5$:
johnston-linear-matrix-algebra-F03278	362	1.000	$1 \leq j \leq 5$	$1 \leq j \leq 5$	a) We just directly compute $\sigma(\tau(j))$ for each $1 \leq j \leq 5$:
johnston-linear-matrix-algebra-F03279	362	1.000	$\sigma \circ \tau = (4\ 3\ 1\ 2\ 5)$	$\sigma \circ \tau = (4\ 3\ 1\ 2\ 5)$	We thus conclude that $\sigma \circ \tau = (4\ 3\ 1\ 2\ 5)$.
johnston-linear-matrix-algebra-F03280	362	1.000	$\tau(\sigma(j))$	$\tau(\sigma(j))$	b) Again, we just directly compute $\tau(\sigma(j))$ for each $1 \leq j \leq 5$:
johnston-linear-matrix-algebra-F03281	362	0.999	$\tau \circ \sigma = (4\ 1\ 3\ 5\ 2)$	$\tau \circ \sigma = (4\ 1\ 3\ 5\ 2)$	It follows that $\tau \circ \sigma = (4\ 1\ 3\ 5\ 2)$.
johnston-linear-matrix-algebra-F03282	362	0.994	$\sigma^{-1} = (4\ 2\ 1\ 5\ 3)$	$\sigma^{-1} = (4\ 2\ 1\ 5\ 3)$	notation, so $\sigma^{-1} = (4\ 2\ 1\ 5\ 3)$.
johnston-linear-matrix-algebra-F03283	362	0.637	τ	τ	d) The numbers 1, 2, 3, 4, 5 appear in positions 2, 4, 5, 3, 1 in τ ’s one-
johnston-linear-matrix-algebra-F03284	362	1.000	$\tau^{-1} = (2\ 4\ 5\ 3\ 1)$	$\tau^{-1} = (2\ 4\ 5\ 3\ 1)$	line notation, so $\tau^{-1} = (2\ 4\ 5\ 3\ 1)$.
johnston-linear-matrix-algebra-F03285	362	1.000	P_σ	P_σ	In other words, the permutation matrix P_σ is the standard matrix of the
johnston-linear-matrix-algebra-F03286	362	1.000	$\mathbf{e}_{\sigma(j)}$	$\mathbf{e}_{\sigma(j)}$	linear transformation that sends $\mathbf{e}_j$ to $\mathbf{e}_{\sigma(j)}$ for each $1 \leq j \leq n$. Geometrically,
johnston-linear-matrix-algebra-F03287	362	1.000	$\sigma = (3\ 4\ 2\ 1)$	$\sigma = (3\ 4\ 2\ 1)$	a) $\sigma = (3\ 4\ 2\ 1)$
johnston-linear-matrix-algebra-F03288	362	1.000	$\sigma = (2\ 5\ 1\ 3\ 4)$	$\sigma = (2\ 5\ 1\ 3\ 4)$	b) $\sigma = (2\ 5\ 1\ 3\ 4)$
johnston-linear-matrix-algebra-F03289	362	0.956	$\iota \in S_6$	$\iota \in S_6$	c) $\iota \in S_6$

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johnston-linear-matrix-algebra-F03290	363	1.000	$\sigma=\left(\begin{array}{lll}2 & 3 & 1\end{array}\right)$	$\sigma = \left(\begin{array}{ccc} 2 & 3 & 1 \end{array} \right)$	Figure 3.26: If $\sigma = (2\ 3\ 1)$ then the permutation matrix P_σ sends $\mathbf{e}_1$ to $\mathbf{e}_2$, $\mathbf{e}_2$ to $\mathbf{e}_3$, and
johnston-linear-matrix-algebra-F03291	363	1.000	$\mathbf{e}_2, \mathbf{e}_2$	$\mathbf{e}_2, \mathbf{e}_2$	Figure 3.26: If $\sigma = (2\ 3\ 1)$ then the permutation matrix P_σ sends $\mathbf{e}_1$ to $\mathbf{e}_2$, $\mathbf{e}_2$ to $\mathbf{e}_3$, and
johnston-linear-matrix-algebra-F03292	363	1.000	$\mathbf{e}_3, \mathbf{e}_4, \mathbf{e}_2$	$\mathbf{e}_3, \mathbf{e}_4, \mathbf{e}_2$	a) As described by Definition 3.A.3, we just place $\mathbf{e}_3$, $\mathbf{e}_4$, $\mathbf{e}_2$, and $\mathbf{e}_1$ as
johnston-linear-matrix-algebra-F03293	363	1.000	$\mathbf{e}_2, \mathbf{e}_5, \mathbf{e}_1, \mathbf{e}_3, \mathbf{e}_4$	$\mathbf{e}_2, \mathbf{e}_5, \mathbf{e}_1, \mathbf{e}_3$	b) Similarly, for this permutation we place $\mathbf{e}_2$, $\mathbf{e}_5$, $\mathbf{e}_1$, $\mathbf{e}_3$, and $\mathbf{e}_4$ as
johnston-linear-matrix-algebra-F03294	363	1.000	$\mathbf{e}_4$	$\mathbf{e}_4$	b) Similarly, for this permutation we place $\mathbf{e}_2$, $\mathbf{e}_5$, $\mathbf{e}_1$, $\mathbf{e}_3$, and $\mathbf{e}_4$ as
johnston-linear-matrix-algebra-F03295	363	1.000	$\{1, 2, 3, 4, 5, 6\}$	$\{1, 2, 3, 4, 5, 6\}$	c) For the identity permutation on $\{1, 2, 3, 4, 5, 6\}$, we place $\mathbf{e}_1, \mathbf{e}_2, \dots$,
johnston-linear-matrix-algebra-F03296	363	1.000	$\mathbf{e}_1, \mathbf{e}_2, \ldots$	$\mathbf{e}_1, \mathbf{e}_2, \dots$	c) For the identity permutation on $\{1, 2, 3, 4, 5, 6\}$, we place $\mathbf{e}_1, \mathbf{e}_2, \dots$,
johnston-linear-matrix-algebra-F03297	363	0.969	$\mathbf{e}_6$	$\mathbf{e}_6$	$\mathbf{e}_6$ as columns into a matrix. Doing so gives us exactly the 6×6
johnston-linear-matrix-algebra-F03298	363	0.969	6×6	6×6	$\mathbf{e}_6$ as columns into a matrix. Doing so gives us exactly the 6×6
johnston-linear-matrix-algebra-F03299	363	0.999	S_6	S_6	associated with the identity permutation in S_6 was the identity matrix in $\mathcal{M}_6$.
johnston-linear-matrix-algebra-F03300	363	0.999	$\mathcal{M}_6$	$\mathcal{M}_6$	associated with the identity permutation in S_6 was the identity matrix in $\mathcal{M}_6$.
johnston-linear-matrix-algebra-F03301	363	1.000	$P_\sigma, P_\tau \in \mathcal{M}_n$	$P_\sigma, P_\tau \in \mathcal{M}_n$	$P_\sigma, P_\tau \in \mathcal{M}_n$. Then
johnston-linear-matrix-algebra-F03302	363	1.000	$P_\sigma P_\tau = P_{\sigma \circ \tau}$	$P_\sigma P_\tau = P_{\sigma \circ \tau}$	a) $P_\sigma P_\tau = P_{\sigma \circ \tau}$
johnston-linear-matrix-algebra-F03303	363	1.000	$P_\sigma^{-1} = P_\sigma^T = P_{\sigma^{-1}}$	$P_\sigma^{-1} = P_\sigma^T = P_{\sigma^{-1}}$	b) P_σ is invertible, and $P_\sigma^{-1} = P_\sigma^T = P_{\sigma^{-1}}$.
johnston-linear-matrix-algebra-F03304	364	1.000	$P_{\sigma \circ \tau}$	$P_{\sigma \circ \tau}$	Recall that $P_{\sigma \circ \tau}$ is
johnston-linear-matrix-algebra-F03305	364	0.843	$\mathbf{e}_{(\sigma \circ \tau)(j)}$	$\mathbf{e}_{(\sigma \circ \tau)(j)}$	$\mathbf{e}_{(\sigma \circ \tau)(j)}$,
johnston-linear-matrix-algebra-F03306	364	1.000	$P_\sigma P_\tau$	$P_\sigma P_\tau$	In particular, this means that, for all $1 \leq j \leq n$, the j -th column of $P_\sigma P_\tau$ is
johnston-linear-matrix-algebra-F03307	364	0.618	$P_\sigma P_{\sigma^{-1}} = P_{\sigma \circ \sigma^{-1}} = P_I = I$	$P_\sigma P_{\sigma^{-1}} = P_{\sigma \circ \sigma^{-1}} = P_I = I$	For part (b), we first note that part (a) tells us that $P_\sigma P_{\sigma^{-1}} = P_{\sigma \circ \sigma^{-1}} = P_I = I$,
johnston-linear-matrix-algebra-F03308	364	1.000	$P_\sigma^{-1} = P_{\sigma^{-1}}$	$P_\sigma^{-1} = P_{\sigma^{-1}}$	so $P_\sigma^{-1} = P_{\sigma^{-1}}$, as claimed. All that remains is to show that $P_\sigma^T = P_\sigma^{-1}$. To this
johnston-linear-matrix-algebra-F03309	364	1.000	$P_\sigma^T = P_\sigma^{-1}$	$P_\sigma^T = P_\sigma^{-1}$	so $P_\sigma^{-1} = P_{\sigma^{-1}}$, as claimed. All that remains is to show that $P_\sigma^T = P_\sigma^{-1}$. To this
johnston-linear-matrix-algebra-F03310	364	1.000	$P_\sigma^T P_\sigma$	$P_\sigma^T P_\sigma$	end, we compute the entries of $P_\sigma^T P_\sigma$:
johnston-linear-matrix-algebra-F03311	364	1.000	$\sigma(i) = \sigma(j)$	$\sigma(i) = \sigma(j)$	If $i = j$ then $\sigma(i) = \sigma(j)$, so the above dot product equals 1 (i.e., the diagonal
johnston-linear-matrix-algebra-F03312	364	0.998	$\mathbf{e}_{\sigma(i)} \cdot \mathbf{e}_{\sigma(j)} = 0$	$\mathbf{e}_{\sigma(i)} \cdot \mathbf{e}_{\sigma(j)} = 0$	$\mathbf{e}_{\sigma(i)} \cdot \mathbf{e}_{\sigma(j)} = 0$ (i.e., the off-diagonal entries of $P_\sigma^T P_\sigma$ all equal 0). It follows
johnston-linear-matrix-algebra-F03313	364	1.000	$P_\sigma^T P_\sigma = I$	$P_\sigma^T P_\sigma = I$	that $P_\sigma^T P_\sigma = I$, so $P_\sigma^T = P_\sigma^{-1}$, which completes the proof.
johnston-linear-matrix-algebra-F03314	364	1.000	$\operatorname{sgn}(\sigma)$	$\operatorname{sgn}(\sigma)$	The sign of a permutation σ , denoted by $\operatorname{sgn}(\sigma)$, is the quantity
johnston-linear-matrix-algebra-F03315	364	0.998	$\sigma = (3\ 4\ 2\ 1)$	$\sigma = \left(\begin{array}{cccc} 3 & 4 & 2 & 1 \end{array} \right)$	a) $\sigma = (3\ 4\ 2\ 1)$
johnston-linear-matrix-algebra-F03316	364	0.804	$\sigma = (2\ 5\ 1\ 3\ 4)$	$\sigma = \left(\begin{array}{ccccc} 2 & 5 & 1 & 3 & 4 \end{array} \right)$	b) $\sigma = (2\ 5\ 1\ 3\ 4)$

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johnston-linear-matrix-algebra-F03317	365	1.000	$(-1)^{\{s\}}$	$(-1)^s$	The sign of a permutation equals $(-1)^s$, where s is the number																					
johnston-linear-matrix-algebra-F03318	365	1.000	$\operatorname{sgn}(\sigma)=(-1)^{\{3\}}=-1$	$\operatorname{sgn}(\sigma)=(-1)^3=-1$	Since we swapped 3 times, we conclude that $\operatorname{sgn}(\sigma)=(-1)^3=-1$.																					
johnston-linear-matrix-algebra-F03319	365	1.000	$\operatorname{sgn}(\sigma)=(-1)^{\{4\}}=1$	$\operatorname{sgn}(\sigma)=(-1)^4=1$	Since we swapped 4 times, we conclude that $\operatorname{sgn}(\sigma)=(-1)^4=1$.																					
johnston-linear-matrix-algebra-F03320	365	0.936	$\operatorname{det}(A)=\sum_{\sigma\in S_n}\operatorname{sgn}(\sigma)a_{\sigma(1),1}a_{\sigma(2),2}\cdots a_{\sigma(n),n}$	$\operatorname{det}(A)=\sum_{\sigma\in S_n}\operatorname{sgn}(\sigma)a_{\sigma(1),1}a_{\sigma(2),2}\cdots a_{\sigma(n),n}$	Suppose $A\in\mathcal{M}_n$. Then $\operatorname{det}(A)=\sum_{\sigma\in S_n}\operatorname{sgn}(\sigma)a_{\sigma(1),1}a_{\sigma(2),2}\cdots a_{\sigma(n),n}$.																					
johnston-linear-matrix-algebra-F03321	365	0.860	$\operatorname{det}(A)=a_{1,1}a_{2,2}-a_{2,1}a_{1,2}$	$\operatorname{det}(A)=a_{1,1}a_{2,2}-a_{2,1}a_{1,2}$	−1. The above theorem thus tells us that $\operatorname{det}(A)=a_{1,1}a_{2,2}-a_{2,1}a_{1,2}$, which is																					
johnston-linear-matrix-algebra-F03322	365	0.998	$n=3$	$n=3$	Similarly, when $n=3$, there are six permutations in S_3 , which we now																					
johnston-linear-matrix-algebra-F03323	366	0.789	$a_{1,1}a_{2,2}a_{3,3}$	$a_{1,1}a_{2,2}a_{3,3}$	<table><thead><tr><th>σ</th><th>$\operatorname{sgn}(\sigma)$</th><th>contribution to $\operatorname{det}(A)$</th></tr></thead><tbody><tr><td>(1 2 3)</td><td>1</td><td>$a_{1,1}a_{2,2}a_{3,3}$</td></tr><tr><td>(1 3 2)</td><td>−1</td><td>$-a_{1,1}a_{3,2}a_{2,3}$</td></tr><tr><td>(2 1 3)</td><td>−1</td><td>$-a_{2,1}a_{1,2}a_{3,3}$</td></tr><tr><td>(2 3 1)</td><td>1</td><td>$a_{2,1}a_{3,2}a_{1,3}$</td></tr><tr><td>(3 1 2)</td><td>1</td><td>$a_{3,1}a_{1,2}a_{2,3}$</td></tr><tr><td>(3 2 1)</td><td>−1</td><td>$-a_{3,1}a_{2,2}a_{1,3}$</td></tr></tbody></table>	σ	$\operatorname{sgn}(\sigma)$	contribution to $\operatorname{det}(A)$	(1 2 3)	1	$a_{1,1}a_{2,2}a_{3,3}$	(1 3 2)	−1	$-a_{1,1}a_{3,2}a_{2,3}$	(2 1 3)	−1	$-a_{2,1}a_{1,2}a_{3,3}$	(2 3 1)	1	$a_{2,1}a_{3,2}a_{1,3}$	(3 1 2)	1	$a_{3,1}a_{1,2}a_{2,3}$	(3 2 1)	−1	$-a_{3,1}a_{2,2}a_{1,3}$
σ	$\operatorname{sgn}(\sigma)$	contribution to $\operatorname{det}(A)$																								
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(2 1 3)	−1	$-a_{2,1}a_{1,2}a_{3,3}$																								
(2 3 1)	1	$a_{2,1}a_{3,2}a_{1,3}$																								
(3 1 2)	1	$a_{3,1}a_{1,2}a_{2,3}$																								
(3 2 1)	−1	$-a_{3,1}a_{2,2}a_{1,3}$																								
johnston-linear-matrix-algebra-F03324	366	0.789	$-a_{1,1}a_{3,2}a_{2,3}$	$-a_{1,1}a_{3,2}a_{2,3}$	<table><thead><tr><th>σ</th><th>$\operatorname{sgn}(\sigma)$</th><th>contribution to $\operatorname{det}(A)$</th></tr></thead><tbody><tr><td>(1 2 3)</td><td>1</td><td>$a_{1,1}a_{2,2}a_{3,3}$</td></tr><tr><td>(1 3 2)</td><td>−1</td><td>$-a_{1,1}a_{3,2}a_{2,3}$</td></tr><tr><td>(2 1 3)</td><td>−1</td><td>$-a_{2,1}a_{1,2}a_{3,3}$</td></tr><tr><td>(2 3 1)</td><td>1</td><td>$a_{2,1}a_{3,2}a_{1,3}$</td></tr><tr><td>(3 1 2)</td><td>1</td><td>$a_{3,1}a_{1,2}a_{2,3}$</td></tr><tr><td>(3 2 1)</td><td>−1</td><td>$-a_{3,1}a_{2,2}a_{1,3}$</td></tr></tbody></table>	σ	$\operatorname{sgn}(\sigma)$	contribution to $\operatorname{det}(A)$	(1 2 3)	1	$a_{1,1}a_{2,2}a_{3,3}$	(1 3 2)	−1	$-a_{1,1}a_{3,2}a_{2,3}$	(2 1 3)	−1	$-a_{2,1}a_{1,2}a_{3,3}$	(2 3 1)	1	$a_{2,1}a_{3,2}a_{1,3}$	(3 1 2)	1	$a_{3,1}a_{1,2}a_{2,3}$	(3 2 1)	−1	$-a_{3,1}a_{2,2}a_{1,3}$
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johnston-linear-matrix-algebra-F03325	366	0.789	$-a_{2,1}a_{1,2}a_{3,3}$	$-a_{2,1}a_{1,2}a_{3,3}$	<table><thead><tr><th>σ</th><th>$\operatorname{sgn}(\sigma)$</th><th>contribution to $\operatorname{det}(A)$</th></tr></thead><tbody><tr><td>(1 2 3)</td><td>1</td><td>$a_{1,1}a_{2,2}a_{3,3}$</td></tr><tr><td>(1 3 2)</td><td>−1</td><td>$-a_{1,1}a_{3,2}a_{2,3}$</td></tr><tr><td>(2 1 3)</td><td>−1</td><td>$-a_{2,1}a_{1,2}a_{3,3}$</td></tr><tr><td>(2 3 1)</td><td>1</td><td>$a_{2,1}a_{3,2}a_{1,3}$</td></tr><tr><td>(3 1 2)</td><td>1</td><td>$a_{3,1}a_{1,2}a_{2,3}$</td></tr><tr><td>(3 2 1)</td><td>−1</td><td>$-a_{3,1}a_{2,2}a_{1,3}$</td></tr></tbody></table>	σ	$\operatorname{sgn}(\sigma)$	contribution to $\operatorname{det}(A)$	(1 2 3)	1	$a_{1,1}a_{2,2}a_{3,3}$	(1 3 2)	−1	$-a_{1,1}a_{3,2}a_{2,3}$	(2 1 3)	−1	$-a_{2,1}a_{1,2}a_{3,3}$	(2 3 1)	1	$a_{2,1}a_{3,2}a_{1,3}$	(3 1 2)	1	$a_{3,1}a_{1,2}a_{2,3}$	(3 2 1)	−1	$-a_{3,1}a_{2,2}a_{1,3}$
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johnston-linear-matrix-algebra-F03326	366	0.789	$a_{2,1}a_{3,2}a_{1,3}$	$a_{2,1}a_{3,2}a_{1,3}$	<table><thead><tr><th>σ</th><th>$\operatorname{sgn}(\sigma)$</th><th>contribution to $\operatorname{det}(A)$</th></tr></thead><tbody><tr><td>(1 2 3)</td><td>1</td><td>$a_{1,1}a_{2,2}a_{3,3}$</td></tr><tr><td>(1 3 2)</td><td>−1</td><td>$-a_{1,1}a_{3,2}a_{2,3}$</td></tr><tr><td>(2 1 3)</td><td>−1</td><td>$-a_{2,1}a_{1,2}a_{3,3}$</td></tr><tr><td>(2 3 1)</td><td>1</td><td>$a_{2,1}a_{3,2}a_{1,3}$</td></tr><tr><td>(3 1 2)</td><td>1</td><td>$a_{3,1}a_{1,2}a_{2,3}$</td></tr><tr><td>(3 2 1)</td><td>−1</td><td>$-a_{3,1}a_{2,2}a_{1,3}$</td></tr></tbody></table>	σ	$\operatorname{sgn}(\sigma)$	contribution to $\operatorname{det}(A)$	(1 2 3)	1	$a_{1,1}a_{2,2}a_{3,3}$	(1 3 2)	−1	$-a_{1,1}a_{3,2}a_{2,3}$	(2 1 3)	−1	$-a_{2,1}a_{1,2}a_{3,3}$	(2 3 1)	1	$a_{2,1}a_{3,2}a_{1,3}$	(3 1 2)	1	$a_{3,1}a_{1,2}a_{2,3}$	(3 2 1)	−1	$-a_{3,1}a_{2,2}a_{1,3}$
σ	$\operatorname{sgn}(\sigma)$	contribution to $\operatorname{det}(A)$																								
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Identifier	Page	Conf.	LaTeX source	Rendered	Image																					
johnston-linear-matrix-algebra-F03327	366	0.789	a_{3,1} a_{1,2} a_{2,3}	a_{3,1}a_{1,2}a_{2,3}	<table><tr><th>σ</th><th>$\text{sgn}(\sigma)$</th><th>contribution to $\det(A)$</th></tr><tr><td>(1 2 3)</td><td>1</td><td>$a_{1,1}a_{2,2}a_{3,3}$</td></tr><tr><td>(1 3 2)</td><td>−1</td><td>$-a_{1,1}a_{3,2}a_{2,3}$</td></tr><tr><td>(2 1 3)</td><td>−1</td><td>$-a_{2,1}a_{1,2}a_{3,3}$</td></tr><tr><td>(2 3 1)</td><td>1</td><td>$a_{2,1}a_{3,2}a_{1,3}$</td></tr><tr><td>(3 1 2)</td><td>1</td><td>$a_{3,1}a_{1,2}a_{2,3}$</td></tr><tr><td>(3 2 1)</td><td>−1</td><td>$-a_{3,1}a_{2,2}a_{1,3}$</td></tr></table>	σ	$\text{sgn}(\sigma)$	contribution to $\det(A)$	(1 2 3)	1	$a_{1,1}a_{2,2}a_{3,3}$	(1 3 2)	−1	$-a_{1,1}a_{3,2}a_{2,3}$	(2 1 3)	−1	$-a_{2,1}a_{1,2}a_{3,3}$	(2 3 1)	1	$a_{2,1}a_{3,2}a_{1,3}$	(3 1 2)	1	$a_{3,1}a_{1,2}a_{2,3}$	(3 2 1)	−1	$-a_{3,1}a_{2,2}a_{1,3}$
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johnston-linear-matrix-algebra-F03328	366	0.789	-a_{3,1} a_{2,2} a_{1,3}	−a_{3,1}a_{2,2}a_{1,3}	<table><tr><th>σ</th><th>$\text{sgn}(\sigma)$</th><th>contribution to $\det(A)$</th></tr><tr><td>(1 2 3)</td><td>1</td><td>$a_{1,1}a_{2,2}a_{3,3}$</td></tr><tr><td>(1 3 2)</td><td>−1</td><td>$-a_{1,1}a_{3,2}a_{2,3}$</td></tr><tr><td>(2 1 3)</td><td>−1</td><td>$-a_{2,1}a_{1,2}a_{3,3}$</td></tr><tr><td>(2 3 1)</td><td>1</td><td>$a_{2,1}a_{3,2}a_{1,3}$</td></tr><tr><td>(3 1 2)</td><td>1</td><td>$a_{3,1}a_{1,2}a_{2,3}$</td></tr><tr><td>(3 2 1)</td><td>−1</td><td>$-a_{3,1}a_{2,2}a_{1,3}$</td></tr></table>	σ	$\text{sgn}(\sigma)$	contribution to $\det(A)$	(1 2 3)	1	$a_{1,1}a_{2,2}a_{3,3}$	(1 3 2)	−1	$-a_{1,1}a_{3,2}a_{2,3}$	(2 1 3)	−1	$-a_{2,1}a_{1,2}a_{3,3}$	(2 3 1)	1	$a_{2,1}a_{3,2}a_{1,3}$	(3 1 2)	1	$a_{3,1}a_{1,2}a_{2,3}$	(3 2 1)	−1	$-a_{3,1}a_{2,2}a_{1,3}$
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johnston-linear-matrix-algebra-F03329	366	1.000	$\mathbf{a}_{\mathbf{j}}$	$\mathbf{a}_j$	<i>Proof of Theorem 3.A.4.</i> Write $A = [\mathbf{a}_1 \mid \mathbf{a}_2 \mid \cdots \mid \mathbf{a}_n]$, so that $\mathbf{a}_j$ denotes the																					
johnston-linear-matrix-algebra-F03330	366	1.000	$i_{\mathbf{1}}, i_{\mathbf{2}}, \ldots, i_{\mathbf{n}}$	$i_1, i_2, \ldots, i_n$	If any of the indices $i_1, i_2, \ldots, i_n$ in the above sum are equal to each other																					
johnston-linear-matrix-algebra-F03331	366	1.000	$\left[\mathbf{e}_{i_{\mathbf{1}}} \mid \mathbf{e}_{i_{\mathbf{2}}} \mid \cdots \mid \mathbf{e}_{i_{\mathbf{n}}}\right]$	$[\mathbf{e}_{i_1} \mid \mathbf{e}_{i_2} \mid \cdots \mid \mathbf{e}_{i_n}]$	then the matrix $[\mathbf{e}_{i_1} \mid \mathbf{e}_{i_2} \mid \cdots \mid \mathbf{e}_{i_n}]$ has a repeated column and is thus not																					
johnston-linear-matrix-algebra-F03332	366	1.000	$\sigma(1)=i_{\mathbf{1}}, \sigma(2)=i_{\mathbf{2}}$	$\sigma(1) = i_1, \sigma(2) = i_2$	$\sigma(1) = i_1, \sigma(2) = i_2$, and so on), so that																					
johnston-linear-matrix-algebra-F03333	367	1.000	$[A \ B]_{i, \ j}$	$[AB]_{i,j}$	Recall that $[AB]_{i,j}$																					
johnston-linear-matrix-algebra-F03334	367	0.996	$\operatorname{det}\left(A_{\mathbf{j}_{\mathbf{1}}}, \ldots, j_{\mathbf{m}}\right)$	$\det(A_{j_1, \ldots, j_m})$	out of $\det(A_{j_1, \ldots, j_m})$																					
johnston-linear-matrix-algebra-F03335	367	0.887	$A_{\mathbf{j}_{\mathbf{1}}}, \ldots, j_{\mathbf{m}}$	$A_{j_1, \ldots, j_m}$	$A_{j_1, \ldots, j_m}$.																					
johnston-linear-matrix-algebra-F03336	367	1.000	$B_{\mathbf{j}_{\mathbf{1}}}, \ldots, j_{\mathbf{m}}$	$B_{j_1, \ldots, j_m}$	Suppose $A \in \mathcal{M}_{m,n}$ and $B \in \mathcal{M}_{n,m}$ with $m \leq n$, and let $A_{j_1, \ldots, j_m}$ and $B_{j_1, \ldots, j_m}$																					
johnston-linear-matrix-algebra-F03337	367	1.000	$m \times m$	$m \times m$	denote the $m \times m$ submatrices of A and B consisting of the $j_1, \ldots, j_m$ -th																					
johnston-linear-matrix-algebra-F03338	367	1.000	$j_{\mathbf{1}}, \ldots, j_{\mathbf{m}}$	$j_1, \ldots, j_m$	denote the $m \times m$ submatrices of A and B consisting of the $j_1, \ldots, j_m$ -th																					
johnston-linear-matrix-algebra-F03339	367	1.000	$\left(j_{\mathbf{1}}, j_{\mathbf{2}}, \ldots, j_{\mathbf{m}}\right)$	$(j_1, j_2, \ldots, j_m)$	From here, we note that this sum is over <i>unordered</i> tuples $(j_1, j_2, \ldots, j_m)$, so																					
johnston-linear-matrix-algebra-F03340	367	1.000	$j_{\mathbf{1}} \leq j_{\mathbf{2}} \leq \cdots \leq j_{\mathbf{m}}$	$j_1 \leq j_2 \leq \cdots \leq j_m$	$j_1 \leq j_2 \leq \cdots \leq j_m$ and an inner sum over permutations of the members of that																					
johnston-linear-matrix-algebra-F03341	367	1.000	$\operatorname{det}\left(A_{\mathbf{j}_{\mathbf{1}}}, \ldots, j_{\mathbf{m}}\right)=0$	$\det(A_{j_1, \ldots, j_m}) = 0$	By recalling that $\det(A_{j_1, \ldots, j_m}) = 0$ whenever two of the “ j ” indices are the																					
johnston-linear-matrix-algebra-F03342	367	1.000	$1 \leq j_{\mathbf{1}} \leq \cdots \leq j_{\mathbf{m}} \leq n$	$1 \leq j_1 \leq \cdots \leq j_m \leq n$	replace the sum over $1 \leq j_1 \leq \cdots \leq j_m \leq n$ above with a sum over $1 \leq j_1 <$																					
johnston-linear-matrix-algebra-F03343	367	1.000	$1 \leq j_{\mathbf{1}} <$	$1 \leq j_1 <$	replace the sum over $1 \leq j_1 \leq \cdots \leq j_m \leq n$ above with a sum over $1 \leq j_1 <$																					
johnston-linear-matrix-algebra-F03344	367	1.000	$\cdots < j_{\mathbf{m}} \leq n$	$\cdots < j_m \leq n$	$\cdots < j_m \leq n$, which completes the proof.																					
johnston-linear-matrix-algebra-F03345	367	0.813	$\left(j_{\mathbf{1}}, j_{\mathbf{2}}, \ldots, j_{\mathbf{n}}\right)$	$(j_1, j_2, \ldots, j_n)$	ply says that $\det(AB) = \det(A) \det(B)$, since the only tuple $(j_1, j_2, \ldots, j_n)$ with																					
johnston-linear-matrix-algebra-F03346	367	1.000	$1 \leq j_{\mathbf{1}} < j_{\mathbf{2}} < \cdots < j_{\mathbf{n}} \leq n$	$1 \leq j_1 < j_2 < \cdots < j_n \leq n$	$1 \leq j_1 < j_2 < \cdots < j_n \leq n$ is $(j_1, j_2, \ldots, j_n) = (1, 2, \ldots, n)$. At the other extreme,																					
johnston-linear-matrix-algebra-F03347	367	1.000	$\left(j_{\mathbf{1}}, j_{\mathbf{2}}, \ldots, j_{\mathbf{n}}\right)=(1, 2, \ldots, n)$	$(j_1, j_2, \ldots, j_n) = (1, 2, \ldots, n)$	$1 \leq j_1 < j_2 < \cdots < j_n \leq n$ is $(j_1, j_2, \ldots, j_n) = (1, 2, \ldots, n)$. At the other extreme,																					

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johnston-linear-matrix-algebra F03348	368	■ 1.000	$m=1$	$m = 1$	if $m = 1$ then $A \in \mathcal{M}_{1,n}$ is a row vector and $B \in \mathcal{M}_{n,1}$ is a column vector, and
johnston-linear-matrix-algebra F03349	368	■ 1.000	$A \in \mathcal{M}_{1,n}$	$A \in \mathcal{M}_{1,n}$	if $m = 1$ then $A \in \mathcal{M}_{1,n}$ is a row vector and $B \in \mathcal{M}_{n,1}$ is a column vector, and
johnston-linear-matrix-algebra F03350	368	■ 1.000	$B \in \mathcal{M}_{n,1}$	$B \in \mathcal{M}_{n,1}$	if $m = 1$ then $A \in \mathcal{M}_{1,n}$ is a row vector and $B \in \mathcal{M}_{n,1}$ is a column vector, and
johnston-linear-matrix-algebra F03351	368	■ 0.998	$\left[\begin{array}{lll}0 & 0 & 1 \\ 0 & 2 & 3 \\ 4 & 5 & 6\end{array}\right]$	$\left[\begin{array}{ccc}0 & 0 & 1 \\ 0 & 2 & 3 \\ 4 & 5 & 6\end{array}\right]$	*(g) $\begin{bmatrix} 0 & 0 & 1 \\ 0 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$
johnston-linear-matrix-algebra F03352	369	■ 0.995	$\left[\begin{array}{llll}3 & 2 & 4 & 3 \\ 1 & 1 & 3 & 2 \\ 4 & 0 & 1 & 1 \\ 3 & 1 & 2 & 4\end{array}\right]$	$\left[\begin{array}{cccc}3 & 2 & 4 & 3 \\ 1 & 1 & 3 & 2 \\ 4 & 0 & 1 & 1 \\ 3 & 1 & 2 & 4\end{array}\right]$	$\begin{bmatrix} 3 & 2 & 4 & 3 \\ 1 & 1 & 3 & 2 \\ 4 & 0 & 1 & 1 \\ 3 & 1 & 2 & 4 \end{bmatrix}$
johnston-linear-matrix-algebra F03353	369	■ 1.000	$\left[\begin{array}{lllll}6 & 2 & 1 & 0 & 4 \\ 4 & 2 & 0 & 4 & 6 \\ 0 & 6 & 3 & 4 & 4 \\ 0 & 3 & 6 & 0 & 3 \\ 2 & 6 & 1 & 6 & 6\end{array}\right]$	$\left[\begin{array}{ccccc}6 & 2 & 1 & 0 & 4 \\ 4 & 2 & 0 & 4 & 6 \\ 0 & 6 & 3 & 4 & 4 \\ 0 & 3 & 6 & 0 & 3 \\ 2 & 6 & 1 & 6 & 6\end{array}\right]$	$\begin{bmatrix} 6 & 2 & 1 & 0 & 4 \\ 4 & 2 & 0 & 4 & 6 \\ 0 & 6 & 3 & 4 & 4 \\ 0 & 3 & 6 & 0 & 3 \\ 2 & 6 & 1 & 6 & 6 \end{bmatrix}$
johnston-linear-matrix-algebra F03354	369	■ 0.264	$\circ$	$\circ$	*(a) $\langle 1\,2\,3 \rangle \circ \langle 3\,2\,1 \rangle$
johnston-linear-matrix-algebra F03355 [conf 0.087]	369	■ 0.087	$\circ(2331)$	$\circ(2331)$	(b) $\langle 3\,1\,2 \rangle \circ \langle 2\,3\,1 \rangle$
johnston-linear-matrix-algebra F03356 [conf 0.079]	369	■ 0.079	$\circ(42.331)$	$\circ(42.331)$	*(c) $\langle 1\,3\,2\,4 \rangle \circ \langle 4\,2\,3\,1 \rangle$
johnston-linear-matrix-algebra F03357	369	■ 0.649	$(24113) \circ (4132)$	$(24113) \circ (4132)$	(d) $\langle 2\,4\,1\,3 \rangle \circ \langle 4\,1\,3\,2 \rangle$
johnston-linear-matrix-algebra F03358	369	■ 0.792	$(241536) \circ (361524)$	$(241536) \circ (361524)$	(h) $\langle 2\,4\,1\,5\,3\,6 \rangle \circ \langle 3\,6\,1\,5\,2\,4 \rangle$
johnston-linear-matrix-algebra F03359	369	■ 0.999	$\operatorname{cof}(I)=I$	$\operatorname{cof}(I) = I$	*(c) $\operatorname{cof}(I) = I$
johnston-linear-matrix-algebra F03360	369	■ 1.000	$\operatorname{cof}(A)=\operatorname{cof}(B)$	$\operatorname{cof}(A) = \operatorname{cof}(B)$	(d) If $A,B \in \mathcal{M}_n$ are such that $\operatorname{cof}(A) = \operatorname{cof}(B)$ then
johnston-linear-matrix-algebra F03361	369	■ 1.000	$\operatorname{rank}(\operatorname{cof}(A))$	$\operatorname{rank}(\operatorname{cof}(A))$	$\operatorname{rank}(\operatorname{cof}(A))$.
johnston-linear-matrix-algebra F03362	369	■ 0.998	$\imath^{-1}=\imath$	$\imath^{-1} = \imath$	$\imath^{-1} = \imath$.
johnston-linear-matrix-algebra F03363	369	■ 1.000	$\sigma \circ$	$\sigma \circ$	(h) Composition of permutations is commutative: $\sigma \circ$
johnston-linear-matrix-algebra F03364	369	■ 1.000	$\tau=\tau \circ \sigma$	$\tau = \tau \circ \sigma$	$\tau = \tau \circ \sigma$ for all $\sigma, \tau \in S_n$.
johnston-linear-matrix-algebra F03365	369	■ 1.000	$S_{\{5\}}$	S_5	*(i) There are exactly 120 permutations in S_5 .
johnston-linear-matrix-algebra F03366	369	■ 0.998	$\operatorname{cof}(A\,B)=\operatorname{cof}(A)\operatorname{cof}(B)$	$\operatorname{cof}(AB) = \operatorname{cof}(A) \operatorname{cof}(B)$	$\operatorname{cof}(AB) = \operatorname{cof}(A) \operatorname{cof}(B)$.
johnston-linear-matrix-algebra F03367	369	■ 1.000	$\operatorname{cof}(A^k)=(\operatorname{cof}(A))^k$	$\operatorname{cof}(A^k) = (\operatorname{cof}(A))^k$	Show that $\operatorname{cof}(A^k) = (\operatorname{cof}(A))^k$. [Side note: This result is
johnston-linear-matrix-algebra F03368	369	■ 1.000	$\operatorname{cof}(c\,A)=$	$\operatorname{cof}(cA) =$	3.A.9 Suppose $A \in \mathcal{M}_n$ and $c \in \mathbb{R}$. Show that $\operatorname{cof}(cA) =$
johnston-linear-matrix-algebra F03369	369	■ 1.000	$c^{n-1}\operatorname{cof}(A)$	$c^{n-1} \operatorname{cof}(A)$	$c^{n-1} \operatorname{cof}(A)$.
johnston-linear-matrix-algebra F03370	369	■ 1.000	$\operatorname{det}(\operatorname{cof}(A))=(\operatorname{det}(A))^{n-1}$	$\operatorname{det}(\operatorname{cof}(A)) = (\operatorname{det}(A))^{n-1}$	$\operatorname{det}(\operatorname{cof}(A)) = (\operatorname{det}(A))^{n-1}$.
johnston-linear-matrix-algebra F03371	369	■ 1.000	$\operatorname{cof}(\operatorname{cof}(A))=(\operatorname{det}(A))^{n-2}A$	$\operatorname{cof}(\operatorname{cof}(A)) = (\operatorname{det}(A))^{n-2}A$	$\operatorname{cof}(\operatorname{cof}(A)) = (\operatorname{det}(A))^{n-2}A$.

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johnston-linear-matrix-algebra-F03372	369	1.000	$\operatorname{cof}(\operatorname{cof}(A))=0$	$\operatorname{cof}(\operatorname{cof}(A)) = O$	vertible, and it shows in particular that $\operatorname{cof}(\operatorname{cof}(A)) = O$																												
johnston-linear-matrix-algebra-F03373	369	1.000	$A \in \mathcal{M}_n, A \mathbf{x}=\mathbf{b}$	$A \in \mathcal{M}_n, A\mathbf{x} = \mathbf{b}$	of Cramer's rule (Theorem 3.A.2). Let $A \in \mathcal{M}_n, A\mathbf{x} = \mathbf{b}$,																												
johnston-linear-matrix-algebra-F03374	369	1.000	$A^{-1} A_{\{j\}}$	$A^{-1}A_j$	(a) Compute the columns of the matrix $A^{-1}A_j$.																												
johnston-linear-matrix-algebra-F03375	369	1.000	$\operatorname{det}(A^{-1} A_{\{j\}})$	$\det(A^{-1}A_j)$	(b) Compute $\det(A^{-1}A_j)$ in two different ways: by using																												
johnston-linear-matrix-algebra-F03376	369	1.000	$\operatorname{sgn}(\sigma \circ \tau)=\operatorname{sgn}(\sigma)(\sigma \circ \tau)$	$\operatorname{sgn}(\sigma \circ \tau) = \operatorname{sgn}(\sigma) \cdot \operatorname{sgn}(\tau)$	(a) Show that $\operatorname{sgn}(\sigma \circ \tau) = \operatorname{sgn}(\sigma) \cdot \operatorname{sgn}(\tau)$.																												
johnston-linear-matrix-algebra-F03377	369	1.000	$m>n$	$m > n$	$\det(AB)$ if $m > n$?																												
johnston-linear-matrix-algebra-F03378	370	1.000	$\operatorname{per}(A)$	$\operatorname{per}(A)$	$\operatorname{per}(A)$, is the quantity defined by																												
johnston-linear-matrix-algebra-F03379	370	1.000	$\left[\begin{array}{cc}1 & 3 \\ 0 & -2\end{array}\right]$	$\left[\begin{array}{cc}1 & 3 \\ 0 & -2\end{array}\right]$	(b) $\left[\begin{array}{cc}1 & 3 \\ 0 & -2\end{array}\right]$																												
johnston-linear-matrix-algebra-F03380	370	1.000	$\left[\begin{array}{ccc}1 & 1 & 3 \\ -4 & 2 & 1 \\ 3 & 1 & 2\end{array}\right]$	$\left[\begin{array}{ccc}1 & 1 & 3 \\ -4 & 2 & 1 \\ 3 & 1 & 2\end{array}\right]$	(f) $\left[\begin{array}{ccc}1 & 1 & 3 \\ -4 & 2 & 1 \\ 3 & 1 & 2\end{array}\right]$																												
johnston-linear-matrix-algebra-F03381	370	1.000	$\operatorname{per}(A^T)=\operatorname{per}(A)$	$\operatorname{per}(A^T) = \operatorname{per}(A)$	satisfies $\operatorname{per}(A^T) = \operatorname{per}(A)$ for all $A \in \mathcal{M}_n$.																												
johnston-linear-matrix-algebra-F03382	370	1.000	$\operatorname{per}(A)=a_{1,1} a_{2,2} \cdots a_{n, n}$	$\operatorname{per}(A) = a_{1,1}a_{2,2} \cdots a_{n,n}$	$\operatorname{per}(A) = a_{1,1}a_{2,2} \cdots a_{n,n}$.																												
johnston-linear-matrix-algebra-F03383	370	0.999	$\operatorname{per}(A B)=$	$\operatorname{per}(AB) =$	Exercise 3.A.16) does <i>not</i> satisfy the property $\operatorname{per}(AB) =$																												
johnston-linear-matrix-algebra-F03384	370	1.000	$\operatorname{per}(A) \operatorname{per}(B)$	$\operatorname{per}(A) \operatorname{per}(B)$	$\operatorname{per}(A)\operatorname{per}(B)$.																												
johnston-linear-matrix-algebra-F03385	370	1.000	$\mathbf{v}_{\{0\}} \in \mathbb{R}^n$	$\mathbf{v}_0 \in \mathbb{R}^n$	starting vector. That is, if we start with a randomly-chosen vector $\mathbf{v}_0 \in \mathbb{R}^n$, we																												
johnston-linear-matrix-algebra-F03386	371	0.997	$A=\left[\begin{array}{ll}4 & 1 \\ 2 & 5\end{array}\right]$	$A = \left[\begin{array}{cc}4 & 1 \\ 2 & 5\end{array}\right]$	Figure 3.27: Repeatedly applying the matrix (linear transformation) $A = \left[\begin{array}{cc}4 & 1 \\ 2 & 5\end{array}\right]$																												
johnston-linear-matrix-algebra-F03387	371	1.000	$\mathbf{v}=(1,2) / \sqrt{5}$	$\mathbf{v} = (1,2)/\sqrt{5}$	squished closer and closer to the line in the direction of $\mathbf{v} = (1,2)/\sqrt{5}$, which is a																												
johnston-linear-matrix-algebra-F03388	371	1.000	$\mathbf{v}_{\{k-1\}}$	$\mathbf{v}_{k-1}$	The idea here is that multiplying $\mathbf{v}_{k-1}$ by A skews it a bit closer toward the																												
johnston-linear-matrix-algebra-F03389	371	1.000	$\mathbf{v}_{\{k\}}^T A \mathbf{v}_{\{k\}}$	$\mathbf{v}_k^T A \mathbf{v}_k$	We claim that, as k gets large, $\mathbf{v}_k^T A \mathbf{v}_k$ typically becomes a better and better																												
johnston-linear-matrix-algebra-F03390	371	1.000	$\mathbf{v}_{\{0\}}$	$\mathbf{v}_0$	We start by picking $\mathbf{v}_0$ arbitrarily. If we select $\mathbf{v}_0 = \mathbf{e}_1$ then power																												
johnston-linear-matrix-algebra-F03391	371	1.000	$\mathbf{v}_{\{0\}}=\mathbf{e}_{\{1\}}$	$\mathbf{v}_0 = \mathbf{e}_1$	We start by picking $\mathbf{v}_0$ arbitrarily. If we select $\mathbf{v}_0 = \mathbf{e}_1$ then power																												
johnston-linear-matrix-algebra-F03392	372	0.546	$A \mathbf{v}_{\{k-1\}}$	$A \mathbf{v}_{k-1}$	<table><tr><th>k</th><th>$A \mathbf{v}_{k-1}$</th><th>$\ A \mathbf{v}_{k-1}\$</th><th>$\mathbf{v}_k = A \mathbf{v}_{k-1} / \ A \mathbf{v}_{k-1}\$</th></tr><tr><td>0</td><td>—</td><td>—</td><td>$\mathbf{v}_0 = \mathbf{e}_1 = (1,0)$</td></tr><tr><td>1</td><td>(1.00,5.00)</td><td>5.10</td><td>(0.20,0.98)</td></tr><tr><td>2</td><td>(2.16,4.90)</td><td>5.36</td><td>(0.40,0.92)</td></tr><tr><td>3</td><td>(2.23,5.68)</td><td>6.10</td><td>(0.37,0.93)</td></tr><tr><td>4</td><td>(2.23,5.55)</td><td>5.98</td><td>(0.37,0.93)</td></tr><tr><td>5</td><td>(2.23,5.57)</td><td>6.00</td><td>(0.37,0.93)</td></tr></table>	k	$A \mathbf{v}_{k-1}$	$\ A \mathbf{v}_{k-1}\ $	$\mathbf{v}_k = A \mathbf{v}_{k-1} / \ A \mathbf{v}_{k-1}\ $	0	—	—	$\mathbf{v}_0 = \mathbf{e}_1 = (1,0)$	1	(1.00,5.00)	5.10	(0.20,0.98)	2	(2.16,4.90)	5.36	(0.40,0.92)	3	(2.23,5.68)	6.10	(0.37,0.93)	4	(2.23,5.55)	5.98	(0.37,0.93)	5	(2.23,5.57)	6.00	(0.37,0.93)
k	$A \mathbf{v}_{k-1}$	$\ A \mathbf{v}_{k-1}\ $	$\mathbf{v}_k = A \mathbf{v}_{k-1} / \ A \mathbf{v}_{k-1}\ $																														
0	—	—	$\mathbf{v}_0 = \mathbf{e}_1 = (1,0)$																														
1	(1.00,5.00)	5.10	(0.20,0.98)																														
2	(2.16,4.90)	5.36	(0.40,0.92)																														
3	(2.23,5.68)	6.10	(0.37,0.93)																														
4	(2.23,5.55)	5.98	(0.37,0.93)																														
5	(2.23,5.57)	6.00	(0.37,0.93)																														

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johnston-linear-matrix-algebra-F03393	372	0.546	$\left\ A \mathbf{v}_{k-1}\right\ $	$\ A\mathbf{v}_{k-1}\ $	<table><tr><th>k</th><th>$A\mathbf{v}_{k-1}$</th><th>$\ A\mathbf{v}_{k-1}\$</th><th>$\mathbf{v}_k = A\mathbf{v}_{k-1}/\ A\mathbf{v}_{k-1}\$</th></tr><tr><td>0</td><td>–</td><td>–</td><td>$\mathbf{v}_0 = \mathbf{e}_1 = (1,0)$</td></tr><tr><td>1</td><td>(1.00,5.00)</td><td>5.10</td><td>(0.20,0.98)</td></tr><tr><td>2</td><td>(2.16,4.90)</td><td>5.36</td><td>(0.40,0.92)</td></tr><tr><td>3</td><td>(2.23,5.68)</td><td>6.10</td><td>(0.37,0.93)</td></tr><tr><td>4</td><td>(2.23,5.55)</td><td>5.98</td><td>(0.37,0.93)</td></tr><tr><td>5</td><td>(2.23,5.57)</td><td>6.00</td><td>(0.37,0.93)</td></tr></table>	k	$A\mathbf{v}_{k-1}$	$\ A\mathbf{v}_{k-1}\ $	$\mathbf{v}_k = A\mathbf{v}_{k-1}/\ A\mathbf{v}_{k-1}\ $	0	–	–	$\mathbf{v}_0 = \mathbf{e}_1 = (1,0)$	1	(1.00,5.00)	5.10	(0.20,0.98)	2	(2.16,4.90)	5.36	(0.40,0.92)	3	(2.23,5.68)	6.10	(0.37,0.93)	4	(2.23,5.55)	5.98	(0.37,0.93)	5	(2.23,5.57)	6.00	(0.37,0.93)
k	$A\mathbf{v}_{k-1}$	$\ A\mathbf{v}_{k-1}\ $	$\mathbf{v}_k = A\mathbf{v}_{k-1}/\ A\mathbf{v}_{k-1}\ $																														
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3	(2.23,5.68)	6.10	(0.37,0.93)																														
4	(2.23,5.55)	5.98	(0.37,0.93)																														
5	(2.23,5.57)	6.00	(0.37,0.93)																														
johnston-linear-matrix-algebra-F03394	372	0.546	$\mathbf{v}_{k}=A \mathbf{v}_{k-1} /\left\ A \mathbf{v}_{k-1}\right\ $	$\mathbf{v}_k = A\mathbf{v}_{k-1}/\ A\mathbf{v}_{k-1}\ $	<table><tr><th>k</th><th>$A\mathbf{v}_{k-1}$</th><th>$\ A\mathbf{v}_{k-1}\$</th><th>$\mathbf{v}_k = A\mathbf{v}_{k-1}/\ A\mathbf{v}_{k-1}\$</th></tr><tr><td>0</td><td>–</td><td>–</td><td>$\mathbf{v}_0 = \mathbf{e}_1 = (1,0)$</td></tr><tr><td>1</td><td>(1.00,5.00)</td><td>5.10</td><td>(0.20,0.98)</td></tr><tr><td>2</td><td>(2.16,4.90)</td><td>5.36</td><td>(0.40,0.92)</td></tr><tr><td>3</td><td>(2.23,5.68)</td><td>6.10</td><td>(0.37,0.93)</td></tr><tr><td>4</td><td>(2.23,5.55)</td><td>5.98</td><td>(0.37,0.93)</td></tr><tr><td>5</td><td>(2.23,5.57)</td><td>6.00</td><td>(0.37,0.93)</td></tr></table>	k	$A\mathbf{v}_{k-1}$	$\ A\mathbf{v}_{k-1}\ $	$\mathbf{v}_k = A\mathbf{v}_{k-1}/\ A\mathbf{v}_{k-1}\ $	0	–	–	$\mathbf{v}_0 = \mathbf{e}_1 = (1,0)$	1	(1.00,5.00)	5.10	(0.20,0.98)	2	(2.16,4.90)	5.36	(0.40,0.92)	3	(2.23,5.68)	6.10	(0.37,0.93)	4	(2.23,5.55)	5.98	(0.37,0.93)	5	(2.23,5.57)	6.00	(0.37,0.93)
k	$A\mathbf{v}_{k-1}$	$\ A\mathbf{v}_{k-1}\ $	$\mathbf{v}_k = A\mathbf{v}_{k-1}/\ A\mathbf{v}_{k-1}\ $																														
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4	(2.23,5.55)	5.98	(0.37,0.93)																														
5	(2.23,5.57)	6.00	(0.37,0.93)																														
johnston-linear-matrix-algebra-F03395	372	0.546	$\mathbf{v}_{0}=\mathbf{e}_{1}=(1,0)$	$\mathbf{v}_0 = \mathbf{e}_1 = (1,0)$	<table><tr><th>k</th><th>$A\mathbf{v}_{k-1}$</th><th>$\ A\mathbf{v}_{k-1}\$</th><th>$\mathbf{v}_k = A\mathbf{v}_{k-1}/\ A\mathbf{v}_{k-1}\$</th></tr><tr><td>0</td><td>–</td><td>–</td><td>$\mathbf{v}_0 = \mathbf{e}_1 = (1,0)$</td></tr><tr><td>1</td><td>(1.00,5.00)</td><td>5.10</td><td>(0.20,0.98)</td></tr><tr><td>2</td><td>(2.16,4.90)</td><td>5.36</td><td>(0.40,0.92)</td></tr><tr><td>3</td><td>(2.23,5.68)</td><td>6.10</td><td>(0.37,0.93)</td></tr><tr><td>4</td><td>(2.23,5.55)</td><td>5.98</td><td>(0.37,0.93)</td></tr><tr><td>5</td><td>(2.23,5.57)</td><td>6.00</td><td>(0.37,0.93)</td></tr></table>	k	$A\mathbf{v}_{k-1}$	$\ A\mathbf{v}_{k-1}\ $	$\mathbf{v}_k = A\mathbf{v}_{k-1}/\ A\mathbf{v}_{k-1}\ $	0	–	–	$\mathbf{v}_0 = \mathbf{e}_1 = (1,0)$	1	(1.00,5.00)	5.10	(0.20,0.98)	2	(2.16,4.90)	5.36	(0.40,0.92)	3	(2.23,5.68)	6.10	(0.37,0.93)	4	(2.23,5.55)	5.98	(0.37,0.93)	5	(2.23,5.57)	6.00	(0.37,0.93)
k	$A\mathbf{v}_{k-1}$	$\ A\mathbf{v}_{k-1}\ $	$\mathbf{v}_k = A\mathbf{v}_{k-1}/\ A\mathbf{v}_{k-1}\ $																														
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1	(1.00,5.00)	5.10	(0.20,0.98)																														
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5	(2.23,5.57)	6.00	(0.37,0.93)																														
johnston-linear-matrix-algebra-F03396	372	0.997	$\mathbf{v}_{6}, \mathbf{v}_{7}, \mathbf{v}_{8}$	$\mathbf{v}_6, \mathbf{v}_7, \mathbf{v}_8$	We could of course keep going and compute $\mathbf{v}_6, \mathbf{v}_7, \mathbf{v}_8$, and so on,																												
johnston-linear-matrix-algebra-F03397	372	1.000	$\mathbf{v} \approx \mathbf{v}_{5} \approx(0.37,0.93)$	$\mathbf{v} \approx \mathbf{v}_5 \approx (0.37,0.93)$	with corresponding unit eigenvector $\mathbf{v} \approx \mathbf{v}_5 \approx (0.37,0.93)$.																												
johnston-linear-matrix-algebra-F03398	372	0.942	$\mathbf{v}=(2,5) / \sqrt{29} \approx(0.37,0.93)$	$\mathbf{v} = (2,5)/\sqrt{29} \approx (0.37,0.93)$	$\mathbf{v} = (2,5)/\sqrt{29} \approx (0.37,0.93)$.																												
johnston-linear-matrix-algebra-F03399	372	1.000	$\mathbf{v}_{0}, \mathbf{v}_{1}, \mathbf{v}_{2}, \ldots$	$\mathbf{v}_0, \mathbf{v}_1, \mathbf{v}_2, \dots$	Also, we can visualize the vectors $\mathbf{v}_0, \mathbf{v}_1, \mathbf{v}_2, \dots$ that we computed via																												
johnston-linear-matrix-algebra-F03400	373	1.000	$\mathbf{v}_{7} \approx(-0.26,-0.41,0.49,0.38,0.49,-0.38,-0.10)$	$\mathbf{v}_7 \approx (-0.26,-0.41,0.49,0.38,0.49,-0.38,-0.10)$	close to $\mathbf{v}_7 \approx (-0.26,-0.41,0.49,0.38,0.49,-0.38,-0.10)$.																												
johnston-linear-matrix-algebra-F03401	373	1.000	$\lambda=-4.92$	$\lambda = -4.92$	located near $\lambda = -4.92$.																												
johnston-linear-matrix-algebra-F03402	373	0.999	$s-1$	$s - 1$	multiplications and $s - 1$ additions, rather than n multiplications and $n - 1$																												
johnston-linear-matrix-algebra-F03403	374	1.000	$p_{\mathcal{C}}(\lambda)=(-1)^n p(\lambda)$	$p_{\mathcal{C}}(\lambda) = (-1)^n p(\lambda)$	$p_{\mathcal{C}}(\lambda) = (-1)^n p(\lambda)$,																												
johnston-linear-matrix-algebra-F03404	374	1.000	$C \mathbf{v}_{k}$	$C\mathbf{v}_k$	$C\mathbf{v}_k$ only requires n																												
johnston-linear-matrix-algebra-F03405	374	0.976	$2 \times 2, 3 \times 3$	$2 \times 2, 3 \times 3$	polynomial. While this method works fine for small (e.g., $2 \times 2, 3 \times 3$,																												
johnston-linear-matrix-algebra-F03406	374	1.000	$p(x)=\lambda^{\{n\}}+a_{\{n-1\}} \lambda^{\{n-1\}}+\cdots+a_{\{1\}} \lambda+a_{\{0\}}$	$p(x) = \lambda^n + a_{n-1}\lambda^{n-1} + \cdots + a_1\lambda + a_0$	Specifically, to find the roots of $p(x) = \lambda^n + a_{n-1}\lambda^{n-1} + \cdots + a_1\lambda + a_0$,																												
johnston-linear-matrix-algebra-F03407	374	0.999	$2 n-1$	$2n - 1$	to this task since C is so sparse (it only has $2n - 1$ non-zero entries, despite																												
johnston-linear-matrix-algebra-F03408	374	1.000	$p(x)=x^{\{6\}}-2 x^{\{5\}}+2 x^{\{4\}}-3 x^{\{3\}}+x^{\{2\}}+x-2$	$p(x) = x^6 - 2x^5 + 2x^4 - 3x^3 + x^2 + x - 2$	nomial $p(x) = x^6 - 2x^5 + 2x^4 - 3x^3 + x^2 + x - 2$, we first construct its																												

Identifier	Page	Conf.	LaTeX source	Rendered	Image																								
johnston-linear-matrix-algebra-F03409	374	1.000	$\mathbf{v}_{k}=C \mathbf{v}_{k-1} /\left C \mathbf{v}_{k-1}\right $	$\mathbf{v}_k = C\mathbf{v}_{k-1}/\ C\mathbf{v}_{k-1}\ $	<table><tr><th>k</th><th>$\mathbf{v}_k = C\mathbf{v}_{k-1}/\ C\mathbf{v}_{k-1}\$</th><th>$\mathbf{v}_k^T C\mathbf{v}_k$</th></tr><tr><td>0</td><td>(1.00,0.00,0.00,0.00,0.00,0.00)</td><td>0.00</td></tr><tr><td>1</td><td>(0.00,0.00,0.00,0.00,0.00,1.00)</td><td>2.00</td></tr><tr><td>2</td><td>(0.00,0.00,0.00,0.00,0.45,0.89)</td><td>1.20</td></tr><tr><td>3</td><td>(0.00,0.00,0.00,0.33,0.67,0.67)</td><td>1.33</td></tr><tr><td></td><td>$\vdots$</td><td></td></tr><tr><td>26</td><td>(0.06,0.10,0.17,0.28,0.48,0.81)</td><td>1.69</td></tr><tr><td>27</td><td>(0.06,0.10,0.17,0.28,0.48,0.81)</td><td>1.68</td></tr></table>	k	$\mathbf{v}_k = C\mathbf{v}_{k-1}/\ C\mathbf{v}_{k-1}\ $	$\mathbf{v}_k^T C\mathbf{v}_k$	0	(1.00,0.00,0.00,0.00,0.00,0.00)	0.00	1	(0.00,0.00,0.00,0.00,0.00,1.00)	2.00	2	(0.00,0.00,0.00,0.00,0.45,0.89)	1.20	3	(0.00,0.00,0.00,0.33,0.67,0.67)	1.33		$\vdots$		26	(0.06,0.10,0.17,0.28,0.48,0.81)	1.69	27	(0.06,0.10,0.17,0.28,0.48,0.81)	1.68
k	$\mathbf{v}_k = C\mathbf{v}_{k-1}/\ C\mathbf{v}_{k-1}\ $	$\mathbf{v}_k^T C\mathbf{v}_k$																											
0	(1.00,0.00,0.00,0.00,0.00,0.00)	0.00																											
1	(0.00,0.00,0.00,0.00,0.00,1.00)	2.00																											
2	(0.00,0.00,0.00,0.00,0.45,0.89)	1.20																											
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	$\vdots$																												
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27	(0.06,0.10,0.17,0.28,0.48,0.81)	1.68																											
johnston-linear-matrix-algebra-F03410	374	1.000	$\mathbf{v}_{k}^T C \mathbf{v}_{k}$	$\mathbf{v}_k^T C\mathbf{v}_k$	<table><tr><th>k</th><th>$\mathbf{v}_k = C\mathbf{v}_{k-1}/\ C\mathbf{v}_{k-1}\$</th><th>$\mathbf{v}_k^T C\mathbf{v}_k$</th></tr><tr><td>0</td><td>(1.00,0.00,0.00,0.00,0.00,0.00)</td><td>0.00</td></tr><tr><td>1</td><td>(0.00,0.00,0.00,0.00,0.00,1.00)</td><td>2.00</td></tr><tr><td>2</td><td>(0.00,0.00,0.00,0.00,0.45,0.89)</td><td>1.20</td></tr><tr><td>3</td><td>(0.00,0.00,0.00,0.33,0.67,0.67)</td><td>1.33</td></tr><tr><td></td><td>$\vdots$</td><td></td></tr><tr><td>26</td><td>(0.06,0.10,0.17,0.28,0.48,0.81)</td><td>1.69</td></tr><tr><td>27</td><td>(0.06,0.10,0.17,0.28,0.48,0.81)</td><td>1.68</td></tr></table>	k	$\mathbf{v}_k = C\mathbf{v}_{k-1}/\ C\mathbf{v}_{k-1}\ $	$\mathbf{v}_k^T C\mathbf{v}_k$	0	(1.00,0.00,0.00,0.00,0.00,0.00)	0.00	1	(0.00,0.00,0.00,0.00,0.00,1.00)	2.00	2	(0.00,0.00,0.00,0.00,0.45,0.89)	1.20	3	(0.00,0.00,0.00,0.33,0.67,0.67)	1.33		$\vdots$		26	(0.06,0.10,0.17,0.28,0.48,0.81)	1.69	27	(0.06,0.10,0.17,0.28,0.48,0.81)	1.68
k	$\mathbf{v}_k = C\mathbf{v}_{k-1}/\ C\mathbf{v}_{k-1}\ $	$\mathbf{v}_k^T C\mathbf{v}_k$																											
0	(1.00,0.00,0.00,0.00,0.00,0.00)	0.00																											
1	(0.00,0.00,0.00,0.00,0.00,1.00)	2.00																											
2	(0.00,0.00,0.00,0.00,0.45,0.89)	1.20																											
3	(0.00,0.00,0.00,0.33,0.67,0.67)	1.33																											
	$\vdots$																												
26	(0.06,0.10,0.17,0.28,0.48,0.81)	1.69																											
27	(0.06,0.10,0.17,0.28,0.48,0.81)	1.68																											
johnston-linear-matrix-algebra-F03411	374	0.992	$1.68,-0.72,-0.17 \pm 1.32 i$	$1.68,-0.72,-0.17 \pm 1.32i$	indeed, it is—the roots of p are approximately $1.68,-0.72,-0.17 \pm 1.32i$,																								
johnston-linear-matrix-algebra-F03412	374	1.000	$0.69 \pm 0.67 i$	$0.69 \pm 0.67i$	and $0.69 \pm 0.67i$).																								
johnston-linear-matrix-algebra-F03413	375	1.000	$(1,1) /\sqrt{2}$	$(1,1) / \sqrt{2}$	$(1,1) / \sqrt{2}$																								
johnston-linear-matrix-algebra-F03414	375	1.000	$(1,3) /\sqrt{10}$	$(1,3) / \sqrt{10}$	$(1,3) / \sqrt{10}$. Similarly,																								
johnston-linear-matrix-algebra-F03415	375	1.000	$k=1,2,3, \ldots$	$k=1,2,3, \ldots$	ting $\mathbf{v}_k=A \mathbf{v}_{k-1} /\left\ A \mathbf{v}_{k-1}\right\ $ for $k=1,2,3, \ldots$, then absolutely nothing																								
johnston-linear-matrix-algebra-F03416	375	1.000	$\mathbf{v}_{0}=(0.80,0.60)$	$\mathbf{v}_0=(0.80,0.60)$	be $\mathbf{v}_0=(0.80,0.60)$ then power iteration does indeed converge to the																								
johnston-linear-matrix-algebra-F03417	375	1.000	$A=\left[\begin{array}{cc} 1 & 1 \\ 3 & -1 \end{array}\right]$	$A=\left[\begin{array}{cc} 1 & 1 \\ 3 & -1 \end{array}\right]$	Apply power iteration to the matrix $A=\left[\begin{array}{cc} 1 & 1 \\ 3 & -1 \end{array}\right]$.																								
johnston-linear-matrix-algebra-F03418	376	1.000	$y=-3 x$	$y=-3 x$	$y=-3 x$ are the																								
johnston-linear-matrix-algebra-F03419	376	1.000	$A^2=4 I$	$A^2=4 I$	“winning” as A is applied more times. In particular, $A^2=4 I$, so applying A																								
johnston-linear-matrix-algebra-F03420	376	1.000	$\mathbf{v}_{1}=A \mathbf{v}_{0} /\left A \mathbf{v}_{0}\right $	$\mathbf{v}_1=A \mathbf{v}_0 /\left\ A \mathbf{v}_0\right\ $	a real vector $\mathbf{v}_0$ then $\mathbf{v}_1=A \mathbf{v}_0 /\left\ A \mathbf{v}_0\right\ $ will also be real, as will $\mathbf{v}_2, \mathbf{v}_3$, and so																								
johnston-linear-matrix-algebra-F03421	376	1.000	$\mathbf{v}_{2}, \mathbf{v}_{3}$	$\mathbf{v}_2, \mathbf{v}_3$	a real vector $\mathbf{v}_0$ then $\mathbf{v}_1=A \mathbf{v}_0 /\left\ A \mathbf{v}_0\right\ $ will also be real, as will $\mathbf{v}_2, \mathbf{v}_3$, and so																								
johnston-linear-matrix-algebra-F03422	376	1.000	$ \lambda =\left \bar{\lambda}\right $	$ \lambda =\left \bar{\lambda}\right $	complex conjugates have the same absolute value of each other: $ \lambda =\left \bar{\lambda}\right $ for																								
johnston-linear-matrix-algebra-F03423	376	0.999	$1 \pm i$	$1 \pm i$	has eigenvalues $1 \pm i$, which have the same absolute value as each other:																								
johnston-linear-matrix-algebra-F03424	377	1.000	$\mathbf{v}_{0} \in \mathbb{C}_n$	$\mathbf{v}_0 \in \mathbb{C}_n$	$\mathbf{v}_0 \in \mathbb{C}_n$ instead.																								
johnston-linear-matrix-algebra-F03425	377	1.000	$ 1+i = 1-i =\sqrt{2}$	$ 1+i = 1-i =\sqrt{2}$	$ 1+i = 1-i =\sqrt{2}$. Power iteration thus does not necessarily converge to																								
johnston-linear-matrix-algebra-F03426	377	1.000	$\left\{\mathbf{w}_{1}, \mathbf{w}_{2}, \ldots, \mathbf{w}_n\right\}$	$\left\{\mathbf{w}_1, \mathbf{w}_2, \ldots, \mathbf{w}_n\right\}$	$\left\{\mathbf{w}_1, \mathbf{w}_2, \ldots, \mathbf{w}_n\right\}$ with corresponding eigenvalues $\lambda_1, \lambda_2, \ldots, \lambda_n$, respec-																								

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johnston-linear-matrix-algebra-F03427	377	1.000	$\left \lambda_1\right >\left \lambda_j\right $	$ \lambda_1 > \lambda_j $	$\lambda_1 > \lambda_j$ for all $2 \leq j \leq n$, and
johnston-linear-matrix-algebra-F03428	377	1.000	$2 \leq j \leq n$	$2 \leq j \leq n$	$\lambda_1 > \lambda_j$ for all $2 \leq j \leq n$, and
johnston-linear-matrix-algebra-F03429	377	1.000	$\mathbf{v}_0 \notin \operatorname{span}\{\mathbf{w}_2, \mathbf{w}_3, \dots, \mathbf{w}_n\}$	$\mathbf{v}_0 \notin \operatorname{span}\{\mathbf{w}_2, \mathbf{w}_3, \dots, \mathbf{w}_n\}$	$\mathbf{v}_0 \notin \operatorname{span}\{\mathbf{w}_2, \mathbf{w}_3, \dots, \mathbf{w}_n\}$.
johnston-linear-matrix-algebra-F03430	377	1.000	$\lim_{k \rightarrow \infty} \mathbf{v}_k^* A \mathbf{v}_k = \lambda_1$	$\lim_{k \rightarrow \infty} \mathbf{v}_k^* A \mathbf{v}_k = \lambda_1$	If we define $\mathbf{v}_k = A \mathbf{v}_{k-1} / \ A \mathbf{v}_{k-1}\ $ for all $k \geq 1$ then $\lim_{k \rightarrow \infty} \mathbf{v}_k^* A \mathbf{v}_k = \lambda_1$.
johnston-linear-matrix-algebra-F03431	377	1.000	$\mathbf{v}_k = A^k \mathbf{v}_0 / \ A^k \mathbf{v}_0\ $	$\mathbf{v}_k = A^k \mathbf{v}_0 / \ A^k \mathbf{v}_0\ $	for $\mathbf{v}_k$ a bit more explicitly as $\mathbf{v}_k = A^k \mathbf{v}_0 / \ A^k \mathbf{v}_0\ $. With this in mind, we begin
johnston-linear-matrix-algebra-F03432	377	1.000	$A^k \mathbf{v}_0$	$A^k \mathbf{v}_0$	by investigating what happens to $A^k \mathbf{v}_0$ as k gets large.
johnston-linear-matrix-algebra-F03433	377	1.000	$\mathbf{w}_j$	$\mathbf{w}_j$	Without loss of generality, we can assume that each eigenvector $\mathbf{w}_j$ is a
johnston-linear-matrix-algebra-F03434	377	1.000	$A \mathbf{w}_j = \lambda_j \mathbf{w}_j$	$A \mathbf{w}_j = \lambda_j \mathbf{w}_j$	Then repeatedly using the fact that $A \mathbf{w}_j = \lambda_j \mathbf{w}_j$ for each $1 \leq j \leq n$ gives
johnston-linear-matrix-algebra-F03435	377	0.985	$ \lambda_j / \lambda_1 < 1$	$ \lambda_j / \lambda_1 < 1$	We then notice that, since $ \lambda_1 > \lambda_j $ for all $2 \leq j \leq n$, we have $ \lambda_j / \lambda_1 < 1$
johnston-linear-matrix-algebra-F03436	377	1.000	$(\lambda_j / \lambda_1)^k \rightarrow 0$	$(\lambda_j / \lambda_1)^k \rightarrow 0$	and so $(\lambda_j / \lambda_1)^k \rightarrow 0$ as $k \rightarrow \infty$ for all $2 \leq j \leq n$ as well. For simplicity, we
johnston-linear-matrix-algebra-F03437	377	1.000	$k \rightarrow \infty$	$k \rightarrow \infty$	and so $(\lambda_j / \lambda_1)^k \rightarrow 0$ as $k \rightarrow \infty$ for all $2 \leq j \leq n$ as well. For simplicity, we
johnston-linear-matrix-algebra-F03438	377	1.000	$\mathbf{r}_k$	$\mathbf{r}_k$	then define $\mathbf{r}_k$ to be the “remainder” vector that consists of terms that become
johnston-linear-matrix-algebra-F03439	378	1.000	$ \lambda_1 ^{2k}$	$ \lambda_1 ^{2k}$	The $ \lambda_1 ^{2k}$ term here
johnston-linear-matrix-algebra-F03440	378	0.994	$\bar{\lambda}_1^k$	$\bar{\lambda}_1^k$	combining $\bar{\lambda}_1^k$ from
johnston-linear-matrix-algebra-F03441	378	1.000	$\mathbf{v}_k^*$	$\mathbf{v}_k^*$	the $\mathbf{v}_k^*$ on the left
johnston-linear-matrix-algebra-F03442	378	1.000	λ_1^k	λ_1^k	with λ_1^k from the $\mathbf{v}_k$
johnston-linear-matrix-algebra-F03443	378	1.000	$\lim_{k \rightarrow \infty} \mathbf{r}_k = \mathbf{0}$	$\lim_{k \rightarrow \infty} \mathbf{r}_k = \mathbf{0}$	In particular, by using the fact that $\lim_{k \rightarrow \infty} \mathbf{r}_k = \mathbf{0}$ we can now compute the desired
johnston-linear-matrix-algebra-F03444	378	1.000	$\mathbf{w}_1$	$\mathbf{w}_1$	where at the end we use the facts that $\mathbf{w}_1$ is a unit vector (twice) and that it is
johnston-linear-matrix-algebra-F03445	378	1.000	$(\lambda_1 / \lambda_1)^k$	$(\lambda_1 / \lambda_1)^k$	scalar $(\lambda_1 / \lambda_1)^k$ can change sign (or in the case of complex matrices, bounce
johnston-linear-matrix-algebra-F03446	378	0.990	$\lambda_1 / \lambda_1 = -1$	$\lambda_1 / \lambda_1 = -1$	being negative, and thus $\lambda_1 / \lambda_1 = -1$).
johnston-linear-matrix-algebra-F03447	378	1.000	$ \lambda_2 / \lambda_1 $	$ \lambda_2 / \lambda_1 $	the ratio $ \lambda_2 / \lambda_1 $ of the two largest (in absolute value) eigenvalues of A —the
johnston-linear-matrix-algebra-F03448	378	1.000	$A = \begin{bmatrix} 8 & -1 & 0 \\ -1 & 2 & 5 \\ 0 & 5 & -6 \end{bmatrix}$	$A = \begin{bmatrix} 8 & -1 & 0 \\ -1 & 2 & 5 \\ 0 & 5 & -6 \end{bmatrix}$	Apply power iteration to the matrix $A = \begin{bmatrix} 8 & -1 & 0 \\ -1 & 2 & 5 \\ 0 & 5 & -6 \end{bmatrix}$.
johnston-linear-matrix-algebra-F03449	379	1.000	$\mathbf{v}_{318}$	$\mathbf{v}_{318}$	The vectors $\mathbf{v}_{318}$ and
johnston-linear-matrix-algebra-F03450	379	1.000	$\mathbf{v}_{320}$	$\mathbf{v}_{320}$	$\mathbf{v}_{320}$ are actually
johnston-linear-matrix-algebra-F03451	379	1.000	$\mathbf{v}_{319}$	$\mathbf{v}_{319}$	(and similarly for $\mathbf{v}_{319}$)

Identifier	Page	Conf.	LaTeX source	Rendered	Image																		
johnston-linear-matrix-algebra-F03452	379	1.000	$\mathbf{v}_{321}$	$\mathbf{v}_{321}$	and $\mathbf{v}_{321}$).																		
johnston-linear-matrix-algebra-F03453	379	0.990	$\lambda_3 \approx 4.19$	$\lambda_3 \approx 4.19$	$\lambda_3 \approx 4.19$.																		
johnston-linear-matrix-algebra-F03454	379	1.000	$A - \lambda_1 \mathbf{w}_1 \mathbf{w}_1^T$	$A - \lambda_1 \mathbf{w}_1 \mathbf{w}_1^T$	$A - \lambda_1 \mathbf{w}_1 \mathbf{w}_1^T$ may																		
johnston-linear-matrix-algebra-F03455	379	1.000	$k=63$	$k = 63$	to be very, very wrong. After $k = 63$ iterations, the eigenvalue estimate																		
johnston-linear-matrix-algebra-F03456	379	0.803	$k=156$	$k = 156$	dips back down below 8.00, after $k = 156$ iterations it dips down below																		
johnston-linear-matrix-algebra-F03457	379	1.000	$k=320$	$k = 320$	$k = 320$:																		
johnston-linear-matrix-algebra-F03458	379	0.999	$\lambda_1 \approx -8.41$	$\lambda_1 \approx -8.41$	$\lambda_1 \approx -8.41$ and $\lambda_2 \approx 8.22$ of A are so close to each other in absolute																		
johnston-linear-matrix-algebra-F03459	379	0.999	$\lambda_2 \approx 8.22$	$\lambda_2 \approx 8.22$	$\lambda_1 \approx -8.41$ and $\lambda_2 \approx 8.22$ of A are so close to each other in absolute																		
johnston-linear-matrix-algebra-F03460	379	0.942	$ \lambda_2/\lambda_1 \approx 0.98$	$ \lambda_2/\lambda_1 \approx 0.98$	value (so the ratio $ \lambda_2/\lambda_1 \approx 0.98$ is close to 1). Furthermore, the starting																		
johnston-linear-matrix-algebra-F03461	379	0.763	$\operatorname{span}\{(0.97, -0.22, -0.08)\}$	$\operatorname{span}\{(0.97, -0.22, -0.08)\}$	is $\operatorname{span}\{(0.97, -0.22, -0.08)\}$, so it starts off by getting pulled toward																		
johnston-linear-matrix-algebra-F03462	379	1.000	$\mathbf{v}_k^* A \mathbf{v}_k$	$\mathbf{v}_k^* A \mathbf{v}_k$	plot of the eigenvalue estimate $\mathbf{v}_k^* A \mathbf{v}_k$, as a function of the iteration k , is																		
johnston-linear-matrix-algebra-F03463	380	1.000	$\mathbf{w}_2$	$\mathbf{w}_2$	Keep in mind that $\mathbf{w}_2$																		
johnston-linear-matrix-algebra-F03464	380	0.993	$A=\left[\begin{array}{lll}1 & 2 & 3 \\ 4 & 5 & 4 \\ 3 & 2 & 1\end{array}\right]$	$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 4 \\ 3 & 2 & 1 \end{bmatrix}$	Use power iteration to find all eigenvalues of $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 4 \\ 3 & 2 & 1 \end{bmatrix}$.																		
johnston-linear-matrix-algebra-F03465	380	1.000	$\lambda_1 \approx 8.53$	$\lambda_1 \approx 8.53$	Indeed, the dominant eigenvalue of A is $\lambda_1 \approx 8.53$ with corresponding																		
johnston-linear-matrix-algebra-F03466	380	1.000	$\mathbf{w}_1 \approx (0.38, 0.85, 0.37)$	$\mathbf{w}_1 \approx (0.38, 0.85, 0.37)$	unit eigenvector $\mathbf{w}_1 \approx (0.38, 0.85, 0.37)$.																		
johnston-linear-matrix-algebra-F03467	380	1.000	$\mathbf{v}_k=B \mathbf{v}_{k-1} /\left\ B \mathbf{v}_{k-1}\right\ $	$\mathbf{v}_k = B \mathbf{v}_{k-1} /\left\ B \mathbf{v}_{k-1}\right\ $	<table><tr><th>k</th><th>$\mathbf{v}_k = B \mathbf{v}_{k-1} /\left\ B \mathbf{v}_{k-1}\right\$</th><th>$\mathbf{v}_k^T B \mathbf{v}_k$</th></tr><tr><td>0</td><td>(1.00, 0.00, 0.00)</td><td>−0.21</td></tr><tr><td>1</td><td>(−0.10, 0.57, 0.81)</td><td>−0.52</td></tr><tr><td>2</td><td>(0.81, 0.22, −0.54)</td><td>−1.81</td></tr><tr><td>3</td><td>(−0.68, 0.04, 0.73)</td><td>−1.99</td></tr><tr><td>4</td><td>(0.72, 0.03, −0.69)</td><td>−2.00</td></tr></table>	k	$\mathbf{v}_k = B \mathbf{v}_{k-1} /\left\ B \mathbf{v}_{k-1}\right\ $	$\mathbf{v}_k^T B \mathbf{v}_k$	0	(1.00, 0.00, 0.00)	−0.21	1	(−0.10, 0.57, 0.81)	−0.52	2	(0.81, 0.22, −0.54)	−1.81	3	(−0.68, 0.04, 0.73)	−1.99	4	(0.72, 0.03, −0.69)	−2.00
k	$\mathbf{v}_k = B \mathbf{v}_{k-1} /\left\ B \mathbf{v}_{k-1}\right\ $	$\mathbf{v}_k^T B \mathbf{v}_k$																					
0	(1.00, 0.00, 0.00)	−0.21																					
1	(−0.10, 0.57, 0.81)	−0.52																					
2	(0.81, 0.22, −0.54)	−1.81																					
3	(−0.68, 0.04, 0.73)	−1.99																					
4	(0.72, 0.03, −0.69)	−2.00																					
johnston-linear-matrix-algebra-F03468	380	1.000	$\mathbf{v}_k^T B \mathbf{v}_k$	$\mathbf{v}_k^T B \mathbf{v}_k$	<table><tr><th>k</th><th>$\mathbf{v}_k = B \mathbf{v}_{k-1} /\left\ B \mathbf{v}_{k-1}\right\$</th><th>$\mathbf{v}_k^T B \mathbf{v}_k$</th></tr><tr><td>0</td><td>(1.00, 0.00, 0.00)</td><td>−0.21</td></tr><tr><td>1</td><td>(−0.10, 0.57, 0.81)</td><td>−0.52</td></tr><tr><td>2</td><td>(0.81, 0.22, −0.54)</td><td>−1.81</td></tr><tr><td>3</td><td>(−0.68, 0.04, 0.73)</td><td>−1.99</td></tr><tr><td>4</td><td>(0.72, 0.03, −0.69)</td><td>−2.00</td></tr></table>	k	$\mathbf{v}_k = B \mathbf{v}_{k-1} /\left\ B \mathbf{v}_{k-1}\right\ $	$\mathbf{v}_k^T B \mathbf{v}_k$	0	(1.00, 0.00, 0.00)	−0.21	1	(−0.10, 0.57, 0.81)	−0.52	2	(0.81, 0.22, −0.54)	−1.81	3	(−0.68, 0.04, 0.73)	−1.99	4	(0.72, 0.03, −0.69)	−2.00
k	$\mathbf{v}_k = B \mathbf{v}_{k-1} /\left\ B \mathbf{v}_{k-1}\right\ $	$\mathbf{v}_k^T B \mathbf{v}_k$																					
0	(1.00, 0.00, 0.00)	−0.21																					
1	(−0.10, 0.57, 0.81)	−0.52																					
2	(0.81, 0.22, −0.54)	−1.81																					
3	(−0.68, 0.04, 0.73)	−1.99																					
4	(0.72, 0.03, −0.69)	−2.00																					
johnston-linear-matrix-algebra-F03469	380	1.000	$\lambda_2 \approx -2.00$	$\lambda_2 \approx -2.00$	in absolute value) is thus $\lambda_2 \approx -2.00$, with corresponding unit eigenvector																		
johnston-linear-matrix-algebra-F03470	380	1.000	$\mathbf{w}_2 \approx (0.72, 0.03, -0.69)$	$\mathbf{w}_2 \approx (0.72, 0.03, -0.69)$	$\mathbf{w}_2 \approx (0.72, 0.03, -0.69)$.																		
johnston-linear-matrix-algebra-F03471	381	1.000	$\lambda_3 \approx 0.47$	$\lambda_3 \approx 0.47$	value) is thus $\lambda_3 \approx 0.47$.																		

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johnston-linear-matrix-algebra-F03472	381	1.000	$-2, (9 + \sqrt{65}) / 2 \approx 8.53$	$-2, (9 + \sqrt{65})/2 \approx 8.53$	which indeed has roots $-2, (9 + \sqrt{65})/2 \approx 8.53$, and $(9 - \sqrt{65})/2 \approx 0.47$.
johnston-linear-matrix-algebra-F03473	381	1.000	$(9 - \sqrt{65}) / 2 \approx 0.47$	$(9 - \sqrt{65})/2 \approx 0.47$	which indeed has roots $-2, (9 + \sqrt{65})/2 \approx 8.53$, and $(9 - \sqrt{65})/2 \approx 0.47$.
johnston-linear-matrix-algebra-F03474	381	1.000	$6.25, -2.55 \pm 1.88 i$	$6.25, -2.55 \pm 1.88i$	has eigenvalues equal to approximately $6.25, -2.55 \pm 1.88i$, and -0.16 . How-
johnston-linear-matrix-algebra-F03475	381	1.000	$-2.55 \pm 1.88 i$	$-2.55 \pm 1.88i$	that the next two eigenvalues $-2.55 \pm 1.88i$ are equal to each other in absolute
johnston-linear-matrix-algebra-F03476	381	0.985	$\left[\begin{array}{ll} 3 & 1 \\ 2 & 7 \end{array} \right]$	$\begin{bmatrix} 3 & 1 \\ 2 & 7 \end{bmatrix}$	Positive Not positive $\begin{bmatrix} 3 & 1 \\ 2 & 7 \end{bmatrix} \quad \begin{bmatrix} 2 & 4 & 1 \\ 8 & 1 & 3 \end{bmatrix} \quad \left \quad \begin{bmatrix} 1 & -1 \\ -2 & 3 \end{bmatrix} \quad \begin{bmatrix} 4 & 3 & 2 \\ 1 & 0 & 1 \end{bmatrix}.$
johnston-linear-matrix-algebra-F03477	381	0.985	$\left[\begin{array}{lll} 2 & 4 & 1 \\ 8 & 1 & 3 \end{array} \right]$	$\begin{bmatrix} 2 & 4 & 1 \\ 8 & 1 & 3 \end{bmatrix}$	Positive Not positive $\begin{bmatrix} 3 & 1 \\ 2 & 7 \end{bmatrix} \quad \begin{bmatrix} 2 & 4 & 1 \\ 8 & 1 & 3 \end{bmatrix} \quad \left \quad \begin{bmatrix} 1 & -1 \\ -2 & 3 \end{bmatrix} \quad \begin{bmatrix} 4 & 3 & 2 \\ 1 & 0 & 1 \end{bmatrix}.$
johnston-linear-matrix-algebra-F03478	381	0.985	$\left[\begin{array}{cc} 1 & -1 \\ -2 & 3 \end{array} \right]$	$\begin{bmatrix} 1 & -1 \\ -2 & 3 \end{bmatrix}$	Positive Not positive $\begin{bmatrix} 3 & 1 \\ 2 & 7 \end{bmatrix} \quad \begin{bmatrix} 2 & 4 & 1 \\ 8 & 1 & 3 \end{bmatrix} \quad \left \quad \begin{bmatrix} 1 & -1 \\ -2 & 3 \end{bmatrix} \quad \begin{bmatrix} 4 & 3 & 2 \\ 1 & 0 & 1 \end{bmatrix}.$
johnston-linear-matrix-algebra-F03479	381	0.985	$\left[\begin{array}{lll} 4 & 3 & 2 \\ 1 & 0 & 1 \end{array} \right]$	$\begin{bmatrix} 4 & 3 & 2 \\ 1 & 0 & 1 \end{bmatrix}$	Positive Not positive $\begin{bmatrix} 3 & 1 \\ 2 & 7 \end{bmatrix} \quad \begin{bmatrix} 2 & 4 & 1 \\ 8 & 1 & 3 \end{bmatrix} \quad \left \quad \begin{bmatrix} 1 & -1 \\ -2 & 3 \end{bmatrix} \quad \begin{bmatrix} 4 & 3 & 2 \\ 1 & 0 & 1 \end{bmatrix}.$
johnston-linear-matrix-algebra-F03480	382	1.000	$[A \mathbf{v}]_j$	$[A\mathbf{v}]_j$	The notation $[A\mathbf{v}]_j$
johnston-linear-matrix-algebra-F03481	382	0.662	$\lambda_1 > \lambda_j $	$\lambda_1 > \lambda_j $	$\lambda_1 > \lambda_j $ for all $2 \leq j \leq n$.
johnston-linear-matrix-algebra-F03482	382	1.000	$\lambda_j (2 \leq j \leq n)$	$\lambda_j (2 \leq j \leq n)$	d) Every eigenvector corresponding to an eigenvalue $\lambda_j (2 \leq j \leq n)$
johnston-linear-matrix-algebra-F03483	382	1.000	$\mathbf{v} \geq \mathbf{w}$	$\mathbf{v} \geq \mathbf{w}$	the proof. For brevity, we write $\mathbf{v} \geq \mathbf{w}$ and $\mathbf{v} > \mathbf{w}$ to indicate that $v_j \geq w_j$ or
johnston-linear-matrix-algebra-F03484	382	1.000	$\mathbf{v} > \mathbf{w}$	$\mathbf{v} > \mathbf{w}$	the proof. For brevity, we write $\mathbf{v} \geq \mathbf{w}$ and $\mathbf{v} > \mathbf{w}$ to indicate that $v_j \geq w_j$ or
johnston-linear-matrix-algebra-F03485	382	1.000	$v_j \geq w_j$	$v_j \geq w_j$	the proof. For brevity, we write $\mathbf{v} \geq \mathbf{w}$ and $\mathbf{v} > \mathbf{w}$ to indicate that $v_j \geq w_j$ or
johnston-linear-matrix-algebra-F03486	382	0.999	$v_j > w_j$	$v_j > w_j$	$v_j > w_j$ for all j , respectively (just like we did in Section 2.B). In particular,
johnston-linear-matrix-algebra-F03487	382	0.874	$\mathbf{v} \geq \mathbf{0}$	$\mathbf{v} \geq \mathbf{0}$	this means that non-negativity of $\mathbf{v}$ can be denoted by $\mathbf{v} \geq \mathbf{0}$, and positivity of $\mathbf{v}$
johnston-linear-matrix-algebra-F03488	382	1.000	$\mathbf{v} > \mathbf{0}$	$\mathbf{v} > \mathbf{0}$	can be denoted by $\mathbf{v} > \mathbf{0}$. We also let $S^n \subset \mathbb{R}^n$ be the set of unit vectors with
johnston-linear-matrix-algebra-F03489	382	1.000	$S^n \subset \mathbb{R}^n$	$S^n \subset \mathbb{R}^n$	can be denoted by $\mathbf{v} > \mathbf{0}$. We also let $S^n \subset \mathbb{R}^n$ be the set of unit vectors with
johnston-linear-matrix-algebra-F03490	382	1.000	S^n	S^n	Figure 3.28: The set S^n is the portion of the unit circle/sphere/hypersphere that is
johnston-linear-matrix-algebra-F03491	382	1.000	$\mathbf{v} \in S^n$	$\mathbf{v} \in S^n$	Note that if $\mathbf{v} \in S^n$ then $A\mathbf{v} > \mathbf{0}$, since it is computed by multiplying together
johnston-linear-matrix-algebra-F03492	382	1.000	$A \mathbf{v} > \mathbf{0}$	$A\mathbf{v} > \mathbf{0}$	Note that if $\mathbf{v} \in S^n$ then $A\mathbf{v} > \mathbf{0}$, since it is computed by multiplying together
johnston-linear-matrix-algebra-F03493	382	1.000	$L: S^n \rightarrow \mathbb{R}$	$L: S^n \rightarrow \mathbb{R}$	positive. We can thus define a function $L: S^n \rightarrow \mathbb{R}$ by

Identifier	Page	Conf.	LaTeX source	Rendered	Image
johnston-linear-matrix-algebra-F03494	382	1.000	$c > 0$	$c > 0$	In particular, there exists a $c > 0$ such that $A\mathbf{v} \geq c\mathbf{v}$ since $A\mathbf{v} > \mathbf{0}$ (so $L(\mathbf{v}) > 0$ for
johnston-linear-matrix-algebra-F03495	382	1.000	$A \, \mathbf{v} \geq c \, \mathbf{v}$	$A\mathbf{v} \geq c\mathbf{v}$	In particular, there exists a $c > 0$ such that $A\mathbf{v} \geq c\mathbf{v}$ since $A\mathbf{v} > \mathbf{0}$ (so $L(\mathbf{v}) > 0$ for
johnston-linear-matrix-algebra-F03496	382	1.000	$L(\mathbf{v}) > 0$	$L(\mathbf{v}) > 0$	In particular, there exists a $c > 0$ such that $A\mathbf{v} \geq c\mathbf{v}$ since $A\mathbf{v} > \mathbf{0}$ (so $L(\mathbf{v}) > 0$ for
johnston-linear-matrix-algebra-F03497	382	1.000	$L(\mathbf{v})$	$L(\mathbf{v})$	such c (i.e., the value of $L(\mathbf{v})$) can be found by increasing c until one of the
johnston-linear-matrix-algebra-F03498	383	1.000	$ [B \, \mathbf{w}]_i $	$ [B\mathbf{w}]_i $	clear, $ [B\mathbf{w}]_i $ is the
johnston-linear-matrix-algebra-F03499	383	1.000	$B \, \mathbf{w}$	$B\mathbf{w}$	the i -th entry of $B\mathbf{w}$,
johnston-linear-matrix-algebra-F03500	383	0.998	$[B \mathbf{w}]_i$	$[B \mathbf{w}]_i$	while $[B \mathbf{w}]_i$ is the
johnston-linear-matrix-algebra-F03501	383	1.000	$B \mathbf{w} $	$B \mathbf{w} $	i -th entry of $B \mathbf{w} $.
johnston-linear-matrix-algebra-F03502	383	1.000	$\mu \in \mathbb{R}$	$\mu \in \mathbb{R}$	is, there is a particular number $\mu \in \mathbb{R}$ and a particular vector $\mathbf{v}_* \in S^n$ with the
johnston-linear-matrix-algebra-F03503	383	1.000	$\mathbf{v}_* \in S^n$	$\mathbf{v}_* \in S^n$	is, there is a particular number $\mu \in \mathbb{R}$ and a particular vector $\mathbf{v}_* \in S^n$ with the
johnston-linear-matrix-algebra-F03504	383	1.000	$L(\mathbf{v}_*) = \mu \geq L(\mathbf{v})$	$L(\mathbf{v}_*) = \mu \geq L(\mathbf{v})$	property that $L(\mathbf{v}_*) = \mu \geq L(\mathbf{v})$ for all $\mathbf{v} \in S^n$.
johnston-linear-matrix-algebra-F03505	383	0.998	$A, \mathbf{v}_*$	$A, \mathbf{v}_*$	between $A, \mathbf{v}_*$, and μ that we will use repeatedly.
johnston-linear-matrix-algebra-F03506	383	0.998	μ	μ	between $A, \mathbf{v}_*$, and μ that we will use repeatedly.
johnston-linear-matrix-algebra-F03507	383	1.000	$L(\mathbf{v}) = \mu$	$L(\mathbf{v}) = \mu$	Claim 1: If $\mathbf{v} \in S^n$ and $L(\mathbf{v}) = \mu$ then $A\mathbf{v} = \mu\mathbf{v}$. To see why this claim holds,
johnston-linear-matrix-algebra-F03508	383	1.000	$A \, \mathbf{v} = \mu \mathbf{v}$	$A\mathbf{v} = \mu\mathbf{v}$	Claim 1: If $\mathbf{v} \in S^n$ and $L(\mathbf{v}) = \mu$ then $A\mathbf{v} = \mu\mathbf{v}$. To see why this claim holds,
johnston-linear-matrix-algebra-F03509	383	1.000	$A \, \mathbf{v} \geq \mu \mathbf{v}$	$A\mathbf{v} \geq \mu\mathbf{v}$	notice that $L(\mathbf{v}) = \mu$ implies $A\mathbf{v} \geq \mu\mathbf{v}$ and thus $A\mathbf{v} - \mu\mathbf{v} \geq \mathbf{0}$. Suppose
johnston-linear-matrix-algebra-F03510	383	1.000	$A \, \mathbf{v} - \mu \mathbf{v} \geq \mathbf{0}$	$A\mathbf{v} - \mu\mathbf{v} \geq \mathbf{0}$	notice that $L(\mathbf{v}) = \mu$ implies $A\mathbf{v} \geq \mu\mathbf{v}$ and thus $A\mathbf{v} - \mu\mathbf{v} \geq \mathbf{0}$. Suppose
johnston-linear-matrix-algebra-F03511	383	0.985	$A \, \mathbf{v} - \mu \mathbf{v} \neq \mathbf{0}$	$A\mathbf{v} - \mu\mathbf{v} \neq \mathbf{0}$	(for the purpose of establishing a contradiction) that $A\mathbf{v} - \mu\mathbf{v} \neq \mathbf{0}$. Since
johnston-linear-matrix-algebra-F03512	383	1.000	$A(A\mathbf{v} - \mu\mathbf{v}) > \mathbf{0}$	$A(A\mathbf{v} - \mu\mathbf{v}) > \mathbf{0}$	A is positive, this implies that $A(A\mathbf{v} - \mu\mathbf{v}) > \mathbf{0}$, and re-arranging then
johnston-linear-matrix-algebra-F03513	383	1.000	$A(A\mathbf{v}) > \mu A\mathbf{v}$	$A(A\mathbf{v}) > \mu A\mathbf{v}$	gives $A(A\mathbf{v}) > \mu A\mathbf{v}$. It follows that
johnston-linear-matrix-algebra-F03514	383	1.000	$\mu \geq L(A\mathbf{v} / \ A\mathbf{v}\)$	$\mu \geq L(A\mathbf{v} / \ A\mathbf{v}\)$	so $\mu \geq L(A\mathbf{v} / \ A\mathbf{v}\)$. It follows that our assumption that $A\mathbf{v} - \mu\mathbf{v} \neq \mathbf{0}$
johnston-linear-matrix-algebra-F03515	383	1.000	$\mathbf{v}^*$	$\mathbf{v}^*$	eigenvector $\mathbf{v}^*$. The remaining claims will help us understand this particular
johnston-linear-matrix-algebra-F03516	383	1.000	$1 \leq i \leq n$	$1 \leq i \leq n$	so for each $1 \leq i \leq n$, the sum $[A\mathbf{v}]_i = a_{i,1}v_1 + a_{i,2}v_2 + \cdots + a_{i,n}v_n$ has at
johnston-linear-matrix-algebra-F03517	383	1.000	$[A\mathbf{v}]_i = a_{i,1}v_1 + a_{i,2}v_2 + \cdots + a_{i,n}v_n$	$[A\mathbf{v}]_i = a_{i,1}v_1 + a_{i,2}v_2 + \cdots + a_{i,n}v_n$	so for each $1 \leq i \leq n$, the sum $[A\mathbf{v}]_i = a_{i,1}v_1 + a_{i,2}v_2 + \cdots + a_{i,n}v_n$ has at
johnston-linear-matrix-algebra-F03518	383	1.000	$\mathbf{v} = (A\mathbf{v}) / \mu > \mathbf{0}$	$\mathbf{v} = (A\mathbf{v}) / \mu > \mathbf{0}$	$\mathbf{v} = (A\mathbf{v}) / \mu > \mathbf{0}$ as well.
johnston-linear-matrix-algebra-F03519	383	1.000	$\mathbf{v}_*$	$\mathbf{v}_*$	Claim 3 above shows that the eigenvector $\mathbf{v}_*$ of A is positive. Since we want the
johnston-linear-matrix-algebra-F03520	383	1.000	$\mu = \lambda_1$	$\mu = \lambda_1$	we must prove that $\mu = \lambda_1$. The fourth and final claim demonstrates this fact:

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johnston-linear-matrix-algebra-F03521	383	1.000	$B \in \mathcal{M}_n(\mathbb{R})$	$B \in \mathcal{M}_n(\mathbb{R})$	Claim 4: If $B \in \mathcal{M}_n(\mathbb{R})$ is such that $O \leq B \leq A$ and λ is an eigenvalue of B
johnston-linear-matrix-algebra-F03522	383	1.000	$0 \leq B \leq A$	$O \leq B \leq A$	Claim 4: If $B \in \mathcal{M}_n(\mathbb{R})$ is such that $O \leq B \leq A$ and λ is an eigenvalue of B
johnston-linear-matrix-algebra-F03523	383	1.000	$ \lambda \leq \mu$	$ \lambda \leq \mu$	then $ \lambda \leq \mu$. To see why this property holds, notice that if $\mathbf{w}$ is a unit
johnston-linear-matrix-algebra-F03524	383	0.997	$ \mathbf{w} $	$ \mathbf{w} $	fact that A is positive. If we let $ \mathbf{w} $ denote the vector whose entries are
johnston-linear-matrix-algebra-F03525	383	0.996	$ \mathbf{w} = (\left w_1\right , \left w_2\right , \ldots, \left w_n\right)$	$ \mathbf{w} = (w_1 , w_2 , \dots, w_n)$	the absolute values of the entries of $\mathbf{w}$ (i.e., $ \mathbf{w} = (w_1 , w_2 , \dots, w_n)$)
johnston-linear-matrix-algebra-F03526	383	1.000	$B \mathbf{w} \geq \lambda \mathbf{w} $	$B \mathbf{w} \geq \lambda \mathbf{w} $	then the above inequalities implies $B \mathbf{w} \geq \lambda \mathbf{w} $. However, since $A - B$
johnston-linear-matrix-algebra-F03527	383	1.000	$(A - B) \mathbf{w} \geq \mathbf{0}$	$(A - B) \mathbf{w} \geq \mathbf{0}$	is entry-wise non-negative, it is also the case that $(A - B) \mathbf{w} \geq \mathbf{0}$, so
johnston-linear-matrix-algebra-F03528	383	1.000	$A \mathbf{w} \geq B \mathbf{w} \geq \lambda \mathbf{w} $	$A \mathbf{w} \geq B \mathbf{w} \geq \lambda \mathbf{w} $	$A \mathbf{w} \geq B \mathbf{w} \geq \lambda \mathbf{w} $. It follows that $L(\mathbf{w}) \geq \lambda $, but since $\mu \geq L(\mathbf{w})$
johnston-linear-matrix-algebra-F03529	383	1.000	$L(\mathbf{w}) \geq \lambda $	$L(\mathbf{w}) \geq \lambda $	$A \mathbf{w} \geq B \mathbf{w} \geq \lambda \mathbf{w} $. It follows that $L(\mathbf{w}) \geq \lambda $, but since $\mu \geq L(\mathbf{w})$
johnston-linear-matrix-algebra-F03530	383	1.000	$\mu \geq L(\mathbf{w})$	$\mu \geq L(\mathbf{w})$	$A \mathbf{w} \geq B \mathbf{w} \geq \lambda \mathbf{w} $. It follows that $L(\mathbf{w}) \geq \lambda $, but since $\mu \geq L(\mathbf{w})$
johnston-linear-matrix-algebra-F03531	383	1.000	$\mu \geq \lambda $	$\mu \geq \lambda $	we then have $\mu \geq \lambda $.
johnston-linear-matrix-algebra-F03532	384	1.000	$\mu \leq \lambda$	$\mu \leq \lambda$	If $\mu \leq \lambda$ then we can
johnston-linear-matrix-algebra-F03533	384	1.000	$\mu \geq \lambda$	$\mu \geq \lambda$	arbitrarily. If $\mu \geq \lambda$
johnston-linear-matrix-algebra-F03534	384	1.000	$\mathbf{v}_1 > c\mathbf{v}$	$\mathbf{v}_1 > c\mathbf{v}$	that $\mathbf{v}_1 > c\mathbf{v}$.
johnston-linear-matrix-algebra-F03535	384	1.000	$B = A$	$B = A$	In particular, if we choose $B = A$ in Claim 4 then we see that μ is indeed
johnston-linear-matrix-algebra-F03536	384	1.000	$\lambda = \mu$	$\lambda = \mu$	goal is to show that this implies $\lambda = \mu$). Then $A\mathbf{v} = \lambda\mathbf{v}$, and since $A\mathbf{v} > \mathbf{0}$ (by
johnston-linear-matrix-algebra-F03537	384	1.000	$\lambda > 0$	$\lambda > 0$	Claim 2), we conclude that $\lambda > 0$. Then there exists some real scalar $c \neq 0$
johnston-linear-matrix-algebra-F03538	384	1.000	$c\mu \geq c\lambda$	$c\mu \geq c\lambda$	such that (i) $c\mu \geq c\lambda$, and (ii) $\mathbf{v}_1 > c\mathbf{v}$, so
johnston-linear-matrix-algebra-F03539	384	1.000	$A(\mathbf{v}_1 - c\mathbf{v}) =$	$A(\mathbf{v}_1 - c\mathbf{v}) =$	ity (3.B.3) is actually an equality. Claim 1 then tells us that $A(\mathbf{v}_1 - c\mathbf{v}) =$
johnston-linear-matrix-algebra-F03540	384	1.000	$\mu(\mathbf{v}_1 - c\mathbf{v})$	$\mu(\mathbf{v}_1 - c\mathbf{v})$	$\mu(\mathbf{v}_1 - c\mathbf{v})$, so $\mathbf{v}_1 - c\mathbf{v}$ is an eigenvector of A corresponding to the eigenvalue
johnston-linear-matrix-algebra-F03541	384	1.000	$\mathbf{v}_1 - c\mathbf{v}$	$\mathbf{v}_1 - c\mathbf{v}$	$\mu(\mathbf{v}_1 - c\mathbf{v})$, so $\mathbf{v}_1 - c\mathbf{v}$ is an eigenvector of A corresponding to the eigenvalue
johnston-linear-matrix-algebra-F03542	384	1.000	$ \lambda = \mu$	$ \lambda = \mu$	goal is to show that if $ \lambda = \mu$ then $\lambda = \mu$.
johnston-linear-matrix-algebra-F03543	384	1.000	$b_{i,j}w_j$	$b_{i,j}w_j$	the terms $b_{i,j}w_j$ are non-negative real multiples of each other. Since $b_{i,j} > 0$
johnston-linear-matrix-algebra-F03544	384	1.000	$b_{i,j} > 0$	$b_{i,j} > 0$	the terms $b_{i,j}w_j$ are non-negative real multiples of each other. Since $b_{i,j} > 0$
johnston-linear-matrix-algebra-F03545	385	1.000	$\mathbf{v}_5 = (0.58, 0.58, 0.58)$	$\mathbf{v}_5 = (0.58, 0.58, 0.58)$	$\mathbf{v}_5 = (0.58, 0.58, 0.58)$.
johnston-linear-matrix-algebra-F03546	385	1.000	$\lambda = 7$	$\lambda = 7$	The calculation above suggests that $\lambda = 7$ is the dominant eigenvalue
johnston-linear-matrix-algebra-F03547	385	1.000	$\lambda\mathbf{v}$	$\lambda\mathbf{v}$	is much easier to just compute $A\mathbf{v}$ and see that it equals $\lambda\mathbf{v}$:

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johnston-linear-matrix-algebra-F03548	385	1.000	$n \approx 5$	$n \approx 5$	web pages to be ranked ($n \approx 5$ billion at the time of this writing) and we let r_i
johnston-linear-matrix-algebra-F03549	385	1.000	$r_{\{i\}}$	r_i	web pages to be ranked ($n \approx 5$ billion at the time of this writing) and we let r_i
johnston-linear-matrix-algebra-F03550	385	1.000	$i, p_{\{j\}}$	i, p_j	denote the rank of page i , p_j denote the number of pages that page j links to,
johnston-linear-matrix-algebra-F03551	386	0.915	$\mathrm{A}, \mathrm{B}, \mathrm{C}, \mathrm{D}$	A, B, C, D	Suppose that 5 web pages, which we call A, B, C, D, and E, link to each
johnston-linear-matrix-algebra-F03552	386	1.000	$r_{\{A\}}, r_{\{B\}}, r_{\{C\}}, r_{\{D\}}$	r_A, r_B, r_C, r_D	system (3.B.4) for determining the ranks r_A, r_B, r_C, r_D , and r_E of these
johnston-linear-matrix-algebra-F03553	386	1.000	$r_{\{E\}}$	r_E	system (3.B.4) for determining the ranks r_A, r_B, r_C, r_D , and r_E of these
johnston-linear-matrix-algebra-F03554	386	0.999	$\left(r_{\{A\}}, r_{\{B\}}, r_{\{C\}}, r_{\{D\}}, r_{\{E\}}\right)=$	$(r_A, r_B, r_C, r_D, r_E) =$	tions of this linear system are exactly the multiples of $(r_A, r_B, r_C, r_D, r_E) =$
johnston-linear-matrix-algebra-F03555	386	1.000	$r_{\{A\}} > r_{\{B\}} = r_{\{C\}} > r_{\{E\}} > r_{\{D\}}$	$r_A > r_B = r_C > r_E > r_D$	ular, we have $r_A > r_B = r_C > r_E > r_D$ as the resulting ranking of how
johnston-linear-matrix-algebra-F03556	386	1.000	$1 / p_{\{j\}}$	$1/p_j$	$1/p_j$, where p_j is the total number of edges coming out of vertex j . For
johnston-linear-matrix-algebra-F03557	386	1.000	$p_{\{j\}}$	p_j	$1/p_j$, where p_j is the total number of edges coming out of vertex j . For
johnston-linear-matrix-algebra-F03558	387	1.000	$A \mathbf{r} = \mathbf{r}$	$A\mathbf{r} = \mathbf{r}$	pages then simply has the form $A\mathbf{r} = \mathbf{r}$, where A is the adjacency matrix
johnston-linear-matrix-algebra-F03559	387	1.000	$\mathbf{r} = \left(r_{\{A\}}, r_{\{B\}}, r_{\{C\}}, r_{\{D\}}, r_{\{E\}}\right)$	$\mathbf{r} = (r_A, r_B, r_C, r_D, r_E)$	It is straightforward to verify that if $\mathbf{r} = (r_A, r_B, r_C, r_D, r_E)$ then the linear
johnston-linear-matrix-algebra-F03560	388	1.000	$\lambda_2 = -0.77$	$\lambda_2 = -0.77$	$\lambda_2 = -0.77$ that is
johnston-linear-matrix-algebra-F03561	388	0.562	$\lambda_1 = 1 \mathrm{in}$	$\lambda_1 = 1$ in	eigenvalue $\lambda_1 = 1$ in
johnston-linear-matrix-algebra-F03562	388	0.998	$(-1 \pm i \sqrt{3}) / 2$	$(-1 \pm i\sqrt{3})/2$	has eigenvalues 1 and $(-1 \pm i\sqrt{3})/2$, each of which have absolute value 1.
johnston-linear-matrix-algebra-F03563	388	1.000	$A + \varepsilon J$	$A + \varepsilon J$	it is typically applied to $A + \varepsilon J$ instead, where J is the matrix of appropriate
johnston-linear-matrix-algebra-F03564	388	1.000	J	J	it is typically applied to $A + \varepsilon J$ instead, where J is the matrix of appropriate
johnston-linear-matrix-algebra-F03565	388	1.000	ε	ε	size with every entry equal to 1 and ε is some very small (but positive) real
johnston-linear-matrix-algebra-F03566	389	1.000	$\left[\begin{array}{cc} 3 & -1 \\ -4 & 5 \end{array}\right]$	$\begin{bmatrix} 3 & -1 \\ -4 & 5 \end{bmatrix}$	(b) $\begin{bmatrix} 3 & -1 \\ -4 & 5 \end{bmatrix}$
johnston-linear-matrix-algebra-F03567	389	0.877	$\left[\begin{array}{ccc} -2 & 1 & 1 \\ 4 & 1 & 2 \\ 1 & 0 & -2 \end{array}\right]$	$\begin{bmatrix} -2 & 1 & 1 \\ 4 & 1 & 2 \\ 1 & 0 & -2 \end{bmatrix}$	*(c) $\begin{bmatrix} -2 & 1 & 1 \\ 4 & 1 & 2 \\ 1 & 0 & -2 \end{bmatrix}$
johnston-linear-matrix-algebra-F03568	389	1.000	$\left[\begin{array}{ccc} 3 & -1 & 2 \\ 0 & 4 & 1 \\ -4 & 1 & 2 \end{array}\right]$	$\begin{bmatrix} 3 & -1 & 2 \\ 0 & 4 & 1 \\ -4 & 1 & 2 \end{bmatrix}$	(d) $\begin{bmatrix} 3 & -1 & 2 \\ 0 & 4 & 1 \\ -4 & 1 & 2 \end{bmatrix}$
johnston-linear-matrix-algebra-F03569	389	0.497	$\left[\begin{array}{llll} 3 & 2 & 4 & 3 \\ 1 & 1 & 1 & 3 \\ 3 & 4 & 2 & 0 \\ 0 & 3 & 1 & 2 \end{array}\right]$	$\begin{bmatrix} 3 & 2 & 4 & 3 \\ 1 & 1 & 1 & 3 \\ 3 & 4 & 2 & 0 \\ 0 & 3 & 1 & 2 \end{bmatrix}$	*(e) $\begin{bmatrix} 3 & 2 & 4 & 3 \\ 1 & 1 & 1 & 3 \\ 3 & 4 & 2 & 0 \\ 0 & 3 & 1 & 2 \end{bmatrix}$

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johnston-linear-matrix-algebra-F03570	389	0.999	$\left[\begin{array}{llll}0 & 5 & 5 & 2 \\ 5 & 4 & 7 & 2 \\ 5 & 4 & 0 & 6 \\ 7 & 5 & 5 & 4\end{array}\right]$	$\begin{bmatrix} 0 & 5 & 5 & 2 \\ 5 & 4 & 7 & 2 \\ 5 & 4 & 0 & 6 \\ 7 & 5 & 5 & 4 \end{bmatrix}$	(f) $\begin{bmatrix} 0 & 5 & 5 & 2 \\ 5 & 4 & 7 & 2 \\ 5 & 4 & 0 & 6 \\ 7 & 5 & 5 & 4 \end{bmatrix}$
johnston-linear-matrix-algebra-F03571	389	1.000	$\left[\begin{array}{cc}-2 & 4 \\ 5 & 5\end{array}\right]$	$\begin{bmatrix} -2 & 4 \\ 5 & 5 \end{bmatrix}$	(b) $\begin{bmatrix} -2 & 4 \\ 5 & 5 \end{bmatrix}$
johnston-linear-matrix-algebra-F03572	389	1.000	$\left[\begin{array}{ccc}-2 & 3 & -1 \\ 5 & 2 & 3 \\ 3 & 3 & 0\end{array}\right]$	$\begin{bmatrix} -2 & 3 & -1 \\ 5 & 2 & 3 \\ 3 & 3 & 0 \end{bmatrix}$	(d) $\begin{bmatrix} -2 & 3 & -1 \\ 5 & 2 & 3 \\ 3 & 3 & 0 \end{bmatrix}$
johnston-linear-matrix-algebra-F03573	389	0.990	$A \in \mathcal{M}_3(\mathbb{R})$	$A \in \mathcal{M}_3(\mathbb{R})$	(b) If $A \in \mathcal{M}_3(\mathbb{R})$ has eigenvalues 3, 2, and 1, then
johnston-linear-matrix-algebra-F03574	389	1.000	$\mathbf{v}_0, \mathbf{v}_1, \mathbf{v}_2$	$\mathbf{v}_0, \mathbf{v}_1, \mathbf{v}_2$	$A \in \mathcal{M}_n(\mathbb{R})$ then the sequence of vectors $\mathbf{v}_0, \mathbf{v}_1, \mathbf{v}_2,$
johnston-linear-matrix-algebra-F03575	389	1.000	$\mathbf{v}_0 = \mathbf{e}_2$	$\mathbf{v}_0 = \mathbf{e}_2$	with $\mathbf{v}_0 = \mathbf{e}_2$ instead of $\mathbf{v}_0 = \mathbf{e}_1$.
johnston-linear-matrix-algebra-F03576	389	0.415	$**3$	$**3$	**3.B.6 Suppose $A \in \mathcal{M}_n$ satisfies the hypotheses of The-
johnston-linear-matrix-algebra-F03577	389	1.000	$A \in \mathcal{M}_2(\mathbb{R})$	$A \in \mathcal{M}_2(\mathbb{R})$	3.B.7 Show that if $A \in \mathcal{M}_2(\mathbb{R})$ is positive then all of its
johnston-linear-matrix-algebra-F03578	390	0.495	$\left[\begin{array}{ll}0 & 1 \\ 1 & 1\end{array}\right]$	$\begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix}$	*(a) $\begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix}$
johnston-linear-matrix-algebra-F03579	390	1.000	$\left[\begin{array}{cc}2 & 1 \\ 1 & -1\end{array}\right]$	$\begin{bmatrix} 2 & 1 \\ 1 & -1 \end{bmatrix}$	(b) $\begin{bmatrix} 2 & 1 \\ 1 & -1 \end{bmatrix}$
johnston-linear-matrix-algebra-F03580	390	0.534	$\left[\begin{array}{cc}i & i \\ i & i\end{array}\right]$	$\begin{bmatrix} i & i \\ i & i \end{bmatrix}$	*(c) $\begin{bmatrix} i & i \\ i & i \end{bmatrix}$
johnston-linear-matrix-algebra-F03581	390	0.798	$\left[\begin{array}{ll}\sqrt{5}+i & \sqrt{5}-i \\ \sqrt{5}-i & \sqrt{5}+i\end{array}\right]$	$\begin{bmatrix} \sqrt{5}+i & \sqrt{5}-i \\ \sqrt{5}-i & \sqrt{5}+i \end{bmatrix}$	(d) $\begin{bmatrix} \sqrt{5}+i & \sqrt{5}-i \\ \sqrt{5}-i & \sqrt{5}+i \end{bmatrix}$
johnston-linear-matrix-algebra-F03582	390	0.809	$*(\mathbf{e})\left[\begin{array}{lll}0 & 1 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 0\end{array}\right]$	$*(\mathbf{e})\begin{bmatrix} 0 & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$	*(e) $\begin{bmatrix} 0 & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$
johnston-linear-matrix-algebra-F03583	390	1.000	$\left[\begin{array}{ccc}0 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & -1\end{array}\right]$	$\begin{bmatrix} 0 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & -1 \end{bmatrix}$	(f) $\begin{bmatrix} 0 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & -1 \end{bmatrix}$
johnston-linear-matrix-algebra-F03584	390	1.000	$\left[A^k\right]_{i,j}>0$	$[A^k]_{i,j} > 0$	k (that may depend on i and j) such that $[A^k]_{i,j} > 0$.
johnston-linear-matrix-algebra-F03585	390	0.972	$\left[\begin{array}{cc}-1 & 1 \\ 1 & 1\end{array}\right]$	$\begin{bmatrix} -1 & 1 \\ 1 & 1 \end{bmatrix}$	*(a) $\begin{bmatrix} -1 & 1 \\ 1 & 1 \end{bmatrix}$
johnston-linear-matrix-algebra-F03586	390	1.000	$\left[\begin{array}{ll}1 & 1 \\ 0 & 1\end{array}\right]$	$\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$	(b) $\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$
johnston-linear-matrix-algebra-F03587	390	0.817	$\left[\begin{array}{ll}0 & 1 \\ 2 & 3\end{array}\right]$	$\begin{bmatrix} 0 & 1 \\ 2 & 3 \end{bmatrix}$	*(c) $\begin{bmatrix} 0 & 1 \\ 2 & 3 \end{bmatrix}$
johnston-linear-matrix-algebra-F03588	390	1.000	$\left[\begin{array}{cc}-2 & 1 \\ 1 & 3\end{array}\right]$	$\begin{bmatrix} -2 & 1 \\ 1 & 3 \end{bmatrix}$	(d) $\begin{bmatrix} -2 & 1 \\ 1 & 3 \end{bmatrix}$

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johnston-linear-matrix-algebra-F03589	390	1.000	$\left[\begin{array}{ccc}1 & -1 & 1 \\ -1 & 1 & 1 \\ 1 & 1 & 1\end{array}\right]$	$\begin{bmatrix} 1 & -1 & 1 \\ -1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$	$\begin{bmatrix} 1 & -1 & 1 \\ -1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$
johnston-linear-matrix-algebra-F03590	390	0.997	$\left[\begin{array}{lll}0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0\end{array}\right]$	$\begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$	(f) $\begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$
johnston-linear-matrix-algebra-F03591	390	1.000	$A=\left[\begin{array}{cc}-3 & -2 \\ 4 & 1\end{array}\right]$	$A = \begin{bmatrix} -3 & -2 \\ 4 & 1 \end{bmatrix}$	Compute bases of the eigenspaces of the matrix $A = \begin{bmatrix} -3 & -2 \\ 4 & 1 \end{bmatrix}$.
johnston-linear-matrix-algebra-F03592	391	1.000	$4+(1-i)(-2-2i)=$	$4 + (1 - i)(-2 - 2i) =$	$4 + (1 - i)(-2 - 2i) =$
johnston-linear-matrix-algebra-F03593	391	1.000	$4+(-2+2i-2i-2)=$	$4 + (-2 + 2i - 2i - 2) =$	$4 + (-2 + 2i - 2i - 2) =$
johnston-linear-matrix-algebra-F03594	391	0.935	$4-4=0$	$4 - 4 = 0$	$4 - 4 = 0.$
johnston-linear-matrix-algebra-F03595	391	1.000	$\overline{B}=B$	$B = B$	real entries, so $\overline{B} = B$.
johnston-linear-matrix-algebra-F03596	391	1.000	$\lambda_{\pm}=-1\pm 2i$	$\lambda_{\pm} = -1 \pm 2i$	We showed in Example 3.3.8 that A has eigenvalues $\lambda_{\pm} = -1 \pm 2i$.
johnston-linear-matrix-algebra-F03597	391	1.000	$\mathbf{v}_{\pm}$	$\mathbf{v}_{\pm}$	We thus want to find the non-zero vectors $\mathbf{v}_{\pm}$ satisfying the linear systems
johnston-linear-matrix-algebra-F03598	391	0.998	$(A-(-1\pm 2i)I)\mathbf{v}_{\pm}=\mathbf{0}$	$(A - (-1 \pm 2i)I)\mathbf{v}_{\pm} = \mathbf{0}$	$(A - (-1 \pm 2i)I)\mathbf{v}_{\pm} = \mathbf{0}$. We start by solving the “+” system:
johnston-linear-matrix-algebra-F03599	391	0.999	$\lambda_{+}=-1+2i$	$\lambda_{+} = -1 + 2i$	It follows that one basis of the eigenspace corresponding to $\lambda_{+} = -1 + 2i$
johnston-linear-matrix-algebra-F03600	391	1.000	$\mathbf{v}_{+}=(1,-1-i)$	$\mathbf{v}_{+} = (1, -1 - i)$	consists of the single vector $\mathbf{v}_{+} = (1, -1 - i)$. A similar calculation shows
johnston-linear-matrix-algebra-F03601	391	1.000	$\lambda_{-}=-1-2i$	$\lambda_{-} = -1 - 2i$	that one basis of the eigenspace corresponding to $\lambda_{-} = -1 - 2i$ consists
johnston-linear-matrix-algebra-F03602	391	1.000	$\mathbf{v}_{-}=(1,-1+i)$	$\mathbf{v}_{-} = (1, -1 + i)$	of the vector $\mathbf{v}_{-} = (1, -1 + i)$.
johnston-linear-matrix-algebra-F03603	391	1.000	$\overline{-1+2i}=$	$\overline{-1 + 2i} =$	this case, its eigenvalues are complex conjugates of each other, since $\overline{-1 + 2i} =$
johnston-linear-matrix-algebra-F03604	391	1.000	$-1-2i$	$-1 - 2i$	$-1 - 2i$, and its eigenvectors are also complex conjugates of each other, since
johnston-linear-matrix-algebra-F03605	391	1.000	$\overline{(1,-1-i)}=(1,-1+i)$	$\overline{(1, -1 - i)} = (1, -1 + i)$	$\overline{(1, -1 - i)} = (1, -1 + i)$.
johnston-linear-matrix-algebra-F03606	391	1.000	$B\mathbf{v}=\lambda\mathbf{v}$	$B\mathbf{v} = \lambda\mathbf{v}$	is an eigenvector of B with corresponding eigenvalue λ . Then $B\mathbf{v} = \lambda\mathbf{v}$, which
johnston-linear-matrix-algebra-F03607	391	1.000	$B\overline{\mathbf{v}}$	$B\overline{\mathbf{v}}$	we can use to compute $B\overline{\mathbf{v}}$ as follows:
johnston-linear-matrix-algebra-F03608	391	1.000	$B\overline{\mathbf{v}}=\bar{\lambda}\overline{\mathbf{v}}$	$B\overline{\mathbf{v}} = \bar{\lambda}\overline{\mathbf{v}}$	Since $B\overline{\mathbf{v}} = \bar{\lambda}\overline{\mathbf{v}}$, we conclude that $\overline{\mathbf{v}}$ is an eigenvector of B with correspond-
johnston-linear-matrix-algebra-F03609	391	0.735	$\mathbf{2} \times \mathbf{2}$	$\mathbf{2} \times \mathbf{2}$	3.C.1 Geometric Interpretation for 2×2 Matrices
johnston-linear-matrix-algebra-F03610	391	1.000	$\lambda=re^{i\theta}$	$\lambda = re^{i\theta}$	$\lambda = re^{i\theta}$, which means that λ is a distance of r from the origin in the complex
johnston-linear-matrix-algebra-F03611	391	1.000	$r=\sqrt{a^2+b^2}$	$r = \sqrt{a^2 + b^2}$	particular, r and θ can be computed via $r = \sqrt{a^2 + b^2}$ and $\theta = \arctan(b/a)$ if
johnston-linear-matrix-algebra-F03612	391	1.000	$\theta=\arctan(b/a)$	$\theta = \arctan(b/a)$	particular, r and θ can be computed via $r = \sqrt{a^2 + b^2}$ and $\theta = \arctan(b/a)$ if
johnston-linear-matrix-algebra-F03613	391	1.000	$a>0$	$a > 0$	$a > 0$ (if $a \leq 0$ then we have to be more careful when computing θ to make

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johnston-linear-matrix-algebra-F03614	391	1.000	$a \leq 0$	$a \leq 0$	$a > 0$ (if $a \leq 0$ then we have to be more careful when computing θ to make
johnston-linear-matrix-algebra-F03615	391	1.000	$e^{i\theta} = \cos(\theta) + i \sin(\theta)$	$e^{i\theta} = \cos(\theta) + i \sin(\theta)$	$e^{i\theta} = \cos(\theta) + i \sin(\theta)$
johnston-linear-matrix-algebra-F03616	391	1.000	$\operatorname{Re}(\mathbf{v})$	$\operatorname{Re}(\mathbf{v})$	Second, we use $\operatorname{Re}(\mathbf{v})$ and $\operatorname{Im}(\mathbf{v})$ to denote the real and imaginary parts of
johnston-linear-matrix-algebra-F03617	391	1.000	$\operatorname{Im}(\mathbf{v})$	$\operatorname{Im}(\mathbf{v})$	Second, we use $\operatorname{Re}(\mathbf{v})$ and $\operatorname{Im}(\mathbf{v})$ to denote the real and imaginary parts of
johnston-linear-matrix-algebra-F03618	391	1.000	$\mathbf{v} = \mathbf{x} + i\mathbf{y}$	$\mathbf{v} = \mathbf{x} + i\mathbf{y}$	a vector, respectively. That is, if $\mathbf{v} = \mathbf{x} + i\mathbf{y}$ for some $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$ then $\operatorname{Re}(\mathbf{v}) = \mathbf{x}$
johnston-linear-matrix-algebra-F03619	391	1.000	$\operatorname{Re}(\mathbf{v}) = \mathbf{x}$	$\operatorname{Re}(\mathbf{v}) = \mathbf{x}$	a vector, respectively. That is, if $\mathbf{v} = \mathbf{x} + i\mathbf{y}$ for some $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$ then $\operatorname{Re}(\mathbf{v}) = \mathbf{x}$
johnston-linear-matrix-algebra-F03620	391	1.000	$\operatorname{Im}(\mathbf{v}) = \mathbf{y}$	$\operatorname{Im}(\mathbf{v}) = \mathbf{y}$	and $\operatorname{Im}(\mathbf{v}) = \mathbf{y}$. Also, by recalling that the complex conjugate of $\mathbf{v}$ is $\bar{\mathbf{v}} =$
johnston-linear-matrix-algebra-F03621	391	1.000	$\overline{\mathbf{v}} =$	$\bar{\mathbf{v}} =$	and $\operatorname{Im}(\mathbf{v}) = \mathbf{y}$. Also, by recalling that the complex conjugate of $\mathbf{v}$ is $\bar{\mathbf{v}} =$
johnston-linear-matrix-algebra-F03622	391	1.000	$\operatorname{Re}(\mathbf{v}) - i \operatorname{Im}(\mathbf{v})$	$\operatorname{Re}(\mathbf{v}) - i \operatorname{Im}(\mathbf{v})$	$\operatorname{Re}(\mathbf{v}) - i \operatorname{Im}(\mathbf{v})$, we can solve for $\operatorname{Re}(\mathbf{v})$ and $\operatorname{Im}(\mathbf{v})$ in terms of $\mathbf{v}$ and $\bar{\mathbf{v}}$:
johnston-linear-matrix-algebra-F03623	392	1.000	$a + ib = r e^{i\theta}$	$a + ib = r e^{i\theta}$	Figure 3.29: Every complex number $a + ib$ can be written in the form $a + ib = r e^{i\theta}$,
johnston-linear-matrix-algebra-F03624	392	1.000	$[\mathbf{x}]_B$	$[\mathbf{x}]_B$	Third, recall from Section 3.1 that if B is a basis of a subspace then $[\mathbf{x}]_B$
johnston-linear-matrix-algebra-F03625	392	0.712	$B = \{\mathbf{v}, \mathbf{w}\}$	$B = \{\mathbf{v}, \mathbf{w}\}$	if $B = \{\mathbf{v}, \mathbf{w}\}$ is a basis of a 2-dimensional subspace (i.e., a plane) and $\mathbf{x} =$
johnston-linear-matrix-algebra-F03626	392	1.000	$c\mathbf{v} + d\mathbf{w}$	$c\mathbf{v} + d\mathbf{w}$	$c\mathbf{v} + d\mathbf{w}$, then $[\mathbf{x}]_B = (c, d)$.
johnston-linear-matrix-algebra-F03627	392	1.000	$[\mathbf{x}]_B = (c, d)$	$[\mathbf{x}]_B = (c, d)$	$c\mathbf{v} + d\mathbf{w}$, then $[\mathbf{x}]_B = (c, d)$.
johnston-linear-matrix-algebra-F03628	392	1.000	π	π	an integer multiple of π , then the upcoming theorem can be thought of as the
johnston-linear-matrix-algebra-F03629	392	0.999	$re^{i\theta} \in \mathbb{C}$	$re^{i\theta} \in \mathbb{C}$	Suppose $A \in \mathcal{M}_2(\mathbb{R})$ has a complex (non-real) eigenvalue $re^{i\theta} \in \mathbb{C}$ with
johnston-linear-matrix-algebra-F03630	392	1.000	$\mathbf{v} \in \mathbb{C}^2$	$\mathbf{v} \in \mathbb{C}^2$	corresponding eigenvector $\mathbf{v} \in \mathbb{C}^2$. If we let $Q = [\operatorname{Re}(\mathbf{v}) \mid -\operatorname{Im}(\mathbf{v})]$ then
johnston-linear-matrix-algebra-F03631	392	1.000	$Q = [\operatorname{Re}(\mathbf{v}) \mid -\operatorname{Im}(\mathbf{v})]$	$Q = [\operatorname{Re}(\mathbf{v}) \mid -\operatorname{Im}(\mathbf{v})]$	corresponding eigenvector $\mathbf{v} \in \mathbb{C}^2$. If we let $Q = [\operatorname{Re}(\mathbf{v}) \mid -\operatorname{Im}(\mathbf{v})]$ then
johnston-linear-matrix-algebra-F03632	392	0.999	$B = \{\operatorname{Re}(\mathbf{v}), -\operatorname{Im}(\mathbf{v})\}$	$B = \{\operatorname{Re}(\mathbf{v}), -\operatorname{Im}(\mathbf{v})\}$	$B = \{\operatorname{Re}(\mathbf{v}), -\operatorname{Im}(\mathbf{v})\}$.
johnston-linear-matrix-algebra-F03633	392	1.000	$\lambda = -1 + 2i$	$\lambda = -1 + 2i$	from Example 3.3.8, we recall that one of its eigenvalues is $\lambda = -1 + 2i =$
johnston-linear-matrix-algebra-F03634	392	1.000	$\sqrt{5}e^{i\theta}$	$\sqrt{5}e^{i\theta}$	$\sqrt{5}e^{i\theta}$, where $\theta \approx \pm(0.6476)\pi$. In Example 3.C.1, we found that one cor-
johnston-linear-matrix-algebra-F03635	392	1.000	$\theta \approx \pm(0.6476)\pi$	$\theta \approx \pm(0.6476)\pi$	$\sqrt{5}e^{i\theta}$, where $\theta \approx \pm(0.6476)\pi$. In Example 3.C.1, we found that one cor-
johnston-linear-matrix-algebra-F03636	392	1.000	$\mathbf{v} = (1, -1 - i)$	$\mathbf{v} = (1, -1 - i)$	responding eigenvector is $\mathbf{v} = (1, -1 - i)$. It follows that this matrix can be
johnston-linear-matrix-algebra-F03637	393	1.000	$r = \sqrt{5}$	$r = \sqrt{5}$	a factor of $r = \sqrt{5}$,
johnston-linear-matrix-algebra-F03638	393	1.000	$B = \{\operatorname{Re}(\mathbf{v}), -\operatorname{Im}(\mathbf{v})\} = \{(1, -1), (0, 1)\}$	$B = \{\operatorname{Re}(\mathbf{v}), -\operatorname{Im}(\mathbf{v})\} = \{(1, -1), (0, 1)\}$	viewed as rotating $\mathbb{R}^2$ in the basis $B = \{\operatorname{Re}(\mathbf{v}), -\operatorname{Im}(\mathbf{v})\} = \{(1, -1), (0, 1)\}$
johnston-linear-matrix-algebra-F03639	393	1.000	$\theta \approx$	$\theta \approx$	counter-clockwise by approximately 0.6476 half-rotations (i.e., an angle of $\theta \approx$
johnston-linear-matrix-algebra-F03640	393	0.936	$(0.6476)\pi$	$(0.6476)\pi$	$(0.6476)\pi$) and stretching it by a factor of $r = \sqrt{5} \approx 2.2361$, as in Figure 3.30.

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johnston-linear-matrix-algebra-F03641	393	■ 0.936	$r=\sqrt{5} \approx 2.2361$	$r = \sqrt{5} \approx 2.2361$	$(0.6476)\pi$) and stretching it by a factor of $r = \sqrt{5} \approx 2.2361$, as in Figure 3.30.
johnston-linear-matrix-algebra-F03642	393	■ 0.561	$A=\left[\begin{array}{rl}-3 & -2 \\ 4 & 1\end{array}\right]$	$A = \begin{bmatrix} -3 & -2 \\ 4 & 1 \end{bmatrix}$	Figure 3.30: A visualization of the geometric effect of the matrix $A = \begin{bmatrix} -3 & -2 \\ 4 & 1 \end{bmatrix}$ from
johnston-linear-matrix-algebra-F03643	393	■ 1.000	$r e^{i \theta}$	$r e^{i \theta}$	<i>Proof Theorem 3.C.1.</i> First, we note that since the eigenvalue $r e^{i \theta}$ is not real,
johnston-linear-matrix-algebra-F03644	393	■ 1.000	$r e^{-i \theta}$	$r e^{-i \theta}$	it does not equal its complex conjugate $r e^{-i \theta}$, so A is diagonalizable via $A =$
johnston-linear-matrix-algebra-F03645	393	■ 1.000	$H \in \mathcal{M}_2(\mathbb{C})$	$H \in \mathcal{M}_2(\mathbb{C})$	with real entries). To this end, let $H \in \mathcal{M}_2(\mathbb{C})$ be the matrix
johnston-linear-matrix-algebra-F03646	394	■ 0.913	$Q H=P$	$Q H = P$	$Q H = P$, so
johnston-linear-matrix-algebra-F03647	394	■ 0.947	$H P^{-1}=Q^{-1}$	$H P^{-1} = Q^{-1}$	$H P^{-1} = Q^{-1}$.
johnston-linear-matrix-algebra-F03648	394	■ 0.927	$\sin (-\theta)=-\sin (\theta)$	$\sin (-\theta) = -\sin (\theta)$	$\sin (-\theta) = -\sin (\theta)$
johnston-linear-matrix-algebra-F03649	394	■ 1.000	$e^{-i \theta}=\cos (\theta)-i \sin (\theta)$	$e^{-i \theta} = \cos (\theta) - i \sin (\theta)$	$e^{-i \theta} = \cos (\theta) - i \sin (\theta)$.
johnston-linear-matrix-algebra-F03650	394	■ 1.000	$Q=P H^{-1}$	$Q = P H^{-1}$	which tells us that Q is invertible since it implies $Q = P H^{-1}$, which is the
johnston-linear-matrix-algebra-F03651	394	■ 1.000	$D=P^{-1} A P$	$D = P^{-1} A P$	relationship between P and Q to transform $D = P^{-1} A P$ into $Q^{-1} A Q$ and see
johnston-linear-matrix-algebra-F03652	394	■ 1.000	$Q^{-1} A Q$	$Q^{-1} A Q$	relationship between P and Q to transform $D = P^{-1} A P$ into $Q^{-1} A Q$ and see
johnston-linear-matrix-algebra-F03653	394	■ 1.000	$H D H^{-1}$	$H D H^{-1}$	theorem. In particular, we now compute $H D H^{-1}$ in two different ways. First,
johnston-linear-matrix-algebra-F03654	394	■ 1.000	$H D H^{-1}=r\left[R^{\theta}\right]$	$H D H^{-1} = r\left[R^{\theta}\right]$	so if we can also show that $H D H^{-1} = r\left[R^{\theta}\right]$ then we will be done. Equivalently,
johnston-linear-matrix-algebra-F03655	394	■ 1.000	$H D=r\left[R^{\theta}\right] H$	$H D = r\left[R^{\theta}\right] H$	we want to show that $H D = r\left[R^{\theta}\right] H$, which can be done simply by multiplying
johnston-linear-matrix-algebra-F03656	394	■ 1.000	H	H	The matrix H from Equation 3.C.1 is quite remarkable for the fact that it
johnston-linear-matrix-algebra-F03657	394	■ 1.000	$\lambda_{\pm}=-1 \pm$	$\lambda_{\pm} = -1 \pm$	We start by recalling that the eigenvalues of this matrix are $\lambda_{\pm} = -1 \pm$
johnston-linear-matrix-algebra-F03658	394	■ 1.000	$2 i$	$2 i$	$2 i$ and a pair of corresponding eigenvectors is given by $\mathbf{v}_{\pm} = (1, -1 \mp i)$.
johnston-linear-matrix-algebra-F03659	394	■ 1.000	$\mathbf{v}_{\pm}=(1,-1 \mp i)$	$\mathbf{v}_{\pm} = (1, -1 \mp i)$	$2 i$ and a pair of corresponding eigenvectors is given by $\mathbf{v}_{\pm} = (1, -1 \mp i)$.
johnston-linear-matrix-algebra-F03660	395	■ 1.000	$\left[\operatorname{Re}\left(\mathbf{v}_{+}\right) \mid \operatorname{Im}\left(\mathbf{v}_{+}\right)\right]$	$[\operatorname{Re}\left(\mathbf{v}_{+}\right) \mid -\operatorname{Im}\left(\mathbf{v}_{+}\right)]$	which is exactly $[\operatorname{Re}\left(\mathbf{v}_{+}\right) \mid -\operatorname{Im}\left(\mathbf{v}_{+}\right)]$, as we would hope (we want our
johnston-linear-matrix-algebra-F03661	395	■ 1.000	$\sqrt{5}\left[R^{\theta}\right]$	$\sqrt{5}\left[R^{\theta}\right]$	(i.e., it equals $\sqrt{5}\left[R^{\theta}\right]$, where $\theta = \arccos (-1 / \sqrt{5}) \approx (0.6476) \pi$).
johnston-linear-matrix-algebra-F03662	395	■ 1.000	$\theta=\arccos (-1 / \sqrt{5}) \approx(0.6476) \pi$	$\theta = \arccos (-1 / \sqrt{5}) \approx (0.6476) \pi$	(i.e., it equals $\sqrt{5}\left[R^{\theta}\right]$, where $\theta = \arccos (-1 / \sqrt{5}) \approx (0.6476) \pi$).
johnston-linear-matrix-algebra-F03663	395	■ 1.000	$\operatorname{diag}(a, b, c, \ldots)$	$\operatorname{diag}(a, b, c, \ldots)$	Before stating the theorem, we note that we use the notation $\operatorname{diag}(a, b, c, \ldots)$
johnston-linear-matrix-algebra-F03664	395	■ 0.999	$a, b, c, \ldots$	$a, b, c, \ldots$	to denote the diagonal matrix with diagonal entries $a, b, c, \ldots$. In fact, we even
johnston-linear-matrix-algebra-F03665	395	■ 1.000	$A=\mathcal{M}_n(\mathbb{R})$	$A = \mathcal{M}_n(\mathbb{R})$	Suppose $A = \mathcal{M}_n(\mathbb{R})$ has ℓ complex conjugate eigenvalue pairs and m
johnston-linear-matrix-algebra-F03666	395	■ 1.000	2ℓ	2ℓ	eigenvalues and 2ℓ
johnston-linear-matrix-algebra-F03667	395	■ 0.951	$2 \ell+m=n$	$2 \ell+m = n$	$2 \ell+m = n$.

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johnston-linear-matrix-algebra-F03668	395	1.000	$r_{\{j\}}, \lambda_{\{j\}}, \theta_{\{j\}} \in \mathbb{R}, \mathbf{w}_{\{j\}} \in \mathbb{R}^n$	$r_j, \lambda_j, \theta_j \in \mathbb{R}, \mathbf{w}_j \in \mathbb{R}^n$	with $r_j, \lambda_j, \theta_j \in \mathbb{R}, \mathbf{w}_j \in \mathbb{R}^n$, and $\mathbf{v}_j \in \mathbb{C}^n$ for all j .
johnston-linear-matrix-algebra-F03669	395	1.000	$\mathbf{v}_{\{j\}} \in \mathbb{C}^n$	$\mathbf{v}_j \in \mathbb{C}^n$	with $r_j, \lambda_j, \theta_j \in \mathbb{R}, \mathbf{w}_j \in \mathbb{R}^n$, and $\mathbf{v}_j \in \mathbb{C}^n$ for all j .
johnston-linear-matrix-algebra-F03670	395	1.000	$A = Q B Q^{-1}$	$A = Q B Q^{-1}$	Then $A = Q B Q^{-1}$, where $B, Q \in \mathcal{M}_n(\mathbb{R})$ are defined by
johnston-linear-matrix-algebra-F03671	395	1.000	$B, Q \in \mathcal{M}_n(\mathbb{R})$	$B, Q \in \mathcal{M}_n(\mathbb{R})$	Then $A = Q B Q^{-1}$, where $B, Q \in \mathcal{M}_n(\mathbb{R})$ are defined by
johnston-linear-matrix-algebra-F03672	395	1.000	$P, D \in \mathcal{M}_n(\mathbb{C})$	$P, D \in \mathcal{M}_n(\mathbb{C})$	as $A = P D P^{-1}$ via some $P, D \in \mathcal{M}_n(\mathbb{C})$, but not via a <i>real</i> P and D (in other
johnston-linear-matrix-algebra-F03673	396	1.000	$\tilde{H}$	$\tilde{H}$	simply because $\tilde{H}$ is
johnston-linear-matrix-algebra-F03674	396	0.993	$(1 \pm i) / 2$	$(1 \pm i)/2$	has distinct eigenvalues 2 and $(1 \pm i)/2$ and is thus diagonalizable over $\mathbb{C}$, but
johnston-linear-matrix-algebra-F03675	396	1.000	$1 \leq j \leq \ell$	$1 \leq j \leq \ell$	for all $1 \leq j \leq \ell$, and we also define
johnston-linear-matrix-algebra-F03676	396	1.000	$Q_j H = P_j$	$Q_j H = P_j$	$Q_j H = P_j$ for all $1 \leq j \leq \ell$, so block matrix multiplication then shows that
johnston-linear-matrix-algebra-F03677	396	1.000	$H D_j =$	$H D_j =$	Similarly, we also showed in the proof of Theorem 3.C.1 that $H D_j =$
johnston-linear-matrix-algebra-F03678	396	1.000	$r_{\{j\}} [R^{\theta_j}] H$	$r_j [R^{\theta_j}] H$	$r_j [R^{\theta_j}] H$ for all $1 \leq j \leq \ell$, so block matrix multiplication shows that
johnston-linear-matrix-algebra-F03679	396	0.999	$\tilde{H} D \tilde{H}^{-1} = B$	$\tilde{H} D \tilde{H}^{-1} = B$	It follows that $\tilde{H} D \tilde{H}^{-1} = B$, so
johnston-linear-matrix-algebra-F03680	397	1.000	$-\operatorname{Im}(\mathbf{v})$	$-\operatorname{Im}(\mathbf{v})$	$-\operatorname{Im}(\mathbf{v})$.
johnston-linear-matrix-algebra-F03681	397	1.000	$p_A(\lambda) = -\lambda^3 + 3\lambda^2 - (5/2)\lambda + 1$	$p_A(\lambda) = -\lambda^3 + 3\lambda^2 - (5/2)\lambda + 1$	• Its characteristic polynomial is $p_A(\lambda) = -\lambda^3 + 3\lambda^2 - (5/2)\lambda + 1$.
johnston-linear-matrix-algebra-F03682	397	0.672	$(1+i) / 2, (1-i) / 2$	$(1+i)/2, (1-i)/2$	• Its eigenvalues are $(1+i)/2, (1-i)/2$, and 2.
johnston-linear-matrix-algebra-F03683	397	1.000	$\mathbf{v} = (1, i, 1), \bar{\mathbf{v}} = (1, -i, 1)$	$\mathbf{v} = (1, i, 1), \bar{\mathbf{v}} = (1, -i, 1)$	are $\mathbf{v} = (1, i, 1), \bar{\mathbf{v}} = (1, -i, 1)$, and $\mathbf{w} = (0, 1, 2)$.
johnston-linear-matrix-algebra-F03684	397	1.000	$\mathbf{w} = (0, 1, 2)$	$\mathbf{w} = (0, 1, 2)$	are $\mathbf{v} = (1, i, 1), \bar{\mathbf{v}} = (1, -i, 1)$, and $\mathbf{w} = (0, 1, 2)$.
johnston-linear-matrix-algebra-F03685	397	0.980	$(1+i) / 2$	$(1+i)/2$	A and write it in polar form. We choose the eigenvalue $(1+i)/2$, so $r =$
johnston-linear-matrix-algebra-F03686	397	0.980	$r =$	$r =$	A and write it in polar form. We choose the eigenvalue $(1+i)/2$, so $r =$
johnston-linear-matrix-algebra-F03687	397	1.000	$\sqrt{(1/2)^2 + (1/2)^2} = 1/\sqrt{2}$	$\sqrt{(1/2)^2 + (1/2)^2} = 1/\sqrt{2}$	$\sqrt{(1/2)^2 + (1/2)^2} = 1/\sqrt{2}$ and $\theta = \arctan((1/2)/(1/2)) = \arctan(1) =$
johnston-linear-matrix-algebra-F03688	397	1.000	$\theta = \arctan((1/2)/(1/2)) = \arctan(1) =$	$\theta = \arctan((1/2)/(1/2)) = \arctan(1) =$	$\sqrt{(1/2)^2 + (1/2)^2} = 1/\sqrt{2}$ and $\theta = \arctan((1/2)/(1/2)) = \arctan(1) =$
johnston-linear-matrix-algebra-F03689	397	1.000	$(1+i) / 2 = \frac{1}{\sqrt{2}} e^{i\pi/4}$	$(1+i)/2 = \frac{1}{\sqrt{2}} e^{i\pi/4}$	$\pi/4$. The polar form of this eigenvalue is thus $(1+i)/2 = \frac{1}{\sqrt{2}} e^{i\pi/4}$.
johnston-linear-matrix-algebra-F03690	397	0.539	$(1+$	$(1+$	If we denote the eigenvectors corresponding to the eigenvalues $(1+$
johnston-linear-matrix-algebra-F03691	397	1.000	$i) / 2$	$i)/2$	$i)/2$ and 2 by $\mathbf{v} = (1, i, 1)$ and $\mathbf{w} = (0, 1, 2)$, respectively, it then follows
johnston-linear-matrix-algebra-F03692	397	1.000	$\mathbf{v} = (1, i, 1)$	$\mathbf{v} = (1, i, 1)$	$i)/2$ and 2 by $\mathbf{v} = (1, i, 1)$ and $\mathbf{w} = (0, 1, 2)$, respectively, it then follows
johnston-linear-matrix-algebra-F03693	397	0.920	$(0, 1, 2), A$	$(0, 1, 2), A$	$(0, 1, 2)$, A stretches by a factor of 2 (this is nothing new—this is exactly

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johnston-linear-matrix-algebra-F03694	397	1.000	$\theta=\pi / 4$	$\theta=\pi / 4$	then A acts on $\mathcal{S}$ like a rotation by an angle of $\theta=\pi / 4$ composed with a
johnston-linear-matrix-algebra-F03695	398	1.000	$r=1 / \sqrt{2}$	$r=1 / \sqrt{2}$	scaling by a factor of $r=1 / \sqrt{2}$, as displayed below:
johnston-linear-matrix-algebra-F03696	398	0.996	$\left[R^{\theta}\right]^k=\left[R^{k \theta}\right]$	$\left[R^{\theta}\right]^k=\left[R^{k \theta}\right]$	recall that $\left[R^{\theta}\right]^k=\left[R^{k \theta}\right]$ since a rotating $\mathbb{R}^2$ by an angle of $k \theta$ is equivalent
johnston-linear-matrix-algebra-F03697	398	0.996	$k \theta$	$k \theta$	recall that $\left[R^{\theta}\right]^k=\left[R^{k \theta}\right]$ since a rotating $\mathbb{R}^2$ by an angle of $k \theta$ is equivalent
johnston-linear-matrix-algebra-F03698	398	1.000	A^{40}	A^{40}	to compute A^{40} for the matrix A from Example 3.C.3 via its diagonalization, we
johnston-linear-matrix-algebra-F03699	398	1.000	$((1+i) / 2)^{40}$	$((1+i) / 2)^{40}$	would have to compute complex numbers like $((1+i) / 2)^{40}$, which is perhaps
johnston-linear-matrix-algebra-F03700	398	1.000	$\left.\left((1+i) / 2\right)^{40}=1 / 2^{20}\right)$	$((1+i) / 2)^{40}=1 / 2^{20})$	this particular case, the answer is in fact real: $((1+i) / 2)^{40}=1 / 2^{20})$.
johnston-linear-matrix-algebra-F03701	399	1.000	$\left[R^{40 \pi / 4}\right]$	$\left[R^{40 \pi / 4}\right]$	$\left[R^{40 \pi / 4}\right]$ is simply the
johnston-linear-matrix-algebra-F03702	399	1.000	$\theta=40 \pi / 4=10 \pi$	$\theta=40 \pi / 4=10 \pi$	$\theta=40 \pi / 4=10 \pi$ is
johnston-linear-matrix-algebra-F03703	399	0.996	$r, \theta \in \mathbb{R}$	$r, \theta \in \mathbb{R}$	That is, find values for $r, \theta \in \mathbb{R}$ and a matrix $Q \in \mathcal{M}_2(\mathbb{R})$
johnston-linear-matrix-algebra-F03704	399	0.996	$Q \in \mathcal{M}_2(\mathbb{R})$	$Q \in \mathcal{M}_2(\mathbb{R})$	That is, find values for $r, \theta \in \mathbb{R}$ and a matrix $Q \in \mathcal{M}_2(\mathbb{R})$
johnston-linear-matrix-algebra-F03705	399	1.000	$Q\left(r\left[R^{\theta}\right]\right) Q^{-1}$	$Q\left(r\left[R^{\theta}\right]\right) Q^{-1}$	such that the given matrix has the form $Q\left(r\left[R^{\theta}\right]\right) Q^{-1}$.
johnston-linear-matrix-algebra-F03706	399	0.725	$\left(\mathbf{a}\right)\left[\begin{array}{cc} 0 & -1 \\ 1 & 0 \end{array}\right]$	$\left(\mathbf{a}\right)\left[\begin{array}{cc} 0 & -1 \\ 1 & 0 \end{array}\right]$	$\left(\mathbf{a}\right)\left[\begin{array}{cc} 0 & -1 \\ 1 & 0 \end{array}\right]$ $\left(\mathbf{c}\right)\left[\begin{array}{cc} 2 & 1 \\ -2 & 0 \end{array}\right]$
johnston-linear-matrix-algebra-F03707	399	0.725	$\left(\mathbf{c}\right)\left[\begin{array}{cc} 2 & 1 \\ -2 & 0 \end{array}\right]$	$\left(\mathbf{c}\right)\left[\begin{array}{cc} 2 & 1 \\ -2 & 0 \end{array}\right]$	$\left(\mathbf{a}\right)\left[\begin{array}{cc} 0 & -1 \\ 1 & 0 \end{array}\right]$ $\left(\mathbf{c}\right)\left[\begin{array}{cc} 2 & 1 \\ -2 & 0 \end{array}\right]$
johnston-linear-matrix-algebra-F03708	399	1.000	$\left[\begin{array}{cc} 0 & 1 \\ -2 & 0 \end{array}\right]$	$\left[\begin{array}{cc} 0 & 1 \\ -2 & 0 \end{array}\right]$	$\left(\mathbf{b}\right)\left[\begin{array}{cc} 0 & 1 \\ -2 & 0 \end{array}\right]$
johnston-linear-matrix-algebra-F03709	399	1.000	$\left[\begin{array}{cc} 1 & -\sqrt{3} \\ \sqrt{3} & 1 \end{array}\right]$	$\left[\begin{array}{cc} 1 & -\sqrt{3} \\ \sqrt{3} & 1 \end{array}\right]$	$\left(\mathbf{d}\right)\left[\begin{array}{cc} 1 & -\sqrt{3} \\ \sqrt{3} & 1 \end{array}\right]$
johnston-linear-matrix-algebra-F03710	399	0.970	$\left[\begin{array}{ccc} 1 & 0 & 0 \\ -1 & 0 & 1 \\ 1 & -1 & 0 \end{array}\right]$	$\left[\begin{array}{ccc} 1 & 0 & 0 \\ -1 & 0 & 1 \\ 1 & -1 & 0 \end{array}\right]$	$\left(\mathbf{a}\right)\left[\begin{array}{ccc} 1 & 0 & 0 \\ -1 & 0 & 1 \\ 1 & -1 & 0 \end{array}\right]$
johnston-linear-matrix-algebra-F03711	399	1.000	$\left[\begin{array}{ccc} 1 & -\sqrt{3} & \sqrt{3} \\ 0 & 3 & 0 \\ -\sqrt{3} & 2 & 1 \end{array}\right]$	$\left[\begin{array}{ccc} 1 & -\sqrt{3} & \sqrt{3} \\ 0 & 3 & 0 \\ -\sqrt{3} & 2 & 1 \end{array}\right]$	$\left(\mathbf{b}\right)\left[\begin{array}{ccc} 1 & -\sqrt{3} & \sqrt{3} \\ 0 & 3 & 0 \\ -\sqrt{3} & 2 & 1 \end{array}\right]$
johnston-linear-matrix-algebra-F03712	399	0.943	$\left[\begin{array}{cccc} 1 & -2 & 2 & 1 \\ 0 & 1 & 0 & 0 \\ -1 & -1 & 2 & 1 \\ 1 & 0 & 0 & 1 \end{array}\right]$	$\left[\begin{array}{cccc} 1 & -2 & 2 & 1 \\ 0 & 1 & 0 & 0 \\ -1 & -1 & 2 & 1 \\ 1 & 0 & 0 & 1 \end{array}\right]$	$\left(\mathbf{c}\right)\left[\begin{array}{cccc} 1 & -2 & 2 & 1 \\ 0 & 1 & 0 & 0 \\ -1 & -1 & 2 & 1 \\ 1 & 0 & 0 & 1 \end{array}\right]$

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johnston-linear-matrix-algebra-F03713	399	1.000	$\left[\begin{array}{cccc} 4 & -3 & -1 & 2 \\ 6 & -5 & 0 & 1 \\ 4 & -5 & 1 & 0 \\ 0 & -1 & 3 & -2 \end{array}\right]$	$\begin{bmatrix} 4 & -3 & -1 & 2 \\ 6 & -5 & 0 & 1 \\ 4 & -5 & 1 & 0 \\ 0 & -1 & 3 & -2 \end{bmatrix}$	(d)
johnston-linear-matrix-algebra-F03714	399	1.000	$A \in \mathcal{M}_4(\mathbb{R})$	$A \in \mathcal{M}_4(\mathbb{R})$	(b) Every matrix $A \in \mathcal{M}_4(\mathbb{R})$ has a real eigenvalue.
johnston-linear-matrix-algebra-F03715	399	1.000	$\left[R^{\theta}\right] \in \mathcal{M}_2(\mathbb{R})$	$\left[R^{\theta}\right] \in \mathcal{M}_2(\mathbb{R})$	(f) Every rotation matrix $\left[R^{\theta}\right] \in \mathcal{M}_2(\mathbb{R})$ has complex
johnston-linear-matrix-algebra-F03716	400	1.000	$x_{\{0\}}$	x_0	sequences at x_0
johnston-linear-matrix-algebra-F03717	400	1.000	$x_{\{1\}}, x_{\{2\}}, x_{\{3\}}, \ldots$	$x_1, x_2, x_3, \ldots$	numbers $x_1, x_2, x_3, \ldots$ that are defined in a similar linear way.
johnston-linear-matrix-algebra-F03718	400	1.000	$x_{\{n\}}$	x_n	A linear recurrence relation is a formula that represents a term x_n in a se-
johnston-linear-matrix-algebra-F03719	400	1.000	$a_{\{0\}}, a_{\{1\}}, \ldots, a_{\{k-1\}}$	$a_0, a_1, \ldots, a_{k-1}$	the sequence. That is, it is a formula for which there exist scalars $a_0, a_1, \ldots, a_{k-1}$
johnston-linear-matrix-algebra-F03720	400	0.994	$F_{\{0\}}, F_{\{1\}}, F_{\{2\}}, \ldots$	$F_0, F_1, F_2, \ldots$	For example, the Fibonacci numbers $F_0, F_1, F_2, \ldots$ from Example 3.4.8
johnston-linear-matrix-algebra-F03721	400	0.907	$F_{\{n\}}=F_{\{n-1\}}+F_{\{n-2\}}$	$F_n = F_{n-1} + F_{n-2}$	satisfy the degree-2 linear recurrence relation $F_n = F_{n-1} + F_{n-2}$, and the initial
johnston-linear-matrix-algebra-F03722	400	1.000	$F_{\{0\}}=0$	$F_0 = 0$	conditions were the requirements that $F_0 = 0$ and $F_1 = 1$.
johnston-linear-matrix-algebra-F03723	400	1.000	$F_{\{1\}}=1$	$F_1 = 1$	conditions were the requirements that $F_0 = 0$ and $F_1 = 1$.
johnston-linear-matrix-algebra-F03724	400	1.000	$x_{\{0\}}=3, x_{\{1\}}=-2$	$x_0 = 3, x_1 = -2$	if $x_0 = 3, x_1 = -2$, and $x_2 = 8$.
johnston-linear-matrix-algebra-F03725	400	1.000	$x_{\{2\}}=8$	$x_2 = 8$	if $x_0 = 3, x_1 = -2$, and $x_2 = 8$.
johnston-linear-matrix-algebra-F03726	400	1.000	$x_{\{6\}}, x_{\{7\}}$	x_6, x_7	By using this same procedure to compute x_6, x_7 , and so on, we see that
johnston-linear-matrix-algebra-F03727	401	1.000	$x_{\{n\}}=x_{\{n-1\}}+x_{\{n-2\}}$	$x_n = x_{n-1} + x_{n-2}$	a) $x_n = x_{n-1} + x_{n-2}$,
johnston-linear-matrix-algebra-F03728	401	1.000	$x_{\{n\}}=4 x_{\{n-1\}}-x_{\{n-2\}}-6 x_{\{n-3\}}$	$x_n = 4x_{n-1} - x_{n-2} - 6x_{n-3}$	b) $x_n = 4x_{n-1} - x_{n-2} - 6x_{n-3}$, and
johnston-linear-matrix-algebra-F03729	401	1.000	$x_{\{n\}}=x_{\{n-1\}}+4 x_{\{n-3\}}+2 x_{\{n-4\}}$	$x_n = x_{n-1} + 4x_{n-3} + 2x_{n-4}$	c) $x_n = x_{n-1} + 4x_{n-3} + 2x_{n-4}$.
johnston-linear-matrix-algebra-F03730	401	1.000	$a_{\{1\}}=a_{\{0\}}=1$	$a_1 = a_0 = 1$	a) This is a degree-2 linear recurrence relation with $a_1 = a_0 = 1$, so its
johnston-linear-matrix-algebra-F03731	401	0.877	$a_{\{2\}}=4, a_{\{1\}}=-1$	$a_2 = 4, a_1 = -1$	b) This is a degree-3 linear recurrence relation with $a_2 = 4, a_1 = -1$,
johnston-linear-matrix-algebra-F03732	401	1.000	$a_{\{0\}}=-6$	$a_0 = -6$	and $a_0 = -6$, so its companion matrix is
johnston-linear-matrix-algebra-F03733	401	0.948	$a_{\{3\}}=1, a_{\{2\}}=0, a_{\{1\}}=4$	$a_3 = 1, a_2 = 0, a_1 = 4$	lation with $a_3 = 1, a_2 = 0, a_1 = 4$, and $a_0 = 2$, so its companion
johnston-linear-matrix-algebra-F03734	401	0.948	$a_{\{0\}}=2$	$a_0 = 2$	lation with $a_3 = 1, a_2 = 0, a_1 = 4$, and $a_0 = 2$, so its companion
johnston-linear-matrix-algebra-F03735	401	0.999	$x_{\{0\}}, x_{\{1\}}, x_{\{2\}}, \ldots$	$x_0, x_1, x_2, \ldots$	Let $x_0, x_1, x_2, \ldots$ be a sequence satisfying a linear recurrence rela-
johnston-linear-matrix-algebra-F03736	401	1.000	$n=0,1,2, \ldots$	$n = 0, 1, 2, \ldots$	tion with companion matrix C . For each $n = 0, 1, 2, \ldots$, define $\mathbf{x}_n =$
johnston-linear-matrix-algebra-F03737	401	1.000	$\mathbf{x}_{\{n\}}=$	$\mathbf{x}_n =$	tion with companion matrix C . For each $n = 0, 1, 2, \ldots$, define $\mathbf{x}_n =$
johnston-linear-matrix-algebra-F03738	401	1.000	$\left(x_{\{n\}}, x_{\{n+1\}}, \ldots, x_{\{n+k-1\}}\right) \in \mathbb{R}^k$	$\left(x_n, x_{n+1}, \ldots, x_{n+k-1}\right) \in \mathbb{R}^k$	$\left(x_n, x_{n+1}, \ldots, x_{n+k-1}\right) \in \mathbb{R}^k$. Then

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johnston-linear-matrix-attrib F03739	402	1.000	$\mathbf{x}_0 = \left(x_0, x_1, \ldots, x_{k-1}\right)$	$\mathbf{x}_0 = (x_0, x_1, \ldots, x_{k-1})$	$\mathbf{x}_0 = (x_0, x_1, \ldots, x_{k-1})$
johnston-linear-matrix-attrib F03740	402	1.000	$n=k$	$n = k$	where we used the given recurrence relation (with $n = k$) to simplify the bottom
johnston-linear-matrix-attrib F03741	402	1.000	$C^n \mathbf{x}_0$	$C^n \mathbf{x}_0$	In particular, since x_n equals the first entry of $C^n \mathbf{x}_0$ (or the last entry of
johnston-linear-matrix-attrib F03742	402	1.000	$C^{n-k+1} \mathbf{x}_0$	$C^{n-k+1} \mathbf{x}_0$	$C^{n-k+1} \mathbf{x}_0$), any explicit formula for the powers of C immediately gives us an
johnston-linear-matrix-attrib F03743	403	1.000	$3^n - 2^n + 3(-1)^n$	$3^n - 2^n + 3(-1)^n$	and the top entry of this matrix-vector product is $3^n - 2^n + 3(-1)^n$. We
johnston-linear-matrix-attrib F03744	403	1.000	$x_n = 3^n - 2^n + 3(-1)^n$	$x_n = 3^n - 2^n + 3(-1)^n$	thus conclude that $x_n = 3^n - 2^n + 3(-1)^n$ for all n .
johnston-linear-matrix-attrib F03745	403	1.000	$p(\lambda) = (-1)^k \left(\lambda^k - a_{k-1} \lambda^{k-1} - \cdots - a_1 \lambda - a_0 \right)$	$p(\lambda) = (-1)^k (\lambda^k - a_{k-1} \lambda^{k-1} - \cdots - a_1 \lambda - a_0)$	is simply $p(\lambda) = (-1)^k (\lambda^k - a_{k-1} \lambda^{k-1} - \cdots - a_1 \lambda - a_0)$. Since the leading
johnston-linear-matrix-attrib F03746	404	1.000	x_{n-j}	x_{n-j}	x_{n-j} with λ^{k-j} .
johnston-linear-matrix-attrib F03747	404	1.000	λ^{k-j}	λ^{k-j}	x_{n-j} with λ^{k-j} .
johnston-linear-matrix-attrib F03748	404	1.000	$c_0, c_1, \ldots, c_{k-1}$	$c_0, c_1, \ldots, c_{k-1}$	choose $c_0, c_1, \ldots, c_{k-1}$
johnston-linear-matrix-attrib F03749	404	1.000	$n=0$	$n = 0$	plugging $n = 0$ into
johnston-linear-matrix-attrib F03750	404	1.000	$\lambda_0, \lambda_1, \ldots, \lambda_{k-1}$	$\lambda_0, \lambda_1, \ldots, \lambda_{k-1}$	whose characteristic polynomial has distinct roots $\lambda_0, \lambda_1, \ldots, \lambda_{k-1}$. Let
johnston-linear-matrix-attrib F03751	404	1.000	$p(\lambda) = \lambda^k - a_{k-1} \lambda^{k-1} - a_{k-2} \lambda^{k-2} - \cdots -$	$p(\lambda) = \lambda^k - a_{k-1} \lambda^{k-1} - a_{k-2} \lambda^{k-2} - \cdots -$	root of the characteristic polynomial $p(\lambda) = \lambda^k - a_{k-1} \lambda^{k-1} - a_{k-2} \lambda^{k-2} - \cdots -$
johnston-linear-matrix-attrib F03752	404	1.000	$a_1 \lambda - a_0$	$a_1 \lambda - a_0$	$a_1 \lambda - a_0$, so there is never a term of the form 0^n in the explicit formula (3.D.2)
johnston-linear-matrix-attrib F03753	404	1.000	0^n	0^n	$a_1 \lambda - a_0$, so there is never a term of the form 0^n in the explicit formula (3.D.2)
johnston-linear-matrix-attrib F03754	404	1.000	$a_0 = 0$	$a_0 = 0$	is simply that it would imply $a_0 = 0$, which means that the recurrence relation
johnston-linear-matrix-attrib F03755	404	1.000	$k-1$	$k - 1$	actually has degree $k - 1$ (or less), so p is not actually its characteristic polyno-
johnston-linear-matrix-attrib F03756	405	1.000	$\lambda_0 = 1, \lambda_1 = 2$	$\lambda_0 = 1, \lambda_1 = 2$	$\lambda_0 = 1, \lambda_1 = 2$, and
johnston-linear-matrix-attrib F03757	405	1.000	$\lambda_2 = 5$	$\lambda_2 = 5$	$\lambda_2 = 5$, while the
johnston-linear-matrix-attrib F03758	405	1.000	$x_0 = 1, x_1 = 4$	$x_0 = 1, x_1 = 4$	$x_0 = 1, x_1 = 4$, and
johnston-linear-matrix-attrib F03759	405	1.000	$x_2 = 22$	$x_2 = 22$	$x_2 = 22$.
johnston-linear-matrix-attrib F03760	405	1.000	x_k	x_k	which is exactly the claimed formula for x_k . Repeating this argument for x_{k+1} ,
johnston-linear-matrix-attrib F03761	405	1.000	x_{k+1}	x_{k+1}	which is exactly the claimed formula for x_k . Repeating this argument for x_{k+1} ,
johnston-linear-matrix-attrib F03762	405	1.000	x_{k+2}	x_{k+2}	x_{k+2} , and so on shows that the result holds for all n .
johnston-linear-matrix-attrib F03763	405	1.000	$(c_0, c_1, c_2) = (1, -1, 1)$	$(c_0, c_1, c_2) = (1, -1, 1)$	The unique solution is $(c_0, c_1, c_2) = (1, -1, 1)$, so we conclude that
johnston-linear-matrix-attrib F03764	405	1.000	$\mathbf{x}_n = C^n \mathbf{x}_0$	$\mathbf{x}_n = C^n \mathbf{x}_0$	$\mathbf{x}_n = C^n \mathbf{x}_0$, where C is the companion matrix, and then diagonalize C . In
johnston-linear-matrix-attrib F03765	406	1.000	$V \mathbf{c} = \mathbf{x}_0$	$V \mathbf{c} = \mathbf{x}_0$	that if $V \mathbf{c} = \mathbf{x}_0$ then
johnston-linear-matrix-attrib F03766	406	0.996	$V^{-1} \mathbf{x}_0 = \mathbf{c}$	$V^{-1} \mathbf{x}_0 = \mathbf{c}$	$V^{-1} \mathbf{x}_0 = \mathbf{c}$.

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johnston-linear-matrix-atlas-0-F03767	406	1.000	$\lambda_1=i$	$\lambda_1 = i$	pairs $(\lambda_1 = i$ and
johnston-linear-matrix-atlas-0-F03768	406	1.000	$\lambda_2=\overline{\lambda_1}=-i$	$\lambda_2 = \overline{i} = -i$	$\lambda_2 = \overline{i} = -i$ here), as
johnston-linear-matrix-atlas-0-F03769	406	1.000	$c_1=1-3i$	$c_1 = 1 - 3i$	$(c_1 = 1 - 3i$ and
johnston-linear-matrix-atlas-0-F03770	406	1.000	$c_2=\overline{1-3i}=1+3i$	$c_2 = \overline{1 - 3i} = 1 + 3i$	$c_2 = \overline{1 - 3i} = 1 + 3i$
johnston-linear-matrix-atlas-0-F03771	406	0.988	$\mathbf{c}=(c_0, c_1, \dots, c_{k-1})$	$\mathbf{c} = (c_0, c_1, \dots, c_{k-1})$	If we let $\mathbf{c} = (c_0, c_1, \dots, c_{k-1})$ be a vector for which $V\mathbf{c} = \mathbf{x}_0$, then Theo-
johnston-linear-matrix-atlas-0-F03772	406	1.000	$V^{-1}D^n\mathbf{c}$	$V D^n \mathbf{c}$	It is straightforward to explicitly compute the top entry of $V D^n \mathbf{c}$ to then
johnston-linear-matrix-atlas-0-F03773	406	1.000	$x_0=3, x_1=8$	$x_0 = 3, x_1 = 8$	if $x_0 = 3, x_1 = 8$, and $x_2 = 2$.
johnston-linear-matrix-atlas-0-F03774	406	1.000	$x_2=2$	$x_2 = 2$	if $x_0 = 3, x_1 = 8$, and $x_2 = 2$.
johnston-linear-matrix-atlas-0-F03775	406	1.000	$\lambda_0=2, \lambda_1=i$	$\lambda_0 = 2, \lambda_1 = i$	which has roots $\lambda_0 = 2, \lambda_1 = i$, and $\lambda_2 = -i$. Since these roots are distinct,
johnston-linear-matrix-atlas-0-F03776	406	1.000	$\lambda_2=-i$	$\lambda_2 = -i$	which has roots $\lambda_0 = 2, \lambda_1 = i$, and $\lambda_2 = -i$. Since these roots are distinct,
johnston-linear-matrix-atlas-0-F03777	406	1.000	$(c_0, c_1, c_2)=(1, 1-3i, 1+3i)$	$(c_0, c_1, c_2) = (1, 1 - 3i, 1 + 3i)$	The unique solution is $(c_0, c_1, c_2) = (1, 1 - 3i, 1 + 3i)$, so we conclude that
johnston-linear-matrix-atlas-0-F03778	406	0.814	$x_n=c_0\lambda_0^n+c_1\lambda_1^n+c_2\lambda_2^n=2^n+(1-3i)i^n+(1+3i)(-i)^n$	$x_n = c_0\lambda_0^n + c_1\lambda_1^n + c_2\lambda_2^n = 2^n + (1 - 3i)i^n + (1 + 3i)(-i)^n$	$x_n = c_0\lambda_0^n + c_1\lambda_1^n + c_2\lambda_2^n = 2^n + (1 - 3i)i^n + (1 + 3i)(-i)^n$ for all $n \geq 0$.
johnston-linear-matrix-atlas-0-F03779	406	0.814	$n \geq 0$	$n \geq 0$	$x_n = c_0\lambda_0^n + c_1\lambda_1^n + c_2\lambda_2^n = 2^n + (1 - 3i)i^n + (1 + 3i)(-i)^n$ for all $n \geq 0$.
johnston-linear-matrix-atlas-0-F03780	407	0.999	$r_0=\dots=r_{m-1}=1$	$r_0 = \dots = r_{m-1} = 1$	$r_0 = \dots = r_{m-1} = 1$.
johnston-linear-matrix-atlas-0-F03781	407	1.000	$\lambda_1=i=e^{i\pi/2}$	$\lambda_1 = i = e^{i\pi/2}$	we can write the complex roots λ_1 and λ_2 in their polar forms $\lambda_1 = i = e^{i\pi/2}$
johnston-linear-matrix-atlas-0-F03782	407	1.000	$\lambda_2=-i=e^{-i\pi/2}$	$\lambda_2 = -i = e^{-i\pi/2}$	and $\lambda_2 = -i = e^{-i\pi/2}$ and simplify:
johnston-linear-matrix-atlas-0-F03783	407	1.000	$e^{i\theta}+e^{-i\theta}=2\cos(\theta)$	$e^{i\theta} + e^{-i\theta} = 2\cos(\theta)$	so $e^{i\theta} + e^{-i\theta} = 2\cos(\theta)$ and $e^{i\theta} - e^{-i\theta} = 2i\sin(\theta)$ (see Appendix A.1).
johnston-linear-matrix-atlas-0-F03784	407	1.000	$e^{i\theta}-e^{-i\theta}=2i\sin(\theta)$	$e^{i\theta} - e^{-i\theta} = 2i\sin(\theta)$	so $e^{i\theta} + e^{-i\theta} = 2\cos(\theta)$ and $e^{i\theta} - e^{-i\theta} = 2i\sin(\theta)$ (see Appendix A.1).
johnston-linear-matrix-atlas-0-F03785	407	1.000	$\lambda_0, \lambda_1, \dots, \lambda_{m-1}$	$\lambda_0, \lambda_1, \dots, \lambda_{m-1}$	whose characteristic polynomial has roots $\lambda_0, \lambda_1, \dots, \lambda_{m-1}$, with multiplic-
johnston-linear-matrix-atlas-0-F03786	407	1.000	$r_0, r_1, \dots, r_{m-1}$	$r_0, r_1, \dots, r_{m-1}$	ities $r_0, r_1, \dots, r_{m-1}$, respectively. Let $q_0, q_1, \dots, q_{m-1}$ be the (necessarily
johnston-linear-matrix-atlas-0-F03787	407	1.000	$q_0, q_1, \dots, q_{m-1}$	$q_0, q_1, \dots, q_{m-1}$	ities $r_0, r_1, \dots, r_{m-1}$, respectively. Let $q_0, q_1, \dots, q_{m-1}$ be the (necessarily
johnston-linear-matrix-atlas-0-F03788	407	1.000	$r_0-1, r_1-1, \dots, r_{m-1}-1$	$r_0 - 1, r_1 - 1, \dots, r_{m-1} - 1$	unique) polynomials with degrees $r_0 - 1, r_1 - 1, \dots, r_{m-1} - 1$, respectively,
johnston-linear-matrix-atlas-0-F03789	408	0.874	x_{n-1}	x_{n-1}	x_{n-1} is 0).
johnston-linear-matrix-atlas-0-F03790	408	1.000	V_j	V_j	Notice that each V_j
johnston-linear-matrix-atlas-0-F03791	408	1.000	$k \times r_j$	$k \times r_j$	has size $k \times r_j$, so $\tilde{V}$
johnston-linear-matrix-atlas-0-F03792	408	1.000	$k \times (r_0+\dots+r_{m-1})=$	$k \times (r_0 + \dots + r_{m-1}) =$	$k \times (r_0 + \dots + r_{m-1}) =$
johnston-linear-matrix-atlas-0-F03793	408	1.000	$r_j=1$	$r_j = 1$	matrix if $r_j = 1$ (i.e., V_j

Identifier	Page	Conf.	LaTeX source	Rendered	Image
johnston-linear-matrix-attrib F03794	408	1.000	$k \times 1$	$k \times 1$	is $k \times 1$ for all j . This
johnston-linear-matrix-attrib F03795	408	1.000	$x_{\{0\}}=6, \ x_{\{1\}}=0$	$x_0 = 6, x_1 = 0$	if $x_0 = 6, x_1 = 0$, and $x_2 = 3$.
johnston-linear-matrix-attrib F03796	408	1.000	$x_{\{2\}}=3$	$x_2 = 3$	if $x_0 = 6, x_1 = 0$, and $x_2 = 3$.
johnston-linear-matrix-attrib F03797	408	1.000	$\lambda_{\{0\}}=2$	$\lambda_0 = 2$	which has roots $\lambda_0 = 2$ and $\lambda_1 = -1$, with multiplicities 1 and 2, respec-
johnston-linear-matrix-attrib F03798	408	1.000	$q_{\{0\}}$	q_0	tively. The polynomials q_0 and q_1 in Theorem 3.D.3 then have degrees 0
johnston-linear-matrix-attrib F03799	408	1.000	$q_{\{1\}}$	q_1	tively. The polynomials q_0 and q_1 in Theorem 3.D.3 then have degrees 0
johnston-linear-matrix-attrib F03800	408	0.979	$q_{\{0\}}(x)=c_{\{0\}}$	$q_0(x) = c_0$	and 1, respectively, so they are of the form $q_0(x) = c_0$ and $q_1(x) = c_1 + c_2x$.
johnston-linear-matrix-attrib F03801	408	0.979	$q_{\{1\}}(x)=c_{\{1\}}+c_{\{2\}} \ x$	$q_1(x) = c_1 + c_2x$	and 1, respectively, so they are of the form $q_0(x) = c_0$ and $q_1(x) = c_1 + c_2x$.
johnston-linear-matrix-attrib F03802	408	1.000	$c_{\{0\}}, \ c_{\{1\}}, \ c_{\{2\}}$	c_0, c_1, c_2	Our goal is thus to find c_0, c_1, c_2 such that
johnston-linear-matrix-attrib F03803	408	1.000	$n=0, 1$	$n = 0, 1$	By plugging $n = 0, 1$, and 2 into this equation, we arrive at the linear
johnston-linear-matrix-attrib F03804	408	1.000	$\left(c_{\{0\}}, \ c_{\{1\}}, \ c_{\{2\}}\right)=(1, 5, -3)$	$(c_0, c_1, c_2) = (1, 5, -3)$	which has unique solution $(c_0, c_1, c_2) = (1, 5, -3)$. It follows that
johnston-linear-matrix-attrib F03805	408	1.000	$q_{\{0\}}, \ q_{\{1\}}, \ \ldots, \ q_{\{k-1\}}$	$q_0, q_1, \dots, q_{k-1}$	<i>Proof of Theorem 3.D.3.</i> We start by showing that $q_0, q_1, \dots, q_{k-1}$ exist and
johnston-linear-matrix-attrib F03806	408	1.000	$q_{\{j\}}(n)=c_{\{j, \ 0\}}+c_{\{j, \ 1\}} \ n+\cdots+c_{\{j, \ r_{\{j\}}-1\}} \ n^{r_{\{j\}}-1}$	$q_j(n) = c_{j,0} + c_{j,1}n + \cdots + c_{j,r_j-1}n^{r_j-1}$	more explicitly as $q_j(n) = c_{j,0} + c_{j,1}n + \cdots + c_{j,r_j-1}n^{r_j-1}$ for $0 \leq j < m$. Then
johnston-linear-matrix-attrib F03807	408	1.000	$0 \ \leq \ j < m$	$0 \leq j < m$	more explicitly as $q_j(n) = c_{j,0} + c_{j,1}n + \cdots + c_{j,r_j-1}n^{r_j-1}$ for $0 \leq j < m$. Then
johnston-linear-matrix-attrib F03808	408	1.000	$c_{\{j, \ \ell\}}$	$c_{j,\ell}$	from $\ell = 0$) corresponds to $c_{j,\ell}$.
johnston-linear-matrix-attrib F03809	409	1.000	$\theta^{\{0\}}=1$	$0^0 = 1$	$0^0 = 1$, so $d_0 0^0 \lambda_j^0 = d_0$,
johnston-linear-matrix-attrib F03810	409	1.000	$d_{\{0\}} \ \theta^{\{0\}} \ \lambda_{\{j\}}^{\{0\}}=d_{\{0\}}$	$d_0 0^0 \lambda_j^0 = d_0$	$0^0 = 1$, so $d_0 0^0 \lambda_j^0 = d_0$,
johnston-linear-matrix-attrib F03811	409	0.999	$d_{\{0\}} \ \theta^{\{\ell\}} \ \lambda_{\{j\}}^{\{0\}}=0$	$d_0 0^\ell \lambda_j^0 = 0$	but $d_0 0^\ell \lambda_j^0 = 0$ when
johnston-linear-matrix-attrib F03812	409	1.000	$\ell > 0$	$\ell > 0$	$\ell > 0$.
johnston-linear-matrix-attrib F03813	409	0.999	$d_{\{0\}}, \ d_{\{1\}}, \ \ldots, \ d_{\{k-1\}}$	$d_0, d_1, \dots, d_{k-1}$	suppose that some linear combination (with coefficients $d_0, d_1, \dots, d_{k-1}$) of
johnston-linear-matrix-attrib F03814	409	1.000	$r_{\{0\}}+r_{\{1\}}+\cdots+r_{\{m-1\}}=k$	$r_0 + r_1 + \cdots + r_{m-1} = k$	enough). Counting multiplicity, we have thus found $r_0 + r_1 + \cdots + r_{m-1} = k$
johnston-linear-matrix-attrib F03815	409	1.000	$d_{\{0\}}=d_{\{1\}}=\cdots=d_{\{k-1\}}=0$	$d_0 = d_1 = \cdots = d_{k-1} = 0$	$d_0 = d_1 = \cdots = d_{k-1} = 0$. The rows of V are thus linearly independent, so it is
johnston-linear-matrix-attrib F03816	409	1.000	$x_{\{0\}}=0, \ x_{\{1\}}=3, \ x_{\{2\}}=8$	$x_0 = 0, x_1 = 3, x_2 = 8$	if $x_0 = 0, x_1 = 3, x_2 = 8$, and $x_3 = 7$.
johnston-linear-matrix-attrib F03817	409	1.000	$x_{\{3\}}=7$	$x_3 = 7$	if $x_0 = 0, x_1 = 3, x_2 = 8$, and $x_3 = 7$.
johnston-linear-matrix-attrib F03818	410	1.000	$1^{\{n\}}=1$	$1^n = 1$	Since $1^n = 1$ for all n ,
johnston-linear-matrix-attrib F03819	410	1.000	$\left(c_{\{0\}}+c_{\{1\}} \ n+c_{\{2\}} \ n^2\right) \ 1^{\{n\}}$	$(c_0 + c_1n + c_2n^2) 1^n$	$(c_0 + c_1n + c_2n^2) 1^n$
johnston-linear-matrix-attrib F03820	410	1.000	$c_{\{0\}}+c_{\{1\}} \ n+c_{\{2\}} \ n^2$	$c_0 + c_1n + c_2n^2$	$c_0 + c_1n + c_2n^2$ here.
johnston-linear-matrix-attrib F03821	410	1.000	$1-\phi \approx -0.6180$	$1 - \phi \approx -0.6180$	Since $1 - \phi \approx -0.6180$,

Identifier	Page	Conf.	LaTeX source	Rendered	Image																					
johnston-linear-matrix-atglr_u F03822	410	■ 1.000	$(1-\phi)^n \rightarrow 0$	$(1-\phi)^n \rightarrow 0$	$(1-\phi)^n \rightarrow 0$ as $n \rightarrow \infty$.																					
johnston-linear-matrix-atglr_u F03823	410	■ 1.000	$n \rightarrow \infty$	$n \rightarrow \infty$	$(1-\phi)^n \rightarrow 0$ as $n \rightarrow \infty$.																					
johnston-linear-matrix-atglr_u F03824	410	■ 0.996	$\lambda_0=1$	$\lambda_0 = 1$	which has roots $\lambda_0 = 1$ and $\lambda_1 = 3$, with multiplicities 3 and 1, respectively.																					
johnston-linear-matrix-atglr_u F03825	410	■ 0.996	$\lambda_1=3$	$\lambda_1 = 3$	which has roots $\lambda_0 = 1$ and $\lambda_1 = 3$, with multiplicities 3 and 1, respectively.																					
johnston-linear-matrix-atglr_u F03826	410	■ 1.000	$q_0(x)=c_0+c_1x+c_2x^2$	$q_0(x) = c_0 + c_1x + c_2x^2$	respectively, so they are of the form $q_0(x) = c_0 + c_1x + c_2x^2$ and $q_1(x) = c_3$.																					
johnston-linear-matrix-atglr_u F03827	410	■ 1.000	$q_1(x)=c_3$	$q_1(x) = c_3$	respectively, so they are of the form $q_0(x) = c_0 + c_1x + c_2x^2$ and $q_1(x) = c_3$.																					
johnston-linear-matrix-atglr_u F03828	410	■ 1.000	c_0, c_1, c_2, c_3	c_0, c_1, c_2, c_3	Our goal is thus to find c_0, c_1, c_2, c_3 such that																					
johnston-linear-matrix-atglr_u F03829	410	■ 1.000	$n=0,1,2$	$n = 0, 1, 2$	By plugging $n = 0, 1, 2$, and 3 into this equation, we arrive at the linear																					
johnston-linear-matrix-atglr_u F03830	410	■ 1.000	$\left(c_0, c_1, c_2, c_3\right)=(1,2,3,-1)$	$(c_0, c_1, c_2, c_3) = (1, 2, 3, -1)$	which has unique solution $(c_0, c_1, c_2, c_3) = (1, 2, 3, -1)$. It follows that																					
johnston-linear-matrix-atglr_u F03831	410	■ 1.000	$F_0=0, F_1=1$	$F_0 = 0, F_1 = 1$	satisfy $F_0 = 0, F_1 = 1$, and $F_n = F_{n-1} + F_{n-2}$ when $n \geq 2$), as well as the ratios																					
johnston-linear-matrix-atglr_u F03832	410	■ 1.000	F_n / F_{n-1}	F_n / F_{n-1}	<table><tr><th>n</th><th>F_n</th><th>F_n / F_{n-1}</th></tr><tr><td>0</td><td>0</td><td>–</td></tr><tr><td>1</td><td>1</td><td>–</td></tr><tr><td>2</td><td>1</td><td>1.0000</td></tr><tr><td>3</td><td>2</td><td>2.0000</td></tr><tr><td>4</td><td>3</td><td>1.5000</td></tr><tr><td>5</td><td>5</td><td>1.6667</td></tr></table>	n	F_n	F_n / F_{n-1}	0	0	–	1	1	–	2	1	1.0000	3	2	2.0000	4	3	1.5000	5	5	1.6667
n	F_n	F_n / F_{n-1}																								
0	0	–																								
1	1	–																								
2	1	1.0000																								
3	2	2.0000																								
4	3	1.5000																								
5	5	1.6667																								
johnston-linear-matrix-atglr_u F03833	410	■ 1.000	ϕ^n	ϕ^n	In particular, since ϕ^n is significantly larger than $(1-\phi)^n$ when n is large, F_n																					
johnston-linear-matrix-atglr_u F03834	410	■ 1.000	$(1-\phi)^n$	$(1-\phi)^n$	In particular, since ϕ^n is significantly larger than $(1-\phi)^n$ when n is large, F_n																					
johnston-linear-matrix-atglr_u F03835	410	■ 1.000	$\phi^n / \sqrt{5}$	$\phi^n / \sqrt{5}$	grows in roughly the same way that $\phi^n / \sqrt{5}$ grows, so the Fibonacci sequence																					
johnston-linear-matrix-atglr_u F03836	411	■ 1.000	$q_0(n) \lambda_0^n$	$q_0(n) \lambda_0^n$	$q_0(n) \lambda_0^n$ here is a																					
johnston-linear-matrix-atglr_u F03837	411	■ 1.000	q	q	q and any number c with $ c < 1$, we have																					
johnston-linear-matrix-atglr_u F03838	411	■ 0.887	$q(n+1)$	$q(n+1)$	The two limits above are hopefully intuitive enough $q(n+1)$ and $q(n)$ have the																					
johnston-linear-matrix-atglr_u F03839	411	■ 0.887	$q(n)$	$q(n)$	The two limits above are hopefully intuitive enough $q(n+1)$ and $q(n)$ have the																					
johnston-linear-matrix-atglr_u F03840	411	■ 1.000	$x_8=6308$	$x_8 = 6308$	that every term is roughly 3 times as large as the previous one (e.g., $x_8 = 6308$,																					
johnston-linear-matrix-atglr_u F03841	411	■ 1.000	$x_7=2056$	$x_7 = 2056$	which is roughly 3 times as large as $x_7 = 2056$). This now makes sense, since																					
johnston-linear-matrix-atglr_u F03842	412	■ 1.000	$x_0=0, x_1=0, x_2=1, x_n=x_{n-1}-4x_{n-2}+4x_{n-3}$	$x_0 = 0, x_1 = 0, x_2 = 1, x_n = x_{n-1} - 4x_{n-2} + 4x_{n-3}$	a) $x_0 = 0, x_1 = 0, x_2 = 1, x_n = x_{n-1} - 4x_{n-2} + 4x_{n-3}$ when $n \geq 3$, and																					
johnston-linear-matrix-atglr_u F03843	412	■ 1.000	$x_0=0, x_1=1, x_2=0, x_n=2x_{n-1}+9x_{n-2}-18x_{n-3}$	$x_0 = 0, x_1 = 1, x_2 = 0, x_n = 2x_{n-1} + 9x_{n-2} - 18x_{n-3}$	b) $x_0 = 0, x_1 = 1, x_2 = 0, x_n = 2x_{n-1} + 9x_{n-2} - 18x_{n-3}$ when $n \geq 3$.																					
johnston-linear-matrix-atglr_u F03844	412	■ 1.000	$1, 2i$	$1, 2i$	which has roots 1, $2i$, and $-2i$. Since the absolute values of these																					
johnston-linear-matrix-atglr_u F03845	412	■ 1.000	$-2i$	$-2i$	which has roots 1, $2i$, and $-2i$. Since the absolute values of these																					

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johnston-linear-matrix-atglu-4 F03846	412	■ 0.999	$x_{\{n\}} / x_{\{n-1\}}=1$	$x_n/x_{n-1} = 1$	back and forth between repeating the previous term (so $x_n/x_{n-1} = 1$)
johnston-linear-matrix-atglu-4 F03847	412	■ 0.979	$x_{\{n\}} / x_{\{n-1\}} \approx -4$	$x_n/x_{n-1} \approx -4$	and growing by roughly a factor of -4 (so $x_n/x_{n-1} \approx -4$). The
johnston-linear-matrix-atglu-4 F03848	412	■ 1.000	$(2\ i)^{\{2\}}=(-2\ i)^{\{2\}}=-4$	$(2i)^2 = (-2i)^2 = -4$	polynomial satisfy $(2i)^2 = (-2i)^2 = -4$.
johnston-linear-matrix-atglu-4 F03849	412	■ 1.000	$x_{\{n\}} / x_{\{n-1\}}$	x_n/x_{n-1}	exist—the ratio x_n/x_{n-1} alternates back and forth between 0 and
johnston-linear-matrix-atglu-4 F03850	412	■ 1.000	$3^{\{2\}}=(-3)^{\{2\}}=9$	$3^2 = (-3)^2 = 9$	its characteristic polynomial satisfy $3^2 = (-3)^2 = 9$.
johnston-linear-matrix-atglu-4 F03851	413	■ 1.000	$x_{\{n-3\}}, x_{\{n-11\}}, x_{\{n-12\}}$	$x_{n-3}, x_{n-11}, x_{n-12}$	Of $x_{n-3}, x_{n-11}, x_{n-12}$,
johnston-linear-matrix-atglu-4 F03852	413	■ 0.915	$x_{\{n-45\}}, x_{\{n-54\}}$	x_{n-45}, x_{n-54}	x_{n-45}, x_{n-54} is 0.
johnston-linear-matrix-atglu-4 F03853	413	■ 1.000	$x_{\{5\}}=6$	$x_5 = 6$	the look-and-say sequence—for example, $x_5 = 6$ since the fifth terms in
johnston-linear-matrix-atglu-4 F03854	413	■ 1.000	$p(\backslash\lambda\mathrm{b}\mathrm{d}\mathrm{a})=(\backslash\lambda\mathrm{b}\mathrm{d}\mathrm{a}+1)(\backslash\lambda\mathrm{b}\mathrm{d}\mathrm{a}-1)^{\{2\}}\ q(\backslash\lambda\mathrm{b}\mathrm{d}\mathrm{a})$	$p(\lambda) = (\lambda + 1)(\lambda - 1)^2q(\lambda)$	as $p(\lambda) = (\lambda + 1)(\lambda - 1)^2q(\lambda)$, where $q(\lambda)$ is the following degree-71
johnston-linear-matrix-atglu-4 F03855	413	■ 1.000	$q(\backslash\lambda\mathrm{b}\mathrm{d}\mathrm{a})$	$q(\lambda)$	as $p(\lambda) = (\lambda + 1)(\lambda - 1)^2q(\lambda)$, where $q(\lambda)$ is the following degree-71
johnston-linear-matrix-atglu-4 F03856	414	■ 0.992	$x_{\{n\}}=x_{\{n-1\}}+6\ x_{\{n-2\}}$	$x_n = x_{n-1} + 6x_{n-2}$	*(a) $x_n = x_{n-1} + 6x_{n-2}$
johnston-linear-matrix-atglu-4 F03857	414	■ 1.000	$x_{\{n\}}=6\ x_{\{n-1\}}-9\ x_{\{n-2\}}$	$x_n = 6x_{n-1} - 9x_{n-2}$	(b) $x_n = 6x_{n-1} - 9x_{n-2}$
johnston-linear-matrix-atglu-4 F03858	414	■ 0.995	$x_{\{n\}}=6\ x_{\{n-1\}}-4\ x_{\{n-2\}}$	$x_n = 6x_{n-1} - 4x_{n-2}$	*(c) $x_n = 6x_{n-1} - 4x_{n-2}$
johnston-linear-matrix-atglu-4 F03859	414	■ 1.000	$x_{\{n\}}=2\ x_{\{n-1\}}-2\ x_{\{n-2\}}$	$x_n = 2x_{n-1} - 2x_{n-2}$	(d) $x_n = 2x_{n-1} - 2x_{n-2}$
johnston-linear-matrix-atglu-4 F03860	414	■ 1.000	$x_{\{n\}}=6\ x_{\{n-1\}}-11\ x_{\{n-2\}}+6\ x_{\{n-3\}}$	$x_n = 6x_{n-1} - 11x_{n-2} + 6x_{n-3}$	*(e) $x_n = 6x_{n-1} - 11x_{n-2} + 6x_{n-3}$
johnston-linear-matrix-atglu-4 F03861	414	■ 1.000	$x_{\{n\}}=2\ x_{\{n-2\}}+x_{\{n-3\}}$	$x_n = 2x_{n-2} + x_{n-3}$	(f) $x_n = 2x_{n-2} + x_{n-3}$
johnston-linear-matrix-atglu-4 F03862	414	■ 0.982	$x_{\{n\}}=6\ x_{\{n-1\}}-12\ x_{\{n-2\}}+8\ x_{\{n-3\}}$	$x_n = 6x_{n-1} - 12x_{n-2} + 8x_{n-3}$	*(g) $x_n = 6x_{n-1} - 12x_{n-2} + 8x_{n-3}$
johnston-linear-matrix-atglu-4 F03863	414	■ 1.000	$x_{\{n\}}=7\ x_{\{n-1\}}-8\ x_{\{n-2\}}-16\ x_{\{n-3\}}$	$x_n = 7x_{n-1} - 8x_{n-2} - 16x_{n-3}$	(h) $x_n = 7x_{n-1} - 8x_{n-2} - 16x_{n-3}$
johnston-linear-matrix-atglu-4 F03864	414	■ 1.000	$x_{\{n\}}=5\ x_{\{n-1\}}-4\ x_{\{n-2\}}-6\ x_{\{n-3\}}$	$x_n = 5x_{n-1} - 4x_{n-2} - 6x_{n-3}$	*(i) $x_n = 5x_{n-1} - 4x_{n-2} - 6x_{n-3}$
johnston-linear-matrix-atglu-4 F03865	414	■ 1.000	$x_{\{n\}}=7\ x_{\{n-1\}}-17\ x_{\{n-2\}}+15\ x_{\{n-3\}}$	$x_n = 7x_{n-1} - 17x_{n-2} + 15x_{n-3}$	(j) $x_n = 7x_{n-1} - 17x_{n-2} + 15x_{n-3}$
johnston-linear-matrix-atglu-4 F03866	414	■ 0.999	$x_{\{0\}}=4,\ x_{\{1\}}=-3$	$x_0 = 4, x_1 = -3$	*(a) $x_0 = 4, x_1 = -3,$
johnston-linear-matrix-atglu-4 F03867	414	■ 1.000	$x_{\{0\}}=1,\ x_{\{1\}}=0$	$x_0 = 1, x_1 = 0$	(b) $x_0 = 1, x_1 = 0,$
johnston-linear-matrix-atglu-4 F03868	414	■ 0.993	$x_{\{0\}}=2,\ x_{\{1\}}=0$	$x_0 = 2, x_1 = 0$	*(c) $x_0 = 2, x_1 = 0,$
johnston-linear-matrix-atglu-4 F03869	414	■ 1.000	$x_{\{0\}}=2,\ x_{\{1\}}=3$	$x_0 = 2, x_1 = 3$	(d) $x_0 = 2, x_1 = 3,$
johnston-linear-matrix-atglu-4 F03870	414	■ 0.997	$x_{\{0\}}=0,\ x_{\{1\}}=2,\ x_{\{2\}}=8$	$x_0 = 0, x_1 = 2, x_2 = 8$	*(e) $x_0 = 0, x_1 = 2, x_2 = 8,$
johnston-linear-matrix-atglu-4 F03871	414	■ 1.000	$x_{\{0\}}=0,\ x_{\{1\}}=2,\ x_{\{2\}}=16$	$x_0 = 0, x_1 = 2, x_2 = 16$	(f) $x_0 = 0, x_1 = 2, x_2 = 16,$
johnston-linear-matrix-atglu-4 F03872	414	■ 1.000	$x_{\{0\}}=-2,\ x_{\{1\}}=1,\ x_{\{2\}}=5$	$x_0 = -2, x_1 = 1, x_2 = 5$	*(g) $x_0 = -2, x_1 = 1, x_2 = 5,$
johnston-linear-matrix-atglu-4 F03873	415	■ 1.000	$x_{\{0\}}=7,\ x_{\{1\}}=8,\ x_{\{2\}}=4$	$x_0 = 7, x_1 = 8, x_2 = 4$	(h) $x_0 = 7, x_1 = 8, x_2 = 4,$
johnston-linear-matrix-atglu-4 F03874	415	■ 1.000	$x_{\{n\}}=4\ x_{\{n-1\}}-6\ x_{\{n-2\}}+4\ x_{\{n-3\}}$	$x_n = 4x_{n-1} - 6x_{n-2} + 4x_{n-3}$	$x_n = 4x_{n-1} - 6x_{n-2} + 4x_{n-3}$ when $n \geq 3$

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johnston-linear-matrix-atgcluro F03875	415	■ 1.000	$x_{\{n\}}=2^{\{n\}}-3^{\{n\}}$	$x_n = 2^n - 3^n$	(b) The sequence defined by the formula $x_n = 2^n - 3^n$
johnston-linear-matrix-atgcluro F03876	415	■ 0.976	$x_{\{n\}}=n^{\{2\}}-n^{\{3\}}$	$x_n = n^2 - n^3$	*(c) The sequence defined by the formula $x_n = n^2 - n^3$
johnston-linear-matrix-atgcluro F03877	415	■ 1.000	$x_{\{n\}}=\sqrt{2\ n}-$	$x_n = \sqrt{2n}-$	(d) The sequence defined by the formula $x_n = \sqrt{2n}-$
johnston-linear-matrix-atgcluro F03878	415	■ 1.000	$\sqrt{3\ n}$	$\sqrt{3n}$	$\sqrt{3n}$ satisfies a linear recurrence relation.
johnston-linear-matrix-atgcluro F03879	415	■ 0.916	$\lim_{n\rightarrow\infty} x_{\{n+1\}} / x_{\{n\}}=4$	$\lim_{n\rightarrow\infty} x_{n+1}/x_n = 4$	2, then $\lim_{n\rightarrow\infty} x_{n+1}/x_n = 4$.
johnston-linear-matrix-atgcluro F03880	415	■ 0.853	$x_{\{n\}}=2^{\{n\}}+3^{\{n\}}$	$x_n = 2^n + 3^n$	*(a) $x_n = 2^n + 3^n$
johnston-linear-matrix-atgcluro F03881	415	■ 1.000	$x_{\{n\}}=2^{\{n\}}-3^{\{n+1\}}$	$x_n = 2^n - 3^{n+1}$	(b) $x_n = 2^n - 3^{n+1}$
johnston-linear-matrix-atgcluro F03882	415	■ 0.977	$x_{\{n\}}=2^{\{n\}}+3^{\{n\}}+4^{\{n\}}$	$x_n = 2^n + 3^n + 4^n$	*(c) $x_n = 2^n + 3^n + 4^n$
johnston-linear-matrix-atgcluro F03883	415	■ 1.000	$x_{\{n\}}=(1+n)\ 2^{\{n\}}+3^{\{n\}}$	$x_n = (1+n)2^n + 3^n$	(d) $x_n = (1+n)2^n + 3^n$
johnston-linear-matrix-atgcluro F03884	415	■ 0.999	$x_{\{n\}}=\left(1+n+n^{\{2\}}\right)2^{\{n\}}+3^{\{n\}}$	$x_n = (1+n+n^2)2^n + 3^n$	*(e) $x_n = (1+n+n^2)2^n + 3^n$
johnston-linear-matrix-atgcluro F03885	415	■ 1.000	$x_{\{n\}}=7+2^{\{n\}}$	$x_n = 7 + 2^n$	(f) $x_n = 7 + 2^n$
johnston-linear-matrix-atgcluro F03886	415	■ 0.975	$x_{\{n\}}=n^{\{2\}}$	$x_n = n^2$	*(g) $x_n = n^2$
johnston-linear-matrix-atgcluro F03887	415	■ 1.000	$x_{\{n\}}=2(1+i)^{\{n\}}+2(1-i)^{\{n\}}$	$x_n = 2(1+i)^n + 2(1-i)^n$	(h) $x_n = 2(1+i)^n + 2(1-i)^n$
johnston-linear-matrix-atgcluro F03888	415	■ 0.808	$x_{\{n\}}=(n-5)\ 3^{\{n\}}+(3+i)(1-2\ i)^{\{n\}}+(3-i)(1+2\ i)^{\{n\}}$	$x_n = (n-5)3^n + (3+i)(1-2i)^n + (3-i)(1+2i)^n$	*(i) $x_n = (n-5)3^n + (3+i)(1-2i)^n + (3-i)(1+2i)^n$
johnston-linear-matrix-atgcluro F03889	415	■ 1.000	$x_{\{n\}}=2^{\{n\}}\ \cos\left(\pi\ n\ /\ 4\right)-2^{\{n\}}\ \sin\left(\pi\ n\ /\ 4\right)$	$x_n = 2^n \cos(\pi n/4) - 2^n \sin(\pi n/4)$	(j) $x_n = 2^n \cos(\pi n/4) - 2^n \sin(\pi n/4)$
johnston-linear-matrix-atgcluro F03890	415	■ 1.000	$x_{\{n\}}=-x_{\{n-2\}}$	$x_n = -x_{n-2}$	$x_n = -x_{n-2}$ when $n \geq 2$
johnston-linear-matrix-atgcluro F03891	415	■ 1.000	$x_{\{0\}}=4,\ x_{\{1\}}=4$	$x_0 = 4, x_1 = 4$	(b) $x_0 = 4, x_1 = 4,$
johnston-linear-matrix-atgcluro F03892	415	■ 1.000	$x_{\{0\}}=3,\ x_{\{1\}}=1,\ x_{\{2\}}=-7$	$x_0 = 3, x_1 = 1, x_2 = -7$	*(c) $x_0 = 3, x_1 = 1, x_2 = -7,$
johnston-linear-matrix-atgcluro F03893	415	■ 1.000	$x_{\{n\}}=x_{\{n-1\}}-4\ x_{\{n-2\}}+4\ x_{\{n-3\}}$	$x_n = x_{n-1} - 4x_{n-2} + 4x_{n-3}$	$x_n = x_{n-1} - 4x_{n-2} + 4x_{n-3}$ when $n \geq 3$
johnston-linear-matrix-atgcluro F03894	415	■ 1.000	$x_{\{0\}}=-1,\ x_{\{1\}}=0,\ x_{\{2\}}=8$	$x_0 = -1, x_1 = 0, x_2 = 8$	(d) $x_0 = -1, x_1 = 0, x_2 = 8,$
johnston-linear-matrix-atgcluro F03895	415	■ 1.000	$x_{\{n\}}=4\ x_{\{n-1\}}-8\ x_{\{n-2\}}+8\ x_{\{n-3\}}$	$x_n = 4x_{n-1} - 8x_{n-2} + 8x_{n-3}$	$x_n = 4x_{n-1} - 8x_{n-2} + 8x_{n-3}$ when $n \geq 3$
johnston-linear-matrix-atgcluro F03896	415	■ 1.000	$x_{\{n\}}=\left\lfloor 4+\sqrt{11}\right\rfloor^{\{n\}}\right\rfloor$	$x_n = \left\lfloor (4 + \sqrt{11})^n \right\rfloor$	$x_n = \lfloor (4 + \sqrt{11})^n \rfloor$, where $\lfloor \cdot \rfloor$ denotes the “floor” func-
johnston-linear-matrix-atgcluro F03897	415	■ 1.000	$\lfloor\cdot\rfloor$	$\lfloor \cdot \rfloor$	$x_n = \lfloor (4 + \sqrt{11})^n \rfloor$, where $\lfloor \cdot \rfloor$ denotes the “floor” func-
johnston-linear-matrix-atgcluro F03898	415	■ 1.000	$4+\sqrt{11}$	$4 + \sqrt{11}$	relation should have $4 + \sqrt{11}$ as a root.]
johnston-linear-matrix-atgcluro F03899	415	■ 1.000	$x_{\{n\}}=x_{\{n-1\}}-$	$x_n = x_{n-1}-$	quence defined by the linear recurrence relation $x_n = x_{n-1}-$
johnston-linear-matrix-atgcluro F03900	415	■ 1.000	$4\ x_{\{n-2\}}+4\ x_{\{n-3\}}$	$4x_{n-2} + 4x_{n-3}$	$4x_{n-2} + 4x_{n-3}$, with initial conditions $x_0 = 0, x_1 = 0$ and
johnston-linear-matrix-atgcluro F03901	415	■ 1.000	$x_{\{0\}}=0,\ x_{\{1\}}=0$	$x_0 = 0, x_1 = 0$	$4x_{n-2} + 4x_{n-3}$, with initial conditions $x_0 = 0, x_1 = 0$ and
johnston-linear-matrix-atgcluro F03902	415	■ 1.000	$x_{\{2\}}=1$	$x_2 = 1$	$x_2 = 1$, repeats every term (for example, $x_0 = x_1, x_2 = x_3,$
johnston-linear-matrix-atgcluro F03903	415	■ 1.000	$x_{\{0\}}=x_{\{1\}},\ x_{\{2\}}=x_{\{3\}}$	$x_0 = x_1, x_2 = x_3$	$x_2 = 1$, repeats every term (for example, $x_0 = x_1, x_2 = x_3,$

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johnston-linear-matrix-atgela_F03904	417	■ 1.000	$a \ b=b \ a, \ a+b=b+a$	$ab = ba, a + b = b + a$	That is, we still have all of the properties like $ab = ba$, $a + b = b + a$, and
johnston-linear-matrix-atgela_F03905	417	■ 1.000	$a, \ b, \ c \ \text{in} \ \mathbb{C}$	$a, b, c \in \mathbb{C}$	$a(b + c) = ab + ac$ for all $a, b, c \in \mathbb{C}$ that we would expect “numbers” to
johnston-linear-matrix-atgela_F03906	417	■ 0.992	$\varepsilon \times 0=1$	$\varepsilon \times 0 = 1$	property that $\varepsilon \times 0 = 1$ (i.e., we are thinking of ε as $1/0$ much like we
johnston-linear-matrix-atgela_F03907	417	■ 0.992	$1 \ / \ 0$	$1/0$	property that $\varepsilon \times 0 = 1$ (i.e., we are thinking of ε as $1/0$ much like we
johnston-linear-matrix-atgela_F03908	417	■ 1.000	$\sqrt{-1}$	$\sqrt{-1}$	think of i as $\sqrt{-1}$), then we have
johnston-linear-matrix-atgela_F03909	417	■ 1.000	$(a+b \ i)+(c+d \ i)=(a+c)+(b+d) \ i$	$(a + bi) + (c + di) = (a + c) + (b + d)i$	add up their real and imaginary parts: $(a + bi) + (c + di) = (a + c) + (b + d)i$,
johnston-linear-matrix-atgela_F03910	417	■ 1.000	$x^{\wedge}\{2\}-2 \ x+5=0$	$x^2 - 2x + 5 = 0$	For example, the equation $x^2 - 2x + 5 = 0$ has no real solutions, but it has
johnston-linear-matrix-atgela_F03911	417	■ 1.000	$x=1+2 \ i$	$x = 1 + 2i$	$x = 1 + 2i$ as a complex solution, since
johnston-linear-matrix-atgela_F03912	418	■ 0.923	$a+b \ i \ \text{in} \ \mathbb{C}$	$a + bi \in \mathbb{C}$	complex number $a + bi \in \mathbb{C}$ as the vector $(a, b) \in \mathbb{R}^2$), but with a multiplica-
johnston-linear-matrix-atgela_F03913	418	■ 0.923	$(a, \ b) \ \text{in} \ \mathbb{R}^{\wedge}\{2\}$	$(a, b) \in \mathbb{R}^2$	complex number $a + bi \in \mathbb{C}$ as the vector $(a, b) \in \mathbb{R}^2$), but with a multiplica-
johnston-linear-matrix-atgela_F03914	419	■ 1.000	$(a+b \ i)(a-b \ i)$	$(a + bi)(a - bi)$	$(a + bi)(a - bi)$ here
johnston-linear-matrix-atgela_F03915	419	■ 0.957	$(a \ b-a \ b) \ i=0$	$(ab - ab)i = 0$	$(ab - ab)i = 0$.
johnston-linear-matrix-atgela_F03916	419	■ 1.000	$z=a+b \ i$	$z = a + bi$	• The magnitude of a complex number $z = a + bi$ can be written in terms
johnston-linear-matrix-atgela_F03917	419	■ 1.000	$ z ^{\wedge}\{2\}=z \ \overline{z}$	$ z ^2 = z\bar{z}$	of the product of z with its complex conjugate: $ z ^2 = z\bar{z}$, since
johnston-linear-matrix-atgela_F03918	419	■ 1.000	$z= z \ u$	$z = z u$	$z = z u$, where $ z $ is the magnitude of z and u is a number on the unit circle in
johnston-linear-matrix-atgela_F03919	419	■ 1.000	$ z $	$ z $	$z = z u$, where $ z $ is the magnitude of z and u is a number on the unit circle in
johnston-linear-matrix-atgela_F03920	419	■ 1.000	u	u	$z = z u$, where $ z $ is the magnitude of z and u is a number on the unit circle in
johnston-linear-matrix-atgela_F03921	419	■ 1.000	$\theta \ \text{in} [0,2 \ \pi)$	$\theta \in [0, 2\pi)$	$\theta \in [0, 2\pi)$ means
johnston-linear-matrix-atgela_F03922	419	■ 1.000	$2 \ \pi$	2π	(inclusive) and 2π
johnston-linear-matrix-atgela_F03923	419	■ 1.000	$\cos \ (\theta)+i \ \sin \ (\theta)$	$\cos(\theta) + i \sin(\theta)$	unit circle in the complex plane can be written in the form $\cos(\theta) + i \sin(\theta)$, as
johnston-linear-matrix-atgela_F03924	419	■ 1.000	$z=$	$z =$	It follows that we can write every complex number in the form $z =$
johnston-linear-matrix-atgela_F03925	419	■ 1.000	$ z (\cos \ (\theta)+i \ \sin \ (\theta))$	$ z (\cos(\theta) + i \sin(\theta))$	$ z (\cos(\theta) + i \sin(\theta))$. However, we can simplify this expression somewhat
johnston-linear-matrix-atgela_F03926	419	■ 1.000	$\operatorname{Re}\!\left(e^{\wedge}\{i \ \theta\}\right)=\cos \ (\theta)$	$\operatorname{Re}\left(e^{i\theta}\right) = \cos(\theta)$	$\operatorname{Re}(e^{i\theta}) = \cos(\theta)$ and
johnston-linear-matrix-atgela_F03927	419	■ 0.995	$\operatorname{Im}\!\left(e^{\wedge}\{i \ \theta\}\right)=\sin \ (\theta)$	$\operatorname{Im}\left(e^{i\theta}\right) = \sin(\theta)$	$\operatorname{Im}(e^{i\theta}) = \sin(\theta)$.
johnston-linear-matrix-atgela_F03928	419	■ 1.000	$e^{\wedge}\{i \ \theta\}$	$e^{i\theta}$	this formula simply as a definition—just think of $e^{i\theta}$ as a shorthand for the
johnston-linear-matrix-atgela_F03929	420	■ 0.980	$ \cos \ (\theta)+i \ \sin \ (\theta) ^{\wedge}\{2\}=$	$ \cos(\theta) + i \sin(\theta) ^2 =$	$ \cos(\theta) + i \sin(\theta) ^2 =$
johnston-linear-matrix-atgela_F03930	420	■ 1.000	$\cos ^{\wedge}\{2\}(\theta)+\sin ^{\wedge}\{2\}(\theta)=1$	$\cos^2(\theta) + \sin^2(\theta) = 1$	$\cos^2(\theta) + \sin^2(\theta) = 1$,

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johnston-linear-matrix-atgkwr_u F03931	420	■ 0.964	$\operatorname{sign}(b) = \pm 1$	$\operatorname{sign}(b) = \pm 1$	$\operatorname{sign}(b) = \pm 1$,
johnston-linear-matrix-atgkwr_u F03932	420	■ 1.000	$b < 0$	$b < 0$	or negative. If $b < 0$
johnston-linear-matrix-atgkwr_u F03933	420	■ 1.000	$-\pi < \theta < 0$	$-\pi < \theta < 0$	$-\pi < \theta < 0$, which
johnston-linear-matrix-atgkwr_u F03934	420	■ 1.000	$[0, 2 \pi)$	$[0, 2\pi)$	interval $[0, 2\pi)$ by
johnston-linear-matrix-atgkwr_u F03935	420	■ 1.000	$e^x, \cos(x)$	$e^x, \cos(x)$	e^x , $\cos(x)$, and $\sin(x)$ have Taylor series
johnston-linear-matrix-atgkwr_u F03936	420	■ 1.000	$x = i \theta$	$x = i\theta$	respectively. Plugging $x = i\theta$ into the Taylor series for e^x then gives
johnston-linear-matrix-atgkwr_u F03937	420	■ 1.000	$z = r e^{i \theta}$	$z = r e^{i\theta}$	number $z \in \mathbb{C}$ in the form $z = r e^{i\theta}$, where r is the magnitude of z (i.e., $r = z $)
johnston-linear-matrix-atgkwr_u F03938	420	■ 1.000	$r = z $	$r = z $	number $z \in \mathbb{C}$ in the form $z = r e^{i\theta}$, where r is the magnitude of z (i.e., $r = z $)
johnston-linear-matrix-atgkwr_u F03939	420	■ 1.000	$\left(r_1 e^{i \theta_1}\right) \left(r_2 e^{i \theta_2}\right) =$	$\left(r_1 e^{i\theta_1}\right) \left(r_2 e^{i\theta_2}\right) =$	in polar form, but multiplication is quite straightforward: $(r_1 e^{i\theta_1})(r_2 e^{i\theta_2}) =$
johnston-linear-matrix-atgkwr_u F03940	420	■ 1.000	$\left(r_1 r_2\right) e^{i\left(\theta_1+\theta_2\right)}$	$\left(r_1 r_2\right) e^{i\left(\theta_1+\theta_2\right)}$	$\left(r_1 r_2\right) e^{i\left(\theta_1+\theta_2\right)}$. We can thus think of complex numbers as stretched rotations—
johnston-linear-matrix-atgkwr_u F03941	420	■ 1.000	$\left(r e^{i \theta}\right)^n = r^n e^{i n \theta}$	$\left(r e^{i\theta}\right)^n = r^n e^{i n \theta}$	multiplying a complex number in polar form by itself gives $(r e^{i\theta})^n = r^n e^{i n \theta}$ for
johnston-linear-matrix-atgkwr_u F03942	421	■ 1.000	$e^{2 \pi i}=e^{0 i}=1$	$e^{2 \pi i}=e^{0 i}=1$	that $e^{2 \pi i}=e^{0 i}=1$.
johnston-linear-matrix-atgkwr_u F03943	421	■ 1.000	$z=i$	$z=i$	Compute all 3 cube roots of the complex number $z=i$.
johnston-linear-matrix-atgkwr_u F03944	421	■ 1.000	$z=e^{\pi i / 2}$	$z=e^{\pi i / 2}$	We start by writing z in its polar form $z=e^{\pi i / 2}$, from which we see
johnston-linear-matrix-atgkwr_u F03945	421	■ 1.000	$e^{\pi i / 6}=(\sqrt{3}+i) / 2$	$e^{\pi i / 6}=(\sqrt{3}+i) / 2$	Geometrically, the cube root $e^{\pi i / 6}=(\sqrt{3}+i) / 2$ of i lies on the unit
johnston-linear-matrix-atgkwr_u F03946	421	■ 1.000	$r^{1 / n} e^{i \theta / n}$	$r^{1 / n} e^{i \theta / n}$	$r^{1 / n} e^{i \theta / n}$ its principal n-th root , which we denote by $z^{1 / n}$. The principal root
johnston-linear-matrix-atgkwr_u F03947	421	■ 1.000	$z^{1 / n}$	$z^{1 / n}$	$r^{1 / n} e^{i \theta / n}$ its principal n-th root , which we denote by $z^{1 / n}$. The principal root
johnston-linear-matrix-atgkwr_u F03948	422	■ 1.000	$p: \mathbb{C} \rightarrow \mathbb{C}$	$p: \mathbb{C} \rightarrow \mathbb{C}$	$p: \mathbb{C} \rightarrow \mathbb{C}$ with
johnston-linear-matrix-atgkwr_u F03949	422	■ 1.000	$\theta=0$	$\theta=0$	z is a positive real number (i.e., $\theta=0$) then its principal roots are positive real
johnston-linear-matrix-atgkwr_u F03950	422	■ 0.996	$-i$	$-i$	root of -1 is i , not $-i$), and we showed in Example A.1.1 that the principal
johnston-linear-matrix-atgkwr_u F03951	422	■ 1.000	$z=e^{\pi i / 2}=i$	$z=e^{\pi i / 2}=i$	cube root of $z=e^{\pi i / 2}=i$ is $z^{1 / 3}=e^{\pi i / 6}=(\sqrt{3}+i) / 2$.
johnston-linear-matrix-atgkwr_u F03952	422	■ 1.000	$z^{1 / 3}=e^{\pi i / 6}=(\sqrt{3}+i) / 2$	$z^{1 / 3}=e^{\pi i / 6}=(\sqrt{3}+i) / 2$	cube root of $z=e^{\pi i / 2}=i$ is $z^{1 / 3}=e^{\pi i / 6}=(\sqrt{3}+i) / 2$.
johnston-linear-matrix-atgkwr_u F03953	422	■ 1.000	$p: \mathbb{R} \rightarrow \mathbb{R}$	$p: \mathbb{R} \rightarrow \mathbb{R}$	A polynomial is a function $p: \mathbb{R} \rightarrow \mathbb{R}$ of the form
johnston-linear-matrix-atgkwr_u F03954	422	■ 1.000	$a_0, a_1, a_2, \ldots, a_{n-1}, a_n \in \mathbb{R}$	$a_0, a_1, a_2, \ldots, a_{n-1}, a_n \in \mathbb{R}$	where $a_0, a_1, a_2, \ldots, a_{n-1}, a_n \in \mathbb{R}$ are constants (called the coefficients of p).
johnston-linear-matrix-atgkwr_u F03955	422	■ 0.995	$a_n \neq 0$	$a_n \neq 0$	example, the polynomial displayed above has degree n , as long as $a_n \neq 0$).
johnston-linear-matrix-atgkwr_u F03956	422	■ 1.000	$p(x)=0$	$p(x)=0$	which are the solutions to the equation $p(x)=0$. Geometrically, the roots of
johnston-linear-matrix-atgkwr_u F03957	422	■ 1.000	$x=0.5$	$x=0.5$	$x=0.5$, and $x=1.5$.

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johnston-linear-matrix-atgkwoe F03958	422	■ 1.000	$x=1.5$	$x = 1.5$	$x = 0.5$, and $x = 1.5$.
johnston-linear-matrix-atgkwoe F03959	423	■ 1.000	$x^{2}-6\ x+8$	$x^2 - 6x + 8$	a) $x^2 - 6x + 8$,
johnston-linear-matrix-atgkwoe F03960	423	■ 1.000	$2\ x^{2}+8\ x+8$	$2x^2 + 8x + 8$	b) $2x^2 + 8x + 8$, and
johnston-linear-matrix-atgkwoe F03961	423	■ 1.000	$x^{2}+2\ x+3$	$x^2 + 2x + 3$	c) $x^2 + 2x + 3$.
johnston-linear-matrix-atgkwoe F03962	423	■ 1.000	$a=1,\ b=-6$	$a = 1, b = -6$	a) This quadratic has $a = 1$, $b = -6$, and $c = 8$, so the quadratic
johnston-linear-matrix-atgkwoe F03963	423	■ 1.000	$c=8$	$c = 8$	a) This quadratic has $a = 1$, $b = -6$, and $c = 8$, so the quadratic
johnston-linear-matrix-atgkwoe F03964	423	■ 1.000	$a=2,\ b=8$	$a = 2, b = 8$	b) This quadratic has $a = 2$, $b = 8$, and $c = 8$, so the quadratic formula
johnston-linear-matrix-atgkwoe F03965	423	■ 1.000	$a=1,\ b=2$	$a = 1, b = 2$	c) This quadratic has $a = 1$, $b = 2$, and $c = 3$, so the quadratic formula
johnston-linear-matrix-atgkwoe F03966	423	■ 1.000	$c=3$	$c = 3$	c) This quadratic has $a = 1$, $b = 2$, and $c = 3$, so the quadratic formula
johnston-linear-matrix-atgkwoe F03967	424	■ 1.000	$b\ /\ c$	b/c	b/c , where b and c
johnston-linear-matrix-atgkwoe F03968	424	■ 1.000	$-1\ \pm\ i\ \sqrt{2}$	$-1 \pm i\sqrt{2}$	are $-1 \pm i\sqrt{2}$. This quadratic is displayed below (notice that it does
johnston-linear-matrix-atgkwoe F03969	424	■ 1.000	$a_{\{0\}},\ a_{\{n\}}\ \neq\ 0$	$a_0, a_n \neq 0$	where $a_0, a_1, a_2, \dots, a_{n-1}, a_n \in \mathbb{R}$ are integers and $a_0, a_n \neq 0$. Then every
johnston-linear-matrix-atgkwoe F03970	424	■ 1.000	$a_{\{0\}}$	a_0	rational root of p has the form b/c , where b is a divisor of a_0 and c is a
johnston-linear-matrix-atgkwoe F03971	424	■ 1.000	$a_{\{n\}}$	a_n	divisor of a_n .
johnston-linear-matrix-atgkwoe F03972	424	■ 1.000	$x^{2}-2\ x-3$	$x^2 - 2x - 3$	a) $x^2 - 2x - 3$,
johnston-linear-matrix-atgkwoe F03973	424	■ 1.000	$2\ x^{3}-9\ x^{2}-6\ x+5$	$2x^3 - 9x^2 - 6x + 5$	b) $2x^3 - 9x^2 - 6x + 5$, and
johnston-linear-matrix-atgkwoe F03974	424	■ 1.000	$3\ x^{3}-14\ x+4$	$3x^3 - 14x + 4$	c) $3x^3 - 14x + 4$.
johnston-linear-matrix-atgkwoe F03975	424	■ 1.000	$a_{\{2\}}=1$	$a_2 = 1$	a) Since $a_2 = 1$ and $a_0 = 3$, the rational root theorem tells us that the
johnston-linear-matrix-atgkwoe F03976	424	■ 1.000	$a_{\{0\}}=3$	$a_0 = 3$	a) Since $a_2 = 1$ and $a_0 = 3$, the rational root theorem tells us that the
johnston-linear-matrix-atgkwoe F03977	424	■ 0.273	± 1	± 1	only possible rational roots of this polynomial are ± 1 and ± 3 (i.e.,
johnston-linear-matrix-atgkwoe F03978	424	■ 0.273	± 3	± 3	only possible rational roots of this polynomial are ± 1 and ± 3 (i.e.,
johnston-linear-matrix-atgkwoe F03979	424	■ 1.000	$x=-1$	$x = -1$	The rational roots of this polynomial are thus $x = -1$ and $x = 3$
johnston-linear-matrix-atgkwoe F03980	424	■ 1.000	$x=3$	$x = 3$	The rational roots of this polynomial are thus $x = -1$ and $x = 3$
johnston-linear-matrix-atgkwoe F03981	425	■ 1.000	$a_{\{3\}}=2$	$a_3 = 2$	b) Since $a_3 = 2$ and $a_0 = 5$, the rational root theorem tells us that the
johnston-linear-matrix-atgkwoe F03982	425	■ 1.000	$a_{\{0\}}=5$	$a_0 = 5$	b) Since $a_3 = 2$ and $a_0 = 5$, the rational root theorem tells us that the
johnston-linear-matrix-atgkwoe F03983	425	■ 0.967	$x=5,\ x=1\ /\ 2$	$x = 5, x = 1/2$	It follows that the rational roots of this polynomial are $x = 5, x = 1/2$,
johnston-linear-matrix-atgkwoe F03984	425	■ 1.000	$a_{\{3\}}=3$	$a_3 = 3$	c) Since $a_3 = 3$ and $a_0 = 4$, the rational root theorem tells us that

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johnston-linear-matrix-atlas- F03985	425	■ 1.000	$a_{\{0\}}=4$	$a_0 = 4$	c) Since $a_3 = 3$ and $a_0 = 4$, the rational root theorem tells us that
johnston-linear-matrix-atlas- F03986	425	■ 0.880	$\pm 4 / 3, \pm 2 / 3$	$\pm 4/3, \pm 2/3$	$\pm 4/3, \pm 2/3$, and $\pm 1/3$. Plugging these values into the polynomial
johnston-linear-matrix-atlas- F03987	425	■ 0.880	$\pm 1 / 3$	$\pm 1/3$	$\pm 4/3, \pm 2/3$, and $\pm 1/3$. Plugging these values into the polynomial
johnston-linear-matrix-atlas- F03988	425	■ 1.000	$p(x)=3 x^{\{3\}}-14 x+4$	$p(x) = 3x^3 - 14x + 4$	the cubic $p(x) = 3x^3 - 14x + 4$. However, this cubic actually has two other
johnston-linear-matrix-atlas- F03989	425	■ 1.000	$\sqrt{2}, \pi$	$\sqrt{2}, \pi$	include $\sqrt{2}, \pi$, and e .
johnston-linear-matrix-atlas- F03990	425	■ 1.000	$p(x)$	$p(x)$	$x = 2$ is one of its roots to try to factor $p(x)$ as
johnston-linear-matrix-atlas- F03991	425	■ 0.699	$\left(p(x)=3 x^{\{3\}}-14 x+4\right)$	$(p(x) = 3x^3 - 14x + 4)$	Step 1: Divide the highest-degree term in the dividend ($p(x) = 3x^3 - 14x + 4$)
johnston-linear-matrix-atlas- F03992	425	■ 0.999	$(x-2)$	$(x - 2)$	by the highest-degree term in the divisor ($x - 2$). In this case, we divide
johnston-linear-matrix-atlas- F03993	425	■ 1.000	$3 x^{\{3\}}$	$3x^3$	$3x^3$ by x , getting $3x^2$ as our result:
johnston-linear-matrix-atlas- F03994	425	■ 1.000	$3 x^{\{2\}}$	$3x^2$	$3x^3$ by x , getting $3x^2$ as our result:
johnston-linear-matrix-atlas- F03995	426	■ 0.999	$0 x^{\{2\}}-\left(-6 x^{\{2\}}\right)=6 x^{\{2\}}$	$0x^2 - (-6x^2) = 6x^2$	$0x^2 - (-6x^2) = 6x^2$.
johnston-linear-matrix-atlas- F03996	426	■ 0.517	$x-2$	$x - 2$	Step 2: Multiply the divisor ($x - 2$) by the quotient from Step 1 ($3x^2$) and subtract
johnston-linear-matrix-atlas- F03997	426	■ 1.000	$2\left(6 x^{\{2\}}-14 x+4\right)$	$2(6x^2 - 14x + 4)$	subtraction in Step 2 ($6x^2 - 14x + 4$) as the new dividend:
johnston-linear-matrix-atlas- F03998	426	■ 0.933	$\left(q(x)=3 x^{\{2\}}+6 x-2\right)$	$(q(x) = 3x^2 + 6x - 2)$	polynomial written at the top of this calculation ($q(x) = 3x^2 + 6x - 2$).
johnston-linear-matrix-atlas- F03999	426	■ 1.000	$q(x)=3 x^{\{2\}}+6 x-2$	$q(x) = 3x^2 + 6x - 2$	find the roots of $q(x) = 3x^2 + 6x - 2$, which are
johnston-linear-matrix-atlas- F04000	426	■ 0.996	$2,-1+\sqrt{5 / 3}$	$2, -1 + \sqrt{5/3}$	We (finally!) have shown that the roots of p are $2, -1 + \sqrt{5/3}$, and $-1 - \sqrt{5/3}$.
johnston-linear-matrix-atlas- F04001	426	■ 0.996	$-1-\sqrt{5 / 3}$	$-1 - \sqrt{5/3}$	We (finally!) have shown that the roots of p are $2, -1 + \sqrt{5/3}$, and $-1 - \sqrt{5/3}$.
johnston-linear-matrix-atlas- F04002	426	■ 0.997	$p(x)=3 x^{\{4\}}+4 x^{\{3\}}-14 x^{\{2\}}-11 x-2$	$p(x) = 3x^4 + 4x^3 - 14x^2 - 11x - 2$	Find all roots of the polynomial $p(x) = 3x^4 + 4x^3 - 14x^2 - 11x - 2$.
johnston-linear-matrix-atlas- F04003	426	■ 1.000	$a_{\{4\}}=3$	$a_4 = 3$	Since $a_4 = 3$ and $a_0 = -2$, the rational root theorem tells us that the
johnston-linear-matrix-atlas- F04004	426	■ 1.000	$a_{\{0\}}=-2$	$a_0 = -2$	Since $a_4 = 3$ and $a_0 = -2$, the rational root theorem tells us that the
johnston-linear-matrix-atlas- F04005	426	■ 1.000	$x=-1 / 3$	$x = -1/3$	$x = -1/3$ are the only ones that are <i>actually</i> roots of p .
johnston-linear-matrix-atlas- F04006	427	■ 1.000	$x^{\{2\}}+3 x+1$	$x^2 + 3x + 1$	to $x^2 + 3x + 1$:
johnston-linear-matrix-atlas- F04007	427	■ 1.000	$2, \frac{-1}{3}, \frac{1}{2}(-3+\sqrt{5})$	$2, \frac{-1}{3}, \frac{1}{2}(-3 + \sqrt{5})$	It follows that the roots of p are $2, \frac{-1}{3}, \frac{1}{2}(-3 + \sqrt{5})$, and $\frac{1}{2}(-3 - \sqrt{5})$.
johnston-linear-matrix-atlas- F04008	427	■ 1.000	$\frac{1}{2}(-3-\sqrt{5})$	$\frac{1}{2}(-3 - \sqrt{5})$	It follows that the roots of p are $2, \frac{-1}{3}, \frac{1}{2}(-3 + \sqrt{5})$, and $\frac{1}{2}(-3 - \sqrt{5})$.
johnston-linear-matrix-atlas- F04009	427	■ 1.000	$x-r$	$x - r$	If r is a root of a polynomial p and we divide $p(x)$ by $x - r$ via polynomial
johnston-linear-matrix-atlas- F04010	427	■ 1.000	$p(x)=(x-r) q(x)$	$p(x) = (x - r)q(x)$	$p(x) = (x - r)q(x)$.

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johnston-linear-matrix-atlas- F04011	428	■ 1.000	$p(x)=x^2+1$	$p(x) = x^2 + 1$	Some polynomials like $p(x) = x^2 + 1$ do not have any real roots at all, so the
johnston-linear-matrix-atlas- F04012	428	■ 0.999	x^n	x^n	where a is the coefficient of x^n in p and $r_1, r_2, \dots, r_n$ are the (potentially com-
johnston-linear-matrix-atlas- F04013	428	■ 0.999	$r_{\{1\}}, r_{\{2\}}, \ldots, r_{\{n\}}$	$r_1, r_2, \dots, r_n$	where a is the coefficient of x^n in p and $r_1, r_2, \dots, r_n$ are the (potentially com-
johnston-linear-matrix-atlas- F04014	428	■ 0.679	$2x^2+8x+8=2(x+2)^2$	$2x^2 + 8x + 8 = 2(x + 2)^2$	For example, the root -2 of the polynomial $2x^2 + 8x + 8 = 2(x + 2)^2$ has
johnston-linear-matrix-atlas- F04015	428	■ 0.922	$x+2$	$x + 2$	multiplicity 2, since the term $x + 2$ is squared, while all of the other roots that
johnston-linear-matrix-atlas- F04016	429	■ 1.000	$xy=yx$	$xy = yx$	properties of the real numbers, like the fact that $xy = yx$ for all real numbers x
johnston-linear-matrix-atlas- F04017	429	■ 1.000	m^2	m^2	If m is an even integer then so is m^2 .
johnston-linear-matrix-atlas- F04018	429	■ 1.000	$m=2k$	$m = 2k$	<i>Proof.</i> If m is an even integer then there is some integer k such that $m = 2k$.
johnston-linear-matrix-atlas- F04019	429	■ 1.000	$2k^2$	$2k^2$	which is also even since it is 2 times the integer $2k^2$.
johnston-linear-matrix-atlas- F04020	429	■ 1.000	$m=2k+1$	$m = 2k + 1$	odd integer m we would write $m = 2k + 1$ for some integer k , and if we were
johnston-linear-matrix-atlas- F04021	429	■ 1.000	$m=7k$	$m = 7k$	trying to prove a theorem about multiples of 7 then we would write $m = 7k$.
johnston-linear-matrix-atlas- F04022	429	■ 1.000	$m^2=(2k)^2$	$m^2 = (2k)^2$	based on that, so we then simply compute $m^2 = (2k)^2$, and we finish the proof
johnston-linear-matrix-atlas- F04023	429	■ 1.000	$(2k)^2=2(2k^2)$	$(2k)^2 = 2(2k^2)$	by again using the definition of “even integer”: we write $(2k)^2 = 2(2k^2)$, which
johnston-linear-matrix-atlas- F04024	429	■ 1.000	$x^2+y^2 \geq 2xy$	$x^2 + y^2 \geq 2xy$	If x and y are real numbers then $x^2 + y^2 \geq 2xy$.
johnston-linear-matrix-atlas- F04025	429	■ 0.997	$x-$	$x-$	<i>Proof.</i> Recall that the square of every real number is non-negative, so $(x-$
johnston-linear-matrix-atlas- F04026	429	■ 1.000	$y)^2 \geq 0$	$y)^2 \geq 0$	$y)^2 \geq 0$. However, using some algebra to expand $(x-y)^2$ shows that
johnston-linear-matrix-atlas- F04027	429	■ 1.000	$(x-y)^2$	$(x - y)^2$	$y)^2 \geq 0$. However, using some algebra to expand $(x-y)^2$ shows that
johnston-linear-matrix-atlas- F04028	429	■ 1.000	$2xy$	$2xy$	Adding $2xy$ to both sides of this inequality shows that $2xy \leq x^2 + y^2$, as desired.
johnston-linear-matrix-atlas- F04029	429	■ 1.000	$2xy \leq x^2+y^2$	$2xy \leq x^2 + y^2$	Adding $2xy$ to both sides of this inequality shows that $2xy \leq x^2 + y^2$, as desired.
johnston-linear-matrix-atlas- F04030	430	■ 0.889	x^2+y^2-2xy	$x^2 + y^2 - 2xy$	all we did in that proof—we factored $x^2 + y^2 - 2xy$ as $(x-y)^2$). This same trick
johnston-linear-matrix-atlas- F04031	430	■ 0.998	a, b, c	a, b, c	If a, b, c , and x are real or complex numbers such that $ax^2 + bx + c = 0$
johnston-linear-matrix-atlas- F04032	430	■ 0.998	$ax^2+bx+c=0$	$ax^2 + bx + c = 0$	If a, b, c , and x are real or complex numbers such that $ax^2 + bx + c = 0$
johnston-linear-matrix-atlas- F04033	430	■ 1.000	bx	bx	groups that linear bx term and quadratic ax^2 term together:
johnston-linear-matrix-atlas- F04034	430	■ 1.000	ax^2	ax^2	groups that linear bx term and quadratic ax^2 term together:
johnston-linear-matrix-atlas- F04035	431	■ 1.000	$m^2=2k$	$m^2 = 2k$	writing $m^2 = 2k$ for
johnston-linear-matrix-atlas- F04036	431	■ 1.000	$Q^{\prime \prime}$	Q''	only if Q'' is
johnston-linear-matrix-atlas- F04037	431	■ 1.000	$2(2k^2+2k)$	$2(2k^2 + 2k)$	which is also odd since it is 1 larger than the even integer $2(2k^2 + 2k)$.

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johnston-linear-matrix-attrib F04038	431	0.597	2×1	2×1	chessboard by 2×1 dominoes. That is, we lay 2×1 dominoes on top of the
johnston-linear-matrix-attrib F04039	432	0.997	8×8	8×8	8×8 chessboard.
johnston-linear-matrix-attrib F04040	432	0.666	$8 \times 8 = 64$	$8 \times 8 = 64$	remove one of the $8 \times 8 = 64$ squares from the chessboard, then the remaining
johnston-linear-matrix-attrib F04041	434	1.000	$\sqrt{2}=a / b$	$\sqrt{2} = a/b$	There do not exist integers a and b such that $\sqrt{2} = a/b$. In other words,
johnston-linear-matrix-attrib F04042	434	1.000	a / b	a/b	form a/b , where a
johnston-linear-matrix-attrib F04043	434	1.000	$\sqrt{2} \ b=a$	$\sqrt{2}b = a$	Multiplying the equation $\sqrt{2} = a/b$ through by b shows that $\sqrt{2}b = a$, and
johnston-linear-matrix-attrib F04044	434	1.000	$2 \ b^{\{2\}}=a^{\{2\}}$	$2b^2 = a^2$	then squaring shows that $2b^2 = a^2$, so a^2 is even. It follows from Theorem A.3.5
johnston-linear-matrix-attrib F04045	434	1.000	$a^{\{2\}}$	a^2	then squaring shows that $2b^2 = a^2$, so a^2 is even. It follows from Theorem A.3.5
johnston-linear-matrix-attrib F04046	434	1.000	$a=2 \ k$	$a = 2k$	that a is even and can be written in the form $a = 2k$ for some integer k . Since
johnston-linear-matrix-attrib F04047	435	1.000	$2 \ b^{\{2\}}=(2 \ k)^{\{2\}}=4 \ k^{\{2\}}$	$2b^2 = (2k)^2 = 4k^2$	Plugging $a = 2k$ into the equation $2b^2 = a^2$ shows that $2b^2 = (2k)^2 = 4k^2$,
johnston-linear-matrix-attrib F04048	435	1.000	$b^{\{2\}}=2 \ k^{\{2\}}$	$b^2 = 2k^2$	so $b^2 = 2k^2$. It follows that b^2 is even, so using Theorem A.3.5 again shows
johnston-linear-matrix-attrib F04049	435	1.000	$b^{\{2\}}$	b^2	so $b^2 = 2k^2$. It follows that b^2 is even, so using Theorem A.3.5 again shows
johnston-linear-matrix-attrib F04050	435	1.000	$1+3+5+\cdots+(2 \ n-1)=n^{\{2\}}$	$1 + 3 + 5 + \cdots + (2n - 1) = n^2$	perfect squares (i.e., we have shown that $1 + 3 + 5 + \cdots + (2n - 1) = n^2$, at
johnston-linear-matrix-attrib F04051	435	1.000	$n \ \leq 5$	$n \leq 5$	least for $n \leq 5$). We expect that this pattern continues forever, but how could
johnston-linear-matrix-attrib F04052	435	0.999	$P(1), \ P(2), \ P(3), \ \ldots$	$P(1), P(2), P(3), \dots$	$P(1), P(2), P(3), \dots$, indexed by the natural numbers 1, 2, 3, ..., then proof by
johnston-linear-matrix-attrib F04053	435	0.999	$1,2,3, \ \ldots$	$1, 2, 3, \dots$	$P(1), P(2), P(3), \dots$, indexed by the natural numbers 1, 2, 3, ..., then proof by
johnston-linear-matrix-attrib F04054	435	1.000	$P(1)$	$P(1)$	• The base case: we start by showing that $P(1)$ is true.
johnston-linear-matrix-attrib F04055	435	1.000	$P(n)$	$P(n)$	$P(n)$ holds for a
johnston-linear-matrix-attrib F04056	435	1.000	$P(n+1)$	$P(n + 1)$	n , then $P(n + 1)$ must also be true.
johnston-linear-matrix-attrib F04057	435	1.000	$P(2)$	$P(2)$	$P(2)$: True by the inductive step: if $P(1)$ is true (it is!) then $P(2)$ is true too.
johnston-linear-matrix-attrib F04058	435	1.000	$P(3)$	$P(3)$	$P(3)$: True by the inductive step: if $P(2)$ is true (it is!) then $P(3)$ is true too.
johnston-linear-matrix-attrib F04059	435	0.988	$P(4)$	$P(4)$	$P(4)$: True because $P(3)$ is true. You get the idea...
johnston-linear-matrix-attrib F04060	435	1.000	$1=1$	$1 = 1$	that $1 = 1$, this step is complete.
johnston-linear-matrix-attrib F04061	436	1.000	$2 \ n+1$	$2n + 1$	The trick that lets us do this is to add $2n + 1$ to both sides of $P(n)$:
johnston-linear-matrix-attrib F04062	436	0.995	$1,3,5, \ \ldots, \ 2 \ n-1$	$1, 3, 5, \dots, 2n - 1$	of 1, 3, 5, ..., $2n - 1$ squares, it follows that $1 + 3 + 5 + \cdots + (2n - 1) = n^2$, as
johnston-linear-matrix-attrib F04063	437	0.992	$\left[A_{\{i, \ j\}}\right]_{\{k, \ \ell\}}$	$[A_{i,j}]_{k,\ell}$	block it is. Specifically, we use $[A_{i,j}]_{k,\ell}$ to refer to the (k, ℓ) -entry of the matrix
johnston-linear-matrix-attrib F04064	437	0.992	$(k, \ \ell)$	(k, ℓ)	block it is. Specifically, we use $[A_{i,j}]_{k,\ell}$ to refer to the (k, ℓ) -entry of the matrix

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johnston-linear-matrix-atgda7a-F04065	437	1.000	$A_{\{i, j\}}$	$A_{i,j}$	$A_{i,j}$, which itself is the matrix that makes up the i -th block row and j -th block
johnston-linear-matrix-atgda7a-F04066	438	1.000	$R, S \in \mathcal{M}_{m,n}$	$R, S \in \mathcal{M}_{m,n}$	matrices $R, S \in \mathcal{M}_{m,n}$, both of which are in reduced row echelon form, then
johnston-linear-matrix-atgda7a-F04067	438	1.000	$R=S$	$R = S$	$R = S$.
johnston-linear-matrix-atgda7a-F04068	438	1.000	$R \neq S$	$R \neq S$	To this end, suppose for the sake of contradiction that $R \neq S$. Let $\widetilde{R}$ and $\widetilde{S}$
johnston-linear-matrix-atgda7a-F04069	438	1.000	$\widetilde{R}$	$\widetilde{R}$	To this end, suppose for the sake of contradiction that $R \neq S$. Let $\widetilde{R}$ and $\widetilde{S}$
johnston-linear-matrix-atgda7a-F04070	438	1.000	$\widetilde{S}$	$\widetilde{S}$	To this end, suppose for the sake of contradiction that $R \neq S$. Let $\widetilde{R}$ and $\widetilde{S}$
johnston-linear-matrix-atgda7a-F04071	439	1.000	$\widetilde{\mathbf{R}}$	$\widetilde{\mathbf{R}}$	Case 1: $\widetilde{\mathbf{R}}$ has the first form. Then the linear system associated with $\widetilde{R}$ has the
johnston-linear-matrix-atgda7a-F04072	439	0.960	$\mathbf{c}=\mathbf{b}$	$\mathbf{c} = \mathbf{b}$	the unique solution $\mathbf{b}$, so it must also have the first form with $\mathbf{c} = \mathbf{b}$ (and
johnston-linear-matrix-atgda7a-F04073	439	1.000	$\widetilde{S}=\widetilde{R}$	$\widetilde{S} = \widetilde{R}$	thus $\widetilde{S} = \widetilde{R}$).
johnston-linear-matrix-atgda7a-F04074	439	1.000	$\widetilde{R}=\widetilde{S}$	$\widetilde{R} = \widetilde{S}$	We have $\widetilde{R} = \widetilde{S}$ in both cases, which contradicts the fact that we chose them
johnston-linear-matrix-atgda7a-F04075	439	1.000	$I_{\{m\}}$	I_m	I_m results in a matrix E , then $EA = B$.
johnston-linear-matrix-atgda7a-F04076	439	1.000	$E \ A=B$	$EA = B$	I_m results in a matrix E , then $EA = B$.
johnston-linear-matrix-atgda7a-F04077	439	0.976	(j, j)	(j, j)	case, E is diagonal with all diagonal entries equal to 1, except for the (j, j) -
johnston-linear-matrix-atgda7a-F04078	440	1.000	$\mathbf{e}_j+c \ \mathbf{e}_i$	$\mathbf{e}_j + c\mathbf{e}_i$	of E is $\mathbf{e}_j + c\mathbf{e}_i$, not
johnston-linear-matrix-atgda7a-F04079	440	1.000	$\mathbf{e}_i+c \ \mathbf{e}_j$	$\mathbf{e}_i + c\mathbf{e}_j$	$\mathbf{e}_i + c\mathbf{e}_j$. The reason
johnston-linear-matrix-atgda7a-F04080	440	1.000	$\mathbf{e}_1 \ \mathbf{a}_1^T$	$\mathbf{e}_1 \mathbf{a}_1^T$	$\mathbf{e}_1 \mathbf{a}_1^T$ are matrices,
johnston-linear-matrix-atgda7a-F04081	440	1.000	$\mathbf{a}_1^T$	$\mathbf{a}_1^T$	is $m \times 1$ and $\mathbf{a}_1^T$ is $1 \times n$.
johnston-linear-matrix-atgda7a-F04082	440	1.000	$\{1,2, \ldots, n\}, \sigma$	$\{1, 2, \dots, n\}, \sigma$	$\{1, 2, \dots, n\}$, σ is a
johnston-linear-matrix-atgda7a-F04083	440	1.000	$\mathbf{e}_i \ \mathbf{a}_j^T$	$\mathbf{e}_i \mathbf{a}_j^T$	Since $\mathbf{e}_i \mathbf{a}_j^T$ is the matrix that contains the entries from the j -th row of A in its
johnston-linear-matrix-atgda7a-F04084	440	1.000	$A+c \ \mathbf{e}_i \ \mathbf{a}_j^T$	$A + c\mathbf{e}_i \mathbf{a}_j^T$	i -th row, it follows that $A + c\mathbf{e}_i \mathbf{a}_j^T$ is exactly the matrix B that is obtained by
johnston-linear-matrix-atgda7a-F04085	441	1.000	$v+c \ \mathbf{w}$	$v + c\mathbf{w}$	the $v + c\mathbf{w}$ column
johnston-linear-matrix-atgda7a-F04086	441	0.996	$\sigma(j)=j$	$\sigma(j) = j$	is the identity permutation (i.e., $\sigma(j) = j$ for all $1 \leq j \leq n$) then
johnston-linear-matrix-atgda7a-F04087	442	1.000	$p_A^{\prime}$	p_A'	The notation p_A'
johnston-linear-matrix-atgda7a-F04088	442	1.000	$\frac{\partial}{\partial \lambda_i}$	$\frac{\partial}{\partial \lambda_i}$	The notation $\frac{\partial}{\partial \lambda_i}$
johnston-linear-matrix-atgda7a-F04089	442	1.000	λ_i	λ_i	respect to λ_i .
johnston-linear-matrix-atgda7a-F04090	442	1.000	$\mu \neq 0$	$\mu \neq 0$	because $\mu \neq 0$ (this
johnston-linear-matrix-atgda7a-F04091	442	1.000	$L(\mathbf{w})=\mu$	$L(\mathbf{w}) = \mu$	so if $ \lambda = \mu$ then in fact we have $L(\mathbf{w}) = \mu$. Claims 1 and 3 then tell us that

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johnston-linear-matrix-algebra- F04092	442	1.000	$A \mathbf{w} =\mu \mathbf{w} $	$A \mathbf{w} =\mu \mathbf{w} $	$A \mathbf{w} =\mu \mathbf{w} $ and $ \mathbf{w} >\mathbf{0}$. It follows that
johnston-linear-matrix-algebra- F04093	442	1.000	$ \mathbf{w} >\mathbf{0}$	$ \mathbf{w} >\mathbf{0}$	$A \mathbf{w} =\mu \mathbf{w} $ and $ \mathbf{w} >\mathbf{0}$. It follows that
johnston-linear-matrix-algebra- F04094	442	0.948	$(A-B) \mathbf{w} =\mathbf{0}$	$(A-B) \mathbf{w} =\mathbf{0}$	but since $A-B$ is entrywise non-negative, this is only possible if $(A-B) \mathbf{w} =\mathbf{0}$.
johnston-linear-matrix-algebra- F04095	442	1.000	$A-B=0$	$A-B=O$	Since $ \mathbf{w} >\mathbf{0}$, this then implies $A-B=O$, so $A=B$, which completes the
johnston-linear-matrix-algebra- F04096	442	1.000	$p_A(\lambda)=\operatorname{det}(A-$	$p_A(\lambda)=\det(A-$	when we take the derivative of the characteristic polynomial $p_A(\lambda)=\det(A-$
johnston-linear-matrix-algebra- F04097	442	0.986	$\lambda I)$	$\lambda I)$	$\lambda I)$ of A . To show that the algebraic multiplicity of μ is 1, it suffices to show
johnston-linear-matrix-algebra- F04098	442	1.000	$p_A'(\mu)\neq 0$	$p_A'(\mu)\neq 0$	that $p_A'(\mu)\neq 0$ (via Theorem B.6.1, for example). To see why this is the case,
johnston-linear-matrix-algebra- F04099	442	1.000	$f:\mathbb{R}^n\rightarrow\mathbb{R}$	$f:\mathbb{R}^n\rightarrow\mathbb{R}$	consider the somewhat uglier multivariable function $f:\mathbb{R}^n\rightarrow\mathbb{R}$ defined by
johnston-linear-matrix-algebra- F04100	442	0.999	Λ	Λ	where Λ is the diagonal matrix whose diagonal entries are $\lambda_1,\lambda_2,\dots,\lambda_n$. If we
johnston-linear-matrix-algebra- F04101	442	1.000	A_i	A_i	let A_i denote the $(n-1)\times(n-1)$ matrix that is obtained by removing the i -th
johnston-linear-matrix-algebra- F04102	442	1.000	Λ_i	Λ_i	row and column from A (and similarly for Λ_i from Λ), and we let $c_{i,j}$ denote
johnston-linear-matrix-algebra- F04103	442	1.000	$A-\Lambda$	$A-\Lambda$	the (i,j) -cofactor of $A-\Lambda$, then applying a cofactor expansion along the i -th
johnston-linear-matrix-algebra- F04104	442	1.000	$a_{i,1},\dots,a_{i,n}$	$a_{i,1},\dots,a_{i,n}$	Since none of $a_{i,1},\dots,a_{i,n}$ or $c_{i,1},\dots,c_{i,n}$ depend on λ_i , it follows that
johnston-linear-matrix-algebra- F04105	442	1.000	$c_{i,1},\dots,c_{i,n}$	$c_{i,1},\dots,c_{i,n}$	Since none of $a_{i,1},\dots,a_{i,n}$ or $c_{i,1},\dots,c_{i,n}$ depend on λ_i , it follows that
johnston-linear-matrix-algebra- F04106	442	1.000	B_i	B_i	If we now let B_i be the matrix obtained by replacing the i -th row and column
johnston-linear-matrix-algebra- F04107	442	1.000	$p_{B_i}(\lambda)=-\lambda p_A(\lambda)$	$p_{B_i}(\lambda)=-\lambda p_A(\lambda)$	column of zeros at the i -th index), then $p_{B_i}(\lambda)=-\lambda p_A(\lambda)$ for all i , so
johnston-linear-matrix-algebra- F04108	442	1.000	$0\leq B_i\leq A$	$0\leq B_i\leq A$	Since $0\leq B_i\leq A$, we know from our earlier claim that the absolute values
johnston-linear-matrix-algebra- F04109	442	1.000	$p_{B_i}(\mu)>0$	$p_{B_i}(\mu)>0$	$p_{B_i}(\mu)>0$ for all i (if n is even) or $p_{B_i}(\mu)<0$ for all i (if n is odd). Either
johnston-linear-matrix-algebra- F04110	442	1.000	$p_{B_i}(\mu)<0$	$p_{B_i}(\mu)<0$	$p_{B_i}(\mu)>0$ for all i (if n is even) or $p_{B_i}(\mu)<0$ for all i (if n is odd). Either
johnston-linear-matrix-algebra- F04111	443	1.000	$p^{(k)}$	$p^{(k)}$	The notation $p^{(k)}$
johnston-linear-matrix-algebra- F04112	443	0.999	$(fg)'(x)=$	$(fg)'(x)=$	that $(fg)'(x)=$
johnston-linear-matrix-algebra- F04113	443	0.715	$f'(x)g(x)+f(x)g'(x)$	$f'(x)g(x)+f(x)g'(x)$	$f'(x)g(x)+f(x)g'(x)$.
johnston-linear-matrix-algebra- F04114	443	1.000	$m-1$	$m-1$	m if and only if r is a root of each of p and its first $m-1$ derivatives:
johnston-linear-matrix-algebra- F04115	443	1.000	$p,p',\dots,p^{(m-1)}$	$p,p',\dots,p^{(m-1)}$	exactly m then is a root of $p,p',\dots,p^{(m-1)}$, but not of $p^{(m)}$.
johnston-linear-matrix-algebra- F04116	443	1.000	$p^{(m)}$	$p^{(m)}$	exactly m then is a root of $p,p',\dots,p^{(m-1)}$, but not of $p^{(m)}$.
johnston-linear-matrix-algebra- F04117	443	0.981	$p(x)=$	$p(x)=$	Recall that the factor theorem (Theorem A.2.2) lets us factor p as $p(x)=$
johnston-linear-matrix-algebra- F04118	443	1.000	$(x-r)^mq(x)$	$(x-r)^mq(x)$	$(x-r)^mq(x)$, where q is some polynomial with $q(r)\neq 0$. We begin by showing

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johnston-linear-matrix-algebra-F04119	443	1.000	$q(r) \neq 0$	$q(r) \neq 0$	$(x-r)^m q(x)$, where q is some polynomial with $q(r) \neq 0$. We begin by showing
johnston-linear-matrix-algebra-F04120	443	1.000	$c_k \neq 0$	$c_k \neq 0$	where $c_k \neq 0$ is a scalar and $s_k(x)$ is a polynomial depending on k with the
johnston-linear-matrix-algebra-F04121	443	1.000	$s_k(x)$	$s_k(x)$	where $c_k \neq 0$ is a scalar and $s_k(x)$ is a polynomial depending on k with the
johnston-linear-matrix-algebra-F04122	443	1.000	$s_k(r)=0$	$s_k(r) = 0$	property that $s_k(r) = 0$ (note that in the $k = 0$ case, we simply have $c_0 = 1$ and
johnston-linear-matrix-algebra-F04123	443	1.000	$c_0=1$	$c_0 = 1$	property that $s_k(r) = 0$ (note that in the $k = 0$ case, we simply have $c_0 = 1$ and
johnston-linear-matrix-algebra-F04124	443	0.999	$s_k(x)=0$	$s_k(x) = 0$	$s_k(x) = 0$).
johnston-linear-matrix-algebra-F04125	443	1.000	$c_1=m$	$c_1 = m$	so the derivative of p does have the claimed form, with $c_1 = m$ and $s_1(x) =$
johnston-linear-matrix-algebra-F04126	443	1.000	$s_1(x)=$	$s_1(x) =$	so the derivative of p does have the claimed form, with $c_1 = m$ and $s_1(x) =$
johnston-linear-matrix-algebra-F04127	443	1.000	$(x-r) q'(\prime{x})$	$(x-r)q'(x)$	$(x-r)q'(x)$.
johnston-linear-matrix-algebra-F04128	443	1.000	$p^{(k+1)}$	$p^{(k+1)}$	so $p^{(k+1)}$ also has the form (B.6.1), with
johnston-linear-matrix-algebra-F04129	443	1.000	$0 \leq k \leq m$	$0 \leq k \leq m$	the form (B.6.1) for all $0 \leq k \leq m$.
johnston-linear-matrix-algebra-F04130	443	1.000	$x=r$	$x = r$	It follows by plugging in $x = r$ that $p^{(k)}(r) = 0$ for all $0 \leq k < m$. Further-
johnston-linear-matrix-algebra-F04131	443	1.000	$p^{(k)}(r)=0$	$p^{(k)}(r) = 0$	It follows by plugging in $x = r$ that $p^{(k)}(r) = 0$ for all $0 \leq k < m$. Further-
johnston-linear-matrix-algebra-F04132	443	1.000	$0 \leq k < m$	$0 \leq k < m$	It follows by plugging in $x = r$ that $p^{(k)}(r) = 0$ for all $0 \leq k < m$. Further-
johnston-linear-matrix-algebra-F04133	443	1.000	$s_m(r)=0$	$s_m(r) = 0$	more, since $s_m(r) = 0$ but $c_m, q(r) \neq 0$, it follows that
johnston-linear-matrix-algebra-F04134	443	1.000	$c_m, q(r) \neq 0$	$c_m, q(r) \neq 0$	more, since $s_m(r) = 0$ but $c_m, q(r) \neq 0$, it follows that
johnston-linear-matrix-algebra-F04135	444	1.000	$n x^{n-1}$	nx^{n-1}	nx^{n-1} .
johnston-linear-matrix-algebra-F04136	444	1.000	$p_0, p_1, \ldots, p_m$	$p_0, p_1, \ldots, p_m$	Consider polynomials $p_0, p_1, \ldots, p_m$ of the form
johnston-linear-matrix-algebra-F04137	444	1.000	$r \neq 0$	$r \neq 0$	Then $r \neq 0$ is a root of p_0 with multiplicity at least $m+1$ if and only if it
johnston-linear-matrix-algebra-F04138	444	1.000	p_0	p_0	Then $r \neq 0$ is a root of p_0 with multiplicity at least $m+1$ if and only if it
johnston-linear-matrix-algebra-F04139	444	1.000	$m=0$	$m = 0$	<i>Proof.</i> If $m = 0$ then the result is trivial (it just says that r is a root of p_0 if and
johnston-linear-matrix-algebra-F04140	444	1.000	$m+1=2$	$m+1 = 2$	p_0 with multiplicity $m+1 = 2$ then Theorem B.6.1 tells us that it is also a root
johnston-linear-matrix-algebra-F04141	444	1.000	$p_0^{\prime}$	p_0'	of its derivative, p_0' , and thus also a root of
johnston-linear-matrix-algebra-F04142	444	1.000	$p_1(x)=x p_0^{\prime}(x)$	$p_1(x) = xp_0'(x)$	is a root of $p_1(x) = xp_0'(x)$ then it is a root of $p_0'(x)$, so Theorem B.6.1 tells us
johnston-linear-matrix-algebra-F04143	444	1.000	$p_0^{\prime}(x)$	$p_0'(x)$	is a root of $p_1(x) = xp_0'(x)$ then it is a root of $p_0'(x)$, so Theorem B.6.1 tells us
johnston-linear-matrix-algebra-F04144	444	1.000	$m>1$	$m > 1$	If $m > 1$ then we can just iterate the argument above, using the facts that
johnston-linear-matrix-algebra-F04145	444	1.000	$p_2(x)=x p_1^{\prime}(x), p_3(x)=x p_2^{\prime}(x)$	$p_2(x) = xp_1'(x), p_3(x) = xp_2'(x)$	$p_2(x) = xp_1'(x)$, $p_3(x) = xp_2'(x)$, and so on.

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johnston-linear-matrix-atgndra-F04146	444	■ 0.931	θ / θ	$0/0$	exists and is an indeterminate form of type “0/0” or “ ∞/∞ ” (along with some
johnston-linear-matrix-atgndra-F04147	444	■ 0.931	∞ / ∞	∞/∞	exists and is an indeterminate form of type “0/0” or “ ∞/∞ ” (along with some
johnston-linear-matrix-atgndra-F04148	444	■ 1.000	$q(x)=c_{\{k\}} x^{\{k\}}+\cdots+c_{\{1\}} x+c_{\{0\}}$	$q(x)=c_kx^k+\cdots+c_1x+c_0$	explicitly as $q(x)=c_kx^k+\cdots+c_1x+c_0$. Then using L’Hôpital’s rule a total of
johnston-linear-matrix-atgndra-F04149	445	■ 1.000	$b^{\{x\}}$	b^x	derivative of b^x is
johnston-linear-matrix-atgndra-F04150	445	■ 1.000	$b^{\{x\}} \ln (b)$	$b^x \ln (b)$	$b^x \ln (b)$.
johnston-linear-matrix-atgndra-F04151	445	■ 0.996	$c_{\{k\}} k!$	$c_k k!$	$c_k k!$), whereas the
johnston-linear-matrix-atgndra-F04152	445	■ 1.000	$b=1 / c$	$b=1/c$	To evaluate the other limit (B.7.1), we define $b=1/c$ and notice that $ b >1$
johnston-linear-matrix-atgndra-F04153	445	■ 1.000	$ b >1$	$ b >1$	To evaluate the other limit (B.7.1), we define $b=1/c$ and notice that $ b >1$
johnston-linear-matrix-atgndra-F04154	446	■ 1.000	$\mathbf{v}=2 \mathbf{e}_{\{3\}}$	$\mathbf{v}=2\mathbf{e}_3$	(a) $\mathbf{v}=2\mathbf{e}_3$
johnston-linear-matrix-atgndra-F04155	446	■ 0.992	$\mathbf{x}=\mathbf{e}_{\{1\}}+2 \mathbf{e}_{\{2\}}$	$\mathbf{x}=\mathbf{e}_1+2\mathbf{e}_2$	(c) $\mathbf{x}=\mathbf{e}_1+2\mathbf{e}_2$
johnston-linear-matrix-atgndra-F04156	446	■ 1.000	$c(1,2)=(3,4)$	$c(1,2)=(3,4)$	1.1.12 Algebraically, we note that $c(1,2)=(3,4)$ implies
johnston-linear-matrix-atgndra-F04157	446	■ 1.000	$2 c=4$	$2c=4$	$c=3$ and $2c=4$, so $c=2$. Since c cannot simulta-
johnston-linear-matrix-atgndra-F04158	446	■ 1.000	$c=2$	$c=2$	$c=3$ and $2c=4$, so $c=2$. Since c cannot simulta-
johnston-linear-matrix-atgndra-F04159	446	■ 0.990	$(3,4)$	$(3,4)$	we notice that $(1,2)$ and $(3,4)$ point in different di-
johnston-linear-matrix-atgndra-F04160	446	■ 1.000	$\mathbf{v}+\mathbf{w}=\mathbf{x}$	$\mathbf{v}+\mathbf{w}=\mathbf{x}$	1.1.9 We want vectors $\mathbf{v}$ and $\mathbf{w}$ such that $\mathbf{v}+\mathbf{w}=\mathbf{x}$ and
johnston-linear-matrix-atgndra-F04161	446	■ 1.000	$\mathbf{v}-\mathbf{w}=\mathbf{y}$	$\mathbf{v}-\mathbf{w}=\mathbf{y}$	$\mathbf{v}-\mathbf{w}=\mathbf{y}$. Adding these two equations together
johnston-linear-matrix-atgndra-F04162	446	■ 1.000	$2 \mathbf{v}=\mathbf{x}+\mathbf{y}$	$2\mathbf{v}=\mathbf{x}+\mathbf{y}$	tells us that $2\mathbf{v}=\mathbf{x}+\mathbf{y}$, and subtracting the second
johnston-linear-matrix-atgndra-F04163	446	■ 1.000	$2 \mathbf{w}=\mathbf{x}-\mathbf{y}$	$2\mathbf{w}=\mathbf{x}-\mathbf{y}$	equation from the first tells us that $2\mathbf{w}=\mathbf{x}-\mathbf{y}$, so
johnston-linear-matrix-atgndra-F04164	446	■ 1.000	$\mathbf{v}=(\mathbf{x}+\mathbf{y}) / 2=(2,1)$	$\mathbf{v}=(\mathbf{x}+\mathbf{y})/2=(2,1)$	$\mathbf{v}=(\mathbf{x}+\mathbf{y})/2=(2,1)$ and $\mathbf{w}=(\mathbf{x}-\mathbf{y})/2=(1,-3)$.
johnston-linear-matrix-atgndra-F04165	446	■ 1.000	$\mathbf{w}=(\mathbf{x}-\mathbf{y}) / 2=(1,-3)$	$\mathbf{w}=(\mathbf{x}-\mathbf{y})/2=(1,-3)$	$\mathbf{v}=(\mathbf{x}+\mathbf{y})/2=(2,1)$ and $\mathbf{w}=(\mathbf{x}-\mathbf{y})/2=(1,-3)$.
johnston-linear-matrix-atgndra-F04166	446	■ 0.996	$\mathbf{x}=(2,2)$	$\mathbf{x}=(2,2)$	(a) $\mathbf{x}=(2,2)$
johnston-linear-matrix-atgndra-F04167	446	■ 1.000	$\mathbf{x}=\frac{1}{2}(\mathbf{v}-\mathbf{w})$	$\mathbf{x}=\frac{1}{2}(\mathbf{v}-\mathbf{w})$	(a) $\mathbf{x}=\frac{1}{2}(\mathbf{v}-\mathbf{w})$
johnston-linear-matrix-atgndra-F04168	447	■ 1.000	$\ \mathbf{v}\ =5, \mathbf{u}=(3 / 5,4 / 5)$	$\ \mathbf{v}\ =5, \mathbf{u}=(3/5,4/5)$	1.2.2 (a) $\ \mathbf{v}\ =5, \mathbf{u}=(3/5,4/5)$
johnston-linear-matrix-atgndra-F04169	447	■ 1.000	$\ \mathbf{v}\ =6, \mathbf{u}=(-\sqrt{2} / 3,-1 / 2, \sqrt{10} / 6,1 / 2)$	$\ \mathbf{v}\ =6, \mathbf{u}=(-\sqrt{2}/3,-1/2,\sqrt{10}/6,1/2)$	(c) $\ \mathbf{v}\ =6, \mathbf{u}=(-\sqrt{2}/3,-1/2,\sqrt{10}/6,1/2)$
johnston-linear-matrix-atgndra-F04170	447	■ 1.000	$\arccos (-1 / \sqrt{2})=3 \pi / 4$	$\arccos (-1/\sqrt{2})=3\pi/4$	(c) $\arccos (-1/\sqrt{2})=3\pi/4$
johnston-linear-matrix-atgndra-F04171	447	■ 1.000	$\mathbf{x}=(1,-1)$	$\mathbf{x}=(1,-1)$	$\mathbf{w}=(1,1)$, and $\mathbf{x}=(1,-1)$. In this case, $\mathbf{v} \cdot \mathbf{w} =$
johnston-linear-matrix-atgndra-F04172	447	■ 1.000	$\mathbf{v} \cdot \mathbf{x}=1$	$\mathbf{v} \cdot \mathbf{x}=1$	$\mathbf{v} \cdot \mathbf{x}=1$, but $\mathbf{w} \neq \mathbf{x}$.
johnston-linear-matrix-atgndra-F04173	447	■ 1.000	$\ \mathbf{v}+\mathbf{w}\ \leq\ \mathbf{v}\ +\ \mathbf{w}\ $	$\ \mathbf{v}+\mathbf{w}\ \leq\ \mathbf{v}\ +\ \mathbf{w}\ $	(c) True. The triangle inequality tells us that $\ \mathbf{v}+\mathbf{w}\ \leq\ \mathbf{v}\ +\ \mathbf{w}\ $
johnston-linear-matrix-atgndra-F04174	447	■ 0.942	$\ \mathbf{w}\ \leq\ \mathbf{v}\ +\ \mathbf{w}\ \leq 2$	$\ \mathbf{w}\ \leq\ \mathbf{v}\ +\ \mathbf{w}\ \leq 2$	$\ \mathbf{w}\ \leq\ \mathbf{v}\ +\ \mathbf{w}\ \leq 2$.

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johnston-linear-matrix-algebra F04175	447	■ 0.995	$(1, 0, 0)$	$(1, 0, 0)$	$(1, 0, 0)$ and $\mathbf{w} = (-1, \sqrt{3}, 0)$.
johnston-linear-matrix-algebra F04176	447	■ 0.995	$\mathbf{w} = (-1, \sqrt{3}, 0)$	$\mathbf{w} = (-1, \sqrt{3}, 0)$	$(1, 0, 0)$ and $\mathbf{w} = (-1, \sqrt{3}, 0)$.
johnston-linear-matrix-algebra F04177	447	■ 1.000	$\mathbf{v} = (2, 0)$	$\mathbf{v} = (2, 0)$	(g) False. A counter-example is given by $\mathbf{v} = (2, 0)$
johnston-linear-matrix-algebra F04178	447	■ 1.000	$\mathbf{w} = (0, 2)$	$\mathbf{w} = (0, 2)$	and $\mathbf{w} = (0, 2)$.
johnston-linear-matrix-algebra F04179	447	■ 1.000	$\mathbf{v} \cdot \mathbf{w} = \cos(\theta) \ \mathbf{v}\ \ \mathbf{w}\ = \cos(\pi/3)(2\sqrt{3})(2) =$ $\ \mathbf{w}\ = \cos(\pi/3)(2\sqrt{3})(2) =$	$\mathbf{v} \cdot \mathbf{w} = \cos(\theta) \ \mathbf{v}\ \ \mathbf{w}\ = \cos(\pi/3)(2\sqrt{3})(2) =$	1.2.6 Since $\mathbf{v} \cdot \mathbf{w} = \cos(\theta) \ \mathbf{v}\ \ \mathbf{w}\ = \cos(\pi/3)(2\sqrt{3})(2) =$
johnston-linear-matrix-algebra F04180	447	■ 1.000	$2\sqrt{3}$	$2\sqrt{3}$	$2\sqrt{3}$, we find that $3w_1 + \sqrt{3}w_2 = 2\sqrt{3}$, so
johnston-linear-matrix-algebra F04181	447	■ 1.000	$3w_1 + \sqrt{3}w_2 = 2\sqrt{3}$	$3w_1 + \sqrt{3}w_2 = 2\sqrt{3}$	$2\sqrt{3}$, we find that $3w_1 + \sqrt{3}w_2 = 2\sqrt{3}$, so
johnston-linear-matrix-algebra F04182	447	■ 1.000	$w_2 = 2 - \sqrt{3}w_1$	$w_2 = 2 - \sqrt{3}w_1$	$w_2 = 2 - \sqrt{3}w_1$. On the other hand, $\ \mathbf{w}\ = 2$,
johnston-linear-matrix-algebra F04183	447	■ 1.000	$w_1^2 + w_2^2 = 4$	$w_1^2 + w_2^2 = 4$	so $w_1^2 + w_2^2 = 4$. Plugging in our formula for w_2
johnston-linear-matrix-algebra F04184	447	■ 1.000	w_2	w_2	so $w_1^2 + w_2^2 = 4$. Plugging in our formula for w_2
johnston-linear-matrix-algebra F04185	447	■ 1.000	$w_1^2 + (2 - \sqrt{3}w_1)^2 = 4$	$w_1^2 + (2 - \sqrt{3}w_1)^2 = 4$	gives $w_1^2 + (2 - \sqrt{3}w_1)^2 = 4$, and simplifying
johnston-linear-matrix-algebra F04186	447	■ 1.000	$w_1(w_1 - \sqrt{3}) = 0$	$w_1(w_1 - \sqrt{3}) = 0$	gives $w_1(w_1 - \sqrt{3}) = 0$, so $w_1 = 0$ or $w_1 = \sqrt{3}$.
johnston-linear-matrix-algebra F04187	447	■ 1.000	$w_1 = 0$	$w_1 = 0$	gives $w_1(w_1 - \sqrt{3}) = 0$, so $w_1 = 0$ or $w_1 = \sqrt{3}$.
johnston-linear-matrix-algebra F04188	447	■ 1.000	$w_1 = \sqrt{3}$	$w_1 = \sqrt{3}$	gives $w_1(w_1 - \sqrt{3}) = 0$, so $w_1 = 0$ or $w_1 = \sqrt{3}$.
johnston-linear-matrix-algebra F04189	447	■ 0.999	$w_2 = 2$	$w_2 = 2$	$w_2 = 2$, and $w_1 = \sqrt{3}$ implies $w_2 = -1$. There
johnston-linear-matrix-algebra F04190	447	■ 0.999	$w_2 = -1$	$w_2 = -1$	$w_2 = 2$, and $w_1 = \sqrt{3}$ implies $w_2 = -1$. There
johnston-linear-matrix-algebra F04191	447	■ 0.925	$\mathbf{w} = (\sqrt{3}, -1)$	$\mathbf{w} = (\sqrt{3}, -1)$	$\mathbf{w} = (\sqrt{3}, -1)$.
johnston-linear-matrix-algebra F04192	447	■ 1.000	$\mathbf{v}^2$	$\mathbf{v}^2$	exponent is not standard notation. Does $\mathbf{v}^2$
johnston-linear-matrix-algebra F04193	447	■ 0.974	$\ \mathbf{v}\ ^2 = \mathbf{v} \cdot \mathbf{v}$	$\ \mathbf{v}\ ^2 = \mathbf{v} \cdot \mathbf{v}$	of the entries of $\mathbf{v}$, or the scalar $\ \mathbf{v}\ ^2 = \mathbf{v} \cdot \mathbf{v}$, or
johnston-linear-matrix-algebra F04194	447	■ 0.996	$\mathbf{w} = (w_1, w_2) \in \mathbb{R}^2$	$\mathbf{w} = (w_1, w_2) \in \mathbb{R}^2$	$\mathbf{w} = (w_1, w_2) \in \mathbb{R}^2$ satisfies $w_1 + 2w_2 = 0$, so
johnston-linear-matrix-algebra F04195	447	■ 0.996	$w_1 + 2w_2 = 0$	$w_1 + 2w_2 = 0$	$\mathbf{w} = (w_1, w_2) \in \mathbb{R}^2$ satisfies $w_1 + 2w_2 = 0$, so
johnston-linear-matrix-algebra F04196	447	■ 1.000	$w_1 =$	$w_1 =$	we can choose w_2 arbitrarily and set $w_1 =$
johnston-linear-matrix-algebra F04197	447	■ 1.000	$-2w_2$	$-2w_2$	$-2w_2$. One possibility is $\mathbf{w} = (-2, 1)$.
johnston-linear-matrix-algebra F04198	447	■ 0.975	$\mathbf{v} \cdot \mathbf{y} = 0$	$\mathbf{v} \cdot \mathbf{y} = 0$	part (a) that the only solutions $\mathbf{y}$ to $\mathbf{v} \cdot \mathbf{y} = 0$ are
johnston-linear-matrix-algebra F04199	447	■ 0.531	$\mathbf{y} = y_2(-2, 1)$	$\mathbf{y} = y_2(-2, 1)$	of the form $\mathbf{y} = y_2(-2, 1)$. However, $\mathbf{y}$ then can-
johnston-linear-matrix-algebra F04200	447	■ 1.000	$\mathbf{w} \cdot \mathbf{y} = y_2(4 + 1)$	$\mathbf{w} \cdot \mathbf{y} = y_2(4 + 1)$	not be orthogonal to $\mathbf{w}$, since $\mathbf{w} \cdot \mathbf{y} = y_2(4 + 1)$,
johnston-linear-matrix-algebra F04201	447	■ 1.000	$\mathbf{y} = \mathbf{0}$	$\mathbf{y} = \mathbf{0}$	which equals 0 if and only if $\mathbf{y} = \mathbf{0}$. However,
johnston-linear-matrix-algebra F04202	447	■ 1.000	$(2, 2, 2) + (-1, -2, -3) + (-3, 0, 1) = (-2, 0, 0)$	$(2, 2, 2) + (-1, -2, -3) + (-3, 0, 1) = (-2, 0, 0)$	$(2, 2, 2) + (-1, -2, -3) + (-3, 0, 1) = (-2, 0, 0)$.

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johnston-linear-matrix-algebra-F04203	447	1.000	$\overline{\overline{\bar{x}}} = x$	$\bar{\bar{x}} = x$	where we used the facts that $\bar{\bar{x}} = x$ and $\bar{x} \cdot \bar{y} =$
johnston-linear-matrix-algebra-F04204	447	1.000	$\overline{\overline{x \cdot y}} =$	$\bar{x} \cdot \bar{y} =$	where we used the facts that $\bar{\bar{x}} = x$ and $\bar{x} \cdot \bar{y} =$
johnston-linear-matrix-algebra-F04205	447	1.000	$\overline{\bar{x} \cdot \bar{y}}$	$\bar{x} \cdot \bar{y}$	$\bar{x} \cdot \bar{y}$ for all $x, y \in \mathbb{C}$.
johnston-linear-matrix-algebra-F04206	447	1.000	$x, y \in \mathbb{C}$	$x, y \in \mathbb{C}$	$\bar{x} \cdot \bar{y}$ for all $x, y \in \mathbb{C}$.
johnston-linear-matrix-algebra-F04207	447	1.000	$\ \mathbf{v}\ \ \mathbf{w}\ = 0$	$\ \mathbf{v}\ \ \mathbf{w}\ = 0$	and $\ \mathbf{v}\ \ \mathbf{w}\ = 0$, so equality holds in the
johnston-linear-matrix-algebra-F04208	447	0.992	$ \mathbf{v} \cdot \mathbf{w} = (c \mathbf{w}) \cdot \mathbf{w} = c \mathbf{w} ^2$	$ \mathbf{v} \cdot \mathbf{w} = (c \mathbf{w}) \cdot \mathbf{w} = c \ \mathbf{w}\ ^2$	then $ \mathbf{v} \cdot \mathbf{w} = (c \mathbf{w}) \cdot \mathbf{w} = c \ \mathbf{w}\ ^2$, whereas
johnston-linear-matrix-algebra-F04209	447	1.000	$\ \mathbf{v}\ \ \mathbf{w}\ = \ c \mathbf{w}\ \ \mathbf{w}\ = c \ \mathbf{w}\ ^2$	$\ \mathbf{v}\ \ \mathbf{w}\ = \ c \mathbf{w}\ \ \mathbf{w}\ = c \ \mathbf{w}\ ^2$	$\ \mathbf{v}\ \ \mathbf{w}\ = \ c \mathbf{w}\ \ \mathbf{w}\ = c \ \mathbf{w}\ ^2$, so equality
johnston-linear-matrix-algebra-F04210	447	1.000	$\mathbf{v} \cdot \mathbf{w} = \ \mathbf{v}\ \ \mathbf{w}\ $	$\mathbf{v} \cdot \mathbf{w} = \ \mathbf{v}\ \ \mathbf{w}\ $	$\mathbf{v} \cdot \mathbf{w} = \ \mathbf{v}\ \ \mathbf{w}\ $ implies (by tracing the proof
johnston-linear-matrix-algebra-F04211	447	0.985	$\ \mathbf{w}\ \mathbf{v} - \ \mathbf{v}\ \mathbf{w}$	$\ \mathbf{w}\ \mathbf{v} - \ \mathbf{v}\ \mathbf{w}$	$\ \mathbf{w}\ \mathbf{v} - \ \mathbf{v}\ \mathbf{w}$ is 0. However, we saw in Theo-
johnston-linear-matrix-algebra-F04212	447	0.555	$\ \mathbf{w}\ \mathbf{v} = \ \mathbf{v}\ \mathbf{w}$	$\ \mathbf{w}\ \mathbf{v} = \ \mathbf{v}\ \mathbf{w}$	0 , so it follows that $\ \mathbf{w}\ \mathbf{v} = \ \mathbf{v}\ \mathbf{w}$, so $\mathbf{v}$ and $\mathbf{w}$
johnston-linear-matrix-algebra-F04213	448	1.000	$\ \mathbf{v}\ = \ \mathbf{v}\ $	$\ \mathbf{v}\ = \ \mathbf{v}\ $	ity just says that $\ \mathbf{v}\ = \ \mathbf{v}\ $, which is true,
johnston-linear-matrix-algebra-F04214	448	1.000	$\ \mathbf{v} + \mathbf{w}\ = \ (c \mathbf{w}) + \mathbf{w}\ =$	$\ \mathbf{v} + \mathbf{w}\ = \ (c \mathbf{w}) + \mathbf{w}\ =$	and if $\mathbf{v} = c \mathbf{w}$ then $\ \mathbf{v} + \mathbf{w}\ = \ (c \mathbf{w}) + \mathbf{w}\ =$
johnston-linear-matrix-algebra-F04215	448	1.000	$\ (c + 1) \mathbf{w}\ = (c + 1) \ \mathbf{w}\ = c \ \mathbf{w}\ + \ \mathbf{w}\ =$	$\ (c + 1) \mathbf{w}\ = (c + 1) \ \mathbf{w}\ = c \ \mathbf{w}\ + \ \mathbf{w}\ =$	$\ (c + 1) \mathbf{w}\ = (c + 1) \ \mathbf{w}\ = c \ \mathbf{w}\ + \ \mathbf{w}\ =$
johnston-linear-matrix-algebra-F04216	448	1.000	$\ c \mathbf{w}\ + \ \mathbf{w}\ = \ \mathbf{v}\ + \ \mathbf{w}\ $	$\ c \mathbf{w}\ + \ \mathbf{w}\ = \ \mathbf{v}\ + \ \mathbf{w}\ $	$\ c \mathbf{w}\ + \ \mathbf{w}\ = \ \mathbf{v}\ + \ \mathbf{w}\ $.
johnston-linear-matrix-algebra-F04217	448	1.000	$\ \mathbf{v} + \mathbf{w}\ =$	$\ \mathbf{v} + \mathbf{w}\ =$	proof of Theorem 1.2.4 that if $\ \mathbf{v} + \mathbf{w}\ =$
johnston-linear-matrix-algebra-F04218	448	1.000	$2(\mathbf{v} \cdot \mathbf{w}) = 2\ \mathbf{v}\ \ \mathbf{w}\ $	$2(\mathbf{v} \cdot \mathbf{w}) = 2\ \mathbf{v}\ \ \mathbf{w}\ $	$\ \mathbf{v}\ + \ \mathbf{w}\ $ then $2(\mathbf{v} \cdot \mathbf{w}) = 2\ \mathbf{v}\ \ \mathbf{w}\ $. It then
johnston-linear-matrix-algebra-F04219	448	1.000	$c \geq 0$	$c \geq 0$	remains is to show that $c \geq 0$. To this end, just
johnston-linear-matrix-algebra-F04220	448	1.000	$2(\mathbf{v} \cdot \mathbf{w}) = 2c \ \mathbf{w}\ ^2$	$2(\mathbf{v} \cdot \mathbf{w}) = 2c \ \mathbf{w}\ ^2$	with $c < 0$ then $2(\mathbf{v} \cdot \mathbf{w}) = 2c \ \mathbf{w}\ ^2$, whereas
johnston-linear-matrix-algebra-F04221	448	0.956	$2\ \mathbf{v}\ \ \mathbf{w}\ = 2 c \ \mathbf{w}\ ^2 = -2c \ \mathbf{w}\ ^2$	$2\ \mathbf{v}\ \ \mathbf{w}\ = 2 c \ \mathbf{w}\ ^2 = -2c \ \mathbf{w}\ ^2$	$2\ \mathbf{v}\ \ \mathbf{w}\ = 2 c \ \mathbf{w}\ ^2 = -2c \ \mathbf{w}\ ^2$, which con-
johnston-linear-matrix-algebra-F04222	448	1.000	$\ \mathbf{x} + \mathbf{y}\ \leq \ \mathbf{x}\ +$	$\ \mathbf{x} + \mathbf{y}\ \leq \ \mathbf{x}\ +$	1.2.21 The triangle inequality tells us that $\ \mathbf{x} + \mathbf{y}\ \leq \ \mathbf{x}\ +$
johnston-linear-matrix-algebra-F04223	448	1.000	$\ \mathbf{y}\ $	$\ \mathbf{y}\ $	$\ \mathbf{y}\ $ for all $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$. If we choose $\mathbf{x} = \mathbf{v} - \mathbf{w}$ and
johnston-linear-matrix-algebra-F04224	448	1.000	$\mathbf{x} = \mathbf{v} - \mathbf{w}$	$\mathbf{x} = \mathbf{v} - \mathbf{w}$	$\ \mathbf{y}\ $ for all $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$. If we choose $\mathbf{x} = \mathbf{v} - \mathbf{w}$ and
johnston-linear-matrix-algebra-F04225	448	1.000	$\mathbf{y} = \mathbf{w}$	$\mathbf{y} = \mathbf{w}$	$\mathbf{y} = \mathbf{w}$, then this says that
johnston-linear-matrix-algebra-F04226	448	0.999	$f(x) = \ \mathbf{v} - x \mathbf{w}\ ^2 = (\mathbf{v} - x \mathbf{w}) \cdot (\mathbf{v} - x \mathbf{w}) =$	$f(x) = \ \mathbf{v} - x \mathbf{w}\ ^2 = (\mathbf{v} - x \mathbf{w}) \cdot (\mathbf{v} - x \mathbf{w}) =$	1.2.23 (a) $f(x) = \ \mathbf{v} - x \mathbf{w}\ ^2 = (\mathbf{v} - x \mathbf{w}) \cdot (\mathbf{v} - x \mathbf{w}) =$
johnston-linear-matrix-algebra-F04227	448	1.000	$x^2 \ \mathbf{w}\ ^2 - 2x(\mathbf{v} \cdot \mathbf{w}) + \ \mathbf{v}\ ^2$	$x^2 \ \mathbf{w}\ ^2 - 2x(\mathbf{v} \cdot \mathbf{w}) + \ \mathbf{v}\ ^2$	$x^2 \ \mathbf{w}\ ^2 - 2x(\mathbf{v} \cdot \mathbf{w}) + \ \mathbf{v}\ ^2$
johnston-linear-matrix-algebra-F04228	448	1.000	$4(\mathbf{v} \cdot \mathbf{w})^2 - 4\ \mathbf{v}\ ^2 \ \mathbf{w}\ ^2$	$4(\mathbf{v} \cdot \mathbf{w})^2 - 4\ \mathbf{v}\ ^2 \ \mathbf{w}\ ^2$	(b) $4(\mathbf{v} \cdot \mathbf{w})^2 - 4\ \mathbf{v}\ ^2 \ \mathbf{w}\ ^2$
johnston-linear-matrix-algebra-F04229	448	1.000	$4(\mathbf{v} \cdot \mathbf{w})^2 -$	$4(\mathbf{v} \cdot \mathbf{w})^2 -$	(d) By parts (b) and (c), we know that $4(\mathbf{v} \cdot \mathbf{w})^2 -$
johnston-linear-matrix-algebra-F04230	448	1.000	$4\ \mathbf{v}\ ^2 \ \mathbf{w}\ ^2 \leq 0$	$4\ \mathbf{v}\ ^2 \ \mathbf{w}\ ^2 \leq 0$	$4\ \mathbf{v}\ ^2 \ \mathbf{w}\ ^2 \leq 0$. By adding $4\ \mathbf{v}\ ^2 \ \mathbf{w}\ ^2$ to both
johnston-linear-matrix-algebra-F04231	448	1.000	$4\ \mathbf{v}\ ^2 \ \mathbf{w}\ ^2$	$4\ \mathbf{v}\ ^2 \ \mathbf{w}\ ^2$	$4\ \mathbf{v}\ ^2 \ \mathbf{w}\ ^2 \leq 0$. By adding $4\ \mathbf{v}\ ^2 \ \mathbf{w}\ ^2$ to both

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johnston-linear-matrix-algebra-F04232	448	1.000	$\left[\begin{array}{cc}1 & 0 \\ -2 & 3\end{array}\right]$	$\begin{bmatrix} 1 & 0 \\ -2 & 3 \end{bmatrix}$	(a) $\begin{bmatrix} 1 & 0 \\ -2 & 3 \end{bmatrix}$
johnston-linear-matrix-algebra-F04233	448	1.000	$\left[\begin{array}{cc}3 & -2 \\ 2 & 3\end{array}\right]$	$\begin{bmatrix} 3 & -2 \\ 2 & 3 \end{bmatrix}$	(c) $\begin{bmatrix} 3 & -2 \\ 2 & 3 \end{bmatrix}$
johnston-linear-matrix-algebra-F04234	448	1.000	$A^{\{5\}}=A^{\{1\}}, A^{\{6\}}=A^{\{2\}}$	$A^5 = A^1, A^6 = A^2$	this point on (i.e., $A^5 = A^1, A^6 = A^2$, and so on). We
johnston-linear-matrix-algebra-F04235	448	0.979	$A^{\{2\}}=\left[\begin{array}{ll}1 & 2 \\ 0 & 1\end{array}\right]$	$A^2 = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$	1.3.7 (a) $A^2 = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$
johnston-linear-matrix-algebra-F04236	448	1.000	$A^{\{3\}}=\left[\begin{array}{ll}1 & 3 \\ 0 & 1\end{array}\right]$	$A^3 = \begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix}$	(b) $A^3 = \begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix}$
johnston-linear-matrix-algebra-F04237	449	1.000	$k=0,1$	$k = 0, 1$	$k = 0, 1$ are clear, and we also already showed
johnston-linear-matrix-algebra-F04238	449	1.000	$k=2,3$	$k = 2, 3$	the $k = 2, 3$ cases. For the inductive step, as-
johnston-linear-matrix-algebra-F04239	449	0.808	$A^{\{\ell\}}=\left[\begin{array}{cc}1 & \ell \\ 0 & 1\end{array}\right]$	$A^\ell = \begin{bmatrix} 1 & \ell \\ 0 & 1 \end{bmatrix}$	sume that $A^\ell = \begin{bmatrix} 1 & \ell \\ 0 & 1 \end{bmatrix}$ for some particular
johnston-linear-matrix-algebra-F04240	449	1.000	$\left[J_n\right]_{i,j}=(1,1,\ldots,1)\cdot(1,1,\ldots,1)=n$	$[J_n]_{i,j} = (1, 1, \dots, 1) \cdot (1, 1, \dots, 1) = n$	is, $[J_n]_{i,j} = (1, 1, \dots, 1) \cdot (1, 1, \dots, 1) = n$ for all i, j ,
johnston-linear-matrix-algebra-F04241	449	1.000	$J_n^2=nJ_n$	$J_n^2 = nJ_n$	so $J_n^2 = nJ_n$.
johnston-linear-matrix-algebra-F04242	449	1.000	$A^C{}^T$	AC^T	but some of these products (like AC^T) do not
johnston-linear-matrix-algebra-F04243	449	1.000	$A^B=\left[\begin{array}{lll}3 & 3 & 3 \\ 3 & 3 & 3\end{array}\right]$	$AB = \left[\begin{array}{cc cc} 3 & 3 & 3 & 3 \\ 3 & 3 & 3 & 3 \end{array}\right]$	1.3.13 (a) $AB = \left[\begin{array}{cc cc} 3 & 3 & 3 & 3 \\ 3 & 3 & 3 & 3 \end{array}\right]$
johnston-linear-matrix-algebra-F04244	449	0.997	$A^B=\left[\begin{array}{rr r}2 & 4 & 6 \\ 3 & 5 & 7 \\ 4 & 6 & 8 \\ \hline 2 & 4 & 6\end{array}\right]$	$AB = \left[\begin{array}{cc cc} 2 & 4 & 6 \\ 3 & 5 & 7 \\ 4 & 6 & 8 \\ \hline 2 & 4 & 6 \end{array}\right]$	(c) $AB = \left[\begin{array}{cc cc} 2 & 4 & 6 \\ 3 & 5 & 7 \\ 4 & 6 & 8 \\ \hline 2 & 4 & 6 \end{array}\right]$
johnston-linear-matrix-algebra-F04245	449	1.000	$r=n$	$r = n$	1.3.15 (a) For AB to exist, we need $r = n$, and for BA to
johnston-linear-matrix-algebra-F04246	449	1.000	$p=m$	$p = m$	exist, we need $p = m$. For both products to ex-
johnston-linear-matrix-algebra-F04247	449	1.000	$m=n=r=p$	$m = n = r = p$	(and thus $m = n = r = p$).
johnston-linear-matrix-algebra-F04248	449	1.000	$\mathbf{v} \cdot \mathbf{w}=\mathbf{v}^T \mathbf{w}$	$\mathbf{v} \cdot \mathbf{w} = \mathbf{v}^T \mathbf{w}$	tors then $\mathbf{v} \cdot \mathbf{w} = \mathbf{v}^T \mathbf{w}$. It follows that $\mathbf{x} \cdot (A\mathbf{y}) =$
johnston-linear-matrix-algebra-F04249	449	1.000	$\mathbf{x} \cdot(A \mathbf{y})=$	$\mathbf{x} \cdot (A\mathbf{y}) =$	tors then $\mathbf{v} \cdot \mathbf{w} = \mathbf{v}^T \mathbf{w}$. It follows that $\mathbf{x} \cdot (A\mathbf{y}) =$
johnston-linear-matrix-algebra-F04250	449	1.000	$\mathbf{x}^T A \mathbf{y}$	$\mathbf{x}^T A\mathbf{y}$	$\mathbf{x}^T A\mathbf{y}$ and $(A^T \mathbf{x}) \cdot \mathbf{y} = (A^T \mathbf{x})^T \mathbf{y} = \mathbf{x}^T A\mathbf{y}$. Since
johnston-linear-matrix-algebra-F04251	449	1.000	$\left(A^T \mathbf{x}\right) \cdot \mathbf{y}=\left(A^T \mathbf{x}\right)^T \mathbf{y}=\mathbf{x}^T A \mathbf{y}$	$(A^T \mathbf{x}) \cdot \mathbf{y} = (A^T \mathbf{x})^T \mathbf{y} = \mathbf{x}^T A\mathbf{y}$	$\mathbf{x}^T A\mathbf{y}$ and $(A^T \mathbf{x}) \cdot \mathbf{y} = (A^T \mathbf{x})^T \mathbf{y} = \mathbf{x}^T A\mathbf{y}$. Since
johnston-linear-matrix-algebra-F04252	449	1.000	$\mathbf{x} \cdot(A \mathbf{y})=\mathbf{x}^T A \mathbf{y}$	$\mathbf{x} \cdot (A\mathbf{y}) = \mathbf{x}^T A\mathbf{y}$	(b) Just like in part (a), $\mathbf{x} \cdot (A\mathbf{y}) = \mathbf{x}^T A\mathbf{y}$ and
johnston-linear-matrix-algebra-F04253	449	1.000	$(B \mathbf{x}) \cdot \mathbf{y}=\mathbf{x}^T B^T \mathbf{y}$	$(B\mathbf{x}) \cdot \mathbf{y} = \mathbf{x}^T B^T \mathbf{y}$	$(B\mathbf{x}) \cdot \mathbf{y} = \mathbf{x}^T B^T \mathbf{y}$. We thus know that $\mathbf{x}^T A\mathbf{y} =$
johnston-linear-matrix-algebra-F04254	449	1.000	$\mathbf{x}^T A \mathbf{y}=$	$\mathbf{x}^T A\mathbf{y} =$	$(B\mathbf{x}) \cdot \mathbf{y} = \mathbf{x}^T B^T \mathbf{y}$. We thus know that $\mathbf{x}^T A\mathbf{y} =$

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johnston-linear-matrix-algebra-F04255	449	1.000	$\mathbf{x}^T \mathbf{B}^T \mathbf{y}$	$\mathbf{x}^T \mathbf{B}^T \mathbf{y}$	$\mathbf{x}^T \mathbf{B}^T \mathbf{y}$ for all $\mathbf{x} \in \mathbb{R}^m$ and $\mathbf{y} \in \mathbb{R}^n$. However, if
johnston-linear-matrix-algebra-F04256	449	1.000	$\mathbf{x} = \mathbf{e}_i$	$\mathbf{x} = \mathbf{e}_i$	we choose $\mathbf{x} = \mathbf{e}_i$ and $\mathbf{y} = \mathbf{e}_j$ then $\mathbf{x}^T \mathbf{A} \mathbf{y} = a_{i,j}$
johnston-linear-matrix-algebra-F04257	449	1.000	$\mathbf{y} = \mathbf{e}_j$	$\mathbf{y} = \mathbf{e}_j$	we choose $\mathbf{x} = \mathbf{e}_i$ and $\mathbf{y} = \mathbf{e}_j$ then $\mathbf{x}^T \mathbf{A} \mathbf{y} = a_{i,j}$
johnston-linear-matrix-algebra-F04258	449	1.000	$\mathbf{x}^T \mathbf{A} \mathbf{y} = a_{i,j}$	$\mathbf{x}^T \mathbf{A} \mathbf{y} = a_{i,j}$	we choose $\mathbf{x} = \mathbf{e}_i$ and $\mathbf{y} = \mathbf{e}_j$ then $\mathbf{x}^T \mathbf{A} \mathbf{y} = a_{i,j}$
johnston-linear-matrix-algebra-F04259	449	1.000	$\mathbf{x}^T \mathbf{B}^T \mathbf{y} = b_{j,i}$	$\mathbf{x}^T \mathbf{B}^T \mathbf{y} = b_{j,i}$	and $\mathbf{x}^T \mathbf{B}^T \mathbf{y} = b_{j,i}$, so we see that $a_{i,j} = b_{j,i}$ for
johnston-linear-matrix-algebra-F04260	449	1.000	$a_{i,j} = b_{j,i}$	$a_{i,j} = b_{j,i}$	and $\mathbf{x}^T \mathbf{B}^T \mathbf{y} = b_{j,i}$, so we see that $a_{i,j} = b_{j,i}$ for
johnston-linear-matrix-algebra-F04261	449	0.994	$[\mathbf{e}_i]_k = 0$	$[\mathbf{e}_i]_k = 0$	$[\mathbf{e}_i]_k = 0$ if $k \neq i$ and $[\mathbf{e}_i]_i = 1$. It follows that
johnston-linear-matrix-algebra-F04262	449	0.994	$k \neq i$	$k \neq i$	$[\mathbf{e}_i]_k = 0$ if $k \neq i$ and $[\mathbf{e}_i]_i = 1$. It follows that
johnston-linear-matrix-algebra-F04263	449	0.994	$[\mathbf{e}_i]_i = 1$	$[\mathbf{e}_i]_i = 1$	$[\mathbf{e}_i]_k = 0$ if $k \neq i$ and $[\mathbf{e}_i]_i = 1$. It follows that
johnston-linear-matrix-algebra-F04264	449	1.000	$\mathbf{a}_i$	$\mathbf{a}_i$	(c) We let $\mathbf{a}_i$ denote the i -th column of A and then
johnston-linear-matrix-algebra-F04265	449	1.000	$(A - B)\mathbf{v} = \mathbf{0}$	$(A - B)\mathbf{v} = \mathbf{0}$	$(A - B)\mathbf{v} = \mathbf{0}$ for all $\mathbf{v} \in \mathbb{R}^n$. In particular then,
johnston-linear-matrix-algebra-F04266	449	1.000	$(A - B)\mathbf{e}_i = \mathbf{0}$	$(A - B)\mathbf{e}_i = \mathbf{0}$	$(A - B)\mathbf{e}_i = \mathbf{0}$ for all i , so part (a) tells us that
johnston-linear-matrix-algebra-F04267	449	1.000	$(A + B) + C$	$(A + B) + C$	1.3.19 (a) We compute the (i, j) -entry of both $(A + B) + C$
johnston-linear-matrix-algebra-F04268	449	1.000	$A + (B + C)$	$A + (B + C)$	and $A + (B + C)$:
johnston-linear-matrix-algebra-F04269	449	1.000	$(A + B) + C =$	$(A + B) + C =$	$A + (B + C)$ are all the same, so $(A + B) + C =$
johnston-linear-matrix-algebra-F04270	449	1.000	$c(A + B)$	$c(A + B)$	(b) We compute the (i, j) -entry of both $c(A + B)$
johnston-linear-matrix-algebra-F04271	449	1.000	$cA + cB$	$cA + cB$	and $cA + cB$:
johnston-linear-matrix-algebra-F04272	449	0.987	$cA +$	$cA +$	It follows that the entries of $c(A + B)$ and $cA +$
johnston-linear-matrix-algebra-F04273	449	1.000	cB	cB	cB are all the same, so $c(A + B) = cA + cB$.
johnston-linear-matrix-algebra-F04274	450	1.000	$(c + d)A$	$(c + d)A$	(c) We compute the (i, j) -entry of both $(c + d)A$
johnston-linear-matrix-algebra-F04275	450	1.000	$cA + dA$	$cA + dA$	and $cA + dA$:
johnston-linear-matrix-algebra-F04276	450	1.000	dA	dA	dA are all the same, so $(c + d)A = cA + dA$.
johnston-linear-matrix-algebra-F04277	450	1.000	$c(dA)$	$c(dA)$	(d) We compute the (i, j) -entry of both $c(dA)$ and
johnston-linear-matrix-algebra-F04278	450	0.936	$(cd)A$	$(cd)A$	$(cd)A$:
johnston-linear-matrix-algebra-F04279	450	1.000	$(AB)C$	$(AB)C$	1.3.20 (a) We compute the (i, j) -entry of both $(AB)C$ and
johnston-linear-matrix-algebra-F04280	450	1.000	$A(BC)$	$A(BC)$	$A(BC)$:
johnston-linear-matrix-algebra-F04281	450	1.000	$AC + BC$	$AC + BC$	and $AC + BC$:
johnston-linear-matrix-algebra-F04282	450	1.000	$A(B + C) =$	$A(B + C) =$	$AB + AC$ are all the same, so $A(B + C) =$
johnston-linear-matrix-algebra-F04283	450	1.000	$c(AB)$	$c(AB)$	(c) We compute the (i, j) -entry of both $c(AB)$ and

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johnston-linear-matrix-attributable_F04284	450	■ 1.000	$(c\ A)\ B$	$(cA)B$	$(cA)B$:
johnston-linear-matrix-attributable_F04285	450	■ 0.734	$I_{\{m\}}\ A$	$I_m A$	1.3.21 We recall that the (i, j) -entry of $I_m A$ is the dot prod-
johnston-linear-matrix-attributable_F04286	450	■ 0.998	$A: a_{\{i, j\}}$	$A: a_{i,j}$	column of A : $a_{i,j}$. All entries of $I_m A$ and A thus
johnston-linear-matrix-attributable_F04287	450	■ 1.000	$A^{\{k\}}\ A^{\{\ell\}} = (A\ A\ \cdots\ A)(A\ A\ \cdots\ A)$	$A^k A^\ell = (AA \cdots A)(AA \cdots A)$	1.3.22 (a) Just note that $A^k A^\ell = (AA \cdots A)(AA \cdots A)$,
johnston-linear-matrix-attributable_F04288	450	■ 1.000	$k + \ell$	$k + \ell$	the second has ℓ copies of A , for a total of $k + \ell$
johnston-linear-matrix-attributable_F04289	450	■ 1.000	$A^{k + \ell}$	$A^{k + \ell}$	copies of A (and is thus equal to $A^{k + \ell}$).
johnston-linear-matrix-attributable_F04290	450	■ 1.000	$k\ \ell$	$k\ell$	there are ℓ parentheses, for a total of $k\ell$ copies
johnston-linear-matrix-attributable_F04291	450	■ 1.000	$A^{k\ \ell}$	$A^{k\ell}$	of A (which is thus equal to $A^{k\ell}$).
johnston-linear-matrix-attributable_F04292	450	■ 1.000	$\left[\left(\left(A^T\right)^T\right)^T\right]_{\{i, j\}} = \left[A^T\right]_{\{j, i\}} = [A]_{\{i, j\}}$	$\left[(A^T)^T\right]_{i,j} = [A^T]_{j,i} = [A]_{i,j}$	$[(A^T)^T]_{i,j} = [A^T]_{j,i} = [A]_{i,j}$, so $(A^T)^T = A$.
johnston-linear-matrix-attributable_F04293	450	■ 1.000	$[(A +$	$[(A +$	(b) Compute the (i, j) -entry of both sides: $[(A +$
johnston-linear-matrix-attributable_F04294	450	■ 0.431	$\left.\right\}^T\right]_{\{i, j\}} = [A + B]_{\{j, i\}} = [A]_{\{j, i\}} + [B]_{\{j, i\}} = \left[A^T\right]_{\{i, j\}} +$	$\left.\right\}^T]_{i,j} = [A + B]_{j,i} = [A]_{j,i} + [B]_{j,i} = [A^T]_{i,j} +$	$[B]^T]_{i,j} = [A + B]_{j,i} = [A]_{j,i} + [B]_{j,i} = [A^T]_{i,j} +$
johnston-linear-matrix-attributable_F04295	450	■ 1.000	$\left[B^T\right]_{\{i, j\}} = \left[A^T + B^T\right]_{\{i, j\}}$	$[B^T]_{i,j} = [A^T + B^T]_{i,j}$	$[B^T]_{i,j} = [A^T + B^T]_{i,j}$, so $(A + B)^T = A^T + B^T$.
johnston-linear-matrix-attributable_F04296	450	■ 1.000	$\left[(c\ A)^T\right]_{\{i, j\}} = [c\ A]_{\{j, i\}} = c[A]_{\{j, i\}} = c\left[A^T\right]_{\{i, j\}} =$	$[(cA)^T]_{i,j} = [cA]_{j,i} = c[A]_{j,i} = c[A^T]_{i,j} =$	$[(cA)^T]_{i,j} = [cA]_{j,i} = c[A]_{j,i} = c[A^T]_{i,j} =$
johnston-linear-matrix-attributable_F04297	450	■ 0.998	$\left[c\ A^T\right]_{\{i, j\}}$	$[cA^T]_{i,j}$	$[cA^T]_{i,j}$, so $(cA)^T = cA^T$.
johnston-linear-matrix-attributable_F04298	450	■ 1.000	$p\ \times\ 1$	$p \times 1$	as a $p \times 1$ block matrix and B as a 1×1 block matrix,
johnston-linear-matrix-attributable_F04299	450	■ 1.000	$\mathbf{a}_1, \mathbf{a}_2, \ldots \mathbf{a}_n$	$\mathbf{a}_1, \mathbf{a}_2, \ldots \mathbf{a}_n$	where $\mathbf{a}_1, \mathbf{a}_2, \ldots \mathbf{a}_n$ are the columns of A . Com-
johnston-linear-matrix-attributable_F04300	450	■ 1.000	$(A\ B)^T =$	$(AB)^T =$	The rows of AB are the columns of $(AB)^T =$
johnston-linear-matrix-attributable_F04301	450	■ 1.000	$\left[\begin{array}{cc} 1 & 2 \\ 3 & -1 \end{array}\right]$	$\left[\begin{array}{cc} 1 & 2 \\ 3 & -1 \end{array}\right]$	(a) $\left[\begin{array}{cc} 1 & 2 \\ 3 & -1 \end{array}\right]$
johnston-linear-matrix-attributable_F04302	450	■ 1.000	$\left[\begin{array}{ccc} 1 & 1 & 0 \\ 1 & 1 & -1 \end{array}\right]$	$\left[\begin{array}{ccc} 1 & 1 & 0 \\ 1 & 1 & -1 \end{array}\right]$	(c) $\left[\begin{array}{ccc} 1 & 1 & 0 \\ 1 & 1 & -1 \end{array}\right]$
johnston-linear-matrix-attributable_F04303	451	■ 1.000	$T(1, 0, 0) =$	$T(1, 0, 0) =$	(e) This time, we must be more clever. $T(1, 0, 0) =$
johnston-linear-matrix-attributable_F04304	451	■ 1.000	$T(0, 1, 0) = T(1, 1, 0) - T(1, 0, 0) =$	$T(0, 1, 0) = T(1, 1, 0) - T(1, 0, 0) =$	column is $T(0, 1, 0) = T(1, 1, 0) - T(1, 0, 0) =$
johnston-linear-matrix-attributable_F04305	451	■ 1.000	$(0, 1, 2) - (1, 2, 3) = (-1, -1, -1)$	$(0, 1, 2) - (1, 2, 3) = (-1, -1, -1)$	$(0, 1, 2) - (1, 2, 3) = (-1, -1, -1)$. Its third
johnston-linear-matrix-attributable_F04306	451	■ 1.000	$T(0, 0, 1) = T(1, 1, 1) - T(1, 1, 0) =$	$T(0, 0, 1) = T(1, 1, 1) - T(1, 1, 0) =$	column is $T(0, 0, 1) = T(1, 1, 1) - T(1, 1, 0) =$
johnston-linear-matrix-attributable_F04307	451	■ 1.000	$(0, 0, 1) - (0, 1, 2) = (0, -1, -1)$	$(0, 0, 1) - (0, 1, 2) = (0, -1, -1)$	$(0, 0, 1) - (0, 1, 2) = (0, -1, -1)$. It follows
johnston-linear-matrix-attributable_F04308	451	■ 1.000	$[T] =$	$[T] =$	(which completely determines T) is $[T] =$
johnston-linear-matrix-attributable_F04309	451	■ 1.000	$\left[T\left(\mathbf{e}_1\right) \mid T\left(\mathbf{e}_2\right)\right]$	$[T(\mathbf{e}_1) \mid T(\mathbf{e}_2)]$	$[T(\mathbf{e}_1) \mid T(\mathbf{e}_2)]$.
johnston-linear-matrix-attributable_F04310	451	■ 1.000	$T\left(\mathbf{e}_1\right)=(2,1)$	$T(\mathbf{e}_1) = (2, 1)$	(e) False. If $T(\mathbf{e}_1) = (2, 1)$ and $T(\mathbf{e}_2) = (1, 3)$

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johnston-linear-matrix-algebra-F04311	451	■ 1.000	$T(\mathbf{e}_2)=(1,3)$	$T(\mathbf{e}_2) = (1,3)$	(e) False. If $T(\mathbf{e}_1) = (2,1)$ and $T(\mathbf{e}_2) = (1,3)$
johnston-linear-matrix-algebra-F04312	451	■ 1.000	$T(1,1)=T(\mathbf{e}_1+\mathbf{e}_2)=T(\mathbf{e}_1)+T(\mathbf{e}_2)=(2,1)+(1,3)=(3,4) \neq (3,3)$	$T(1,1) = T(\mathbf{e}_1 + \mathbf{e}_2) = T(\mathbf{e}_1) + T(\mathbf{e}_2) =$	then $T(1,1) = T(\mathbf{e}_1 + \mathbf{e}_2) = T(\mathbf{e}_1) + T(\mathbf{e}_2) =$
johnston-linear-matrix-algebra-F04313	451	■ 1.000	$(2,1)+(1,3)=(3,4) \neq (3,3)$	$(2,1) + (1,3) = (3,4) \neq (3,3)$	$(2,1) + (1,3) = (3,4) \neq (3,3)$.
johnston-linear-matrix-algebra-F04314	451	■ 1.000	$\mathbf{v}=(0,1,0)$	$\mathbf{v} = (0,1,0)$	and $\mathbf{v} = (0,1,0)$. Then $\hat{R}_{xz}^\theta(\mathbf{v}) = \mathbf{v}$, but
johnston-linear-matrix-algebra-F04315	451	■ 1.000	$R_{xz}^\theta(\mathbf{v})=\mathbf{v}$	$R_{xz}^\theta(\mathbf{v}) = \mathbf{v}$	and $\mathbf{v} = (0,1,0)$. Then $\hat{R}_{xz}^\theta(\mathbf{v}) = \mathbf{v}$, but
johnston-linear-matrix-algebra-F04316	451	■ 0.623	$(R_{xy}^\theta \circ R_{yz}^\theta)(\mathbf{v})=R_{xy}^\theta(0,0,1)=(0,0,1) \neq \mathbf{v}$	$(R_{xy}^\theta \circ R_{yz}^\theta)(\mathbf{v}) = R_{xy}^\theta(0,0,1) = (0,0,1) \neq \mathbf{v}$	$(R_{xy}^\theta \circ R_{yz}^\theta)(\mathbf{v}) = R_{xy}^\theta(0,0,1) = (0,0,1) \neq \mathbf{v}$.
johnston-linear-matrix-algebra-F04317	451	■ 1.000	$R_{xy}^\theta \circ R_{yz}^\theta \neq R_{xz}^\theta$	$R_{xy}^\theta \circ R_{yz}^\theta \neq R_{xz}^\theta$	It follows that $R_{xy}^\theta \circ R_{yz}^\theta \neq R_{xz}^\theta$.
johnston-linear-matrix-algebra-F04318	451	■ 1.000	$T(2(1,1))=(4,2)$	$T(2(1,1)) = (4,2)$	$T(2(1,1)) = (4,2)$, but $2T(1,1) = (2,2)$.
johnston-linear-matrix-algebra-F04319	451	■ 1.000	$2T(1,1)=(2,2)$	$2T(1,1) = (2,2)$	$T(2(1,1)) = (4,2)$, but $2T(1,1) = (2,2)$.
johnston-linear-matrix-algebra-F04320	451	■ 0.996	$T(0,0)=(0,-1) \neq \mathbf{0}$	$T(0,0) = (0,-1) \neq \mathbf{0}$	cally, $T(0,0) = (0,-1) \neq \mathbf{0}$.
johnston-linear-matrix-algebra-F04321	451	■ 1.000	$T(2(1,1))=(2\sqrt{2},2)$	$T(2(1,1)) = (2\sqrt{2},2)$	cally, $T(2(1,1)) = (2\sqrt{2},2)$, but $2T(1,1) =$
johnston-linear-matrix-algebra-F04322	451	■ 1.000	$2T(1,1)=$	$2T(1,1) =$	cally, $T(2(1,1)) = (2\sqrt{2},2)$, but $2T(1,1) =$
johnston-linear-matrix-algebra-F04323	451	■ 1.000	$(4,2\sqrt{2})$	$(4,2\sqrt{2})$	$(4,2\sqrt{2})$.
johnston-linear-matrix-algebra-F04324	451	■ 1.000	$T(1,1)+T(-1,-1)=(2,2)$	$T(1,1) + T(-1,-1) = (2,2)$	Specifically, $T(1,1) + T(-1,-1) = (2,2)$, but
johnston-linear-matrix-algebra-F04325	451	■ 1.000	$T((1,1)+(-1,-1))=(0,0)$	$T((1,1) + (-1,-1)) = (0,0)$	$T((1,1) + (-1,-1)) = (0,0)$.
johnston-linear-matrix-algebra-F04326	451	■ 1.000	$\frac{1}{10}\begin{bmatrix} 1 & 3 \\ 3 & 9 \end{bmatrix}$	$\frac{1}{10} \begin{bmatrix} 1 & 3 \\ 3 & 9 \end{bmatrix}$	is $\frac{1}{10} \begin{bmatrix} 1 & 3 \\ 3 & 9 \end{bmatrix}$.
johnston-linear-matrix-algebra-F04327	451	■ 0.999	$\frac{1}{5}\begin{bmatrix} -3 & 4 \\ 4 & 3 \end{bmatrix}$	$\frac{1}{5} \begin{bmatrix} -3 & 4 \\ 4 & 3 \end{bmatrix}$	is $\frac{1}{5} \begin{bmatrix} -3 & 4 \\ 4 & 3 \end{bmatrix}$.
johnston-linear-matrix-algebra-F04328	451	■ 0.798	$\begin{bmatrix} \cos(\pi/5) & -\sin(\pi/5) \\ \sin(\pi/5) & \cos(\pi/5) \end{bmatrix}$	$\begin{bmatrix} \cos(\pi/5) & -\sin(\pi/5) \\ \sin(\pi/5) & \cos(\pi/5) \end{bmatrix}$	is $\begin{bmatrix} \cos(\pi/5) & -\sin(\pi/5) \\ \sin(\pi/5) & \cos(\pi/5) \end{bmatrix}$.
johnston-linear-matrix-algebra-F04329	451	■ 1.000	$\mathbf{u}=\frac{1}{\sqrt{2}}(1,1)$	$\mathbf{u} = \frac{1}{\sqrt{2}}(1,1)$	direction of $\mathbf{u} = \frac{1}{\sqrt{2}}(1,1)$ and rotation are
johnston-linear-matrix-algebra-F04330	451	■ 0.902	$T(\mathbf{0})=0$ $T(\mathbf{0})=\mathbf{0}$	$T(\mathbf{0}) = OT(\mathbf{0}) = \mathbf{0}$	$c = 0$ and $\mathbf{v} = \mathbf{0}$ then this says that $T(\mathbf{0}) = 0T(\mathbf{0}) = \mathbf{0}$,
johnston-linear-matrix-algebra-F04331	451	■ 1.000	$T(c_1\mathbf{v}_1+\cdots$	$T(c_1\mathbf{v}_1 + \cdots +$	1.4.9 For the “if” direction, assume that $T(c_1\mathbf{v}_1 + \cdots +$
johnston-linear-matrix-algebra-F04332	451	■ 1.000	$c_k\mathbf{v}_k)=c_1T(\mathbf{v}_1)+\cdots+c_kT(\mathbf{v}_k)$	$c_k\mathbf{v}_k) = c_1T(\mathbf{v}_1) + \cdots + c_kT(\mathbf{v}_k)$	$c_k\mathbf{v}_k) = c_1T(\mathbf{v}_1) + \cdots + c_kT(\mathbf{v}_k)$ for all $\mathbf{v}_1, \dots, \mathbf{v}_k \in$
johnston-linear-matrix-algebra-F04333	451	■ 1.000	$\mathbf{v}_1, \dots, \mathbf{v}_k \in$	$\mathbf{v}_1, \dots, \mathbf{v}_k \in$	$c_k\mathbf{v}_k) = c_1T(\mathbf{v}_1) + \cdots + c_kT(\mathbf{v}_k)$ for all $\mathbf{v}_1, \dots, \mathbf{v}_k \in$
johnston-linear-matrix-algebra-F04334	451	■ 1.000	$T(c_1\mathbf{v}_1)=c_1T(\mathbf{v}_1)$	$T(c_1\mathbf{v}_1) = c_1T(\mathbf{v}_1)$	then this says that $T(c_1\mathbf{v}_1) = c_1T(\mathbf{v}_1)$, and if we
johnston-linear-matrix-algebra-F04335	451	■ 1.000	$c_1=c_2=1$	$c_1 = c_2 = 1$	choose $k = 2$ and $c_1 = c_2 = 1$ then it says that
johnston-linear-matrix-algebra-F04336	451	■ 1.000	$T(\mathbf{v}_1+\mathbf{v}_2)=T(\mathbf{v}_1)+T(\mathbf{v}_2)$	$T(\mathbf{v}_1 + \mathbf{v}_2) = T(\mathbf{v}_1) + T(\mathbf{v}_2)$	$T(\mathbf{v}_1 + \mathbf{v}_2) = T(\mathbf{v}_1) + T(\mathbf{v}_2)$. These are the two

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johnston-linear-matrix-attached F04337	451	1.000	$T(\mathbf{c}_1 + \cdots + \mathbf{c}_k) = T(\mathbf{c}_1) + \cdots + T(\mathbf{c}_k)$	$T(c_1\mathbf{v}_1 + \cdots + c_k\mathbf{v}_k) = T(c_1\mathbf{v}_1) + \cdots + T(c_k\mathbf{v}_k)$	see that $T(c_1\mathbf{v}_1 + \cdots + c_k\mathbf{v}_k) = T(c_1\mathbf{v}_1) + \cdots + T(c_k\mathbf{v}_k)$
johnston-linear-matrix-attached F04338	451	1.000	$T(\mathbf{c}_1 + \cdots + \mathbf{c}_k) = T(\mathbf{c}_1) + \cdots + T(\mathbf{c}_k)$	$\cdots + c_k\mathbf{v}_k) = c_1T(\mathbf{v}_1) + \cdots + c_kT(\mathbf{v}_k) = \cdots =$	$\cdots + c_k\mathbf{v}_k) = c_1T(\mathbf{v}_1) + \cdots + c_kT(\mathbf{v}_k) = \cdots =$
johnston-linear-matrix-attached F04339	451	1.000	$T(\mathbf{c}_1 + \cdots + \mathbf{c}_k) = T(\mathbf{c}_1) + \cdots + T(\mathbf{c}_k)$	$c_1T(\mathbf{v}_1) + \cdots + c_kT(\mathbf{v}_k)$	$c_1T(\mathbf{v}_1) + \cdots + c_kT(\mathbf{v}_k)$ for all $\mathbf{v}_1, \dots, \mathbf{v}_k \in \mathbb{R}^n$ and
johnston-linear-matrix-attached F04340	452	1.000	$[S+T]\mathbf{v} = (S+T)\mathbf{v}$	$[S+T]\mathbf{v} = (S+T)\mathbf{v}$	for all $\mathbf{v} \in \mathbb{R}^n$ it is the case that $[S+T]\mathbf{v} = (S+T)\mathbf{v}$
johnston-linear-matrix-attached F04341	452	0.977	$T(\mathbf{v}) = S(\mathbf{v}) + T(\mathbf{v}) = [S]\mathbf{v} + [T]\mathbf{v} = ([S] + [T])\mathbf{v}$	$T(\mathbf{v}) = S(\mathbf{v}) + T(\mathbf{v}) = [S]\mathbf{v} + [T]\mathbf{v} = ([S] + [T])\mathbf{v}$	$T(\mathbf{v}) = S(\mathbf{v}) + T(\mathbf{v}) = [S]\mathbf{v} + [T]\mathbf{v} = ([S] + [T])\mathbf{v}$
johnston-linear-matrix-attached F04342	452	1.000	$[T]\mathbf{v}$	$[T]\mathbf{v}$	$[T]\mathbf{v}$. Since $[S+T]\mathbf{v} = ([S] + [T])\mathbf{v}$ for all
johnston-linear-matrix-attached F04343	452	1.000	$[S+T]\mathbf{v} = ([S] + [T])\mathbf{v}$	$[S+T]\mathbf{v} = ([S] + [T])\mathbf{v}$	$[T]\mathbf{v}$. Since $[S+T]\mathbf{v} = ([S] + [T])\mathbf{v}$ for all
johnston-linear-matrix-attached F04344	452	1.000	$[cS]\mathbf{v} = (cS)(\mathbf{v}) = cS(\mathbf{v}) = c[S]\mathbf{v} = (c[S])\mathbf{v}$	$[cS]\mathbf{v} = (cS)(\mathbf{v}) = cS(\mathbf{v}) = c[S]\mathbf{v} = (c[S])\mathbf{v}$	that $[cS]\mathbf{v} = (cS)(\mathbf{v}) = cS(\mathbf{v}) = c[S]\mathbf{v} = (c[S])\mathbf{v}$.
johnston-linear-matrix-attached F04345	452	1.000	$[cS]\mathbf{v} = (c[S])\mathbf{v}$	$[cS]\mathbf{v} = (c[S])\mathbf{v}$	Since $[cS]\mathbf{v} = (c[S])\mathbf{v}$ for all $\mathbf{v} \in \mathbb{R}^n$, it follows
johnston-linear-matrix-attached F04346	452	1.000	$[cS] = c[S]$	$[cS] = c[S]$	from Exercise 1.3.18(d) that $[cS] = c[S]$.
johnston-linear-matrix-attached F04347	452	1.000	$160\pi/4 = 40\pi$	$160\pi/4 = 40\pi$	$160\pi/4 = 40\pi$. Since 40π is a multiple of 2π , this
johnston-linear-matrix-attached F04348	452	1.000	40π	40π	$160\pi/4 = 40\pi$. Since 40π is a multiple of 2π , this
johnston-linear-matrix-attached F04349	452	1.000	$A^{160} = I$	$A^{160} = I$	$A^{160} = I$.
johnston-linear-matrix-attached F04350	452	0.974	$\cos(\theta)\sin(\theta) - \cos(\theta)\sin(\theta) = 0$	$\cos(\theta)\sin(\theta) - \cos(\theta)\sin(\theta) = 0$	$\cos(\theta)\sin(\theta) - \cos(\theta)\sin(\theta) = 0$. We thus con-
johnston-linear-matrix-attached F04351	452	1.000	$\mathbf{u} = (1, m)$	$\mathbf{u} = (1, m)$	$y = mx$ is $\mathbf{u} = (1, m)$, so we know that the standard
johnston-linear-matrix-attached F04352	452	1.000	$E_{i,j} = \mathbf{e}_i\mathbf{e}_j^T$	$E_{i,j} = \mathbf{e}_i\mathbf{e}_j^T$	(b) First, we notice that $E_{i,j} = \mathbf{e}_i\mathbf{e}_j^T$, so $E_{i,j}^2 =$
johnston-linear-matrix-attached F04353	452	1.000	$E_{i,j}^2 =$	$E_{i,j}^2 =$	(b) First, we notice that $E_{i,j} = \mathbf{e}_i\mathbf{e}_j^T$, so $E_{i,j}^2 =$
johnston-linear-matrix-attached F04354	452	0.978	$\mathbf{e}_i\mathbf{e}_j^T\mathbf{e}_i\mathbf{e}_j^T = \mathbf{e}_i(\mathbf{e}_j \cdot \mathbf{e}_i)\mathbf{e}_j^T = \mathbf{0}$	$\mathbf{e}_i\mathbf{e}_j^T\mathbf{e}_i\mathbf{e}_j^T = \mathbf{e}_i(\mathbf{e}_j \cdot \mathbf{e}_i)\mathbf{e}_j^T = \mathbf{0}$	$\mathbf{e}_i\mathbf{e}_j^T\mathbf{e}_i\mathbf{e}_j^T = \mathbf{e}_i(\mathbf{e}_j \cdot \mathbf{e}_i)\mathbf{e}_j^T = \mathbf{0}$, since $i \neq j$. Since
johnston-linear-matrix-attached F04355	452	1.000	$u_1^2 = u_3^2 = 3/7$	$u_1^2 = u_3^2 = 3/7$	each other first, we find that $u_1^2 = u_3^2 = 3/7$,
johnston-linear-matrix-attached F04356	452	1.000	$u_2^2 = 1/7$	$u_2^2 = 1/7$	$u_2^2 = 1/7$, and $\cos(\theta) = 1/8$. Then focusing
johnston-linear-matrix-attached F04357	452	1.000	$\cos(\theta) = 1/8$	$\cos(\theta) = 1/8$	$u_2^2 = 1/7$, and $\cos(\theta) = 1/8$. Then focusing
johnston-linear-matrix-attached F04358	452	1.000	$(\sqrt{3}, 1, \sqrt{3})/\sqrt{7}$	$(\sqrt{3}, 1, \sqrt{3})/\sqrt{7}$	$(\sqrt{3}, 1, \sqrt{3})/\sqrt{7}$ and $\theta = -\arccos(1/8) \approx$
johnston-linear-matrix-attached F04359	452	1.000	$\theta = -\arccos(1/8) \approx$	$\theta = -\arccos(1/8) \approx$	$(\sqrt{3}, 1, \sqrt{3})/\sqrt{7}$ and $\theta = -\arccos(1/8) \approx$
johnston-linear-matrix-attached F04360	452	1.000	$\theta = \arccos(1/8) \approx 1.4455$	$\theta = \arccos(1/8) \approx 1.4455$	you choose $\theta = \arccos(1/8) \approx 1.4455$ then
johnston-linear-matrix-attached F04361	452	1.000	$\sin(\theta)$	$\sin(\theta)$	the $\sin(\theta)$ terms will have the wrong sign,
johnston-linear-matrix-attached F04362	453	1.000	$(AB)^3 = (AB)(AB)(AB)$	$(AB)^3 = (AB)(AB)(AB)$	(c) False. $(AB)^3 = (AB)(AB)(AB)$ and in general
johnston-linear-matrix-attached F04363	453	1.000	$A^3B^3 = AAABBB$	$A^3B^3 = AAABBB$	necessarily equal $A^3B^3 = AAABBB$.

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johnston-linear-matrix-attempt-453 F04364	453	1.000	$R(S(T(\mathbf{v})))$	$R(S(T(\mathbf{v})))$	in $R(S(T(\mathbf{v})))$.
johnston-linear-matrix-attempt-453 F04365	453	0.774	$\mathbf{u}=(1,0)$	$\mathbf{u} = (1, 0)$	$\mathbf{u} = (1, 0)$.
johnston-linear-matrix-attempt-453 F04366	453	1.000	$n \ \theta$	$n\theta$	of R^θ together results in a rotation by an angle of $n\theta$,
johnston-linear-matrix-attempt-453 F04367	453	1.000	$\ (2,1,0) \times (-2,3,0)\ =\ (0,0,8)\ =8$	$\ (2,1,0) \times (-2,3,0)\ = \ (0,0,8)\ = 8$	1.A.2 (a) $\ (2,1,0) \times (-2,3,0)\ = \ (0,0,8)\ = 8$
johnston-linear-matrix-attempt-453 F04368	453	1.000	$\ (1,2,3) \times (3,-1,2)\ =\ (7,7,-7)\ =7 \sqrt{3}$	$\ (1,2,3) \times (3,-1,2)\ = \ (7,7,-7)\ = 7\sqrt{3}$	(c) $\ (1,2,3) \times (3,-1,2)\ = \ (7,7,-7)\ = 7\sqrt{3}$
johnston-linear-matrix-attempt-453 F04369	453	1.000	$\frac{1}{2}\ (0,4,0) \times (1,1,0)\ =\frac{1}{2}\ (0,0,-4)\ =2$	$\frac{1}{2}\ (0,4,0) \times (1,1,0)\ = \frac{1}{2}\ (0,0,-4)\ = 2$	(a) $\frac{1}{2}\ (0,4,0) \times (1,1,0)\ = \frac{1}{2}\ (0,0,-4)\ = 2$
johnston-linear-matrix-attempt-453 F04370	453	1.000	$\frac{1}{2}\ (-1,1,-1) \times (3,2,1)\ =\frac{1}{2}\ (3,-2,-5)\ =$ $\ (3,-2,-5)\ =$	$\frac{1}{2}\ (-1,1,-1) \times (3,2,1)\ = \frac{1}{2}\ (3,-2,-5)\ =$	(c) $\frac{1}{2}\ (-1,1,-1) \times (3,2,1)\ = \frac{1}{2}\ (3,-2,-5)\ =$
johnston-linear-matrix-attempt-453 F04371	453	1.000	$\frac{\sqrt{38}}{2}$	$\frac{\sqrt{38}}{2}$	$\frac{\sqrt{38}}{2}$
johnston-linear-matrix-attempt-453 F04372	453	0.826	$1 \cdot 2 \cdot 3=6$	$1 \cdot 2 \cdot 3 = 6$	lengths 1, 2, and 3, so its volume is $1 \cdot 2 \cdot 3 = 6$.
johnston-linear-matrix-attempt-453 F04373	453	1.000	$ \mathbf{v} \cdot \mathbf{w} \times \mathbf{x} = (1,1,1) \cdot (3,4,-2) =\mid 3+4-$	$ \mathbf{v} \cdot (\mathbf{w} \times \mathbf{x}) = (1,1,1) \cdot (3,4,-2) = 3 + 4 -$	(c) $ \mathbf{v} \cdot (\mathbf{w} \times \mathbf{x}) = (1,1,1) \cdot (3,4,-2) = 3 + 4 -$
johnston-linear-matrix-attempt-453 F04374	453	0.958	$2 \ \mid =5$	$2 \mid = 5$	$2 \mid = 5$
johnston-linear-matrix-attempt-453 F04375	453	1.000	$\ (0,2,0) \times (-2,3,0)\ =\ (0,0,4)\ =4$	$\ (0,2,0) \times (-2,3,0)\ = \ (0,0,4)\ = 4$	(c) $\ (0,2,0) \times (-2,3,0)\ = \ (0,0,4)\ = 4$.
johnston-linear-matrix-attempt-453 F04376	453	0.996	$\mid (1,0,0)$	$\mid (1,0,0)$	(c) Now we use Theorem 1.A.3: $\mid (1,0,0) \cdot$
johnston-linear-matrix-attempt-453 F04377	453	0.998	$((-3,2,0) \times (2,-7,3)) \ \mid =6$	$((-3,2,0) \times (2,-7,3)) \mid = 6$	$((-3,2,0) \times (2,-7,3)) \mid = 6$.
johnston-linear-matrix-attempt-453 F04378	453	0.999	$\ \mathbf{v} \times \mathbf{w}\ =\ \mathbf{v}\ \ \mathbf{w}\ \sin(\theta)$	$\ \mathbf{v} \times \mathbf{w}\ = \ \mathbf{v}\ \ \mathbf{w}\ \sin(\theta)$	1.A.8 Recall that $\ \mathbf{v} \times \mathbf{w}\ = \ \mathbf{v}\ \ \mathbf{w}\ \sin(\theta)$. We are told
johnston-linear-matrix-attempt-453 F04379	453	1.000	$\ \mathbf{v}\ =\ \mathbf{w}\ =1$	$\ \mathbf{v}\ = \ \mathbf{w}\ = 1$	so $\ \mathbf{v}\ = \ \mathbf{w}\ = 1$. Similarly, we can compute
johnston-linear-matrix-attempt-453 F04380	453	1.000	$\ \mathbf{v} \times \mathbf{w}\ =\sqrt{(1/3)^2+(2/3)^2+(2/3)^2}=1$	$\ \mathbf{v} \times \mathbf{w}\ = \sqrt{(1/3)^2 + (2/3)^2 + (2/3)^2} = 1$	$\ \mathbf{v} \times \mathbf{w}\ = \sqrt{(1/3)^2 + (2/3)^2 + (2/3)^2} = 1$. It fol-
johnston-linear-matrix-attempt-453 F04381	453	1.000	$1=\sin(\theta)$	$1 = \sin(\theta)$	lows that $1 = \sin(\theta)$, so $\theta = \pi/2$.
johnston-linear-matrix-attempt-454 F04382	454	1.000	$\mathbf{v}=\mathbf{e}_1, \ \mathbf{w}=\mathbf{e}_2$	$\mathbf{v} = \mathbf{e}_1, \mathbf{w} = \mathbf{e}_2$	$\mathbf{v} = \mathbf{e}_1, \mathbf{w} = \mathbf{e}_2$, and $\mathbf{x} = \mathbf{e}_2$. Then $(\mathbf{v} \times \mathbf{w}) \times \mathbf{x} =$
johnston-linear-matrix-attempt-454 F04383	454	1.000	$\mathbf{x}=\mathbf{e}_2$	$\mathbf{x} = \mathbf{e}_2$	$\mathbf{v} = \mathbf{e}_1, \mathbf{w} = \mathbf{e}_2$, and $\mathbf{x} = \mathbf{e}_2$. Then $(\mathbf{v} \times \mathbf{w}) \times \mathbf{x} =$
johnston-linear-matrix-attempt-454 F04384	454	1.000	$(\mathbf{v} \times \mathbf{w}) \times \mathbf{x} =$	$(\mathbf{v} \times \mathbf{w}) \times \mathbf{x} =$	$\mathbf{v} = \mathbf{e}_1, \mathbf{w} = \mathbf{e}_2$, and $\mathbf{x} = \mathbf{e}_2$. Then $(\mathbf{v} \times \mathbf{w}) \times \mathbf{x} =$
johnston-linear-matrix-attempt-454 F04385	454	1.000	$(\mathbf{e}_1 \times \mathbf{e}_2) \times \mathbf{e}_2 = \mathbf{e}_3 \times \mathbf{e}_2 = -\mathbf{e}_1$	$(\mathbf{e}_1 \times \mathbf{e}_2) \times \mathbf{e}_2 = \mathbf{e}_3 \times \mathbf{e}_2 = -\mathbf{e}_1$	$(\mathbf{e}_1 \times \mathbf{e}_2) \times \mathbf{e}_2 = \mathbf{e}_3 \times \mathbf{e}_2 = -\mathbf{e}_1$, but $\mathbf{v} \times (\mathbf{w} \times \mathbf{x}) =$
johnston-linear-matrix-attempt-454 F04386	454	1.000	$\mathbf{v} \times (\mathbf{w} \times \mathbf{x}) =$	$\mathbf{v} \times (\mathbf{w} \times \mathbf{x}) =$	$(\mathbf{e}_1 \times \mathbf{e}_2) \times \mathbf{e}_2 = \mathbf{e}_3 \times \mathbf{e}_2 = -\mathbf{e}_1$, but $\mathbf{v} \times (\mathbf{w} \times \mathbf{x}) =$
johnston-linear-matrix-attempt-454 F04387	454	1.000	$\mathbf{e}_1 \times (\mathbf{e}_2 \times \mathbf{e}_2) = \mathbf{e}_1 \times \mathbf{0} = \mathbf{0}$	$\mathbf{e}_1 \times (\mathbf{e}_2 \times \mathbf{e}_2) = \mathbf{e}_1 \times \mathbf{0} = \mathbf{0}$	$\mathbf{e}_1 \times (\mathbf{e}_2 \times \mathbf{e}_2) = \mathbf{e}_1 \times \mathbf{0} = \mathbf{0}$.
johnston-linear-matrix-attempt-454 F04388	454	1.000	$\mathbf{v} \cdot (\mathbf{w} \times \mathbf{x}) = \mathbf{w} \cdot (\mathbf{x} \times \mathbf{v})$	$\mathbf{v} \cdot (\mathbf{w} \times \mathbf{x}) = \mathbf{w} \cdot (\mathbf{x} \times \mathbf{v})$	1.A.16 (a) To show that $\mathbf{v} \cdot (\mathbf{w} \times \mathbf{x}) = \mathbf{w} \cdot (\mathbf{x} \times \mathbf{v})$ we explic-
johnston-linear-matrix-attempt-454 F04389	454	1.000	$\mathbf{v} \cdot (\mathbf{w} \times \mathbf{x}) =$	$\mathbf{v} \cdot (\mathbf{w} \times \mathbf{x}) =$	to see that they are equal, so $\mathbf{v} \cdot (\mathbf{w} \times \mathbf{x}) =$
johnston-linear-matrix-attempt-454 F04390	454	1.000	$\mathbf{w} \cdot (\mathbf{x} \times \mathbf{v})$	$\mathbf{w} \cdot (\mathbf{x} \times \mathbf{v})$	$\mathbf{w} \cdot (\mathbf{x} \times \mathbf{v})$. The fact that $\mathbf{w} \cdot (\mathbf{x} \times \mathbf{v}) = \mathbf{x} \cdot (\mathbf{v} \times \mathbf{w})$
johnston-linear-matrix-attempt-454 F04391	454	1.000	$\mathbf{w} \cdot (\mathbf{x} \times \mathbf{v}) = \mathbf{x} \cdot (\mathbf{v} \times \mathbf{w})$	$\mathbf{w} \cdot (\mathbf{x} \times \mathbf{v}) = \mathbf{x} \cdot (\mathbf{v} \times \mathbf{w})$	$\mathbf{w} \cdot (\mathbf{x} \times \mathbf{v})$. The fact that $\mathbf{w} \cdot (\mathbf{x} \times \mathbf{v}) = \mathbf{x} \cdot (\mathbf{v} \times \mathbf{w})$

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johnston-linear-matrix-atlas- F04392	455	■ 1.000	$\left[A^{\{2\}}\right]_{\{1,2\}}=1$	$[A^2]_{1,2} = 1$	There is thus $[A^2]_{1,2} = 1$ path of length 2 from
johnston-linear-matrix-atlas- F04393	455	■ 1.000	$\left[A^{\{4\}}\right]_{\{1,4\}}=9$	$[A^4]_{1,4} = 9$	There are thus $[A^4]_{1,4} = 9$ paths of length 4
johnston-linear-matrix-atlas- F04394	455	■ 1.000	$0 \leq k \leq n-1$	$0 \leq k \leq n-1$	length n : for any $0 \leq k \leq n-1$, the self-loop
johnston-linear-matrix-atlas- F04395	455	■ 1.000	$n-k-1$	$n-k-1$	another $n-k-1$ times.
johnston-linear-matrix-atlas- F04396	455	■ 1.000	$k \geq 1, \left[A^{\{k\}}\right]_{i,j}$	$k \geq 1, [A^k]_{i,j}$	teger $k \geq 1$, $[A^k]_{i,j}$ counts the number of paths of
johnston-linear-matrix-atlas- F04397	455	■ 0.983	$\left[A^{\{k\}}\right]_{\{i,1\}} a_{1,j}$	$[A^k]_{i,1} a_{1,j}$	The first term in the sum above ($[A^k]_{i,1} a_{1,j}$) equals
johnston-linear-matrix-atlas- F04398	455	■ 1.000	$\left[A^{\{k\}}\right]_{\{i,1\}}$	$[A^k]_{i,1}$	$[A^k]_{i,1}$ exactly if there is a path of length k edge be-
johnston-linear-matrix-atlas- F04399	455	■ 1.000	$k+1$	$k+1$	counts the number of paths of length $k+1$ between
johnston-linear-matrix-atlas- F04400	455	■ 1.000	$\left[A^{\{k\}}\right]_{\{i,2\}} a_{2,j}$	$[A^k]_{i,2} a_{2,j}$	Similarly, $[A^k]_{i,2} a_{2,j}$ counts the number of paths of
johnston-linear-matrix-atlas- F04401	455	■ 1.000	$\left[A^{\{k+1\}}\right]_{i,j}$	$[A^{k+1}]_{i,j}$	adding these terms up, we see that $[A^{k+1}]_{i,j}$ counts
johnston-linear-matrix-atlas- F04402	455	■ 1.000	$\left[\begin{array}{ll} 1 & 2 \\ 0 & 0 \\ 0 & 0 \end{array}\right]$	$\begin{bmatrix} 1 & 2 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}$	(c) $\begin{bmatrix} 1 & 2 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}$
johnston-linear-matrix-atlas- F04403	455	■ 1.000	$\left[\begin{array}{lll} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{array}\right]$	$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$	(e) $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$
johnston-linear-matrix-atlas- F04404	455	■ 1.000	$\left[\begin{array}{lll} 1 & 0 & 2 \\ 0 & 1 & 3 \\ 0 & 0 & 0 \end{array}\right]$	$\begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 3 \\ 0 & 0 & 0 \end{bmatrix}$	(g) $\begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 3 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$
johnston-linear-matrix-atlas- F04405	455	■ 1.000	$\left[\begin{array}{cccccc} 1 & 0 & 0 & 0 & -4/3 & 2/5 \\ 0 & 1 & 0 & 0 & -4/3 & -4/5 \\ 0 & 0 & 1 & 0 & 7/3 & 9/20 \\ 0 & 0 & 0 & 1 & 2 & 1/10 \end{array}\right]$	$\begin{bmatrix} 1 & 0 & 0 & 0 & -4/3 & 2/5 \\ 0 & 1 & 0 & 0 & -4/3 & -4/5 \\ 0 & 0 & 1 & 0 & 7/3 & 9/20 \\ 0 & 0 & 0 & 1 & 2 & 1/10 \end{bmatrix}$	$\begin{bmatrix} 1 & 0 & 0 & 0 & -4/3 & 2/5 \\ 0 & 1 & 0 & 0 & -4/3 & -4/5 \\ 0 & 0 & 1 & 0 & 7/3 & 9/20 \\ 0 & 0 & 0 & 1 & 2 & 1/10 \end{bmatrix}$
johnston-linear-matrix-atlas- F04406	455	■ 1.000	$y: x$	$y: x$	ables x and y : x is free and $y = 2$).
johnston-linear-matrix-atlas- F04407	455	■ 1.000	$(x, y)=(1,1)$	$(x,y) = (1,1)$	2.1.5 (a) $(x,y) = (1,1)$.
johnston-linear-matrix-atlas- F04408	456	■ 1.000	v, w	v, w	(g) Infinitely many solutions: The variables v, w ,
johnston-linear-matrix-atlas- F04409	456	■ 0.998	v, w, x	v, w, x	2.1.6 (a) Infinitely many solutions: v, w, x , and y
johnston-linear-matrix-atlas- F04410	456	■ 1.000	$(v, w, x, y, z)=$	$(v, w, x, y, z) =$	solutions are of the form $(v, w, x, y, z) =$
johnston-linear-matrix-atlas- F04411	456	■ 0.999	$(-30,3,60,7,0)+z(-51,3,106,5,6)/6$	$(-30,3,60,7,0) + z(-51,3,106,5,6)/6$	$(-30,3,60,7,0) + z(-51,3,106,5,6)/6$.
johnston-linear-matrix-atlas- F04412	456	■ 0.978	$(x, y$	$(x, y$	2.1.7 (a) True. Each of the variables $(x, y$, and $z)$ are just
johnston-linear-matrix-atlas- F04413	456	■ 0.976	$(\sin(1), \sqrt[3]{5})$	$(\sin(1), \sqrt[3]{5})$	fact that the scalars $(\sin(1), \sqrt[3]{5})$ are ugly does
johnston-linear-matrix-atlas- F04414	456	■ 0.848	$(0,1,1)$	$(0,1,1)$	through the points $(0,0,1)$ and $(0,1,1)$ (i.e.,

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johnston-linear-matrix-atlas- F04415	456	■ 1.000	$x=1, y=0$	$x = 1, y = 0$	is $x = 1, y = 0$.
johnston-linear-matrix-atlas- F04416	456	■ 1.000	$t \neq 1 / 2$	$t \neq 1/2$	if and only if $t \neq 1/2$. This linear system thus has
johnston-linear-matrix-atlas- F04417	456	■ 1.000	$t=1 / 2$	$t = 1/2$	$t = 1/2$, and (iii) never has infinitely many solutions.
johnston-linear-matrix-atlas- F04418	456	■ 1.000	$(w, x, y, z)=(3,-24,30,0)$	$(w, x, y, z) = (3, -24, 30, 0)$	and $h = 1.05$ are $(w, x, y, z) = (3, -24, 30, 0)$,
johnston-linear-matrix-atlas- F04419	456	■ 0.999	$(w, x, y, z)=(-4,60,-180,140)$	$(w, x, y, z) = (-4, 60, -180, 140)$	$(w, x, y, z) = (-4, 60, -180, 140)$, and $(w, x, y, z) =$
johnston-linear-matrix-atlas- F04420	456	■ 1.000	$v_{\{2\}}=-2$	$v_2 = -2$	$v_3 = 1$ then gives us $v_2 = -2$ and $v_1 = 1$, so
johnston-linear-matrix-atlas- F04421	456	■ 1.000	$v_{\{1\}}=1$	$v_1 = 1$	$v_3 = 1$ then gives us $v_2 = -2$ and $v_1 = 1$, so
johnston-linear-matrix-atlas- F04422	456	■ 1.000	$\mathbf{v}=(1,-2,1)$	$\mathbf{v} = (1, -2, 1)$	the vector $\mathbf{v} = (1, -2, 1)$ is orthogonal to both
johnston-linear-matrix-atlas- F04423	456	■ 1.000	$v_{\{4\}}=9$	$v_4 = 9$	$v_4 = 9$ then gives us $v_3 = -5, v_2 = -8$, and
johnston-linear-matrix-atlas- F04424	456	■ 1.000	$v_{\{3\}}=-5, v_{\{2\}}=-8$	$v_3 = -5, v_2 = -8$	$v_4 = 9$ then gives us $v_3 = -5, v_2 = -8$, and
johnston-linear-matrix-atlas- F04425	456	■ 1.000	$v_{\{1\}}=7$	$v_1 = 7$	$v_1 = 7$, so the vector $\mathbf{v} = (7, -8, -5, 9)$ is or-
johnston-linear-matrix-atlas- F04426	456	■ 1.000	$\mathbf{v}=(7,-8,-5,9)$	$\mathbf{v} = (7, -8, -5, 9)$	$v_1 = 7$, so the vector $\mathbf{v} = (7, -8, -5, 9)$ is or-
johnston-linear-matrix-atlas- F04427	456	■ 1.000	$(x, y)=(1,2)$	$(x, y) = (1, 2)$	so the unique solution is $(x, y) = (1, 2)$.
johnston-linear-matrix-atlas- F04428	456	■ 1.000	$\mathbf{b}=(1 / 2) \mathbf{v}_{\{1\}}$	$\mathbf{b} = (1/2)\mathbf{v}_1$	2.1.17 (a) Yes, $\mathbf{b} = (1/2)\mathbf{v}_1$.
johnston-linear-matrix-atlas- F04429	456	■ 1.000	$\mathbf{b}=\mathbf{v}_{\{1\}}+5 \mathbf{v}_{\{2\}}-6 \mathbf{v}_{\{3\}}$	$\mathbf{b} = \mathbf{v}_1 + 5\mathbf{v}_2 - 6\mathbf{v}_3$	(c) Yes, $\mathbf{b} = \mathbf{v}_1 + 5\mathbf{v}_2 - 6\mathbf{v}_3$.
johnston-linear-matrix-atlas- F04430	456	■ 1.000	$\mathbf{b}=\mathbf{v}_{\{1\}}+3 \mathbf{v}_{\{2\}}$	$\mathbf{b} = \mathbf{v}_1 + 3\mathbf{v}_2$	(e) Yes, $\mathbf{b} = \mathbf{v}_1 + 3\mathbf{v}_2$.
johnston-linear-matrix-atlas- F04431	457	■ 1.000	$\mathbf{b}=-\mathbf{v}_{\{1\}}+\mathbf{v}_{\{2\}}-2 \mathbf{v}_{\{3\}}$	$\mathbf{b} = -\mathbf{v}_1 + \mathbf{v}_2 - 2\mathbf{v}_3$	(g) Yes, $\mathbf{b} = -\mathbf{v}_1 + \mathbf{v}_2 - 2\mathbf{v}_3$.
johnston-linear-matrix-atlas- F04432	457	■ 1.000	$R_{\{1\}}$	R_1	By replacing R_1 with $w_1 R_1$ and then replacing R_2
johnston-linear-matrix-atlas- F04433	457	■ 1.000	$w_{\{1\}} R_{\{1\}}$	$w_1 R_1$	By replacing R_1 with $w_1 R_1$ and then replacing R_2
johnston-linear-matrix-atlas- F04434	457	■ 1.000	$R_{\{2\}}$	R_2	By replacing R_1 with $w_1 R_1$ and then replacing R_2
johnston-linear-matrix-atlas- F04435	457	■ 1.000	$R_{\{1\}}-v_{\{1\}} R_{\{2\}}$	$R_1 - v_1 R_2$	with $R_1 - v_1 R_2$, the system becomes
johnston-linear-matrix-atlas- F04436	457	■ 1.000	$w_{\{1\}} \neq 0$	$w_1 \neq 0$	can only replace R_1 by $w_1 R_1$ if $w_1 \neq 0$. However,
johnston-linear-matrix-atlas- F04437	457	■ 1.000	$\mathbf{w} \neq \mathbf{0}$	$\mathbf{w} \neq \mathbf{0}$	since $\mathbf{w} \neq \mathbf{0}$, we can always find <i>some</i> non-zero entry
johnston-linear-matrix-atlas- F04438	457	■ 0.999	$x_{\{1\}}, x_{\{2\}}, x_{\{3\}}$	x_1, x_2, x_3	where x_1, x_2, x_3 are the variables we are trying to
johnston-linear-matrix-atlas- F04439	457	■ 1.000	$x_{\{3\}}=c\left(v_{\{1\}} w_{\{2\}}-\right) .$	$x_3 = c\left(v_1 w_2 -\right)$	If $c \in \mathbb{R}$ is any scalar, and we choose $x_3 = c\left(v_1 w_2 -\right)$
johnston-linear-matrix-atlas- F04440	457	■ 0.865	$v_{\{2\}} w_{\{1\}}$	$v_2 w_1$	$v_2 w_1$) then we find that $x_1 = c\left(v_2 w_3 - v_3 w_2\right)$ and
johnston-linear-matrix-atlas- F04441	457	■ 0.865	$x_{\{1\}}=c\left(v_{\{2\}} w_{\{3\}}-v_{\{3\}} w_{\{2\}}\right)$	$x_1 = c\left(v_2 w_3 - v_3 w_2\right)$	$v_2 w_1$) then we find that $x_1 = c\left(v_2 w_3 - v_3 w_2\right)$ and
johnston-linear-matrix-atlas- F04442	457	■ 1.000	$x_{\{2\}}=c\left(v_{\{3\}} w_{\{1\}}-v_{\{1\}} w_{\{3\}}\right)$	$x_2 = c\left(v_3 w_1 - v_1 w_3\right)$	$x_2 = c\left(v_3 w_1 - v_1 w_3\right)$, so
johnston-linear-matrix-atlas- F04443	457	■ 0.950	$x=45$	$x = 45$	$x = 45$ kilograms of the 13% salt water and $y = 75$

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johnston-linear-matrix-atlas- F04444	457	■ 0.950	y=75	$y = 75$	$x = 45$ kilograms of the 13% salt water and $y = 75$
johnston-linear-matrix-atlas- F04445	457	■ 1.000	w=z	$w = z$	Solving this linear system shows that z is free, $w = z$,
johnston-linear-matrix-atlas- F04446	457	■ 1.000	x=3 z / 2	$x = 3z/2$	$x = 3z/2$, and $y = z$. Choosing $z = 2$ so that all of
johnston-linear-matrix-atlas- F04447	457	■ 1.000	y=z	$y = z$	$x = 3z/2$, and $y = z$. Choosing $z = 2$ so that all of
johnston-linear-matrix-atlas- F04448	457	■ 1.000	v_{1}=18-2 v_{2}-2 v_{3}	$v_1 = 18 - 2v_2 - 2v_3$	leading variable with $v_1 = 18 - 2v_2 - 2v_3$. The vec-
johnston-linear-matrix-atlas- F04449	457	■ 1.000	\mathbf{v}=\left(18-2 v_{2}-2 v_{3}, v_{2}, v_{3}\right)	$\mathbf{v} = (18 - 2v_2 - 2v_3, v_2, v_3)$	form $\mathbf{v} = (18 - 2v_2 - 2v_3, v_2, v_3)$, where $v_2, v_3 \in \mathbb{R}$
johnston-linear-matrix-atlas- F04450	457	■ 1.000	v_{2}, v_{3} \in \mathbb{R}	$v_2, v_3 \in \mathbb{R}$	form $\mathbf{v} = (18 - 2v_2 - 2v_3, v_2, v_3)$, where $v_2, v_3 \in \mathbb{R}$
johnston-linear-matrix-atlas- F04451	457	■ 0.999	0=-1 / 3	$0 = -1/3$	to the equation $0 = -1/3$, we see that there
johnston-linear-matrix-atlas- F04452	457	■ 1.000	T(1,0)	$T(1, 0)$	2.1.26 (a) If we can find $T(1, 0)$ and $T(0, 1)$ then we can
johnston-linear-matrix-atlas- F04453	457	■ 1.000	T(0,1)	$T(0, 1)$	2.1.26 (a) If we can find $T(1, 0)$ and $T(0, 1)$ then we can
johnston-linear-matrix-atlas- F04454	457	■ 1.000	c_{1}=c_{2}=c_{3}=1 / 2	$c_1 = c_2 = c_3 = 1/2$	easily solved to get $c_1 = c_2 = c_3 = 1/2$ and
johnston-linear-matrix-atlas- F04455	457	■ 1.000	c_{4}=-1 / 2	$c_4 = -1/2$	$c_4 = -1/2$. It follows that
johnston-linear-matrix-atlas- F04456	457	■ 1.000	T(0,1)=	$T(0, 1) =$	A similar calculation shows that $T(0, 1) =$
johnston-linear-matrix-atlas- F04457	458	■ 1.000	u=1 / x, v=1 / y	$u = 1/x, v = 1/y$	(c) Let $u = 1/x$, $v = 1/y$, and $w = 1/z$. Then the
johnston-linear-matrix-atlas- F04458	458	■ 1.000	w=1 / z	$w = 1/z$	(c) Let $u = 1/x$, $v = 1/y$, and $w = 1/z$. Then the
johnston-linear-matrix-atlas- F04459	458	■ 1.000	(u, v, w)=(2,-1,3)	$(u, v, w) = (2, -1, 3)$	us the unique solution $(u, v, w) = (2, -1, 3)$.
johnston-linear-matrix-atlas- F04460	458	■ 1.000	(x, y, z)=(1 / 2,-1,1 / 3)	$(x, y, z) = (1/2, -1, 1/3)$	$w = 1/z$ then gives us $(x, y, z) = (1/2, -1, 1/3)$
johnston-linear-matrix-atlas- F04461	458	■ 0.995	\left[\begin{array}{ll} a & b \\ c & d \end{array}\right]	$\begin{bmatrix} a & b \\ c & d \end{bmatrix}$	2.1.28 (a) We need to find all matrices $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$ such that
johnston-linear-matrix-atlas- F04462	458	■ 1.000	c=b, d=a	$c = b, d = a$	$c = b, d = a$. The matrices that commute with
johnston-linear-matrix-atlas- F04463	458	■ 0.963	\left[\begin{array}{cc} a & b \\ b & a \end{array}\right]	$\begin{bmatrix} a & b \\ b & a \end{bmatrix}$	A are thus those of the form $\begin{bmatrix} a & b \\ b & a \end{bmatrix}$.
johnston-linear-matrix-atlas- F04464	458	■ 1.000	c=b, d=a-b	$c = b, d = a - b$	$c = b, d = a - b$. The matrices that commute
johnston-linear-matrix-atlas- F04465	458	■ 0.998	\left[\begin{array}{cc} a & b \\ b & a-b \end{array}\right]	$\begin{bmatrix} a & b \\ b & a - b \end{bmatrix}$	with C are thus those of the form $\begin{bmatrix} a & b \\ b & a - b \end{bmatrix}$.
johnston-linear-matrix-atlas- F04466	458	■ 1.000	B C=C B	$BC = CB$	in particular $AB = BA$ and $BC = CB$, where
johnston-linear-matrix-atlas- F04467	458	■ 1.000	B=\left[\begin{array}{cc} a & b \\ b & a-b \end{array}\right]	$B = \begin{bmatrix} a & b \\ b & a - b \end{bmatrix}$	and (b) of this question, so $B = \begin{bmatrix} a & b \\ b & a - b \end{bmatrix}$
johnston-linear-matrix-atlas- F04468	458	■ 1.000	a-b=a	$a - b = a$	(by part (b)), and $a - b = a$ (by part (a)), so
johnston-linear-matrix-atlas- F04469	458	■ 0.689	B=\left[\begin{array}{ll} a & 0 \\ 0 & a \end{array}\right]=a I	$B = \begin{bmatrix} a & 0 \\ 0 & a \end{bmatrix} = aI$	$b = 0$, so $B = \begin{bmatrix} a & 0 \\ 0 & a \end{bmatrix} = aI$. On the other hand,

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johnston-linear-matrix-attached F04470	458	■ 0.941	$a \ I$	aI	it is clear that the matrix aI commutes with
johnston-linear-matrix-attached F04471	458	■ 0.901	$(a \ I) \ A=a \ A=A(a \ I)$	$(aI)A = aA = A(aI)$	everything in $\mathcal{M}_2$ since $(aI)A = aA = A(aI)$.
johnston-linear-matrix-attached F04472	458	■ 1.000	$(b, \ m)=(-1,3)$	$(b,m) = (-1,3)$	has unique solution $(b,m) = (-1,3)$, so the
johnston-linear-matrix-attached F04473	458	■ 1.000	$y=3 \ x-1$	$y = 3x - 1$	unique line that works is $y = 3x - 1$.
johnston-linear-matrix-attached F04474	458	■ 1.000	$(a, \ b, \ c)=(2,-3,4)$	$(a,b,c) = (2,-3,4)$	has unique solution $(a,b,c) = (2,-3,4)$, so the
johnston-linear-matrix-attached F04475	458	■ 1.000	$y=2 \ x^{\{2\}}-3 \ x+4$	$y = 2x^2 - 3x + 4$	unique parabola that works is $y = 2x^2 - 3x + 4$.
johnston-linear-matrix-attached F04476	458	■ 0.999	$d=4$	$d = 4$	choose $d = 4$, we get exactly the parabola that
johnston-linear-matrix-attached F04477	459	■ 0.999	$\left[\begin{array}{lll}0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0\end{array}\right]$	$\begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$	(a) $\begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$
johnston-linear-matrix-attached F04478	459	■ 1.000	$\left[\begin{array}{lll}1 & 3 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1\end{array}\right]$	$\begin{bmatrix} 1 & 3 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$	(c) $\begin{bmatrix} 1 & 3 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$
johnston-linear-matrix-attached F04479	459	■ 1.000	$\left[\begin{array}{lll}3 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1\end{array}\right]$	$\begin{bmatrix} 3 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$	(e) $\begin{bmatrix} 3 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$
johnston-linear-matrix-attached F04480	459	■ 1.000	$\left[\begin{array}{cc}-5 / 3 & 2 / 3 \\ 4 / 3 & -1 / 3\end{array}\right]$	$\begin{bmatrix} -5/3 & 2/3 \\ 4/3 & -1/3 \end{bmatrix}$	(c) $\begin{bmatrix} -5/3 & 2/3 \\ 4/3 & -1/3 \end{bmatrix}$
johnston-linear-matrix-attached F04481	459	■ 1.000	$\left[\begin{array}{ccc}0 & -1 / 3 & 2 / 3 \\ 0 & 1 / 3 & 1 / 3 \\ 1 & 1 / 3 & -5 / 3\end{array}\right]$	$\begin{bmatrix} 0 & -1/3 & 2/3 \\ 0 & 1/3 & 1/3 \\ 1 & 1/3 & -5/3 \end{bmatrix}$	(e) $\begin{bmatrix} 0 & -1/3 & 2/3 \\ 0 & 1/3 & 1/3 \\ 1 & 1/3 & -5/3 \end{bmatrix}$ (not unique)
johnston-linear-matrix-attached F04482	459	■ 1.000	$\left[\begin{array}{ccc}0 & 0 & 1 \\ 0 & -1 / 2 & 3 / 2 \\ 1 & 1 / 2 & -3 / 2\end{array}\right]$	$\begin{bmatrix} 0 & 0 & 1 \\ 0 & -1/2 & 3/2 \\ 1 & 1/2 & -3/2 \end{bmatrix}$	(g) $\begin{bmatrix} 0 & 0 & 1 \\ 0 & -1/2 & 3/2 \\ 1 & 1/2 & -3/2 \end{bmatrix}$ (not unique)
johnston-linear-matrix-attached F04483	459	■ 1.000	$B=-I$	$B = -I$	(g) False. For example, if $A = I$ and $B = -I$ then
johnston-linear-matrix-attached F04484	459	■ 1.000	$A+B=0$	$A + B = O$	$A + B = O$, which is not invertible.
johnston-linear-matrix-attached F04485	459	■ 1.000	$A^{\{-1\} \ 7}$	$A^{-1}7$	tiply the equation $A^7 = O$ on the left by $A^{-1} \ 7$
johnston-linear-matrix-attached F04486	459	■ 1.000	$I=0$	$I = O$	which simplifies to $I = O$, which is false. It
johnston-linear-matrix-attached F04487	459	■ 1.000	$X=B \ A^{\{-1\}}$	$X = BA^{-1}$	A^{-1} gives $X = BA^{-1}$, which in general is not
johnston-linear-matrix-attached F04488	459	■ 1.000	$A^{\{-1\} \ B}$	$A^{-1}B$	the same as $A^{-1}B$ since matrix multiplication is
johnston-linear-matrix-attached F04489	459	■ 1.000	$A=(a-b) \ I_{\{n\}}+b \ J_{\{n\}}$	$A = (a - b)I_n + bJ_n$	ery entry equal to 1, then $A = (a - b)I_n + bJ_n$.
johnston-linear-matrix-attached F04490	459	■ 1.000	$B=c \ I_{\{n\}}+d \ J_{\{n\}}$	$B = cI_n + dJ_n$	trix of the form $B = cI_n + dJ_n$ for some $c, d \in \mathbb{R}$
johnston-linear-matrix-attached F04491	460	■ 1.000	$c=1 \ /(a-b)$	$c = 1/(a - b)$	choose $c = 1/(a - b)$ and then solve $bc +$
johnston-linear-matrix-attached F04492	460	■ 1.000	$b \ c+$	$bc+$	choose $c = 1/(a - b)$ and then solve $bc +$
johnston-linear-matrix-attached F04493	460	■ 0.995	$d(a-b)+n \ b \ d=0$	$d(a - b) + nbd = 0$	$d(a - b) + nbd = 0$ to get $d = -b(nd + 1/(a -$

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johnston-linear-matrix-atg6b1 F04494	460	■ 0.995	$d = -b(n-d+1)/(a-$	$d = -b(nd+1)/(a-$	$d(a-b)+nbd=0$ to get $d = -b(nd+1)/(a-$
johnston-linear-matrix-atg6b1 F04495	460	■ 0.992	$b))/ (a-b)$	$b))/ (a-b)$	$b))/ (a-b)$. It follows that A is invertible, and
johnston-linear-matrix-atg6b1 F04496	460	■ 0.985	$A^2 = \left[\begin{array}{cc} 7 & 4 \\ 12 & 7 \end{array} \right]$	$A^2 = \left[\begin{array}{cc} 7 & 4 \\ 12 & 7 \end{array} \right]$	2.2.9 (a) $A^2 = \left[\begin{array}{cc} 7 & 4 \\ 12 & 7 \end{array} \right]$
johnston-linear-matrix-atg6b1 F04497	460	■ 1.000	$A^{-2} = \left[\begin{array}{cc} 7 & -4 \\ -12 & 7 \end{array} \right]$	$A^{-2} = \left[\begin{array}{cc} 7 & -4 \\ -12 & 7 \end{array} \right]$	(c) $A^{-2} = \left[\begin{array}{cc} 7 & -4 \\ -12 & 7 \end{array} \right]$
johnston-linear-matrix-atg6b1 F04498	460	■ 1.000	$m \times n, A^T A$	$m \times n, A^T A$	2.2.11 (a) Since A is $m \times n$, $A^T A$ is $n \times n$, so $(A^T A)^{-1}$ is
johnston-linear-matrix-atg6b1 F04499	460	■ 1.000	$\left(A^T A \right)^{-1}$	$(A^T A)^{-1}$	2.2.11 (a) Since A is $m \times n$, $A^T A$ is $n \times n$, so $(A^T A)^{-1}$ is
johnston-linear-matrix-atg6b1 F04500	460	■ 1.000	$P^T = \left(A \left(A^T A \right)^{-1} A^T \right)^T = (A^T)^T \left(\left(A^T A \right)^{-1} \right)^T A^T =$	$P^T = \left(A \left(A^T A \right)^{-1} A^T \right)^T = (A^T)^T \left(\left(A^T A \right)^{-1} \right)^T A^T =$	(b) $P^T = (A(A^T A)^{-1} A^T)^T = (A^T)^T ((A^T A)^{-1})^T A^T =$
johnston-linear-matrix-atg6b1 F04501	460	■ 0.968	$A \left(\left(A^T A \right)^T \right)^{-1} A^T = A \left(A^T A \right)^{-1} A^T = P$	$A \left(\left(A^T A \right)^T \right)^{-1} A^T = A \left(A^T A \right)^{-1} A^T = P$	$A \left(\left(A^T A \right)^T \right)^{-1} A^T = A \left(A^T A \right)^{-1} A^T = P$.
johnston-linear-matrix-atg6b1 F04502	460	■ 1.000	$E_k \cdots E_2 E_1 P = I$	$E_k \cdots E_2 E_1 P = I$	then $E_k \cdots E_2 E_1 P = I$, so $E_k \cdots E_2 E_1 = P^{-1}$, so the
johnston-linear-matrix-atg6b1 F04503	460	■ 1.000	$E_k \cdots E_2 E_1 = P^{-1}$	$E_k \cdots E_2 E_1 = P^{-1}$	then $E_k \cdots E_2 E_1 P = I$, so $E_k \cdots E_2 E_1 = P^{-1}$, so the
johnston-linear-matrix-atg6b1 F04504	460	■ 1.000	$P^{-1} Q$	$P^{-1} Q$	matrix on the right is $P^{-1} Q$, as claimed.
johnston-linear-matrix-atg6b1 F04505	460	■ 1.000	$X =$	$X =$	2.2.13 (a) Multiplying on the left by A^{-1} gives $X =$
johnston-linear-matrix-atg6b1 F04506	460	■ 1.000	$X = A^{-1} I B^{-1} = A^{-1} B^{-1}$	$X = A^{-1} I B^{-1} = A^{-1} B^{-1}$	by B^{-1} gives $X = A^{-1} I B^{-1} = A^{-1} B^{-1}$.
johnston-linear-matrix-atg6b1 F04507	460	■ 1.000	$a_{n,n} x_n = 0$	$a_{n,n} x_n = 0$	substitution. The final equation says $a_{n,n} x_n = 0$,
johnston-linear-matrix-atg6b1 F04508	460	■ 1.000	$x_n = 0$	$x_n = 0$	which implies $x_n = 0$ as long as $a_{n,n} \neq 0$. Plug-
johnston-linear-matrix-atg6b1 F04509	460	■ 1.000	$a_{n,n} \neq 0$	$a_{n,n} \neq 0$	which implies $x_n = 0$ as long as $a_{n,n} \neq 0$. Plug-
johnston-linear-matrix-atg6b1 F04510	460	■ 0.997	$a_{n-1,n-1} x_{n-1} = 0$	$a_{n-1,n-1} x_{n-1} = 0$	$a_{n-1,n-1} x_{n-1} = 0$, which implies $x_{n-1} = 0$ as
johnston-linear-matrix-atg6b1 F04511	460	■ 0.997	$x_{n-1} = 0$	$x_{n-1} = 0$	$a_{n-1,n-1} x_{n-1} = 0$, which implies $x_{n-1} = 0$ as
johnston-linear-matrix-atg6b1 F04512	460	■ 1.000	$a_{n-1,n-1} \neq 0$	$a_{n-1,n-1} \neq 0$	long as $a_{n-1,n-1} \neq 0$. Continuing in this way
johnston-linear-matrix-atg6b1 F04513	460	■ 1.000	$a_{j,j} = 0$	$a_{j,j} = 0$	entries equals 0 (let's say $a_{j,j} = 0$) then the
johnston-linear-matrix-atg6b1 F04514	460	■ 1.000	$a_{j,j} x_j = 0$	$a_{j,j} x_j = 0$	equation $a_{j,j} x_j = 0$ gives no restriction on x_j ,
johnston-linear-matrix-atg6b1 F04515	460	■ 1.000	x_j	x_j	equation $a_{j,j} x_j = 0$ gives no restriction on x_j ,
johnston-linear-matrix-atg6b1 F04516	460	■ 1.000	$A \mathbf{x}_j = \mathbf{e}_j$	$A \mathbf{x}_j = \mathbf{e}_j$	so $A \mathbf{x}_j = \mathbf{e}_j$ for all $1 \leq j \leq n$. Since the last
johnston-linear-matrix-atg6b1 F04517	460	■ 0.994	$n-j$	$n-j$	$n-j$ entries of $\mathbf{e}_j$ all equal 0, the last $n-j$
johnston-linear-matrix-atg6b1 F04518	460	■ 1.000	$\mathbf{x}_j$	$\mathbf{x}_j$	entries of $\mathbf{x}_j$ must equal 0 as well, so A^{-1} is
johnston-linear-matrix-atg6b1 F04519	460	■ 0.999	$b_1, b_2, \dots, b_n$	$b_1, b_2, \dots, b_n$	entries of A^{-1} by $b_1, b_2, \dots, b_n$, and use aster-
johnston-linear-matrix-atg6b1 F04520	460	■ 1.000	$b_1 = 1/a_1, b_2 = 1/a_2$	$b_1 = 1/a_1, b_2 = 1/a_2$	$b_1 = 1/a_1, b_2 = 1/a_2$, and so on.

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johnston-linear-matrix-algebra-F04521	460	■ 0.999	$\left(A^{-1}\right)^{-1}$	$(A^{-1})^{-1}$	2.2.17 (a) An inverse of A^{-1} is a matrix $(A^{-1})^{-1}$ such
johnston-linear-matrix-algebra-F04522	460	■ 1.000	$A^{-1}\left(A^{-1}\right)^{-1}=\left(A^{-1}\right)^{-1}A^{-1}=I$ $A^{-1}=I$	$A^{-1}\left(A^{-1}\right)^{-1}=\left(A^{-1}\right)^{-1}A^{-1}=I$	that $A^{-1}\left(A^{-1}\right)^{-1}=\left(A^{-1}\right)^{-1}A^{-1}=I$. Replac-
johnston-linear-matrix-algebra-F04523	460	■ 1.000	$\left(A^{-1}\right)^T A^T=A^T\left(A^{-1}\right)^T=I$ $A^T=A^T\left(A^{-1}\right)^T=I$	$(A^{-1})^T A^T = A^T (A^{-1})^T = I$	show that $(A^{-1})^T A^T = A^T (A^{-1})^T = I$. Well,
johnston-linear-matrix-algebra-F04524	460	■ 0.997	$I=I^T=\left(A A^{-1}\right)^T=\left(A^{-1}\right)^T A^T$ $A^T=A^T$	$I=I^T=\left(A A^{-1}\right)^T=\left(A^{-1}\right)^T A^T$	$I=I^T=\left(A A^{-1}\right)^T=\left(A^{-1}\right)^T A^T$, and similarly
johnston-linear-matrix-algebra-F04525	460	■ 1.000	$I=I^T=\left(A^{-1} A\right)^T=A^T\left(A^{-1}\right)^T$ $A^T=A^T$	$I=I^T=\left(A^{-1} A\right)^T=A^T\left(A^{-1}\right)^T$	$I=I^T=\left(A^{-1} A\right)^T=A^T\left(A^{-1}\right)^T$, as desired.
johnston-linear-matrix-algebra-F04526	460	■ 1.000	$A^{-1} k$	$A^{-1} k$	(c) Notice that if we multiply A^k by A^{-1} k times
johnston-linear-matrix-algebra-F04527	460	■ 1.000	$A A^{-1}$	$A A^{-1}$	The central $A A^{-1}$ cancels to give an identity
johnston-linear-matrix-algebra-F04528	460	■ 1.000	$A^k\left(A^{-1}\right)^k=I$	$A^k\left(A^{-1}\right)^k=I$	matrix left. It follows that $A^k\left(A^{-1}\right)^k=I$ (and
johnston-linear-matrix-algebra-F04529	460	■ 1.000	$\left(A^{-1}\right)^k A^k=I$	$(A^{-1})^k A^k=I$	the equation $(A^{-1})^k A^k=I$ is proved similarly),
johnston-linear-matrix-algebra-F04530	461	■ 1.000	$k, \ell \geq 0$	$k, \ell \geq 0$	2.2.18 (a) We already proved this claim when $k, \ell \geq 0$ in
johnston-linear-matrix-algebra-F04531	461	■ 1.000	$k, \ell < 0$	$k, \ell < 0$	Exercise 1.3.22, and it follows when $k, \ell < 0$
johnston-linear-matrix-algebra-F04532	461	■ 1.000	$\ell < 0$	$\ell < 0$	the case when $k \geq 0$ and $\ell < 0$. In this case, we
johnston-linear-matrix-algebra-F04533	461	■ 1.000	$A^k A^\ell=(A A \cdots A)\left(A^{-1} A^{-1} \cdots A^{-1}\right)$ $A^k A^\ell=(A A \cdots A)\left(A^{-1} A^{-1} \cdots A^{-1}\right)$	$A^k A^\ell=(A A \cdots A)\left(A^{-1} A^{-1} \cdots A^{-1}\right)$	notice that $A^k A^\ell=(A A \cdots A)\left(A^{-1} A^{-1} \cdots A^{-1}\right)$,
johnston-linear-matrix-algebra-F04534	461	■ 1.000	$ \ell =-\ell$	$ \ell =-\ell$	the second has $ \ell =-\ell$ copies of A^{-1} . The
johnston-linear-matrix-algebra-F04535	461	■ 1.000	$k- \ell =k+\ell$	$k- \ell =k+\ell$	ing just $k- \ell =k+\ell$ copies of A (if $k \geq \ell $)
johnston-linear-matrix-algebra-F04536	461	■ 1.000	$k \geq \ell $	$k \geq \ell $	ing just $k- \ell =k+\ell$ copies of A (if $k \geq \ell $)
johnston-linear-matrix-algebra-F04537	461	■ 1.000	$ \ell -k=-(k+\ell)$	$ \ell -k=-(k+\ell)$	or $ \ell -k=-(k+\ell)$ copies of A^{-1} (if $ \ell > k$).
johnston-linear-matrix-algebra-F04538	461	■ 1.000	$ \ell >k$	$ \ell >k$	or $ \ell -k=-(k+\ell)$ copies of A^{-1} (if $ \ell > k$).
johnston-linear-matrix-algebra-F04539	461	■ 1.000	$k \geq 0, \ell < 0$	$k \geq 0, \ell < 0$	(b) Again, we just explicitly prove the $k \geq 0, \ell < 0$
johnston-linear-matrix-algebra-F04540	461	■ 1.000	$\left(A^k\right)^\ell=$	$(A^k)^\ell=$	similarly. In this case, notice that $(A^k)^\ell=$
johnston-linear-matrix-algebra-F04541	461	■ 0.999	$(A A \cdots A)^{-1}(A A \cdots A)^{-1} \cdots(A A \cdots A)^{-1}$ $(A A \cdots A)^{-1}$	$(A A \cdots A)^{-1}(A A \cdots A)^{-1} \cdots(A A \cdots A)^{-1}$	$(A A \cdots A)^{-1}(A A \cdots A)^{-1} \cdots(A A \cdots A)^{-1}$,
johnston-linear-matrix-algebra-F04542	461	■ 1.000	$(A A \cdots A)^{-1}=$	$(A A \cdots A)^{-1}=$	are $ \ell =-\ell$ brackets. Since $(A A \cdots A)^{-1}=$
johnston-linear-matrix-algebra-F04543	461	■ 1.000	$A^{-1} A^{-1} \cdots A^{-1}$	$A^{-1} A^{-1} \cdots A^{-1}$	$A^{-1} A^{-1} \cdots A^{-1}$, it follows that there are a total
johnston-linear-matrix-algebra-F04544	461	■ 1.000	$k \ell =-k \ell$	$k \ell =-k \ell$	of $k \ell =-k \ell$ copies of A^{-1} , so this product
johnston-linear-matrix-algebra-F04545	461	■ 1.000	$A^{- k \ell }=A^{k \ell}$	$A^{- k \ell }=A^{k \ell}$	equals $A^{- k \ell }=A^{k \ell}$.
johnston-linear-matrix-algebra-F04546	461	■ 1.000	$B^T A^T=(A B)^T=I^T=I$	$B^T A^T=(A B)^T=I^T=I$	gives $B^T A^T=(A B)^T=I^T=I$. It follows from the
johnston-linear-matrix-algebra-F04547	461	■ 1.000	$\left(A^T\right)^{-1}=B^T$	$(A^T)^{-1}=B^T$	A^T is invertible and $(A^T)^{-1}=B^T$. Using the fact
johnston-linear-matrix-algebra-F04548	461	■ 0.992	$A_{11} B_{1,1}, A_{22} B_{2,2}, \ldots, A_{nn} B_{n,n}$	$A_1 B_{1,1}, A_2 B_{2,2}, \ldots, A_n B_{n,n}$	then the products $A_1 B_{1,1}, A_2 B_{2,2}, \ldots, A_n B_{n,n}$

Identifier	Page	Conf.	LaTeX source	Rendered	Image
johnston-linear-matrix-algebra-F04549	461	1.000	$\left[\begin{array}{cc} A & I \\ 0 & I \end{array}\right] = \left[\begin{array}{cc} I & A-A \\ 0 & I \end{array}\right]$	$\begin{bmatrix} A & I \\ I & O \end{bmatrix} \begin{bmatrix} O & I \\ I & -A \end{bmatrix} = \begin{bmatrix} I & A-A \\ O & I \end{bmatrix} =$	(a) $\begin{bmatrix} A & I \\ I & O \end{bmatrix} \begin{bmatrix} O & I \\ I & -A \end{bmatrix} = \begin{bmatrix} I & A-A \\ O & I \end{bmatrix} =$
johnston-linear-matrix-algebra-F04550	462	0.977	$(\pi, 0)$	$(\pi, 0)$	(c) Not a subspace, since (for example) $(\pi, 0)$ is in
johnston-linear-matrix-algebra-F04551	462	1.000	$\frac{1}{2}(\pi, 0) = (\pi/2, 0)$	$\frac{1}{2}(\pi, 0) = (\pi/2, 0)$	the set but $\frac{1}{2}(\pi, 0) = (\pi/2, 0)$ is not in the set.
johnston-linear-matrix-algebra-F04552	462	1.000	$\operatorname{span}\{(-2, 1)\}$	$\operatorname{span}\{(-2, 1)\}$	is $\operatorname{span}\{(-2, 1)\}$, and spans are always sub-
johnston-linear-matrix-algebra-F04553	462	0.955	$(0, -1)$	$(0, -1)$	$(0, -1)$ are in the set, but $(1, 0) + (0, -1) =$
johnston-linear-matrix-algebra-F04554	462	0.955	$(1, 0) + (0, -1) =$	$(1, 0) + (0, -1) =$	$(0, -1)$ are in the set, but $(1, 0) + (0, -1) =$
johnston-linear-matrix-algebra-F04555	462	1.000	$x - y = 0$	$x - y = 0$	line has equation $y = x$, or $x - y = 0$.
johnston-linear-matrix-algebra-F04556	462	1.000	$-3x + y + z = 0$	$-3x + y + z = 0$	$-3x + y + z = 0$, which is the equation of a
johnston-linear-matrix-algebra-F04557	462	1.000	$y = -x$	$y = -x$	$(1, -1)$ (i.e., the line with equation $y = -x$).
johnston-linear-matrix-algebra-F04558	462	0.523	$z)$	$z)$	$z)$ and one leading variable x with $x + y + 2z =$
johnston-linear-matrix-algebra-F04559	462	0.523	$x + y + 2z =$	$x + y + 2z =$	$z)$ and one leading variable x with $x + y + 2z =$
johnston-linear-matrix-algebra-F04560	462	1.000	$(1, 0), \mathbf{v}_2 = (0, 1)$	$(1, 0), \mathbf{v}_2 = (0, 1)$	$(1, 0), \mathbf{v}_2 = (0, 1)$, and $\mathbf{v}_3 = (1, 1)$.
johnston-linear-matrix-algebra-F04561	462	1.000	$\mathbf{v}_3 = (1, 1)$	$\mathbf{v}_3 = (1, 1)$	$(1, 0), \mathbf{v}_2 = (0, 1)$, and $\mathbf{v}_3 = (1, 1)$.
johnston-linear-matrix-algebra-F04562	462	1.000	$\mathbf{v} = \mathbf{w} = \mathbf{0}$	$\mathbf{v} = \mathbf{w} = \mathbf{0}$	$\mathbf{v} = \mathbf{w} = \mathbf{0}$.
johnston-linear-matrix-algebra-F04563	463	1.000	$3 - 3k \neq 0$	$3 - 3k \neq 0$	If $3 - 3k \neq 0$ (i.e., $k \neq 1$), then this system of
johnston-linear-matrix-algebra-F04564	463	1.000	$k \neq 1$	$k \neq 1$	If $3 - 3k \neq 0$ (i.e., $k \neq 1$), then this system of
johnston-linear-matrix-algebra-F04565	463	1.000	$y + z - 3x = 0$	$y + z - 3x = 0$	$y + z - 3x = 0$. This is the equation of a plane,
johnston-linear-matrix-algebra-F04566	463	1.000	$\left(c_0, c_1\right) = (-3, 2)$	$(c_0, c_1) = (-3, 2)$	$(c_0, c_1) = (-3, 2)$, so the interpolating polyno-
johnston-linear-matrix-algebra-F04567	463	1.000	$p(x) = 2x - 3$	$p(x) = 2x - 3$	mial is $p(x) = 2x - 3$.
johnston-linear-matrix-algebra-F04568	463	1.000	$\left(c_0, c_1, c_2\right) = (2, -2, 1)$	$(c_0, c_1, c_2) = (2, -2, 1)$	The unique solution is $(c_0, c_1, c_2) = (2, -2, 1)$,
johnston-linear-matrix-algebra-F04569	463	1.000	$x^2 - 2x + 2$	$x^2 - 2x + 2$	$x^2 - 2x + 2$.
johnston-linear-matrix-algebra-F04570	463	1.000	$A\mathbf{0} = \mathbf{0} \neq \mathbf{b}$	$A\mathbf{0} = \mathbf{0} \neq \mathbf{b}$	2.3.13 Since $A\mathbf{0} = \mathbf{0} \neq \mathbf{b}$, the solution set does not contain
johnston-linear-matrix-algebra-F04571	463	1.000	$A_1 = 1$	$A_1 = 1$	2.3.14 It is straightforward to check that $A_1 = 1$ so $A_1^{-1} = 1$,
johnston-linear-matrix-algebra-F04572	463	1.000	$A_1^{-1} = 1$	$A_1^{-1} = 1$	2.3.14 It is straightforward to check that $A_1 = 1$ so $A_1^{-1} = 1$,
johnston-linear-matrix-algebra-F04573	463	0.996	$(1, 2, \ldots, n)$	$(1, 2, \ldots, n)$	tice that the first row of A_n is the vector $(1, 2, \ldots, n)$
johnston-linear-matrix-algebra-F04574	463	0.998	$(n+1, n+2, \ldots, 2n)$	$(n+1, n+2, \ldots, 2n)$	and its second row is the vector $(n+1, n+2, \ldots, 2n)$.
johnston-linear-matrix-algebra-F04575	463	1.000	$\mathbf{v}_1 + \mathbf{v}_2 \in \mathcal{S}$	$\mathbf{v}_1 + \mathbf{v}_2 \in \mathcal{S}$	$\mathbf{v}_1 + \mathbf{v}_2 \in \mathcal{S}$ whenever $\mathbf{v}_1, \mathbf{v}_2 \in \mathcal{S}$. Similarly, if we

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johnston-linear-matrix-attrib F04576	463	1.000	$\mathbf{v}_1, \mathbf{v}_2 \in \mathcal{S}$	$\mathbf{v}_1, \mathbf{v}_2 \in \mathcal{S}$	$\mathbf{v}_1 + \mathbf{v}_2 \in \mathcal{S}$ whenever $\mathbf{v}_1, \mathbf{v}_2 \in \mathcal{S}$. Similarly, if we
johnston-linear-matrix-attrib F04577	463	1.000	$c_1 \mathbf{v}_1 \in \mathcal{S}$	$c_1 \mathbf{v}_1 \in \mathcal{S}$	choose $k = 1$ then we see that $c_1 \mathbf{v}_1 \in \mathcal{S}$ whenever
johnston-linear-matrix-attrib F04578	463	1.000	$\mathbf{v}_1 \in \mathcal{S}$	$\mathbf{v}_1 \in \mathcal{S}$	$\mathbf{v}_1 \in \mathcal{S}$, so both of the defining properties of sub-
johnston-linear-matrix-attrib F04579	463	0.998	$c_2 \mathbf{v}_2 \in \mathcal{S}, \dots, c_k \mathbf{v}_k \in \mathcal{S}$	$c_2 \mathbf{v}_2 \in \mathcal{S}, \dots, c_k \mathbf{v}_k \in \mathcal{S}$	$c_2 \mathbf{v}_2 \in \mathcal{S}, \dots, c_k \mathbf{v}_k \in \mathcal{S}$ and thus $c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 \in \mathcal{S}$,
johnston-linear-matrix-attrib F04580	463	0.998	$c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 \in \mathcal{S}$	$c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 \in \mathcal{S}$	$c_2 \mathbf{v}_2 \in \mathcal{S}, \dots, c_k \mathbf{v}_k \in \mathcal{S}$ and thus $c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 \in \mathcal{S}$,
johnston-linear-matrix-attrib F04581	463	1.000	$(c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2) + c_3 \mathbf{v}_3 \in \mathcal{S}$	$(c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2) + c_3 \mathbf{v}_3 \in \mathcal{S}$	so $(c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2) + c_3 \mathbf{v}_3 \in \mathcal{S}$, and so on until we get
johnston-linear-matrix-attrib F04582	463	0.999	$c_1 \mathbf{v}_1 + \dots + c_k \mathbf{v}_k \in \mathcal{S}$	$c_1 \mathbf{v}_1 + \dots + c_k \mathbf{v}_k \in \mathcal{S}$	$c_1 \mathbf{v}_1 + \dots + c_k \mathbf{v}_k \in \mathcal{S}$.
johnston-linear-matrix-attrib F04583	463	1.000	$(A - I) \mathbf{v} = \mathbf{0}$	$(A - I) \mathbf{v} = \mathbf{0}$	$(A - I) \mathbf{v} = \mathbf{0}$. In other words, the fixed-point sub-
johnston-linear-matrix-attrib F04584	463	0.508	$\operatorname{null}(A - I)$	$\operatorname{null}(A - I)$	space of A is equal to $\operatorname{null}(A - I)$, and we already
johnston-linear-matrix-attrib F04585	463	1.000	$c_1 \mathbf{v} + c_2 \mathbf{w} = \mathbf{0}$	$c_1 \mathbf{v} + c_2 \mathbf{w} = \mathbf{0}$	such that $c_1 \mathbf{v} + c_2 \mathbf{w} = \mathbf{0}$. If $c_1 = 0$ then $c_2 \neq 0$
johnston-linear-matrix-attrib F04586	463	1.000	$c_1 = 0$	$c_1 = 0$	such that $c_1 \mathbf{v} + c_2 \mathbf{w} = \mathbf{0}$. If $c_1 = 0$ then $c_2 \neq 0$
johnston-linear-matrix-attrib F04587	463	1.000	$c_2 \neq 0$	$c_2 \neq 0$	such that $c_1 \mathbf{v} + c_2 \mathbf{w} = \mathbf{0}$. If $c_1 = 0$ then $c_2 \neq 0$
johnston-linear-matrix-attrib F04588	463	1.000	$c_1 \neq 0$	$c_1 \neq 0$	then $\mathbf{w} = \mathbf{0}$. On the other hand, if $c_1 \neq 0$ then
johnston-linear-matrix-attrib F04589	463	1.000	$\mathbf{v} = (-c_2 / c_1) \mathbf{w}$	$\mathbf{v} = (-c_2 / c_1) \mathbf{w}$	$\mathbf{v} = (-c_2 / c_1) \mathbf{w}$.
johnston-linear-matrix-attrib F04590	463	1.000	$0 \mathbf{v} + 1 \mathbf{w} = \mathbf{0}$	$0 \mathbf{v} + 1 \mathbf{w} = \mathbf{0}$	In the other direction, if $\mathbf{w} = \mathbf{0}$ then $0 \mathbf{v} + 1 \mathbf{w} = \mathbf{0}$,
johnston-linear-matrix-attrib F04591	463	1.000	$1 \mathbf{v} + (-c) \mathbf{w} = \mathbf{0}$	$1 \mathbf{v} + (-c) \mathbf{w} = \mathbf{0}$	$1 \mathbf{v} + (-c) \mathbf{w} = \mathbf{0}$, so $\{\mathbf{v}, \mathbf{w}\}$ is linearly dependent.
johnston-linear-matrix-attrib F04592	463	1.000	$S = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}$	$S = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}$	2.3.21 Let $S = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}$ be the set of vectors. If S is
johnston-linear-matrix-attrib F04593	463	1.000	$c_i \neq 0$	$c_i \neq 0$	Suppose that $c_i \neq 0$ (which we can do since not all
johnston-linear-matrix-attrib F04594	463	0.990	$\mathbf{v}_i = \frac{-c_1}{c_i} \mathbf{v}_1 + \dots + \frac{-c_{i-1}}{c_i} \mathbf{v}_{i-1} + \frac{-c_{i+1}}{c_i} \mathbf{v}_{i+1} + \dots + \frac{-c_k}{c_i} \mathbf{v}_k$	$\mathbf{v}_i = \frac{-c_1}{c_i} \mathbf{v}_1 + \dots + \frac{-c_{i-1}}{c_i} \mathbf{v}_{i-1} + \frac{-c_{i+1}}{c_i} \mathbf{v}_{i+1} + \dots + \frac{-c_k}{c_i} \mathbf{v}_k$	$\mathbf{v}_i = \frac{-c_1}{c_i} \mathbf{v}_1 + \dots + \frac{-c_{i-1}}{c_i} \mathbf{v}_{i-1} + \frac{-c_{i+1}}{c_i} \mathbf{v}_{i+1} + \dots + \frac{-c_k}{c_i} \mathbf{v}_k$.
johnston-linear-matrix-attrib F04595	463	1.000	$B = \{\mathbf{0}, \mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}$	$B = \{\mathbf{0}, \mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}$	2.3.23 Suppose $B = \{\mathbf{0}, \mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}$. Then
johnston-linear-matrix-attrib F04596	463	1.000	$C = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k, \mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_m\}$	$C = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k, \mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_m\}$	If $C = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k, \mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_m\}$ then we
johnston-linear-matrix-attrib F04597	463	1.000	$\mathbf{x} \in \operatorname{range}(A B)$	$\mathbf{x} \in \operatorname{range}(AB)$	2.3.27 (a) If $\mathbf{x} \in \operatorname{range}(AB)$ then there exists $\mathbf{y} \in \mathbb{R}^p$
johnston-linear-matrix-attrib F04598	463	1.000	$\mathbf{y} \in \mathbb{R}^p$	$\mathbf{y} \in \mathbb{R}^p$	2.3.27 (a) If $\mathbf{x} \in \operatorname{range}(AB)$ then there exists $\mathbf{y} \in \mathbb{R}^p$
johnston-linear-matrix-attrib F04599	463	1.000	$\mathbf{x} = (AB) \mathbf{y} = A(B \mathbf{y})$	$\mathbf{x} = (AB) \mathbf{y} = A(B \mathbf{y})$	such that $\mathbf{x} = (AB) \mathbf{y} = A(B \mathbf{y})$. It follows that
johnston-linear-matrix-attrib F04600	463	0.999	$\mathbf{x} \in \operatorname{range}(A)$	$\mathbf{x} \in \operatorname{range}(A)$	$\mathbf{x} \in \operatorname{range}(A)$ as well (since it can be written as
johnston-linear-matrix-attrib F04601	463	1.000	$\mathbf{x} \in \operatorname{null}(B)$	$\mathbf{x} \in \operatorname{null}(B)$	(b) If $\mathbf{x} \in \operatorname{null}(B)$ then $B \mathbf{x} = \mathbf{0}$, so $AB \mathbf{x} = A \mathbf{0} =$
johnston-linear-matrix-attrib F04602	463	1.000	$B \mathbf{x} = \mathbf{0}$	$B \mathbf{x} = \mathbf{0}$	(b) If $\mathbf{x} \in \operatorname{null}(B)$ then $B \mathbf{x} = \mathbf{0}$, so $AB \mathbf{x} = A \mathbf{0} =$
johnston-linear-matrix-attrib F04603	463	1.000	$AB \mathbf{x} = A \mathbf{0} =$	$AB \mathbf{x} = A \mathbf{0} =$	(b) If $\mathbf{x} \in \operatorname{null}(B)$ then $B \mathbf{x} = \mathbf{0}$, so $AB \mathbf{x} = A \mathbf{0} =$

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johnston-linear-matrix-atg604-F04604	463	■ 0.998	$\mathbf{x} \in \operatorname{null}(A \ B)$	$\mathbf{x} \in \operatorname{null}(AB)$	$\mathbf{0}$, so $\mathbf{x} \in \operatorname{null}(AB)$. It follows that $\operatorname{null}(B) \subseteq$
johnston-linear-matrix-atg605-F04605	463	■ 0.998	$\operatorname{null}(B) \subseteq$	$\operatorname{null}(B) \subseteq$	$\mathbf{0}$, so $\mathbf{x} \in \operatorname{null}(AB)$. It follows that $\operatorname{null}(B) \subseteq$
johnston-linear-matrix-atg606-F04606	464	■ 0.997	$\operatorname{range}(A \ B)=\operatorname{span}((1,0))$	$\operatorname{range}(AB) = \operatorname{span}((1,0))$	which has $\operatorname{range}(AB) = \operatorname{span}((1,0))$, which is
johnston-linear-matrix-atg607-F04607	464	■ 0.999	$\operatorname{range}(B)=\operatorname{span}((0,1))$	$\operatorname{range}(B) = \operatorname{span}((0,1))$	not contained within $\operatorname{range}(B) = \operatorname{span}((0,1))$.
johnston-linear-matrix-atg608-F04608	464	■ 1.000	$\operatorname{null}(A)=\operatorname{span}((1,0))$	$\operatorname{null}(A) = \operatorname{span}((1,0))$	Similarly, $\operatorname{null}(A) = \operatorname{span}((1,0))$, which is not
johnston-linear-matrix-atg609-F04609	464	■ 0.992	$\operatorname{null}(A \ B)=\operatorname{span}((0,1))$	$\operatorname{null}(AB) = \operatorname{span}((0,1))$	contained within $\operatorname{null}(AB) = \operatorname{span}((0,1))$.
johnston-linear-matrix-atg610-F04610	464	■ 1.000	$(x, \ y)=(1,3)$	$(x, y) = (1, 3)$	any non-zero vector on the line. $(x,y) = (1,3)$
johnston-linear-matrix-atg611-F04611	464	■ 1.000	$B=\{(1,3)\}$	$B = \{(1, 3)\}$	works, so $B = \{(1,3)\}$ is a basis of this subspace.
johnston-linear-matrix-atg612-F04612	464	■ 1.000	$B=\{(1,1,-3),(1,0,-2)\}$	$B = \{(1, 1, -3), (1, 0, -2)\}$	$B = \{(1, 1, -3), (1, 0, -2)\}$ is a basis of this
johnston-linear-matrix-atg613-F04613	464	■ 1.000	$z=3$	$z = 3$	choosing $z = 3$ and solving by back substitution
johnston-linear-matrix-atg614-F04614	464	■ 1.000	$y=4$	$y = 4$	gives $y = 4$ and $x = -1$. The line points thus in
johnston-linear-matrix-atg615-F04615	464	■ 1.000	$(x, \ y, \ z)=(-1,4,3)$	$(x, y, z) = (-1, 4, 3)$	the direction of the vector $(x,y,z) = (-1,4,3)$,
johnston-linear-matrix-atg616-F04616	464	■ 1.000	$B=\{(-1,4,3)\}$	$B = \{(-1, 4, 3)\}$	so $B = \{(-1,4,3)\}$ is a basis of this line.
johnston-linear-matrix-atg617-F04617	464	■ 0.999	$\{(1,2,3,4),(3,5,7,9)\}$	$\{(1, 2, 3, 4), (3, 5, 7, 9)\}$	2.4.3 (a) $\{(1,2,3,4),(3,5,7,9)\}$
johnston-linear-matrix-atg618-F04618	464	■ 1.000	$\{(1,1,4,5,1),(2,-7,4,2,2),(5,2,4,5,3)\}$	$\{(1, 1, 4, 5, 1), (2, -7, 4, 2, 2), (5, 2, 4, 5, 3)\}$	(c) $\{(1,1,4,5,1),(2,-7,4,2,2),(5,2,4,5,3)\}$
johnston-linear-matrix-atg619-F04619	464	■ 0.997	$\{(2,3,1,1),(3,3,2,3),(2,1,2,3),(1,0,0,0)\}$	$\{(2, 3, 1, 1), (3, 3, 2, 3), (2, 1, 2, 3), (1, 0, 0, 0)\}$	(a) $\{(2,3,1,1),(3,3,2,3),(2,1,2,3),(1,0,0,0)\}$
johnston-linear-matrix-atg620-F04620	464	■ 1.000	$d_1, d_2, \dots, d_n \in \mathbb{R}$	$d_1, d_2, \dots, d_n \in \mathbb{R}$	2.3.28 Suppose $d_1, d_2, \dots, d_n \in \mathbb{R}$ are such that
johnston-linear-matrix-atg621-F04621	464	■ 1.000	$d_1 c_1 = \dots = d_n c_n = 0$	$d_1 c_1 = \dots = d_n c_n = 0$	equation implies $d_1 c_1 = \dots = d_n c_n = 0$. Since c_1 ,
johnston-linear-matrix-atg622-F04622	464	■ 1.000	$\dots, c_n$	$\dots, c_n$	$\dots, c_n$ are non-zero, this implies $d_1 = \dots = d_n = 0$,
johnston-linear-matrix-atg623-F04623	464	■ 1.000	$d_1 = \dots = d_n = 0$	$d_1 = \dots = d_n = 0$	$\dots, c_n$ are non-zero, this implies $d_1 = \dots = d_n = 0$,
johnston-linear-matrix-atg624-F04624	464	■ 1.000	$\mathbf{v}_j = (1/c_j)(c_j \mathbf{v}_j)$	$\mathbf{v}_j = (1/c_j)(c_j \mathbf{v}_j)$	$\mathbf{v}_j = (1/c_j)(c_j \mathbf{v}_j)$ for each $1 \leq j \leq n$ and then using
johnston-linear-matrix-atg625-F04625	464	■ 0.999	$\{(2,4,3,4),(3,2,-1,-1),(1,0,0,0),(0,1,0,0)\}$	$\{(2, 4, 3, 4), (3, 2, -1, -1), (1, 0, 0, 0), (0, 1, 0, 0)\}$	(c) $\{(2,4,3,4),(3,2,-1,-1),(1,0,0,0),(0,1,0,0)\}$
johnston-linear-matrix-atg626-F04626	464	■ 0.988	$(A): \{(1,0),(0,1)\}$	$(A): \{(1,0), (0,1)\}$	Basis of $\operatorname{range}(A)$: $\{(1,0), (0,1)\}$
johnston-linear-matrix-atg627-F04627	464	■ 1.000	$\operatorname{null}(A): \{ \} (\operatorname{null}(A) = \{\mathbf{0}\})$	$\operatorname{null}(A): \{ \} (\operatorname{null}(A) = \{\mathbf{0}\})$	Basis of $\operatorname{null}(A)$: $\{ \} (\operatorname{null}(A) = \{\mathbf{0}\})$
johnston-linear-matrix-atg628-F04628	464	■ 0.894	$\operatorname{range}(A^T): \{(1,0),(0,1)\}$	$\operatorname{range}(A^T): \{(1,0), (0,1)\}$	Basis of $\operatorname{range}(A^T)$: $\{(1,0), (0,1)\}$
johnston-linear-matrix-atg629-F04629	464	■ 0.994	$\operatorname{null}(A^T): \{ \} (\operatorname{null}(A^T) = \{\mathbf{0}\})$	$\operatorname{null}(A^T): \{ \} (\operatorname{null}(A^T) = \{\mathbf{0}\})$	Basis of $\operatorname{null}(A^T)$: $\{ \} (\operatorname{null}(A^T) = \{\mathbf{0}\})$
johnston-linear-matrix-atg630-F04630	464	■ 0.813	$\operatorname{range}(A): \{(0,1,0)\}$	$\operatorname{range}(A): \{(0,1,0)\}$	Basis of $\operatorname{range}(A)$: $\{(0,1,0)\}$
johnston-linear-matrix-atg631-F04631	464	■ 0.999	$\operatorname{null}(A): \{(1,0,0),(0,-2,1)\}$	$\operatorname{null}(A): \{(1,0,0), (0,-2,1)\}$	Basis of $\operatorname{null}(A)$: $\{(1,0,0), (0,-2,1)\}$
johnston-linear-matrix-atg632-F04632	464	■ 0.966	$(A^T): \{(0,1,2)\}$	$(A^T): \{(0,1,2)\}$	Basis of $\operatorname{range}(A^T)$: $\{(0,1,2)\}$

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johnston-linear-matrix-atlas- F04633	464	1.000	$\operatorname{null}(A^T): \{(1,0,0), (0,0,1)\}$	$\operatorname{null}(A^T) : \{(1,0,0), (0,0,1)\}$	Basis of $\operatorname{null}(A^T)$: $\{(1,0,0), (0,0,1)\}$
johnston-linear-matrix-atlas- F04634	464	0.608	$\operatorname{range}(A): \{(1,0,0), (0,2,1)\}$	$\operatorname{range}(A) : \{(1,0,0), (0,2,1)\}$	(e) Basis of $\operatorname{range}(A)$: $\{(1,0,0), (0,2,1)\}$
johnston-linear-matrix-atlas- F04635	464	1.000	$\operatorname{null}(A^T): \{(0,-1,2)\}$	$\operatorname{null}(A^T) : \{(0,-1,2)\}$	Basis of $\operatorname{null}(A^T)$: $\{(0,-1,2)\}$
johnston-linear-matrix-atlas- F04636	464	0.928	$(A): \{(1,0,0), (0,1,0), (0,0,1)\}$	$(A) : \{(1,0,0), (0,1,0), (0,0,1)\}$	Basis of $\operatorname{range}(A)$: $\{(1,0,0), (0,1,0), (0,0,1)\}$
johnston-linear-matrix-atlas- F04637	464	0.699	$\operatorname{null}(A): \{(-2,1,0,0,0), (-3,0,1,1,0)\}$	$\operatorname{null}(A) : \{(-2,1,0,0,0), (-3,0,1,1,0)\}$	Basis of $\operatorname{null}(A)$: $\{(-2,1,0,0,0), (-3,0,1,1,0)\}$
johnston-linear-matrix-atlas- F04638	464	0.865	$\operatorname{null}(A^T): \{(1,2,0,3,0), (0,0,1,-1,0)\}$	$(A^T) : \{(1,2,0,3,0), (0,0,1,-1,0)\}$	Basis of $\operatorname{range}(A^T)$: $\{(1,2,0,3,0), (0,0,1,-1,0),$
johnston-linear-matrix-atlas- F04639	464	0.706	$\operatorname{range}(A): \{(0,-1,-2), (-4,2,0)\}$	$\operatorname{range}(A) : \{(0,-1,-2), (-4,2,0)\}$	Basis of $\operatorname{range}(A)$: $\{(0,-1,-2), (-4,2,0)\}$
johnston-linear-matrix-atlas- F04640	464	0.725	$\operatorname{null}(A): \{(1,0,1,0,0), (3,1/2,0,1,0)\}$	$\operatorname{null}(A) : \{(1,0,1,0,0), (3,1/2,0,1,0)\}$	Basis of $\operatorname{null}(A)$: $\{(1,0,1,0,0), (3,1/2,0,1,0),$
johnston-linear-matrix-atlas- F04641	464	0.546	$\operatorname{range}(A^T): \{(1,0,-1,-3,-3/2)\}$	$(A^T) : \{(1,0,-1,-3,-3/2)\}$	Basis of $\operatorname{range}(A^T)$: $\{(1,0,-1,-3,-3/2),$
johnston-linear-matrix-atlas- F04642	464	1.000	$\operatorname{null}(A^T): \{(-1,-2,1)\}$	$\operatorname{null}(A^T) : \{(-1,-2,1)\}$	Basis of $\operatorname{null}(A^T)$: $\{(-1,-2,1)\}$
johnston-linear-matrix-atlas- F04643	465	0.999	$\operatorname{rank}(A)=4$	$\operatorname{rank}(A) = 4$	2.4.7 (a) If we call this matrix A then $\operatorname{rank}(A) = 4$,
johnston-linear-matrix-atlas- F04644	465	1.000	$\operatorname{nullity}(A)=4$	$\operatorname{nullity}(A) = 4$	$\operatorname{nullity}(A) = 4$, and we have:
johnston-linear-matrix-atlas- F04645	465	0.998	$(A): \{(0,-6,4,-1), (2,-1,-2,0)$	$(A) : \{(0,-6,4,-1), (2,-1,-2,0)$	Basis of $\operatorname{range}(A)$: $\{(0,-6,4,-1), (2,-1,-2,0),$
johnston-linear-matrix-atlas- F04646	465	1.000	$(-1,1,1,0), (-1,6,-2,1)\}$	$(-1,1,1,0), (-1,6,-2,1)\}$	$(-1,1,1,0), (-1,6,-2,1)\}$ (or the standard
johnston-linear-matrix-atlas- F04647	465	0.904	$\operatorname{null}(A): \{(2,-1,0,0,0,0,0,0)\}$	$\operatorname{null}(A) : \{(2,-1,0,0,0,0,0,0)\}$	Basis of $\operatorname{null}(A)$: $\{(2,-1,0,0,0,0,0,0),$
johnston-linear-matrix-atlas- F04648	465	0.826	$\operatorname{range}(A^T): \{(1,2,0,1,0,-1,1,0)$	$\operatorname{range}(A^T) : \{(1,2,0,1,0,-1,1,0)$	Basis of $\operatorname{range}(A^T)$: $\{(1,2,0,1,0,-1,1,0),$
johnston-linear-matrix-atlas- F04649	465	1.000	$\{(1,1), (2,2)\}$	$\{(1,1), (2,2)\}$	set $\{(1,1), (2,2)\}$, which is not even linearly
johnston-linear-matrix-atlas- F04650	465	0.978	$\operatorname{rank}(A+B) \leq$	$\operatorname{rank}(A+B) \leq$	(i) False. All that we can say is that $\operatorname{rank}(A+B) \leq$
johnston-linear-matrix-atlas- F04651	465	0.855	$A=B=I_n$	$A = B = I_n$	$A = B = I_n$ then $\operatorname{rank}(A+B) = \operatorname{rank}(2I_n) = n$,
johnston-linear-matrix-atlas- F04652	465	0.855	$(A+B)=\operatorname{rank}(2I_n)=n$	$(A+B) = \operatorname{rank}(2I_n) = n$	$A = B = I_n$ then $\operatorname{rank}(A+B) = \operatorname{rank}(2I_n) = n$,
johnston-linear-matrix-atlas- F04653	465	1.000	$\operatorname{rank}(A)+\operatorname{rank}(B)=n+n=2n$	$\operatorname{rank}(A) + \operatorname{rank}(B) = n + n = 2n$	but $\operatorname{rank}(A) + \operatorname{rank}(B) = n + n = 2n$.
johnston-linear-matrix-atlas- F04654	465	0.992	$\operatorname{rank}(AB) \leq$	$\operatorname{rank}(AB) \leq$	(k) False. All that we can say is that $\operatorname{rank}(AB) \leq$
johnston-linear-matrix-atlas- F04655	465	1.000	$\min\{\operatorname{rank}(A), \operatorname{rank}(B)\}$	$\min\{\operatorname{rank}(A), \operatorname{rank}(B)\}$	$\min\{\operatorname{rank}(A), \operatorname{rank}(B)\}$, thanks to Theo-
johnston-linear-matrix-atlas- F04656	465	0.983	$\operatorname{rank}(A_2)=2$	$\operatorname{rank}(A_2) = 2$	2.4.12 It is straightforward to check that $\operatorname{rank}(A_2) = 2$, so
johnston-linear-matrix-atlas- F04657	465	1.000	$k \geq 3$	$k \geq 3$	nation of those two rows. In particular, if $k \geq 3$ then
johnston-linear-matrix-atlas- F04658	465	1.000	$R_2 - 2R_1$	$R_2 - 2R_1$	2.4.13 (a) By performing the row operation $R_2 - 2R_1$, we
johnston-linear-matrix-atlas- F04659	465	1.000	$x \neq 2$	$x \neq 2$	(i.e., when $x \neq 2$).
johnston-linear-matrix-atlas- F04660	465	0.586	$x \neq \pm 1$	$x \neq \pm 1$	the rank is 2. If $x \neq \pm 1$ then the rank is 3.
johnston-linear-matrix-atlas- F04661	465	1.000	$A^{-1}(AB)=B$	$A^{-1}(AB) = B$	that $A^{-1}(AB) = B$, so using Theorem 2.4.11

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johnston-linear-matrix-atlas- F04662	465	■ 1.000	$\operatorname{rank}(B)=\operatorname{rank}\left(A^{-1}(A\right.\left.B)\right)\leq$	$\operatorname{rank}(B)=\operatorname{rank}\left(A^{-1}(AB)\right)\leq$	again tells us that $\operatorname{rank}(B)=\operatorname{rank}(A^{-1}(AB))\leq$
johnston-linear-matrix-atlas- F04663	466	■ 0.523	$\operatorname{span}\left(\mathbf{v}_{\{1\}},\ldots,\mathbf{v}_{\{k\}}\right)=\operatorname{span}\left(\mathbf{v}_{\{1\}},\ldots,\mathbf{v}_{\{i-1\}},\mathbf{v}_{\{i+1\}},\ldots,\mathbf{v}_{\{k\}}\right)$	$\operatorname{span}\left(\mathbf{v}_1,\ldots,\mathbf{v}_k\right)=\operatorname{span}\left(\mathbf{v}_1,\ldots,\mathbf{v}_{i-1},\mathbf{v}_{i+1},\ldots,\mathbf{v}_k\right)$	$\operatorname{span}\left(\mathbf{v}_1,\ldots,\mathbf{v}_k\right)=\operatorname{span}\left(\mathbf{v}_1,\ldots,\mathbf{v}_{i-1},\mathbf{v}_{i+1},\ldots,\mathbf{v}_k\right)$.
johnston-linear-matrix-atlas- F04664	466	■ 1.000	$\left(\mathbf{v}_{\{1\}},\ldots,\mathbf{v}_{\{i-1\}},\mathbf{v}_{\{i+1\}},\ldots,\mathbf{v}_{\{k\}}\right)$	$\left\{\mathbf{v}_1,\ldots,\mathbf{v}_{i-1},\mathbf{v}_{i+1},\ldots,\mathbf{v}_k\right\}$	We then repeat: if $\left\{\mathbf{v}_1,\ldots,\mathbf{v}_{i-1},\mathbf{v}_{i+1},\ldots,\mathbf{v}_k\right\}$ is lin-
johnston-linear-matrix-atlas- F04665	466	■ 1.000	$\{(1,-1,2),(2,-2,4)\}$	$\{(1,-1,2),(2,-2,4)\}$	with the set $\{(1,-1,2),(2,-2,4)\}$, and the
johnston-linear-matrix-atlas- F04666	466	■ 1.000	$\mathbf{v}=A\mathbf{x}$	$\mathbf{v}=A\mathbf{x}$	2.4.26 (a) If $\mathbf{v}\in\operatorname{range}(A)$ then we can write it as $\mathbf{v}=A\mathbf{x}$
johnston-linear-matrix-atlas- F04667	466	■ 1.000	$\mathbf{w}=$	$\mathbf{w}=$	(b) If $\mathbf{w}\in\operatorname{range}(A^T)$ then we can write it as $\mathbf{w}=$
johnston-linear-matrix-atlas- F04668	466	■ 1.000	$A^T\mathbf{x}$	$A^T\mathbf{x}$	$A^T\mathbf{x}$ for some $\mathbf{x}\in\mathbb{R}^m$. Then
johnston-linear-matrix-atlas- F04669	466	■ 0.998	$[B\mid\mathbf{0}]$	$[B\mid\mathbf{0}]$	$[B\mid\mathbf{0}]$ have the same RREF $[R\mid\mathbf{0}]$, so the lin-
johnston-linear-matrix-atlas- F04670	466	■ 1.000	$S\mathbf{x}\neq\mathbf{0}$	$S\mathbf{x}\neq\mathbf{0}$	that $R\mathbf{x}=\mathbf{0}$ and $S\mathbf{x}\neq\mathbf{0}$, so $\operatorname{null}(A)\neq\operatorname{null}(B)$.
johnston-linear-matrix-atlas- F04671	466	■ 1.000	$\operatorname{null}(A)\neq\operatorname{null}(B)$	$\operatorname{null}(A)\neq\operatorname{null}(B)$	that $R\mathbf{x}=\mathbf{0}$ and $S\mathbf{x}\neq\mathbf{0}$, so $\operatorname{null}(A)\neq\operatorname{null}(B)$.
johnston-linear-matrix-atlas- F04672	466	■ 1.000	$\operatorname{range}\left(A^T\right)\neq\operatorname{range}\left(B^T\right)$	$\operatorname{range}\left(A^T\right)\neq\operatorname{range}\left(B^T\right)$	of R). It follows that $\operatorname{range}(A^T)\neq\operatorname{range}(B^T)$.
johnston-linear-matrix-atlas- F04673	466	■ 1.000	$\operatorname{rank}(A)=\operatorname{rank}\left(A^T\right)$	$\operatorname{rank}(A)=\operatorname{rank}\left(A^T\right)$	$\operatorname{rank}(A)=\operatorname{rank}(A^T)$.
johnston-linear-matrix-atlas- F04674	466	■ 1.000	$\operatorname{rank}\left(\mathbf{v}\right)=\operatorname{rank}\left(\mathbf{w}^T\right)=1$	$\operatorname{rank}(\mathbf{v})=\operatorname{rank}\left(\mathbf{w}^T\right)=1$	$\operatorname{rank}(\mathbf{v})=\operatorname{rank}(\mathbf{w}^T)=1$. Then Theorem 2.4.11(b)
johnston-linear-matrix-atlas- F04675	466	■ 0.997	$\operatorname{rank}(A)=\operatorname{rank}\left(\mathbf{v}\mathbf{w}^T\right)\leq 1$	$\operatorname{rank}(A)=\operatorname{rank}\left(\mathbf{v}\mathbf{w}^T\right)\leq 1$	tells us that $\operatorname{rank}(A)=\operatorname{rank}(\mathbf{v}\mathbf{w}^T)\leq 1$. Since $\mathbf{v}$ and
johnston-linear-matrix-atlas- F04676	466	■ 1.000	$\operatorname{rank}(A)\neq 0$	$\operatorname{rank}(A)\neq 0$	$\mathbf{w}$ are non-zero, $A\neq O$ so $\operatorname{rank}(A)\neq 0$. It follows that
johnston-linear-matrix-atlas- F04677	466	■ 0.748	$w_2,w_3,\ldots,w_n$	$w_2,w_3,\ldots,w_n$	is $\mathbf{v}$ then there are constants $w_2,w_3,\ldots,w_n$ such that
johnston-linear-matrix-atlas- F04678	466	■ 1.000	$w_2\mathbf{v}$	$w_2\mathbf{v}$	the 2nd column is $w_2\mathbf{v}$, the 3rd column is $w_3\mathbf{v}$, and
johnston-linear-matrix-atlas- F04679	466	■ 1.000	$w_3\mathbf{v}$	$w_3\mathbf{v}$	the 2nd column is $w_2\mathbf{v}$, the 3rd column is $w_3\mathbf{v}$, and
johnston-linear-matrix-atlas- F04680	466	■ 1.000	$\mathbf{w}^T=(1,w_2,w_3,\ldots,w_n)$	$\mathbf{w}^T=(1,w_2,w_3,\ldots,w_n)$	If $\mathbf{w}^T=(1,w_2,w_3,\ldots,w_n)$ then the above expres-
johnston-linear-matrix-atlas- F04681	467	■ 0.999	$[R\mid\mathbf{c}]$	$[R\mid\mathbf{c}]$	of the form $[R\mid\mathbf{c}]$, where $\mathbf{c}$ in general may be
johnston-linear-matrix-atlas- F04682	467	■ 0.974	$R\mathbf{x}=\mathbf{c}$	$R\mathbf{x}=\mathbf{c}$	same solution set as $R\mathbf{x}=\mathbf{c}$ for <i>some</i> vector $\mathbf{c}$,
johnston-linear-matrix-atlas- F04683	467	■ 1.000	$\mathbf{c}\neq\mathbf{b}$	$\mathbf{c}\neq\mathbf{b}$	but $\mathbf{c}\neq\mathbf{b}$ in general.
johnston-linear-matrix-atlas- F04684	467	■ 0.637	$\operatorname{range}(A)=\{\mathbf{0}\}$	$\operatorname{range}(A)=\{\mathbf{0}\}$	(g) False. If $\operatorname{range}(A)=\{\mathbf{0}\}$ then $A=O$, which is
johnston-linear-matrix-atlas- F04685	467	■ 1.000	$R=E_k\cdots E_2E_1A$	$R=E_k\cdots E_2E_1A$	In particular, $R=E_k\cdots E_2E_1A$, so $A=$
johnston-linear-matrix-atlas- F04686	467	■ 1.000	$E_1^{-1}E_2^{-1}\cdots E_k^{-1}R$	$E_1^{-1}E_2^{-1}\cdots E_k^{-1}R$	$E_1^{-1}E_2^{-1}\cdots E_k^{-1}R$ (since elementary matrices are
johnston-linear-matrix-atlas- F04687	467	■ 1.000	$P=E_1^{-1}E_2^{-1}\cdots E_k^{-1}$	$P=E_1^{-1}E_2^{-1}\cdots E_k^{-1}$	invertible). We can thus choose $P=E_1^{-1}E_2^{-1}\cdots E_k^{-1}$,
johnston-linear-matrix-atlas- F04688	467	■ 1.000	$P_1,P_2\in\mathcal{M}_m$	$P_1,P_2\in\mathcal{M}_m$	vertible matrices $P_1,P_2\in\mathcal{M}_m$ such that $A=P_1R$
johnston-linear-matrix-atlas- F04689	467	■ 1.000	$A=P_1R$	$A=P_1R$	vertible matrices $P_1,P_2\in\mathcal{M}_m$ such that $A=P_1R$

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johnston-linear-matrix-atlas- F04690	467	■ 1.000	$B=P_{\{2\}} R$	$B = P_2 R$	and $B = P_2 R$. Then multiplying on the left by $P_1 P_2^{-1}$
johnston-linear-matrix-atlas- F04691	467	■ 1.000	$P_{\{1\}} P_{\{2\}}^{-1}$	$P_1 P_2^{-1}$	and $B = P_2 R$. Then multiplying on the left by $P_1 P_2^{-1}$
johnston-linear-matrix-atlas- F04692	467	■ 1.000	$P=P_{\{1\}} P_{\{2\}}^{-1}$	$P = P_1 P_2^{-1}$	so we can choose $P = P_1 P_2^{-1}$.
johnston-linear-matrix-atlas- F04693	467	■ 1.000	$P=E_{\{1\}} E_{\{2\}} \cdots E_{\{k\}}$	$P = E_1 E_2 \cdots E_k$	elementary matrices: $P = E_1 E_2 \cdots E_k$. It follows that
johnston-linear-matrix-atlas- F04694	467	■ 1.000	$A=E_{\{1\}} E_{\{2\}} \cdots E_{\{k\}} B$	$A = E_1 E_2 \cdots E_k B$	$A = E_1 E_2 \cdots E_k B$, so A can be reached from B via
johnston-linear-matrix-atlas- F04695	467	■ 1.000	$(x, y, z)=(0,1,0)$	$(x,y,z) = (0,1,0)$	2.A.1 (a) $(x,y,z) = (0,1,0)$.
johnston-linear-matrix-atlas- F04696	467	■ 0.999	$(w, x, y, z)=(0,1,0,0)$	$(w,x,y,z) = (0,1,0,0)$	(c) $(w,x,y,z) = (0,1,0,0)$ and $(w,x,y,z) =$
johnston-linear-matrix-atlas- F04697	467	■ 0.998	$(v, w, x, y, z)=(1,1,0,0,0)$	$(v,w,x,y,z) = (1,1,0,0,0)$	(e) $(v,w,x,y,z) = (1,1,0,0,0)$.
johnston-linear-matrix-atlas- F04698	467	■ 1.000	$(x, y, z)=(1,0,1)$	$(x,y,z) = (1,0,1)$	2.A.2 (a) $(x,y,z) = (1,0,1)$.
johnston-linear-matrix-atlas- F04699	468	■ 1.000	$(v, w, x, y, z)=(0,1,1,0,0)$	$(v,w,x,y,z) = (0,1,1,0,0)$	$(v,w,x,y,z) = (0,1,1,0,0)$, though the original
johnston-linear-matrix-atlas- F04700	468	■ 0.982	$(v, w, x, y, z)=(4,-1,3,2,8)$	$(v,w,x,y,z) = (4,-1,3,2,8)$	$(v,w,x,y,z) = (4,-1,3,2,8)$ is another solution).
johnston-linear-matrix-atlas- F04701	468	■ 1.000	$\mathrm{\mathbf{v}}_{\mathrm{\{v\}}} - \mathrm{\mathbf{v}}_{\mathrm{\{s\}}} =$	$\mathbf{v}_e - \mathbf{v}_s =$	need to solve is $A\mathbf{x} = \mathbf{v}_e - \mathbf{v}_s$, where $\mathbf{v}_e - \mathbf{v}_s =$
johnston-linear-matrix-atlas- F04702	468	■ 1.000	$\mathrm{\mathbf{x}}=(1,0,0,1,0,1,1,0,1)$	$\mathbf{x} = (1,0,0,1,0,1,1,0,1)$	part (a) is $\mathbf{x} = (1,0,0,1,0,1,1,0,1)$, which
johnston-linear-matrix-atlas- F04703	468	■ 1.000	$m_{\{j\}}$	m_j	the other hand, if the j -th row has m_j ones in
johnston-linear-matrix-atlas- F04704	468	■ 1.000	$B-I$	$B - I$	(b) Just apply part (a) to the matrix $B - I$.
johnston-linear-matrix-atlas- F04705	468	■ 1.000	$\left(x_{\{1\}}, x_{\{2\}}\right)=(3,0)$	$(x_1,x_2) = (3,0)$	$(x_1,x_2) = (3,0)$.
johnston-linear-matrix-atlas- F04706	468	■ 1.000	$\left(x_{\{1\}}, x_{\{2\}}\right)=(21,1) / 8$	$(x_1,x_2) = (21,1)/8$	$(x_1,x_2) = (21,1)/8$.
johnston-linear-matrix-atlas- F04707	468	■ 1.000	$\left(x_{\{1\}}, x_{\{2\}}, x_{\{3\}}\right)=(3,11,23) / 18$	$(x_1,x_2,x_3) = (3,11,23)/18$	at $(x_1,x_2,x_3) = (3,11,23)/18$.
johnston-linear-matrix-atlas- F04708	468	■ 1.000	$\left(x_{\{1\}}, x_{\{2\}}, x_{\{3\}}, x_{\{4\}}\right)=(6,2,0,23) / 30$	$(x_1,x_2,x_3,x_4) = (6,2,0,23)/30$	at $(x_1,x_2,x_3,x_4) = (6,2,0,23)/30$.
johnston-linear-matrix-atlas- F04709	468	■ 1.000	$\left(x_{\{1\}}, x_{\{2\}}\right)=(3,1) / 2$	$(x_1,x_2) = (3,1)/2$	$(x_1,x_2) = (3,1)/2$.
johnston-linear-matrix-atlas- F04710	468	■ 1.000	$\left(x_{\{1\}}, x_{\{2\}}, x_{\{3\}}\right)=(1,0,1)$	$(x_1,x_2,x_3) = (1,0,1)$	$(x_1,x_2,x_3) = (1,0,1)$.
johnston-linear-matrix-atlas- F04711	468	■ 0.995	$7 / 4$	$7/4$	(e) The optimal value is $7/4$, which is attained at
johnston-linear-matrix-atlas- F04712	468	■ 0.882	$\left(x_{\{1\}}, x_{\{2\}}, x_{\{3\}}, x_{\{4\}}\right)=(8,2,0,1) / 8$	$(x_1,x_2,x_3,x_4) = (8,2,0,1)/8$	$(x_1,x_2,x_3,x_4) = (8,2,0,1)/8$.
johnston-linear-matrix-atlas- F04713	468	■ 1.000	$\mathrm{\mathbf{x}}=(1,0), \mathrm{\mathbf{b}}=(2,2)$	$\mathbf{x} = (1,0), \mathbf{b} = (2,2)$	2.B.5 (a) False. For example, if $\mathbf{x} = (1,0)$, $\mathbf{b} = (2,2)$,
johnston-linear-matrix-atlas- F04714	469	■ 1.000	$x_{\{j\}} \geq y_{\{j\}}$	$x_j \geq y_j$	(c) True. If $\mathbf{x} \geq \mathbf{y}$ then $x_j \geq y_j$ for each $1 \leq j \leq n$,
johnston-linear-matrix-atlas- F04715	469	■ 1.000	$c_{\{j\}} x_{\{j\}} \geq c_{\{j\}} y_{\{j\}}$	$c_j x_j \geq c_j y_j$	so $c_j x_j \geq c_j y_j$ for each j , so
johnston-linear-matrix-atlas- F04716	469	■ 1.000	$\left(x_{\{1\}}, x_{\{2\}}\right)=(0.6,0.8)$	$(x_1,x_2) = (0.6,0.8)$	which is attained at $(x_1,x_2) = (0.6,0.8)$. To see
johnston-linear-matrix-atlas- F04717	469	■ 1.000	$x_{\{2\}} \geq 1 / 2$	$x_2 \geq 1/2$	$x_2 \geq 1/2$, whereas adding the first constraint to -2
johnston-linear-matrix-atlas- F04718	469	■ 0.999	$x_{\{2\}} \leq 4 / 5$	$x_2 \leq 4/5$	times the third constraint shows that $x_2 \leq 4/5$. In

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johnston-linear-matrix-attempt-F04719	469	■ 1.000	$2x_2 \leq c$	$2x_2 \leq c$	2.B.8 (a) Adding the two constraints shows that $2x_2 \leq c$,
johnston-linear-matrix-attempt-F04720	469	■ 1.000	$x_2 \leq c/2$	$x_2 \leq c/2$	so $x_2 \leq c/2$, so certainly the optimal value of
johnston-linear-matrix-attempt-F04721	469	■ 1.000	$c/2$	$c/2$	the linear program cannot exceed $c/2$. On the
johnston-linear-matrix-attempt-F04722	469	■ 0.381	$1/2, c/2$	$1/2, c/2$	other hand, $(1/2, c/2)$ is a feasible point of
johnston-linear-matrix-attempt-F04723	469	■ 1.000	$x_1 \leq 1 - x_2/c$	$x_1 \leq 1 - x_2/c$	(b) The first constraint tells us that $x_1 \leq 1 - x_2/c$
johnston-linear-matrix-attempt-F04724	469	■ 1.000	$x_1 \geq$	$x_1 \geq$	and the second constraint tells us that $x_1 \geq$
johnston-linear-matrix-attempt-F04725	469	■ 1.000	x_2/c	x_2/c	x_2/c . Since $x_2 \geq 0$, it follows that $0 \leq x_1 \leq x_2$,
johnston-linear-matrix-attempt-F04726	469	■ 1.000	$0 \leq x_1 \leq x_2$	$0 \leq x_1 \leq x_2$	x_2/c . Since $x_2 \geq 0$, it follows that $0 \leq x_1 \leq x_2$,
johnston-linear-matrix-attempt-F04727	469	■ 1.000	$A^T \mathbf{1} = \mathbf{1}$	$A^T \mathbf{1} = \mathbf{1}$	Notice that the rows of A^T sum to 1 so $A^T \mathbf{1} = \mathbf{1}$
johnston-linear-matrix-attempt-F04728	469	■ 1.000	$\mathbf{1} = (1, 1, \dots, 1)$	$\mathbf{1} = (1, 1, \dots, 1)$	(where $\mathbf{1} = (1, 1, \dots, 1)$). It follows that the latter
johnston-linear-matrix-attempt-F04729	469	■ 1.000	$\mathbf{y} = \mathbf{1}$	$\mathbf{y} = \mathbf{1}$	problem is feasible, since we can choose $\mathbf{y} = \mathbf{1}$ in it.
johnston-linear-matrix-attempt-F04730	469	■ 1.000	$\left[\begin{array}{c} 1 \\ -1 \end{array} \right] \left[\begin{array}{cc} 1 & -1 \end{array} \right]$	$\left[\begin{array}{c} 1 \\ -1 \end{array} \right] \left[\begin{array}{cc} 1 & -1 \end{array} \right]$	(a) $\left[\begin{array}{c} 1 \\ -1 \end{array} \right] \left[\begin{array}{cc} 1 & -1 \end{array} \right]$
johnston-linear-matrix-attempt-F04731	469	■ 0.924	$\left[\begin{array}{ccc} 2 & 4 & 0 \\ 1 & -2 & 0 \\ 2 & 0 & -1 \end{array} \right] \left[\begin{array}{ccc} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{array} \right]$	$\left[\begin{array}{ccc} 2 & 4 & 0 \\ 1 & -2 & 0 \\ 2 & 0 & -1 \end{array} \right] \left[\begin{array}{ccc} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{array} \right]$	(c) $\left[\begin{array}{ccc} 2 & 4 & 0 \\ 1 & -2 & 0 \\ 2 & 0 & -1 \end{array} \right] \left[\begin{array}{ccc} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{array} \right]$
johnston-linear-matrix-attempt-F04732	469	■ 1.000	$\left[\begin{array}{c} 3 \\ 1 \end{array} \right] \left[\begin{array}{cc} 2 & 1 \end{array} \right]$	$\left[\begin{array}{c} 3 \\ 1 \end{array} \right] \left[\begin{array}{cc} 2 & 1 \end{array} \right]$	(e) $\left[\begin{array}{c} 3 \\ 1 \end{array} \right] \left[\begin{array}{cc} 2 & 1 \end{array} \right]$
johnston-linear-matrix-attempt-F04733	469	■ 1.000	$\left[\begin{array}{cc} 2 & 6 \\ 5 & 4 \\ 3 & -2 \end{array} \right] \left[\begin{array}{cccc} 1 & 0 & 2 & 5/11 \\ 0 & 1 & -1/2 & -7/22 \end{array} \right]$	$\left[\begin{array}{cc} 2 & 6 \\ 5 & 4 \\ 3 & -2 \end{array} \right] \left[\begin{array}{cccc} 1 & 0 & 2 & 5/11 \\ 0 & 1 & -1/2 & -7/22 \end{array} \right]$	(g) $\left[\begin{array}{cc} 2 & 6 \\ 5 & 4 \\ 3 & -2 \end{array} \right] \left[\begin{array}{cccc} 1 & 0 & 2 & 5/11 \\ 0 & 1 & -1/2 & -7/22 \end{array} \right]$
johnston-linear-matrix-attempt-F04734	470	■ 0.994	$\tilde{C} \in \mathcal{M}_{m,r}$	$\tilde{C} \in \mathcal{M}_{m,r}$	2.C.6 It is clear that $\tilde{C} \in \mathcal{M}_{m,r}$ and $\tilde{R} \in \mathcal{M}_{r,n}$, so we just
johnston-linear-matrix-attempt-F04735	470	■ 0.994	$\tilde{R} \in \mathcal{M}_{r,n}$	$\tilde{R} \in \mathcal{M}_{r,n}$	2.C.6 It is clear that $\tilde{C} \in \mathcal{M}_{m,r}$ and $\tilde{R} \in \mathcal{M}_{r,n}$, so we just
johnston-linear-matrix-attempt-F04736	470	■ 1.000	$\tilde{C} \tilde{R} = CP^{-1}PR = CR = A$	$\tilde{C} \tilde{R} = CP^{-1}PR = CR = A$	pute $\tilde{C} \tilde{R} = CP^{-1}PR = CR = A$.
johnston-linear-matrix-attempt-F04737	470	■ 1.000	PR	PR	PR , where $P \in \mathcal{M}_m$ is invertible and $R \in \mathcal{M}_{m,n}$
johnston-linear-matrix-attempt-F04738	470	■ 1.000	P_1, P_2, Q_1	P_1, P_2, Q_1	find invertible matrices P_1, P_2, Q_1 , and Q_2 such
johnston-linear-matrix-attempt-F04739	470	■ 1.000	Q_2	Q_2	find invertible matrices P_1, P_2, Q_1 , and Q_2 such
johnston-linear-matrix-attempt-F04740	470	■ 1.000	P_2^{-1}	P_2^{-1}	Multiplying on the left by P_2^{-1} and on the right
johnston-linear-matrix-attempt-F04741	470	■ 1.000	Q_2^{-1}	Q_2^{-1}	by Q_2^{-1} shows that
johnston-linear-matrix-attempt-F04742	470	■ 1.000	$Q = Q_2^{-1}Q_1$	$Q = Q_2^{-1}Q_1$	so we can choose $P = P_1P_2^{-1}$ and $Q = Q_2^{-1}Q_1$

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johnston-linear-matrix-atgevrn F04743	470	■ 0.951	$L=\left[\begin{array}{ll}1 & 0 \\ 2 & 1\end{array}\right], U=\left[\begin{array}{ll}1 & 2 \\ 0 & 1\end{array}\right]$	$L = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}, U = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$	2.D.1 (a) $L = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}, U = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}.$
johnston-linear-matrix-atgevrn F04744	470	■ 0.994	$L=\left[\begin{array}{cc}1 & 0 \\ -1 & 1\end{array}\right], U=\left[\begin{array}{ccc}3 & 1 & 2 \\ 0 & -2 & 1\end{array}\right]$	$L = \begin{bmatrix} 1 & 0 \\ -1 & 1 \end{bmatrix}, U = \begin{bmatrix} 3 & 1 & 2 \\ 0 & -2 & 1 \end{bmatrix}$	(c) $L = \begin{bmatrix} 1 & 0 \\ -1 & 1 \end{bmatrix}, U = \begin{bmatrix} 3 & 1 & 2 \\ 0 & -2 & 1 \end{bmatrix}.$
johnston-linear-matrix-atgevrn F04745	470	■ 1.000	$L=\left[\begin{array}{lll}1 & 0 & 0 \\ 2 & 1 & 0 \\ 1 & 3 & 1\end{array}\right], U=\left[\begin{array}{cc}1 & 2 \\ 0 & 0\end{array}\right]$	$L = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 1 & 3 & 1 \end{bmatrix}, U = \begin{bmatrix} 1 & 2 \\ 0 & -1 \\ 0 & 0 \end{bmatrix}$	(e) $L = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 1 & 3 & 1 \end{bmatrix}, U = \begin{bmatrix} 1 & 2 \\ 0 & -1 \\ 0 & 0 \end{bmatrix}.$
johnston-linear-matrix-atgevrn F04746	470	■ 1.000	$L=\left[\begin{array}{ccc}1 & 0 & 0 \\ 3 & 1 & 0 \\ -2 & -1 & 1\end{array}\right], U=\left[\begin{array}{ccc}1 & -4 & 5 \\ 0 & 3 & -7 \\ 0 & 0 & 1\end{array}\right]$	$L = \begin{bmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ -2 & -1 & 1 \end{bmatrix}, U = \begin{bmatrix} 1 & -4 & 5 \\ 0 & 3 & -7 \\ 0 & 0 & 1 \end{bmatrix}$	(g) $L = \begin{bmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ -2 & -1 & 1 \end{bmatrix}, U = \begin{bmatrix} 1 & -4 & 5 \\ 0 & 3 & -7 \\ 0 & 0 & 1 \end{bmatrix}.$
johnston-linear-matrix-atgevrn F04747	470	■ 0.995	$P=\left[\begin{array}{ll}0 & 1 \\ 1 & 0\end{array}\right], L=\left[\begin{array}{ll}1 & 0 \\ 0 & 1\end{array}\right], U=\left[\begin{array}{ll}1 & 3 \\ 0 & 2\end{array}\right]$	$P = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, L = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, U = \begin{bmatrix} 1 & 3 \\ 0 & 2 \end{bmatrix}$	2.D.2 (a) $P = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, L = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, U = \begin{bmatrix} 1 & 3 \\ 0 & 2 \end{bmatrix}.$
johnston-linear-matrix-atgevrn F04748	470	■ 0.999	$P=\left[\begin{array}{lll}1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0\end{array}\right], L=\left[\begin{array}{lll}1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 0 & 1\end{array}\right], U=$	$P = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}, L = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}, U =$	(c) $P = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}, L = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}, U =$
johnston-linear-matrix-atgevrn F04749	470	■ 0.760	$\left[\begin{array}{lll}1 & 1 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & 1\end{array}\right]$	$\begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix}.$
johnston-linear-matrix-atgevrn F04750	470	■ 1.000	$(x, y)=(5, -3)$	$(x, y) = (5, -3)$	2.D.3 (a) $(x, y) = (5, -3).$
johnston-linear-matrix-atgevrn F04751	470	■ 1.000	$(x, y, z)=(-3, 6, -2)$	$(x, y, z) = (-3, 6, -2)$	(c) $(x, y, z) = (-3, 6, -2).$
johnston-linear-matrix-atgevrn F04752	470	■ 1.000	$A=[B \mid C]$	$A = [B \mid C]$	wider than it is tall). If we write $A = [B \mid C]$ and let
johnston-linear-matrix-atgevrn F04753	470	■ 1.000	$\widetilde{U}=\left[U \mid L^{-1} C\right]$	$\widetilde{U} = [U \mid L^{-1}C]$	$\widetilde{U} = [U \mid L^{-1}C]$ (note that L is invertible since its
johnston-linear-matrix-atgevrn F04754	471	■ 1.000	$A=L_{\{1\}} U_{\{1\}}=L_{\{2\}} U_{\{2\}}$	$A = L_1U_1 = L_2U_2$	sitions $A = L_1U_1 = L_2U_2$. Since A is invert-
johnston-linear-matrix-atgevrn F04755	471	■ 1.000	$L_{\{1\}}, L_{\{2\}}, U_{\{1\}}$	L_1, L_2, U_1	of L_1, L_2, U_1 , and U_2 must be invertible as
johnston-linear-matrix-atgevrn F04756	471	■ 1.000	$U_{\{2\}}$	U_2	of L_1, L_2, U_1 , and U_2 must be invertible as
johnston-linear-matrix-atgevrn F04757	471	■ 1.000	$L_{\{1\}}^{-1} L_{\{2\}}=U_{\{1\}} U_{\{2\}}^{-1}$	$L_1^{-1}L_2 = U_1U_2^{-1}$	$L_1^{-1}L_2 = U_1U_2^{-1}.$
johnston-linear-matrix-atgevrn F04758	471	■ 1.000	$L_{\{1\}}^{-1} L_{\{2\}}$	$L_1^{-1}L_2$	lar, it follows that $L_1^{-1}L_2$ and $U_1U_2^{-1}$ must in
johnston-linear-matrix-atgevrn F04759	471	■ 1.000	$U_{\{1\}} U_{\{2\}}^{-1}$	$U_1U_2^{-1}$	lar, it follows that $L_1^{-1}L_2$ and $U_1U_2^{-1}$ must in
johnston-linear-matrix-atgevrn F04760	471	■ 1.000	$L_{\{1\}}^{-1}$	L_1^{-1}	entries of L_1^{-1} and L_2 all equal 1, it follows
johnston-linear-matrix-atgevrn F04761	471	■ 1.000	$L_{\{2\}}$	L_2	entries of L_1^{-1} and L_2 all equal 1, it follows

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johnston-linear-matrix-algebra-F04762	471	1.000	$L_{\{1\}}^{-1} L_{\{2\}}=I$	$L_1^{-1} L_2 = I$	that $L_1^{-1} L_2 = I$, so $L_1 = L_2$. Multiplying the
johnston-linear-matrix-algebra-F04763	471	1.000	$L_{\{1\}}=L_{\{2\}}$	$L_1 = L_2$	that $L_1^{-1} L_2 = I$, so $L_1 = L_2$. Multiplying the
johnston-linear-matrix-algebra-F04764	471	1.000	$L_{\{1\}} U_{\{1\}}=L_{\{2\}} U_{\{2\}}$	$L_1 U_1 = L_2 U_2$	equation $L_1 U_1 = L_2 U_2$ on the left by L_1^{-1} then
johnston-linear-matrix-algebra-F04765	471	1.000	$U_{\{1\}}=U_{\{2\}}$	$U_1 = U_2$	shows that $U_1 = U_2$ as well, so these two unit
johnston-linear-matrix-algebra-F04766	471	1.000	$\underset{\sim}{\sim}\{k\} \times k$	$\tilde{k} \times k$	matrix multiplication shows that the top-left $\tilde{k} \times k$
johnston-linear-matrix-algebra-F04767	471	1.000	$\widetilde{L} \widetilde{U}$	$\tilde{L} \tilde{U}$	submatrix of A has the form $\tilde{L} \tilde{U}$, where $\tilde{L}$ and $\tilde{U}$ are
johnston-linear-matrix-algebra-F04768	471	1.000	$\widetilde{L}$	$\tilde{L}$	submatrix of A has the form $\tilde{L} \tilde{U}$, where $\tilde{L}$ and $\tilde{U}$ are
johnston-linear-matrix-algebra-F04769	471	1.000	$\widetilde{U}$	$\tilde{U}$	submatrix of A has the form $\tilde{L} \tilde{U}$, where $\tilde{L}$ and $\tilde{U}$ are
johnston-linear-matrix-algebra-F04770	471	1.000	$\frac{1}{275} \left[\begin{array}{cccccc} 171 & 67 & 16 & -3 & -28 & -14 \\ 67 & 134 & 32 & -6 & -56 & -28 \\ 17 & 34 & 82 & 19 & -6 & -3 \\ 34 & 82 & 19 & -6 & -3 & 2 \\ 2 & 21 & 82 & 32 & 16 & -13 \\ -13 & -26 & 2 & 34 & 134 & 67 \\ -40 & -80 & -15 & 20 & 95 & 185 \end{array} \right]$	$\frac{1}{275} \left[\begin{array}{cccccc} 171 & 67 & 16 & -3 & -28 & -14 \\ 67 & 134 & 32 & -6 & -56 & -28 \\ 17 & 34 & 82 & 19 & -6 & -3 \\ 34 & 82 & 19 & -6 & -3 & 2 \\ 2 & 21 & 82 & 32 & 16 & -13 \\ -13 & -26 & 2 & 34 & 134 & 67 \\ -40 & -80 & -15 & 20 & 95 & 185 \end{array} \right]$	(b) $\frac{1}{275} \left[\begin{array}{cccccc} 171 & 67 & 16 & -3 & -28 & -14 \\ 67 & 134 & 32 & -6 & -56 & -28 \\ 17 & 34 & 82 & 19 & -6 & -3 \\ 34 & 82 & 19 & -6 & -3 & 2 \\ 2 & 21 & 82 & 32 & 16 & -13 \\ -13 & -26 & 2 & 34 & 134 & 67 \\ -40 & -80 & -15 & 20 & 95 & 185 \end{array} \right]$
johnston-linear-matrix-algebra-F04771	471	1.000	$\widetilde{U}=D U$	$\tilde{U} = D U$	2.D.10 If $A = L D U$ then setting $\tilde{U} = D U$ gives $A = L \tilde{U}$ as a
johnston-linear-matrix-algebra-F04772	471	1.000	$A=L \widetilde{U}$	$A = L \tilde{U}$	2.D.10 If $A = L D U$ then setting $\tilde{U} = D U$ gives $A = L \tilde{U}$ as a
johnston-linear-matrix-algebra-F04773	471	1.000	$\widetilde{U}=D^{-1} U$	$\tilde{U} = D^{-1} U$	$\tilde{U} = D^{-1} U$ to get $A = L \tilde{D} \tilde{U}$ as an LDU decomposi-
johnston-linear-matrix-algebra-F04774	471	1.000	$A=L D \widetilde{U}$	$A = L D \tilde{U}$	$\tilde{U} = D^{-1} U$ to get $A = L \tilde{D} \tilde{U}$ as an LDU decomposi-
johnston-linear-matrix-algebra-F04775	471	1.000	$L D \widetilde{U}=L D D^{-1} U=L U=A$	$L D \tilde{U} = L D D^{-1} U = L U = A$	tion of A (since $L D \tilde{U} = L D D^{-1} U = L U = A$). This
johnston-linear-matrix-algebra-F04776	471	1.000	$L=\left[\begin{array}{ll} 1 & 0 \\ 2 & 1 \end{array}\right], D=\left[\begin{array}{ll} 1 & 0 \\ 0 & 1 \end{array}\right], U=\left[\begin{array}{ll} 1 & 2 \\ 0 & 1 \end{array}\right]$	$L = \left[\begin{array}{cc} 1 & 0 \\ 2 & 1 \end{array} \right], D = \left[\begin{array}{cc} 1 & 0 \\ 0 & 1 \end{array} \right], U = \left[\begin{array}{cc} 1 & 2 \\ 0 & 1 \end{array} \right]$	2.D.11 (a) $L = \left[\begin{array}{cc} 1 & 0 \\ 2 & 1 \end{array} \right], D = \left[\begin{array}{cc} 1 & 0 \\ 0 & 1 \end{array} \right], U = \left[\begin{array}{cc} 1 & 2 \\ 0 & 1 \end{array} \right]$.
johnston-linear-matrix-algebra-F04777	471	1.000	$R_{\{1\}}+c R_{\{2\}}$	$R_1 + c R_2$	top-left corner are $R_1 \leftrightarrow R_2$ or $R_1 + c R_2$, so
johnston-linear-matrix-algebra-F04778	472	1.000	$u_{\{1,1\}}=0$	$u_{1,1} = 0$	shows that $u_{1,1} = 0$. However, this then im-
johnston-linear-matrix-algebra-F04779	472	1.000	$[\mathbf{v}]_B=(2,1 / 2)$	$[\mathbf{v}]_B = (2, 1/2)$	3.1.1 (a) $[\mathbf{v}]_B = (2, 1/2)$, since $2(3,0) + \frac{1}{2}(0,2) = (6,1)$.
johnston-linear-matrix-algebra-F04780	472	1.000	$2(3,0)+\frac{1}{2}(0,2)=(6,1)$	$2(3,0) + \frac{1}{2}(0,2) = (6,1)$	3.1.1 (a) $[\mathbf{v}]_B = (2, 1/2)$, since $2(3,0) + \frac{1}{2}(0,2) = (6,1)$.
johnston-linear-matrix-algebra-F04781	472	1.000	$(6,1)=c_{\{1\}}(3,0)+c_{\{2\}}(0,2)$	$(6,1) = c_1(3,0) + c_2(0,2)$	$(6,1) = c_1(3,0) + c_2(0,2)$.
johnston-linear-matrix-algebra-F04782	472	1.000	$[\mathbf{v}]_B=(1 / 2,3 / 2)$	$[\mathbf{v}]_B = (1/2, 3/2)$	(c) $[\mathbf{v}]_B = (1/2, 3/2)$ since $(2,3,1) =$
johnston-linear-matrix-algebra-F04783	472	1.000	$(2,3,1)=$	$(2,3,1) =$	(c) $[\mathbf{v}]_B = (1/2, 3/2)$ since $(2,3,1) =$
johnston-linear-matrix-algebra-F04784	472	1.000	$\frac{1}{2}(1,0,-1)+\frac{3}{2}(1,2,1)$	$\frac{1}{2}(1,0,-1) + \frac{3}{2}(1,2,1)$	$\frac{1}{2}(1,0,-1) + \frac{3}{2}(1,2,1)$.

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johnston-linear-matrix-atgevr2-F04785	472	1.000	$B=\{(2,3),(2,-1)\}$	$B = \{(2,3), (2,-1)\}$	3.1.2 (a) $B = \{(2,3), (2,-1)\}$ works.
johnston-linear-matrix-atgevr2-F04786	472	1.000	$\left[\begin{array}{ll}1 & 3 \\ 2 & 4\end{array}\right]$	$\begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix}$	(a) $\begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix}$
johnston-linear-matrix-atgevr2-F04787	472	0.999	$\left[\begin{array}{cc}11 / 10 & 5 / 2 \\ 3 / 10 & 1 / 2\end{array}\right]$	$\begin{bmatrix} 11/10 & 5/2 \\ 3/10 & 1/2 \end{bmatrix}$	(c) $\begin{bmatrix} 11/10 & 5/2 \\ 3/10 & 1/2 \end{bmatrix}$
johnston-linear-matrix-atgevr2-F04788	472	1.000	$\left[\begin{array}{cc}2 & 0 \\ -5 & 2\end{array}\right]$	$\begin{bmatrix} 2 & 0 \\ -5 & 2 \end{bmatrix}$	(a) $\begin{bmatrix} 2 & 0 \\ -5 & 2 \end{bmatrix}$
johnston-linear-matrix-atgevr2-F04789	472	1.000	$\left[\begin{array}{cc}2 & 0 \\ 0 & -3\end{array}\right]$	$\begin{bmatrix} 2 & 0 \\ 0 & -3 \end{bmatrix}$	(c) $\begin{bmatrix} 2 & 0 \\ 0 & -3 \end{bmatrix}$
johnston-linear-matrix-atgevr2-F04790	472	1.000	$\left[\begin{array}{ccc}10 & -5 & 0 \\ 5 & -6 & -4 \\ -11 & 5 & -1\end{array}\right]$	$\begin{bmatrix} 10 & -5 & 0 \\ 5 & -6 & -4 \\ -11 & 5 & -1 \end{bmatrix}$	$\begin{bmatrix} 10 & -5 & 0 \\ 5 & -6 & -4 \\ -11 & 5 & -1 \end{bmatrix}$
johnston-linear-matrix-atgevr2-F04791	472	1.000	$\operatorname{tr}(A)=\operatorname{tr}(B)=2$	$\operatorname{tr}(A) = \operatorname{tr}(B) = 2$	$\operatorname{tr}(A) = \operatorname{tr}(B) = 2$.
johnston-linear-matrix-atgevr2-F04792	472	1.000	$\operatorname{tr}(A)=6$	$\operatorname{tr}(A) = 6$	(c) $\operatorname{tr}(A) = 6$ and $\operatorname{tr}(B) = 3$, so they are not similar
johnston-linear-matrix-atgevr2-F04793	472	1.000	$\operatorname{tr}(B)=3$	$\operatorname{tr}(B) = 3$	(c) $\operatorname{tr}(A) = 6$ and $\operatorname{tr}(B) = 3$, so they are not similar
johnston-linear-matrix-atgevr2-F04794	472	0.995	$n!$	$n!$	unit PLU decomposition for each of the $n!$ dif-
johnston-linear-matrix-atgevr2-F04795	472	0.999	$P^T P$	$P^T P$	$\langle i, j \rangle$ -entry of $P^T P$ is
johnston-linear-matrix-atgevr2-F04796	472	0.549	$\mathbf{p}_i \cdot \mathbf{p}_j = 1$	$\mathbf{p}_i \cdot \mathbf{p}_j = 1$	other entries equal to 0, we conclude that $\mathbf{p}_i \cdot \mathbf{p}_i = 1$
johnston-linear-matrix-atgevr2-F04797	472	1.000	$\mathbf{p}_i$	$\mathbf{p}_i$	the other hand, if $i \neq j$ then the columns $\mathbf{p}_i$ and $\mathbf{p}_j$
johnston-linear-matrix-atgevr2-F04798	472	1.000	$\mathbf{p}_i \cdot \mathbf{p}_j = 0$	$\mathbf{p}_i \cdot \mathbf{p}_j = 0$	would be a row of P with two 1s in it), so $\mathbf{p}_i \cdot \mathbf{p}_i = 0$
johnston-linear-matrix-atgevr2-F04799	472	0.953	$P^T P = I$	$P^T P = I$	$P^T P$ equal 0. It follows that $P^T P = I$, so $P^T = P^{-1}$.
johnston-linear-matrix-atgevr2-F04800	472	0.953	$P^T = P^{-1}$	$P^T = P^{-1}$	$P^T P$ equal 0. It follows that $P^T P = I$, so $P^T = P^{-1}$.
johnston-linear-matrix-atgevr2-F04801	472	1.000	$p_{1,2}$	$p_{1,2}$	its solutions have $p_{1,2}$ and $p_{2,2}$ free, with
johnston-linear-matrix-atgevr2-F04802	472	1.000	$p_{2,2}$	$p_{2,2}$	its solutions have $p_{1,2}$ and $p_{2,2}$ free, with
johnston-linear-matrix-atgevr2-F04803	472	1.000	$p_{1,1} = -p_{1,2} / 2$	$p_{1,1} = -p_{1,2}/2$	$p_{1,1} = -p_{1,2}/2$ and $p_{2,1} = -p_{2,2}$. We just need
johnston-linear-matrix-atgevr2-F04804	472	1.000	$p_{2,1} = -p_{2,2}$	$p_{2,1} = -p_{2,2}$	$p_{1,1} = -p_{1,2}/2$ and $p_{2,1} = -p_{2,2}$. We just need
johnston-linear-matrix-atgevr2-F04805	472	1.000	$P_{B \leftarrow B}$	$P_{B \leftarrow B}$	of $P_{B \leftarrow B}$ is thus $\mathbf{e}_j$, so $P_{B \leftarrow B} = I$.
johnston-linear-matrix-atgevr2-F04806	472	1.000	$P^{-1} P = I$	$P^{-1} P = I$	(c) True. This follows from the fact that $P^{-1} P = I$
johnston-linear-matrix-atgevr2-F04807	472	1.000	$P P^{-1} = I$	$P P^{-1} = I$	$P P^{-1} = I$ for all invertible matrices P .
johnston-linear-matrix-atgevr2-F04808	472	1.000	$P A^{-1} = A$	$P A^{-1} = A$	$P A^{-1} = A$ for all invertible P (i.e., A is only
johnston-linear-matrix-atgevr2-F04809	472	1.000	$A=B \in \mathcal{M}_2$	$A = B = I \in \mathcal{M}_2$	(g) False. For example, if $A = B = I \in \mathcal{M}_2$ then
johnston-linear-matrix-atgevr2-F04810	472	1.000	$\operatorname{tr}(A B)=2$	$\operatorname{tr}(A B) = 2$	$\operatorname{tr}(A B) = 2$ but $\operatorname{tr}(A) \operatorname{tr}(B) = 4$.

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johnston-linear-matrix-atgevr2 F04811	472	■ 1.000	$\operatorname{tr}(A) \operatorname{tr}(B)=4$	$\operatorname{tr}(A) \operatorname{tr}(B) = 4$	$\operatorname{tr}(AB) = 2$ but $\operatorname{tr}(A)\operatorname{tr}(B) = 4$.
johnston-linear-matrix-atgevr2 F04812	472	■ 1.000	$\operatorname{tr}(A)=\operatorname{tr}(B)=$	$\operatorname{tr}(A) = \operatorname{tr}(B) =$	(a) $\operatorname{rank}(A) = \operatorname{rank}(B) = 2$ and $\operatorname{tr}(A) = \operatorname{tr}(B) =$
johnston-linear-matrix-atgevr2 F04813	472	■ 1.000	$P \in \mathcal{M}_2$	$P \in \mathcal{M}_2$	matrix $P \in \mathcal{M}_2$ such that
johnston-linear-matrix-atgevr2 F04814	473	■ 1.000	$d_{\{j\}}$	d_j	Since not all of the d_j 's are zero, this is a different
johnston-linear-matrix-atgevr2 F04815	473	■ 1.000	$[\mathbf{w}]_B : [\mathbf{v}]_B =$	$[\mathbf{w}]_B : [\mathbf{v}]_B =$	to the entries of $[\mathbf{v}]_B$ and $[\mathbf{w}]_B$: $[\mathbf{v}]_B =$
johnston-linear-matrix-atgevr2 F04816	473	■ 0.999	$\left(c_{\{1\}}, c_{\{2\}}, \ldots, c_{\{k\}}\right)$	$(c_1, c_2, \ldots, c_k)$	$(c_1, c_2, \ldots, c_k)$ and $[\mathbf{w}]_B = (d_1, d_2, \ldots, d_k)$.
johnston-linear-matrix-atgevr2 F04817	473	■ 0.999	$[\mathbf{w}]_B=\left(d_{\{1\}}, d_{\{2\}}, \ldots, d_{\{k\}}\right)$	$[\mathbf{w}]_B = (d_1, d_2, \ldots, d_k)$	$(c_1, c_2, \ldots, c_k)$ and $[\mathbf{w}]_B = (d_1, d_2, \ldots, d_k)$.
johnston-linear-matrix-atgevr2 F04818	473	■ 1.000	$[\mathbf{v}+\mathbf{w}]_B=\left(c_{\{1\}}+d_{\{1\}}, c_{\{2\}}+\right.$ $\left.\right.$	$[\mathbf{v}+\mathbf{w}]_B = (c_1 + d_1, c_2 +$	which means that $[\mathbf{v}+\mathbf{w}]_B = (c_1 + d_1, c_2 +$
johnston-linear-matrix-atgevr2 F04819	473	■ 0.566	$\left.d_{\{2\}}, \ldots, c_{\{k\}}+d_{\{k\}}\right)$	$d_2, \ldots, c_k + d_k)$	$d_2, \ldots, c_k + d_k)$, which is the same as $[\mathbf{v}]_B +$
johnston-linear-matrix-atgevr2 F04820	473	■ 0.566	$[\mathbf{v}]_B+$	$[\mathbf{v}]_B+$	$d_2, \ldots, c_k + d_k)$, which is the same as $[\mathbf{v}]_B +$
johnston-linear-matrix-atgevr2 F04821	473	■ 1.000	$[\mathbf{v}]_B=\left(d_{\{1\}}, d_{\{2\}}, \ldots, d_{\{k\}}\right)$	$[\mathbf{v}]_B = (d_1, d_2, \ldots, d_k)$	$[\mathbf{v}]_B = (d_1, d_2, \ldots, d_k)$. This means that
johnston-linear-matrix-atgevr2 F04822	473	■ 0.956	$[c \mathbf{v}]_B=\left(c d_{\{1\}}, c d_{\{2\}}, \ldots, c\right.$ $\left.d_{\{k\}}\right)$	$[c\mathbf{v}]_B = (cd_1, cd_2, \ldots, cd_k)$	which means that $[c\mathbf{v}]_B = (cd_1, cd_2, \ldots, cd_k)$,
johnston-linear-matrix-atgevr2 F04823	473	■ 1.000	$c[\mathbf{v}]_B$	$c[\mathbf{v}]_B$	which is the same as $c[\mathbf{v}]_B$.
johnston-linear-matrix-atgevr2 F04824	473	■ 1.000	$[\mathbf{v}]_B=\mathbf{0}$	$[\mathbf{v}]_B = \mathbf{0}$	We just notice that $[\mathbf{v}]_B = \mathbf{0}$ if and only if $\mathbf{v} = \mathbf{0}$,
johnston-linear-matrix-atgevr2 F04825	473	■ 1.000	$c_{\{1\}}=\cdots=c_{\{m\}}=0$	$c_1 = \cdots = c_m = 0$	In particular, $c_1 = \cdots = c_m = 0$ is the only so-
johnston-linear-matrix-atgevr2 F04826	473	■ 0.997	$[\mathbf{v}]_B \in \mathbb{R}^k$	$[\mathbf{v}]_B \in \mathbb{R}^k$	nate vector $[\mathbf{v}]_B \in \mathbb{R}^k$, but conversely every vector
johnston-linear-matrix-atgevr2 F04827	473	■ 1.000	$\mathcal{S}: \mathbf{x}=[\mathbf{v}]_B$	$\mathcal{S}: \mathbf{x} = [\mathbf{v}]_B$	some vector from $\mathcal{S}$: $\mathbf{x} = [\mathbf{v}]_B$ for some $\mathbf{v} \in \mathcal{S}$.
johnston-linear-matrix-atgevr2 F04828	473	■ 0.998	$C=\left\{\mathbf{v}_{\{1\}}, \mathbf{v}_{\{2\}}, \ldots,\right.$ $\left.\mathbf{v}_{\{k\}}\right\}$	$C = \{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_k\}$	that $C = \{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_k\}$ is a basis of $\mathcal{S}$, and (by
johnston-linear-matrix-atgevr2 F04829	473	■ 1.000	$P_{C \leftarrow E} P_{E \leftarrow B}=P_{C \leftarrow B}$	$P_{C \leftarrow E} P_{E \leftarrow B} = P_{C \leftarrow B}$	$P_{C \leftarrow E} P_{E \leftarrow B} = P_{C \leftarrow B}$.
johnston-linear-matrix-atgevr2 F04830	473	■ 1.000	$[T]_B[\mathbf{v}]_B=[T(\mathbf{v})]_B$	$[T]_B[\mathbf{v}]_B = [T(\mathbf{v})]_B$	To see that $[T]_B[\mathbf{v}]_B = [T(\mathbf{v})]_B$, first write $[\mathbf{v}]_B =$
johnston-linear-matrix-atgevr2 F04831	473	■ 1.000	$\left(c_{\{1\}}, \ldots, c_{\{n\}}\right)$	$(c_1, \ldots, c_n)$	$(c_1, \ldots, c_n)$ and then use block matrix multiplication:
johnston-linear-matrix-atgevr2 F04832	473	■ 1.000	$[T(\mathbf{v})]_B=A[\mathbf{v}]_B$	$[T(\mathbf{v})]_B = A[\mathbf{v}]_B$	any matrix such that $[T(\mathbf{v})]_B = A[\mathbf{v}]_B$ for all $\mathbf{v} \in \mathbb{R}^n$.
johnston-linear-matrix-atgevr2 F04833	473	■ 1.000	$[T]_B[\mathbf{v}]_B=A[\mathbf{v}]_B$	$[T]_B[\mathbf{v}]_B = A[\mathbf{v}]_B$	Then $[T]_B[\mathbf{v}]_B = A[\mathbf{v}]_B$ for all $\mathbf{v} \in \mathbb{R}^n$. It follows from
johnston-linear-matrix-atgevr2 F04834	473	■ 1.000	$A=[T] \ B$	$A = [T]B$	Exercise 1.3.18(d) that $A = [T]B$, which completes
johnston-linear-matrix-atgevr2 F04835	473	■ 1.000	$\operatorname{nullity}(A)=n-\operatorname{rank}(A)$	$\operatorname{nullity}(A) = n - \operatorname{rank}(A)$	have $\operatorname{nullity}(A) = n - \operatorname{rank}(A)$. Since we already
johnston-linear-matrix-atgevr2 F04836	474	■ 1.000	$\operatorname{tr}(A+B)$	$\operatorname{tr}(A+B)$	3.1.19 (a) We compute $\operatorname{tr}(A+B)$ directly from the defini-
johnston-linear-matrix-atgevr2 F04837	474	■ 1.000	$\operatorname{tr}(c A)$	$\operatorname{tr}(cA)$	(b) Similarly, we now compute $\operatorname{tr}(cA)$ directly
johnston-linear-matrix-atgevr2 F04838	474	■ 1.000	$\mathbf{e}_{\{i\}} \mathbf{e}_{\{j\}}^T$	$\mathbf{e}_i \mathbf{e}_j^T$	$\mathbf{e}_i \mathbf{e}_j^T$ and $C = \mathbf{e}_j \mathbf{e}_k^T$ for some i, j, k with $i \neq k$:
johnston-linear-matrix-atgevr2 F04839	474	■ 1.000	$C=\mathbf{e}_{\{j\}} \mathbf{e}_{\{k\}}^T$	$C = \mathbf{e}_j \mathbf{e}_k^T$	$\mathbf{e}_i \mathbf{e}_j^T$ and $C = \mathbf{e}_j \mathbf{e}_k^T$ for some i, j, k with $i \neq k$:

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johnston-linear-matrix-atgevrn- F04840	474	1.000	i, j, k	i, j, k	$\mathbf{e}_i \mathbf{e}_j^T$ and $C = \mathbf{e}_j \mathbf{e}_k^T$ for some i, j, k with $i \neq k$:
johnston-linear-matrix-atgevrn- F04841	474	1.000	$i \neq k$	$i \neq k$	$\mathbf{e}_i \mathbf{e}_j^T$ and $C = \mathbf{e}_j \mathbf{e}_k^T$ for some i, j, k with $i \neq k$:
johnston-linear-matrix-atgevrn- F04842	474	1.000	$f(A \ C \ B)=f(A \ B \ C)$	$f(ACB) = f(ABC)$	Since $f(ACB) = f(ABC)$, it follows that $f(A\mathbf{e}_i \mathbf{e}_k^T) =$
johnston-linear-matrix-atgevrn- F04843	474	1.000	$f\left(A \ \mathbf{e}_{\{i\}} \ \mathbf{e}_{\{k\}}^T\right)=$	$f\left(A\mathbf{e}_i \mathbf{e}_k^T\right) =$	Since $f(ACB) = f(ABC)$, it follows that $f(A\mathbf{e}_i \mathbf{e}_k^T) =$
johnston-linear-matrix-atgevrn- F04844	474	0.999	$A=\mathbf{e}_{\{j\}} \ \mathbf{e}_{\{i\}}^T$	$A = \mathbf{e}_j \mathbf{e}_i^T$	0. By now choosing $A = \mathbf{e}_j \mathbf{e}_i^T$, we see that
johnston-linear-matrix-atgevrn- F04845	474	1.000	j, k	j, k	for all j, k . Since f is linear and every matrix $A \in \mathcal{M}_n$
johnston-linear-matrix-atgevrn- F04846	474	1.000	$A=\sum_{j, k=1}^n a_{j, k} \ \mathbf{e}_{\{j\}} \ \mathbf{e}_{\{k\}}^T$	$A = \sum_{j, k=1}^n a_{j, k} \mathbf{e}_j \mathbf{e}_k^T$	can be written in the form $A = \sum_{j, k=1}^n a_{j, k} \mathbf{e}_j \mathbf{e}_k^T$, it fol-
johnston-linear-matrix-atgevrn- F04847	474	0.999	$B=\left\{\mathbf{v}_{\{1\}}, \ \mathbf{v}_{\{2\}}, \ \ldots, \ \mathbf{v}_{\{k\}}\right\}, [\mathbf{v}]_B=\left(c_{\{1\}}, \ c_{\{2\}}, \ \ldots, \ c_{\{k\}}\right)$	$B = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}, [\mathbf{v}]_B = (c_1, c_2, \dots, c_k)$	$B = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}, [\mathbf{v}]_B = (c_1, c_2, \dots, c_k)$ and
johnston-linear-matrix-atgevrn- F04848	474	1.000	$\mathbf{v}_{\{i\}} \cdot \mathbf{v}_{\{j\}}=1$	$\mathbf{v}_i \cdot \mathbf{v}_j = 1$	that $\mathbf{v}_i \cdot \mathbf{v}_j = 1$ if $i = j$ and $\mathbf{v}_i \cdot \mathbf{v}_j = 0$ if $i \neq j$
johnston-linear-matrix-atgevrn- F04849	474	1.000	$\mathbf{v}_{\{i\}} \cdot \mathbf{v}_{\{j\}}=0$	$\mathbf{v}_i \cdot \mathbf{v}_j = 0$	that $\mathbf{v}_i \cdot \mathbf{v}_j = 1$ if $i = j$ and $\mathbf{v}_i \cdot \mathbf{v}_j = 0$ if $i \neq j$
johnston-linear-matrix-atgevrn- F04850	474	1.000	$B=\{(1, 1), (0, 1)\}$	$B = \{(1, 1), (0, 1)\}$	For example, if $B = \{(1, 1), (0, 1)\}$ and $\mathbf{v} =$
johnston-linear-matrix-atgevrn- F04851	474	1.000	$(1, 0), \ \mathbf{w}=(0, 1)$	$(1, 0), \mathbf{w} = (0, 1)$	$(1, 0), \mathbf{w} = (0, 1)$ then $[\mathbf{v}]_B = (1, -1)$ and
johnston-linear-matrix-atgevrn- F04852	474	1.000	$[\mathbf{v}]_B=(1, -1)$	$[\mathbf{v}]_B = (1, -1)$	$(1, 0), \mathbf{w} = (0, 1)$ then $[\mathbf{v}]_B = (1, -1)$ and
johnston-linear-matrix-atgevrn- F04853	474	1.000	$\mathbf{v}_{\{1\}}, \ \mathbf{v}_{\{2\}}, \ \ldots, \ \mathbf{v}_{\{k\}} \in B$	$\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k \in B$	3.1.22 We start by supposing that $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k \in B$ and
johnston-linear-matrix-atgevrn- F04854	474	1.000	$\mathbf{v}_{\{1\}} \cdot \mathbf{0}$	$\mathbf{v}_1 \cdot \mathbf{0}$	this end, we start by computing $\mathbf{v}_1 \cdot \mathbf{0}$ in two different
johnston-linear-matrix-atgevrn- F04855	474	1.000	$\left \mathbf{v}_{\{1\}}\right \neq 0$	$\ \mathbf{v}_1\ \neq 0$	that $\ \mathbf{v}_1\ \neq 0$, so this implies $c_1 = 0$.
johnston-linear-matrix-atgevrn- F04856	474	1.000	$\mathbf{v}_{\{2\}} \cdot \mathbf{0}$	$\mathbf{v}_2 \cdot \mathbf{0}$	A similar computation involving $\mathbf{v}_2 \cdot \mathbf{0}$ shows that
johnston-linear-matrix-atgevrn- F04857	474	1.000	$c_{\{2\}}=0$	$c_2 = 0$	$c_2 = 0$, and so on up to $\mathbf{v}_k \cdot \mathbf{0}$ showing that $c_k = 0$,
johnston-linear-matrix-atgevrn- F04858	474	1.000	$\mathbf{v}_{\{k\}} \cdot \mathbf{0}$	$\mathbf{v}_k \cdot \mathbf{0}$	$c_2 = 0$, and so on up to $\mathbf{v}_k \cdot \mathbf{0}$ showing that $c_k = 0$,
johnston-linear-matrix-atgevrn- F04859	474	1.000	$c_{\{k\}}=0$	$c_k = 0$	$c_2 = 0$, and so on up to $\mathbf{v}_k \cdot \mathbf{0}$ showing that $c_k = 0$,
johnston-linear-matrix-atgevrn- F04860	474	0.976	$\sqrt{\sin ^2(\theta)+\cos ^2(\theta)}=\sqrt{1}=$	$\sqrt{\sin^2(\theta) + \cos^2(\theta)} = \sqrt{1} =$	which has length $\sqrt{\sin^2(\theta) + \cos^2(\theta)} = \sqrt{1} =$
johnston-linear-matrix-atgevrn- F04861	474	1.000	$\left\{\mathbf{u}_{\{1\}}, \ \mathbf{u}_{\{2\}}\right\}$	$\{\mathbf{u}_1, \mathbf{u}_2\}$	(b) Suppose that $\{\mathbf{u}_1, \mathbf{u}_2\}$ is an orthonormal ba-
johnston-linear-matrix-atgevrn- F04862	474	1.000	$\mathbf{u}_{\{1\}}=(a, b)$	$\mathbf{u}_1 = (a, b)$	sis of $\mathbb{R}^2$. Since $\mathbf{u}_1 = (a, b)$ is a unit vector,
johnston-linear-matrix-atgevrn- F04863	474	1.000	$ a \leq 1$	$ a \leq 1$	we know that $ a \leq 1$, so we can find θ such
johnston-linear-matrix-atgevrn- F04864	474	0.999	$\cos (\theta)=a$	$\cos(\theta) = a$	that $\cos(\theta) = a$. Then $b = \pm \sqrt{1 - \cos^2(\theta)} =$
johnston-linear-matrix-atgevrn- F04865	474	0.999	$b=\pm \sqrt{1-\cos ^2(\theta)}=$	$b = \pm \sqrt{1 - \cos^2(\theta)} =$	that $\cos(\theta) = a$. Then $b = \pm \sqrt{1 - \cos^2(\theta)} =$
johnston-linear-matrix-atgevrn- F04866	474	1.000	$\pm \sin (\theta) $	$\pm \sin(\theta) $	$\pm \sin(\theta) $. If b is positive, we have writ-
johnston-linear-matrix-atgevrn- F04867	474	1.000	$\mathbf{u}_{\{1\}}=(\cos (\theta), \sin (\theta))$	$\mathbf{u}_1 = (\cos(\theta), \sin(\theta))$	ten $\mathbf{u}_1 = (\cos(\theta), \sin(\theta))$. If b is negative,

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johnston-linear-matrix-algebra-F04868	474	1.000	$-\theta$	$-\theta$	then replace θ by $-\theta$ so that we have $\mathbf{u}_1 =$
johnston-linear-matrix-algebra-F04869	474	1.000	$\mathbf{u}_1 =$	$\mathbf{u}_1 =$	then replace θ by $-\theta$ so that we have $\mathbf{u}_1 =$
johnston-linear-matrix-algebra-F04870	475	1.000	$\mathbf{u}_2$	$\mathbf{u}_2$	Next, since $\mathbf{u}_2$ is orthogonal to $\mathbf{u}_1$, we know
johnston-linear-matrix-algebra-F04871	475	1.000	$\mathbf{u}_1$	$\mathbf{u}_1$	Next, since $\mathbf{u}_2$ is orthogonal to $\mathbf{u}_1$, we know
johnston-linear-matrix-algebra-F04872	475	0.953	$(-\sin(\theta), \cos(\theta))$	$(-\sin(\theta), \cos(\theta))$	$\mathbf{u}_2$ is a multiple of $(-\sin(\theta), \cos(\theta))$. Since
johnston-linear-matrix-algebra-F04873	475	0.817	$\mathbf{u}_2 =$	$\mathbf{u}_2 =$	are 1 and -1 , so we know that either $\mathbf{u}_2 =$
johnston-linear-matrix-algebra-F04874	475	1.000	$2^7 3^3 = 3456$	$2^7 3^3 = 3456$	(g) $2^7 3^3 = 3456$
johnston-linear-matrix-algebra-F04875	475	1.000	$\operatorname{det}(-A) = (-1)^n \operatorname{det}(A)$	$\det(-A) = (-1)^n \det(A)$	(a) False. $\det(-A) = (-1)^n \det(A)$, so the given
johnston-linear-matrix-algebra-F04876	475	1.000	$A = I_2$	$A = I_2$	(c) False. For example, if $A = I_2$ and $B = -I_2$
johnston-linear-matrix-algebra-F04877	475	1.000	$B = -I_2$	$B = -I_2$	(c) False. For example, if $A = I_2$ and $B = -I_2$
johnston-linear-matrix-algebra-F04878	475	1.000	$\operatorname{det}(A+B) = \operatorname{det}(O) = 0$	$\det(A+B) = \det(O) = 0$	then $\det(A+B) = \det(O) = 0$, but $\det(A) +$
johnston-linear-matrix-algebra-F04879	475	1.000	$\operatorname{det}(A) +$	$\det(A) +$	then $\det(A+B) = \det(O) = 0$, but $\det(A) +$
johnston-linear-matrix-algebra-F04880	475	1.000	$\operatorname{det}(B) = 1 + 1 = 2$	$\det(B) = 1 + 1 = 2$	$\det(B) = 1 + 1 = 2$.
johnston-linear-matrix-algebra-F04881	475	0.980	$\operatorname{det}(\left[\begin{smallmatrix} P_{\mathbf{u}} \end{smallmatrix}\right]) = 0$	$\det([P_{\mathbf{u}}]) = 0$	$\det([P_{\mathbf{u}}]) = 0$.
johnston-linear-matrix-algebra-F04882	475	1.000	$\mathbf{u}_2 = (\sin(\theta), -\cos(\theta))$	$\mathbf{u}_2 = (\sin(\theta), -\cos(\theta))$	$(-\sin(\theta), \cos(\theta))$ or $\mathbf{u}_2 = (\sin(\theta), -\cos(\theta))$.
johnston-linear-matrix-algebra-F04883	475	1.000	$\mathbf{u}_1 = R^\theta(\mathbf{e}_1)$	$\mathbf{u}_1 = R^\theta(\mathbf{e}_1)$	In the former case, we are done: $\mathbf{u}_1 = R^\theta(\mathbf{e}_1)$
johnston-linear-matrix-algebra-F04884	475	1.000	$\mathbf{u}_2 = R^\theta(\mathbf{e}_2)$	$\mathbf{u}_2 = R^\theta(\mathbf{e}_2)$	and $\mathbf{u}_2 = R^\theta(\mathbf{e}_2)$. In the latter case, we use
johnston-linear-matrix-algebra-F04885	475	1.000	$\sin(\theta) = \cos(\theta - \pi/2)$	$\sin(\theta) = \cos(\theta - \pi/2)$	the facts that $\sin(\theta) = \cos(\theta - \pi/2)$ and
johnston-linear-matrix-algebra-F04886	475	0.650	$-\cos(\theta) = \sin(\theta - \pi/2)$	$-\cos(\theta) = \sin(\theta - \pi/2)$	$-\cos(\theta) = \sin(\theta - \pi/2)$ to see that $\mathbf{u}_1 =$
johnston-linear-matrix-algebra-F04887	475	1.000	$R_{\theta - \pi/2}(\mathbf{e}_2)$	$R_{\theta - \pi/2}(\mathbf{e}_2)$	$R_{\theta - \pi/2}(\mathbf{e}_2)$ and $\mathbf{u}_2 = R^{\theta - \pi/2}(\mathbf{e}_1)$, so again we
johnston-linear-matrix-algebra-F04888	475	1.000	$\mathbf{u}_2 = R^{\theta - \pi/2}(\mathbf{e}_1)$	$\mathbf{u}_2 = R^{\theta - \pi/2}(\mathbf{e}_1)$	$R_{\theta - \pi/2}(\mathbf{e}_2)$ and $\mathbf{u}_2 = R^{\theta - \pi/2}(\mathbf{e}_1)$, so again we
johnston-linear-matrix-algebra-F04889	475	1.000	$\operatorname{det}(A^T) = 0$	$\det(A^T) = 0$	must also not be invertible, so $\det(A^T) = 0$ as
johnston-linear-matrix-algebra-F04890	475	1.000	$E_1 E_2 \cdots E_k$	$E_1 E_2 \cdots E_k$	$E_1 E_2 \cdots E_k$. Then $\det(A) = \det(E_1 E_2 \cdots E_k) =$
johnston-linear-matrix-algebra-F04891	475	1.000	$\operatorname{det}(A) = \operatorname{det}(\left(E_1 E_2 \cdots E_k\right)) =$	$\det(A) = \det(E_1 E_2 \cdots E_k) =$	$E_1 E_2 \cdots E_k$. Then $\det(A) = \det(E_1 E_2 \cdots E_k) =$
johnston-linear-matrix-algebra-F04892	475	1.000	$\operatorname{det}(\left(E_1\right)) \cdots \operatorname{det}(\left(E_k\right))$	$\det(E_1) \cdots \det(E_k)$	$\det(E_1) \cdots \det(E_k)$. Also, $A^T = E_{k\ldots 1}^T \cdots E_2^T E_1^T$,
johnston-linear-matrix-algebra-F04893	475	1.000	$A^T = E_k^T \cdots E_2^T E_1^T$	$A^T = E_k^T \cdots E_2^T E_1^T$	$\det(E_1) \cdots \det(E_k)$. Also, $A^T = E_{k\ldots 1}^T \cdots E_2^T E_1^T$,
johnston-linear-matrix-algebra-F04894	475	1.000	$\operatorname{det}(\left(A^T\right)) = \operatorname{det}(\left(E_k^T \cdots E_2^T E_1^T\right)) =$	$\det(A^T) = \det(E_k^T \cdots E_2^T E_1^T) =$	so $\det(A^T) = \det(E_k^T \cdots E_2^T E_1^T) =$
johnston-linear-matrix-algebra-F04895	475	1.000	$\operatorname{det}(\left(E_k^T\right)) \cdots \operatorname{det}(\left(E_2^T\right)) \operatorname{det}(\left(E_1^T\right))$	$\det(E_k^T) \cdots \det(E_2^T) \det(E_1^T)$	$\det(E_k^T) \cdots \det(E_2^T) \det(E_1^T)$, so it suffices to

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johnston-linear-matrix-atgevr F04896	475	1.000	$\operatorname{det}(E)=\operatorname{det}(\left(E^{\mathrm{T}}\right.\left.\right)$	$\det(E)=\det(E^T)$	show that $\det(E)=\det(E^T)$ whenever E is an
johnston-linear-matrix-atgevr F04897	475	1.000	$E^{\mathrm{T}}=E$	$E^T=E$	operation cR_i then $E^T=E$, so it is trivially
johnston-linear-matrix-atgevr F04898	475	1.000	$\operatorname{det}(\left(E^{\mathrm{T}}\right)\operatorname{det}(E)$	$\det(E^T)=\det(E)$	the case that $\det(E^T)=\det(E)$. For the one
johnston-linear-matrix-atgevr F04899	475	1.000	$\operatorname{det}(C)$	$\det(C)$	on the right have determinants equal to $\det(C)$
johnston-linear-matrix-atgevr F04900	476	1.000	$\widetilde{B}$	$\widetilde{B}$	where $\widetilde{B}$ is the matrix obtained from B by re-
johnston-linear-matrix-atgevr F04901	476	1.000	$\operatorname{det}(\left(I_m+A\right.\left.B\right)=\operatorname{det}(\left(I_n+B\right.\left.A\right)$	$\det(I_m+AB)=\det(I_n+BA)$	We thus conclude that $\det(I_m+AB)=\det(I_n+BA)$,
johnston-linear-matrix-atgevr F04902	476	1.000	$\operatorname{det}(\left(I_n+\mathbf{v}\right.\mathbf{w}^{\mathrm{T}}\right)=1+\mathbf{w}^{\mathrm{T}}\mathbf{v}$	$\det(I_n+\mathbf{v}\mathbf{w}^T)=1+\mathbf{w}^T\mathbf{v}$	so $\det(I_n+\mathbf{v}\mathbf{w}^T)=1+\mathbf{w}^T\mathbf{v}$.
johnston-linear-matrix-atgevr F04903	477	1.000	$\mathbf{v}=(5,-4)$	$\mathbf{v}=(5,-4)$	3.3.2 (a) A corresponding eigenvector is $\mathbf{v}=(5,-4)$.
johnston-linear-matrix-atgevr F04904	477	1.000	$\mathbf{v}=(3,2,0)$	$\mathbf{v}=(3,2,0)$	(c) $\mathbf{v}=(3,2,0)$
johnston-linear-matrix-atgevr F04905	477	1.000	$\mathbf{v}=(-1,1,0,1)$	$\mathbf{v}=(-1,1,0,1)$	(c) $\mathbf{v}=(-1,1,0,1)$
johnston-linear-matrix-atgevr F04906	477	1.000	$(A-\mathbf{0}\ I)\mathbf{v}=\mathbf{0}$	$(A-0I)\mathbf{v}=\mathbf{0}$	$(A-0I)\mathbf{v}=\mathbf{0}$:
johnston-linear-matrix-atgevr F04907	477	1.000	$v_1=-2\ v_2$	$v_1=-2v_2$	$v_1=-2v_2$, so all eigenvectors are of the form
johnston-linear-matrix-atgevr F04908	477	1.000	$\mathbf{v}=v_2(-2,1)$	$\mathbf{v}=v_2(-2,1)$	$\mathbf{v}=v_2(-2,1)$. It follows that $\{(-2,1)\}$ is a ba-
johnston-linear-matrix-atgevr F04909	477	1.000	$\{(-2,1)\}$	$\{(-2,1)\}$	$\mathbf{v}=v_2(-2,1)$. It follows that $\{(-2,1)\}$ is a ba-
johnston-linear-matrix-atgevr F04910	477	1.000	$\mathbf{v}=v_2(-1,1)$	$\mathbf{v}=v_2(-1,1)$	$\mathbf{v}=v_2(-1,1)$. It follows that $\{(-1,1)\}$ is a
johnston-linear-matrix-atgevr F04911	477	1.000	$\{(-1,1)\}$	$\{(-1,1)\}$	$\mathbf{v}=v_2(-1,1)$. It follows that $\{(-1,1)\}$ is a
johnston-linear-matrix-atgevr F04912	477	1.000	$\{(1,0)\}$	$\{(1,0)\}$	eigenspace is $\{(1,0)\}$, so this eigenvalue has
johnston-linear-matrix-atgevr F04913	477	1.000	$\left\{\mathbf{e}_1\right\},\left\{\mathbf{e}_2\right\}$	$\{\mathbf{e}_1\},\{\mathbf{e}_2\}$	are $\{\mathbf{e}_1\},\{\mathbf{e}_2\}$, and $\{\mathbf{e}_3\}$, respectively.
johnston-linear-matrix-atgevr F04914	477	1.000	$\{(1,0,0,0)\},\{(1,-5,0,0)\}$	$\{(1,0,0,0)\},\{(1,-5,0,0)\}$	eigenspaces are $\{(1,0,0,0)\},\{(1,-5,0,0)\}$,
johnston-linear-matrix-atgevr F04915	477	0.998	$\{(1,-1,-2,0)\}$	$\{(1,-1,-2,0)\}$	and $\{(1,-1,-2,0)\}$, respectively, so all of the
johnston-linear-matrix-atgevr F04916	477	1.000	$\{(1,1,-1,-1)\},\{(2,-1,2,-1)\}$	$\{(1,1,-1,-1)\},\{(2,-1,2,-1)\}$	are $\{(1,1,-1,-1)\},\{(2,-1,2,-1)\}$, and
johnston-linear-matrix-atgevr F04917	477	1.000	$\{(1,1,0,-1),(0,1,-2,0)\}$	$\{(1,1,0,-1),(0,1,-2,0)\}$	$\{(1,1,0,-1),(0,1,-2,0)\}$, respectively.
johnston-linear-matrix-atgevr F04918	477	1.000	$\{(0,0,1,0,-1),(1,0,0,2,0)\}$	$\{(0,0,1,0,-1),(1,0,0,2,0)\}$	eigenspaces are $\{(0,0,1,0,-1),(1,0,0,2,0)\}$,
johnston-linear-matrix-atgevr F04919	477	0.998	$\{(1,-1,-1,0,1),(2,-2,0,1,0)\}$	$\{(1,-1,-1,0,1),(2,-2,0,1,0)\}$	$\{(1,-1,-1,0,1),(2,-2,0,1,0)\}$, and
johnston-linear-matrix-atgevr F04920	477	0.999	$\{(0,2,2,-1,0)\}$	$\{(0,2,2,-1,0)\}$	$\{(0,2,2,-1,0)\}$, respectively.
johnston-linear-matrix-atgevr F04921	477	1.000	$\operatorname{det}(A-\lambda I)=(-1)^n\operatorname{det}(\lambda I-A)$	$\det(A-\lambda I)=(-1)^n\det(\lambda I-A)$	made is $\det(A-\lambda I)=(-1)^n\det(\lambda I-A)$.
johnston-linear-matrix-atgevr F04922	477	1.000	$\lambda=\frac{1}{2}(1+k)\pm\sqrt{(1+k)^2-4k-4}$	$\lambda=\frac{1}{2}(1+k)\pm\frac{1}{2}\sqrt{(1+k)^2-4k-4}=$	λ , we get $\lambda=\frac{1}{2}(1+k)\pm\frac{1}{2}\sqrt{(1+k)^2-4k-4}=$
johnston-linear-matrix-atgevr F04923	477	1.000	$\frac{1}{2}(1+k)\pm\sqrt{k^2-2k-3}$	$\frac{1}{2}(1+k)\pm\frac{1}{2}\sqrt{k^2-2k-3}$	$\frac{1}{2}(1+k)\pm\frac{1}{2}\sqrt{k^2-2k-3}$. We get two distinct
johnston-linear-matrix-atgevr F04924	477	1.000	$k^2-2k-3>0$	$k^2-2k-3>0$	$k^2-2k-3>0$. Similarly, we get only one real

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johnston-linear-matrix-atgevro F04925	477	■ 1.000	k^2-2k-3	k^2-2k-3	The discriminant k^2-2k-3 factors as $(k-3)(k+1)$,
johnston-linear-matrix-atgevro F04926	477	■ 1.000	$(k-3)(k+1)$	$(k-3)(k+1)$	The discriminant k^2-2k-3 factors as $(k-3)(k+1)$,
johnston-linear-matrix-atgevro F04927	477	■ 1.000	$k>3$	$k>3$	when $k>3$ or $k<-1$, (ii) we get exactly one distinct
johnston-linear-matrix-atgevro F04928	477	■ 1.000	$k<-1$	$k<-1$	when $k>3$ or $k<-1$, (ii) we get exactly one distinct
johnston-linear-matrix-atgevro F04929	477	■ 1.000	$k=3$	$k=3$	eigenvalue when $k=3$ or $k=-1$, and (iii) we get
johnston-linear-matrix-atgevro F04930	477	■ 1.000	$k=-1$	$k=-1$	eigenvalue when $k=3$ or $k=-1$, and (iii) we get
johnston-linear-matrix-atgevro F04931	477	■ 1.000	$-1<k<3$	$-1<k<3$	no real eigenvalues when $-1<k<3$.
johnston-linear-matrix-atgevro F04932	477	■ 1.000	$p_A(\lambda)=\lambda^2-\operatorname{tr}(A)\lambda+\operatorname{det}(A)$	$p_A(\lambda)=\lambda^2-\operatorname{tr}(A)\lambda+\operatorname{det}(A)$	matrix A is $p_A(\lambda)=\lambda^2-\operatorname{tr}(A)\lambda+\operatorname{det}(A)$. Applying
johnston-linear-matrix-atgevro F04933	478	■ 1.000	$\lambda=\pm 1$	$\lambda=\pm 1$	3.3.12 The only possible eigenvalues are $\lambda=\pm 1$ and
johnston-linear-matrix-atgevro F04934	478	■ 1.000	$\lambda=\pm i$	$\lambda=\pm i$	$\lambda=\pm i$. The fact that these eigenvalues are all pos-
johnston-linear-matrix-atgevro F04935	478	■ 1.000	$A^4\mathbf{v}=\lambda^4\mathbf{v}$	$A^4\mathbf{v}=\lambda^4\mathbf{v}$	shows that $A^4\mathbf{v}=\lambda^4\mathbf{v}$. On the other hand, $A^4=I$, so
johnston-linear-matrix-atgevro F04936	478	■ 0.999	$A^4\mathbf{v}=\mathbf{v}$	$A^4\mathbf{v}=\mathbf{v}$	$A^4\mathbf{v}=\mathbf{v}$, so $\lambda^4\mathbf{v}=\mathbf{v}$, so $\lambda^4=1$.
johnston-linear-matrix-atgevro F04937	478	■ 0.999	$\lambda^4\mathbf{v}=\mathbf{v}$	$\lambda^4\mathbf{v}=\mathbf{v}$	$A^4\mathbf{v}=\mathbf{v}$, so $\lambda^4\mathbf{v}=\mathbf{v}$, so $\lambda^4=1$.
johnston-linear-matrix-atgevro F04938	478	■ 0.999	$\lambda^4=1$	$\lambda^4=1$	$A^4\mathbf{v}=\mathbf{v}$, so $\lambda^4\mathbf{v}=\mathbf{v}$, so $\lambda^4=1$.
johnston-linear-matrix-atgevro F04939	478	■ 1.000	$\operatorname{det}(\left[R^\theta\right]-\lambda I)=0$	$\operatorname{det}([R^\theta]-\lambda I)=0$	3.3.13 (a) Setting $\operatorname{det}([R^\theta]-\lambda I)=0$ gives
johnston-linear-matrix-atgevro F04940	478	■ 1.000	$\lambda=\cos(\theta)\pm\sqrt{\cos^2(\theta)-1}$	$\lambda=\cos(\theta)\pm\sqrt{\cos^2(\theta)-1}$	mula) gives $\lambda=\cos(\theta)\pm\sqrt{\cos^2(\theta)-1}$.
johnston-linear-matrix-atgevro F04941	478	■ 1.000	$\lambda=\cos(\theta)\pm$	$\lambda=\cos(\theta)\pm$	(b) We found the eigenvalues to be $\lambda=\cos(\theta)\pm$
johnston-linear-matrix-atgevro F04942	478	■ 0.978	$\sqrt{\cos^2(\theta)-1}$	$\sqrt{\cos^2(\theta)-1}$	$\sqrt{\cos^2(\theta)-1}$, which are real if and only if
johnston-linear-matrix-atgevro F04943	478	■ 0.999	$\cos^2(\theta)-1\geq 0$	$\cos^2(\theta)-1\geq 0$	$\cos^2(\theta)-1\geq 0$, i.e., $\cos(\theta)=\pm 1$. In other
johnston-linear-matrix-atgevro F04944	478	■ 0.999	$\cos(\theta)=\pm 1$	$\cos(\theta)=\pm 1$	$\cos^2(\theta)-1\geq 0$, i.e., $\cos(\theta)=\pm 1$. In other
johnston-linear-matrix-atgevro F04945	478	■ 1.000	$\theta=k\pi$	$\theta=k\pi$	words, $\theta=k\pi$ for some integer k .
johnston-linear-matrix-atgevro F04946	478	■ 1.000	$\theta=2k\pi$	$\theta=2k\pi$	$\theta=2k\pi$ then all vectors are being rotated back
johnston-linear-matrix-atgevro F04947	478	■ 0.999	$\theta=(2k+1)\pi$	$\theta=(2k+1)\pi$	value 1). If $\theta=(2k+1)\pi$ then every vector is
johnston-linear-matrix-atgevro F04948	478	■ 1.000	$\theta\neq k\pi$	$\theta\neq k\pi$	if $\theta\neq k\pi$, then every vector is rotated so that
johnston-linear-matrix-atgevro F04949	478	■ 1.000	$\mathbf{v}\cdot\mathbf{w}=\mathbf{v}^*\mathbf{w}$	$\mathbf{v}\cdot\mathbf{w}=\mathbf{v}^*\mathbf{w}$	tors then $\mathbf{v}\cdot\mathbf{w}=\mathbf{v}^*\mathbf{w}$. It follows that $\mathbf{x}\cdot(A\mathbf{y})=$
johnston-linear-matrix-atgevro F04950	478	■ 1.000	$\mathbf{x}^*A\mathbf{y}$	$\mathbf{x}^*A\mathbf{y}$	$\mathbf{x}^*A\mathbf{y}$ and $(A^*\mathbf{x})\cdot\mathbf{y}=(A^*\mathbf{x})^*\mathbf{y}=\mathbf{x}^*A\mathbf{y}$. Since
johnston-linear-matrix-atgevro F04951	478	■ 1.000	$(A^*\mathbf{x})\cdot\mathbf{y}=(A^*\mathbf{x})^*\mathbf{y}=\mathbf{x}^*A\mathbf{y}$	$(A^*\mathbf{x})\cdot\mathbf{y}=(A^*\mathbf{x})^*\mathbf{y}=\mathbf{x}^*A\mathbf{y}$	$\mathbf{x}^*A\mathbf{y}$ and $(A^*\mathbf{x})\cdot\mathbf{y}=(A^*\mathbf{x})^*\mathbf{y}=\mathbf{x}^*A\mathbf{y}$. Since
johnston-linear-matrix-atgevro F04952	478	■ 1.000	$\mathbf{x}\cdot(A\mathbf{y})=(A^*\mathbf{x})\cdot\mathbf{y}$	$\mathbf{x}\cdot(A\mathbf{y})=(A^*\mathbf{x})\cdot\mathbf{y}$	$\mathbf{x}\cdot(A\mathbf{y})=(A^*\mathbf{x})\cdot\mathbf{y}$.

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johnston-linear-matrix-atgenvr0 F04953	478	■ 1.000	$\mathbf{x} \cdot (A \mathbf{y}) = \mathbf{x}^* A \mathbf{y}$	$\mathbf{x} \cdot (A\mathbf{y}) = \mathbf{x}^* A\mathbf{y}$	(b) Just like in part (a), $\mathbf{x} \cdot (A\mathbf{y}) = \mathbf{x}^* A\mathbf{y}$ and
johnston-linear-matrix-atgenvr0 F04954	478	■ 1.000	$(B \mathbf{x}) \cdot \mathbf{y} = \mathbf{x}^* B^* \mathbf{y}$	$(B\mathbf{x}) \cdot \mathbf{y} = \mathbf{x}^* B^* \mathbf{y}$	$(B\mathbf{x}) \cdot \mathbf{y} = \mathbf{x}^* B^* \mathbf{y}$. We thus know that $\mathbf{x}^* A\mathbf{y} =$
johnston-linear-matrix-atgenvr0 F04955	478	■ 1.000	$\mathbf{x}^* A \mathbf{y} =$	$\mathbf{x}^* A\mathbf{y} =$	$(B\mathbf{x}) \cdot \mathbf{y} = \mathbf{x}^* B^* \mathbf{y}$. We thus know that $\mathbf{x}^* A\mathbf{y} =$
johnston-linear-matrix-atgenvr0 F04956	478	■ 0.999	$\mathbf{x}^* B^* \mathbf{y}$	$\mathbf{x}^* B^* \mathbf{y}$	$\mathbf{x}^* B^* \mathbf{y}$ for all $\mathbf{x} \in \mathbb{C}^m$ and $\mathbf{y} \in \mathbb{C}^n$. However, if
johnston-linear-matrix-atgenvr0 F04957	478	■ 0.999	$\mathbf{x} \in \mathbb{C}^m$	$\mathbf{x} \in \mathbb{C}^m$	$\mathbf{x}^* B^* \mathbf{y}$ for all $\mathbf{x} \in \mathbb{C}^m$ and $\mathbf{y} \in \mathbb{C}^n$. However, if
johnston-linear-matrix-atgenvr0 F04958	478	■ 0.999	$\mathbf{y} \in \mathbb{C}^n$	$\mathbf{y} \in \mathbb{C}^n$	$\mathbf{x}^* B^* \mathbf{y}$ for all $\mathbf{x} \in \mathbb{C}^m$ and $\mathbf{y} \in \mathbb{C}^n$. However, if
johnston-linear-matrix-atgenvr0 F04959	478	■ 1.000	$\mathbf{x}^* A \mathbf{y} = a_{i,j}$	$\mathbf{x}^* A\mathbf{y} = a_{i,j}$	we choose $\mathbf{x} = \mathbf{e}_i$ and $\mathbf{y} = \mathbf{e}_j$ then $\mathbf{x}^* A\mathbf{y} = a_{i,j}$
johnston-linear-matrix-atgenvr0 F04960	478	■ 1.000	$\mathbf{x}^* B^* \mathbf{y} = b_{j,i}$	$\mathbf{x}^* B^* \mathbf{y} = b_{j,i}$	and $\mathbf{x}^* B^* \mathbf{y} = b_{j,i}$, so we see that $a_{i,j} = \overline{b_{j,i}}$ for
johnston-linear-matrix-atgenvr0 F04961	478	■ 1.000	$a_{i,j} = \overline{b_{j,i}}$	$a_{i,j} = \overline{b_{j,i}}$	and $\mathbf{x}^* B^* \mathbf{y} = b_{j,i}$, so we see that $a_{i,j} = \overline{b_{j,i}}$ for
johnston-linear-matrix-atgenvr0 F04962	478	■ 0.975	$A \mathbf{v} = 0 \mathbf{v} = \mathbf{0}$	$A\mathbf{v} = 0\mathbf{v} = \mathbf{0}$	exists a non-zero vector $\mathbf{v}$ such that $A\mathbf{v} = 0\mathbf{v} = \mathbf{0}$,
johnston-linear-matrix-atgenvr0 F04963	478	■ 1.000	$(A^*)^* = A$	$(A^*)^* = A$	lows from the fact that $(A^*)^* = A$ so if $\bar{\lambda}$ is an
johnston-linear-matrix-atgenvr0 F04964	478	■ 1.000	$\overline{\bar{\lambda}} = \lambda$	$\bar{\bar{\lambda}} = \lambda$	eigenvalue of A^* then $\bar{\bar{\lambda}} = \lambda$ is an eigenvalue
johnston-linear-matrix-atgenvr0 F04965	478	■ 1.000	$\mathbf{w} \cdot (A\mathbf{v})$	$\mathbf{w} \cdot (A\mathbf{v})$	(b) We compute $\mathbf{w} \cdot (A\mathbf{v})$ in two different ways:
johnston-linear-matrix-atgenvr0 F04966	478	■ 1.000	$\mathbf{w} \cdot \mathbf{v} = 0$	$\mathbf{w} \cdot \mathbf{v} = 0$	is if $\mathbf{w} \cdot \mathbf{v} = 0$ (or equivalently, $\mathbf{v} \cdot \mathbf{w} = 0$).
johnston-linear-matrix-atgenvr0 F04967	478	■ 1.000	$\lambda_2, \lambda_3, \dots, \lambda_n$	$\lambda_2, \lambda_3, \dots, \lambda_n$	its other eigenvalues are $\lambda_2, \lambda_3, \dots, \lambda_n$. To this
johnston-linear-matrix-atgenvr0 F04968	478	■ 1.000	$\overline{\lambda_j}$	$\overline{\lambda_j}$	ing to the eigenvalue $\overline{\lambda_j}$ for $2 \leq j \leq n$. Then
johnston-linear-matrix-atgenvr0 F04969	478	■ 1.000	$\lambda_1 \neq$	$\lambda_1 \neq$	Exercise 3.3.21 and the assumption that $\lambda_1 \neq$
johnston-linear-matrix-atgenvr0 F04970	478	■ 1.000	$\lambda_1 = \lambda_j$	$\lambda_1 = \lambda_j$	λ_j . The statement is still true even if $\lambda_1 = \lambda_j$,
johnston-linear-matrix-atgenvr0 F04971	478	■ 1.000	B^*	B^*	It follows that B^* has $\overline{\lambda_2}, \overline{\lambda_3}, \dots, \overline{\lambda_n}$ as eigen-
johnston-linear-matrix-atgenvr0 F04972	478	■ 1.000	$\overline{\lambda_2}, \overline{\lambda_3}, \dots, \overline{\lambda_n}$	$\overline{\lambda_2}, \overline{\lambda_3}, \dots, \overline{\lambda_n}$	It follows that B^* has $\overline{\lambda_2}, \overline{\lambda_3}, \dots, \overline{\lambda_n}$ as eigen-
johnston-linear-matrix-atgenvr0 F04973	478	■ 1.000	$\lambda_1 = 1$	$\lambda_1 = 1$	which has eigenvalues $\lambda_1 = 1$ and $\lambda_2 = 2$ with
johnston-linear-matrix-atgenvr0 F04974	478	■ 1.000	$\lambda_2 = 2$	$\lambda_2 = 2$	which has eigenvalues $\lambda_1 = 1$ and $\lambda_2 = 2$ with
johnston-linear-matrix-atgenvr0 F04975	478	■ 1.000	$\mathbf{v}_1 = (1, 0)$	$\mathbf{v}_1 = (1, 0)$	corresponding unit eigenvectors $\mathbf{v}_1 = (1, 0)$
johnston-linear-matrix-atgenvr0 F04976	478	■ 1.000	$\mathbf{v}_2 = (1, 1) / \sqrt{2}$	$\mathbf{v}_2 = (1, 1) / \sqrt{2}$	and $\mathbf{v}_2 = (1, 1) / \sqrt{2}$, respectively. Then
johnston-linear-matrix-atgenvr0 F04977	478	■ 1.000	$\mathbf{v}_2 = (1, 2)$	$\mathbf{v}_2 = (1, 2)$	multiples of $\mathbf{v}_2 = (1, 2)$.
johnston-linear-matrix-atgenvr0 F04978	479	■ 1.000	$A \mathbf{1} = \mathbf{1}$	$A\mathbf{1} = \mathbf{1}$	$A\mathbf{1} = \mathbf{1}$, since $A\mathbf{1}$ is simply the vector whose
johnston-linear-matrix-atgenvr0 F04979	479	■ 1.000	$A \mathbf{1}$	$A\mathbf{1}$	$A\mathbf{1} = \mathbf{1}$, since $A\mathbf{1}$ is simply the vector whose
johnston-linear-matrix-atgenvr0 F04980	479	■ 0.973	$ v_j \leq 1$	$ v_j \leq 1$	fact that we scaled $\mathbf{v}$ so that $ v_j \leq 1$ for all j .
johnston-linear-matrix-atgenvr0 F04981	479	■ 1.000	$\lambda = i$	$\lambda = i$	(g) Not diagonalizable, since its eigenvalue $\lambda = i$

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johnston-linear-matrix-atlas- F04982	480	■ 1.000	2+i, 2+i, 2-i, 2-i	$2+i, 2+i, 2-i, 2-i$	with diagonal entries $2+i, 2+i, 2-i, 2-i$,
johnston-linear-matrix-atlas- F04983	480	■ 0.999	$\left[\begin{array}{ll}\sin(2)/2 & \sin(2)/2 \\ \sin(2)/2 & \sin(2)/2\end{array}\right]$	$\begin{bmatrix} \sin(2)/2 & \sin(2)/2 \\ \sin(2)/2 & \sin(2)/2 \end{bmatrix}$	(b) $\begin{bmatrix} \sin(2)/2 & \sin(2)/2 \\ \sin(2)/2 & \sin(2)/2 \end{bmatrix}$
johnston-linear-matrix-atlas- F04984	480	■ 1.000	P=D=0	$P = D = O$	choose $P = D = O$. To make this statement
johnston-linear-matrix-atlas- F04985	480	■ 1.000	A=P D P^{-1}=P(\lambda I) P^{-1}=\lambda P P^{-1}=\lambda I	$A = PDP^{-1} = P(\lambda I)P^{-1} = \lambda PP^{-1} = \lambda I$	$A = PDP^{-1} = P(\lambda I)P^{-1} = \lambda PP^{-1} = \lambda I$.
johnston-linear-matrix-atlas- F04986	480	■ 1.000	L_{0}=2	$L_0 = 2$	except for the initial condition $L_0 = 2$ instead of
johnston-linear-matrix-atlas- F04987	480	■ 0.999	2 \phi-1=\sqrt{5}	$2\phi - 1 = \sqrt{5}$	where we used the fact that $2\phi - 1 = \sqrt{5}$ in the fourth
johnston-linear-matrix-atlas- F04988	480	■ 1.000	$\operatorname{rank}(A)=\operatorname{rank}\left(P D P^{-1}\right)=\operatorname{rank}\left(D P^{-1}\right)=$	$\operatorname{rank}(A) = \operatorname{rank}\left(PDP^{-1}\right) = \operatorname{rank}\left(DP^{-1}\right) =$	$\operatorname{rank}(A) = \operatorname{rank}(PDP^{-1}) = \operatorname{rank}(DP^{-1}) =$
johnston-linear-matrix-atlas- F04989	480	■ 0.996	$\operatorname{rank}(D)$	$\operatorname{rank}(D)$	$\operatorname{rank}(D)$. Since D is diagonal, its rank is the
johnston-linear-matrix-atlas- F04990	480	■ 1.000	B=P D P^{-1}	$B = PDP^{-1}$	is, $B = PDP^{-1}$ too, so $\tilde{A} = B$.
johnston-linear-matrix-atlas- F04991	481	■ 0.913	$(\{(1,0)\})$	$(\{(1,0)\})$	tors $(\{(1,0)\})$ is a basis of the corresponding
johnston-linear-matrix-atlas- F04992	481	■ 1.000	P \widetilde{D} P^{-1}	$\widetilde{PDP^{-1}}$	k^n distinct k -th roots of the form $\widetilde{PDP^{-1}}$, where
johnston-linear-matrix-atlas- F04993	481	■ 1.000	$k^n k$	$k^n k$	there are <i>exactly</i> k^n k -th roots), notice that if
johnston-linear-matrix-atlas- F04994	481	■ 1.000	P=	$P =$	we will call λ , and we partition P as $P =$
johnston-linear-matrix-atlas- F04995	481	■ 1.000	$\left[\mathbf{v}_1\left \mathbf{v}_2\right P_2\right] P_2$	$[\mathbf{v}_1 \mathbf{v}_2 P_2]$	$[\mathbf{v}_1 \mathbf{v}_2 P_2]$. If we let
johnston-linear-matrix-atlas- F04996	481	■ 1.000	$\left\{c_1 \mathbf{v}_1+c_2 \mathbf{v}_2, d_1 \mathbf{v}_1+d_2 \mathbf{v}_2\right\}, d_1 \mathbf{v}_1+\mathbf{v}_2, d_2 \mathbf{v}_1+\mathbf{v}_2\right\}$	$\left\{c_1 \mathbf{v}_1+c_2 \mathbf{v}_2, d_1 \mathbf{v}_1+d_2 \mathbf{v}_2\right\}$	be such that $\left\{c_1 \mathbf{v}_1+c_2 \mathbf{v}_2, d_1 \mathbf{v}_1+d_2 \mathbf{v}_2\right\}$ is
johnston-linear-matrix-atlas- F04997	481	■ 1.000	A=Q D Q^{-1}	$A = QDQ^{-1}$	$A = QDQ^{-1}$ as well. If we let $\tilde{D}$ be a k -th root
johnston-linear-matrix-atlas- F04998	481	■ 1.000	$\widetilde{D}$	$\tilde{D}$	$A = QDQ^{-1}$ as well. If we let $\tilde{D}$ be a k -th root
johnston-linear-matrix-atlas- F04999	481	■ 1.000	Q \widetilde{D} Q^{-1}	$Q\tilde{D}Q^{-1}$	matrices of the form $Q\tilde{D}Q^{-1}$ (by varying the
johnston-linear-matrix-atlas- F05000	481	■ 1.000	c_1, c_2, d_1, d_2	c_1, c_2, d_1, d_2	coefficients c_1, c_2, d_1, d_2 in the definition of Q),
johnston-linear-matrix-atlas- F05001	481	■ 1.000	$A^2 \mathbf{v}=\lambda A \mathbf{v}=\lambda^2 \mathbf{v}$	$A^2 \mathbf{v} = \lambda A \mathbf{v} = \lambda^2 \mathbf{v}$	that $A^2 \mathbf{v} = \lambda A \mathbf{v} = \lambda^2 \mathbf{v}$.
johnston-linear-matrix-atlas- F05002	481	■ 1.000	$\lambda^0=1$	$\lambda^0 = 1$	$A^0 = I$ with eigenvalue $\lambda^0 = 1$.
johnston-linear-matrix-atlas- F05003	481	■ 1.000	$p(x)=c_k x^k+\cdots+c_1 x+c_0$	$p(x) = c_k x^k + \cdots + c_1 x + c_0$	$p(x) = c_k x^k + \cdots + c_1 x + c_0$ then
johnston-linear-matrix-atlas- F05004	481	■ 1.000	$x^r x^s=x^{r+s}$	$x^r x^s = x^{r+s}$	works entrywise and $x^r x^s = x^{r+s}$ for all scalars
johnston-linear-matrix-atlas- F05005	481	■ 1.000	$\left(x^r\right)^s=x^{r s}$	$\left(x^r\right)^s = x^{rs}$	works entrywise and $\left(x^r\right)^s = x^{rs}$ for all scalars
johnston-linear-matrix-atlas- F05006	481	■ 1.000	$\operatorname{det}\left(e^{A+B}\right)=e^{\operatorname{tr}(A+B)}$	$\det\left(e^{A+B}\right) = e^{\operatorname{tr}(A+B)}$	similarly $\det(e^{A+B}) = e^{\operatorname{tr}(A+B)}$. We thus conclude
johnston-linear-matrix-atlas- F05007	481	■ 1.000	$-a_0-a_1 \lambda_j-a_2 \lambda_j^2-\cdots-a_{n-1} \lambda_j^{n-1}-\lambda_j^n$	$-a_0 - a_1 \lambda_j - a_2 \lambda_j^2 - \cdots - a_{n-1} \lambda_j^{n-1} - \lambda_j^n$	where the final entry $-a_0 - a_1 \lambda_j - a_2 \lambda_j^2 - \cdots - a_{n-1} \lambda_j^{n-1}$ equals λ_j^n , since λ_j is a root
johnston-linear-matrix-atlas- F05008	481	■ 1.000	$\cdots-a_{n-1} \lambda_j^{n-1}$	$\cdots - a_{n-1} \lambda_j^{n-1}$	$\cdots - a_{n-1} \lambda_j^{n-1}$ equals λ_j^n , since λ_j is a root
johnston-linear-matrix-atlas- F05009	481	■ 1.000	λ_j^n	λ_j^n	$\cdots - a_{n-1} \lambda_j^{n-1}$ equals λ_j^n , since λ_j is a root

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johnston-linear-matrix-atlas-02-F05010	482	■ 1.000	$C \, \mathbf{v} = \lambda \mathbf{v}$	$C\mathbf{v} = \lambda\mathbf{v}$	value λ . Then $C\mathbf{v} = \lambda\mathbf{v}$ implies
johnston-linear-matrix-atlas-02-F05011	482	■ 1.000	$\operatorname{tr}(B) =$	$\operatorname{tr}(B) =$	(c) Not similar. For example, $\operatorname{tr}(A) = 6$ but $\operatorname{tr}(B) =$
johnston-linear-matrix-atlas-02-F05012	482	■ 1.000	$(\lambda - 2)(\lambda - 3)$	$(\lambda - 2)(\lambda - 3)$	istic polynomials $(\lambda - 2)(\lambda - 3)$. Explicitly,
johnston-linear-matrix-atlas-02-F05013	482	■ 1.000	$(\lambda - 1)(\lambda - 3)^2$	$(\lambda - 1)(\lambda - 3)^2$	istic polynomials $(\lambda - 1)(\lambda - 3)^2$. Explicitly,
johnston-linear-matrix-atlas-02-F05014	482	■ 1.000	$A^2 \, \mathbf{v} = 3 \, A \, \mathbf{v} = 9 \, \mathbf{v}$	$A^2\mathbf{v} = 3A\mathbf{v} = 9\mathbf{v}$	by A gives $A^2\mathbf{v} = 3A\mathbf{v} = 9\mathbf{v}$.
johnston-linear-matrix-atlas-02-F05015	482	■ 1.000	$p_A(\lambda) = p_B(\lambda) = \lambda^4$	$p_A(\lambda) = p_B(\lambda) = \lambda^4$	is $p_A(\lambda) = p_B(\lambda) = \lambda^4$.
johnston-linear-matrix-atlas-02-F05016	482	■ 0.728	$\operatorname{span}\{(1, 0, 0, 0), (0, 1, 0, 0)\}$	$\operatorname{span}\{(1, 0, 0, 0), (0, 1, 0, 0)\}$	$\operatorname{span}\{(1, 0, 0, 0), (0, 1, 0, 0)\}$. In particular, for
johnston-linear-matrix-atlas-02-F05017	482	■ 1.000	$v_2 = \lambda v_1$	$v_2 = \lambda v_1$	$v_2 = \lambda v_1$, the next entry tells us that $v_3 = \lambda v_2$,
johnston-linear-matrix-atlas-02-F05018	482	■ 1.000	$v_3 = \lambda v_2$	$v_3 = \lambda v_2$	$v_2 = \lambda v_1$, the next entry tells us that $v_3 = \lambda v_2$,
johnston-linear-matrix-atlas-02-F05019	482	■ 1.000	$v_n = \lambda v_{n-1}$	$v_n = \lambda v_{n-1}$	and so on up to $v_n = \lambda v_{n-1}$. It follows that
johnston-linear-matrix-atlas-02-F05020	482	■ 1.000	$v_j = \lambda^{j-1} v_1$	$v_j = \lambda^{j-1} v_1$	$v_j = \lambda^{j-1} v_1$ for $1 \leq j \leq n - 1$. Then the bot-
johnston-linear-matrix-atlas-02-F05021	482	■ 1.000	$1 \leq j \leq n - 1$	$1 \leq j \leq n - 1$	$v_j = \lambda^{j-1} v_1$ for $1 \leq j \leq n - 1$. Then the bot-
johnston-linear-matrix-atlas-02-F05022	482	■ 1.000	$v_n = \lambda^{n-1} v_1$	$v_n = \lambda^{n-1} v_1$	so $v_n = \lambda^{n-1} v_1$. We have thus shown
johnston-linear-matrix-atlas-02-F05023	482	■ 1.000	$v_1(1, \lambda, \lambda^2, \dots, \lambda^{n-1})$	$v_1(1, \lambda, \lambda^2, \dots, \lambda^{n-1})$	$v_1(1, \lambda, \lambda^2, \dots, \lambda^{n-1})$. This set of vectors is
johnston-linear-matrix-atlas-02-F05024	482	■ 1.000	$P \in \mathcal{M}_4$	$P \in \mathcal{M}_4$	(d) Following the hint, let $P \in \mathcal{M}_4$ and write the
johnston-linear-matrix-atlas-02-F05025	482	■ 1.000	$a + d = 0$	$a + d = 0$	that either $c = 0$ or $a + d = 0$. However, if $a + d = 0$
johnston-linear-matrix-atlas-02-F05026	482	■ 1.000	$b(0 + 0) = 1$	$b(0 + 0) = 1$	gives us a contradiction since it says $b(0 + 0) = 1$. It
johnston-linear-matrix-atlas-02-F05027	483	■ 0.979	$\operatorname{det}(A) = -3, \operatorname{det}(A_1) = -3, \operatorname{det}(A_2) = -3$	$\operatorname{det}(A) = -3, \operatorname{det}(A_1) = -3, \operatorname{det}(A_2) = -3$	(a) $\operatorname{det}(A) = -3, \operatorname{det}(A_1) = -3, \operatorname{det}(A_2) = -3$,
johnston-linear-matrix-atlas-02-F05028	483	■ 0.971	$\operatorname{det}(A) = 4, \operatorname{det}(A_1) = 2, \operatorname{det}(A_2) = 8$	$\operatorname{det}(A) = 4, \operatorname{det}(A_1) = 2, \operatorname{det}(A_2) = 8$	(c) $\operatorname{det}(A) = 4, \operatorname{det}(A_1) = 2, \operatorname{det}(A_2) = 8$,
johnston-linear-matrix-atlas-02-F05029	483	■ 1.000	$\operatorname{det}(A_3) = 6, (x, y, z) = (1/2, 2, 3/2)$	$\operatorname{det}(A_3) = 6, (x, y, z) = (1/2, 2, 3/2)$	$\operatorname{det}(A_3) = 6, (x, y, z) = (1/2, 2, 3/2)$.
johnston-linear-matrix-atlas-02-F05030	483	■ 0.998	$\operatorname{det}(A) = -2, \operatorname{det}(A_1) = 1, \operatorname{det}(A_2) = -2$	$\operatorname{det}(A) = -2, \operatorname{det}(A_1) = 1, \operatorname{det}(A_2) = -2$	(e) $\operatorname{det}(A) = -2, \operatorname{det}(A_1) = 1, \operatorname{det}(A_2) = -2$,
johnston-linear-matrix-atlas-02-F05031	483	■ 1.000	$\operatorname{det}(A_3) = -1, (x, y, z) = (-1/2, 1, 1/2)$	$\operatorname{det}(A_3) = -1, (x, y, z) = (-1/2, 1, 1/2)$	$\operatorname{det}(A_3) = -1, (x, y, z) = (-1/2, 1, 1/2)$.
johnston-linear-matrix-atlas-02-F05032	483	■ 0.999	$\operatorname{rank}(\operatorname{cof}(A)) = n$	$\operatorname{rank}(\operatorname{cof}(A)) = n$	then $\operatorname{rank}(\operatorname{cof}(A)) = n$ by part (a) of this exer-
johnston-linear-matrix-atlas-02-F05033	483	■ 0.741	$\lVert \mathbf{l} \rVert = \lVert \mathbf{l} \rVert$	(not rendered)	(g) True. This follows from the fact that $\mathbf{l} \circ \mathbf{l} = \mathbf{l}$.
johnston-linear-matrix-atlas-02-F05034	483	■ 1.000	$n = 5$	$n = 5$	$n = 5$, there are $5! = 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1 = 120$ per-
johnston-linear-matrix-atlas-02-F05035	483	■ 1.000	$5! = 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1 = 120$	$5! = 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1 = 120$	$n = 5$, there are $5! = 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1 = 120$ per-
johnston-linear-matrix-atlas-02-F05036	483	■ 1.000	$\operatorname{cof}(A^2) = \operatorname{cof}(AA) = \operatorname{cof}(A) \operatorname{cof}(A) = (\operatorname{cof}(A))^2$	$\operatorname{cof}(A^2) = \operatorname{cof}(AA) = \operatorname{cof}(A) \operatorname{cof}(A) = (\operatorname{cof}(A))^2$	that $\operatorname{cof}(A^2) = \operatorname{cof}(AA) = \operatorname{cof}(A) \operatorname{cof}(A) = (\operatorname{cof}(A))^2$.
johnston-linear-matrix-atlas-02-F05037	483	■ 1.000	$\operatorname{cof}(A^{-1}) =$	$\operatorname{cof}(A^{-1}) =$	If $k = -1$ then we notice that $\operatorname{cof}(A^{-1}) =$

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johnston-linear-matrix-atgsoa-F05038	483	1.000	$A^T / \operatorname{det}(A) = (\operatorname{cof}(A))^{-1}$	$A^T / \det(A) = (\operatorname{cof}(A))^{-1}$	$A^T / \det(A) = (\operatorname{cof}(A))^{-1}$. For $k < -1$ we just re-
johnston-linear-matrix-atgsoa-F05039	483	1.000	$\operatorname{cof}(A) =$	$\operatorname{cof}(A) =$	3.A.10 We use Theorem 3.A.1 to see that $\operatorname{cof}(A) =$
johnston-linear-matrix-atgsoa-F05040	483	1.000	$\operatorname{det}(A) \operatorname{left}(A^T \operatorname{right})^{-1}$	$\det(A) (A^T)^{-1}$	$\det(A) (A^T)^{-1}$. Taking the determinant of both sides
johnston-linear-matrix-atgsoa-F05041	483	1.000	$\tau = (21345 \cdots n)$	$\tau = (21345 \cdots n)$	(b) Let $\tau = (2\ 1\ 3\ 4\ 5 \cdots n)$ be the permutation
johnston-linear-matrix-atgsoa-F05042	483	1.000	$\operatorname{sgn}(\tau) = -1$	$\operatorname{sgn}(\tau) = -1$	alone. Then $\operatorname{sgn}(\tau) = -1$ and as σ ranges over
johnston-linear-matrix-atgsoa-F05043	483	0.993	$\operatorname{rank}(A) \leq n < m$	$\operatorname{rank}(A) \leq n < m$	3.A.14 If $m > n$ then $\operatorname{rank}(A) \leq n < m$ so $\operatorname{rank}(AB) \leq n < m$
johnston-linear-matrix-atgsoa-F05044	483	0.993	$\operatorname{rank}(A\ B) \leq n < m$	$\operatorname{rank}(AB) \leq n < m$	3.A.14 If $m > n$ then $\operatorname{rank}(A) \leq n < m$ so $\operatorname{rank}(AB) \leq n < m$
johnston-linear-matrix-atgsoa-F05045	483	1.000	$\operatorname{det}(A\ B) = 0$	$\det(AB) = 0$	so it is thus not invertible and $\det(AB) = 0$.
johnston-linear-matrix-atgsoa-F05046	483	0.999	$\operatorname{per}(A) = 2, \operatorname{per}(B) = 10$	$\operatorname{per}(A) = 2, \operatorname{per}(B) = 10$	then $\operatorname{per}(A) = 2, \operatorname{per}(B) = 10$, but $\operatorname{per}(AB) = 48$.
johnston-linear-matrix-atgsoa-F05047	483	0.999	$\operatorname{per}(A\ B) = 48$	$\operatorname{per}(AB) = 48$	then $\operatorname{per}(A) = 2, \operatorname{per}(B) = 10$, but $\operatorname{per}(AB) = 48$.
johnston-linear-matrix-atgsoa-F05048	484	0.881	$1 \quad$	1	3.B.1 (a) $\lambda \approx 5.37, \mathbf{v} \approx (0.42, 0.91)$
johnston-linear-matrix-atgsoa-F05049	484	0.881	$\quad \lambda \approx 5.37, \mathbf{v} \approx (0.42, 0.91)$	$\lambda \approx 5.37, \mathbf{v} \approx (0.42, 0.91)$	3.B.1 (a) $\lambda \approx 5.37, \mathbf{v} \approx (0.42, 0.91)$
johnston-linear-matrix-atgsoa-F05050	484	1.000	$\lambda \approx -3.33, \mathbf{v} \approx (0.73, -0.42, -0.55)$	$\lambda \approx -3.33, \mathbf{v} \approx (0.73, -0.42, -0.55)$	(c) $\lambda \approx -3.33, \mathbf{v} \approx (0.73, -0.42, -0.55)$
johnston-linear-matrix-atgsoa-F05051	484	1.000	$\lambda \approx 7.92, \mathbf{v} \approx (0.73, 0.29, 0.57, 0.24)$	$\lambda \approx 7.92, \mathbf{v} \approx (0.73, 0.29, 0.57, 0.24)$	(e) $\lambda \approx 7.92, \mathbf{v} \approx (0.73, 0.29, 0.57, 0.24)$
johnston-linear-matrix-atgsoa-F05052	484	1.000	$2 \quad$	2	3.B.2 (a) $\lambda_1 \approx 5.37, \mathbf{v}_1 \approx (0.42, 0.91)$
johnston-linear-matrix-atgsoa-F05053	484	1.000	$\quad \lambda_1 \approx 5.37, \mathbf{v}_1 \approx (0.42, 0.91)$	$\lambda_1 \approx 5.37, \mathbf{v}_1 \approx (0.42, 0.91)$	3.B.2 (a) $\lambda_1 \approx 5.37, \mathbf{v}_1 \approx (0.42, 0.91)$
johnston-linear-matrix-atgsoa-F05054	484	1.000	$\lambda_2 \approx -0.37, \mathbf{v}_2 \approx (0.82, -0.57)$	$\lambda_2 \approx -0.37, \mathbf{v}_2 \approx (0.82, -0.57)$	$\lambda_2 \approx -0.37, \mathbf{v}_2 \approx (0.82, -0.57)$
johnston-linear-matrix-atgsoa-F05055	484	1.000	$\lambda_1 \approx 16.12, \mathbf{v}_1 \approx (0.23, 0.53, 0.82)$	$\lambda_1 \approx 16.12, \mathbf{v}_1 \approx (0.23, 0.53, 0.82)$	(c) $\lambda_1 \approx 16.12, \mathbf{v}_1 \approx (0.23, 0.53, 0.82)$
johnston-linear-matrix-atgsoa-F05056	484	1.000	$\lambda_2 \approx -1.12, \mathbf{v}_2 \approx (0.79, 0.09, -0.61)$	$\lambda_2 \approx -1.12, \mathbf{v}_2 \approx (0.79, 0.09, -0.61)$	$\lambda_2 \approx -1.12, \mathbf{v}_2 \approx (0.79, 0.09, -0.61)$
johnston-linear-matrix-atgsoa-F05057	484	1.000	$\lambda_3 \approx 0.00, \mathbf{v}_2 \approx (0.41, -0.82, 0.41)$	$\lambda_3 \approx 0.00, \mathbf{v}_2 \approx (0.41, -0.82, 0.41)$	$\lambda_3 \approx 0.00, \mathbf{v}_2 \approx (0.41, -0.82, 0.41)$
johnston-linear-matrix-atgsoa-F05058	484	0.520	$3 \quad$	3	3.B.3 (a) $\lambda \approx 11.58,$
johnston-linear-matrix-atgsoa-F05059	484	0.520	$\lambda \approx 11.58$	$\lambda \approx 11.58$	3.B.3 (a) $\lambda \approx 11.58,$
johnston-linear-matrix-atgsoa-F05060	484	1.000	$\mathbf{v} \approx (0.15, 0.25, 0.01, 0.54, -0.27, 0.46, 0.58)$	$\mathbf{v} \approx (0.15, 0.25, 0.01, 0.54, -0.27, 0.46, 0.58)$	$\mathbf{v} \approx (0.15, 0.25, 0.01, 0.54, -0.27, 0.46, 0.58)$
johnston-linear-matrix-atgsoa-F05061	484	1.000	$A \mathbf{v}_k \approx \lambda_1 \mathbf{v}_k$	$A \mathbf{v}_k \approx \lambda_1 \mathbf{v}_k$	ing to the dominant eigenvalue of A, so $A \mathbf{v}_k \approx \lambda_1 \mathbf{v}_k$
johnston-linear-matrix-atgsoa-F05062	484	1.000	$\operatorname{left} \ A \mathbf{v}_k\ \operatorname{right} \approx \operatorname{left} \lambda_1 \mathbf{v}_k \operatorname{right} = \lambda_1$	$\ A \mathbf{v}_k\ \approx \ \lambda_1 \mathbf{v}_k\ = \lambda_1$	when k is large, so $\ A \mathbf{v}_k\ \approx \ \lambda_1 \mathbf{v}_k\ = \lambda_1$ when k is
johnston-linear-matrix-atgsoa-F05063	484	0.994	$k - 2 \cdot 4^2 -$	$k - 2 \cdot 4^2 -$	$k - 2 \cdot 4^2 -$
johnston-linear-matrix-atgsoa-F05064	484	1.000	$k = 4: A^4 = \begin{bmatrix} 8 & 8 \\ 8 & 8 \end{bmatrix}$	$k = 4: A^4 = \begin{bmatrix} 8 & 8 \\ 8 & 8 \end{bmatrix}$	(c) $k = 4: A^4 = \begin{bmatrix} 8 & 8 \\ 8 & 8 \end{bmatrix}$

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johnston-linear-matrix-atlas-04-F05065	484	0.993	$k=4: A^{\{4\}}=\left[\begin{array}{lll}5 & 4 & 1 \\ 1 & 3 & 2\end{array}\right]$	$k = 4: A^4 = \begin{bmatrix} 5 & 4 & 1 \\ 4 & 6 & 3 \\ 1 & 3 & 2 \end{bmatrix}$	(e) $k = 4: A^4 = \begin{bmatrix} 5 & 4 & 1 \\ 4 & 6 & 3 \\ 1 & 3 & 2 \end{bmatrix}$
johnston-linear-matrix-atlas-04-F05066	484	0.445	$\lambda \approx 1.62, \mathbf{v} \approx (0.53, 0.85)$	$\lambda \approx 1.62, \mathbf{v} \approx (0.53, 0.85)$	3.B.10 (a) $\lambda \approx 1.62, \mathbf{v} \approx (0.53, 0.85)$
johnston-linear-matrix-atlas-04-F05067	484	1.000	$\lambda \approx 2.00i, \mathbf{v} \approx (0.71, 0.71)$	$\lambda \approx 2.00i, \mathbf{v} \approx (0.71, 0.71)$	(c) $\lambda \approx 2.00i, \mathbf{v} \approx (0.71, 0.71)$
johnston-linear-matrix-atlas-04-F05068	484	1.000	$\lambda \approx 1.80, \mathbf{v} \approx (0.59, 0.74, 0.33)$	$\lambda \approx 1.80, \mathbf{v} \approx (0.59, 0.74, 0.33)$	(e) $\lambda \approx 1.80, \mathbf{v} \approx (0.59, 0.74, 0.33)$
johnston-linear-matrix-atlas-04-F05069	484	1.000	$\left[R^{\pi/2}\right]$	$\left[R^{\pi/2}\right]$	3.C.1 (a) We recognize this as the rotation matrix $\left[R^{\pi/2}\right]$,
johnston-linear-matrix-atlas-04-F05070	484	1.000	$Q=I, r=1$	$Q = I, r = 1$	just from choosing $Q = I, r = 1$, and $\theta = \pi/2$.
johnston-linear-matrix-atlas-04-F05071	484	1.000	$1 \pm i = \sqrt{2} e^{i \pi/4}$	$1 \pm i = \sqrt{2} e^{\pm i \pi/4}$	(c) This matrix has eigenvalues $1 \pm i = \sqrt{2} e^{\pm i \pi/4}$
johnston-linear-matrix-atlas-04-F05072	484	0.999	$(1, -1 \pm i)$	$(1, -1 \pm i)$	with corresponding eigenvectors $(1, -1 \pm i)$, so
johnston-linear-matrix-atlas-04-F05073	484	1.000	$r = \sqrt{2}, \theta = \pi/4$	$r = \sqrt{2}, \theta = \pi/4$	we can choose $r = \sqrt{2}, \theta = \pi/4$, and
johnston-linear-matrix-atlas-04-F05074	484	0.922	$B=\operatorname{diag}\left(1,\left[R^{\pi/2}\right]\right), Q=\left[\begin{array}{ccc}1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & -1\end{array}\right]$	$B = \operatorname{diag}\left(1,\left[R^{\pi/2}\right]\right), Q = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & -1 \end{bmatrix}$	3.C.2 (a) $B = \operatorname{diag}\left(1,\left[R^{\pi/2}\right]\right), Q = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & -1 \end{bmatrix}$.
johnston-linear-matrix-atlas-04-F05075	484	1.000	$B=\operatorname{diag}\left(1,2,\sqrt{2}\left[R^{\pi/4}\right]\right)$	$B = \operatorname{diag}\left(1,2,\sqrt{2}\left[R^{\pi/4}\right]\right)$	(c) $B = \operatorname{diag}\left(1,2,\sqrt{2}\left[R^{\pi/4}\right]\right)$,
johnston-linear-matrix-atlas-04-F05076	484	0.998	$B=\operatorname{diag}\left(2\sqrt{2}\left[R^{\pi/4}\right],2\sqrt{2}\left[R^{\pi/4}\right],4\right)$	$B = \operatorname{diag}\left(2\sqrt{2}\left[R^{\pi/4}\right],2\sqrt{2}\left[R^{\pi/4}\right],4\right)$	3.C.3 (a) $B = \operatorname{diag}\left(2\sqrt{2}\left[R^{\pi/4}\right],2\sqrt{2}\left[R^{\pi/4}\right],4\right)$,
johnston-linear-matrix-atlas-04-F05077	485	0.391	$2^k(\cos(k\pi/6) - \sin(k\pi/6))$	$2^k(\cos(k\pi/6) - \sin(k\pi/6))$	$2^k(\cos(k\pi/6) - \sin(k\pi/6))$, top-right
johnston-linear-matrix-atlas-04-F05078	485	1.000	$2^k \sin(k\pi/6)$	$2^k \sin(k\pi/6)$	entry $2^k \sin(k\pi/6)$, bottom-left entry
johnston-linear-matrix-atlas-04-F05079	485	1.000	$-2^{k+1} \sin(k\pi/6)$	$-2^{k+1} \sin(k\pi/6)$	$-2^{k+1} \sin(k\pi/6)$, and bottom-right entry
johnston-linear-matrix-atlas-04-F05080	485	1.000	$2^k(\cos(k\pi/6) + \sin(k\pi/6))$	$2^k(\cos(k\pi/6) + \sin(k\pi/6))$	$2^k(\cos(k\pi/6) + \sin(k\pi/6))$.
johnston-linear-matrix-atlas-04-F05081	485	1.000	$\lambda^2 - \lambda - 6$	$\lambda^2 - \lambda - 6$	currence relation is $\lambda^2 - \lambda - 6$, which has roots
johnston-linear-matrix-atlas-04-F05082	485	0.998	$\lambda_0=3$	$\lambda_0 = 3$	$\lambda_0 = 3$ and $\lambda_1 = -2$, each with multiplicity 1.
johnston-linear-matrix-atlas-04-F05083	485	0.998	$\lambda_1=-2$	$\lambda_1 = -2$	$\lambda_0 = 3$ and $\lambda_1 = -2$, each with multiplicity 1.
johnston-linear-matrix-atlas-04-F05084	485	1.000	$3 \pm \sqrt{5}$	$3 \pm \sqrt{5}$	(c) $3 \pm \sqrt{5}$, each with multiplicity 1.
johnston-linear-matrix-atlas-04-F05085	485	0.999	$1 \pm \sqrt{3}$	$1 \pm \sqrt{3}$	(i) 3 and $1 \pm \sqrt{3}$, each with multiplicity 1.
johnston-linear-matrix-atlas-04-F05086	485	1.000	$\left[\begin{array}{ll}0 & 1 \\ 6 & 1\end{array}\right]$	$\begin{bmatrix} 0 & 1 \\ 6 & 1 \end{bmatrix}$	(a) $\begin{bmatrix} 0 & 1 \\ 6 & 1 \end{bmatrix}$
johnston-linear-matrix-atlas-04-F05087	485	1.000	$\left[\begin{array}{cc}0 & 1 \\ -4 & 6\end{array}\right]$	$\begin{bmatrix} 0 & 1 \\ -4 & 6 \end{bmatrix}$	(c) $\begin{bmatrix} 0 & 1 \\ -4 & 6 \end{bmatrix}$
johnston-linear-matrix-atlas-04-F05088	485	1.000	$\left[\begin{array}{ccc}0 & 1 & 0 \\ 0 & 0 & 1 \\ 6 & -11 & 6\end{array}\right]$	$\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 6 & -11 & 6 \end{bmatrix}$	(e) $\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 6 & -11 & 6 \end{bmatrix}$

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johnston-linear-matrix-atlas- F05089	485	■ 1.000	$\left[\begin{array}{ccc}0 & 1 & 0 \\ 0 & 0 & 1 \\ 8 & -12 & 6\end{array}\right]$	$\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 8 & -12 & 6 \end{bmatrix}$	(g) $\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 8 & -12 & 6 \end{bmatrix}$
johnston-linear-matrix-atlas- F05090	485	■ 1.000	$\left[\begin{array}{ccc}0 & 1 & 0 \\ 0 & 0 & 1 \\ -6 & -4 & 5\end{array}\right]$	$\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -6 & -4 & 5 \end{bmatrix}$	(i) $\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -6 & -4 & 5 \end{bmatrix}$
johnston-linear-matrix-atlas- F05091	485	■ 1.000	$\lambda^2 z - \lambda - 6$	$\lambda^2 z - \lambda - 6$	urrence relation is $\lambda^2 z - \lambda - 6$, which has roots
johnston-linear-matrix-atlas- F05092	485	■ 1.000	$c_{\{0\}}$	c_0	scalars c_0 and c_1 such that $x_n = c_0 3^n + c_1 (-2)^n$
johnston-linear-matrix-atlas- F05093	485	■ 1.000	$x_{\{n\}} = c_{\{0\}} 3^n + c_{\{1\}} (-2)^n$	$x_n = c_0 3^n + c_1 (-2)^n$	scalars c_0 and c_1 such that $x_n = c_0 3^n + c_1 (-2)^n$
johnston-linear-matrix-atlas- F05094	485	■ 1.000	$x_{\{0\}} = 4$	$x_0 = 4$	for all n . Using the fact that $x_0 = 4$ and $x_1 = -3$
johnston-linear-matrix-atlas- F05095	485	■ 1.000	$x_{\{1\}} = -3$	$x_1 = -3$	for all n . Using the fact that $x_0 = 4$ and $x_1 = -3$
johnston-linear-matrix-atlas- F05096	485	■ 1.000	$x_{\{n\}} = 3^n +$	$x_n = 3^n +$	shows that $c_0 = 1$ and $c_1 = 3$, so $x_n = 3^n +$
johnston-linear-matrix-atlas- F05097	485	■ 1.000	$3(-2)^n$	$3(-2)^n$	$3(-2)^n$.
johnston-linear-matrix-atlas- F05098	485	■ 1.000	$x_{\{n\}} = i^n + (-i)^n$	$x_n = i^n + (-i)^n$	(c) $x_n = i^n + (-i)^n$. Alternatively, a formula that
johnston-linear-matrix-atlas- F05099	485	■ 1.000	$x_{\{n\}} =$	$x_n =$	does not involve complex numbers is $x_n =$
johnston-linear-matrix-atlas- F05100	485	■ 1.000	$2 \cos(\pi n / 2)$	$2 \cos(\pi n / 2)$	$2 \cos(\pi n / 2)$.
johnston-linear-matrix-atlas- F05101	485	■ 1.000	$x_{\{n\}} = 3^n - 1$	$x_n = 3^n - 1$	(e) $x_n = 3^n - 1$.
johnston-linear-matrix-atlas- F05102	485	■ 1.000	$x_{\{n\}} = (2n - 3) + 2^n$	$x_n = (2n - 3) + 2^n$	(g) $x_n = (2n - 3) + 2^n$.
johnston-linear-matrix-atlas- F05103	485	■ 1.000	$\lambda^2 + 1$	$\lambda^2 + 1$	mial $\lambda^2 + 1$, which has roots $\pm i$.
johnston-linear-matrix-atlas- F05104	485	■ 1.000	$\pm i$	$\pm i$	mial $\lambda^2 + 1$, which has roots $\pm i$.
johnston-linear-matrix-atlas- F05105	485	■ 1.000	$n^2 - n^3$	$n^2 - n^3$	$n^2 - n^3$ has degree $4 - 1 = 3$). One such lin-
johnston-linear-matrix-atlas- F05106	485	■ 1.000	$4 - 1 = 3$	$4 - 1 = 3$	$n^2 - n^3$ has degree $4 - 1 = 3$). One such lin-
johnston-linear-matrix-atlas- F05107	485	■ 1.000	$x_{\{n\}} = 4x_{\{n-1\}} - 6x_{\{n-2\}} +$	$x_n = 4x_{n-1} - 6x_{n-2} +$	ear recurrence relation is $x_n = 4x_{n-1} - 6x_{n-2} +$
johnston-linear-matrix-atlas- F05108	485	■ 1.000	$4x_{\{n-3\}} - x_{\{n-4\}}$	$4x_{n-3} - x_{n-4}$	$4x_{n-3} - x_{n-4}$, which has characteristic polyno-
johnston-linear-matrix-atlas- F05109	485	■ 1.000	$(\lambda - 1)^4$	$(\lambda - 1)^4$	mial $(\lambda - 1)^4$.
johnston-linear-matrix-atlas- F05110	485	■ 0.998	$p(\lambda) = (\lambda - 2)(\lambda - 3) = \lambda^2 - 5\lambda + 6$	$p(\lambda) = (\lambda - 2)(\lambda - 3) = \lambda^2 - 5\lambda + 6$	it is $p(\lambda) = (\lambda - 2)(\lambda - 3) = \lambda^2 - 5\lambda + 6$, so
johnston-linear-matrix-atlas- F05111	485	■ 1.000	$x_{\{n\}} = 9x_{\{n-1\}} - 26x_{\{n-2\}} + 24x_{\{n-3\}}$	$x_n = 9x_{n-1} - 26x_{n-2} + 24x_{n-3}$	(c) $x_n = 9x_{n-1} - 26x_{n-2} + 24x_{n-3}$
johnston-linear-matrix-atlas- F05112	485	■ 1.000	$x_{\{n\}} = 9x_{\{n-1\}} - 30x_{\{n-2\}} + 44x_{\{n-3\}} - 24x_{\{n-4\}}$	$x_n = 9x_{n-1} - 30x_{n-2} + 44x_{n-3} - 24x_{n-4}$	(e) $x_n = 9x_{n-1} - 30x_{n-2} + 44x_{n-3} - 24x_{n-4}$
johnston-linear-matrix-atlas- F05113	485	■ 1.000	$x_{\{n\}} = 3x_{\{n-1\}} - 3x_{\{n-2\}} + x_{\{n-3\}}$	$x_n = 3x_{n-1} - 3x_{n-2} + x_{n-3}$	(g) $x_n = 3x_{n-1} - 3x_{n-2} + x_{n-3}$
johnston-linear-matrix-atlas- F05114	485	■ 1.000	$x_{\{n\}} = 8x_{\{n-1\}} - 26x_{\{n-2\}} + 48x_{\{n-3\}} - 45x_{\{n-4\}}$	$x_n = 8x_{n-1} - 26x_{n-2} + 48x_{n-3} - 45x_{n-4}$	(i) $x_n = 8x_{n-1} - 26x_{n-2} + 48x_{n-3} - 45x_{n-4}$
johnston-linear-matrix-atlas- F05115	485	■ 1.000	$\pm i =$	$\pm i =$	rence relation is $\lambda^2 + 1$, which has roots $\pm i =$

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johnston-linear-matrix-attrib F05116	485	1.000	$e^{\pm i \pi / 2}$	$e^{\pm i \pi / 2}$	$e^{\pm i \pi / 2}$. After solving for the coefficients in the
johnston-linear-matrix-attrib F05117	485	1.000	$(\pm i)^n = \left(e^{\pm i \pi / 2} \right)^n =$	$(\pm i)^n = \left(e^{\pm i \pi / 2} \right)^n =$	tuting in the polar forms $(\pm i)^n = (e^{\pm i \pi / 2})^n =$
johnston-linear-matrix-attrib F05118	485	1.000	$e^{\pm i n \pi / 2}$	$e^{\pm i n \pi / 2}$	$e^{\pm i n \pi / 2}$, we see that
johnston-linear-matrix-attrib F05119	485	1.000	$x_n = 1 + 2^n \cos(n \pi / 2)$	$x_n = 1 + 2^{n+1} \cos(n \pi / 2)$	(c) $x_n = 1 + 2^{n+1} \cos(n \pi / 2)$.
johnston-linear-matrix-attrib F05120	485	1.000	$x_0 = x_1 = 0$	$x_0 = x_1 = 0$	trivial since we were told that $x_0 = x_1 = 0$. For the in-
johnston-linear-matrix-attrib F05121	485	1.000	$x_{n-1} = x_{n-2}$	$x_{n-1} = x_{n-2}$	ductive step, suppose $x_{n-1} = x_{n-2}$. Then using the re-
johnston-linear-matrix-attrib F05122	485	1.000	$x_{n+1} = x_n - 4x_{n-1} + 4x_{n-2} =$	$x_{n+1} = x_n - 4x_{n-1} + 4x_{n-2} =$	currence relation gives $x_{n+1} = x_n - 4x_{n-1} + 4x_{n-2} =$
johnston-linear-matrix-attrib F05123	485	1.000	$x_n - 4x_{n-1} + 4x_{n-1} = x_n$	$x_n - 4x_{n-1} + 4x_{n-1} = x_n$	$x_n - 4x_{n-1} + 4x_{n-1} = x_n$, which completes the induc-
johnston-linear-matrix-attrib F05124	492	1.000	$O_{m,n}$ $m \times n$	$O_{m,n}$ $m \times n$	$O_{m,n}$ $m \times n$ zero matrix
johnston-linear-matrix-attrib F05125	492	1.000	O_n $n \times n$	O_n $n \times n$	O_n $n \times n$ zero matrix
johnston-linear-matrix-attrib F05126	492	1.000	$[A]_{i,j}$ (i,j)	$[A]_{i,j}$ (i,j)	$[A]_{i,j}$ (i,j) -entry of the matrix A
johnston-linear-matrix-attrib F05127	492	0.788	$[T]$	$[T]$	$[T]$ standard matrix of T (with respect to the standard basis)
johnston-linear-matrix-attrib F05128	492	1.000	$[T]_B$	$[T]_B$	$[T]_B$ standard matrix of T with respect to basis B
johnston-linear-matrix-attrib F05129	492	0.623	$\ \cdot\ $	$\ \cdot\ $	$\ \cdot\ $ the length of a vector
johnston-linear-matrix-attrib F05130	492	1.000	$\bar{A}$	$\bar{A}$	$\bar{A}$ entrywise complex conjugate of the matrix or number A
johnston-linear-matrix-attrib F05131	492	0.957	$\operatorname{det}(A)$ $\operatorname{determinant}$	$\det(A)$ $\operatorname{determinant}$	$\det(A)$ $\operatorname{determinant}$ of the matrix A
johnston-linear-matrix-attrib F05132	492	0.629	$\mathbf{e}_j$ j	$\mathbf{e}_j$ j	$\mathbf{e}_j$ j -th standard basis vector (1 in its j -th entry, 0 elsewhere)
johnston-linear-matrix-attrib F05133	492	1.000	I_n $n \times n$	I_n $n \times n$	I_n $n \times n$ identity matrix
johnston-linear-matrix-attrib F05134	492	1.000	$\mathcal{M}_{m,n}$ $m \times n$	$\mathcal{M}_{m,n}$ $m \times n$	$\mathcal{M}_{m,n}$ $m \times n$ matrices
johnston-linear-matrix-attrib F05135	492	1.000	$\mathcal{M}_n$ $n \times n$	$\mathcal{M}_n$ $n \times n$	$\mathcal{M}_n$ $n \times n$ matrices
johnston-linear-matrix-attrib F05136	492	1.000	$\mathcal{M}_n(\mathbb{C})$ $n \times n$	$\mathcal{M}_n(\mathbb{C})$ $n \times n$	$\mathcal{M}_n(\mathbb{C})$ $n \times n$ matrices with complex entries
johnston-linear-matrix-attrib F05137	492	1.000	$\mathcal{M}_n(\mathbb{R})$ $n \times n$	$\mathcal{M}_n(\mathbb{R})$ $n \times n$	$\mathcal{M}_n(\mathbb{R})$ $n \times n$ matrices with real entries
johnston-linear-matrix-attrib F05138	492	1.000	$\mathbb{R}^n$	$\mathbb{R}^n$	$\mathbb{R}^n$ vectors/tuples with n real entries
johnston-linear-matrix-attrib F05139	492	1.000	$\operatorname{tr}(A)$	$\operatorname{tr}(A)$	$\operatorname{tr}(A)$ trace of the matrix A
johnston-linear-matrix-attrib F05140	492	0.644	$\mathbb{Z}_p$	$\mathbb{Z}_p$	field of integers $\{0, 1, \dots, p-1\}$ under arithmetic mod p
johnston-linear-matrix-attrib F05141	492	0.644	$\{0, 1, \dots, p-1\}$	$\{0, 1, \dots, p-1\}$	field of integers $\{0, 1, \dots, p-1\}$ under arithmetic mod p
johnston-linear-matrix-attrib F05142	492	0.644	$\bmod p$	$\bmod p$	field of integers $\{0, 1, \dots, p-1\}$ under arithmetic mod p
johnston-linear-matrix-attrib F05143	492	1.000	$\mathbb{Z}_p^n$	$\mathbb{Z}_p^n$	$\mathbb{Z}_p^n$ vectors/tuples with n entries from $\mathbb{Z}_p$